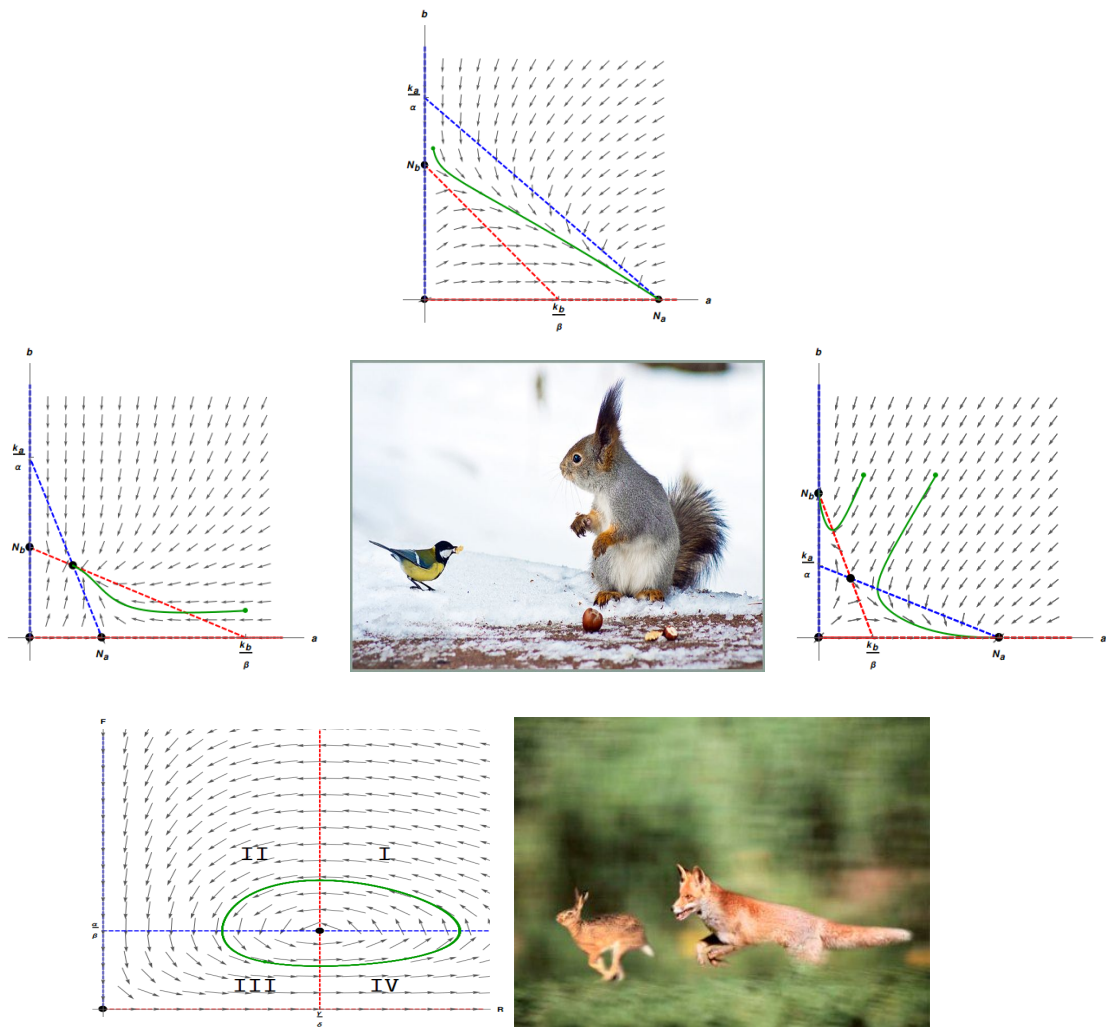


Calculus and Modeling for the Biological, Health, and Earth Sciences

Brent Young



Version 4.1

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Prerequisites

This text was developed from lecture notes for a one semester course in applied mathematics focusing on the use of mathematics to describe and predict the behavior of systems that occur in nature – especially systems arising in the biological sciences. As such, this text is intended for an audience that has completed a traditional one semester course in differential and integral calculus. In particular, the text assumes familiarity with basic function concepts, properties of the usual transcendental functions (exponential, logarithmic, and trigonometric functions), as well as standard results about differentiation and integration (up to and including integration by substitution).

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Chapter 0

Introduction

The unifying topic in this text is the concept of **dynamical system**. A dynamical system is any mathematical model which attempts to predict the changes in some object (or objects) over time. The object under study is assumed to have some number of **states** that it can be in at any given time (typically labeled by some set of numbers), and the job of the dynamical system is to predict the future state of the object given its past history.

As an example, suppose we are interested in modeling the population size of some group of animals. The first thing to do is to give a name to the quantity we want to model. So, we let $P(t)$ represent the size of the population at time t . In the real world, the population of a group of animals is a *natural number*, $\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$, while the time t can be any real number (the real numbers are given the symbol $\mathbb{R}$). Quite often, we are interested in the size of the population after some initial time which we typically (and arbitrarily) label as $t = 0$. In such a case, the time is a non-negative real number (the set of which is represented by $\mathbb{R}_{\geq 0}$), and our dynamical system is modeled by a function of the form

$$P : \mathbb{R}_{\geq 0} \rightarrow \mathbb{N}.$$

In words, it is a function taking a non-negative real number (the current time) as an argument and returning a natural number (the size of the population). Quite often, we make the simplifying assumption that the population size can also be any positive real number; this allows us to think of our population model as a function

$$P : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}.$$

While it would be silly to say there are 410,605.73 animals in a group, this convenient fiction will allow us to apply some powerful mathematical tools to the problem.

As another example, suppose we want to model the speed of a skydiver falling from an airplane. Let $v(t)$ be the function which returns the speed of the skydiver t seconds after leaving the plane. Since speeds are non-negative real numbers, our function in this case is clearly

$$v : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}.$$

In other words, our speed function takes a non-negative real number (the current time) as an argument and returns a non-negative real number (the speed of the skydiver).

Notice that in both cases we need two types of information to make any progress. First, we need to know how the current state and the past history of the system determines its state for future times. We will have much more to say about this later. We also need to know the actual state of the system for some amount of time in the past. In many cases, the future state of the system is completely determined once we know its state at one arbitrary time (typically the state at time $t = 0$). This type of information is referred to as **initial data** for the dynamical system.

So far, all we have done is give a name to the quantity we want to study. Quite often, we only have indirect information about how the system will evolve with time. For instance, we may know that a certain population doubles in size every 5 hours (think bacteria). In the case of the skydiver, we know that gravity near the surface of the earth causes an object to *accelerate* at a nearly constant rate; in other words, we have information about the derivative $v'(t)$ rather than having direct information about $v(t)$. Our goal will be to develop tools that allow us to predict the future state of a system given these indirect sorts of information.

Before diving in, we need to discuss some basic terminology. Dynamical systems can be divided into categories depending on how future states are influenced by past states and on how we view time. If the future state of the system is completely determined by its past history, we say that the dynamical system is **deterministic**. If the past history only tells us the probability that the system will be in some future state, we say that the system is **stochastic** (think of stochastic as a fancy word for random). Our previous examples, the population which doubles every 5 hours and the skydiver falling from an airplane, were both deterministic. An example of a

stochastic dynamical system would be the outcome of throwing a fair, six-sided die. We will focus on deterministic dynamical systems in the early part of the book. After a discussion of the basics of probability, the text ends with a discussion of some basic stochastic systems. As a note, we usually refer to deterministic dynamical systems simply as dynamical systems (only adding the word “deterministic” for emphasis). If we wish to talk about systems with some degree of randomness, we always add the descriptor “stochastic.”

The next natural division of dynamical systems depends on how we incorporate time into the system. In many systems, time can take on any value from the non-negative real numbers. In such a case, we say that the system is a **continuous-time dynamical system**. The model of a skydiver falling from an airplane is clearly a continuous dynamical system as the time, t , since the diver left the plane can be any real number in the range $0 \leq t < \infty$. On the other hand, in many systems it is natural to treat time as coming in discrete, integer amounts, i.e. $t = 0, 1, 2, 3, 4, \dots$. For the population that doubles every 5 hours, it would be natural to mark time in 5 hour periods. So, $t = 1$ would indicate the end of the first 5 hour period, $t = 2$ would indicate the end of the second 5 hour period (so 10 hours since the start of the experiment), etc. If time is treated as an integer quantity, the system is said to be a **discrete-time dynamical system**.

These basic divisions give us four types of dynamical systems:

- 1.) **Discrete-Time (Deterministic) Dynamical Systems:** dynamical systems which are deterministic and for which time comes in integer amounts.
- 2.) **Continuous-Time (Deterministic) Dynamical Systems:** dynamical systems which are deterministic and for which time can be any positive real number.
- 3.) **Discrete-Time Stochastic Dynamical Systems:** dynamical systems which have some degree of randomness and for which time comes in integer amounts.
- 4.) **Continuous-Time Stochastic Dynamical Systems:** dynamical systems which have some degree of randomness and for which time can be any positive real number.

Stochastic dynamical systems (discrete or continuous) are also referred to as **stochastic processes**. In this text, we will have something to say about the

first three of these types of dynamical systems. The last type (continuous-time stochastic processes) requires the use of some rather sophisticated mathematical tools and is beyond the scope of this introductory text.

The material in this book can be covered in many different orders. To make charting a course to particular material easier, the chapter dependencies are diagrammed below.

Discrete Dynamical Systems

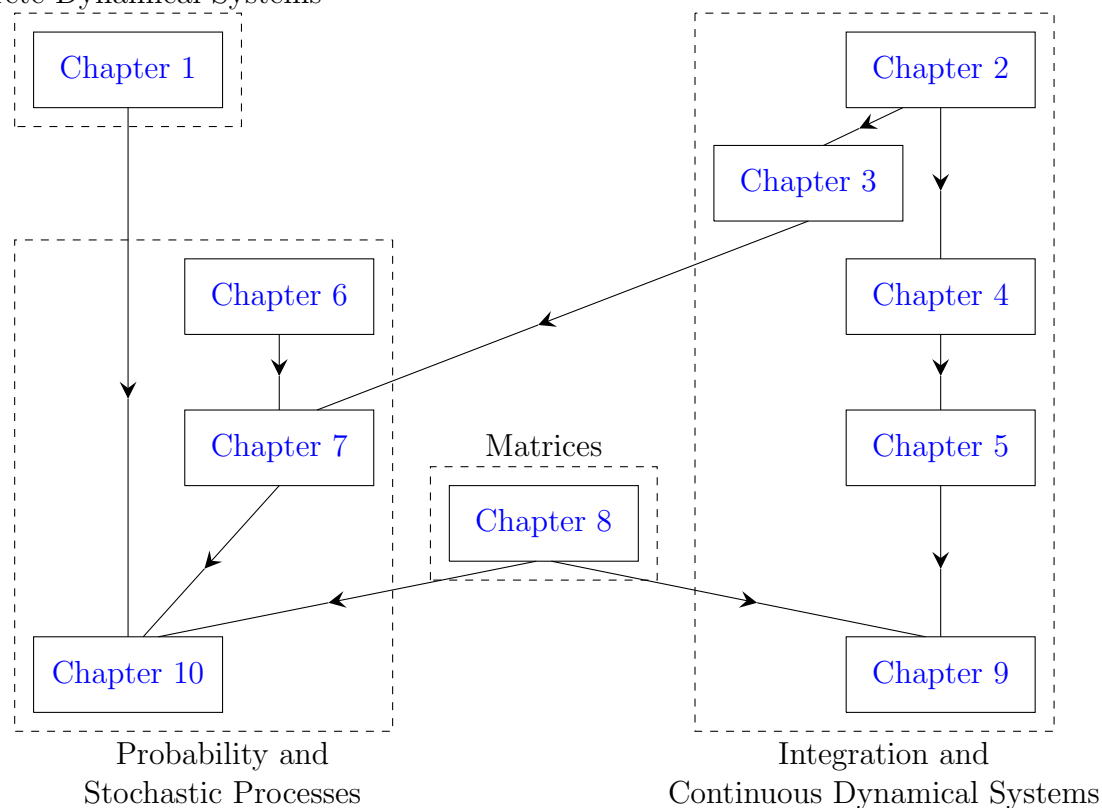


Figure 0.0.1: Chapter Dependencies

Chapter 1

Discrete Dynamical Systems

1.1 Introduction to Discrete Dynamical Systems

Recall that a **discrete-time dynamical system** is a deterministic dynamical system where time is assumed to come in integer amounts. In other words, we can assume that t is a natural number ($t = 0, 1, 2, 3, 4, \dots$). It is customary to write q_t for the state of the system at time t instead of $q(t)$ when t is an integer. Also, since t is assumed to be a natural number, we often write q_n where $n = 0, 1, 2, 3, \dots$ rather than q_t .

Since the system is deterministic, it must be possible to predict the future state in terms of its past history. This means that the state one step in the future, q_{n+1} , must be a function of the present state and possibly some of the states that came before: $q_n, q_{n-1}, q_{n-2}, \dots, q_0$. It could also be a function of the time n . As such, there must be an **updating function** which tells us what the next state will be:

$$q_{n+1} = f(n, q_n, q_{n-1}, q_{n-2}, \dots, q_{n-m}).$$

Notice that the updating function only depends on the current state, q_n , together with a fixed number of past states ($q_{n-1}, \dots, q_{n-m}$). While we could imagine a circumstance where the next state depends on the *entire past history*, that would make the mathematical description far more challenging.

Example 1.1.1 (The Fibonacci Numbers). Consider the discrete-time dynamical system with updating function

$$q_{n+1} = q_n + q_{n-1}$$

and initial data

$$q_0 = 0 \text{ and } q_1 = 1.$$

We get the following list of values

$$\begin{aligned} q_0 &= 0 \\ q_1 &= 1 \\ q_2 &= q_1 + q_0 = 1 + 0 = 1 \\ q_3 &= q_2 + q_1 = 1 + 1 = 2 \\ q_4 &= q_3 + q_2 = 2 + 1 = 3 \\ q_5 &= q_4 + q_3 = 3 + 2 = 5 \\ q_6 &= q_5 + q_4 = 5 + 3 = 8 \\ q_7 &= q_6 + q_5 = 8 + 5 = 13 \\ &\vdots \end{aligned}$$

Note that the updating function for the Fibonacci numbers does not directly depend on the time n , but the next state depends on the current state as well as the state one unit of time in the past. ■

Example 1.1.2. Consider the discrete-time dynamical system with updating function

$$q_{n+1} = n + q_n + q_{n-1},$$

and initial data

$$q_0 = 0 \text{ and } q_1 = 1.$$

We get the following list of values

$$\begin{aligned} q_0 &= 0 \\ q_1 &= 1 \\ q_2 &= 1 + q_1 + q_0 = 1 + 1 + 0 = 2 \\ q_3 &= 2 + q_2 + q_1 = 2 + 2 + 1 = 5 \\ q_4 &= 3 + q_3 + q_2 = 3 + 5 + 2 = 10 \\ q_5 &= 4 + q_4 + q_3 = 4 + 10 + 5 = 19 \\ q_6 &= 5 + q_5 + q_4 = 5 + 19 + 10 = 34 \\ q_7 &= 6 + q_6 + q_5 = 6 + 34 + 19 = 59 \\ &\vdots \end{aligned}$$

Just as with the Fibonacci numbers, the updating function depends on the current state as well as the state one unit of time in the past. However, this updating function directly depends on the current time! ■

In order to keep the mathematics tractable, we will only consider a special class of discrete-time dynamical systems. First, we will assume that our dynamical system is **autonomous** which means that the time variable t or n does not appear explicitly in the updating function. This restricts the updating function to the form

$$f = f(q_n, q_{n-1}, \dots, q_{n-m}).$$

The dynamical system giving the Fibonacci Numbers is an example of an autonomous system while the second example above is not.

Second, we will assume that our discrete-time dynamical system is **memoryless**. This means that the future state q_{n+1} can only depend on the current state q_n (and so the updating function can only depend on q_n). Memoryless systems are often described as systems where the future depends on the past only through the present. Notice that neither of our first two examples is memoryless (since they depend on both the current and previous value of the system).

Example 1.1.3. Consider the dynamical system given by

$$\begin{aligned} q_{n+1} &= 3q_n + n, \\ q_0 &= 1. \end{aligned}$$

Notice that this system is memoryless but not autonomous. The system takes on the following values:

$$\begin{aligned} q_0 &= 1 \\ q_1 &= 3q_0 + 0 = 3(1) + 0 = 3 \\ q_2 &= 3q_1 + 1 = 3(3) + 1 = 10 \\ q_3 &= 3q_2 + 2 = 3(10) + 2 = 32 \\ q_4 &= 3q_3 + 3 = 3(32) + 3 = 99 \\ &\vdots \quad \blacksquare \end{aligned}$$

In general, an **autonomous, memoryless discrete-time dynamical system** is a dynamical system of the form

$$\begin{aligned} q_{n+1} &= f(q_n) \\ q_0 &= \text{some initial value} \end{aligned}$$

Since this is the only type of discrete-time dynamical system we will consider in this text, we will simply call them discrete-time dynamical systems from this point forward.

Example 1.1.4. Consider the dynamical system given by

$$\begin{aligned} L_{n+1} &= 2L_n - 1 \\ L_0 &= 3. \end{aligned}$$

The system takes on the following values:

$$\begin{aligned} L_0 &= 3 \\ L_1 &= 2L_0 - 1 = 2(3) - 1 = 5 \\ L_2 &= 2L_1 - 1 = 2(5) - 1 = 9 \\ L_3 &= 2L_2 - 1 = 2(9) - 1 = 17 \\ L_4 &= 2L_3 - 1 = 2(17) - 1 = 33 \\ &\vdots \blacksquare \end{aligned}$$

For some discrete-time dynamical systems, it is possible to find a direct formula for q_{n+1} without referencing the updating function. This can be useful for answering certain questions about future values of the system.

Example 1.1.5. Consider the following dynamical system.

$$\begin{aligned} q_{n+1} &= 3q_n, \\ q_0 &= 1. \end{aligned}$$

Looking at the values of the system, we see a pattern:

$$\begin{aligned} q_0 &= 1 = 3^0 \\ q_1 &= 3q_0 = 3(1) = 3 = 3^1 \\ q_2 &= 3q_1 = 3(3) = 9 = 3^2 \\ q_3 &= 3q_2 = 3(9) = 27 = 3^3 \\ q_4 &= 3q_3 = 3(27) = 81 = 3^4 \\ &\vdots \end{aligned}$$

This suggests (and it can be shown to be true) that

$$q_n = 3^n. \blacksquare$$

Example 1.1.6. Consider the dynamical system from Example 1.1.4:

$$\begin{aligned} L_{n+1} &= 2L_n - 1 \\ L_0 &= 3. \end{aligned}$$

To see a pattern, we try to get a formula using L_0

$$\begin{aligned} L_0 &= 3 \\ L_1 &= 2L_0 - 1 = 2(3) - 1 = 5 \\ L_2 &= 2L_1 - 1 = 2(2L_0 - 1) - 1 = 4L_0 - 3 = 9 \\ L_3 &= 2L_2 - 1 = 2(4L_0 - 3) - 1 = 8L_0 - 7 = 17 \\ L_4 &= 2L_3 - 1 = 2(8L_0 - 7) - 1 = 16L_0 - 15 = 33 \\ &\vdots \end{aligned}$$

What we see is

$$L_n = 2^n L_0 - (2^n - 1). \blacksquare$$

So far, all of our dynamical systems have been **time reversible**. This means that given q_{n+1} we can recover q_n . In other words, given the current value of the system, we can figure out the value of the system in the previous time step.

Example 1.1.7. Consider again the dynamical system from Example 1.1.5. If we know that $q_5 = 54$, we can determine q_4 by solving $q_5 = 3q_4$ for q_4 :

$$q_4 = \frac{1}{3}q_5 = \frac{1}{3}(54) = 18.$$

So q_4 must have been 18. In general, we see that

$$q_n = \frac{1}{3}q_{n+1}. \blacksquare$$

In this example, the fact that we can solve for q_n in terms of q_{n+1} shows that the system is time reversible.

Example 1.1.8. Suppose in the dynamical system from Example 1.1.4 we know that $L_6 = 67$. Solving the equation $L_6 = 2L_5 - 1$ for L_5 gives

$$L_5 = \frac{1}{2}(L_6 + 1) = \frac{1}{2}(67 + 1) = 34.$$

We can see that in general

$$L_n = \frac{1}{2}(L_{n+1} + 1). \blacksquare$$

The next example is a famous dynamical system that is not time reversible (as well as the source of an unsolved mathematical conjecture).

Example 1.1.9. (The Collatz Function) Consider the function on the positive integers given by

$$C(x) = \begin{cases} \frac{1}{2}x, & \text{if } x \text{ is even} \\ 3x + 1, & \text{if } x \text{ is odd} \end{cases}$$

So, even numbers are divided by two while odd numbers are multiplied by three and then have a one added.

Consider the dynamical system whose states are positive integers and whose updating function and initial data are given by

$$\begin{aligned} q_{n+1} &= C(q_n) \\ q_0 &= 7. \end{aligned}$$

Then we get the following sequence of numbers

$$\begin{aligned}
 q_0 &= 7 \\
 q_1 &= C(7) = 3(7) + 1 = 22 \\
 q_2 &= C(22) = \frac{1}{2}(22) = 11 \\
 q_3 &= C(11) = 3(11) + 1 = 34 \\
 q_4 &= C(34) = \frac{1}{2}(34) = 17 \\
 q_5 &= C(17) = 3(17) + 1 = 52 \\
 q_6 &= C(52) = \frac{1}{2}(52) = 26 \\
 q_7 &= C(26) = \frac{1}{2}(26) = 13 \\
 q_8 &= C(13) = 3(13) + 1 = 40 \\
 q_9 &= C(40) = \frac{1}{2}(40) = 20 \\
 q_{10} &= C(20) = \frac{1}{2}(20) = 10 \\
 q_{11} &= C(10) = \frac{1}{2}(10) = 5 \\
 q_{12} &= C(5) = 3(5) + 1 = 16 \\
 q_{13} &= C(16) = \frac{1}{2}(16) = 8 \\
 q_{14} &= C(8) = \frac{1}{2}(8) = 4 \\
 q_{15} &= C(4) = \frac{1}{2}(4) = 2 \\
 q_{16} &= C(2) = \frac{1}{2}(2) = 1
 \end{aligned}$$

Note that this system is not time reversible! Suppose that we know $q_{n+1} = 16$. Is it possible to determine what q_n (the previous state) must have been? The answer is no – all we can say is $q_n = 32$ or $q_n = 5$ since these are the only numbers that solve $C(n) = 16$. Without more information, we have no idea which of these two possibilities was the actual previous state.

In an unrelated note, the (in)famous Collatz Conjecture claims that no matter what positive integer you choose for initial data, the dynamical system above will eventually reach the number 1. While this conjecture has been studied intensely since it was first posed in 1937 and a lot of evidence points to it being true¹, no one has been able to completely prove or disprove it!

■

1.2 Cobweb Plots

Before we study some specific discrete-time dynamical systems, we need a convenient way to visualize the evolution of the system. Perhaps the easiest way to display the dynamical system is simply to plot the points (n, q_n) as dots in the xy -plane. Such a graph is referred to as a **sequential plot**.

Example 1.2.1. Consider the discrete-time dynamical system given by

$$\begin{aligned} q_{n+1} &= 2q_n - 1 \\ q_0 &= 2 \end{aligned}$$

The first few values of the system are

$$\begin{aligned} q_0 &= 2, \\ q_1 &= 3, \\ q_2 &= 5, \\ q_3 &= 9, \\ q_4 &= 17, \\ q_5 &= 33. \end{aligned}$$

This gives us the following list of ordered pairs:

$$(0, 2); (1, 3); (2, 5); (3, 9); (4, 17); (5, 33).$$

If we plot these points, we obtain the plot below.

¹Every number between 2 and roughly 10^{20} has been tried an initial datum. In every case, the system eventually reaches 1.

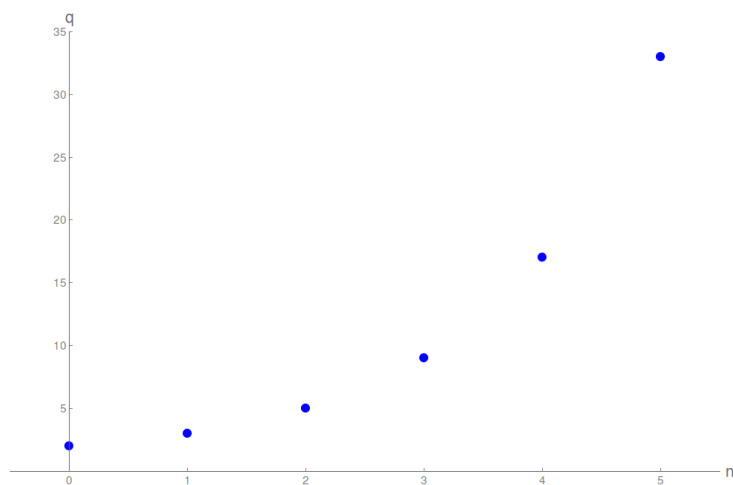


Figure 1.2.1: Sequential Plot for Example 1.2.1

While a sequential plot seems the most natural way to display the evolution of a discrete-time dynamical system, it turns out there is a better method – a **cobweb plot**.

Cobweb Plots

To produce a cobweb plot of the system $q_{n+1} = f(q_n)$ with some initial data q_0 , we go through the following steps:

- 1.) Produce a plot displaying both $y = f(x)$ (a plot of the updating function) and the line $y = x$.
- 2.) Mark the value of q_0 on the x -axis.
- 3.) Move vertically from q_0 in the direction of the graph $y = f(x)$, and stop once you intersect it. The y -value of the point you end at will be $q_1 = f(q_0)$.
- 4.) Move horizontally in the direction of the graph $y = x$, and stop once you intersect it. Both the x - and y -values of the point you end at will be q_1 .
- 5.) Starting at the point from step 4.) (which will be (q_1, q_1)), move vertically towards $y = f(x)$ and then horizontally to $y = x$. You will end up at the point (q_2, q_2) .
- 6.) Repeat the vertical and horizontal moves from step 5.) to build up $q_3, q_4, q_5, \dots$ for as many steps as needed or required.

Example 1.2.2. Consider the discrete-time dynamical system given by

$$\begin{aligned} q_{n+1} &= 2q_n - 1 \\ q_0 &= 2. \end{aligned}$$

To produce a cobweb plot of this dynamical system, we begin by graphing $y = 2x - 1$ (the updating function) together with $y = x$. We plot the updating function in blue below.

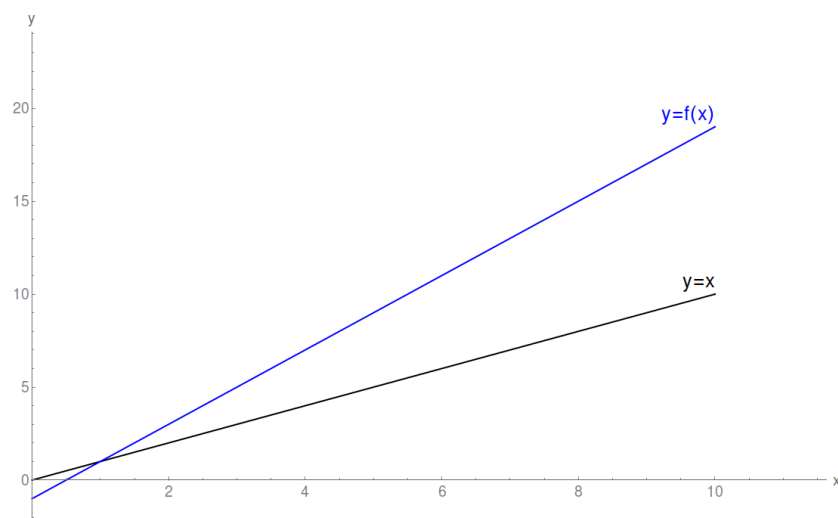


Figure 1.2.2: First Step of Cobwebbing

Starting from $x = 2$ (the value of q_0), we move vertically until we intersect the graph of the updating function, and then horizontally until we intersect $y = x$.

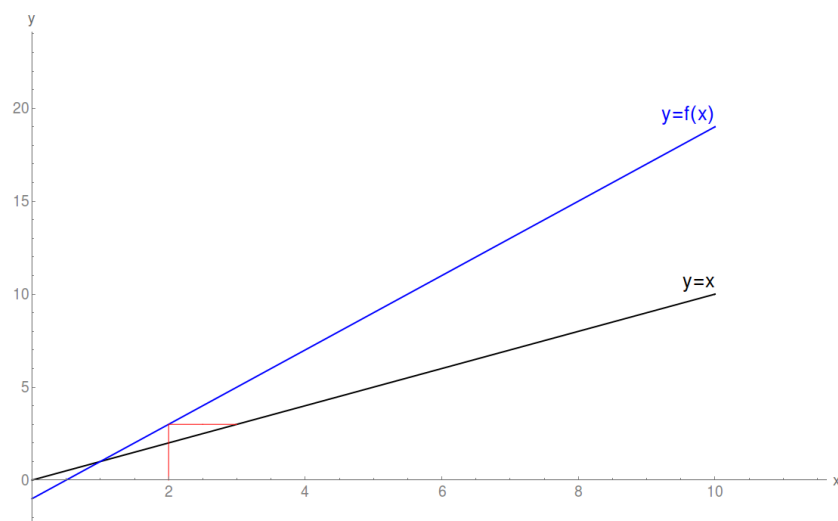


Figure 1.2.3: Steps 2, 3, and 4 of Cobwebbing

Notice that the point we have stopped at is $(3, 3)$. This tells us that $q_1 = 3$. Repeating the horizontal and vertical moves gives us the point $(5, 5)$,

and so $q_2 = 5$.

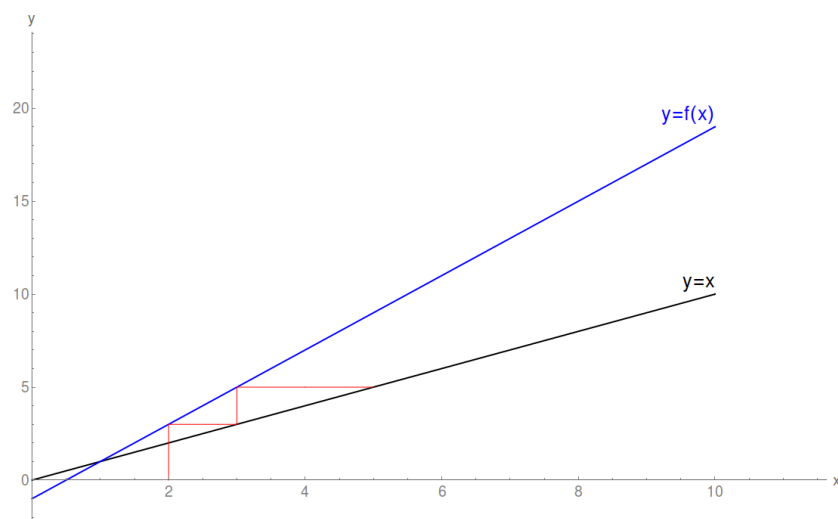


Figure 1.2.4: Cobweb Plot up to q_2

Repeating again, we wind up at the point $(9, 9)$ which gives $q_3 = 9$.

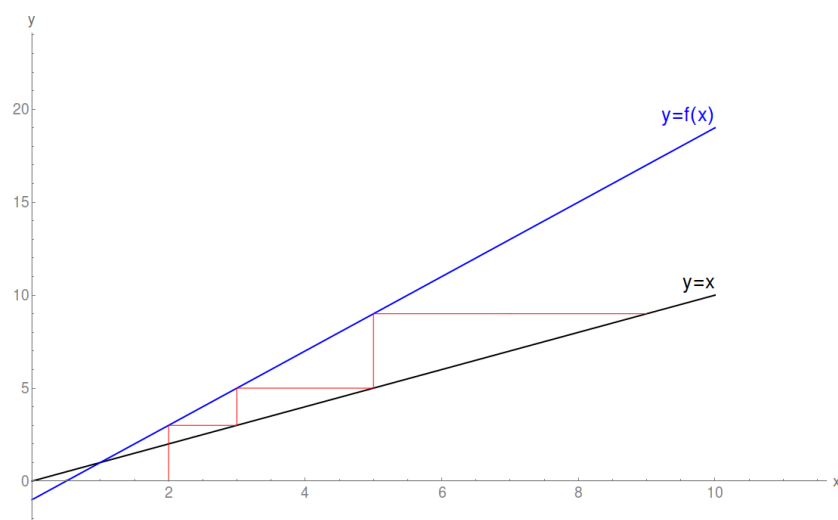


Figure 1.2.5: Continuation of Cobweb Plot

We could continue this process to find q_4, q_5 , etc.

Example 1.2.3. As a second example, consider the discrete-time dynamical system

$$\begin{aligned} q_{n+1} &= \frac{1}{5}q_n^3 - q_n + 1 \\ q_0 &= 2. \end{aligned}$$

First, we plot $y = \frac{1}{5}x^3 - x + 1$ and $y = x$.

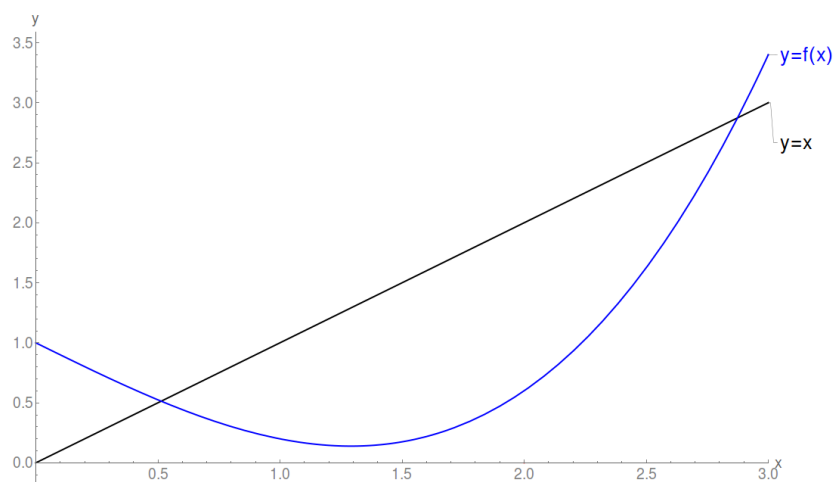


Figure 1.2.6: Setup for Cobweb Plot of Example 1.2.3

Starting at $x = 2$, we move vertically to the graph of the updating function and then horizontally to $y = x$.

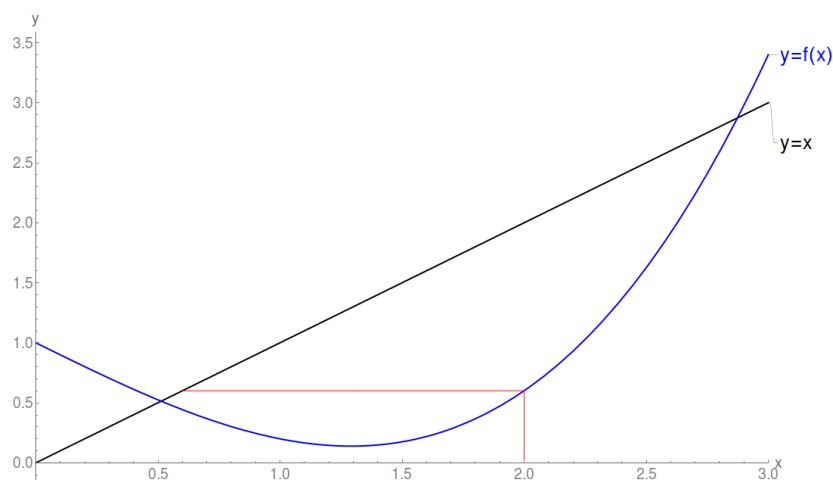


Figure 1.2.7: First Step in Cobweb Plot of Example 1.2.3

This moves us to the point with x - and y -coordinates equal to $q_1 = \frac{3}{5}$. Moving from this point takes us to the point with x - and y -coordinates equal to $q_2 = \frac{277}{625}$.

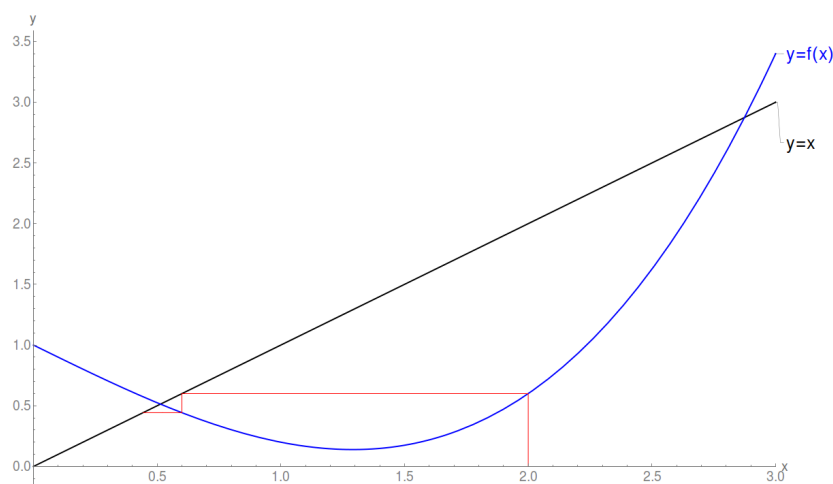


Figure 1.2.8: Second Step in Cobweb Plot of Example 1.2.3

As we continue to build up the cobweb plots, the values of q_3, q_4 , etc. continue to become more complicated rational expressions. However, the cobweb plot reveals some interesting behavior.

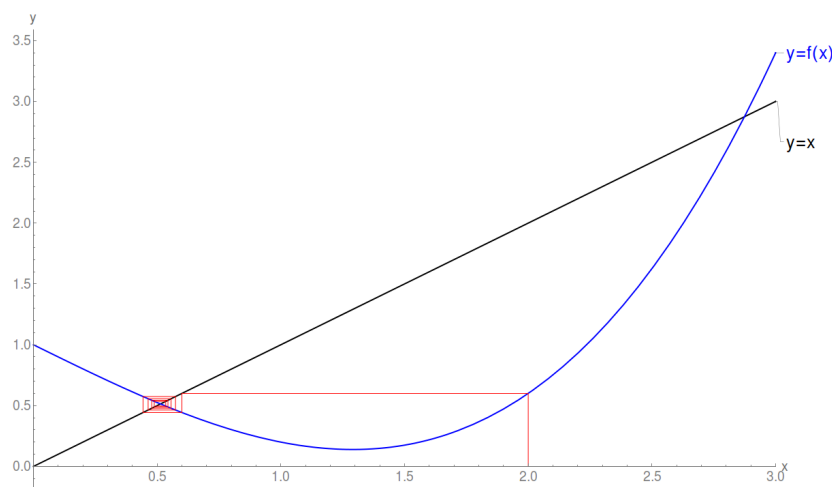


Figure 1.2.9: Cobweb Plot of Example 1.2.3

It appears that the system is closing in on one of the points where the updating function intersects $y = x$. We will have more to say about this in the next section!

Building up an accurate cobweb plot depends heavily on having a precise graph of the updating function and $y = x$. To avoid faulty conclusions from inaccurate graphs, cobweb plots are often generated by graphing calculators or mathematics software packages. Even with technology, care must be taken in interpreting results (especially with graphs where the axes have different scales).

1.3 Equilibria and Stability

As we saw in Example 1.2.3, there is a special role played by points where the updating function crosses the line $y = x$. These points are known as **equilibria**. More precisely, an equilibrium value of a dynamical system is any initial datum for the system such that the system remains at this value for all future times.

Example 1.3.1. Consider the discrete-time dynamical system given by

$$\begin{aligned} q_{n+1} &= 2q_n - 1 \\ q_0 &= 1. \end{aligned}$$

Notice that

$$\begin{aligned} q_0 &= 1, \\ q_1 &= 2q_0 - 1 = 2(1) - 1 = 1, \\ q_2 &= 2q_1 - 1 = 2(1) - 1 = 1, \\ q_3 &= 2q_2 - 1 = 2(1) - 1 = 1, \\ &\vdots \end{aligned}$$

This shows that 1 is an equilibrium value of the system described by $q_{n+1} = 2q_n - 1$ ■.

It turns out there is a very simple way to find the equilibria of discrete-time dynamical systems.

Theorem 1.3.1 (Equilibria of Discrete-Time Dynamical Systems). *A value q^* is an equilibrium of the discrete-time dynamical system $q_{n+1} = f(q_n)$ if and only if q^* satisfies*

$$q^* = f(q^*).$$

In other words, the equilibria of a discrete-time dynamical system are precisely the solutions to the equation $x = f(x)$. The solutions to this equation are also known as **fixed points** of the function f .

Proof:

Suppose that q^* is a solution to $x = f(x)$. Taking $q_0 = q^*$, we find

$$\begin{aligned} q_0 &= q^*, \\ q_1 &= f(q_0) = f(q^*) = q^*, \\ q_2 &= f(q_1) = f(q^*) = q^*, \\ q_3 &= f(q_2) = f(q^*) = q^*, \\ &\vdots \end{aligned}$$

and we see that q^* is an equilibrium of the dynamical system.

Now suppose that q^* is an equilibrium value for the dynamical system. Then for initial data $q_0 = q^*$, we must have all subsequent values q_1, q_2 , etc. equal to q^* . In particular $q_1 = f(q_0)$ gives

$$q^* = f(q^*),$$

which means that q^* satisfies the equation $x = f(x)$. ■

Example 1.3.2. The dynamical system

$$q_{n+1} = 2q_n - 1$$

has updating function $f(x) = 2x - 1$. Setting this equal to x and solving gives

$$2x - 1 = x$$

$$2x - x = 1$$

$$x = 1,$$

which shows that the only equilibrium value for this system is 1. ■

Example 1.3.3. The dynamical system

$$q_{n+1} = \frac{1}{4}q_n^2$$

has updating function $f(x) = \frac{1}{4}x^2$. Setting this equal to x and solving gives

$$\frac{1}{4}x^2 = x$$

$$\frac{1}{4}x^2 - x = 0$$

$$x \left(\frac{1}{4}x - 1 \right) = 0,$$

giving equilibria values of $q^* = 0$ and $q^* = 4$. ■

As we noted in the theorem, the equilibria of a dynamical system are precisely the solutions to $x = f(x)$. Graphically, these are the x -coordinates of the intersection points of $y = f(x)$ and the line $y = x$ which we can clearly see in the cobweb plot!

Consider the cobweb plot for the dynamical system in the example above.

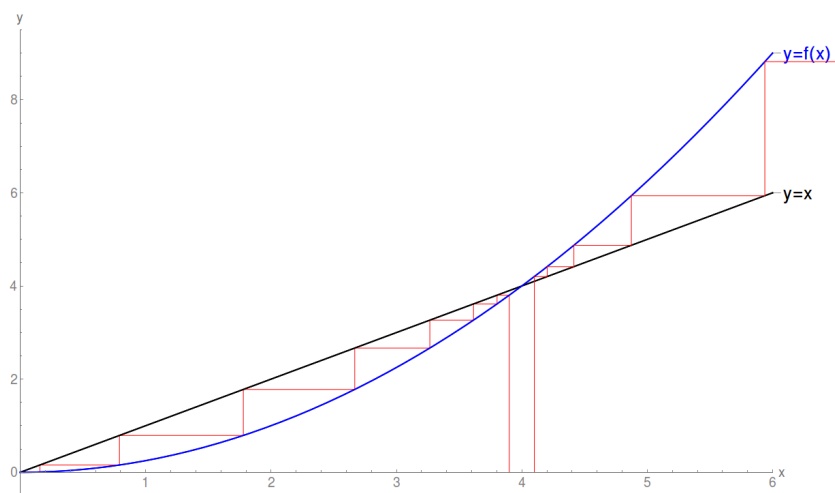


Figure 1.3.1: Cobweb Plot for Example 1.3.3

We have shown the plots for two different initial data – one just smaller than the equilibrium at 4 and one just larger. Notice that both trajectories lead away from the equilibrium. In this case, we say that the equilibrium $q^* = 4$ is **unstable**. The exact definition is a bit technical, but essentially an equilibrium point is unstable if there is a small interval centered on the equilibrium such that we can find initial data points arbitrarily close to the equilibrium (and on either side of it) whose trajectories eventually leave the interval.

If we look at the equilibrium at $q^* = 0$, we see different behavior.

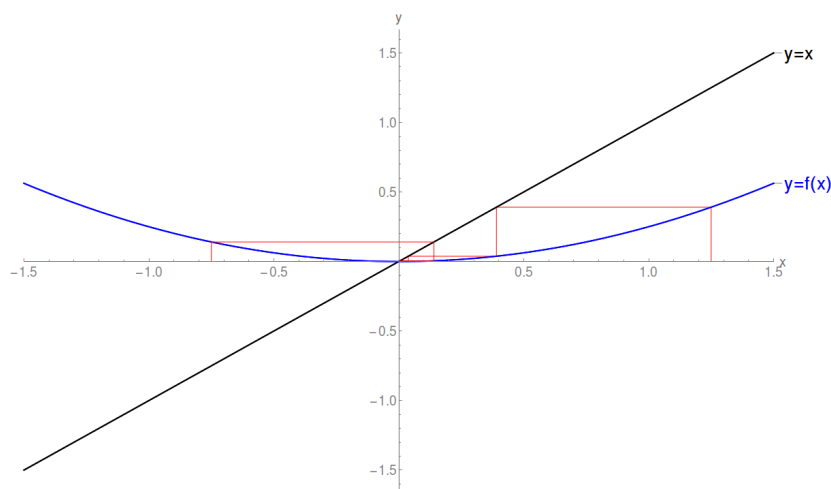


Figure 1.3.2: Cobweb Plot for Example 1.3.3

The equilibrium at 0 is said to be **asymptotically stable**. In general, if initial data sufficiently close² to and on either side of an equilibrium point launch trajectories that move closer to it, then the equilibrium is asymptotically stable. In other words, all initial data relatively close to an asymptotically stable equilibrium give us sequences that converge to the equilibrium.

For both of the equilibria above, nearby initial data points moved either directly away or directly toward the equilibrium. Some equilibria have more complicated behavior. Consider again the equilibrium from Example 1.2.3.

²“Sufficiently close” is more than a little vague. It can be made precise using ϵ - δ definitions similar to the formal definition of limits or more generally using the notion of *neighborhood* from point-set topology.

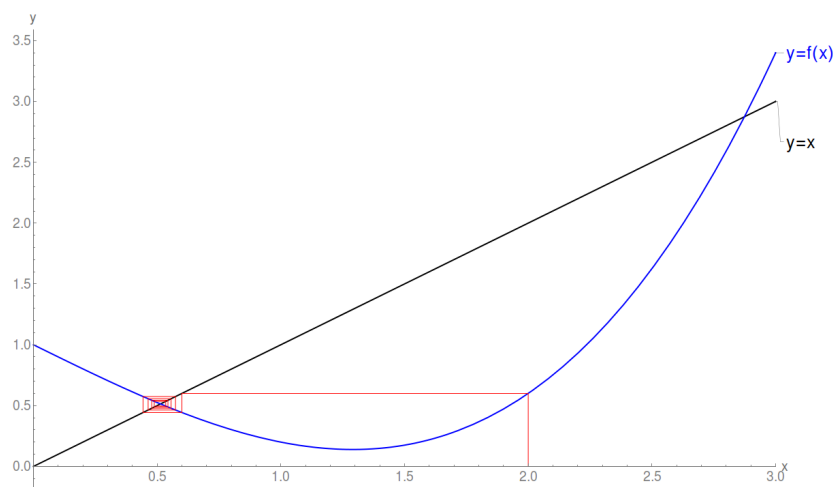


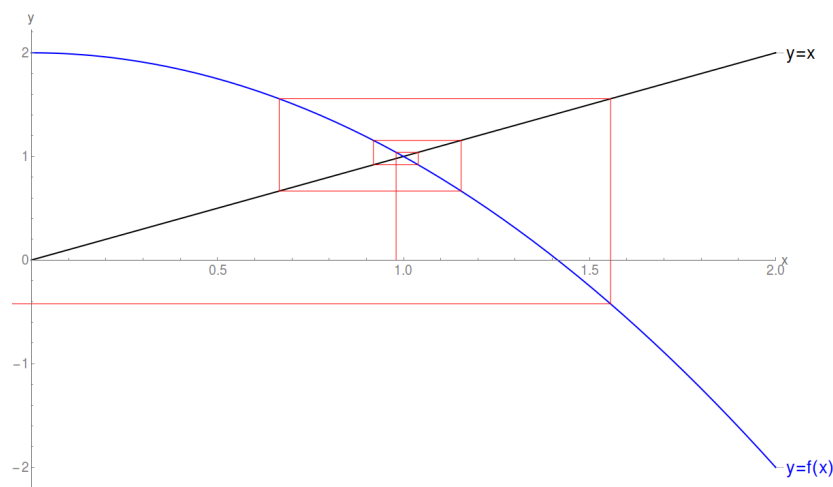
Figure 1.3.3: Cobweb Plot for Example 1.2.3

The equilibrium near 0.5 is asymptotically stable, but instead of moving directly toward the equilibrium value, the trajectory circles around it. We say that this equilibrium is an **asymptotically stable spiral point**.

The opposite behavior is also possible. Consider the dynamical system given by

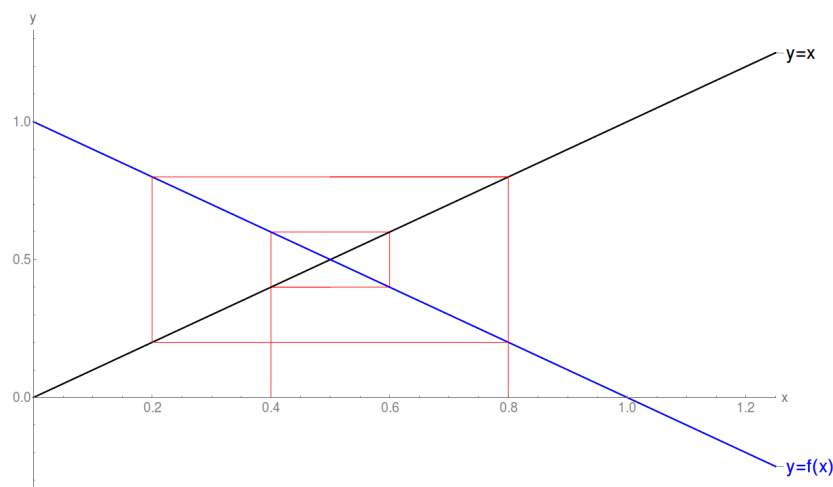
$$q_{n+1} = 2 - q_n^2.$$

By solving the equation $x = 2 - x^2$, we find that the equilibria of the system are $q^* = -2, 1$. Looking at the equilibrium at 1 shows the following behavior.

Figure 1.3.4: Cobweb Plot for $q_{n+1} = 2 - q_n^2$

This type of equilibrium is an **unstable spiral point**.

There are a few other types of equilibria to mention. If we consider the dynamical system given by $q_{n+1} = 1 - q_n$, we see the following behavior around the only equilibrium at $q^* = \frac{1}{2}$.

Figure 1.3.5: Cobweb Plot for $q_{n+1} = 1 - q_n$

Notice that for any choice of initial data q_0 , the system cycles between q_0 and $1 - q_0$, and the trajectory circles the equilibrium for all time without

getting any closer to it. Such equilibria are said to be **stable**. More precisely, an equilibrium point is stable if for any interval centered on it all initial data sufficiently close to and on either side of it launch trajectories that stay in that interval for all future times. Technically speaking, asymptotically stable equilibria are also stable equilibria. However, we should always use asymptotically stable in cases where all trajectories in some region around the equilibrium get progressively closer to it.

Finally, the dynamical system $q_{n+1} = -q_n^2 + 3q_n - 1$ has a single equilibrium at $q^* = 1$. If we look at trajectories on either side of the equilibrium, we find that the system seems to be asymptotically stable from one direction but unstable from another.

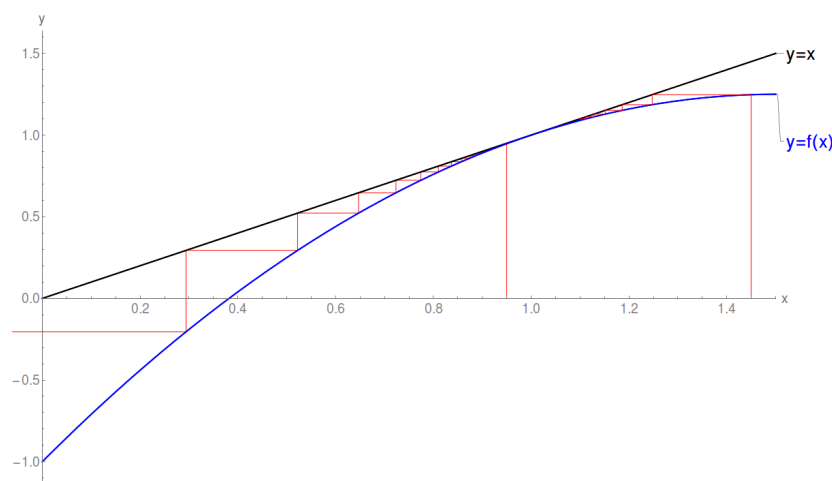


Figure 1.3.6: Cobweb Plot for $q_{n+1} = -q_n^2 + 3q_n - 1$

An equilibrium is said to be **semi-stable** (or is sometimes called a **node**) if all initial data on one side of the equilibrium that are sufficiently close launch trajectories that leave the vicinity of the equilibrium while all sufficiently close initial data on the other side launch trajectories that approach the equilibrium. For the example above, we could be more specific and say that $q^* = 1$ is a semi-stable equilibrium which is asymptotically stable from the right and unstable from the left.

1.4 Stability and the Derivative

Now that we have defined the various kinds of stability that equilibria can enjoy, we need to develop a test that will identify the type of equilibrium directly from the updating function. That is accomplished by the following theorem.

Theorem 1.4.1 (Stability of Equilibria in discrete-time dynamical Systems). *Suppose that q^* is an equilibrium of the discrete-time dynamical system*

$$q_{n+1} = f(q_n)$$

(that is $q^ = f(q^*)$), and suppose that the updating function, f , is differentiable in some open interval containing q^* . Then the number $f'(q^*)$ can be used to classify the behavior near q^* . Specifically,*

- *if $-1 < f'(q^*) < 1$, q^* is an asymptotically stable equilibrium;*
- *if $f'(q^*) < -1$ or $f'(q^*) > 1$, q^* is an unstable equilibrium;*
- *if $f'(q^*) = \pm 1$, the test is inconclusive.*

Moreover if $f'(q^) < 0$, the system will exhibit spiraling behavior in the vicinity of the equilibrium. If $f'(q^*) > 0$, the values of the dynamical system will move directly toward or away from the equilibrium depending on the size of the derivative. Note that equilibria that are semi-stable or stable without being asymptotically stable can only occur when the derivative of the updating function is ± 1 .*

A proof of this theorem can be found in [HK91, pp.73 – 75], and we will not give a rigorous treatment here. Instead, we will offer a few diagrams that will hopefully convince you that the theorem is true. While we will not look at every possible scenario, we will examine the four primary cases of the theorem.

From basic results in calculus, we know that near the equilibrium point, the updating function will be well approximated by its tangent line:

$$f(x) \approx f(q^*) + f'(q^*)(x - q^*).$$

So, we can convince ourselves that the theorem is correct by looking at linear functions with various slopes.

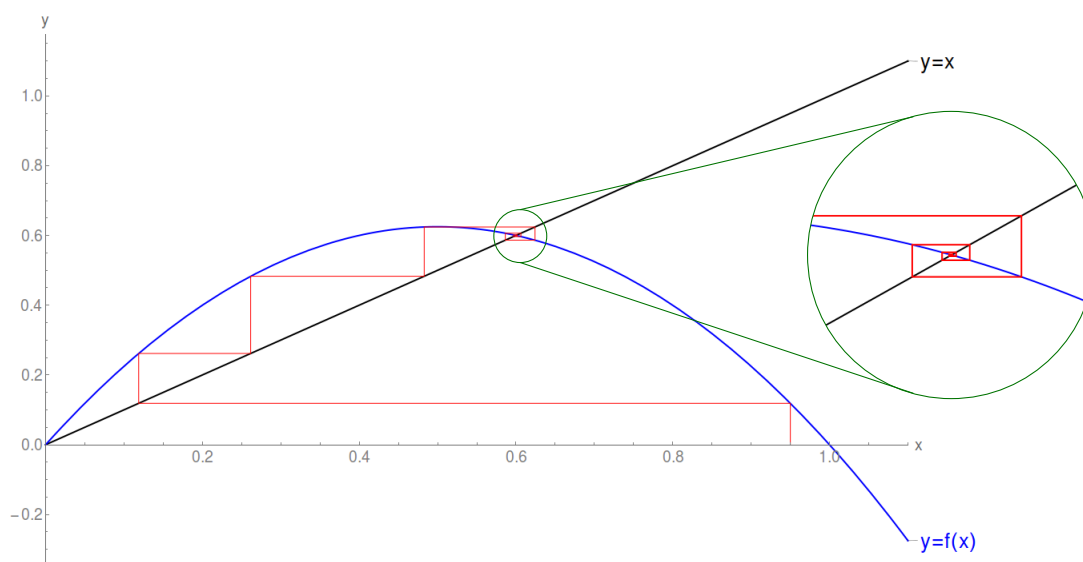
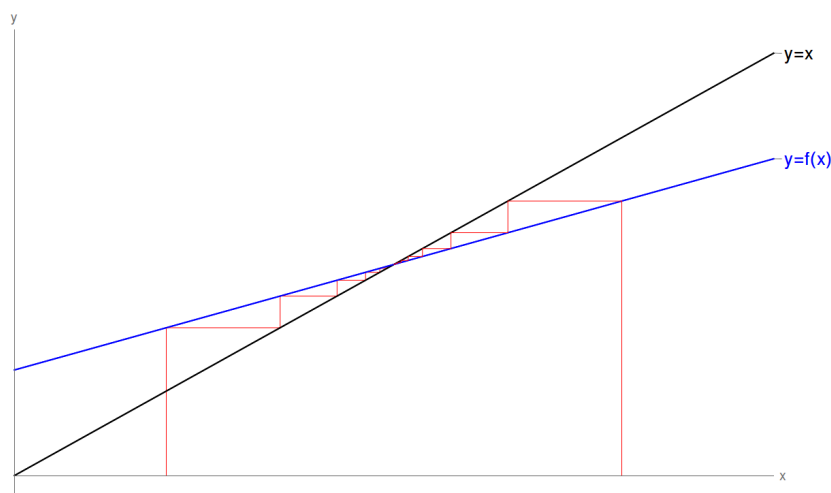


Figure 1.4.1: Nearly Linear Behavior Near Equilibrium

Starting with a slope that is between 0 and 1 (i.e. $0 < f'(q^*) < 1$) we can see that the behavior of the equilibrium will be asymptotically stable without spiraling.

Figure 1.4.2: Asymptotic Stability for $0 < f'(q^*) < 1$

Similarly, if the slope is greater than 1 ($1 < f'(q^*)$), we can see that the

equilibrium is unstable (without spiraling).

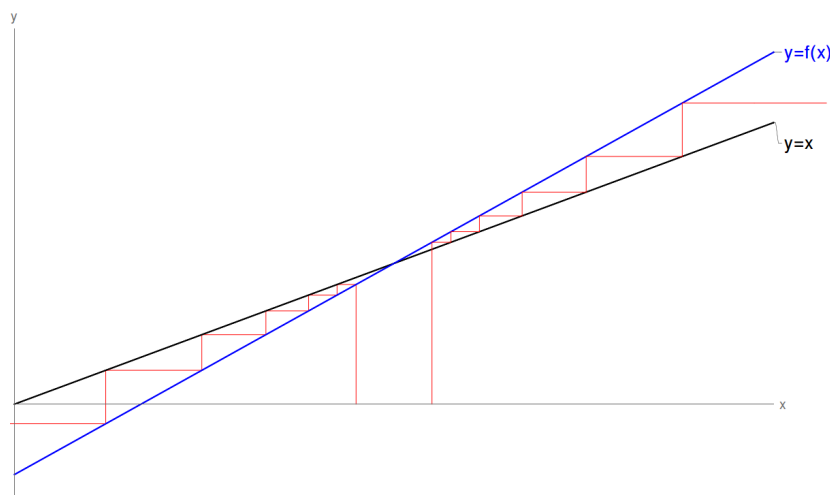


Figure 1.4.3: Instability for $f'(q^*) > 1$

When $-1 < f'(q^*) < 0$, we can clearly see the asymptotically stable spiraling behavior.

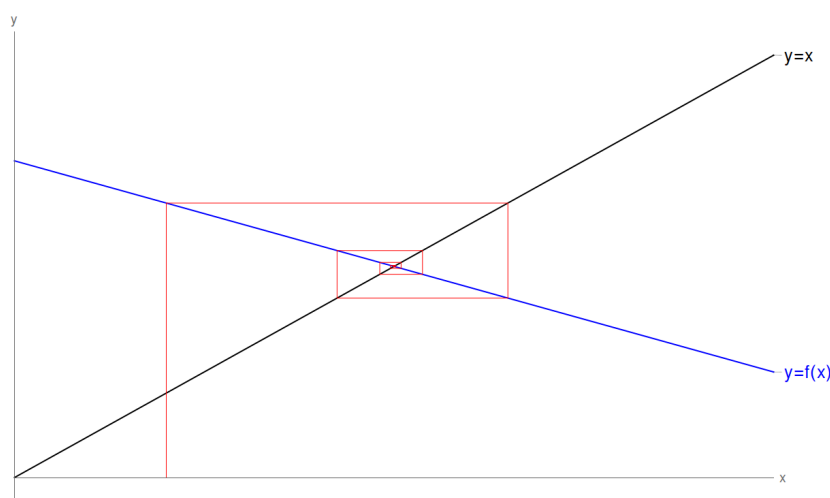
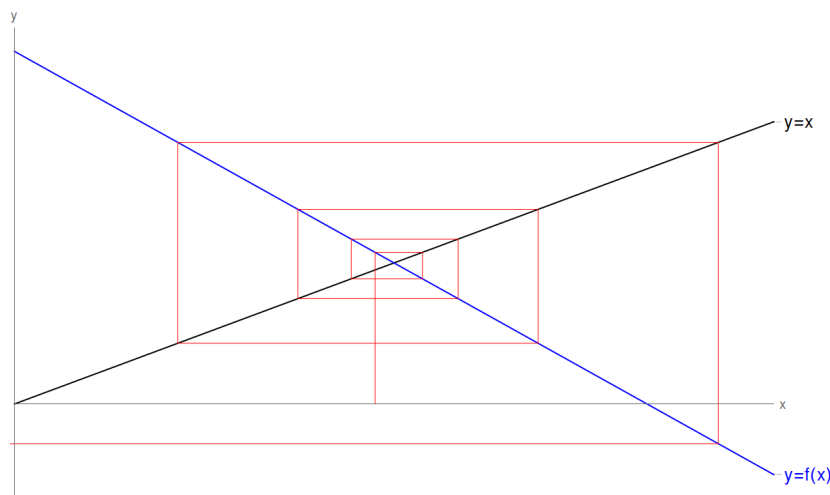


Figure 1.4.4: Asymptotic Stability for $-1 < f'(q^*) < 0$

Similarly, the behavior when $f'(q^*) < -1$ is clearly an unstable spiral.

Figure 1.4.5: Instability for $f'(q^*) < -1$

Note that when $f'(q^*) = 0$, the equilibrium will certainly be asymptotically stable, but trajectories could either spiral inwards or move in directly.

We will now revisit a few of the dynamical systems from the previous section.

Example 1.4.1. The dynamical system

$$q_{n+1} = 2q_n - 1$$

has updating function $f(x) = 2x - 1$ and a single equilibrium at $q^* = 1$. Taking the derivative yields $f'(x) = 2$ and so in particular $f'(1) = 2$. Thus, $q^* = 1$ is an unstable equilibrium. ■

Example 1.4.2. The dynamical system

$$q_{n+1} = \frac{1}{4}q_n^2$$

has updating function $f(x) = \frac{1}{4}x^2$ and equilibria $q^* = 0, 4$. The derivative of the updating function is $f'(x) = \frac{1}{2}x$. Since $f'(0) = 0$, this equilibrium must be asymptotically stable. For $q^* = 4$, $f'(4) = 2$ and this equilibrium is unstable. Note that these results confirm what we saw in the cobweb plot. ■

Example 1.4.3. The dynamical system

$$q_{n+1} = 2 - q_n^2$$

has updating function $f(x) = 2 - x^2$ and equilibria $q^* = -2, 1$. The derivative of the updating function is $f'(x) = -2x$. Since $f'(-2) = 4$, this equilibrium must be unstable. For $q^* = 1$, $f'(1) = -2$ and this equilibrium is an unstable spiral point. ■

Example 1.4.4. The dynamical system

$$q_{n+1} = -q_n^2 + 3q_n - 1$$

has updating function $f(x) = -x^2 + 3x - 1$ and an equilibrium at $q^* = 1$. The derivative of the updating function is $f'(x) = -2x + 3$. Since $f'(1) = 1$, the derivative test is inconclusive. Fortunately, our earlier cobweb plot showed us that this equilibrium is semi-stable. ■

1.5 Examples of Discrete-Time Dynamical Systems

Now that we have developed the tools to discuss and analyze discrete-time dynamical systems, we will examine a few systems that have some real world significance.

1.5.1 The Mutant vs. Wild-Type Model

Example 1.5.1. We are interested in the population growth of two species of bacteria both living in the same culture. One group is the wild-type (the bacteria that are typically found in nature) which double in number every 6 hours. The second group is a mutant strain which reproduce at a faster rate – tripling in size during the same 6 hour period. We will let b_n be the size of the wild-type population and m_n the size of the mutant population. Given our data, we have

$$\begin{aligned} b_{n+1} &= 2b_n \\ m_{n+1} &= 3m_n \end{aligned}$$

where n measures the number of 6 hour blocks of time since the beginning of the experiment.

It is rather unlikely that we will know exactly how many of each kind of bacteria we have. What we can more readily estimate is the fraction of

bacteria that are mutants. Let p_n be the fraction of the bacteria that are mutants at time n . That is

$$p_n = \frac{\text{number of mutants at time } n}{\text{total bacteria at time } n} = \frac{m_n}{b_n + m_n}.$$

Since p_n is the fraction of the total population that are mutants, it is naturally restricted to the range $0 \leq p_n \leq 1$ (and so we can naturally think of it as measuring the percentage of the total population that are mutants). Note that the fraction of wild-type bacteria must be $1 - p_n$. Our job is to determine the dynamical system describing p_n , and the following calculations do just that. In the first step, we use the separate dynamical systems for b_{n+1} and m_{n+1} above.

$$\begin{aligned} p_{n+1} &= \frac{m_{n+1}}{b_{n+1} + m_{n+1}} \\ &= \frac{3m_n}{2b_n + 3m_n} \\ &= \frac{3m_n}{2b_n + 3m_n} \cdot \frac{1/(b_n + m_n)}{1/(b_n + m_n)} \\ &= \frac{3(m_n/(b_n + m_n))}{2(b_n/(b_n + m_n)) + 3(m_n/(b_n + m_n))} \\ &= \frac{3p_n}{2(1 - p_n) + 3p_n} \\ &= \frac{3p_n}{2 + p_n} \end{aligned}$$

We see that the updating function for p_n is given by

$$f(x) = \frac{3x}{2 + x}.$$

Solving for the equilibria gives

$$\begin{aligned} \frac{3x}{2 + x} &= x \\ 3x &= (2 + x)x \\ 3x &= 2x + x^2 \\ x^2 - x &= 0 \\ x(x - 1) &= 0 \\ x &= 0, 1. \end{aligned}$$

This means that the only equilibria are $p^* = 0$ (the wild-type make up all of the population) or $p^* = 1$ (the mutants make up the entire culture). The derivative of the updating function is

$$f'(x) = \frac{6}{(2+x)^2}.$$

Since $f'(0) = 3/2$, $p^* = 0$ is an unstable equilibrium. The other equilibrium, $p^* = 1$, gives $f'(1) = 2/3$ and so is asymptotically stable. Hence, the population will tend to be overwhelmed by the mutant strain as time goes on.

Note that this result is obvious given the circumstances of the problem. Namely, the mutants are reproducing at a faster rate than wild-type bacteria. Given that fact, we would imagine that given enough time the number of mutants will be far larger than the number of wild-type. Note that this DOES NOT imply that the wild-type bacteria die out. In fact, both populations are growing rapidly. It is simply that the mutants will be far more numerous in the long run. ■

Having looked at a specific example, we want to examine this model in more generality. Suppose now that

$$\begin{aligned} b_{n+1} &= r b_n \\ m_{n+1} &= s m_n \end{aligned}$$

where r and s are the (non-negative) growth rates of the two populations. These numbers are **parameters**, that is quantities which may vary from situation to situation but are considered to be constant in any particular version of the system.

As before, let p_n be the fraction of mutants in the total population (and

again $0 \leq p_n \leq 1$). If we repeat our calculations, we find

$$\begin{aligned}
 p_{n+1} &= \frac{m_{n+1}}{b_{n+1} + m_{n+1}} \\
 &= \frac{sm_n}{rb_n + sm_n} \\
 &= \frac{sm_n}{rb_n + sm_n} \cdot \frac{1/(b_n + m_n)}{1/(b_n + m_n)} \\
 &= \frac{s(m_n/(b_n + m_n))}{r(b_n/(b_n + m_n)) + s(m_n/(b_n + m_n))} \\
 &= \frac{sp_n}{r(1 - p_n) + sp_n} \\
 &= \frac{sp_n}{r + (s - r)p_n}.
 \end{aligned}$$

The updating function we have found is

$$f(x) = \frac{sx}{r + (s - r)x}.$$

Finding the fixed points of this function gives

$$\begin{aligned}
 \frac{sx}{r + (s - r)x} &= x \\
 sx &= (r + (s - r)x)x \\
 sx &= rx + (s - r)x^2 \\
 -(s - r)x^2 + (s - r)x &= 0 \\
 (s - r)x(-x + 1) &= 0.
 \end{aligned}$$

Notice that if $r = s$, EVERY value of x is an equilibrium! In this case, the mutant bacteria and the wild-type reproduce at exactly the same rate, and so the fraction of mutants stays constant for all time: $p_n = p_0$.

Suppose that $r \neq s$. In this case, there are only two equilibria for the system: $p^* = 0, 1$. As in our original example, the only possible equilibria are the cases where the population is entirely wild-type ($p^* = 0$) or entirely mutant ($p^* = 1$). To determine stability, we need the derivative

$$f'(x) = \frac{rs}{(r + (s - r)x)^2}.$$

Remember that r and s are parameters (and should be considered constants when taking the derivative). Substituting the equilibria values $x = 0$ and $x = 1$ gives

$$\begin{aligned} f'(0) &= \frac{s}{r} \\ f'(1) &= \frac{r}{s}. \end{aligned}$$

Suppose that $s > r$ which means the mutants reproduce faster than the wild-type. In this case we see that 0 is unstable while 1 is asymptotically stable. Again, this makes perfect sense. If the mutants reproduce faster, they will eventually overwhelm the population.

If $s < r$, the mutants reproduce slower than the wild-type. Now 0 is asymptotically stable and 1 is unstable. In this case, the faster reproducing wild-type bacteria will out perform the mutants, and the population will eventually be dominated by wild-types.

1.5.2 A Simple Disease Model

There are a number of sophisticated disease models, and we will explore several more throughout the text. For a first example, we will make the simplifying assumptions that the size of the population being studied is not changing and that there can only be healthy people and infected people. More complex models can account for changing population sizes as well as the existence of people who are immune to the disease.

Example 1.5.2. There is a disease afflicting a population. While the numbers of healthy and infected individuals change each month, we will assume the total population size is constant. Each month, 25% of the infected recover while 20% of the healthy population become infected. We wish to model the total fraction P_n of infected people in the population.

In this simple model, there are two types of people, healthy and infected. Let H_n be the total number of healthy people in the population and I_n be the total number of infected people in a given month n . The fraction of people that are infected is therefore

$$P_n = \frac{I_n}{H_n + I_n}.$$

As in the previous section, $0 \leq P_n \leq 1$ (and once again we can naturally think of it as the percentage of people who are infected). Also, since the total population is assumed to be constant, we have

$$H_{n+1} + I_{n+1} = H_n + I_n.$$

From the given information, we know that

$$I_{n+1} = 0.75I_n + 0.2H_n$$

$$H_{n+1} = 0.25I_n + 0.8H_n.$$

Note that these equations are consistent with the total population being constant.

Using this equation, we find

$$\begin{aligned} P_{n+1} &= \frac{I_{n+1}}{H_{n+1} + I_{n+1}} \\ &= \frac{0.75I_n + 0.2H_n}{H_n + I_n} \\ &= 0.75 \left(\frac{I_n}{H_n + I_n} \right) + 0.2 \left(\frac{H_n}{H_n + I_n} \right) \\ &= 0.75P_n + 0.2(1 - P_n) \\ &= 0.2 + 0.55P_n. \end{aligned}$$

In the next to last step, we used the fact that the fraction of healthy people must be $1 - P_n$.

So, the updating function for this dynamical system is $f(x) = 0.2 + 0.55x$ giving an equilibrium value of

$$\begin{aligned} 0.2 + 0.55x &= x \\ 0.2 &= 0.45x \\ x &= \frac{0.2}{0.45} = \frac{4}{9}. \end{aligned}$$

Since the derivative of the updating function is $f'(x) = 0.55$, we know that this equilibrium is asymptotically stable. This tells us that in the long run, we expect $4/9 \approx 44.44\%$ of the population to be infected each month. A disease that is present in a stable fraction of a population is said to be *endemic* to the population. ■

As before, we want to generalize this example. Suppose now that r is the fraction of infected people who recover from the disease while s is the fraction of healthy people who become infected. Both r and s are parameters in our model that lie between 0 and 1. We have

$$\begin{aligned} I_{n+1} &= (1-r)I_n + sH_n \\ H_{n+1} &= rI_n + (1-s)H_n, \end{aligned}$$

and the total population is constant

$$I_{n+1} + H_{n+1} = I_n + H_n.$$

Letting P_n be the fraction of the population that is infected and repeating our calculation from above leads to

$$\begin{aligned} P_{n+1} &= \frac{I_{n+1}}{H_{n+1} + I_{n+1}} \\ &= \frac{(1-r)I_n + sH_n}{H_n + I_n} \\ &= (1-r) \left(\frac{I_n}{H_n + I_n} \right) + s \left(\frac{H_n}{H_n + I_n} \right) \\ &= (1-r)P_n + s(1-P_n) \\ &= s + (1-r-s)P_n. \end{aligned}$$

The updating function is $f(x) = s + (1-r-s)x$ and the only equilibrium is

$$P^* = \frac{s}{s+r}.$$

The derivative of the updating function is $f'(x) = 1-r-s$. At one extreme, when $r = s = 0$, we have $f'(x) = 1$ and our derivative test is inconclusive. In this case, infected people never recover but healthy people never become infected. At the other extreme, $r = s = 1$ implies $f'(x) = -1$ (and we have an inconclusive case again). In this extreme case, EVERY infected person recovers each time period. Simultaneously, EVERY healthy person becomes infected.

In any reasonable case, $1-r-s$ will lie between -1 and 1 and the equilibrium will be asymptotically stable. If $1 < r+s < 2$, the asymptotically stable equilibrium will be a spiral point. In any interesting case, the disease will always be endemic to the population in this simple model.

1.5.3 A Simple Model of the Heart

In this section, we will examine a fairly simplistic model of the heart which nonetheless can predict certain cardiac conditions seen in patients. See [Ad13, pp. 126 – 131] for a more thorough discussion.

The heart’s pumping action is controlled by a complex electrical system. Part of the system, the *sinoatrial node*, provides “regular” pulses to the heart – acting as the body’s natural pacemaker. The actual pace set by the sinoatrial node (SA node) changes in response to a number of physiological conditions. For our purposes, we will assume the SA node is sending out regularly spaced pulses.

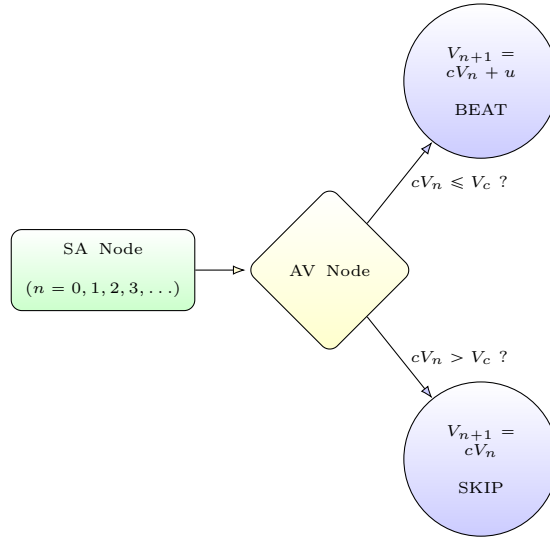
The second major part of the system is the atrioventricular node (AV node). This nerve cluster has several jobs, but most importantly for us, it monitors the condition of the heart and decides if it is ready to beat. In order for the heart muscle to contract, the voltage of the heart must be increased to a sufficiently high level.³ In between beats, the heart needs to relax (i.e. the voltage must subside), and the AV node decides if the heart has relaxed enough to undergo the next beat. If the voltage has decreased enough by the time of the next pulse from the SA node, the AV node allows the heart to beat. Otherwise, the AV node stops the heart from beating in order to prevent damage to the heart muscle.

Our simple model of the heart is as follows.

- 1.) The pulses from the SA node are assumed to be regular, and they will correspond to our discrete time steps $n = 0, 1, 2, 3, 4, \dots$
- 2.) We will model V_n : the overall voltage of the heart muscle at the n -th pulse from the SA node.
- 3.) There is a *critical voltage*, V_c , such that if the voltage of the heart muscle is below this value, the AV node will allow the heart to beat. Otherwise, the AV node prevents the heart from beating.
- 4.) In between beats, the voltage of the heart muscle decreases by a factor c , the *relaxation rate*, which is a number in the range $0 < c < 1$. The closer this number is to 0, the faster the heart recovers. The closer it is to 1, the slower the heart is to relax.

³For our purposes, we will assume the heart simply contracts. The heart’s motion is actually more complex than this, but that will not matter for our purposes.

- 5.) If the heart has relaxed enough to beat, the AV node causes the voltage of the heart to increase by an amount u , the *beat voltage*.



These assumptions give us the following dynamical system

$$V_{n+1} = \begin{cases} cV_n, & \text{if } cV_n > V_c \\ cV_n + u, & \text{if } cV_n \leq V_c \end{cases}.$$

Example 1.5.3. Consider our simple heart model with parameters $c = \frac{2}{5}$, $V_c = 1$, and $u = 1$. Suppose that the initial voltage is $V_0 = 1$. This gives us the dynamical system

$$V_{n+1} = \begin{cases} \frac{2}{5}V_n, & \text{if } \frac{2}{5}V_n > 1 \\ \frac{2}{5}V_n + 1, & \text{if } \frac{2}{5}V_n \leq 1 \end{cases}$$

$$V_0 = 1.$$

To determine V_1 , we first check the condition on $\frac{2}{5}V_0$:

$$\frac{2}{5}V_0 = \frac{2}{5}(1) = \frac{2}{5} < 1.$$

Since this is less than the critical voltage, $V_c = 1$, the heart will beat and

$$V_1 = \frac{2}{5}V_0 + 1 = \frac{7}{5}.$$

To determine V_2 , we check the condition on $\frac{2}{5}V_1$:

$$\frac{2}{5}V_1 = \frac{2}{5} \left(\frac{7}{5} \right) = \frac{14}{25} < 1.$$

Since this is less than the critical voltage, the heart will beat and

$$V_2 = \frac{2}{5}V_1 + 1 = \frac{39}{25}.$$

Repeating this process, we can determine the voltage for as many pulses of the SA node as we wish. For example, since

$$\frac{2}{5}V_2 = \frac{2}{5} \left(\frac{39}{25} \right) = \frac{78}{125} < 1,$$

the heart will beat. Thus

$$V_3 = \frac{2}{5}V_2 + 1 = \frac{203}{125}.$$

If we look at the cobweb plot for this dynamical system, we find

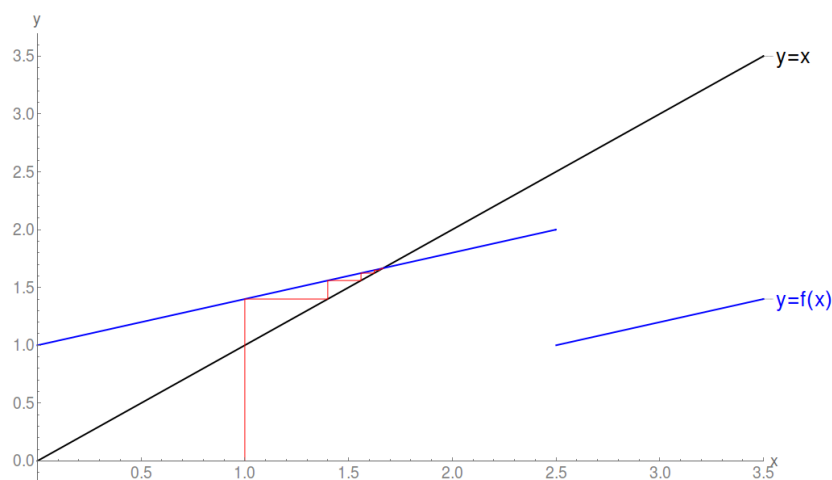


Figure 1.5.1: Cobweb Plot for Example 1.5.3

The upper branch of the updating function corresponds to the heart beating ($V_{n+1} = cV_n + u$) while the branch in the lower right indicates that the heart skips a beat ($V_{n+1} = cV_n$). Also notice that there is a single equilibrium. We will revisit the existence of equilibria for this model a little later.

■

Since

$$\frac{2}{5}V_0 = \frac{2}{5}(3) = \frac{6}{5} > 1,$$

the heart will skip a beat and

$$V_1 = \frac{2}{5}V_0 = \frac{6}{5}.$$

For the next step,

$$\frac{2}{5}V_1 = \frac{2}{5} \binom{6}{5} = \frac{12}{25} < 1,$$

and so the heart will beat. This gives

$$V_2 = \frac{2}{5}V_1 + 1 = \frac{37}{25}.$$

Examining the cobweb plot shows that the heart will beat for every pulse of the SA node after the first one.

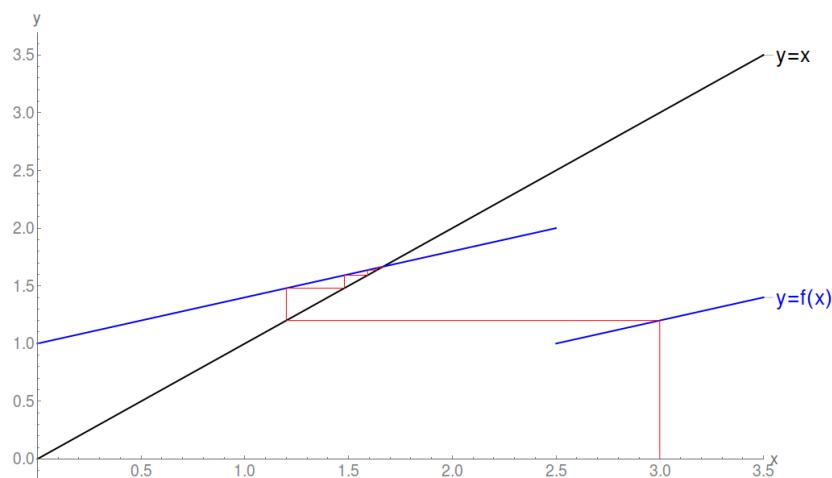


Figure 1.5.2: Cobweb Plot for Example 1.5.4

We can clearly see that the heart skips the first beat since it first hits the lower right branch of the updating function. Thereafter, it always hits the upper left branch – which indicates regular beating. ■

Example 1.5.5. Consider the heart model with parameters $c = 0.65$, $V_c = 1$, and $u = 1$ with initial data $V_0 = 1.75$. We will present the values of the system in a table.

Condition Check	V_n	Beat/Skip
	$V_0 = 1.75$	
$0.65(V_0) = 1.1375 > 1$	$V_1 = 0.65(V_0) = 1.1375$	SKIP
$0.65(V_1) \approx 0.7394 < 1$	$V_2 = 0.65(V_1) + 1 \approx 1.7394$	BEAT
$0.65(V_2) \approx 1.1306 > 1$	$V_3 = 0.65(V_2) \approx 1.1306$	SKIP
$0.65(V_3) \approx 0.7349 < 1$	$V_4 = 0.65(V_3) + 1 \approx 1.7349$	BEAT
$0.65(V_4) \approx 1.1277 > 1$	$V_5 = 0.65(V_4) \approx 1.1277$	SKIP
$0.65(V_5) \approx 0.7330 < 1$	$V_6 = 0.65(V_5) + 1 \approx 1.7330$	BEAT

From the table, we can see that the heart seems to be skipping every other beat! The cobweb plot confirms this.

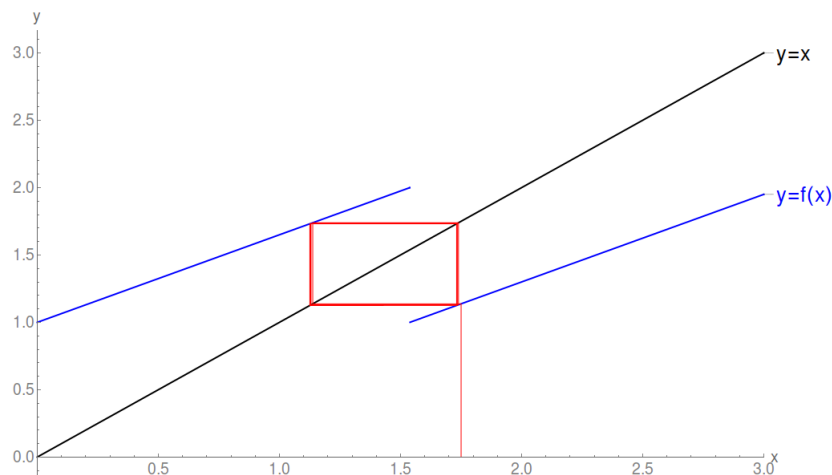


Figure 1.5.3: Cobweb Plot for Example 1.5.5

When the heart is in a rhythm where every other pulse results in a skipped beat, we say the heart is experiencing a **2:1 AV Block**. ■

Before moving on, we should discuss equilibria for the heart model. The updating function is

$$f(x) = \begin{cases} cx, & \text{if } cx > V_c \\ cx + u, & \text{if } cx \leq V_c \end{cases}.$$

In order to solve $f(x) = x$, we have to examine each possible branch of the updating function.

If $cx > V_c$, then we would need

$$cx = x,$$

but this is impossible unless $x = 0$ (recall that $0 < c < 1$). However $x = 0$ will certainly belong to the other branch! Hence, *there is no equilibrium associated to the branch where the heart skips a beat.*

If $cx \leq V_c$, we would then need $cx + u = x$. This is easily solved to give

$$x = \frac{u}{1 - c}.$$

However, this will only be an equilibrium if this value of x satisfies the condition for this branch (i.e. the condition that the heart will beat). So, we will need to check that

$$\frac{cu}{1 - c} \leq V_c.$$

Lemma 1.5.1. *For the heart model with updating function*

$$f(x) = \begin{cases} cx, & \text{if } cx > V_c \\ cx + u, & \text{if } cx \leq V_c \end{cases},$$

there will be an equilibrium at

$$V^* = \frac{u}{1 - c}$$

provided that

$$\frac{cu}{1 - c} \leq V_c.$$

Since the derivative of the updating function will be $f'(x) = c$ on the branch where the equilibrium occurs, the equilibrium must be asymptotically stable if it exists.

For our first two examples above, there is an equilibrium at

$$V^* = \frac{u}{1-c} = \frac{1}{1-2/5} = \frac{5}{3}$$

because

$$\frac{cu}{1-c} = \frac{2}{5} \left(\frac{5}{3} \right) = \frac{2}{3} \leq 1 = V_c.$$

However, for the third model

$$V^* = \frac{u}{1-c} = \frac{1}{1-0.65} = \frac{20}{7}$$

but

$$\frac{cu}{1-c} = 0.65 \left(\frac{20}{7} \right) \approx 1.8571 > 1 = V_c.$$

This means that there is no equilibrium for this set of parameters. Note that any set of parameters will produce a possible equilibrium value, but you must check the condition to ensure that the potential equilibrium value is actually present in the system.

For our simple model of the heart, there are three generic types of behavior we can discern:

- 1.) If the system has an equilibrium, then eventually the heart will settle into a steady rhythm where it beats on every pulse from the SA node. We will refer to this as **steady rhythm**.
- 2.) The system has no equilibrium, and eventually the heart settles into a rhythm where it skips every other beat from the SA node. This is referred to as a **2:1 AV Block**.
- 3.) The system has no equilibrium. The heart eventually settles into a rhythm where it will beat regularly for some number of beats and then suddenly skip a beat. This is referred to as the **Wenckebach Phenomenon**.

We have already seen cases of steady rhythm and the 2:1 AV Block. Our final example exhibits the Wenckebach Phenomenon.

Example 1.5.6. Consider the heart model with $c = 0.5$, $u = 1.5$, and $V_c = 1.4$. So, we have

$$V_{n+1} = \begin{cases} 0.5V_n, & \text{if } 0.5V_n > 1.4 \\ 0.5V_n + 1.5, & \text{if } 0.5V_n \leq 1.4 \end{cases}.$$

If we check for an equilibrium, we find

$$\frac{u}{1-c} = \frac{1.5}{1-0.5} = 3.$$

However, when we check the condition for the equilibrium to exist we find

$$\frac{cu}{1-c} = \frac{0.5(1.5)}{1-0.5} = 1.5 > 1.4 = V_c.$$

So there is no equilibrium, but the condition is *almost* satisfied.

Consider the cobweb plot for this system with $V_0 = 2$.

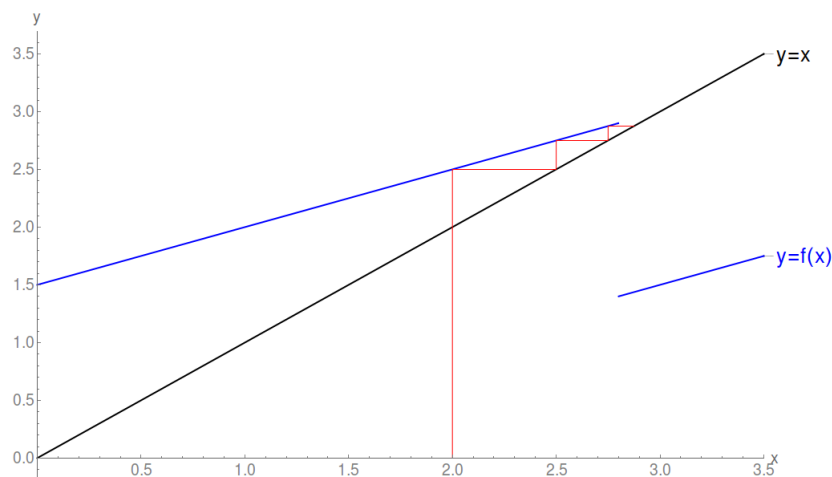


Figure 1.5.4: Cobweb Plot for Example 1.5.6

Notice that the updating function nearly intersects $y = x$, but the jump to the lower branch occurs just before the point of intersection. After the third step in the cobweb plot, the system looks as though it is trying to limit to the point of intersection. However, the fourth step will result in a skipped beat.

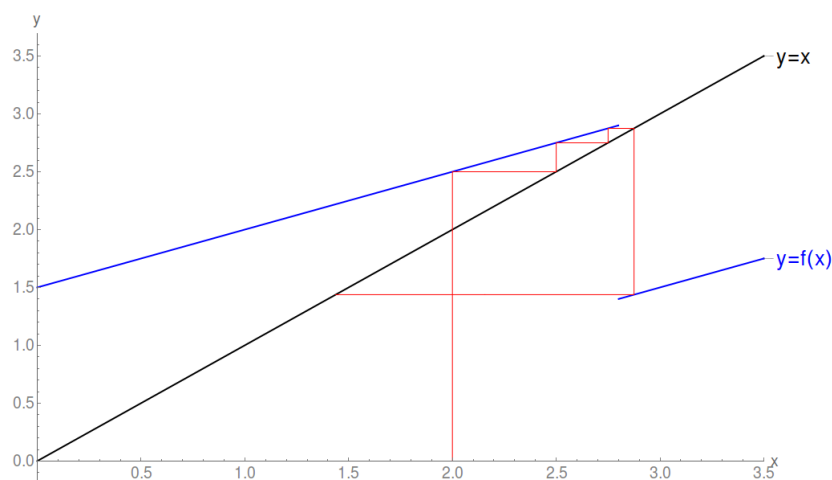


Figure 1.5.5: Cobweb Plot for Example 1.5.6, continued

If we follow the system longer, we see that the heart will beat 3 times in a row and then skip a beat, and it will continue this pattern indefinitely.

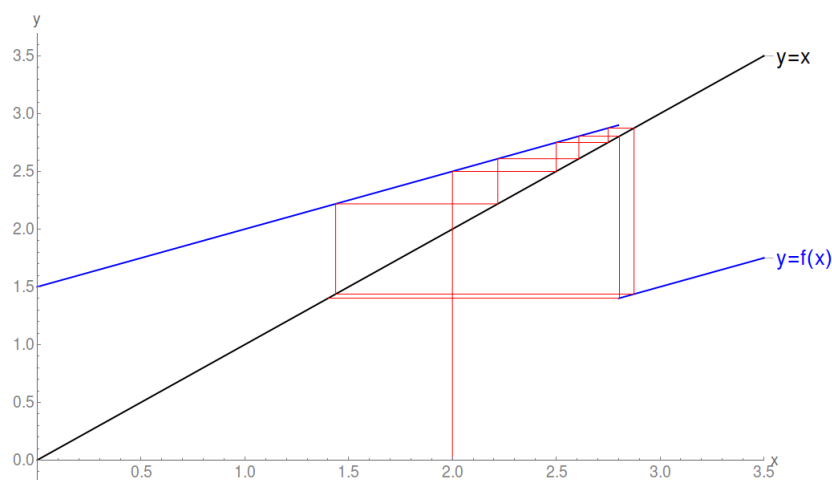


Figure 1.5.6: Cobweb Plot for Example 1.5.6, continued

This is a typical example exhibiting the Wenckebach Phenomenon; a system that nearly satisfies the condition for having an equilibrium can beat normally for several pulses of the SA node but then skip a single beat when it overshoots the discontinuity at $V = V_c/c$ which is just before the value where the equilibrium would exist.

1.5.4 The Discrete Logistic Map

The final discrete-time dynamical system we wish to discuss is the **discrete logistic model** also known as the **logistic map**. This is the dynamical system with updating function

$$f(x) = \lambda x(1 - x)$$

where λ is a parameter in the range $0 < \lambda \leq 4$. This system is a discretized version of the continuous logistic growth model which is discussed in Section 4.8.

For the logistic map, we think of a population living in some region which can support a maximum population size. The variable x in the range $0 \leq x \leq 1$ represents the ratio of the current population size to the maximum size. The restriction on the size of x is the primary reason that λ is restricted to the range given above. Note that the quadratic function $p(x) = x(1 - x)$ is non-negative over $0 \leq x \leq 1$, that its graph is concave down over this interval, and that the function has an absolute maximum at $x = 1/2$. Moreover, $p(1/2) = 1/4$.

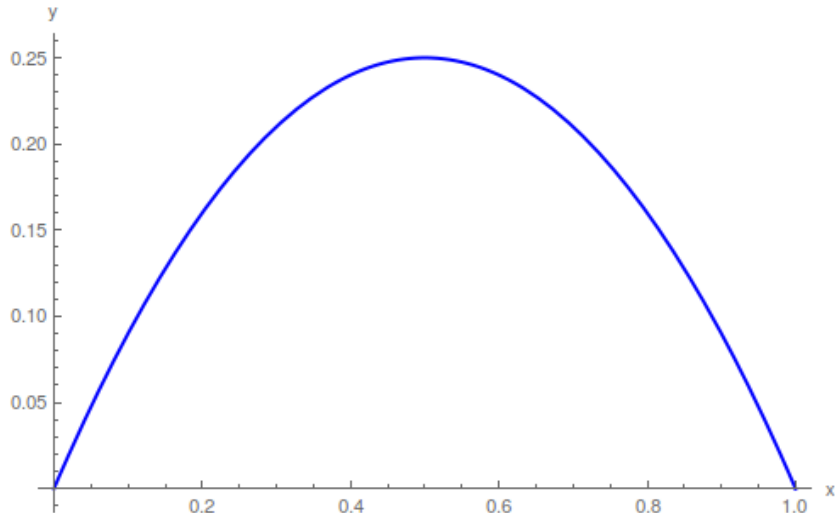


Figure 1.5.7: Plot of $y = x(1 - x)$

So, if we multiply p by any factor in the range $0 < \lambda \leq 4$, we are guaranteed that the output of the function will lie between 0 and 1 if the input is also between 0 and 1. Once we allow a factor that is negative or

greater than 4, an input of $x = 1/2$ will lead to an output that is no longer in the range of interest.

Example 1.5.7. Consider the logistic map with parameter $\lambda = 2$:

$$q_{n+1} = 2q_n(1 - q_n).$$

If we look for equilibria, we find

$$2x(1 - x) = x$$

$$2x - 2x^2 = x$$

$$x - 2x^2 = 0$$

$$x(1 - 2x) = 0$$

$$x = 0, \frac{1}{2}.$$

The first derivative of the updating function is $f'(x) = 2 - 4x$. Since $f'(0) = 2$, we know that $q^* = 0$ is an unstable equilibrium. The equilibrium at $q^* = 1/2$ is asymptotically stable since $f'(1/2) = 0$. The cobweb plot below (using $q_0 = 0.05$) clearly shows the behavior of both of these equilibria. ■

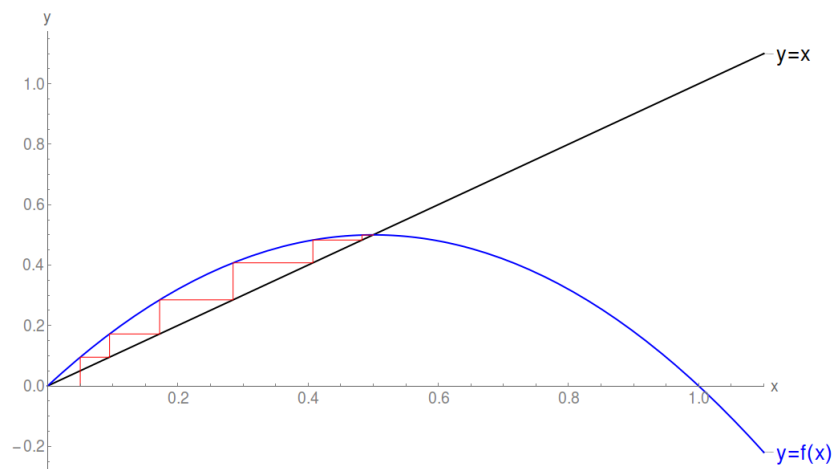


Figure 1.5.8: Cobweb Plot for the Logistic Map with $\lambda = 2$

Example 1.5.8. Consider the logistic map with parameter $\lambda = 5/2$:

$$q_{n+1} = \frac{5}{2}q_n(1 - q_n).$$

If we look for equilibria, we find

$$\begin{aligned}\frac{5}{2}x(1-x) &= x \\ \frac{5}{2}x - \frac{5}{2}x^2 &= x \\ \frac{3}{2}x - \frac{5}{2}x^2 &= 0 \\ \frac{1}{2}x(3-5x) &= 0 \\ x &= 0, \frac{3}{5}.\end{aligned}$$

The first derivative of the updating function is $f'(x) = 5/2 - 5x$. Since $f'(0) = 5/2$, we know that $q^* = 0$ is an unstable equilibrium. The equilibrium at $q^* = 3/5$ is an asymptotically stable spiral point since $f'(3/5) = -1/2$. The cobweb plot below (using $q_0 = 0.05$) clearly shows the behavior of both of these equilibria. ■

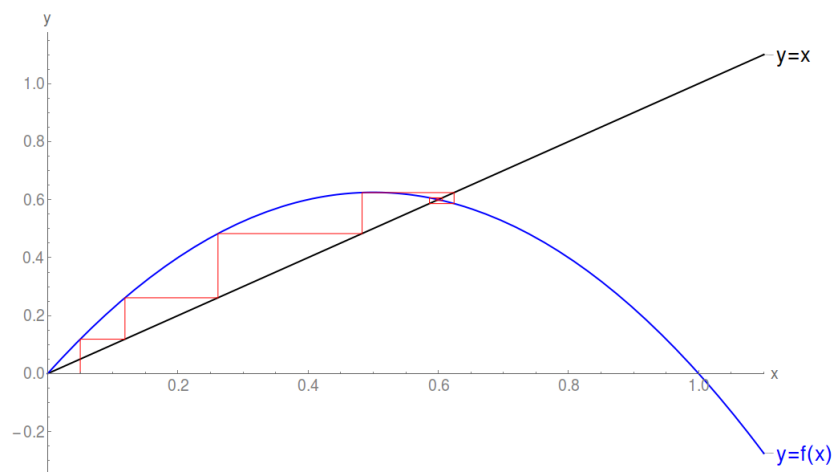


Figure 1.5.9: Cobweb Plot for the Logistic Map with $\lambda = 5/2$

What we would like to do is to classify the behavior of the system for any value of the parameter λ . We will only present partial results about this really fascinating dynamical system. For a more thorough discussion, see [Ro95, Chapter 2] and [HK91, pp. 92 – 100].

First, we find the equilibria for the logistic map.

Lemma 1.5.2 (Equilibria of the Logistic Map). *For the logistic map*

$$q_{n+1} = \lambda q_n(1 - q_n)$$

with $0 < \lambda \leq 4$, $q^ = 0$ is an equilibrium value for any choice of λ . For the parameter range $0 < \lambda \leq 1$, 0 is the only equilibrium in the range of interest ($0 \leq x \leq 1$). For the range $1 < \lambda \leq 4$, there is a second equilibrium in the range of interest at*

$$q^* = \frac{\lambda - 1}{\lambda}.$$

Proof: Since the updating function is $f(x) = \lambda x(1 - x)$, the equilibria are

$$\begin{aligned}\lambda x(1 - x) &= x \\ \lambda x - \lambda x^2 &= x \\ (\lambda - 1)x - \lambda x^2 &= 0 \\ x &= 0, \frac{\lambda - 1}{\lambda}.\end{aligned}$$

Since we only want to consider the range $0 \leq x \leq 1$, the second equilibrium will only be of interest when it is a positive number (note that it is necessarily less than 1). So, the second equilibrium is of interest only when $\lambda > 1$. ■

Now that we have the equilibria, we need to classify them.

Lemma 1.5.3 (Stability of $q^* = 0$). *For $0 < \lambda < 1$, $q^* = 0$ is asymptotically stable for the logistic map. For $\lambda = 1$, $q^* = 0$ is semi-stable, but it is stable from the right (i.e. it is asymptotically stable over the interval of interest $0 \leq x \leq 1$). For $\lambda > 1$, $q^* = 0$ is unstable for the logistic map.*

Proof: The derivative of the updating function is $f'(x) = \lambda - 2\lambda x$, and so $f'(0) = \lambda$. When $0 < \lambda < 1$, the first derivative test for discrete-time dynamical systems tells us that 0 is asymptotically stable, and when $\lambda > 1$, the test tells us it is unstable. The test is inconclusive for $\lambda = 1$, but checking the cobweb plot in this case verifies the assertion of the lemma. ■

Lemma 1.5.4 (Stability of $q^* = (\lambda - 1)/\lambda$). *For the logistic map, the equilibrium at $q^* = (\lambda - 1)/\lambda$ is asymptotically stable for the parameter range $1 < \lambda \leq 3$. Moreover, it is an asymptotically stable spiral point for $2 < \lambda \leq 3$. For $3 < \lambda \leq 4$, this equilibrium is an unstable spiral point.*

Proof: The first derivative evaluated at this equilibrium value is $f'(q^*) = 2 - \lambda$. In order to have an asymptotically stable equilibrium, this value needs to be between -1 and 1.

$$\begin{aligned} -1 &< 2 - \lambda < 1 \\ -3 &< -\lambda < -1 \\ 1 &< \lambda < 3. \end{aligned}$$

The behavior at $\lambda = 3$ can be confirmed by cobweb plot. As soon as $\lambda > 3$, the equilibrium is definitely an unstable spiral point. To determine the range where the equilibrium is an asymptotically stable spiral point, we need $2 - \lambda$ to be between -1 and 0.

$$\begin{aligned} -1 &< 2 - \lambda < 0 \\ -3 &< -\lambda < -2 \\ 2 &< \lambda < 3. \end{aligned}$$

The behavior at $\lambda = 2$ can be confirmed by cobweb plot. ■

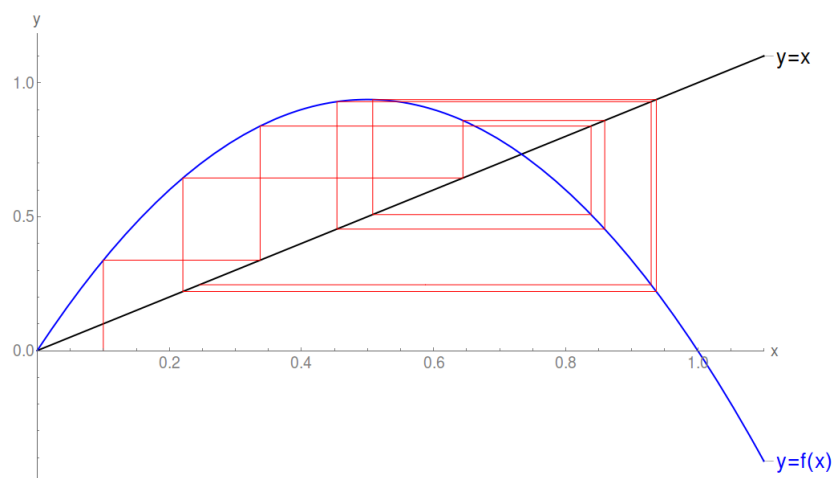
We can characterize these results in the table below:

Equilibria of the Logistic Map

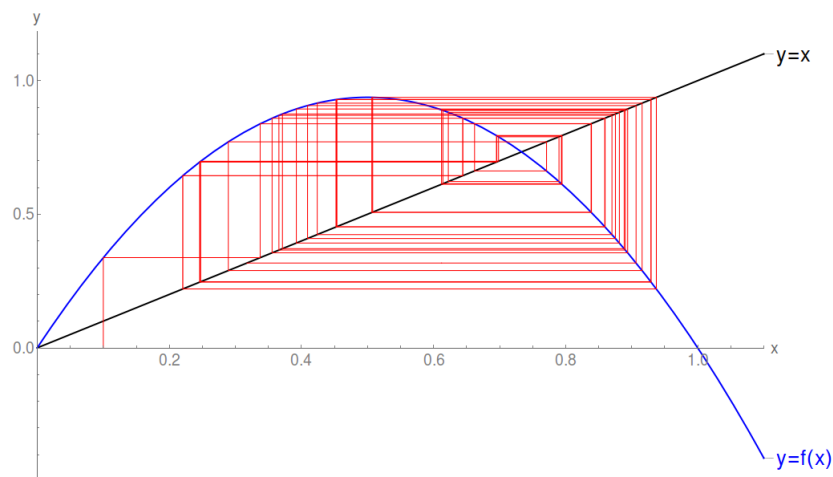
Range of λ	$q^* = 0$	$q^* = \frac{\lambda-1}{\lambda}$
$0 < \lambda \leq 1$	Asymptotically Stable	NONE
$1 < \lambda \leq 2$	Unstable	Asymptotically Stable
$2 < \lambda \leq 3$	Unstable	Asymptotically Stable Spiral
$3 < \lambda \leq 4$	Unstable	Unstable Spiral

The really interesting behavior in this system is in the parameter range $3 < \lambda \leq 4$. For these values of λ , there are no stable equilibria, but all q_n are trapped in the range $0 \leq q_n \leq 1$. This means the values of the dynamical system can bounce around the interval $[0, 1]$ in complex ways. In fact, when $\lambda > 3.57$, the system exhibits *chaotic behavior*.

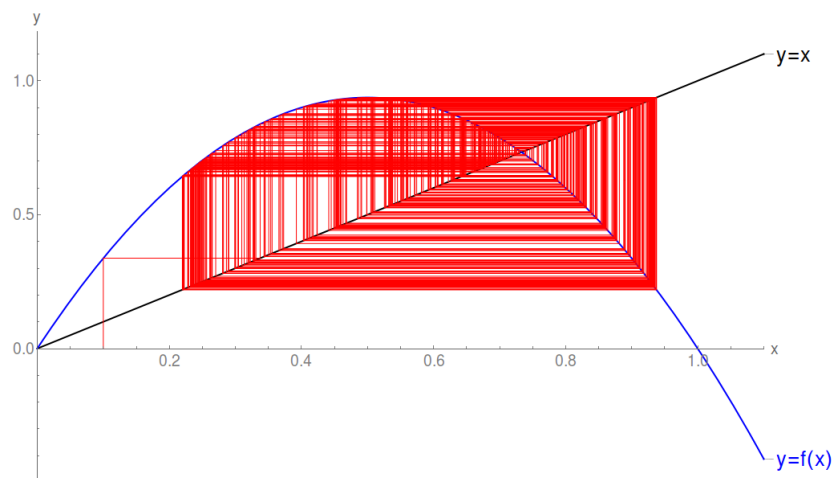
Example 1.5.9. Consider the logistic map with $\lambda = 3.75$. From our results, we know that $q^* = 0$ is unstable and $q^* = 11/15$ is an unstable spiral point. Consider the cobweb plot with $q_0 = 0.1$ out to 10 steps.

Figure 1.5.10: Cobweb Plot for the Logistic Map with $\lambda = 3.75$

We can see that there is a tension between the unstable equilibrium at 0 and the one at $11/15$. If we extend the plot to 50 steps, we can see that the dynamics are quite complicated.

Figure 1.5.11: Cobweb Plot for the Logistic Map with $\lambda = 3.75$, continued

As we extend the cobweb plot for more and more steps, the dynamics begins to fill up a solid region of the plot.

Figure 1.5.12: Cobweb Plot for the Logistic Map with $\lambda = 3.75$, continued

This is a hallmark of chaotic dynamics. ■

1.6 Problems

- 1.) Suppose that a discrete-time dynamical system is given by

$$q_{n+1} = \frac{3}{2}q_n.$$

- a.) If $q_0 = 16$, compute q_1 through q_4 .
 - b.) Find a direct formula for q_n in terms of q_0 . Use this formula to predict q_{15} when $q_0 = 16$.
 - c.) This system is time reversible. Find a formula for q_n in terms of q_{n+1} .
 - d.) If $q_4 = 81$, find q_0 .
- 2.) Suppose that a discrete-time dynamical system is given by

$$L_{n+1} = 3L_n - 2.$$

- a.) If $L_0 = 2$, compute L_1 through L_4 .
- b.) Find a direct formula for L_n in terms of L_0 . Use this formula to predict L_{10} if $L_0 = 2$.

c.) This system is time reversible. Find a formula for L_n in terms of L_{n+1} .

d.) If $L_4 = 163$, find L_0 .

3.) Consider the dynamical system given by

$$\begin{aligned} p_{n+1} &= p_n^2 \\ p_0 &= \frac{1}{2} \end{aligned}$$

a.) Compute p_1 through p_4 .

b.) Find a direct formula for p_n in terms of n . Use this formula to predict p_{10} .

c.) Is this system time reversible? If so, find a formula for p_n in terms of p_{n+1} .

4.) Consider the dynamical system whose updating function is the Collatz function (c.f. Example 1.1.9).

a.) If the initial data is $q_0 = 4$, find q_1 through q_5 . Do you spot a pattern? (You should find an example of what is known as a 3-cycle.)

b.) Show that with initial data $q_0 = 23$, the system eventually enters the 3-cycle you found in part a.). How many steps did it take to reach 1 the first time?

c.) Suppose the system is currently in state $q_{n+1} = 7$. Is it possible to uniquely determine q_n ? What if $q_{n+1} = 8$?

5.) A population of bacteria is doubling every 10 hours. If we measure time in 10 hour blocks, and there are initially 500 organisms in the petri dish, how many 10 hour blocks until there are more than 1,000,000 of the bacteria? (Remember that the time must be an integer in this model.)

6.) In the absence of fishing, the population of trout in a lake will increase by 50% every 2 years. In the same time period, 15,000 fish are removed by fishing. If the lake is stocked with 35,000 trout, what will the population be in 20 years? What would happen to the population if the lake was stocked with only 25,000 fish?

- 7.) Find and determine the stability of all equilibria for the dynamical system given by

$$q_{n+1} = 3q_n - 1.$$

Produce a cobweb plot to verify the stability of each equilibrium.

- 8.) Find and determine the stability of all equilibria for the dynamical system in problem 6.) above. Produce a cobweb plot to verify the stability of each equilibrium.

- 9.) Find and determine the stability of all equilibria for the dynamical system given by

$$L_{n+1} = \frac{1}{2}L_n(1 - L_n).$$

Produce a cobweb plot to verify the stability of each equilibrium.

- 10.) Find and determine the stability of all equilibria for the dynamical system given by

$$M_{n+1} = M_n + \frac{5}{10 + M_n}M_n + 1.$$

Produce a cobweb plot to verify the stability of each equilibrium.

- 11.) Find and determine the stability of all equilibria for the dynamical system given by

$$P_{n+1} = \frac{1}{4}P_n^3 - \frac{1}{8}P_n^2 + \frac{3}{4}P_n + \frac{1}{8}.$$

Produce a cobweb plot to verify the stability of each equilibrium.

- 12.) Consider the mutant vs. wild-type model. Suppose the mutants are doubling every 5 hours while the wild-type population grows to 2.5 times their size in the same period. If originally the population is made up of 20% mutants, how long until the population is only 5% mutant.
- 13.) Consider the mutant vs. wild-type model. If the mutants are reproducing at half the rate of the wild-type, determine any equilibria for the fraction of mutants and classify their stability.

- 14.) Consider our simple disease model. Suppose that each time period 80% of infected people recover while 25% of healthy people become infected. At what fraction should the population be infected in the long run? Do you expect the population to limit directly to this value or oscillate around it?
- 15.) Consider our simple disease model. If the rate people recover from the disease is double the rate healthy people become infected, what fraction of the population should be infected in the long run? For what values of s will the equilibrium be an asymptotically stable spiral point? (HINT: What is the maximum allowable value of s ?)
- 16.) A certain species of bird lives on two neighboring islands. Each year, 30% of the birds on Island 1 migrate to Island 2 while 35% of the birds on Island 2 migrate to Island 1. Let P_t be the fraction of birds living on Island 1 in year t . Assume that the total number of birds each year is constant. Find the updating function for P_t and determine if there are any equilibrium fractions for the system. Be sure to examine the stability of any equilibria you find.
- 17.) Consider the heart model with parameters $c = 0.75$, $u = 0.75$, and $V_c = 2$. Determine the long-term behavior of the system (i.e. classify it as one of the three possibilities discussed for this model).
- 18.) Consider the heart model with parameters $c = 0.5$, $u = 1$, and $V_c = 2$. Determine the long-term behavior of the system (i.e. classify it as one of the three possibilities discussed for this model).
- 19.) Consider the heart model with parameters $c = 0.5$, $u = 1.5$, and $V_c = 1$. Determine the long-term behavior of the system (i.e. classify it as one of the three possibilities discussed for this model).
- 20.) One way to account for a change in the pace set by the SA node in the heart model is to allow the relaxation rate c to change over time. In fact, the relaxation time is given by

$$c = e^{-\alpha\tau}$$

where α is a positive parameter and τ is the time between pulses from the SA node.

Suppose that for a given heart, the increase in voltage when the heart beats is $u = 10$ and the critical voltage is $V_c = 15$. When the SA node is sending out pulses every 0.85 seconds, the relaxation rate is $c = 0.5$. What is the shortest time between pulses from the SA node for which the system can exhibit a steady rhythm?

- 21.) Find and classify all equilibria in the range $0 \leq x \leq 1$ for the logistic map with parameter $\lambda = 3/2$.
- 22.) Find and classify all equilibria in the range $0 \leq x \leq 1$ for the logistic map with parameter $\lambda = 18/5$.
- 23.) The behavior of equilibria should reverse when time is reversed. Consider the dynamical system given by

$$q_{n+1} = \frac{2q_n}{1 + 3q_n}$$

- a.) Find all equilibria for this system and determine their stability.
- b.) Find the time-reversed dynamical system. That is solve the system above for q_n to find

$$q_n = f^{-1}(q_{n+1}).$$

- c.) Consider the discrete-time dynamical system with updating function f^{-1} where f^{-1} was found in part b.). Verify that this system has the same equilibria as the original but that the stability results have been reversed.
- 24.) Consider the Ricker model which has updating function

$$f(x) = rxe^{-x}.$$

The parameter r is strictly positive.

- a.) Find the equilibria for the system. You will get different answers for different ranges of the parameter r .
- b.) Classify the stability of the equilibria you found above (again, this will depend on the range of the parameter r). For any indeterminate cases, use cobweb plots to justify your answer!

- 25.) **Challenge:** Consider the discrete-time dynamical system with updating function

$$f(x) = \alpha x^2(1 - x).$$

- a.) Determine the appropriate range of the parameter α that will guarantee an input in the range $0 \leq x \leq 1$ will generate an output in the same range.
- b.) Find the equilibria for the system. Note that you will get different results for different ranges of the parameter α !
- c.) Classify the stability of the equilibria you found above. For any indeterminate cases, use cobweb plots to justify your answer!

Chapter 2

Techniques of Integration

Before we can move on to continuous dynamical systems, we need to introduce a couple of integration techniques not commonly taught in first semester calculus courses. We begin with a brief review of integration by substitution (in case that was omitted from your first calculus course). See [Appendix B](#) for a list of common derivative and integration rules that you should be familiar with.

The new integration techniques are integration by parts and integration by partial fraction decomposition. Integration by parts is the integral rule associated to the product rule for derivatives and allows us to find the anti-derivatives of a great variety of functions. Integration by partial fractions allows us to integrate any rational function (at least in principle). The primary limitation with this method is that it requires us to factor polynomials, and polynomials of degree larger than 5 can be hard to manage!

Integration is quite difficult in general. While we can differentiate most any combination of polynomials, exponential and logarithmic functions, trigonometric functions, etc., there are examples of fairly simple functions whose anti-derivatives cannot be written in terms of elementary functions. For instance, the anti-derivative

$$\int e^{-x^2} dx$$

exists (the area under $y = e^{-x^2}$ is certainly a sensible quantity), but it is not any combination of functions you are likely to have encountered before!

Since integration can be so challenging, it is important to develop good techniques and strategies to attack them. When you encounter an integral, go through the following thought process:

- 1.) Is this a basic form I should know? (Brush up on these in Appendix [B](#).)
- 2.) Is there a u -substitution that will work?
- 3.) Is the integrand a rational function? If so, try partial fraction decomposition.
- 4.) Try integration by parts.

There are many more integration techniques than the ones we cover, but the checklist above should get you through any integrals you encounter in this text.

2.1 Integration by Substitution

Since integration is the reverse process of differentiation (thanks to the Fundamental Theorem of Calculus), there is a rule for integration corresponding to every differentiation rule. **Integration by substitution** (often called **u -substitution**) corresponds to the chain rule for derivatives:

$$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x) \iff \int f(g(x))g'(x)dx = \int f(u)du$$

where $u = g(x)$ in the final integral on the right. This method hinges on being able to identify appropriate substitutions, and the only way to become comfortable with it is to practice!

Example 2.1.1. Find the anti-derivative

$$\int xe^{x^2} dx.$$

Answer: We try the substitution $u = x^2$ for which $du = 2x dx$. Since we have $x dx$ appearing in the integral, we only need to supply a factor of 2

(remember to balance it with a factor of $1/2$ outside the integral).

$$\begin{aligned}\int x e^{x^2} dx &= \frac{1}{2} \int e^{x^2} 2x dx \\ &= \frac{1}{2} \int e^u du \\ &= \frac{1}{2} e^u + C \\ &= \frac{1}{2} e^{x^2} + C.\end{aligned}$$

Notice in the final step that we replaced all u 's with x 's. ■

Quite often, functions appearing as exponents, functions inside of trigonometric functions or logarithms, or functions in the denominator of a rational expression make excellent choices for u .

Example 2.1.2. Find the anti-derivative

$$\int \frac{\cos(\ln(x))}{x} dx.$$

Answer: We try $u = \ln(x)$ giving $du = 1/x dx$.

$$\begin{aligned}\int \frac{\cos(\ln(x))}{x} dx &= \int \cos(\ln(x)) \frac{1}{x} dx \\ &= \int \cos(u) du \\ &= \sin(u) + C \\ &= \sin(\ln(x)) + C. \quad \blacksquare\end{aligned}$$

Example 2.1.3. Find the anti-derivative

$$\int \frac{x^2 + 2}{x^3 + 6x + 1} dx.$$

Answer: Trying $u = x^3 + 6x + 1$ gives $du = (3x^2 + 6)dx = 3(x^2 + 2)dx$.

Since we are only missing a factor of 3, this substitution will work.

$$\begin{aligned}
 \int \frac{x^2 + 2}{x^3 + 6x + 1} dx &= \frac{1}{3} \int \frac{1}{x^3 + 6x + 1} \cdot 3(x^2 + 2) dx \\
 &= \frac{1}{3} \int \frac{1}{u} du \\
 &= \frac{1}{3} \ln |u| + C \\
 &= \frac{1}{3} \ln |x^3 + 6x + 1| + C. \blacksquare
 \end{aligned}$$

Sometimes, a substitution requires us to replace leftover x -variables with corresponding u -variables.

Example 2.1.4. Find the anti-derivative

$$\int x\sqrt{x+1} \, dx.$$

Answer: If we try $u = x + 1$ (to make the argument under the square root more tractable), we find $du = dx$. In order to get rid of the x in front of the square root, we solve for x to find $x = u - 1$.

$$\begin{aligned}
 \int x\sqrt{x+1} \, dx &= \int (u-1)\sqrt{u} \, du \\
 &= \int (u^{3/2} - u^{1/2}) \, du \\
 &= \frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2} + C \\
 &= \frac{2}{5}(x+1)^{5/2} - \frac{2}{3}(x+1)^{3/2} + C. \blacksquare
 \end{aligned}$$

Example 2.1.5. Find the anti-derivative

$$\int \frac{1}{x + \sqrt{x}} \, dx.$$

Answer: We could try making the entire denominator equal to u , but that would lead to several difficulties. Since the square root in the denominator is the primary difficulty, we try $u = \sqrt{x}$. This gives $du = dx/(2\sqrt{x})$. If we

rearrange this, we find $2\sqrt{x}du = dx$ which looks strange until we remember that $u = \sqrt{x}$. That gives us $2u du = dx$. Similarly, $x = u^2$.

$$\begin{aligned}\int \frac{1}{x + \sqrt{x}} dx &= \int \frac{1}{u^2 + u} 2u du \\ &= 2 \int \frac{u}{u^2 + u} du \\ &= 2 \int \frac{1}{u + 1} du \\ &= 2 \ln |u + 1| + C \\ &= 2 \ln |\sqrt{x} + 1| + C. \blacksquare\end{aligned}$$

For definite integrals, we can avoid having to rewrite the integral in terms of the original variable *if we change the limits of integration according to the u -substitution*.

Example 2.1.6. Compute

$$\int_0^3 \frac{x}{x^2 + 1} dx.$$

Answer: With the substitution $u = x^2 + 1$, we have $du = 2x dx$ and so we only need to supply a factor of 2. When we rewrite the integral in terms of u , we change the limits of integration as follows:

$$\begin{aligned}x = 0 &\rightarrow u = (0)^2 + 1 = 1, \\ x = 3 &\rightarrow u = (3)^2 + 1 = 10.\end{aligned}$$

Putting all of this together gives:

$$\begin{aligned}\int_0^3 \frac{x}{x^2 + 1} dx &= \frac{1}{2} \int_0^3 \frac{1}{x^2 + 1} \color{red}{2x} dx \\ &= \frac{1}{2} \int_1^{10} \frac{1}{u} du \\ &= \frac{1}{2} \ln |u| \Big|_1^{10} \\ &= \frac{1}{2} \ln(10) - \frac{1}{2} \ln(1) \\ &= \frac{\ln(10)}{2}. \blacksquare\end{aligned}$$

Example 2.1.7. Compute

$$\int_0^{\pi/2} \sin(x) e^{\cos(x)} dx.$$

Answer: The obvious substitution here is $u = \cos(x)$ giving $du = -\sin(x) dx$. For the limits of integration, we have

$$\begin{aligned} x = 0 &\rightarrow u = \cos(0) = 1, \\ x = \frac{\pi}{2} &\rightarrow u = \cos\left(\frac{\pi}{2}\right) = 0. \end{aligned}$$

This gives

$$\begin{aligned} \int_0^{\pi/2} \sin(x) e^{\cos(x)} dx &= - \int_1^0 e^{\cos(x)} (-\sin(x)) dx \\ &= - \int_1^0 e^u du \\ &= -e^u \Big|_1^0 \\ &= -e^0 - (-e^1) \\ &= e - 1. \blacksquare \end{aligned}$$

2.2 Integration by Parts

Integration by parts is the integration rule corresponding to the product rule for derivatives. If we take the product rule

$$\frac{d}{dx} [f(x)g(x)] = f'(x)g(x) + f(x)g'(x),$$

and integrate both sides, we get

$$\begin{aligned} \int \frac{d}{dx} [f(x)g(x)] dx &= \int (f'(x)g(x) + f(x)g'(x)) dx \\ f(x)g(x) &= \int f'(x)g(x) dx + \int f(x)g'(x) dx \end{aligned}$$

In the second line, we first use the fact that an anti-derivative is the inverse operation of differentiation (up to a constant). For the right-hand side, we use the fact that the integral of a sum is the sum of the integrals.

Notice that we did not list an arbitrary constant (a “ $+C$ ”). The reason is that there are still indefinite integrals on the right-hand side. Any arbitrary constant we would get from the left-hand side can be combined with the arbitrary constant from the remaining integral. In the same vein, we only need a single $+C$ on the right-hand side.

So far, this does not look like a very useful identity. The integration by parts formula comes from solving for one of the two integrals on the right.

Theorem 2.2.1 (Integration by Parts). *If f and g are continuously differentiable functions, then*

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx.$$

The reason this is useful is that it often allows us to exchange a more difficult integral (the one on the left) for an integral that is hopefully simpler (the one on the right).

Example 2.2.1. Find the anti-derivative

$$\int x \ln(x) dx.$$

Answer: Since this is not a basic form or rational function and there seems to be no reasonable substitution to make, we try integration by parts. The key to using integration by parts is selecting f and g' in the original integral in such a way that f' is simpler than f and g is not much more complicated than g' . For this integral, the appropriate choices are $f(x) = \ln(x)$ and $g'(x) = x$. We need to compute f' and g :

$$\begin{aligned} f(x) &= \ln(x) & g'(x) &= x \\ f'(x) &= \frac{1}{x} & g(x) &= \int x dx = \frac{1}{2}x^2 \end{aligned}$$

Notice that we do not need to supply a “ $+C$ ” until the very last integral has been computed (so no constant appears on g). Using the integration by

parts formula, we find

$$\begin{aligned}
 \int x \ln(x) \, dx &= \ln(x) \left(\frac{1}{2}x^2 \right) - \int \left(\frac{1}{x} \right) \left(\frac{1}{2}x^2 \right) \, dx \\
 &= \frac{1}{2}x^2 \ln(x) - \frac{1}{2} \int x \, dx \\
 &= \frac{1}{2}x^2 \ln(x) - \frac{1}{2} \left(\frac{1}{2}x^2 \right) + C \\
 &= \frac{1}{2}x^2 \ln(x) - \frac{1}{4}x^2 + C.
 \end{aligned}$$

Notice that the arbitrary constant only appeared after the final integral was computed!

If we want to check that our answer is correct, we simply take the derivative of our answer:

$$\begin{aligned}
 \frac{d}{dx} \left[\frac{1}{2}x^2 \ln(x) - \frac{1}{4}x^2 + C \right] &= x \ln(x) + \frac{1}{2}x^2 \left(\frac{1}{x} \right) - \frac{1}{2}x \\
 &= x \ln(x). \quad \blacksquare
 \end{aligned}$$

The trick to using integration by parts successfully is identifying the best choice for f and g' . We will develop some intuition for that shortly (along with a more convenient form for remembering the formula). Before that, a second example.

Example 2.2.2. Find the anti-derivative

$$\int x \sin(x) \, dx.$$

Answer: Again, this is not a basic form, no simple substitution seems appropriate, and the integrand is not a rational function; hence, integration by parts is our last resort. The appropriate choice for f is $f(x) = x$ which leaves $g'(x) = \sin(x)$.

$$\begin{aligned}
 f(x) &= x & g'(x) &= \sin(x) \\
 f'(x) &= 1 & g(x) &= \int \sin(x) \, dx = -\cos(x)
 \end{aligned}$$

This gives

$$\begin{aligned}\int x \sin(x) \, dx &= x(-\cos(x)) - \int 1(-\cos(x)) \, dx \\ &= -x \cos(x) + \int \cos(x) \, dx \\ &= -x \cos(x) + \sin(x) + C. \quad \blacksquare\end{aligned}$$

The integration by parts formula is usually remembered in a slightly different form – one reminiscent of the formula for u -substitution. The idea is to replace $f(x)$ by $u(x)$ and $g(x)$ by $v(x)$. In addition, we use the short-hand $du = u'(x) \, dx$ and $dv = v'(x) \, dx$. This gives:

Theorem 2.2.2 (Integration by Parts – Alternate Form). *For two continuously differentiable functions u and v , we have*

$$\int u \, dv = uv - \int v \, du.$$

This is NOT a different formula! This is equivalent to the original formula, but it has been rewritten in a form that is easier to remember.

Example 2.2.3. Find the anti-derivative

$$\int x e^{2x} \, dx.$$

Answer: We make the following choices:

$$\begin{aligned}u &= x & dv &= e^{2x} \, dx \\ du &= dx & v &= \int e^{2x} \, dx = \frac{1}{2} e^{2x}\end{aligned}$$

This gives

$$\begin{aligned}\int x e^{2x} \, dx &= x \left(\frac{1}{2} e^{2x} \right) - \int \frac{1}{2} e^{2x} \, dx \\ &= \frac{1}{2} x e^{2x} - \frac{1}{2} \int e^{2x} \, dx \\ &= \frac{1}{2} x e^{2x} - \frac{1}{2} \left(\frac{1}{2} e^{2x} \right) + C \\ &= \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + C. \quad \blacksquare\end{aligned}$$

The main question for us to address is how to pick u and dv . There is no rule that is guaranteed to work in every possible situation, but there is a mnemonic for picking u that works in a large majority of cases.

The best choice for the function u when using integration by parts is very often given by the acronym **LIATE** where

L is for **L**ogarithmic functions,
I is for **I**nverse trigonometric functions,
A is for **A**lgebraic functions,
T is for **T**rigonometric functions,
E is for **E**xponential functions.

The logic behind this rule of thumb is that logarithmic and inverse trigonometric functions often become simpler when you take their derivatives (they typically become algebraic functions). Algebraic functions (like powers of x) stay algebraic functions under differentiation. The last two classes, trigonometric and exponential functions, do not become simpler when you differentiate them and are typically not great choices for u .

Example 2.2.4. Find the anti-derivative

$$\int x^2 \cos(x) \, dx.$$

Answer: Based on our mnemonic, we should choose $u = x^2$.

$$\begin{aligned} u &= x^2 & dv &= \cos(x) \, dx \\ du &= 2x \, dx & v &= \int \cos(x) \, dx = \sin(x) \end{aligned}$$

So,

$$\begin{aligned} \int x^2 \cos(x) \, dx &= x^2 \sin(x) - \int \sin(x) \cdot 2x \, dx \\ &= x^2 \sin(x) - 2 \int x \sin(x) \, dx. \end{aligned}$$

How should we attack this new integral? Integration by parts, of course! We choose $u = x$ giving

$$\begin{aligned} u &= x & dv &= \sin(x) \, dx \\ du &= dx & v &= \int \sin(x) \, dx = -\cos(x) \end{aligned}$$

Be careful to distribute the -2 to every term from the second integration by parts.

$$\begin{aligned} \int x^2 \cos(x) \, dx &= x^2 \sin(x) - 2 \int x \sin(x) \, dx \\ &= x^2 \sin(x) - 2 \left[-x \cos(x) - \int -\cos(x) \, dx \right] \\ &= x^2 \sin(x) + 2x \cos(x) - 2 \int \cos(x) \, dx \\ &= x^2 \sin(x) + 2x \cos(x) - 2 \sin(x) + C. \quad \blacksquare \end{aligned}$$

Finally, definite integrals can be handled with integration by parts in the obvious way.

Theorem 2.2.3 (Integration by Parts – Definite Integral Form). *For two continuously differentiable functions u and v , we have*

$$\int_a^b u \, dv = uv \Big|_a^b - \int_a^b v \, du.$$

Example 2.2.5. Evaluate the definite integral

$$\int_1^2 x^2 \ln(x) \, dx.$$

Answer: Choosing $u = \ln(x)$ gives

$$\begin{aligned} u &= \ln(x) & dv &= x^2 \, dx \\ du &= \frac{1}{x} \, dx & v &= \int x^2 \, dx = \frac{1}{3}x^3 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \int_1^2 x^2 \ln(x) \, dx &= \left. \frac{1}{3} x^3 \ln(x) \right|_1^2 - \int_1^2 \frac{1}{3} x^3 \left(\frac{1}{x} \right) \, dx \\
 &= \left(\frac{8}{3} \ln(2) - \frac{1}{3} \ln(1) \right) - \frac{1}{3} \int_1^2 x^2 \, dx \\
 &= \frac{8}{3} \ln(2) - \frac{1}{9} x^3 \Big|_1^2 \\
 &= \frac{8}{3} \ln(2) - \left(\frac{8}{9} - \frac{1}{9} \right) \\
 &= \frac{8}{3} \ln(2) - \frac{7}{9}. \quad \blacksquare
 \end{aligned}$$

On occasion, you need to make a u -substitution before attempting integration by parts.

Example 2.2.6. Evaluate the definite integral

$$\int_0^4 \sqrt{x} e^{\sqrt{x}} \, dx.$$

Answer: If we try to integrate by parts directly, our mnemonic would indicate $u = \sqrt{x}$ giving

$$\begin{aligned}
 u &= \sqrt{x} & dv &= e^{\sqrt{x}} \, dx \\
 du &= \frac{1}{2\sqrt{x}} \, dx & v &= \int e^{\sqrt{x}} \, dx = ???
 \end{aligned}$$

This presents two problems. First, integrating $e^{\sqrt{x}}$ is not easy (it turns out that it can be done by the same substitution we are about to discuss). Second, it is far from clear that $\int v \, du$ will be easy to integrate thanks to the $\sqrt{x}$ that will appear in the denominator. Hence, we try a different approach.

We try the u -substitution $u = \sqrt{x}$ which gives $du = \frac{1}{2\sqrt{x}} \, dx$. Notice that

$$\begin{aligned}
 dx &= 2\sqrt{x} \, du \\
 &= 2u \, du.
 \end{aligned}$$

This transforms our integral to

$$\int_0^4 \sqrt{x} e^{\sqrt{x}} \, dx = 2 \int_0^2 u^2 e^u \, du$$

which we can certainly attack by parts. We should not re-use u for a different purpose, so we will use r and s for integration by parts.

Instead of keeping the limits throughout the computation, we can find the anti-derivative and evaluate the limits at the end of the computation. Choosing $r = u^2$, we have

$$\begin{aligned} r &= u^2 & ds &= e^u du \\ dr &= 2u du & s &= \int e^u du = e^u \end{aligned}$$

This gives

$$\begin{aligned} 2 \int u^2 e^u du &= 2u^2 e^u - 2 \int 2ue^u du \\ &= 2u^2 e^u - 4 \int ue^u du. \end{aligned}$$

We choose $r = u$ for a second round of integration by parts:

$$\begin{aligned} r &= u & ds &= e^u du \\ dr &= du & s &= \int e^u du = e^u \end{aligned}$$

which gives

$$\begin{aligned} 2 \int u^2 e^u du &= 2u^2 e^u - 4 \int ue^u du \\ &= 2u^2 e^u - 4 \left[ue^u - \int e^u du \right] \\ &= 2u^2 e^u - 4ue^u + 4e^u + C. \end{aligned}$$

We can now evaluate the limits:

$$\begin{aligned} 2 \int_0^2 u^2 e^u du &= (2u^2 e^u - 4ue^u + 4e^u) \Big|_0^2 \\ &= (8e^2 - 8e^2 + 4e^2) - (4e^0) \\ &= 4e^2 - 4. \blacksquare \end{aligned}$$

Finally, there are instances where integration by parts does not seem applicable but turns out to be the appropriate method.

Example 2.2.7. Find the anti-derivative

$$\int \ln(x) \, dx.$$

Answer: Considering how important the natural logarithm is, it may strike you as strange that we have not seen its anti-derivative until now. That is because you need integration by parts to get it! If we choose $u = \ln(x)$, what is left for dv ? The only thing left is $dv = dx$ which is completely fine!

$$\begin{aligned} u &= \ln(x) & dv &= dx \\ du &= \frac{1}{x} dx & v &= \int dx = x \end{aligned}$$

This gives

$$\begin{aligned} \int \ln(x) \, dx &= x \ln(x) - \int x \left(\frac{1}{x} \right) dx \\ &= x \ln(x) - \int dx \\ &= x \ln(x) - x + C. \quad \blacksquare \end{aligned}$$

2.3 Integration by Partial Fraction Decomposition

Our final method, **integration by partial fraction decomposition**, is limited to the integration of *proper* rational functions. For a function of the form

$$f(x) = \frac{p(x)}{q(x)}$$

to be a proper rational function, p and q must be polynomials, and the degree of p must be strictly smaller than the degree of q . If you need to integrate an improper rational function, *you must first perform long division to rewrite the function as a polynomial plus a proper rational function.*

For instance, if we need to integrate

$$f(x) = \frac{x^4}{x^2 - 4},$$

we first carry out the long-division to get

$$\begin{array}{r}
 x^2 + 4 \\
 x^2 - 4 \overline{) x^4} \\
 \underline{- x^4 + 4x^2} \\
 4x^2 \\
 \underline{- 4x^2 + 16} \\
 16
 \end{array}$$

This tells us that

$$f(x) = \frac{x^4}{x^2 - 4} = x^2 + 4 + \frac{16}{x^2 - 4}.$$

The polynomial portion on the right is easy to integrate, and partial fraction decomposition will handle the proper rational function.

We begin with an example.

Example 2.3.1. Find the anti-derivative

$$\int \frac{1}{x^2 - 4} dx.$$

Answer: This not a basic form and no substitution seems appropriate. The idea is to try and decompose the proper rational function into a sum of simpler terms. In this case,

$$\frac{1}{x^2 - 4} = \frac{1}{(x - 2)(x + 2)} = \frac{A}{x - 2} + \frac{B}{x + 2},$$

for some numbers A and B . The two terms on the right are the *partial fractions* we hope to *decompose* our original fraction into. The question is whether or not there are two numbers A and B such that the two simpler functions on the right add up to our original function. If we add those two rational terms together, we get

$$\frac{1}{(x - 2)(x + 2)} = \frac{A(x + 2)}{(x - 2)(x + 2)} + \frac{B(x - 2)}{(x + 2)(x - 2)} = \frac{A(x + 2) + B(x - 2)}{(x - 2)(x + 2)}.$$

In order for these two rational expressions to be identical, their numerators must be identical *for any value of x* . This leads us to

$$\begin{aligned}
 1 &= A(x + 2) + B(x - 2) \\
 &= (A + B)x + (2A - 2B).
 \end{aligned}$$

Since the left-hand side is independent of x , we need $A + B = 0$ so that the right-hand side will not have an x term. Likewise, since the left-hand side has a constant term of 1, we must have $2A - 2B = 1$. The first equation tells us that $B = -A$, and substituting this into the second gives

$$\begin{aligned}2A - 2B &= 1 \\4A &= 1 \\A &= \frac{1}{4}.\end{aligned}$$

Which immediately gives $B = -1/4$. Thus, we have shown that

$$\frac{1}{x^2 - 4} = \frac{1}{4} \left(\frac{1}{x - 2} \right) - \frac{1}{4} \left(\frac{1}{x + 2} \right).$$

The advantage of this is that the right-hand side is now easy to integrate!

$$\begin{aligned}\int \frac{1}{x^2 - 4} dx &= \frac{1}{4} \int \frac{1}{x - 2} dx - \frac{1}{4} \int \frac{1}{x + 2} dx \\&= \frac{1}{4} \ln |x - 2| - \frac{1}{4} \ln |x + 2| + C. \blacksquare\end{aligned}$$

This method works in general so long as the rational function is proper and the denominator factors into *distinct linear terms*.

Suppose that the denominator of a *proper* rational function

$$f(x) = \frac{p(x)}{q(x)}$$

factors into distinct linear terms:

$$q(x) = (a_1x + b_1)(a_2x + b_2) \cdots (a_nx + b_n).$$

Then the rational function can be decomposed as

$$\frac{p(x)}{q(x)} = \frac{C_1}{a_1x + b_1} + \frac{C_2}{a_2x + b_2} + \cdots + \frac{C_n}{a_nx + b_n}$$

for some coefficients $C_1, C_2, \dots, C_n$.

Note that

$$\int \frac{1}{a_ix + b_i} dx = \frac{1}{a_i} \ln |a_ix + b_i| + C.$$

If there are repeated linear terms or irreducible quadratic terms in the denominator, then the partial fraction decomposition is a little different. We will discuss those modifications shortly, but first a couple of examples.

Example 2.3.2. Find the anti-derivative

$$\int \frac{x}{x^2 + 3x - 4} dx.$$

Answer: The first method you should attempt is u -substitution. If we try $u = x^2 + 3x - 4$, we would need $du = (2x + 3) dx$ which is not very similar to what we have in the numerator. Since this is a proper rational function, we try partial fractions.

$$\frac{x}{x^2 + 3x - 4} = \frac{x}{(x - 1)(x + 4)} = \frac{A}{x - 1} + \frac{B}{x + 4}.$$

Summing up the rational terms on the right and equating numerators gives

$$x = A(x + 4) + B(x - 1) = (A + B)x + (4A - B).$$

In order for these two expressions to be identical, $A + B = 1$ (since the coefficient of x must be 1) and $4A - B = 0$ (since there should be no constant term). The second equation tells us that $B = 4A$ which combines with the first to give

$$\begin{aligned} A + B &= 1 \\ 5A &= 1 \\ A &= \frac{1}{5}. \end{aligned}$$

Thus, $B = 4/5$ and we have the decomposition

$$\frac{x}{x^2 + 3x - 4} = \frac{1}{5} \left(\frac{1}{x-1} \right) + \frac{4}{5} \left(\frac{1}{x+4} \right).$$

Using this in the integral gives

$$\begin{aligned} \int \frac{x}{x^2 + 3x - 4} dx &= \frac{1}{5} \int \frac{1}{x-1} dx + \frac{4}{5} \int \frac{1}{x+4} dx \\ &= \frac{1}{5} \ln |x-1| + \frac{4}{5} \ln |x+4| + C. \blacksquare \end{aligned}$$

At this point, it is worth mentioning that there is a second way to find the coefficients in the decomposition: substituting various values into the variable! In the example above, we needed

$$x = A(x+4) + B(x-1)$$

to be a true statement *for all values of x* . In particular, the statement should be true at $x = 1$. Substituting this into both sides gives

$$1 = A(5) + B(0),$$

which clearly gives $A = 1/5$! Similarly, substituting $x = -4$ gives

$$-4 = A(0) + B(-5),$$

which gives $B = 4/5$. This method is often faster when the denominator consists of distinct linear factors!

Example 2.3.3. Find the anti-derivative

$$\int \frac{x^4 + 2}{x^3 - x} dx.$$

Answer: Because this is an improper rational function, we have to perform long division before proceeding:

$$\frac{x^4 + 2}{x^3 - x} = x + \frac{x^2 + 2}{x^3 - x}.$$

Since integrating x presents no difficulty, we focus on the remaining proper rational term:

$$\frac{x^2 + 2}{x^3 - x} = \frac{x^2 + 2}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}.$$

So, we need

$$x^2 + 2 = A(x-1)(x+1) + Bx(x+1) + Cx(x-1).$$

If we substitute $x = 0$, we must have $2 = A(-1) + B(0) + C(0)$ which gives $A = -2$. Substituting $x = 1$ gives $3 = A(0) + B(2) + C(0)$, which gives $B = 3/2$. Finally, $x = -1$ gives $3 = A(0) + B(0) + C(2)$, and so $C = 3/2$. Thus,

$$\frac{x^2 + 2}{x^3 - x} = -2 \left(\frac{1}{x} \right) + \frac{3}{2} \left(\frac{1}{x-1} \right) + \frac{3}{2} \left(\frac{1}{x+1} \right).$$

So, our original integral becomes

$$\begin{aligned} \int \frac{x^4 + 2}{x^3 - x} dx &= \int x dx - 2 \int \frac{1}{x} dx + \frac{3}{2} \int \frac{1}{x-1} dx + \frac{3}{2} \int \frac{1}{x+1} dx \\ &= \frac{1}{2}x^2 - 2 \ln |x| + \frac{3}{2} \ln |x-1| + \frac{3}{2} \ln |x+1| + C. \blacksquare \end{aligned}$$

When there are repeated linear factors or irreducible quadratic factors in the denominator, we have to modify the partial fraction expansion for that particular factor. While this is not a major concern for us in this text, it is worth mentioning just for the sake of completeness.

Suppose that the denominator of a *proper* rational function

$$f(x) = \frac{p(x)}{q(x)}$$

has a linear factor to some power $n > 1$; that is, a term like

$$(ax + b)^n$$

appears in the factorization of $q(x)$. Then the contribution to the partial fraction decomposition for this factor should be

$$\frac{C_1}{ax + b} + \frac{C_2}{(ax + b)^2} + \cdots + \frac{C_n}{(ax + b)^n}$$

Note that

$$\int \frac{1}{(ax + b)^i} dx = - \left(\frac{1}{(i-1)a} \right) \frac{1}{(ax + b)^{i-1}} + C$$

for $i > 1$.

Example 2.3.4. Find the anti-derivative

$$\int \frac{x + 2}{x^3 + 2x^2 + x} dx.$$

Answer: The denominator factors as $x^3 + 2x^2 + x = x(x + 1)^2$. So our partial fraction decomposition must be

$$\frac{x + 2}{x^3 + 2x^2 + x} = \frac{A}{x} + \frac{B}{x + 1} + \frac{C}{(x + 1)^2}.$$

This gives us the equality

$$x + 2 = A(x + 1)^2 + Bx(x + 1) + Cx.$$

If we substitute $x = 0$, we get $2 = A(1) + B(0) + C(0)$, which gives $A = 2$. Substituting $x = -1$, we get $1 = A(0) + B(0) + C(-1)$, which gives $C = -1$. There is no simple substitution that will give us B , but we can substitute any value for x we have not currently used! For instance, we can substitute $x = 1$; $3 = A(4) + B(2) + C(1)$ giving

$$2B = 3 - 4A - C = 3 - 8 + 1 = -4.$$

So, $B = -2$. We finally get

$$\frac{x+2}{x^3+2x^2+x} = 2\left(\frac{1}{x}\right) - 2\left(\frac{1}{x+1}\right) - \frac{1}{(x+1)^2}.$$

So, the integral becomes

$$\begin{aligned} \int \frac{x+2}{x^3+2x^2+x} dx &= 2 \int \frac{1}{x} dx - 2 \int \frac{1}{x+1} dx - \int \frac{1}{(x+1)^2} dx \\ &= 2 \ln|x| - 2 \ln|x+1| + \frac{1}{x+1} + C. \blacksquare \end{aligned}$$

A polynomial has an *irreducible quadratic factor* if the polynomial has complex roots. For instance, $x^3 + x = x(x^2 + 1)$, and $x^2 + 1$ cannot be factored over the set of real numbers (its roots are $\pm i$).

Suppose that the denominator of a *proper* rational function

$$f(x) = \frac{p(x)}{q(x)}$$

has an irreducible quadratic factor to some power n ; that is, a term like

$$(ax^2 + bx + c)^n$$

appears in the factorization of $q(x)$, and this quadratic polynomial does not factor over the real numbers. Then the contribution to the partial fraction decomposition for this factor should be

$$\frac{B_1x + C_1}{ax^2 + bx + c} + \frac{B_2x + C_2}{(ax^2 + bx + c)^2} + \cdots + \frac{B_nx + C_n}{(ax^2 + bx + c)^n}$$

This can become quite complicated for repeated irreducible quadratic factors. We will be content with a simpler example.

Example 2.3.5. Find the anti-derivative

$$\int \frac{x+1}{x^3+x} dx.$$

Answer: Since the denominator factors as $x^3 + x = x(x^2 + 1)$, we must look for a partial fraction decomposition of the form

$$\frac{x+1}{x^3+x} = \frac{A}{x} + \frac{Bx+C}{x^2+1}.$$

This gives us the following equality:

$$x+1 = A(x^2+1) + (Bx+C)x.$$

Substituting $x = 0$ will give us A : $1 = A(1) + (B(0) + C)(0) = A$. No simple choices of x will directly give us B or C . We could substitute any two x -values of our choice (except 0) and get a pair of equations for B and C , or we could expand the right-hand side and match coefficients (remembering that $A = 1$):

$$\begin{aligned} x+1 &= 1(x^2+1) + Bx^2 + Cx \\ &= (B+1)x^2 + Cx + 1. \end{aligned}$$

Since the left-hand side has no x^2 term, we must have $B = -1$. Since the coefficient of x on the left-hand side is 1, $C = 1$. This gives us our partial fraction decomposition:

$$\frac{x+1}{x^3+x} = \frac{1}{x} + \frac{-x+1}{x^2+1} = \frac{1}{x} - \frac{x}{x^2+1} + \frac{1}{x^2+1}.$$

The final splitting of the irreducible quadratic term into two pieces is to facilitate integration.

$$\begin{aligned} \int \frac{x+1}{x^3+x} dx &= \int \frac{1}{x} dx - \int \frac{x}{x^2+1} dx + \int \frac{1}{x^2+1} dx \\ &= \ln|x| - \frac{1}{2} \int \frac{2x}{x^2+1} dx + \arctan(x) \\ &= \ln|x| - \frac{1}{2} \ln(x^2+1) + \arctan(x) + C. \end{aligned}$$

Notice that the integral of $(x^2+1)^{-1}$ is a basic form you should be familiar with (i.e. it integrates to $\arctan(x)$). In the second line, we set up a u -substitution to handle the middle integral. You should fill in the details to make sure you understand how the integral was computed. ■

2.4 Problems

Compute the following definite and indefinite integrals.

$$1) \quad \int \frac{x}{1-x^2} dx$$

$$2) \quad \int \frac{1}{1+y^2} dy$$

$$3) \quad \int_0^{\pi/4} x \cos(x) dx$$

$$4) \quad \int \frac{x}{x^2+x-2} dx$$

$$5) \quad \int_{4/\pi}^{2/\pi} \frac{1}{s^2} \sin\left(\frac{1}{s}\right) ds$$

$$6) \quad \int_2^{10} \frac{1}{x^2-x} dx$$

$$7) \quad \int \frac{x+1}{(x+2)(x+3)^2} dx$$

$$8) \quad \int_1^2 x \ln(x) dx$$

$$9) \quad \int \frac{1}{\cos^2(x)} dx$$

$$10) \quad \int r^2 e^{-r} dr$$

$$11) \quad \int \frac{2x^2 + 1}{2x^2 + x - 1} dx$$

$$12) \quad \int x^2 \cos(2x) dx$$

$$13) \quad \int_2^{14} z \sqrt{z + 2} dz$$

$$14) \quad \int \frac{(\ln(x))^{10}}{x} dx$$

$$15) \quad \int_0^2 t^2 e^{2t} dt$$

$$16) \quad \int_0^3 \frac{1}{3x^2 + 4x + 1} dx$$

$$17) \quad \int \frac{x + 1}{x^3 + 4x} dx$$

$$18) \quad \int \frac{1}{y^2 + 4y + 5} dy \quad (\text{HINT: Try completing the square in the denominator!})$$

$$19) \quad \int \arctan(x) dx \quad (\text{HINT: Try integrating by parts followed by } u\text{-substitution!})$$

$$20) \quad \textbf{Challenge: } \int e^{\sqrt{x}} dx \quad (\text{HINT: Try a } u\text{-substitution then integration by parts!})$$

Chapter 3

Applications of Integration

Before we return to dynamical systems, we first discuss some applications of integrals that are often useful in practice. The first two involve finding areas of shapes and the volumes of a special class of solid figures. The final application, improper integration, extends integration to infinite intervals and/or to functions which are unbounded over their range of integration. This topic will be important when we discuss continuous probability distributions in Chapter 7.

3.1 Areas Between Curves

As you learned in a first calculus course, the definite integral

$$\int_a^b f(x) \, dx$$

computes the *signed area* between the curve $y = f(x)$ and the x -axis over the interval $a \leq x \leq b$.

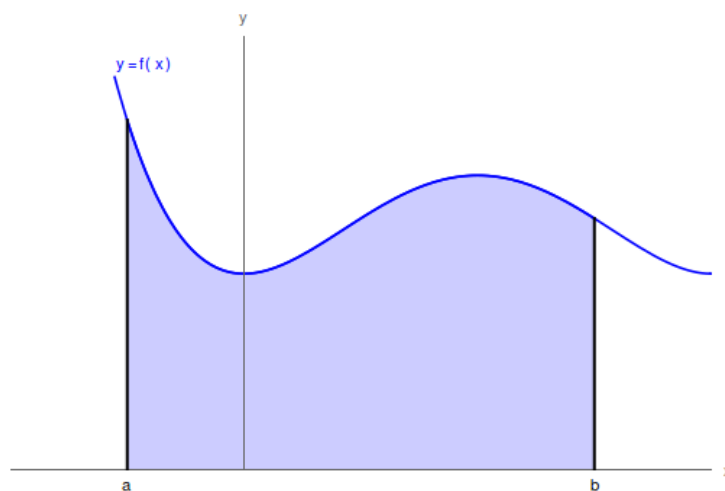


Figure 3.1.1: Area Under a Curve

The reason the definite integral gives us signed area is that regions below the x -axis are counted as negative area.

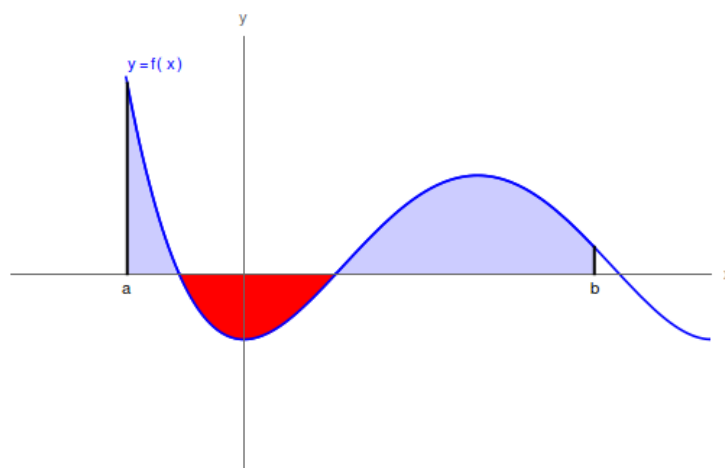


Figure 3.1.2: Signed Area

So, the portion of the figure above that is shaded in red will count as negative area in the integral

$$\int_a^b f(x) \, dx.$$

If we want the *total area* between this curve and the x -axis, we have to split the definite integral into three pieces.

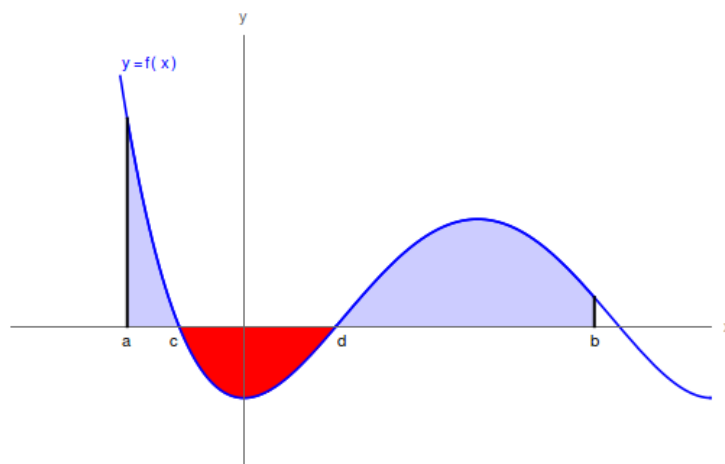


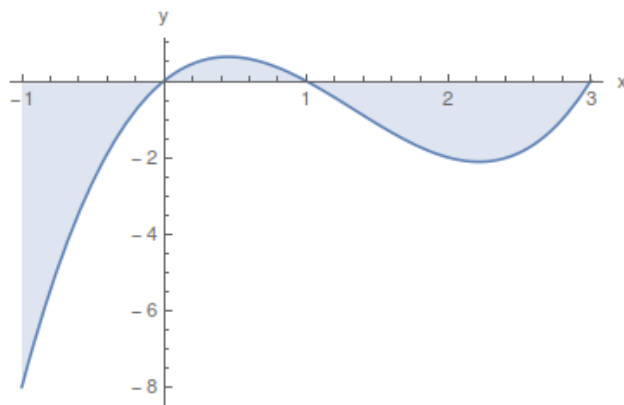
Figure 3.1.3: Signed Area, continued

This gives

$$\text{Total Area} = \int_a^c f(x) \, dx - \int_c^d f(x) \, dx + \int_d^b f(x) \, dx$$

where the x -values c and d are the solutions to $f(x) = 0$ that lie in the range $a \leq x \leq b$.

Example 3.1.1. Find the total area between the x -axis and the curve $y = x^3 - 4x^2 + 3x$ over the interval $-1 \leq x \leq 3$.

Figure 3.1.4: Graph of $y = x^3 - 4x^2 + 3x$ over $-1 \leq x \leq 3$.

Answer: In order to compute the total area (rather than the signed area), we need to compute separate integrals over the different regions where the area is above or below the x -axis. To find the correct limits of integration, we solve

$$\begin{aligned} x^3 - 4x^2 + 3x &= 0 \\ x(x^2 - 4x + 3) &= 0 \\ x(x - 1)(x - 3) &= 0 \\ x &= 0, 1, 3. \end{aligned}$$

This means the total area is given by

$$\begin{aligned} \text{Total Area} &= - \int_{-1}^0 (x^3 - 4x^2 + 3x) \, dx + \int_0^1 (x^3 - 4x^2 + 3x) \, dx \\ &\quad - \int_1^3 (x^3 - 4x^2 + 3x) \, dx. \end{aligned}$$

Note that the signs in front of the integrals correspond to which portions of the graph are above or below the x -axis. The first integral is

$$\begin{aligned} - \int_{-1}^0 (x^3 - 4x^2 + 3x) \, dx &= - \left(\frac{x^4}{4} - \frac{4x^3}{3} + \frac{3x^2}{2} \right) \Big|_{-1}^0 \\ &= - \left[0 - \left(\frac{1}{4} - \frac{-4}{3} + \frac{3}{2} \right) \right] \\ &= \frac{37}{12}. \end{aligned}$$

The second integral gives

$$\begin{aligned}\int_0^1 (x^3 - 4x^2 + 3x) \, dx &= \left(\frac{x^4}{4} - \frac{4x^3}{3} + \frac{3x^2}{2} \right) \Big|_0^1 \\ &= \left(\frac{1}{4} - \frac{4}{3} + \frac{3}{2} \right) - 0 \\ &= \frac{5}{12}.\end{aligned}$$

This third integral is

$$\begin{aligned}-\int_1^3 (x^3 - 4x^2 + 3x) \, dx &= -\left(\frac{x^4}{4} - \frac{4x^3}{3} + \frac{3x^2}{2} \right) \Big|_1^3 \\ &= -\left[\left(\frac{81}{4} - \frac{108}{3} + \frac{27}{2} \right) - \left(\frac{1}{4} - \frac{4}{3} + \frac{3}{2} \right) \right] \\ &= \frac{9}{4} + \frac{5}{12} \\ &= \frac{8}{3}.\end{aligned}$$

Adding these together gives us the total enclosed area.

$$\text{Total Area} = \frac{37}{12} + \frac{5}{12} + \frac{8}{3} = \frac{37}{6}. \quad \blacksquare$$

In many situations, we do not simply want the area between a curve and the x -axis but rather the *area between curves*.

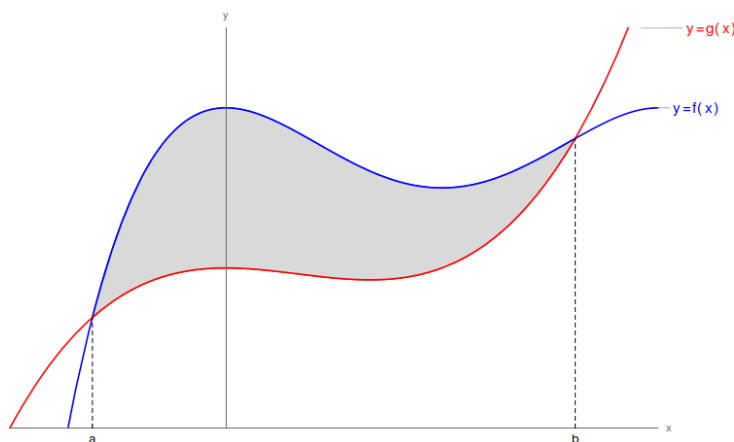


Figure 3.1.5: Area Between Two Curves

The key insight is that we can compute this area by taking the area below $y = f(x)$ and subtracting the area below $y = g(x)$.

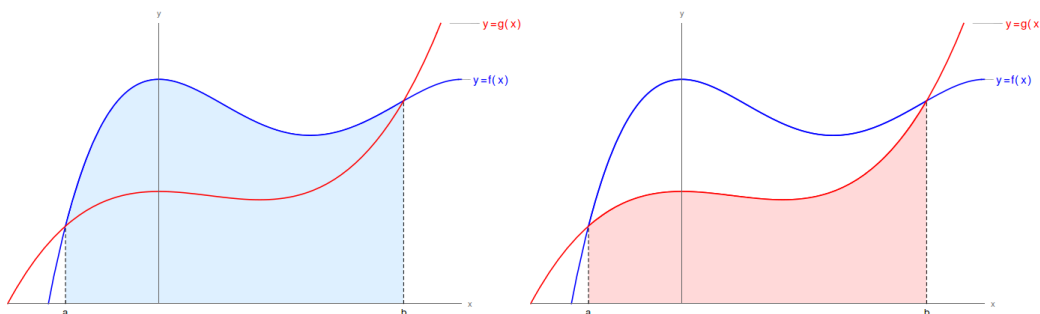


Figure 3.1.6: Area Between Two Curves by Subtraction

This means that the area between these two curves is given by

$$\text{Area} = \int_a^b f(x) \, dx - \int_a^b g(x) \, dx = \int_a^b (f(x) - g(x)) \, dx.$$

In other words, we identify which of the two curves lies underneath the other one over the range of integration, subtract the functions giving the curves in the appropriate order (“top minus bottom”), and then integrate over the appropriate range. Quite often, the limits of integration are the points of intersection of the two curves: $f(x) = g(x)$.

Example 3.1.2. Find the area contained between the curves $y = x^2$ and $y = 8 - x^2$.

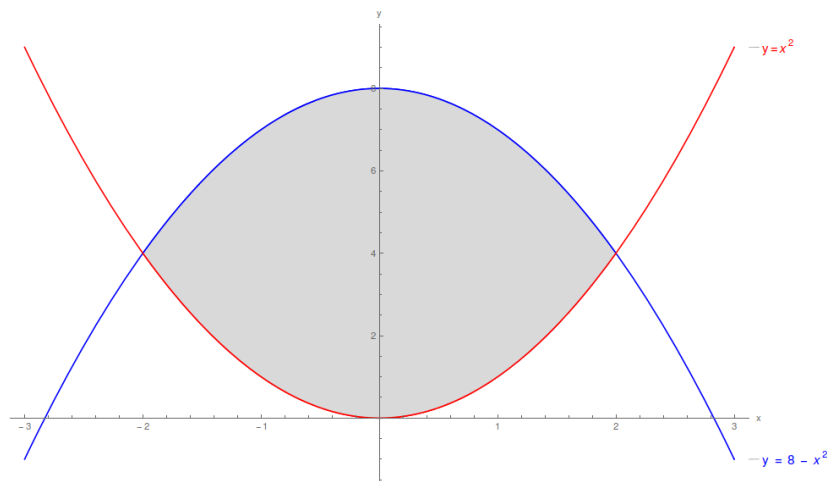


Figure 3.1.7: Area for Example 3.1.2

Answer: First, we find the points of intersection of the two curves.

$$\begin{aligned} x^2 &= 8 - x^2 \\ 2x^2 &= 8 \\ x^2 &= 4 \\ x &= \pm 2. \end{aligned}$$

For the interval $-2 \leq x \leq 2$, we see that the curve $y = 8 - x^2$ lies over $y = x^2$. So the area enclosed by these two curves is

$$\begin{aligned} \text{Area} &= \int_{-2}^2 [(8 - x^2) - x^2] \, dx \\ &= \int_{-2}^2 (8 - 2x^2) \, dx \\ &= \left(8x - \frac{2}{3}x^3 \right) \Big|_{-2}^2 \\ &= \left(16 - \frac{16}{3} \right) - \left(-16 + \frac{16}{3} \right) \\ &= \frac{32}{3} - \left(-\frac{32}{3} \right) \\ &= \frac{64}{3}. \quad \blacksquare \end{aligned}$$

Example 3.1.3. Find the area contained between the curves $y = x^2$ and $y = \sqrt{x}$.

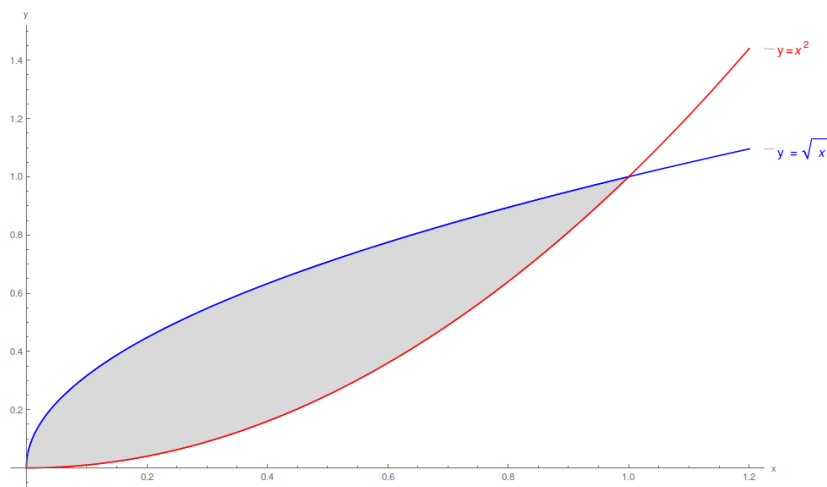


Figure 3.1.8: Area for Example 3.1.3

Answer: The intersection points of these two curves are

$$\begin{aligned}x^2 &= \sqrt{x} \\x^4 &= x \\x^4 - x &= 0 \\x(x^3 - 1) &= 0 \\x &= 0, 1.\end{aligned}$$

Over the interval $0 \leq x \leq 1$, $y = \sqrt{x}$ lies over $y = x^2$. So, the area is given by

$$\begin{aligned}\text{Area} &= \int_0^1 (\sqrt{x} - x^2) \, dx \\&= \left(\frac{2}{3}x^{3/2} - \frac{1}{3}x^3 \right) \Big|_0^1 \\&= \left(\frac{2}{3} - \frac{1}{3} \right) - 0 \\&= \frac{1}{3}. \blacksquare\end{aligned}$$

Example 3.1.4. Find the area bounded by the curves $y = 4$, $y = e^x$, and the y -axis.

Answer: In this example, we have a region bounded by three different curves.

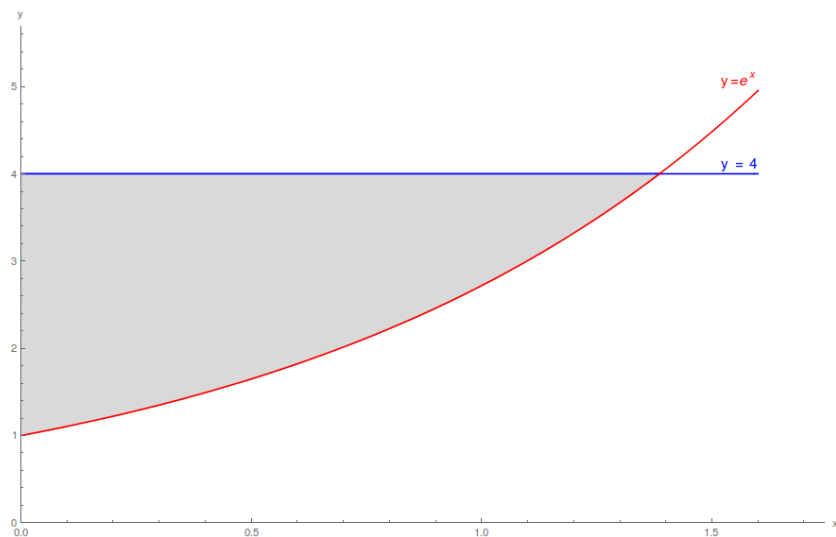


Figure 3.1.9: Area for Example 3.1.4

Clearly, the left boundary is $x = 0$ (i.e. the y -axis). For the right boundary, we need to find where $y = 4$ intersects $y = e^x$:

$$\begin{aligned}e^x &= 4 \\ \ln(e^x) &= \ln(4) \\ x &= \ln(4).\end{aligned}$$

The area of this region is given by

$$\begin{aligned}\text{Area} &= \int_0^{\ln(4)} (4 - e^x) \, dx \\ &= (4x - e^x) \Big|_0^{\ln(4)} \\ &= (4\ln(4) - e^{\ln(4)}) - (0 - e^0) \\ &= 4\ln(4) - 4 + 1 \\ &= 8\ln(2) - 3. \blacksquare\end{aligned}$$

Example 3.1.5. Find the area contained between the curves $y = x^3 - 4x$ and $y = x^2 + 2x$.

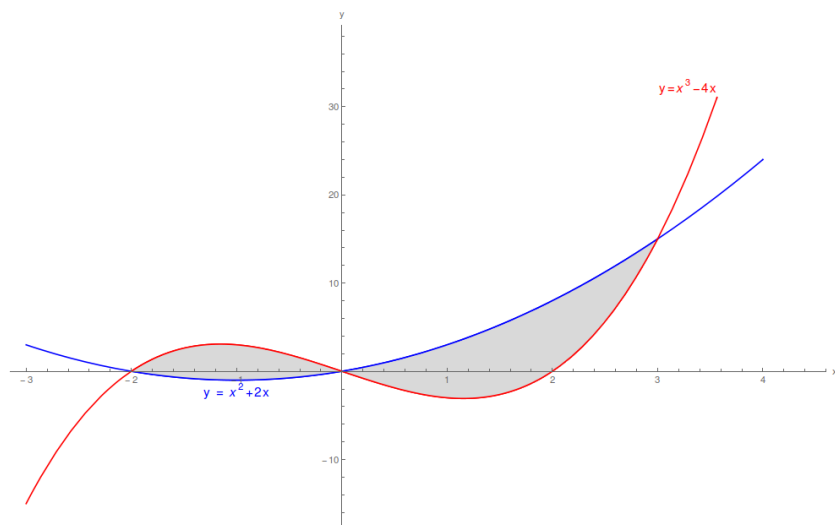


Figure 3.1.10: Area for Example 3.1.5

Answer: As usual, we first find the points of intersection of these two curves.

$$\begin{aligned}x^3 - 4x &= x^2 + 2x \\x^3 - x^2 - 6x &= 0 \\x(x^2 - x - 6) &= 0 \\x(x + 2)(x - 3) &= 0 \\x &= -2, 0, 3.\end{aligned}$$

Notice that on the interval $-2 \leq x \leq 0$, $y = x^3 - 4x$ lies over $y = x^2 + 2x$, but on $0 \leq x \leq 3$, the curves switch their relative positions. So, we need to integrate $(x^3 - 4x) - (x^2 + 2x)$ over $-2 \leq x \leq 0$, while on $0 \leq x \leq 3$ we

should integrate $(x^2 + 2x) - (x^3 - 4x)$.

$$\begin{aligned}
 \text{Area} &= \int_{-2}^0 [(x^3 - 4x) - (x^2 + 2x)] \, dx + \int_0^3 [(x^2 + 2x) - (x^3 - 4x)] \, dx \\
 &= \int_{-2}^0 (x^3 - x^2 - 6x) \, dx + \int_0^3 (-x^3 + x^2 + 6x) \, dx \\
 &= \left(\frac{1}{4}x^4 - \frac{1}{3}x^3 - 3x^2 \right) \Big|_{-2}^0 + \left(-\frac{1}{4}x^4 + \frac{1}{3}x^3 + 3x^2 \right) \Big|_0^3 \\
 &= 0 - \left(\frac{16}{4} + \frac{8}{3} - 12 \right) + \left(-\frac{81}{4} + \frac{27}{3} + 27 \right) - 0 \\
 &= \frac{16}{3} + \frac{63}{4} \\
 &= \frac{253}{12}. \blacksquare
 \end{aligned}$$

Example 3.1.6. Find the area contained between the curves $y = x^4 - 7x^2 + 7$ and $y = 3x^2 - 2$.



Figure 3.1.11: Area for Example 3.1.6

Answer: The points of intersection of these two curves are

$$\begin{aligned}x^4 - 7x^2 + 7 &= 3x^2 - 2 \\x^4 - 10x^2 + 9 &= 0 \\(x^2 - 1)(x^2 - 9) &= 0 \\(x + 1)(x - 1)(x + 3)(x - 3) &= 0 \\x &= -3, -1, 1, 3.\end{aligned}$$

The area should be given by the following three integrals.

$$\begin{aligned}\text{Area} &= \int_{-3}^{-1} [(3x^2 - 2) - (x^4 - 7x^2 + 7)] \, dx \\&\quad + \int_{-1}^1 [(x^4 - 7x^2 + 7) - (3x^2 - 2)] \, dx \\&\quad + \int_1^3 [(3x^2 - 2) - (x^4 - 7x^2 + 7)] \, dx \\&= \int_{-3}^{-1} (-x^4 + 10x^2 - 9) \, dx + \int_{-1}^1 (x^4 - 10x^2 + 9) \, dx \\&\quad + \int_1^3 (-x^4 + 10x^2 - 9) \, dx.\end{aligned}$$

A shortcut for this particular problem is to notice that the area is symmetric about the line $x = 0$. This allows us to compute the area by

$$\begin{aligned}\text{Area} &= 2 \left[\int_0^1 (x^4 - 10x^2 + 9) \, dx + \int_1^3 (-x^4 + 10x^2 - 9) \, dx \right] \\&= 2 \left(\frac{1}{5}x^5 - \frac{10}{3}x^3 + 9x \right) \Big|_0^1 + 2 \left(-\frac{1}{5}x^5 + \frac{10}{3}x^3 - 9x \right) \Big|_1^3 \\&= 2 \left(\frac{1}{5} - \frac{10}{3} + 9 - 0 \right) + 2 \left[-\frac{243}{5} + \frac{270}{3} - 27 - \left(-\frac{1}{5} + \frac{10}{3} - 9 \right) \right] \\&= \frac{176}{15} + \frac{608}{15} \\&= \frac{784}{15}. \blacksquare\end{aligned}$$

3.2 Volumes of Solids with Predictable Cross-Sections

It turns out that integration can be used to compute much more than just areas! One of the most natural extensions of integration is to the computation of volumes. To see how this works, we begin by reviewing the conceptual idea behind computing areas under curves.

Since we know how to compute the areas of simple shapes (e.g. rectangles), we subdivide our given area into N smaller rectangles of width Δx (which means $\Delta x = (b - a)/N$).

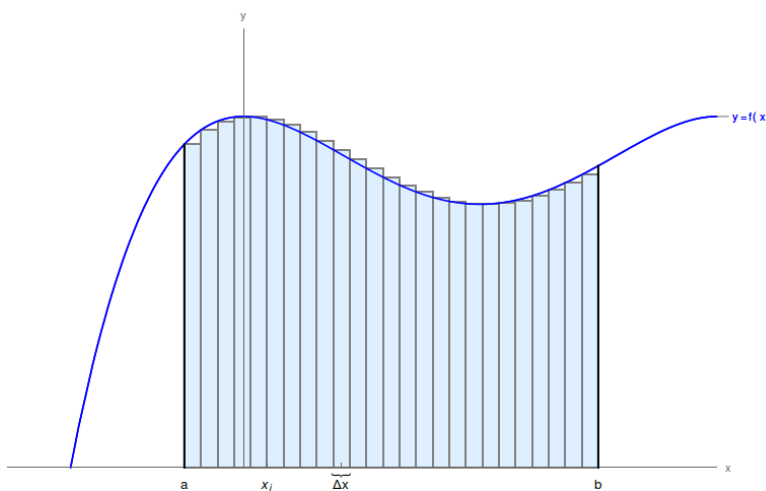


Figure 3.2.1: Subdivision of Area into Rectangles

The points x_i along the x -axis mark the lower-left vertices of each rectangle ($x_i = a + i \cdot \Delta x$), and we can take the height of each rectangle as $f(x_i)$. Even though this will not give us the *exact* area, we can feel confident that

$$\text{Area} \approx f(a)\Delta x + f(x_1)\Delta x + f(x_2)\Delta x + \cdots + f(x_{N-1})\Delta x = \sum_{i=0}^{N-1} f(x_i)\Delta x$$

so long as Δx is rather small (which means $N = (b - a)/\Delta x$ is quite large). If we take the limit $\Delta x \rightarrow 0$, then our approximation becomes exact and the

summation becomes an integral:

$$\sum_{i=0}^{N-1} f(x_i) \Delta x \xrightarrow{\Delta x \rightarrow 0} \int_a^b f(x) dx.$$

This gives us an intuitive way to think about the integral. We can imagine that we stack a huge number of extremely thin rectangles of width dx . The integral then “adds” the content of these thin slices to produce the overall area of the figure. This heuristic idea of integration can be extended to a large class of problems.

Example 3.2.1. Find the volume of a right pyramid with square base of side length 2 units and an overall height of 3 units.

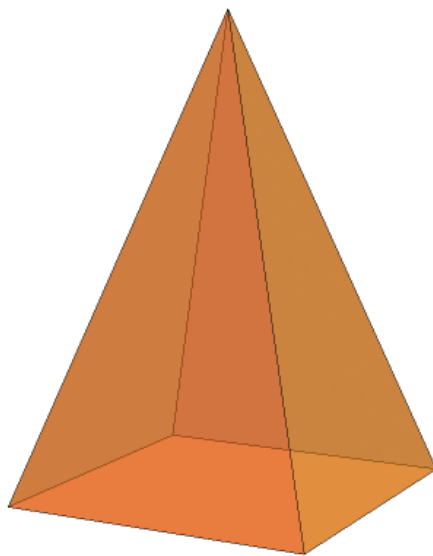


Figure 3.2.2: A Square Pyramid of Side-length 2 and Height 3

Answer: We can think of the pyramid as being stacks of thin, rectangular slabs.

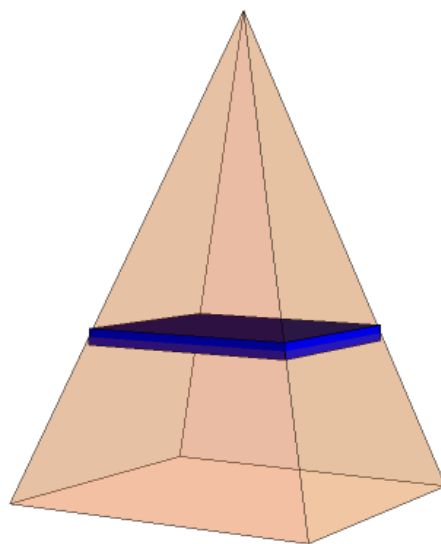
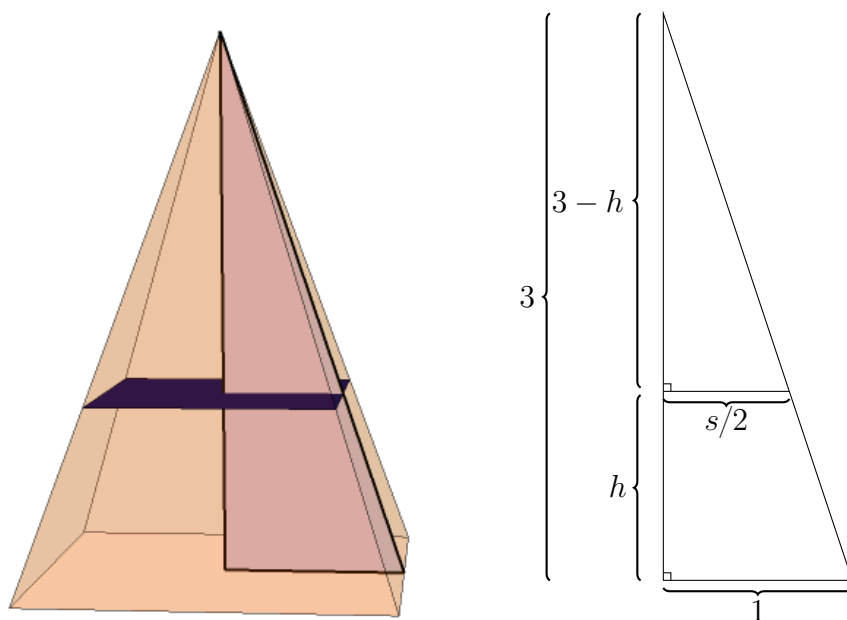


Figure 3.2.3: Pyramid with Cross Section

If this slab is at a height h from the base, the cross section is a smaller square of side-length s (the thickness will be the infinitesimal dh). So, we would have a slab of volume $s^2 dh$. In order to add up the contents of these slabs as the height h goes from $h = 0$ to $h = 3$, we need to determine how the side-length s varies with h . That is, we need to find s as a function of h , $s = s(h)$.

In order to do this, we consider the triangle running from the center line of the pyramid down the center of one of the oblique faces and back through the base.

Figure 3.2.4: Similar Triangle for s

As we can see from the similar triangles above,

$$\frac{3-h}{3} = \frac{s/2}{1}.$$

This gives us the formula for side-length in terms of h :

$$s = \frac{2}{3}(3-h).$$

Notice that at the bottom of the pyramid ($h = 0$) we get $s = 2$ as expected. Likewise, we get $s = 0$ at the top of the pyramid (when $h = 3$). So, the area of a square cross section is

$$\text{cross sectional area} = s^2 = \frac{4}{9}(3-h)^2 = \frac{4}{9}(h-3)^2.$$

To get the volume of the pyramid we simply integrate the cross sectional area

over the entire height of the pyramid.

$$\begin{aligned}
 \text{Volume} &= \int_0^3 \frac{4}{9}(h-3)^2 dh \\
 &= \frac{4}{27}(h-3)^3 \Big|_0^3 \\
 &= 0 - \frac{4}{27}(-3)^3 \\
 &= 4. \blacksquare
 \end{aligned}$$

Example 3.2.2. Find the volume of the right pyramid whose base is an equilateral triangle of side-length 3 units and having a height of 4 units.

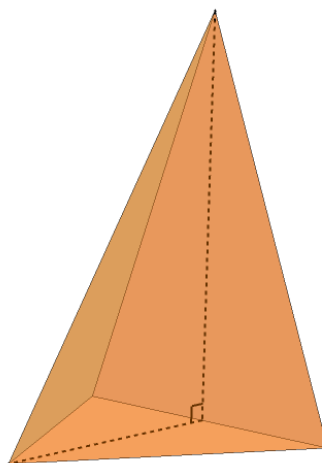
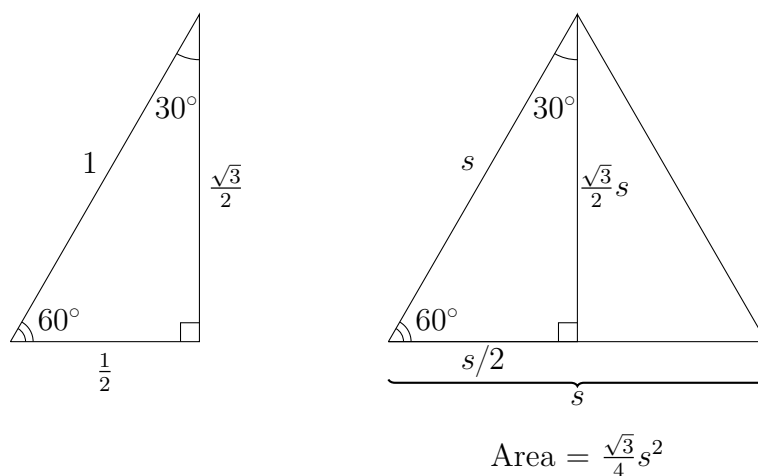


Figure 3.2.5: Pyramid with Equilateral Triangle Base of Side-length 3 and Height 4

Answer: The cross sections of this pyramid will be equilateral triangles whose side-lengths will get progressively smaller. As a matter of review, the $30^\circ - 60^\circ - 90^\circ$ right triangle is pictured below along with the area of a generic equilateral triangle.

Figure 3.2.6: $30^\circ - 60^\circ - 90^\circ$ Triangles and Equilateral Triangles

Looking at a particular cross section of the pyramid, we see the following.

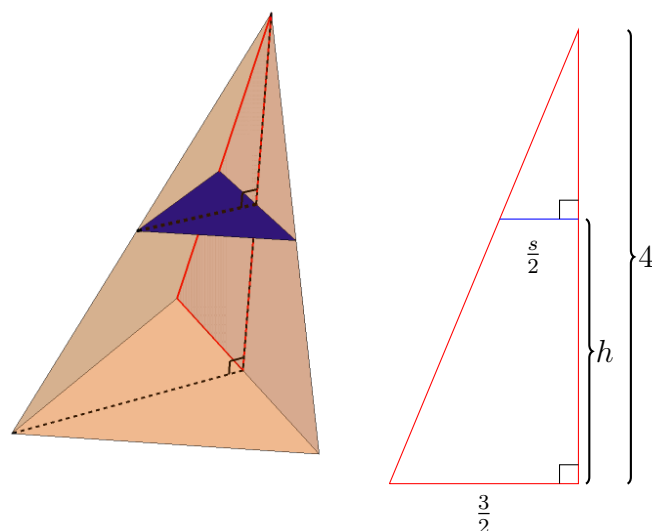


Figure 3.2.7: Cross Section of Pyramid in Example 3.2.2

From the right triangle on the right above, we see by similar triangles that

$$\frac{4-h}{4} = \frac{s/2}{3/2}.$$

This gives

$$s = \frac{3}{4}(4 - h),$$

and from our area formula, we have the cross sectional area at height h is given by

$$\text{cross sectional area} = \frac{\sqrt{3}}{4}s^2 = \frac{9\sqrt{3}}{64}(4 - h)^2 = \frac{9\sqrt{3}}{64}(h - 4)^2.$$

To find the total volume of the pyramid, we only need to integrate this cross sectional area over the height of the pyramid ($0 \leq h \leq 4$).

$$\begin{aligned} \text{Volume} &= \int_0^4 \frac{9\sqrt{3}}{64}(h - 4)^2 dh \\ &= \frac{9\sqrt{3}}{64} \frac{(h - 4)^3}{3} \Big|_0^4 \\ &= 0 - \frac{9\sqrt{3}}{64} \left(\frac{-64}{3} \right) \\ &= 3\sqrt{3}. \quad \blacksquare \end{aligned}$$

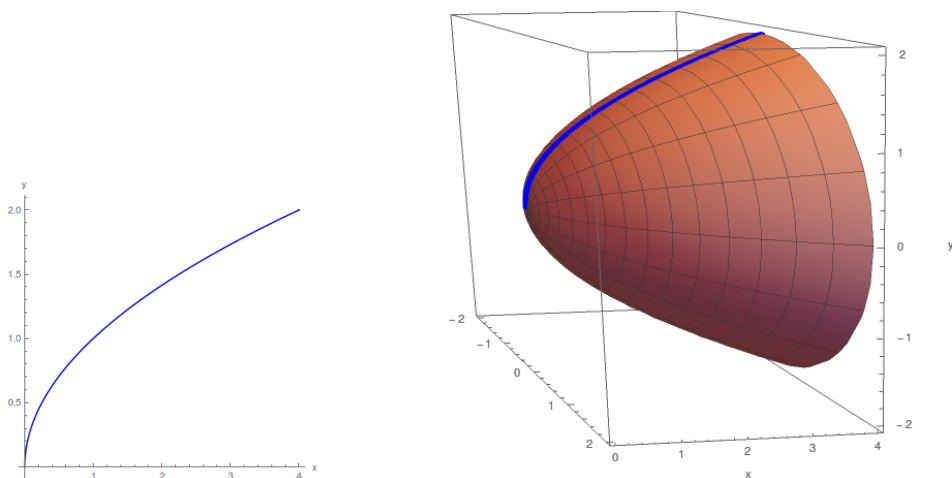
The method of this example will work *in general* if we can build-up a given volume by stacking thin cross sections in a consistent way. Here, “consistent” means in a way we can easily predict from the length, width, or height of the solid figure (as we did for side-length s).

We will not focus on pyramids here since there is a very nice formula for the volume of any pyramid *no matter what kind of base it has!* For a pyramid of height h above a base of area A (regardless of shape), the volume V is given by

$$V = \frac{1}{3}Ah,$$

(i.e. one-third $\times$ base area $\times$ height). Notice that this includes the area of a cone with radius r and height h (whose base is a circle $A = \pi r^2$).

For the remainder of this section, we focus on **solids of revolution**. These are three-dimensional figures obtained by rotating a curve in the xy -plane about an axis in the plane. As an example, consider the curve $y = \sqrt{x}$ on the interval $0 \leq x \leq 4$ rotated about the x -axis.

Figure 3.2.8: $y = \sqrt{x}$ and its Solid of Revolution about the x -axis

The nice thing about solids of revolution is that their cross sections are either circles (also called *disks*) or annuli (also called *washers*).

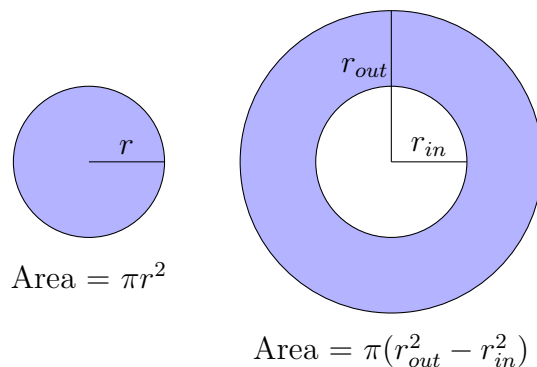


Figure 3.2.9: Disks and Washers

A solid disk has area πr^2 . For a washer, we subtract the area of the smaller disk (radius r_{in}) from the area of the larger disk (radius r_{out}). This gives a total area of $\pi(r_{out}^2 - r_{in}^2)$.

Example 3.2.3. Find the volume of the solid obtained by rotating the region between $y = \sqrt{x}$ and the x -axis on the interval $0 \leq x \leq 4$ about the x -axis.

Answer: This is the solid pictured in Figure 3.2.8. The cross sections of this solid are disks with radius $r = \sqrt{x}$ over the interval $0 \leq x \leq 4$.

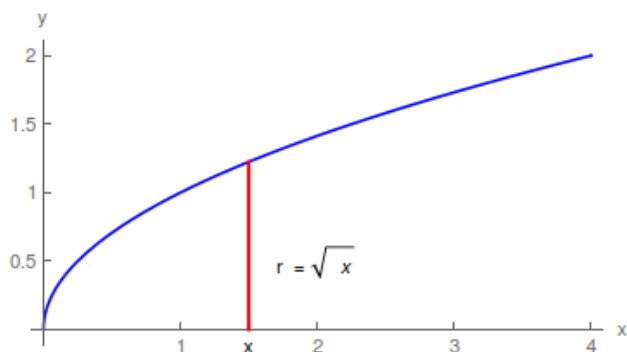


Figure 3.2.10: Radius for Example 3.2.3

This means that the cross sectional area for any particular value of x is

$$\text{cross sectional area} = \pi r^2 = \pi (\sqrt{x})^2 = \pi x.$$

To compute the volume, we simply integrate this cross sectional area over the length of the solid (from $x = 0$ to $x = 4$):

$$\begin{aligned} \text{Volume} &= \int_0^4 \pi x \, dx \\ &= \pi \frac{x^2}{2} \Big|_0^4 \\ &= \pi \frac{4^2}{2} \\ &= 8\pi. \quad \blacksquare \end{aligned}$$

Example 3.2.4. Find the volume of the solid obtained by rotating the region contained between $y = \sqrt{x}$ and $y = x^2$ about the x -axis.

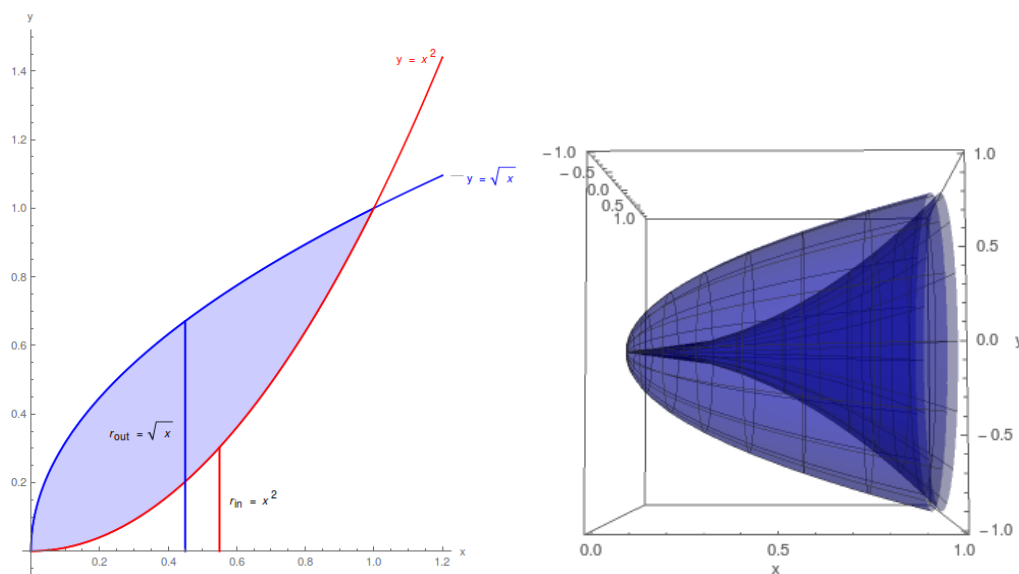


Figure 3.2.11: Radius for Example 3.2.4

Answer: As the figure above shows, the cross sections will be washers with an outer radius of $r_{out} = \sqrt{x}$ and an inner radius of $r_{in} = x^2$. This gives a total cross sectional area of

$$\text{cross sectional area} = \pi \left[(\sqrt{x})^2 - (x^2)^2 \right] = \pi (x - x^4).$$

So, the total volume of this solid will be

$$\begin{aligned} \text{Volume} &= \int_0^1 \pi (x - x^4) \, dx \\ &= \pi \left(\frac{x^2}{2} - \frac{x^5}{5} \right) \Big|_0^1 \\ &= \pi \left(\frac{1}{2} - \frac{1}{5} \right) \\ &= \frac{3\pi}{10}. \quad \blacksquare \end{aligned}$$

While we are not limited to rotating about the coordinate axes, for the purposes of this text we will only consider rotation about the x - and y -axes.

Example 3.2.5. Find the volume of the solid obtained by rotating the region contained between $y = \sqrt{x}$ and the y -axis on the interval $0 \leq x \leq 4$ about the y -axis.

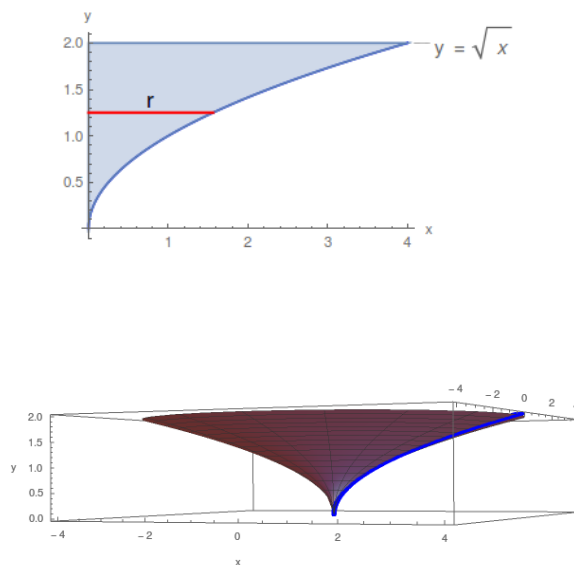


Figure 3.2.12: Area and Solid for Example 3.2.5

Answer: In this example, the cross sections are once again disks. But the disks are now being stacked vertically rather than horizontally (because we rotated about the y -axis). As a result, the radii of the disks are x -values which change as y increases from $y = 0$ to $y = 2$ (the top of the solid figure). In order to find the volume, we have to express the radius r in terms of y (since we will be integrating over the range $0 \leq y \leq 2$). If we take the boundary curve $y = \sqrt{x}$ and solve for x , we find that

$$r = x = y^2.$$

So, the volume of this solid is given by

$$\begin{aligned}
 \text{Volume} &= \int_0^2 \pi (y^2)^2 dy \\
 &= \pi \int_0^2 y^4 dy \\
 &= \pi \left. \frac{y^5}{5} \right|_0^2 \\
 &= \frac{32\pi}{5}. \blacksquare
 \end{aligned}$$

Example 3.2.6. Find the volume of the solid obtained by rotating the region contained between $y = \sqrt[3]{x-1}$ and $y = (x-1)^2$ about the y -axis.

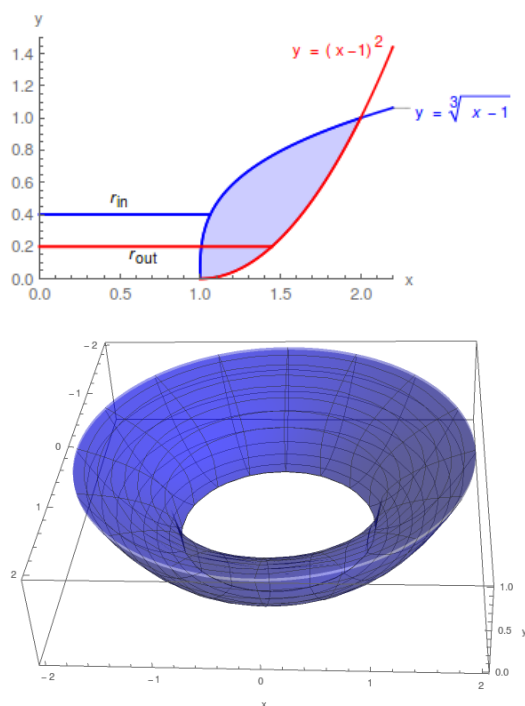


Figure 3.2.13: Area and Solid for Example 3.2.6

Answer: First, we need to determine where these two curves intersect.

$$\begin{aligned}
 \sqrt[3]{x-1} &= (x-1)^2 \\
 x-1 &= (x-1)^6 \\
 0 &= (x-1)^6 - (x-1) \\
 0 &= (x-1)[(x-1)^5 - 1] \\
 x=1 \text{ or } (x-1)^5 &= 1 \\
 x &= 1, 2.
 \end{aligned}$$

Since the cross sections are washers stacked vertically, we need to express the inner and outer radii in terms of y . For the outer radii,

$$\begin{aligned}
 y &= (x-1)^2 \\
 \sqrt{y} &= x-1 \\
 x &= \sqrt{y} + 1, \\
 r_{out} &= \sqrt{y} + 1, \\
 r_{out}^2 &= y + 2\sqrt{y} + 1.
 \end{aligned}$$

The inner radii are given by

$$\begin{aligned}
 y &= \sqrt[3]{x-1} \\
 y^3 &= x-1 \\
 x &= y^3 + 1, \\
 r_{in} &= y^3 + 1, \\
 r_{in}^2 &= y^6 + 2y^3 + 1.
 \end{aligned}$$

So, the cross sectional area at height y is given by

$$\text{cross sectional area} = \pi (y + 2\sqrt{y} - y^6 - 2y^3)$$

For the limits of integration, note that when $x = 1$, both curves are at height

$y = 0$ while at $x = 2$, both are at $y = 1$. So

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^1 (y + 2\sqrt{y} - y^6 - 2y^3) \, dy \\
 &= \pi \left(\frac{y^2}{2} + \frac{4y^{3/2}}{3} - \frac{y^7}{7} - \frac{y^4}{2} \right) \Big|_0^1 \\
 &= \pi \left(\frac{1}{2} + \frac{4}{3} - \frac{1}{7} - \frac{1}{2} \right) \\
 &= \frac{25\pi}{21}. \quad \blacksquare
 \end{aligned}$$

3.3 Improper Integrals

While it may not have been heavily emphasized in your first semester calculus course, the definition of the Riemann integral

$$\int_a^b f(x) \, dx$$

implicitly required the following two important properties:

- 1.) The interval $a \leq x \leq b$ must be a *finite* interval. In other words, the length of the interval, $b - a$, should be finite.
- 2.) The function f must be *bounded* over the interval of integration. That is, f cannot diverge to $\pm\infty$ over $a \leq x \leq b$.

It turns out that we can relax these requirements if we interpret the resulting integral in an appropriate way. Any integral where the two properties above are not satisfied are called **improper integrals**. In general, an improper integral may violate one or both of the requirements above. We will focus on the cases where the integral violates exactly one of the two. The more general case can be handled by splitting up the integral into two or more integrals each of which violates exactly one of the conditions above.

3.3.1 Type I Improper Integrals

We begin with an illustrative example.

Example 3.3.1. Compute the integral

$$\int_1^{\infty} \frac{1}{x^2} dx.$$

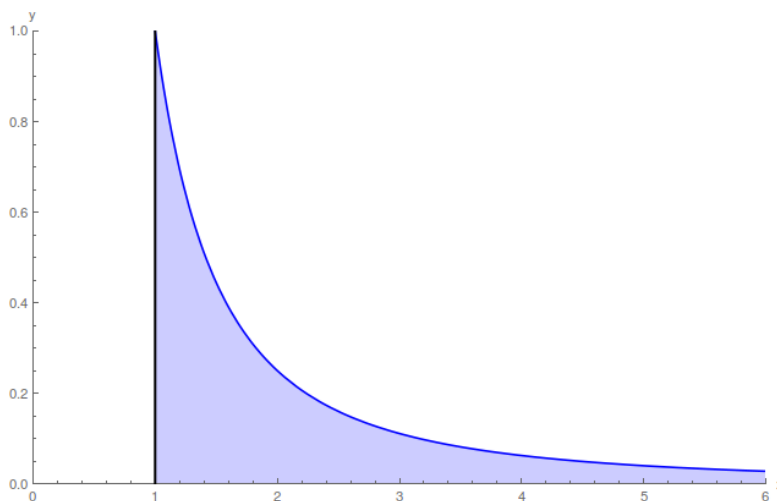


Figure 3.3.1: Graph of $y = \frac{1}{x^2}$

If we consider our two requirements for Riemann integrals, we clearly see that the first is violated: the interval of integration has infinite length. The second requirement (that f is bounded over the interval of integration) is satisfied since $f(x) = 1/x^2$ only diverges at $x = 0$ (which is outside our interval).

An improper integral where a bounded function is to be integrated over an infinite interval of the real line is said to be a **Type I Improper Integral**. In order to make sense of such an interval, we employ a limit!

Answer: Since we cannot explicitly compute a Riemann integral over an infinite interval, we use a limit to do the job:

$$\int_1^{\infty} \frac{1}{x^2} dx = \lim_{R \rightarrow \infty} \int_1^R \frac{1}{x^2} dx.$$

Since R is always a *finite* real number, the integral on the right makes perfect sense. Once we compute the integral, we can then take the limit to “push R

out to infinity.”

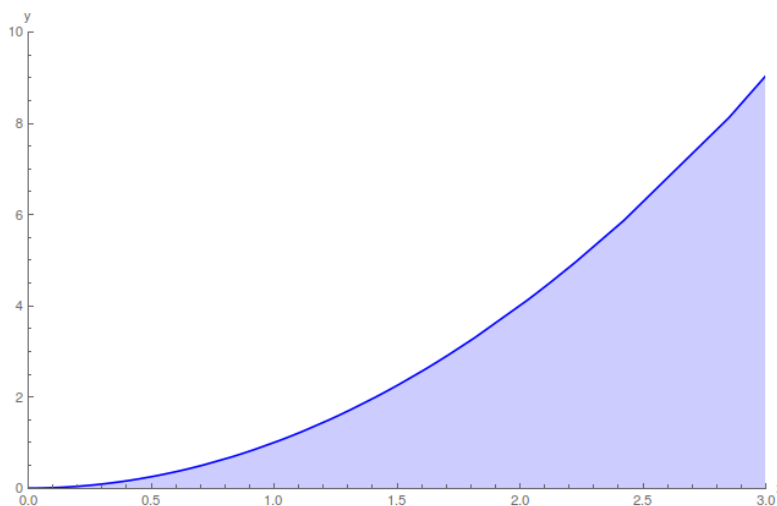
$$\begin{aligned}\int_1^\infty \frac{1}{x^2} dx &= \lim_{R \rightarrow \infty} \int_1^R \frac{1}{x^2} dx \\&= \lim_{R \rightarrow \infty} \left. \frac{-1}{x} \right|_1^R \\&= \lim_{R \rightarrow \infty} \left(\frac{-1}{R} - \frac{-1}{1} \right) \\&= \lim_{R \rightarrow \infty} \left(1 - \frac{1}{R} \right) \\&= 1 - \lim_{R \rightarrow \infty} \frac{1}{R} \\&= 1. \blacksquare\end{aligned}$$

So, the total area under $y = 1/x^2$ on the interval $1 \leq x < \infty$ is equal to 1, a finite value! If an improper integral of either type returns a finite value, we say that it *converges*.

Be aware that an improper integral need not converge! If the resulting limit returns $\pm\infty$, we say the improper integral *diverges*. The other possibility is that the improper integral *does not exist* (when the limit fails to exist). Some authors use “diverge” for both of these cases.

Example 3.3.2. Compute the integral

$$\int_0^\infty x^2 dx.$$

Figure 3.3.2: Graph of $y = x^2$

Answer: Using a limit to make sense of the integral, we find

$$\begin{aligned}
 \int_0^{\infty} x^2 \, dx &= \lim_{R \rightarrow \infty} \int_0^R x^2 \, dx \\
 &= \lim_{R \rightarrow \infty} \left. \frac{1}{3} x^3 \right|_0^R \\
 &= \lim_{R \rightarrow \infty} \frac{R^3}{3} \\
 &= \infty. \quad \blacksquare
 \end{aligned}$$

This improper integral diverges. Of course, this was clear from the graph! *If a function does not limit to 0 as $x \rightarrow \infty$, then its integral over $a \leq x < \infty$ will diverge.* The same is true for integrals out to $-\infty$.

Before we look at several more examples, we give the precise definition of Type I Improper Integrals.

Type I Improper Integrals

Suppose that f is continuous over the intervals of integration below.

$$\begin{aligned}\int_a^\infty f(x) \, dx &= \lim_{R \rightarrow \infty} \int_a^R f(x) \, dx, \\ \int_{-\infty}^b f(x) \, dx &= \lim_{r \rightarrow -\infty} \int_r^b f(x) \, dx, \\ \int_{-\infty}^\infty f(x) \, dx &= \lim_{r \rightarrow -\infty} \int_r^a f(x) \, dx + \lim_{R \rightarrow \infty} \int_a^R f(x) \, dx,\end{aligned}$$

where a in the last integral is an arbitrary real number (often taken to be 0).

Example 3.3.3. Compute

$$\int_2^\infty e^{-x} \, dx.$$

Answer:

$$\begin{aligned}\int_2^\infty e^{-x} \, dx &= \lim_{R \rightarrow \infty} \int_2^R e^{-x} \, dx \\ &= \lim_{R \rightarrow \infty} -e^{-x} \Big|_2^R \\ &= \lim_{R \rightarrow \infty} -\frac{1}{e^R} + e^{-2} \\ &= e^{-2}. \quad \blacksquare\end{aligned}$$

Example 3.3.4. Compute

$$\int_{10}^\infty \frac{1}{\sqrt{x}} \, dx$$

Answer:

$$\begin{aligned}\int_{10}^\infty \frac{1}{\sqrt{x}} \, dx &= \lim_{R \rightarrow \infty} \int_{10}^R \frac{1}{\sqrt{x}} \, dx \\ &= \lim_{R \rightarrow \infty} 2\sqrt{x} \Big|_{10}^R \\ &= \lim_{R \rightarrow \infty} 2\sqrt{R} - 2\sqrt{10} \\ &= \infty. \quad \blacksquare\end{aligned}$$

This last example raises a very important issue. If we look at the graph of the integrand, we find the following.

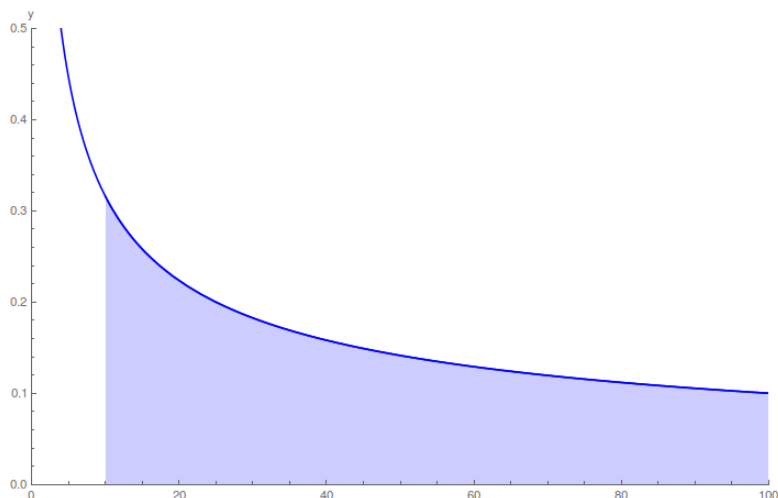


Figure 3.3.3: Graph of $y = 1/\sqrt{x}$

Notice that the function is converging to 0 as $x \rightarrow \infty$, but *it is not converging fast enough to keep the improper integral from diverging*. How fast a function needs to converge to zero for its integral to converge is addressed by the following theorem.

Theorem 3.3.1. *The Type I Improper Integrals*

$$\int_1^{\infty} \frac{1}{x^n} dx$$

- **converge** when $n > 1$,
- **diverge** when $n \leq 1$.

Notice that there is nothing special about our choice of lower limit for these integrals – any *positive* lower limit would give the same results!

Proof: The indefinite integral above is different when $n = 1$, so we split that case from the rest.

$$\begin{aligned}
\int_1^\infty \frac{1}{x^n} dx &= \lim_{R \rightarrow \infty} \int_1^R x^{-n} dx \\
&= \lim_{R \rightarrow \infty} \begin{cases} \left. \frac{x^{-n+1}}{-n+1} \right|_1^R, & n \neq 1 \\ \ln |x| \Big|_1^R, & n = 1 \end{cases} \\
&= \lim_{R \rightarrow \infty} \begin{cases} \frac{R^{-n+1}}{-n+1} - \frac{1}{-n+1}, & n \neq 1 \\ \ln(R), & n = 1 \end{cases}
\end{aligned}$$

When $n > 1$, there will be some power of R in the denominator of the leftmost fraction above. The limit of that term will be zero and we see

$$\int_1^\infty \frac{1}{x^n} dx = \frac{1}{n-1} \text{ when } n > 1.$$

For $n \leq 1$, the limit above will definitely diverge. ■

Example 3.3.5. Compute

$$\int_{-\infty}^{-2} \frac{1}{x^2 - 1} dx.$$

Answer: Notice that the integrand diverges at $x = \pm 1$. Fortunately, this is outside of our interval of integration! Since this integral is a little more complicated, we tackle it first before looking at the limit. By partial fraction decomposition

$$\frac{1}{x^2 - 1} = \frac{1}{2} \left(\frac{1}{x - 1} - \frac{1}{x + 1} \right),$$

and so

$$\begin{aligned}
\int \frac{1}{x^2 - 1} dx &= \frac{1}{2} \int \left(\frac{1}{x - 1} - \frac{1}{x + 1} \right) dx \\
&= \frac{1}{2} (\ln |x - 1| - \ln |x + 1|) + C \\
&= \frac{1}{2} \ln \left| \frac{x - 1}{x + 1} \right| + C.
\end{aligned}$$

Our improper integral becomes

$$\begin{aligned}
 \int_{-\infty}^{-2} \frac{1}{x^2 - 1} dx &= \lim_{r \rightarrow -\infty} \int_r^{-2} \frac{1}{x^2 - 1} dx \\
 &= \frac{1}{2} \lim_{r \rightarrow -\infty} \ln \left| \frac{x-1}{x+1} \right|_r^{-2} \\
 &= \frac{1}{2} \ln \left| \frac{-2-1}{-2+1} \right| - \frac{1}{2} \lim_{r \rightarrow -\infty} \ln \left| \frac{r-1}{r+1} \right| \\
 &= \frac{1}{2} \ln(3) - \frac{1}{2} \ln \left(\lim_{r \rightarrow -\infty} \left| \frac{r-1}{r+1} \right| \right) \\
 &= \frac{1}{2} \ln(3) - \frac{1}{2} \ln(1) \\
 &= \frac{1}{2} \ln(3).
 \end{aligned}$$

Notice that in the fourth step above, we used the continuity of the logarithm (for positive real numbers) to move the limit inside this function. ■

Example 3.3.6. Compute

$$\int_{-\infty}^{\infty} x e^{-x^2} dx.$$

Answer: The indefinite integral is an easy u -substitution: $u = -x^2$ giving $du = -2x dx$.

$$\begin{aligned}
 \int x e^{-x^2} dx &= -\frac{1}{2} \int e^{-x^2} (-2x) dx \\
 &= -\frac{1}{2} \int e^u du \\
 &= -\frac{1}{2} e^u + C \\
 &= -\frac{1}{2} e^{-x^2} + C.
 \end{aligned}$$

In order to compute the given improper integral, we have to split the integral

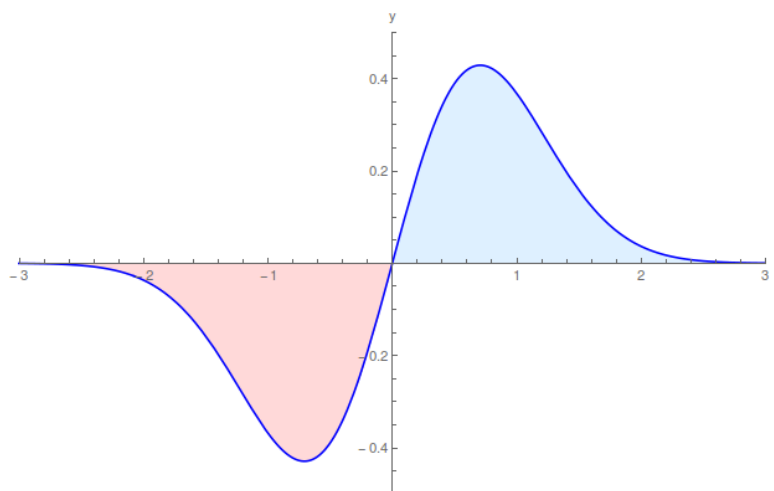
into two parts *at any arbitrary real number*. Often, 0 is an excellent choice.

$$\begin{aligned}
 \int_{-\infty}^{\infty} x e^{-x^2} dx &= \lim_{r \rightarrow -\infty} \int_r^0 x e^{-x^2} dx + \lim_{R \rightarrow \infty} \int_0^R x e^{-x^2} dx \\
 &= \lim_{r \rightarrow -\infty} \left(-\frac{1}{2} e^{-x^2} \Big|_r^0 \right) + \lim_{R \rightarrow \infty} \left(-\frac{1}{2} e^{-x^2} \Big|_0^R \right) \\
 &= \lim_{r \rightarrow -\infty} \left(-\frac{1}{2} + \frac{1}{2} e^{-r^2} \right) + \lim_{R \rightarrow \infty} \left(-\frac{1}{2} e^{-R^2} + \frac{1}{2} \right) \\
 &= -\frac{1}{2} + \frac{1}{2} \\
 &= 0.
 \end{aligned}$$

Just to emphasize that the choice of 0 makes no difference to the final answer, consider what would have happened had we chosen 1:

$$\begin{aligned}
 \int_{-\infty}^{\infty} x e^{-x^2} dx &= \lim_{r \rightarrow -\infty} \int_r^1 x e^{-x^2} dx + \lim_{R \rightarrow \infty} \int_1^R x e^{-x^2} dx \\
 &= \lim_{r \rightarrow -\infty} \left(-\frac{1}{2} e^{-x^2} \Big|_r^1 \right) + \lim_{R \rightarrow \infty} \left(-\frac{1}{2} e^{-x^2} \Big|_1^R \right) \\
 &= \lim_{r \rightarrow -\infty} \left(-\frac{1}{2} e^{-1} + \frac{1}{2} e^{-r^2} \right) + \lim_{R \rightarrow \infty} \left(-\frac{1}{2} e^{-R^2} + \frac{1}{2} e^{-1} \right) \\
 &= -\frac{1}{2} e^{-1} + \frac{1}{2} e^{-1} \\
 &= 0.
 \end{aligned}$$

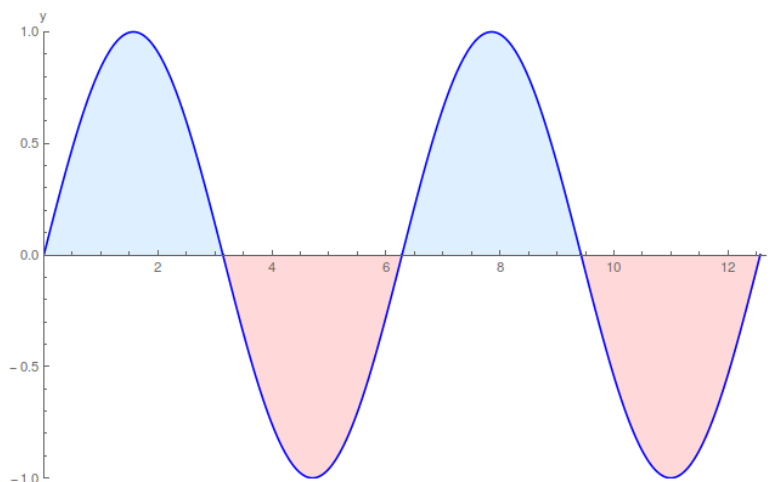
If we had first considered the graph of this function, we could have anticipated that the answer would be zero.

Figure 3.3.4: Graph of $y = xe^{-x^2}$

It certainly seems that the negative area on the left will exactly cancel the positive area on the right (by symmetry). This turns out to be correct in this case, but great care must be taken with this kind of reasoning! ■

Example 3.3.7. Compute

$$\int_0^{\infty} \sin(x) \, dx.$$

Figure 3.3.5: Graph of $y = \sin(x)$

Answer: From the figure above, you might suspect that the value of this integral would be zero, but that is not the case!

$$\begin{aligned}
 \int_0^\infty \sin(x) \, dx &= \lim_{R \rightarrow \infty} \int_0^R \sin(x) \, dx \\
 &= \lim_{R \rightarrow \infty} -\cos(x) \Big|_0^R \\
 &= \cos(0) - \lim_{R \rightarrow \infty} \cos(R) \\
 &= 1 - \lim_{R \rightarrow \infty} \cos(R).
 \end{aligned}$$

But $\lim_{R \rightarrow \infty} \cos(R)$ *does not exist* because cosine has no definite limiting behavior at infinity (it oscillates between -1 and 1 forever). Thus,

$$\int_0^\infty \sin(x) \, dx \text{ does not exist. } \blacksquare$$

3.3.2 Type II Improper Integrals

Type II Improper Integrals are integrals where the integrand diverges at some point over its interval of integration. By breaking integrals apart, we can always assume that the point of divergence is one of the endpoints of the interval of integration.

Type II Improper Integrals

Suppose that $a \leq x \leq b$ is a finite interval and f is continuous over this interval except at one of the endpoints.

If f diverges at the left endpoint a , then

$$\int_a^b f(x) \, dx = \lim_{r \rightarrow a^+} \int_r^b f(x) \, dx.$$

If f diverges at the right endpoint b , then

$$\int_a^b f(x) \, dx = \lim_{R \rightarrow b^-} \int_a^R f(x) \, dx.$$

Notice that the definitions involve either a *left-handed* or a *right-handed* limit. In the case where f diverges at the left endpoint, we take a right-handed limit $r \rightarrow a^+$ because we only need to consider values of x bigger than the lower

limit $x = a$. Likewise, we use a left-handed limit $R \rightarrow b^-$ in the second case since we only need to consider values of x smaller than $x = b$ when computing the integral.

Example 3.3.8. Compute

$$\int_0^1 \frac{1}{\sqrt{x}} dx.$$

Answer: Since the integrand diverges at the left endpoint $x = 0$, we have

$$\begin{aligned} \int_0^1 \frac{1}{\sqrt{x}} dx &= \lim_{r \rightarrow 0^+} \int_r^1 \frac{1}{\sqrt{x}} dx \\ &= \lim_{r \rightarrow 0^+} 2\sqrt{x} \Big|_r^1 \\ &= 2 - 2 \lim_{r \rightarrow 0^+} \sqrt{r} \\ &= 2. \blacksquare \end{aligned}$$

Example 3.3.9. Compute

$$\int_0^2 \frac{1}{(x-2)^2} dx.$$

Answer: This integrand diverges at the right endpoint of the integral.

$$\begin{aligned} \int_0^2 \frac{1}{(x-2)^2} dx &= \lim_{R \rightarrow 2^-} \int_0^R \frac{1}{(x-2)^2} dx \\ &= \lim_{R \rightarrow 2^-} \frac{-1}{x-2} \Big|_0^R \\ &= - \lim_{R \rightarrow 2^-} \frac{1}{R-2} - \frac{1}{2} \\ &= \infty. \end{aligned}$$

So, this improper integral diverges. $\blacksquare$

These two examples are specific cases of a more general trend.

Theorem 3.3.2. *The Type II Improper Integrals*

$$\int_0^1 \frac{1}{x^n} dx$$

- *converge* when $n < 1$,
- *diverge* when $n \geq 1$.

Proof: As above, we split the case $n = 1$ from the rest.

$$\begin{aligned}
 \int_0^1 \frac{1}{x^n} dx &= \lim_{r \rightarrow 0^+} \int_r^1 x^{-n} dx \\
 &= \lim_{r \rightarrow 0^+} \begin{cases} \left. \frac{x^{-n+1}}{-n+1} \right|_r^1, & n \neq 1 \\ \ln |x| \Big|_r^1, & n = 1 \end{cases} \\
 &= \lim_{r \rightarrow 0^+} \begin{cases} \frac{1}{-n+1} - \frac{r^{-n+1}}{-n+1}, & n \neq 1 \\ -\ln(r), & n = 1 \end{cases}
 \end{aligned}$$

When $n < 1$, there will be some power of r in the numerator of the rightmost fraction above. The limit of that term will be zero and we see

$$\int_0^1 \frac{1}{x^n} dx = \frac{1}{1-n} \text{ when } n < 1.$$

For $n \geq 1$, the limit above will definitely diverge. ■

If f diverges in the interior of the interval of integration, split the integral into two parts at the point where f diverges.

Example 3.3.10. Compute

$$\int_{-1}^1 \left(\frac{1}{\sqrt[3]{x}} \right)^2 dx.$$

Answer: The integrand diverges at $x = 0$. So, we split the integral into two pieces:

$$\int_{-1}^1 \left(\frac{1}{\sqrt[3]{x}} \right)^2 dx = \int_{-1}^0 \left(\frac{1}{\sqrt[3]{x}} \right)^2 dx + \int_0^1 \left(\frac{1}{\sqrt[3]{x}} \right)^2 dx.$$

The first integral on the right gives

$$\begin{aligned}
 \int_{-1}^0 \left(\frac{1}{\sqrt[3]{x}} \right)^2 dx &= \lim_{R \rightarrow 0^-} \int_{-1}^R x^{-2/3} dx \\
 &= \lim_{R \rightarrow 0^-} 3x^{1/3} \Big|_{-1}^R \\
 &= \lim_{R \rightarrow 0^-} 3R^{1/3} - 3(-1)^{1/3} \\
 &= 3.
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 \int_0^1 \left(\frac{1}{\sqrt[3]{x}} \right)^2 dx &= \lim_{r \rightarrow 0^+} \int_r^1 x^{-2/3} dx \\
 &= \lim_{r \rightarrow 0^+} 3x^{1/3} \Big|_r^1 \\
 &= 3 - \lim_{r \rightarrow 0^+} 3r^{1/3} \\
 &= 3.
 \end{aligned}$$

Combining these results gives

$$\int_{-1}^1 \left(\frac{1}{\sqrt[3]{x}} \right)^2 dx = 3 + 3 = 6. \quad \blacksquare$$

3.4 Problems

- 1.) Compute

$$\int_{-2}^3 (x^2 - 1) dx.$$

Find the *total area* between the curve $y = x^2 - 1$ and the x -axis on the interval $-2 \leq x \leq 3$. Why are these two numbers different?

- 2.) Find the *total area* between $y = \cos(x)$ and the x -axis on the interval $0 \leq x \leq 2\pi$.
- 3.) Find the area contained between the curves $y = x^2 - 10$ and $y = 8 - x^2$.

- 4.) Find the area contained between the curves $y = 4x$ and $y = x^2 + 3$.
- 5.) Find the area contained between the curves $y = x^3 + 3x^2 + 3$ and $y = x^2 + 3x + 3$.
- 6.) Find the area bounded by the curves $y = 3$, $x = 0$, and

$$y = \frac{8x}{x^2 - 1}.$$

- 7.) Find the area bounded by the curves $y = \sqrt{x}$, $y = \sqrt{2x - 1}$, and $y = 0$.
(Hint: This will require you to compute two integrals!)
- 8.) Find the volume of the solid obtained by rotating the region between $y = \sqrt{2x + 1}$ and the x -axis on the interval $0 \leq x \leq 4$ about the x -axis.
- 9.) Find the volume of the solid obtained by rotating the region between $y = \sqrt{2x + 1}$ and the y -axis on the interval $0 \leq x \leq 4$ about the y -axis.
- 10.) Find the volume of the solid obtained by rotating the region contained between $y = \sqrt[3]{x}$ and $y = \sqrt{x}$ about the x -axis.
- 11.) Find the volume of the solid obtained by rotating the region contained between $y = \sqrt[3]{x}$ and $y = \sqrt{x}$ about the y -axis.
- 12.) Find the volume of the solid obtained by rotating the region contained between $y = x$ and $y = 2\sqrt{x}$ about the x -axis.
- 13.) Find the volume of the solid obtained by rotating the region contained between $y = x$ and $y = 2\sqrt{x}$ about the y -axis.
- 14.) Find the volume of the solid obtained by rotating the region bounded by $y = 1 - e^{-x^2}$, $y = \frac{1}{2}$, and $x = 0$ about the y -axis.
- 15.) Prove that the volume of a right circular cone of radius r and height h is given by

$$V = \frac{1}{3}\pi r^2 h$$

by rotating an appropriate line through the origin about the y -axis.

- 16.) Compute

$$\int_0^\infty e^{-2x} dx.$$

17.) Compute

$$\int_{-\infty}^1 e^{3x} dx.$$

18.) Compute

$$\int_{-\infty}^{-1} \frac{1}{\sqrt[3]{x}} dx$$

19.) Compute

$$\int_1^{\infty} x e^{-x} dx.$$

20.) Compute

$$\int_2^{\infty} \frac{1}{x^3 - 2x^2 + x} dx$$

21.) Compute

$$\int_{-1}^0 \frac{1}{\sqrt[3]{x}} dx$$

22.) Compute

$$\int_2^{10} \frac{1}{(x-2)^3} dx$$

23.) Compute

$$\int_3^6 \frac{x}{\sqrt{x-3}} dx$$

24.) Find the volume of the solid obtained by rotating the region between $y = 1/x$ and the x -axis on the interval $1 \leq x < \infty$ about the x -axis.¹

25.) **Challenge:** Compute

$$\int_1^{\infty} \frac{\ln(x)}{x^2} dx.$$

(HINT: Remember L'Hôpital's Rule!)

¹The solid in Problem 24.) is known as Gabriel's Horn. You should find that it has finite volume, but it turns out that it has *infinite* surface area. Loosely speaking, you could fill the horn with a finite volume of paint, but that amount of paint is not nearly enough to cover its surface!

Chapter 4

Continuous Dynamical Systems

4.1 Introduction to Differential Equations

For a continuous-time dynamical system, we often know something about the *rate of change* of the quantity (or quantities) we are studying rather than having direct knowledge about its actual value. In other words, we have information about the quantity's *derivative*, and our task is to deduce the value of the quantity itself from this information. Since anti-differentiation is the process of “undoing” differentiation, it will come as no surprise that indefinite integrals play a key role.

As an example, consider an object at some particular temperature sitting in a large environment that is held at a fixed temperature. We will make the simplifying assumption that the temperature of the object is uniform throughout its body. We also assume that the environment is large enough that our object does not significantly change the ambient temperature as it heats up or cools down.

Suppose that the temperature of the object at time t is the function $T(t)$. Time here can be any real number, so this is clearly a continuous-time dynamical system. The fixed environmental temperature is T_e (e for environment). In this system, the time t is called the **independent variable**. This indicates that t is the quantity upon which all other variables in the problem will depend. In contrast, the temperature $T = T(t)$ is the **dependent variable**. Dependent variables are any quantities in the problem which are functions of the independent variable.

Isaac Newton is attributed with giving the first universal law of cooling:

the rate at which an object's temperature changes is proportional to its difference in temperature with the environment.¹ In terms, of $T = T(t)$, we have

$$\frac{dT}{dt} = k(T_e - T),$$

where $k > 0$ is the cooling rate of the object. The cooling rate depends on many things in general, but it is most directly affected by the size, shape, and composition of the object as well as the type of environment.²

The equation above is a **differential equation**: an equation for an unknown function in terms of one (or more) of its derivatives. In this case, $T = T(t)$ is the unknown function, and we have information about its first derivative. In such a case, we say that the equation above is a **first order differential equation**. If we had information about the second derivative, we would say that we had a *second order differential equation*. In general, the **order** of a differential equation is the order of the highest derivative appearing in the equation. In this chapter, we will only deal with first order equations. Notice that the differential equation for dT/dt involves the unknown function T itself! This is a common occurrence in differential equations that arise in practice.

There are two constants appearing in the differential equation above: the environmental temperature T_e and the cooling rate k . These are both **parameters**: quantities appearing in a dynamical system which are fixed numbers in any particular version of the problem. In this case, the environmental temperature and cooling rate are fixed once we decide what sort of object we are modeling and what kind of environment the object will be sitting in.

There is an additional piece of information that we need in order to guarantee a unique answer. Thinking about our example of an object that is cooling off (or heating up) in some environment, we would need to know the actual temperature of the object at some particular time to make a definite prediction about the temperature in the future. Typically, we assume that we know the temperature at time $t = 0$, and our task is to predict the

¹While Newton's Law of Cooling is extremely useful in many situations, it has definite limitations. Obviously it does not directly model differences of temperature inside of an object. It also suffers from a confusion between the concepts of heat energy and temperature – a confusion that was only cleared up well after Newton's day.

²For example, a hot piece of metal will cool off much quicker in a water bath than in air at the same temperature.

temperature for future times $t > 0$. This extra information is called the **initial condition** (or *initial datum*) for the differential equation: a given value of the function at some particular time t_0 (often $t_0 = 0$).

Our model of the cooling object is now an instance of an **initial value problem**: a differential equation together with appropriate initial data.

$$\begin{aligned}\frac{dT}{dt} &= k(T_e - T) \\ T(0) &= T_0\end{aligned}$$

Initial value problems (IVPs) are very common in applied mathematics.

4.2 Solutions to Differential Equations

Now that we have discussed the basic terminology of differential equations, our next goal is to understand what it means for a function to be a solution. A **solution to a differential equation** is any function such that substituting it for the dependent variable in the differential equation yields a true statement.

Consider the following example.

Example 4.2.1. Show that $T(t) = 10 - 5e^{-t/2}$ is a solution to the differential equation

$$\frac{dT}{dt} = \frac{1}{2}(10 - T).$$

Answer: We first compute the derivative of T :

$$\begin{aligned}\frac{dT}{dt} &= \frac{d}{dt} [10 - 5e^{-t/2}] \\ &= -5e^{-t/2} \left(-\frac{1}{2}\right) \\ &= \frac{5}{2}e^{-t/2}.\end{aligned}$$

If we look at the right-hand side of the differential equation, we find

$$\begin{aligned}\frac{1}{2}(10 - T) &= \frac{1}{2} [10 - (10 - 5e^{-t/2})] \\ &= \frac{1}{2} (5e^{-t/2}) \\ &= \frac{5}{2}e^{-t/2}.\end{aligned}$$

Since $T(t) = 10 - 5e^{-t/2}$ gives the same function when substituted into both the right- and left-hand sides of the differential equation, it is a solution. ■

Without initial data, typical differential equations will have *infinite families of solutions*. Usually this family of solutions has a definite form with one or more arbitrary constants appearing in the formula. The formula specifying the entire family of solutions to a differential equation is called the **general solution** of the differential equation.

Example 4.2.2. Show that $T(t) = 10 + Ce^{-t/2}$ is a solution to

$$\frac{dT}{dt} = \frac{1}{2}(10 - T)$$

for any choice of the arbitrary constant C .

Answer: As before, we verify that the function $T = T(t)$ given above produces the same result on both sides of the differential equation. Remember that C represents an *arbitrary real number* (and so is not a function of t).

$$\begin{aligned}\frac{dT}{dt} &= \frac{d}{dt} [10 + Ce^{-t/2}] \\ &= Ce^{-t/2} \left(-\frac{1}{2}\right) \\ &= -\frac{C}{2}e^{-t/2}.\end{aligned}$$

For the right-hand side:

$$\begin{aligned}\frac{1}{2}(10 - T) &= \frac{1}{2} [10 - (10 + Ce^{-t/2})] \\ &= \frac{1}{2} (-Ce^{-t/2}) \\ &= -\frac{C}{2}e^{-t/2}.\end{aligned}$$

Since the given function produces the same quantity on both sides of the differential equation, we know that $T(t) = 10 + Ce^{-t/2}$ is a solution of the differential equation *for any choice* of the constant C . It turns out that this

T is the general solution to the differential equation above. We will be able to prove this in later sections! ■

Example 4.2.3. Show that

$$f(x) = Ce^{x^2} - \frac{5}{2}$$

is a solution to the differential equation

$$\frac{df}{dx} = 2xf + 5x$$

for any choice of the arbitrary constant C .

Answer: Taking the derivative of the given function gives

$$\begin{aligned} \frac{df}{dx} &= \frac{d}{dx} \left[Ce^{x^2} - \frac{5}{2} \right] \\ &= 2Cxe^{x^2}. \end{aligned}$$

For the right-hand side, we find

$$\begin{aligned} 2xf + 5x &= 2x \left[Ce^{x^2} - \frac{5}{2} \right] + 5x \\ &= 2Cxe^{x^2} - 5x + 5x \\ &= 2Cxe^{x^2}. \quad \blacksquare \end{aligned}$$

To verify that a given function is a solution to an initial value problem, we have to verify that it satisfies both the differential equation and the given initial condition.

Example 4.2.4. Verify that

$$x(t) = 6e^{t^2-1} - \frac{5}{2}t^2 - \frac{5}{2}$$

is a solution to the initial value problem

$$\begin{aligned} t \frac{dx}{dt} &= 2t^2x + 5t^4 \\ x(1) &= 1. \end{aligned}$$

Answer: Substituting $x = x(t)$ into the left-hand side of the differential equation gives

$$\begin{aligned} t \frac{dx}{dt} &= t \frac{d}{dt} \left(6e^{t^2-1} - \frac{5}{2}t^2 - \frac{5}{2} \right) \\ &= t \left(6e^{t^2-1}(2t) - \frac{5}{2}(2t) \right) \\ &= 12t^2e^{t^2-1} - 5t^2. \end{aligned}$$

For the right-hand side, we find

$$\begin{aligned} 2t^2x + 5t^4 &= 2t^2 \left(6e^{t^2-1} - \frac{5}{2}t^2 - \frac{5}{2} \right) + 5t^4 \\ &= 12t^2e^{t^2-1} - 5t^4 - 5t^2 + 5t^4 \\ &= 12t^2e^{t^2-1} - 5t^2. \end{aligned}$$

This proves that the given function is a solution to the differential equation. We only need to check the initial datum.

$$\begin{aligned} x(1) &= 6e^{(1)^2-1} - \frac{5}{2}(1)^2 - \frac{5}{2} \\ &= 6e^0 - \frac{5}{2} - \frac{5}{2} \\ &= 6 - 5 \\ &= 1. \quad \blacksquare \end{aligned}$$

While it is always worthwhile to know how to verify that a solution is correct, what we really need to understand is how to produce the general solution to a given differential equation. In general, this is a VERY difficult problem (just as integration is difficult in general). We will focus on two special classes of differential equations that are typically easy to solve and useful in many applied problems.

4.3 Separable Differential Equations

The most general first order differential equation is an equation of the form

$$F \left(t, x, \frac{dx}{dt} \right) = 0.$$

This generic equation represents *any combination* of t , $x = x(t)$, and $x'(t)$ set equal to zero. Such an equation can be quite difficult to solve, so most treatments of first order differential equations discuss the slightly less general equation

$$\frac{dx}{dt} = f(t, x).$$

Even this simplified form can be difficult to solve in general, but at the very least there are several techniques that can produce numeric approximations to the actual solutions of this type of differential equation. We will begin by focusing on an even narrower class of differential equations - **separable first order differential equations**.

A separable first order differential equation is any differential equation of the form

$$\frac{dx}{dt} = f(t)g(x).$$

As the name suggests, the key thing about this class is that the right-hand side of the differential equation is a product of separate functions of the independent variable t and the dependent variable $x = x(t)$.

It turns out that there is a simple process for finding the solution to separable differential equations. We will first develop this general process and then apply it to a number of examples.

Theorem 4.3.1. *The general solution to the separable equation*

$$\frac{dx}{dt} = f(t)g(x)$$

is given by the integrals

$$\int \frac{dx}{g(x)} = \int f(t) dt + C.$$

Proof: In order to understand the proof, it will help to explicitly display the functional dependence $x = x(t)$ in the differential equation:

$$x'(t) = f(t)g(x(t)).$$

We begin by bringing all terms involving $x(t)$ to one side of the equation

$$\frac{x'(t)}{g(x(t))} = f(t)$$

(we should technically assume that $g(x(t))$ is non-zero, but this will not make any difference in the proof). Next, we integrate both sides with respect to t .

$$\int \frac{x'(t)}{g(x(t))} dt = \int f(t) dt + C.$$

Notice that we only need one unknown constant on the right-hand side of the equation! We cannot do anything more with the right-hand integral without knowing the specific form of the function $f(t)$. The left-hand side can be simplified by a u -substitution: $u = x(t)$ and $du = x'(t) dt$.

$$\int \frac{x'(t)}{g(x(t))} dt = \int \frac{du}{g(u)}.$$

Since the variable u was an arbitrary choice, we could just as easily use x (committing a slight abuse of notation). This gives us

$$\int \frac{dx}{g(x)} = \int f(t) dt = C. \blacksquare$$

What this theorem suggests is that we can treat dx/dt as if it were a fraction (which it definitely is not). For the equation

$$\frac{dx}{dt} = f(t)g(x),$$

we solve by dividing both sides by $g(x)$ and then multiplying both sides by dt . We then integrate both sides and attempt to solve for $x = x(t)$.

Example 4.3.1. Find the general solution to

$$\frac{dx}{dt} = tx.$$

Answer: We first separate the t and x variables and integrate to get

$$\int \frac{dx}{x} = \int t dt + C.$$

Integrating both sides gives

$$\ln |x| = \frac{1}{2}t^2 + C.$$

Notice that we only need one arbitrary constant for both integrals. We can solve for x by first taking the natural exponential of both sides:

$$\begin{aligned} e^{\ln|x|} &= e^{\frac{1}{2}t^2+C} \\ |x| &= e^C e^{\frac{1}{2}t^2}. \end{aligned}$$

Finally, we undo the absolute value by allowing the right-hand side to be either positive or negative:

$$\begin{aligned} |x| &= e^C e^{\frac{1}{2}t^2} \\ x(t) &= \pm e^C e^{\frac{1}{2}t^2} \\ x(t) &= \mathcal{C} e^{\frac{1}{2}t^2}. \end{aligned}$$

Since $\pm e^C$ can be any arbitrary real number (positive or negative)³, we simply replaced it with a new arbitrary constant $\mathcal{C}$ in the last step. This gives us the general solution to the differential equation. ■

Example 4.3.2. Find the general solution to

$$\frac{dx}{dt} = e^{3t}(1+x^2).$$

Separating and integrating gives

$$\begin{aligned} \int \frac{dx}{1+x^2} &= \int e^{3t} dt + C \\ \arctan(x) &= \frac{1}{3}e^{3t} + C \\ x(t) &= \tan\left(\frac{1}{3}e^{3t} + C\right). \quad \blacksquare \end{aligned}$$

Notice in the last example that the “ $+C$ ” is part of the function argument of the tangent. This is critical to the solution, and the functions $\tan\left(\frac{1}{3}e^{3t}\right) +$

³Technically, $\pm e^C$ can be any value *except* 0. However, allowing $\mathcal{C} = 0$ yields a correct solution (namely $x(t) = 0$). The reason this solution was not found by our calculations was that in dividing by $x = x(t)$, we implicitly assumed it was non-zero! This is a very common happenstance with separable DEs.

C are completely different (and not the general solution to the differential equation above).

For initial value problems, we first find the general solution as above and then force the initial condition (which will determine the arbitrary constant C).

Example 4.3.3. Find the solution to

$$\begin{aligned}\frac{dx}{dt} &= t(x+1) \\ x(0) &= 2.\end{aligned}$$

Answer: We begin by finding the general solution to the separable differential equation as usual.

$$\begin{aligned}\int \frac{dx}{x+1} &= \int t \, dt + C \\ \ln|x+1| &= \frac{1}{2}t^2 + C \\ |x+1| &= e^{\frac{1}{2}t^2 + C} \\ |x+1| &= e^C e^{\frac{1}{2}t^2} \\ x+1 &= \pm e^C e^{\frac{1}{2}t^2} \\ x(t) &= \mathcal{C} e^{\frac{1}{2}t^2} - 1.\end{aligned}$$

Now that we have the general solution, finding the specific solution to our initial value problem is simple:

$$\begin{aligned}x(0) &= \mathcal{C} e^{\frac{1}{2}(0)^2} - 1 \\ &= \mathcal{C} - 1 = 2 \\ \mathcal{C} &= 3.\end{aligned}$$

So, the solution to the initial value problem is

$$x(t) = 3e^{\frac{1}{2}t^2} - 1. \quad \blacksquare$$

Example 4.3.4. Find the solution to

$$\begin{aligned}\frac{dx}{dt} &= \frac{x(1-x)}{t} \\ x(1) &= 5.\end{aligned}$$

Answer:

$$\int \frac{dx}{x(1-x)} = \int \frac{dt}{t} + C = \ln |t| + C.$$

For the integral on the left-hand side, we use partial fraction decomposition:

$$\frac{1}{x(1-x)} = \frac{1}{x} + \frac{1}{1-x}.$$

This allows us to compute the integral above:

$$\begin{aligned}\int \frac{dx}{x(1-x)} &= \int \frac{1}{x} dx + \int \frac{1}{1-x} dx \\ &= \ln |x| - \ln |1-x| + D \\ &= \ln \left| \frac{x}{1-x} \right| + D.\end{aligned}$$

We will not need the second arbitrary constant, “+D,” in the differential equation, but since we were doing a separate indefinite integral it technically needs to be there!

This gives the general solution of the differential equation as

$$\ln \left| \frac{x}{1-x} \right| = \ln |t| + C.$$

Taking the natural exponential of both sides gives

$$\begin{aligned}\left| \frac{x}{1-x} \right| &= e^{\ln |t| + C} \\ &= e^C e^{\ln |t|} \\ &= e^C |t| \\ \frac{x}{1-x} &= \pm e^C |t| \\ \frac{x}{1-x} &= C |t|.\end{aligned}$$

The next question is whether or not we can solve for x . It turns out that we can by multiplying both sides by $1 - x$.

$$\begin{aligned}\frac{x}{1-x} &= \mathcal{C}|t| \\ x &= (1-x)\mathcal{C}|t| \\ x &= \mathcal{C}|t| - \mathcal{C}|t|x \\ x + \mathcal{C}|t|x &= \mathcal{C}|t| \\ x(1 + \mathcal{C}|t|) &= \mathcal{C}|t| \\ x(t) &= \frac{\mathcal{C}|t|}{1 + \mathcal{C}|t|}.\end{aligned}$$

Notice that the constant $\mathcal{C}$ must be the same constant in both the numerator and denominator of the general solution! Finally, we use the initial condition to find the solution to the initial value problem.

$$\begin{aligned}x(1) &= \frac{\mathcal{C}}{1 + \mathcal{C}} = 5 \\ \mathcal{C} &= 5(1 + \mathcal{C}) \\ \mathcal{C} &= 5 + 5\mathcal{C} \\ -4\mathcal{C} &= 5 \\ \mathcal{C} &= -\frac{5}{4}.\end{aligned}$$

This gives

$$\begin{aligned}x(t) &= \frac{-\frac{5}{4}|t|}{1 - \frac{5}{4}|t|} \\ &= \frac{-5|t|}{4 - 5|t|} \\ &= \frac{5|t|}{5|t| - 4}. \quad \blacksquare\end{aligned}$$

4.4 First Order Linear Differential Equations

The second type of differential equations we discuss is the class of **first order linear differential equations**. These are differential equations of the form

$$\frac{dx}{dt} + P(t)x = Q(t).$$

The method used to solve these equations involves finding an *integrating factor*. This is a function $\mu(t)$ which converts the left-hand side of the differential equation into a perfect derivative. In other words, we want to find a function $\mu = \mu(t)$ that has the following property:

$$\mu \frac{dx}{dt} + \mu P(t)x = \frac{d}{dt} [\mu x].$$

Since

$$\frac{d}{dt} [\mu x] = x \frac{d\mu}{dt} + \mu \frac{dx}{dt},$$

our integrating factor μ must satisfy

$$x \frac{d\mu}{dt} = \mu P(t)x.$$

So long as $x \neq 0$, we must have

$$\frac{d\mu}{dt} = \mu P(t).$$

But this is just a separable differential equation for μ ! The solution is given by

$$\begin{aligned} \int \frac{d\mu}{\mu} &= \int P(t) dt + C \\ \ln \mu &= \int P(t) dt + C \\ \mu(t) &= Ce^{\int P(t) dt} \end{aligned}$$

Since we only need a single integrating factor, we take $C = 1$.

Once we have the integrating factor, we find

$$\begin{aligned} \frac{dx}{dt} + P(t)x &= Q(t) \\ \mu(t) \frac{dx}{dt} + \mu(t)P(t)x &= \mu(t)Q(t) \\ \frac{d}{dt} [\mu(t)x] &= \mu(t)Q(t) \\ \mu(t)x(t) &= \int \mu(t)Q(t) dt + C \\ x(t) &= \frac{\int \mu(t)Q(t) dt + C}{\mu(t)}. \end{aligned}$$

We summarize our computations above in the following theorem

Theorem 4.4.1. *The general solution to a differential equation of the form*

$$\frac{dx}{dt} + P(t)x = Q(t)$$

is given by

$$x(t) = \frac{\int \mu(t)Q(t) dt + C}{\mu(t)}$$

where the integrating factor μ is given by

$$\mu(t) = e^{\int P(t) dt}.$$

Example 4.4.1. Find the general solution to

$$\frac{dx}{dt} + 2tx = t.$$

Answer: Since $P(t) = 2t$, we can take

$$\mu(t) = e^{\int 2t dt} = e^{t^2}$$

for the integrating factor. Since $Q(t) = t$, the general solution is given by

$$\begin{aligned} x(t) &= \frac{\int te^{t^2} dt + C}{e^{t^2}} \\ &= \frac{\frac{1}{2}e^{t^2} + C}{e^{t^2}} = \frac{1}{2} + Ce^{-t^2}. \quad \blacksquare \end{aligned}$$

Example 4.4.2. Solve the following initial value problem.

$$\begin{aligned} t \frac{dx}{dt} + 2x &= \frac{\cos(t)}{t} \\ x(\pi) &= 2 \end{aligned}$$

Answer: Before we can use our general solution, we have to write the differential equation in the correct form. Dividing both sides of the equation by t , we find

$$\frac{dx}{dt} + \frac{2}{t}x = \frac{\cos(t)}{t^2}.$$

This means

$$P(t) = \frac{2}{t},$$

$$Q(t) = \frac{\cos(t)}{t^2}.$$

Since

$$\int P(t) dt = \int \frac{2}{t} dt = 2 \ln(t) = \ln(t^2),$$

we can take

$$\mu(t) = e^{\ln(t^2)} = t^2.$$

The general solution to our differential equation is

$$\begin{aligned} x(t) &= \frac{\int \mu(t)Q(t) dt + C}{\mu(t)} \\ &= \frac{\int t^2 \frac{\cos(t)}{t^2} dt + C}{t^2} \\ &= \frac{\int \cos(t) dt + C}{t^2} \\ &= \frac{\sin(t) + C}{t^2}. \end{aligned}$$

Now we can fit the initial condition.

$$\begin{aligned} x(\pi) &= \frac{\sin(\pi) + C}{\pi^2} = 2 \\ \frac{C}{\pi^2} &= 2 \\ C &= 2\pi^2. \end{aligned}$$

So, the solution to our initial value problem is

$$x(t) = \frac{\sin(t) + 2\pi^2}{t^2}. \blacksquare$$

4.5 Application: Newton's Law of Cooling

As we mentioned in Section 4.1, Newton's Law of Cooling states that the rate at which an object's temperature changes is proportional to its difference in temperature with the environment. This leads to the differential equation

$$\frac{dT}{dt} = k(T_e - T)$$

where $T = T(t)$ is the temperature of the object at time t , T_e is the environmental temperature, and k is the cooling rate of the object (assumed to be a positive parameter). Note the order in which the temperatures are subtracted. If the object is colder than the environment, $T = T(t)$ is smaller than T_e . In this case, $k(T_e - T) > 0$ and the temperature of the object increases. Similarly, $k(T_e - T) < 0$ when the object is hotter than the environment. As a result, $T = T(t)$ will decrease.

Theorem 4.5.1. *The solution to the initial value problem*

$$\begin{aligned}\frac{dT}{dt} &= k(T_e - T) \\ T(0) &= T_0\end{aligned}$$

(where k and T_e are parameters) is given by

$$T(t) = T_e - (T_e - T_0)e^{-kt}.$$

Proof: Newton's Law of Cooling is certainly a separable differential equation. The general solution is found as follows.

$$\begin{aligned}\int \frac{dT}{T_e - T} &= \int k \, dt + C \\ -\ln |T_e - T| &= kt + C \\ \ln |T_e - T| &= -kt - C \\ |T_e - T| &= e^{-C}e^{-kt} \\ T_e - T &= \pm e^{-C}e^{-kt} \\ T_e - T &= \mathcal{C}e^{-kt} \\ T(t) &= T_e - \mathcal{C}e^{-kt}.\end{aligned}$$

To find the arbitrary constant $\mathcal{C}$, we use the initial condition $T(0) = T_0$. This gives

$$\begin{aligned} T(0) &= T_e - \mathcal{C}e^0 \\ &= T_e - \mathcal{C} = T_0 \\ \mathcal{C} &= T_e - T_0. \end{aligned}$$

So, the solution to Newton's Law of Cooling is

$$T(t) = T_e - (T_e - T_0)e^{-kt},$$

where T_e is the environmental temperature, T_0 is the initial temperature of the object, and k is the cooling rate of the object. ■

Notice that

$$\lim_{t \rightarrow \infty} T(t) = T_e$$

regardless of the initial temperature (because $k > 0$). This means that the temperature of the object approaches the environmental temperature just as we would expect from experience.

There is no need to remember the formula we just derived since you can reconstruct it for any particular problem by the usual methods for separable differential equations. We give it here just to illustrate the typical way these problems are solved.

Example 4.5.1. Suppose an object initially at a temperature of $T_0 = 150^\circ\text{C}$ is placed in a large water bath at 10°C . If the cooling rate of the object is $k = \frac{1}{20} \text{ sec}^{-1}$, how long until the object reaches 25°C ?

Answer: The given information leads us to the following initial value problem.

$$\begin{aligned} \frac{dT}{dt} &= \frac{1}{20}(10 - T) \\ T(0) &= 150. \end{aligned}$$

Solving the differential equation yields

$$\begin{aligned}\int \frac{dT}{10 - T} &= \int \frac{1}{20} dt + C \\ -\ln |10 - T| &= \frac{1}{20}t + C \\ |10 - T| &= e^{-C}e^{-t/20} \\ 10 - T &= \pm e^{-C}e^{-t/20} \\ T(t) &= 10 - \mathcal{C}e^{-t/20}.\end{aligned}$$

Using the given initial datum gives

$$\begin{aligned}T(0) &= 10 - \mathcal{C} = 150 \\ \mathcal{C} &= 10 - 150 = -140.\end{aligned}$$

So, the temperature of the object is given by

$$T(t) = 10 + 140e^{-t/20}.$$

We are being asked for the time t such that $T(t) = 25$. This means we need to solve

$$T(t) = 10 + 140e^{-t/20} = 25.$$

$$\begin{aligned}10 + 140e^{-t/20} &= 25 \\ 140e^{-t/20} &= 25 - 10 \\ e^{-t/20} &= \frac{15}{140} = \frac{3}{28} \\ -\frac{1}{20}t &= \ln\left(\frac{3}{28}\right) \\ t &= -20 \ln\left(\frac{3}{28}\right) = 20 \ln\left(\frac{28}{3}\right) \\ &\approx 44.7 \text{ seconds. } \blacksquare\end{aligned}$$

Notice in the last step we used

$$-\ln(x) = \ln\left(\frac{1}{x}\right)$$

to simplify the expression for t . This is a common tool for simplifying expressions in these types of problems.

Typically, we are not given the constant k directly. Instead, we usually have additional temperature information at some time after the initial temperature reading. This allows us to find k and predict the temperature at future times as well.

Example 4.5.2. A can of soda is taken from a refrigerator at a temperature of 35°F and is left in room held at a temperature of 70°F . If the can warms to a temperature of 45°F after 20 minutes, how long until the soda is at a temperature of 60°F ?

Answer: We are given the environmental temperature, $T_e = 70^\circ\text{F}$, and the initial temperature of the soda, $T_0 = 35^\circ\text{F}$. Notice that we are not given the constant k . The data we have so far leads us to the initial value problem

$$\begin{aligned}\frac{dT}{dt} &= k(70 - T) \\ T(0) &= 35.\end{aligned}$$

Separating and integrating the differential equation gives

$$\begin{aligned}\int \frac{dT}{70 - T} &= \int k \, dt + C \\ -\ln |70 - T| &= kt + C \\ 70 - T &= \mathcal{C}e^{-kt} \\ T(t) &= 70 - \mathcal{C}e^{-kt}.\end{aligned}$$

Using the initial condition gives

$$\begin{aligned}T(0) &= 70 - \mathcal{C} = 35 \\ \mathcal{C} &= 35.\end{aligned}$$

So far, we know that

$$T(t) = 70 - 35e^{-kt}.$$

The other piece of information we are given is that $T(20) = 45$. Using this,

we can find k .

$$\begin{aligned}
 T(20) &= 70 - 35e^{-k(20)} = 45 \\
 70 - 45 &= 35e^{-20k} \\
 25 &= 35e^{-20k} \\
 e^{-20k} &= \frac{25}{35} = \frac{5}{7} \\
 -20k &= \ln\left(\frac{5}{7}\right) \\
 k &= \frac{-\ln(5/7)}{20} = \frac{\ln(7/5)}{20}.
 \end{aligned}$$

We are now ready to answer the question posed in the problem: find the time t so that $T(t) = 60$. Even though we know k , the expression for it is messy enough that we will not explicitly substitute for it until the end of the problem. Notice that the steps below are exactly like the steps used to find k .

$$\begin{aligned}
 T(t) &= 70 - 35e^{-kt} = 60 \\
 70 - 60 &= 35e^{-kt} \\
 e^{-kt} &= \frac{10}{35} = \frac{2}{7} \\
 -kt &= \ln\left(\frac{2}{7}\right) \\
 t &= \frac{-\ln(2/7)}{k} = \frac{20 \ln(7/2)}{\ln(7/5)} \\
 &\approx 74.5 \text{ minutes. } \blacksquare
 \end{aligned}$$

4.6 Application: Selective Diffusion

A model related to Newton's Law of Cooling is a simplistic model of selective diffusion across a membrane. Consider the diagram below showing a cell sitting inside a ambient solution of some chemical.

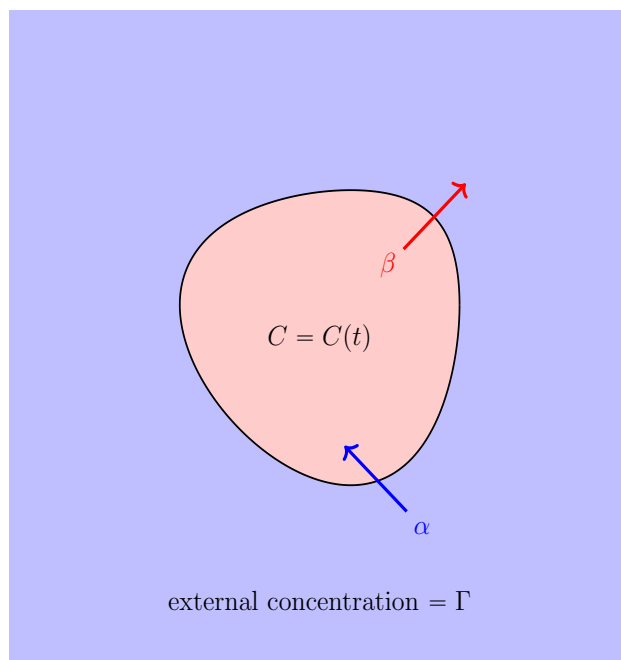


Figure 4.6.1: A Schematic of Selective Diffusion

The concentration of the chemical outside the cell membrane is Γ (assumed to be constant) while the concentration inside the cell is $C = C(t)$. The chemical can cross into the cell at a rate α (a positive parameter) while it can leave at a rate β (also positive). Note that these two rates could be different in general (hence the name *selective* diffusion). Assuming the volume of the cell is constant, we arrive at the following differential equation

$$\frac{dC}{dt} = \alpha\Gamma - \beta C.$$

The units for the rates α and β are “per time” (so “per second,” “per minute,” etc.). If we factor out the rate β , we get the following equation

$$\frac{dC}{dt} = \beta \left(\frac{\alpha}{\beta} \Gamma - C \right)$$

which is equivalent to Newton’s Law of Cooling with k replaced by β and T_e replaced by the concentration $\alpha\Gamma/\beta$. Since this equation has the same form, the problem solving techniques are the same as the previous section!

Example 4.6.1. Suppose that a cell is immersed in a salt solution with a concentration of potassium ions of 5 mmol per liter. The flow rate of ions into the cell is $1/15$ per second while the flow rate out is only $1/25$ per second. If the concentration of potassium ions inside the cell is 1 mmol per liter, how long until the concentration reaches 4 mmol per liter? If the cell remains in the solution, what value will the concentration of potassium ions inside the cell approach in the long run?

Answer: We have $\Gamma = 5$, $\alpha = 1/15$, and $\beta = 1/25$. The differential equation for this problem is

$$\frac{dC}{dt} = \frac{1}{15}(5) - \frac{1}{25}C = \frac{1}{3} - \frac{1}{25}C = \frac{1}{25} \left(\frac{25}{3} - C \right).$$

This gives us the initial value problem

$$\begin{aligned} \frac{dC}{dt} &= \frac{1}{25} \left(\frac{25}{3} - C \right) \\ C(0) &= 1. \end{aligned}$$

Separating and integrating the differential equation gives

$$\begin{aligned} \int \frac{dC}{25/3 - C} &= \int \frac{1}{25} dt + D \\ -\ln \left| \frac{25}{3} - C \right| &= \frac{t}{25} + D \\ \frac{25}{3} - C &= \mathcal{D}e^{-t/25} \\ C(t) &= \frac{25}{3} - \mathcal{D}e^{-t/25}. \end{aligned}$$

Note that we used “ $+D$ ” for the arbitrary constant since C is being used for concentration. Fitting the initial data gives

$$C(t) = \frac{25}{3} - \frac{22}{3}e^{-t/25}.$$

Solving $C(t) = 4$ gives

$$\begin{aligned}\frac{25}{3} - \frac{22}{3}e^{-t/25} &= 4 \\ \frac{22}{3}e^{-t/25} &= \frac{25}{3} - 4 = \frac{13}{3} \\ e^{-t/25} &= \frac{13/3}{22/3} = \frac{13}{22} \\ -\frac{1}{25}t &= \ln\left(\frac{13}{22}\right) \\ t &= -25 \ln\left(\frac{13}{22}\right) = 25 \ln\left(\frac{22}{13}\right) \\ &\approx 13.2 \text{ seconds.}\end{aligned}$$

For the second question, note that

$$\lim_{t \rightarrow \infty} C(t) = \lim_{t \rightarrow \infty} \left(\frac{25}{3} - \frac{22}{3}e^{-t/25} \right) = \frac{25}{3}.$$

So, the concentration will limit to $25/3$ mmol per liter. ■

Notice that the limiting concentration of ions inside the cell ($8\frac{1}{3}$ mmol/L) is higher than the ambient concentration. The slightly slower rate of diffusion out of the cell allows the concentration of ions inside the cell to be higher than the environment.

Example 4.6.2. Suppose that a certain drug leaves a cell three times faster than it enters. The concentration in the bloodstream is currently at $3 \mu\text{g/L}$ while the concentration inside the cell is $8 \mu\text{g/L}$. After 30 minutes, the concentration in the cell has fallen to $5 \mu\text{g/L}$. Once the drug falls below a concentration of $2 \mu\text{g/L}$, it ceases to be effective. How much longer will the drug have an effect inside the cell.

Answer: We know that $\beta = 3\alpha$ and $\Gamma = 3$. This give us the differential equation

$$\frac{dC}{dt} = \alpha(3) - 3\alpha C.$$

Factoring out 3α and noting the initial condition give us the problem

$$\begin{aligned}\frac{dC}{dt} &= 3\alpha(1 - C) \\ C(0) &= 8.\end{aligned}$$

Solving the differential equation gives

$$\begin{aligned}\int \frac{dC}{1 - C} &= \int 3\alpha dt + D \\ -\ln|1 - C| &= 3\alpha t + D \\ 1 - C &= \mathcal{D}e^{-3\alpha t} \\ C(t) &= 1 - \mathcal{D}e^{-3\alpha t}.\end{aligned}$$

Fitting the initial data gives us

$$C(t) = 1 + 7e^{-3\alpha t}.$$

Notice that the limiting concentration inside the cell will be $1 \mu\text{g/L}$.

We use the data $C(30) = 5$ to determine α .

$$\begin{aligned}C(30) &= 1 + 7e^{-3\alpha(30)} = 5 \\ 7e^{-90\alpha} &= 4 \\ e^{-90\alpha} &= \frac{4}{7} \\ -90\alpha &= \ln\left(\frac{4}{7}\right) \\ \alpha &= \frac{-\ln(4/7)}{90} = \frac{\ln(7/4)}{90}.\end{aligned}$$

Now that we know α , we can easily solve $C(t) = 2$.

$$\begin{aligned}1 + 7e^{-3\alpha t} &= 2 \\ 7e^{-3\alpha t} &= 1 \\ e^{-3\alpha t} &= \frac{1}{7} \\ -3\alpha t &= \ln\left(\frac{1}{7}\right) \\ t &= \frac{-\ln(1/7)}{3\alpha} = \frac{30\ln(7)}{\ln(7/4)} \\ &\approx 104.3 \text{ minutes. } \blacksquare\end{aligned}$$

4.7 Application: Exponential Growth and Decay

The simplest of all population models is **exponential growth**. *The rate at which a population is increasing is proportional to the size of the population.* If we let $P = P(t)$ be the size of the population at time t , then we have

$$\frac{dP}{dt} = kP,$$

where $k > 0$ is the growth rate of the population. If we know the size of the population at $t = 0$, we get the initial value problem

$$\begin{aligned}\frac{dP}{dt} &= kP \\ P(0) &= P_0.\end{aligned}$$

Theorem 4.7.1. *The solution to any exponential growth problem is*

$$P(t) = P_0 e^{kt}.$$

Proof: Separating and integrating the differential equation gives

$$\begin{aligned}\int \frac{dP}{P} &= \int k \, dt + C \\ \ln |P| &= kt + C \\ |P| &= e^C e^{kt} \\ P &= \pm e^C e^{kt} \\ P(t) &= C e^{kt}.\end{aligned}$$

Fitting the initial condition gives

$$P(0) = C = P_0. \quad \blacksquare$$

Example 4.7.1. A culture of bacteria is growing exponentially. Initially, there are 200 organisms, and after 20 hours, the population has increased to 1800 organisms. How many bacteria will there be after 2 days? How long until there are 1,000,000 organisms?

Answer: From our result above, we know that

$$P(t) = 200e^{kt}.$$

Our first job is to find k . Using $P(20) = 1800$, we find

$$P(20) = 200e^{20k} = 1800$$

$$e^{20k} = 9$$

$$20k = \ln(9)$$

$$k = \frac{\ln(9)}{20}.$$

For the first question, since 2 days is 48 hours, we have

$$\begin{aligned} P(48) &= 200e^{k(48)} \\ &= 200e^{\frac{48}{20} \ln(9)} \\ &= 200e^{\ln(9^{12/5})} \\ &= 200(9^{12/5}) \approx 39013. \end{aligned}$$

So there will be just over 39,000 bacteria after 48 hours. For the second question, we solve $P(t) = 1,000,000$.

$$\begin{aligned} 200e^{kt} &= 1000000 \\ e^{kt} &= 5000 \\ kt &= \ln(5000) \\ t &= \frac{\ln(5000)}{k} = \frac{20 \ln(5000)}{\ln(9)} \\ &\approx 77.5 \text{ hours. } \blacksquare \end{aligned}$$

Exponential decay of radioactive elements is essentially the same differential equation. In words, *the rate of decay of a radioactive isotope is proportional to the total amount of the element*. If $A = A(t)$ is the amount of the element at time t , we have

$$\begin{aligned} \frac{dA}{dt} &= -kA \\ A(t) &= A_0, \end{aligned}$$

where $k > 0$ is the decay rate of the radioactive element (also called its *activity*) and A_0 is the initial amount of the element.

Theorem 4.7.2. *The solution to any exponential decay problem is*

$$A(t) = A_0 e^{-kt}.$$

Proof: The proof is the same as for exponential growth – just replace k by $-k$. ■

Example 4.7.2. Ru-261 is an isotope of ruthenium which undergoes α -decay into rhodium-257 (among other products). Initially, we have 100 μg of the isotope, and only 35 μg remain after 120 seconds. How long until there is less than 1 μg remaining? How long did it take for half of the sample to decay?

Answer: We know that the amount of the isotope at time t is given by

$$A(t) = 100e^{-kt}.$$

Using $A(120) = 35$ gives

$$\begin{aligned} 100e^{-k(120)} &= 35 \\ e^{-120k} &= \frac{35}{100} = \frac{7}{20} \\ -120k &= \ln\left(\frac{7}{20}\right) = -\ln\left(\frac{20}{7}\right) \\ k &= \frac{\ln(20/7)}{120}. \end{aligned}$$

To find out how long until there is only 1 μg , we solve $A(t) = 1$.

$$\begin{aligned} 100e^{-kt} &= 1 \\ e^{-kt} &= \frac{1}{100} \\ -kt &= \ln\left(\frac{1}{100}\right) = -\ln(100) \\ t &= \frac{120 \ln(100)}{\ln(20/7)} \\ &\approx 526.4 \text{ seconds.} \end{aligned}$$

To find out the time when there was half the original amount, we solve $A(t) = 50$.

$$\begin{aligned} 100e^{-kt} &= 50 \\ e^{-kt} &= \frac{1}{2} \\ -kt &= \ln\left(\frac{1}{2}\right) = -\ln(2) \\ t &= \frac{120 \ln(2)}{\ln(20/7)} \approx 79.2 \text{ seconds. } \blacksquare \end{aligned}$$

Instead of listing the activity of a radioactive element directly, usually the isotope's **half-life** is given. This is defined to be the amount of time until only half the original amount of the isotope remains. In other words, the half-life, $t_{1/2}$, is a solution to

$$A(t_{1/2}) = \frac{1}{2}A_0.$$

If we try to find a relationship between the half-life and the activity k , the first thing we find is

$$\begin{aligned} A_0 e^{-kt_{1/2}} &= \frac{1}{2}A_0 \\ e^{-kt_{1/2}} &= \frac{1}{2}. \end{aligned}$$

Notice that the actual initial amount of the isotope (A_0) is completely irrelevant to the half-life. This means that the half-life is a characteristic of the isotope itself (and not the specific sample of the isotope being studied).

Lemma 4.7.3. *The half-life of a radioactive element, $t_{1/2}$, is related to its activity k by*

$$k = \frac{\ln(2)}{t_{1/2}}.$$

Proof: Finishing the computation above

$$\begin{aligned} A_0 e^{-kt_{1/2}} &= \frac{1}{2} A_0 \\ e^{-kt_{1/2}} &= \frac{1}{2} \\ -kt_{1/2} &= \ln\left(\frac{1}{2}\right) = -\ln(2) \\ k &= \frac{\ln(2)}{t_{1/2}}. \blacksquare \end{aligned}$$

Example 4.7.3. The half-life of plutonium-238 is 87.7 years. If we have a sample containing 30 g of the isotope, how long until there is less than 1 g of plutonium in the sample?

Answer: We know that the activity is

$$k = \frac{\ln(2)}{87.7}.$$

We need to solve $A(t) = 1$.

$$\begin{aligned} 30e^{-kt} &= 1 \\ e^{-kt} &= \frac{1}{30} \\ -kt &= \ln\left(\frac{1}{30}\right) = -\ln(30) \\ t &= \frac{\ln(30)}{k} = \frac{87.7 \ln(30)}{\ln(2)} \\ &\approx 430.3 \text{ years. } \blacksquare \end{aligned}$$

There is a related notion for exponential growth – **doubling time**. The doubling time, t_2 , of a population undergoing exponential growth is the amount of time it takes for the population to double in size. That is,

$$P(t_2) = 2P_0.$$

Lemma 4.7.4. *The doubling time, t_2 , of a population experiencing exponential growth is related to the growth rate k by*

$$k = \frac{\ln(2)}{t_2}.$$

Proof:

$$\begin{aligned} P_0 e^{kt_2} &= 2P_0 \\ e^{kt_2} &= 2 \\ kt_2 &= \ln(2) \\ k &= \frac{\ln(2)}{t_2}. \blacksquare \end{aligned}$$

Example 4.7.4. A certain population is experiencing exponential growth. Initially there are 500 organisms, and the population increased to 1600 organisms after 36 hours. What is the doubling time for this population?

Answer: To find the doubling time, we first need the growth rate. We know $P(t) = 500e^{kt}$ and $P(36) = 1600$. Solving for k gives

$$\begin{aligned} 500e^{36k} &= 1600 \\ e^{36k} &= \frac{16}{5} \\ 36k &= \ln\left(\frac{16}{5}\right) \\ k &= \frac{\ln(16/5)}{36}. \end{aligned}$$

So, the doubling time is given by

$$t_2 = \frac{\ln(2)}{k} = \frac{36 \ln(2)}{\ln(16/5)} \approx 21.5 \text{ hours.}$$

So, the population doubles in size roughly every 21.5 hours. $\blacksquare$

4.8 Application: Logistic Growth

While exponential growth is a decent model for relatively small populations, it fails to account for a number of environmental factors that can limit the growth of larger populations. Most importantly, environments where resources are limited can only support some maximum number of individuals

in the long-term. The simplest population model which can account for limited resources is the **logistic growth** model. *The rate of change of a population's size is roughly proportional to the population size for relatively small populations. However, when the population nears a critical size, the population's growth approaches zero.*

Mathematically speaking, there are many models we could develop that have this behavior. The best advice to give in such situations is to start with as simple a model as possible. In that spirit, logistic growth is given by the differential equation

$$\frac{dP}{dt} = kP \left(1 - \frac{P}{N}\right),$$

where $P = P(t)$ is the population size at time t , $k > 0$ is the logistic growth rate of the population, and $N > 0$ is the critical population size – usually called the **carrying capacity** of the environment.

Just as before, we give the general solution and then examine some particular cases.

Theorem 4.8.1. *The solution to the logistic differential equation*

$$\begin{aligned} \frac{dP}{dt} &= kP \left(1 - \frac{P}{N}\right) \\ P(0) &= P_0, \end{aligned}$$

is given by

$$P(t) = \frac{N}{Ce^{-kt} + 1}$$

where the constant C is related to the initial datum and carrying capacity by

$$C = \frac{N - P_0}{P_0}.$$

Proof: Solving this differential equation is similar to Example 4.3.4. Separating and integrating the differential equation gives

$$\int \frac{dP}{P(1 - P/N)} = \int k \, dt + C = kt + C.$$

In order to attack the left-hand integral, we use partial fraction decomposition:

$$\frac{1}{P(1 - P/N)} = \frac{1}{P} + \frac{1/N}{1 - P/N} = \frac{1}{P} + \frac{1}{N - P}.$$

So, the integral on the left-hand side above is

$$\begin{aligned}\int \frac{dP}{P(1 - P/N)} &= \int \frac{dP}{P} + \int \frac{dP}{N - P} \\ &= \ln |P| - \ln |N - P| + D \\ &= \ln \left| \frac{P}{N - P} \right| + D.\end{aligned}$$

Dropping the extraneous “+D” we have

$$\begin{aligned}\ln \left| \frac{P}{N - P} \right| &= kt + C \\ \left| \frac{P}{N - P} \right| &= e^C e^{kt} \\ \frac{P}{N - P} &= \pm e^C e^{kt} \\ \frac{P}{N - P} &= \mathcal{C} e^{kt} \\ P &= \mathcal{C} e^{kt} (N - P) \\ P + P\mathcal{C} e^{kt} &= N\mathcal{C} e^{kt} \\ P(t) &= \frac{N\mathcal{C} e^{kt}}{1 + \mathcal{C} e^{kt}}.\end{aligned}$$

Fitting the initial datum gives

$$\begin{aligned}P(0) &= \frac{N\mathcal{C}}{1 + \mathcal{C}} = P_0 \\ N\mathcal{C} &= P_0(1 + \mathcal{C}) \\ N\mathcal{C} - P_0\mathcal{C} &= P_0 \\ \mathcal{C} &= \frac{P_0}{N - P_0}.\end{aligned}$$

Substituting this value for $\mathcal{C}$ and simplifying gives

$$\begin{aligned} P(t) &= \frac{N[P_0/(N - P_0)]e^{kt}}{1 + [P_0/(N - P_0)]e^{kt}} \\ &= \frac{N[P_0/(N - P_0)]e^{kt}}{1 + [P_0/(N - P_0)]e^{kt}} \cdot \frac{N - P_0}{N - P_0} \\ &= \frac{NP_0e^{kt}}{N - P_0 + P_0e^{kt}}. \end{aligned}$$

The form of the solution given in the theorem is obtained from this one by dividing both numerator and denominator by P_0e^{kt} . ■

Two typical plots of logistic growth functions are shown below.

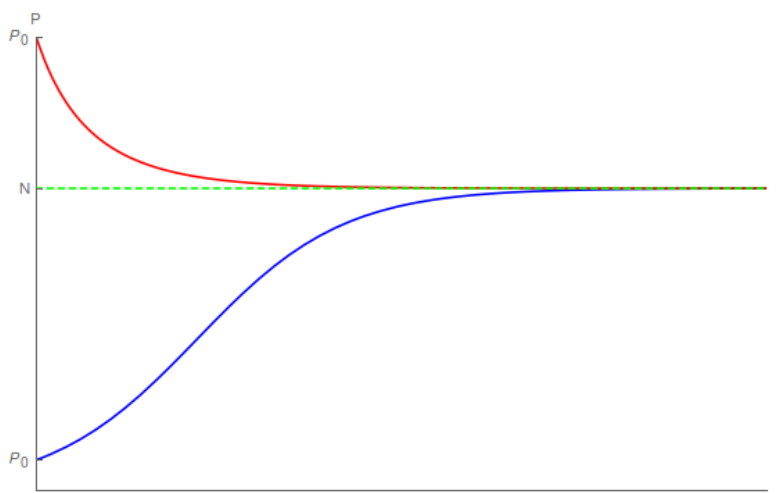


Figure 4.8.1: Typical Logistic Curves

Notice that for initial data significantly smaller than the carrying capacity N , the population starts out growing at an almost exponential rate. Once the population nears the carrying capacity, the size of the population levels off. On the other hand, for initial data that is larger than this critical size the system rapidly decreases to the carrying capacity. The form of the solution given in the theorem clearly indicates that for any reasonable condition (that is, $P_0 > 0$)

$$\lim_{t \rightarrow \infty} P(t) = N.$$

Notice that except for the exceptional case $P_0 = N$, the solution $P(t)$ never actually equals to N ; it merely approaches the value as time goes on.

Example 4.8.1. A population is experiencing logistic growth in a region where the carrying capacity for the species is 25,000 organisms. If there were 1500 individuals initially and 5000 after 5 years, how long until there are 20,000 organisms?

Answer: As usual, we are not given the growth rate directly. Using the given information $P_0 = 1500$ and $N = 25000$, we have

$$P(t) = \frac{25000}{\frac{47}{3}e^{-kt} + 1} = \frac{75000}{47e^{-kt} + 3}.$$

Using $P(5) = 5000$ gives

$$\begin{aligned}\frac{75000}{47e^{-k(5)} + 3} &= 5000 \\ 75000 &= 5000(47e^{-5k} + 3) \\ 15 &= 47e^{-5k} + 3 \\ e^{-5k} &= \frac{12}{47} \\ -5k &= \ln\left(\frac{12}{47}\right) \\ k &= -\frac{1}{5}\ln\left(\frac{12}{47}\right) = \frac{1}{5}\ln\left(\frac{47}{12}\right).\end{aligned}$$

Now that we have k , we can answer the question posed in the problem.

$$\begin{aligned}
 P(t) &= 20000 \\
 \frac{75000}{47e^{-kt} + 3} &= 20000 \\
 75000 &= 20000(47e^{-kt} + 3) \\
 \frac{15}{4} &= 47e^{-kt} + 3 \\
 47e^{-kt} &= \frac{3}{4} \\
 e^{-kt} &= \frac{3}{188} \\
 -kt &= \ln\left(\frac{3}{188}\right) \\
 t &= -\frac{\ln(3/188)}{k} = \frac{5 \ln(188/3)}{\ln(47/12)} \\
 &\approx 15.2 \text{ years. } \blacksquare
 \end{aligned}$$

Determining both the growth rate and the carrying capacity for a population requires 3 observations of the population: the initial population size and two observations at later times. However, it is NOT the case that any 3 observations of increasing population sizes can be made to fit the logistic model. In particular, the proportional growth rate, P'/P , needs to be a linear function of P with negative slope:

$$\frac{dP/dt}{P} = k \left(1 - \frac{P}{N}\right).$$

In actual applications, we would have a large number of observations of the population size (more than three, at any rate). Since such observations come with associated error, we can try to find the logistic curve which best fits the data. This statistical approach would then give us estimates of the carrying capacity, N , and growth rate, k . There are various tests which can be used to determine whether a logistic growth model is appropriate for a given set of data.

4.9 Application: Single Compartment Mixing

A type of problem similar to the Selective Diffusion model is a simple model for a solution being mixed in a large tank. Consider the following situation.

Example 4.9.1. We have a tank that initially holds 50 liters of a saline solution at a concentration of 30 grams per liter. A salt solution of concentration 10 grams per liter is flowing into the tank at a rate of 5 liters per minute. The well-mixed solution is flowing out of the tank at a matching rate of 5 liters per minute. Determine the concentration of salt in the tank as a function of time.

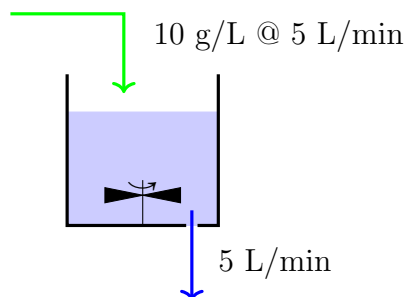


Figure 4.9.1: Tank from Example 4.9.1

Answer: Since the flow rate in matches the flow rate out (both are 5 L/min), the volume of fluid in the tank does not change. Let $C = C(t)$ be the concentration of salt in the tank at time t . Notice that C has units grams per liter. It turns out that the easiest thing to model is the *total amount* of salt in the tank at time t , $A = A(t)$ which has units grams. Since the volume in the tank is a constant 50 L, we have

$$C(t) = \frac{A(t)}{50}.$$

To model A , we need to consider dA/dt which has units grams per minute.

$$\begin{aligned}\frac{dA}{dt} &= (\text{total salt in per minute}) - (\text{total salt out per minute}) \\ (\text{total salt in per minute}) &= \left(10 \frac{\text{g}}{\text{L}}\right) \left(5 \frac{\text{L}}{\text{min}}\right) = 50 \frac{\text{g}}{\text{min}} \\ (\text{total salt out per minute}) &= \left(\frac{A}{50} \frac{\text{g}}{\text{L}}\right) \left(5 \frac{\text{L}}{\text{min}}\right) = \frac{A}{10} \frac{\text{g}}{\text{min}}\end{aligned}$$

Notice that the amount of salt leaving the tank at time t is proportional to the concentration of salt in the tank at that time! This gives us the following initial value problem

$$\begin{aligned}\frac{dA}{dt} &= 50 - \frac{A}{10} \\ A(0) &= \left(30 \frac{\text{g}}{\text{L}}\right) (50 \text{ L}) = 1500 \text{ g}.\end{aligned}$$

The differential equation for A is both separable and first order linear. We solve it here as a first order linear differential equation.

$$\begin{aligned}\frac{dA}{dt} + \frac{1}{10}A &= 50 \\ \mu(t) &= e^{\int 1/10 dt} = e^{t/10} \\ A(t) &= \frac{\int 50e^{t/10} dt + D}{e^{t/10}} \\ &= \frac{500e^{t/10} + D}{e^{t/10}} \\ &= 500 + De^{-t/10}\end{aligned}$$

Fitting the initial condition gives

$$\begin{aligned}A(0) &= 500 + D = 1500 \\ D &= 1000.\end{aligned}$$

So,

$$\begin{aligned}A(t) &= 500 + 1000e^{-t/10} \\ C(t) &= \frac{A(t)}{50} = 10 + 20e^{-t/10}.\end{aligned}$$

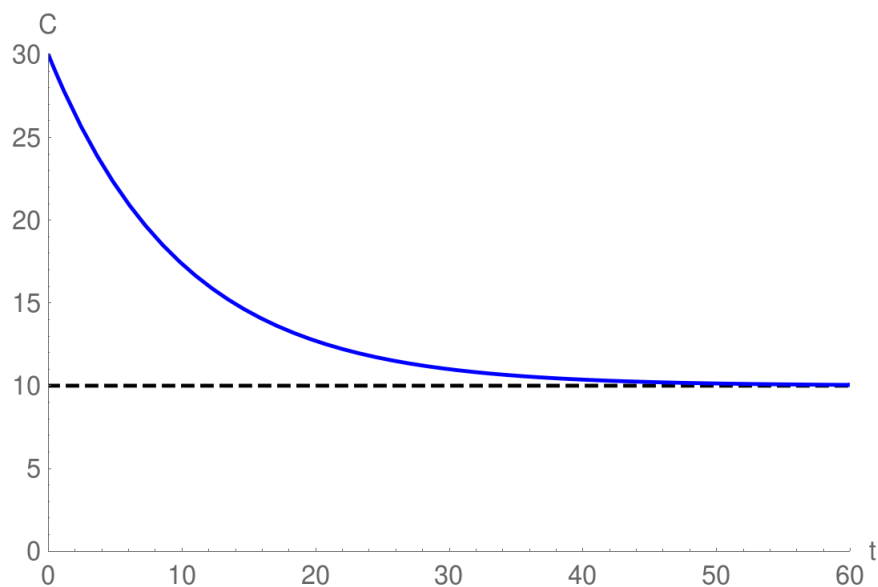


Figure 4.9.2: Solution to Example 4.9.1

Notice that the concentration of salt inside the tank starts at 30 g/L as it should. We also see that

$$\lim_{t \rightarrow \infty} C(t) = 10 \text{ g/L.}$$

This makes sense as that is the concentration of the external salt solution flowing into the tank. Basically, as the solution in the tank is drained it is slowly replaced by the solution flowing in. As a result, the concentration of salt in the tank should limit to the incoming concentration. ■

Analyzing the total amount of the compound in the tank is always the best way to approach these mixing problems. The crucial fact to remember is that the amount of salt flowing out of the tank should be proportional to the concentration of salt currently in the tank (and so will involve the unknown quantity A).

The assumption that the solution is *well-mixed* is fairly suspect. Essentially, we are assuming that the solution that flows into the tank from the outside is fully mixed throughout the tank *instantaneously*. One way of thinking about this assumption is that the mixing process should be very

fast on the time-scale of the problem. Since our previous problem used minutes and salt diffuses fairly rapidly, we are probably okay assuming the solution is always well-mixed. Had we wanted to predict the concentration on the scale of seconds, the well-mixed assumption becomes more suspect. On the flip-side, switching the time scale to hours versus minutes should make the well-mixed assumption even more justifiable. We will simply use this assumption for the remaining problems without further comment.

Example 4.9.2. Suppose the set-up from Example 4.9.1 is modified so that the external solution flows into the tank at rate of 7 L/min. If the tank can hold 100 L of fluid and the solution remains well-mixed for all time, find the concentration inside the tank for all times until the tank is full.

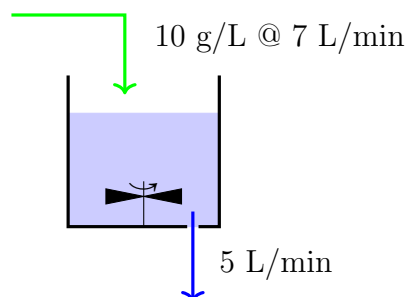


Figure 4.9.3: Tank from Example 4.9.2

Answer: First, note that since the flow rates no longer match, the volume of fluid in the tank is increasing at a rate of 2 liters every minute. This means that the volume of fluid in the tank is given by

$$V(t) = 50 + 2t.$$

Since the tank can hold a maximum of 100 liters, we can only model the concentration of salt inside the tank up to $t = 25$ minutes. After this time, the salt solution will start to overflow the top of the tank (and so more will flow out of the tank than 5 L/min). Now, the concentration, $C(t)$, and the total amount of salt in the tank, $A(t)$, are related by

$$C(t) = \frac{A(t)}{2t + 50} \quad \text{for } 0 \leq t \leq 25.$$

Analyzing the total salt flowing into and out of the tank gives

$$\begin{aligned}\frac{dA}{dt} &= (\text{total salt in per minute}) - (\text{total salt out per minute}) \\ (\text{total salt in per minute}) &= \left(10 \frac{\text{g}}{\text{L}}\right) \left(7 \frac{\text{L}}{\text{min}}\right) = 70 \frac{\text{g}}{\text{min}} \\ (\text{total salt out per minute}) &= \left(\frac{A}{2t+50} \frac{\text{g}}{\text{L}}\right) \left(5 \frac{\text{L}}{\text{min}}\right) = \frac{5A}{2t+50} \frac{\text{g}}{\text{min}}\end{aligned}$$

This gives us the differential equation

$$\begin{aligned}\frac{dA}{dt} + \left(\frac{5}{2t+50}\right)A &= 70 \\ A(0) &= 1500.\end{aligned}$$

Notice that when the volume of fluid in the tank changes, the differential equation is still first order linear but no longer separable.

The integrating factor for this differential equation requires the following integral.

$$\begin{aligned}\int P(t) dt &= \int \frac{5}{2t+50} dt \\ &= \frac{5}{2} \ln(2t+50) \\ &= \ln(2t+50)^{5/2}\end{aligned}$$

This means the integrating factor is

$$\begin{aligned}\mu(t) &= e^{\int P(t) dt} \\ &= e^{\ln(2t+50)^{5/2}} \\ &= (2t+50)^{5/2}.\end{aligned}$$

For the total amount of salt, we find

$$\begin{aligned}A(t) &= \frac{\int 70(2t+50)^{5/2} dt + D}{(2t+50)^{5/2}} \\ &= \frac{70\left(\frac{2}{7}\right)\left(\frac{1}{2}\right)(2t+50)^{7/2} + D}{(2t+50)^{5/2}} \\ &= 10(2t+50) + D(2t+50)^{-5/2}.\end{aligned}$$

For the initial condition, we have

$$\begin{aligned} A(0) &= 10(50) + D(50)^{-5/2} = 1500 \\ \frac{D}{12500\sqrt{2}} &= 1000 \\ D &= 12500000\sqrt{2}. \end{aligned}$$

This gives us

$$\begin{aligned} A(t) &= 10(2t + 50) + 12500000\sqrt{2}(2t + 50)^{-5/2}, \\ C(t) &= 10 + 12500000\sqrt{2}(2t + 50)^{-7/2}. \end{aligned}$$

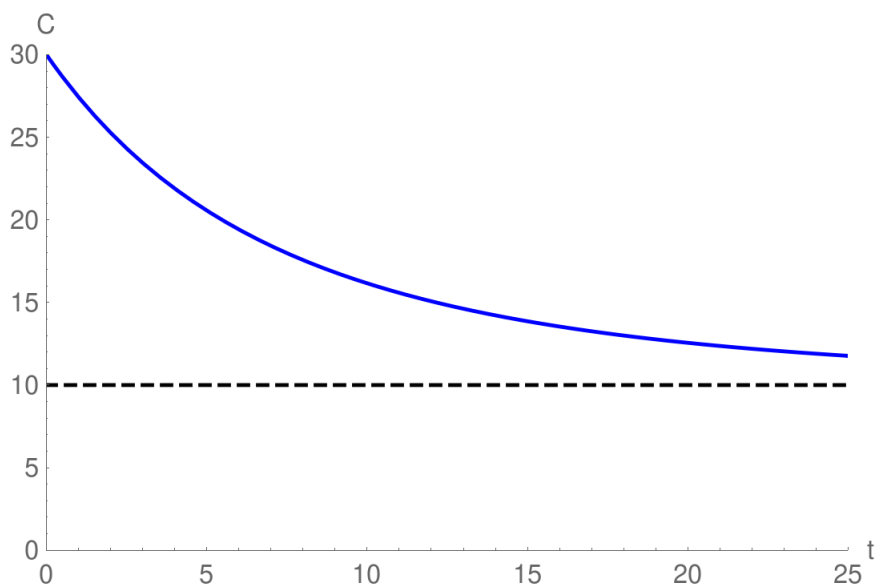


Figure 4.9.4: Solution to Example 4.9.2

Notice that the concentration still limits to 10 g/L just as in the first example. However, our differential equation ceases to be valid after $t = 25$ minutes since the tank begins to overflow at that time. ■

Example 4.9.3. We have a tank containing 100 gallons of a sugar solution at concentration 2 ounces per gallon. A sugar solution of concentration 5

ounces per gallon is flowing into the tank at a rate of 3 gallons per minute. The well-mixed solution is flowing out of the tank at a rate of 4 gallons per minute. Model the concentration of sugar in the tank up until the time when the tank is empty.

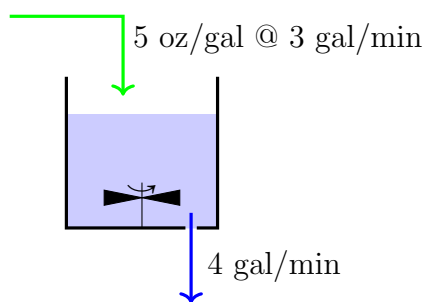


Figure 4.9.5: Tank from Example 4.9.3

Answer: Clearly the tank is losing 1 gallon each minute. This means the volume of fluid in the tank is given by

$$V(t) = 100 - t \text{ for } 0 \leq t \leq 100.$$

After $t = 100$ minutes, the tank is empty and our model will no longer be valid. The relationship between concentration and amount of sugar is given by

$$C(t) = \frac{A(t)}{100 - t}.$$

Running through the same series of computations as before gives

$$\frac{dA}{dt} = (\text{total sugar in per minute}) - (\text{total sugar out per minute})$$

$$(\text{total sugar in per minute}) = \left(5 \frac{\text{oz}}{\text{gal}}\right) \left(3 \frac{\text{gal}}{\text{min}}\right) = 15 \frac{\text{oz}}{\text{min}}$$

$$(\text{total sugar out per minute}) = \left(\frac{A}{100 - t} \frac{\text{oz}}{\text{gal}}\right) \left(4 \frac{\text{gal}}{\text{min}}\right) = \frac{4A}{100 - t} \frac{\text{oz}}{\text{min}}$$

This gives us the following initial value problem

$$\begin{aligned} \frac{dA}{dt} + \left(\frac{4}{100 - t}\right) A &= 15 \\ A(0) &= \left(2 \frac{\text{oz}}{\text{gal}}\right) (100 \text{ gal}) = 200 \text{ oz.} \end{aligned}$$

For the integrating factor, we have

$$\begin{aligned}\mu(t) &= e^{\int 4/(100-t) dt} \\ &= e^{-4 \ln(100-t)} \\ &= e^{\ln(100-t)^{-4}} \\ &= (100-t)^{-4}.\end{aligned}$$

This gives us

$$\begin{aligned}A(t) &= \frac{\int 15(100-t)^{-4} dt + D}{(100-t)^{-4}} \\ &= \frac{5(100-t)^{-3} + D}{(100-t)^{-4}} \\ &= 5(100-t) + D(100-t)^4.\end{aligned}$$

For the initial condition, we find

$$\begin{aligned}A(0) &= 5(100) + D(100)^4 = 200 \\ 100000000c &= -300 \\ D &= -\frac{3}{1000000}\end{aligned}$$

which gives

$$\begin{aligned}A(t) &= 5(100-t) - \frac{3(100-t)^4}{1000000}, \\ C(t) &= 5 - \frac{3(100-t)^3}{1000000}\end{aligned}$$

for $0 \leq t \leq 100$.

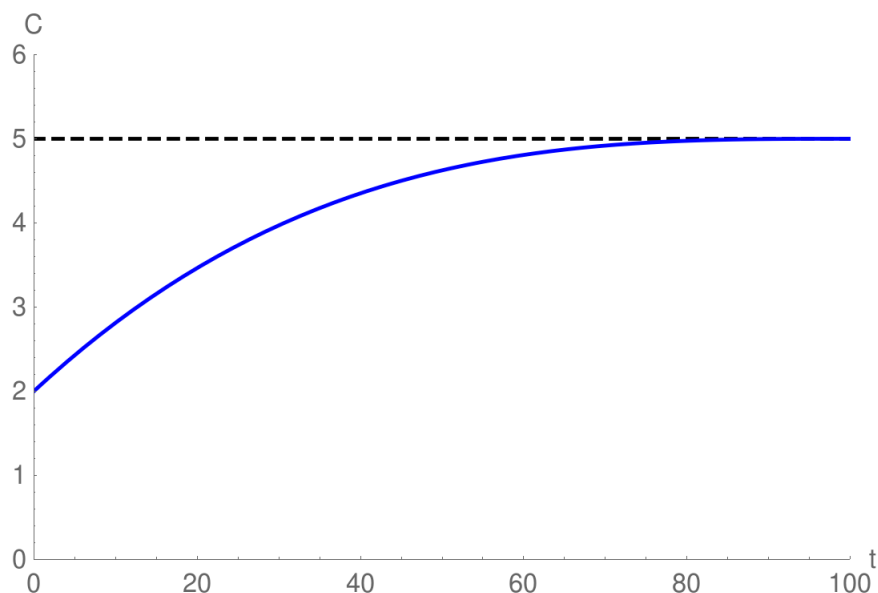


Figure 4.9.6: Solution to Example 4.9.3

Notice that at $t = 100$, the concentration actually becomes 5 oz/gal. This makes sense since in the final moments before the tank empties, the solution in the tank should be almost completely made up of the external solution.

■

4.10 Autonomous Differential Equations, Equilibria, and Stability

The most important examples of differential equations we have seen so far have been of the form

$$\frac{dx}{dt} = g(x).$$

That is, there is no explicit time-dependence on the right-hand side of the differential equation. Such equations are called **autonomous differential equations**.

While we can solve autonomous differential equations using the techniques we have developed in the previous sections (they are just a special type of

separable differential equation), it turns out we can predict the *qualitative behavior* of such systems with very little effort.

4.10.1 Equilibria of Autonomous Differential Equations

An **equilibrium** for an autonomous differential equation

$$\frac{dx}{dt} = g(x)$$

is any value x^* of the dependent variable such that

$$g(x^*) = 0.$$

What make equilibria significant for these differential equations is that they are the only constant solutions to the differential equation.

Theorem 4.10.1. *Suppose that x^* satisfies $g(x^*) = 0$. Then the solution to the initial value problem*

$$\begin{aligned} \frac{dx}{dt} &= g(x) \\ x(0) &= x^* \end{aligned}$$

is the constant function $x(t) = x^$. Moreover, the only constant solutions to this differential equation are given by the equilibria.*

Proof: If x^* is an equilibrium, then for the function $x(t) = x^*$ we have $x'(t) = 0$ and $g(x(t)) = g(x^*) = 0$. So, $x(t) = x^*$ is a solution to the differential equation. On the other hand, if $x(t) = x^*$ is a solution to the differential equation, then since $x'(t) = 0$ we must have

$$g(x(t)) = g(x^*) = 0.$$

Hence, x^* must be an equilibrium value. ■

Example 4.10.1. Find the equilibria of the differential equation

$$\frac{dx}{dt} = 2x^3 - x^2 - x.$$

Answer: To find the equilibria, we need to solve $2x^3 - x^2 - x = 0$.

$$\begin{aligned} 2x^3 - x^2 - x &= 0 \\ x(2x^2 - x - 1) &= 0 \\ x(2x + 1)(x - 1) &= 0 \\ x &= -\frac{1}{2}, 0, 1. \quad \blacksquare \end{aligned}$$

An autonomous differential equation can have any number of equilibria, from none at all to infinitely many.

Example 4.10.2. Find the equilibria of the differential equation

$$\frac{dx}{dt} = x^2 + 2.$$

Answer: Since $x^2 + 2 = 0$ has no real solutions, this differential equation has no equilibria. $\blacksquare$

Example 4.10.3. Find the equilibria of the differential equation

$$\frac{dx}{dt} = \sin(2x).$$

Answer: Since the solutions of $\sin(\theta) = 0$ are $\theta = n\pi$ for any integer n , the equilibria of our differential equation are

$$\begin{aligned} \sin(2x) &= 0 \\ 2x &= n\pi \\ x &= \frac{n\pi}{2}, \end{aligned}$$

where n is any integer. So, this differential equation has an infinite number of equilibria! $\blacksquare$

4.10.2 Phase Lines and Stability

After identifying the equilibria for an autonomous differential equation, we can then develop its **phase line**. The phase line for an autonomous differential equation

$$\frac{dx}{dt} = g(x)$$

is a number line (typically oriented vertically) displaying the equilibria of the differential equation. The regions in between the equilibria are marked with arrows indicating whether g is positive or negative in those regions. Note that we are assuming that g is a *continuous* function of x so that the sign of g is consistent in the regions between equilibria!

Steps to Sketch a Phase Line:

- 1.) Find all equilibria by solving $g(x) = 0$.
- 2.) Draw a vertical number line and mark all equilibria in their appropriate places on it.
- 3.) Pick a test point for each region between equilibria. Substitute the test x -values into g .
- 4.) If $g(x) > 0$ at a test point, mark the corresponding region with up arrows. If $g(x) < 0$, mark the region with down arrows.

Example 4.10.4. Sketch the phase line for the differential equation

$$\frac{dx}{dt} = 2x^3 - x^2 - x.$$

Answer: We have already found the equilibria in Example 4.10.1. We first sketch a vertical number line with the equilibria marked in the appropriate locations.

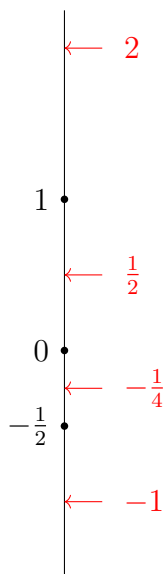


Figure 4.10.1: Phase Line for Example 4.10.4, Steps 1 – 3

Note that we have already selected a number of test points in the regions between the equilibria (shown in red). These points can be chosen arbitrarily so long as we have a representative value in each region. The idea is that since the right-hand side of the differential equation is continuous, the sign of this function must be consistently positive or negative throughout these regions. So, any test point in the region can reveal the sign of the right-hand side of the equation for the entire region. Since we only care about the sign for these test points, it certainly makes life easier to use the factored form

$$g(x) = 2x^3 - x^2 - x = x(2x + 1)(x - 1).$$

$$\begin{aligned} g(-1) &= (-1)(-2+1)(-1-1) \\ &= (-1)(-1)(-2) < 0, \end{aligned}$$

$$\begin{aligned} g\left(-\frac{1}{4}\right) &= \left(-\frac{1}{4}\right)\left(-\frac{1}{2}+1\right)\left(-\frac{1}{4}-1\right) \\ &= \left(-\frac{1}{4}\right)\left(\frac{1}{2}\right)\left(-\frac{5}{4}\right) > 0, \end{aligned}$$

$$\begin{aligned} g\left(\frac{1}{2}\right) &= \left(\frac{1}{2}\right)(1+1)\left(\frac{1}{2}-1\right) \\ &= \left(\frac{1}{2}\right)(2)\left(-\frac{1}{2}\right) < 0, \end{aligned}$$

$$\begin{aligned} g(2) &= (2)(4+1)(2-1) \\ &= (2)(5)(1) > 0. \end{aligned}$$

Adding the arrows according to the signs found above, we find.



Figure 4.10.2: Phase Line for Example [4.10.4](#)

Drawing the phase line was fairly simple, but why is this useful information? Consider the phase line next to a plot of typical solutions to the differential equation.

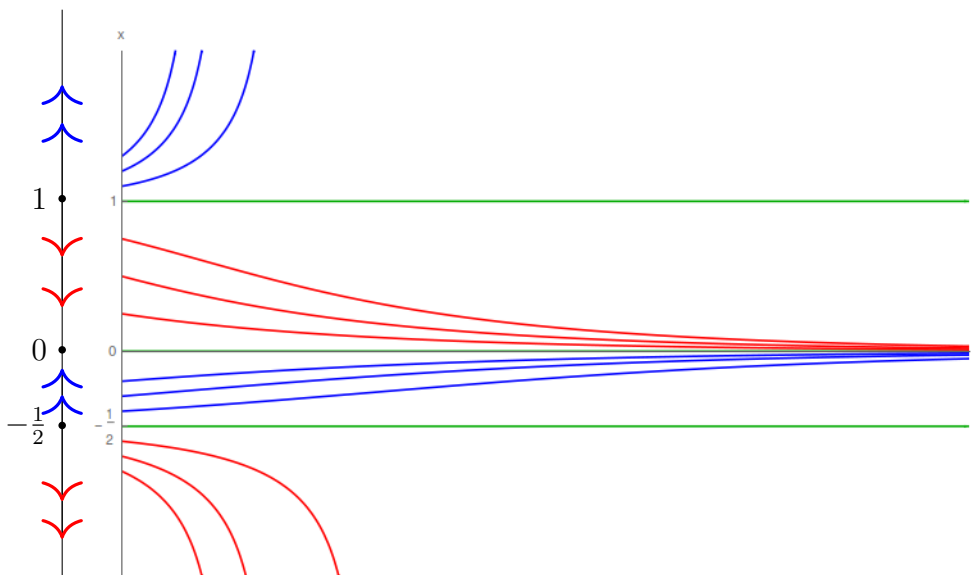


Figure 4.10.3: Phase Line and Typical Solutions for Example 4.10.4

Notice that the arrows correspond to whether solutions are increasing or decreasing. The reason for this is the right-hand side of

$$\frac{dx}{dt} = g(x).$$

controls whether the solution increases or decreases. ■

In order to make sense of the results above, we need a theoretical result about solutions to differential equations.

Theorem 4.10.2 (Existence and Uniqueness of Solutions to Differential Equations). *Suppose that the function g and its derivative g' are both continuous in some open interval containing x_0 . Then, there is a unique solution to the initial value problem*

$$\begin{aligned}\frac{dx}{dt} &= g(x) \\ x(0) &= x_0\end{aligned}$$

for some range of t -values around $t = 0$.

Proof: A proof of this result is beyond the scope of this text. See [Ro95, pp. 141 – 143] for the proof of a slightly more general result. ■

For all of the examples we will consider, g and its derivative will be continuous everywhere. That means any initial datum we happen to choose will lead to a *unique* solution for some range of t -values. The most important consequence of this is that solutions from different initial data *cannot intersect one another*. Consider again the solution curves for Example 4.10.4.

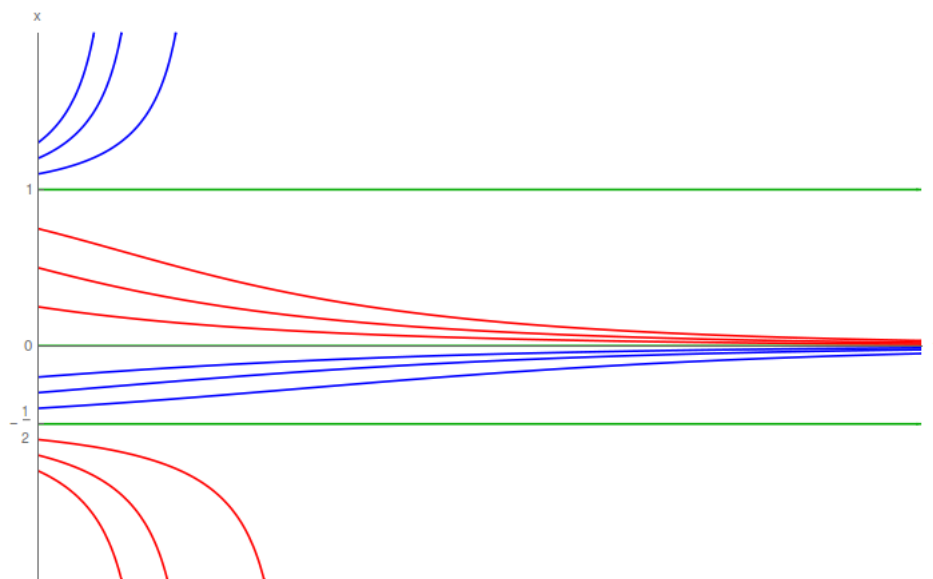


Figure 4.10.4: Typical Solutions for Example 4.10.4

Notice that the three solutions in green are the only constant solutions ($x(t) = -1/2$, $x(t) = 0$, and $x(t) = 1$). Since no two solutions can intersect, no solution curves can cross the equilibria solutions. If we consider the solutions starting from initial data between $-1/2$ and 0 , these solutions must be increasing for all t (because $x'(t) > 0$). However, they cannot cross $x = 0$. Therefore all these solution curves increase forever but are bounded above by $x = 0$ (never quite reaching this value). Similarly, the solutions starting with initial data between 0 and 1 must be decreasing for all t , but cannot cross $x = 0$. So, they decrease forever and are bounded below by $x = 0$. Notice that solutions starting with initial data less than $-1/2$ or

greater than 1 have no equilibria to approach. As such, they will decrease or increase without bound.

Example 4.10.5. Sketch the phase line for

$$\frac{dx}{dt} = (x + 2)(x + 1)^2(x - 1)(2 - x).$$

Answer: The equilibria are fairly easy to find in this case: $x = -2, -1, 1, 2$.

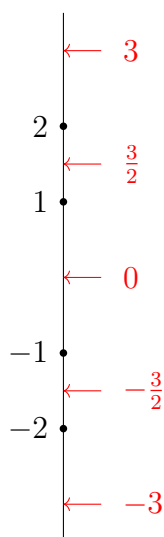


Figure 4.10.5: Phase Line for Example 4.10.5, Steps 1 – 3

Next, we check the sign of $g(x) = (x + 2)(x + 1)^2(x - 1)(2 - x)$ for each

of the values indicated in red above.

$$\begin{aligned} g(-3) &= (-3+2)(-3+1)^2(-3-1)(2+3) \\ &= (-1)(-2)^2(-4)(5) > 0, \end{aligned}$$

$$\begin{aligned} g(-3/2) &= (-3/2+2)(-3/2+1)^2(-3/2-1)(2+3/2) \\ &= (1/2)(-1/2)^2(-5/2)(7/2) < 0, \end{aligned}$$

$$g(0) = (2)(1)^2(-1)(2) < 0,$$

$$\begin{aligned} g(3/2) &= (3/2+2)(3/2+1)^2(3/2-1)(2-3/2) \\ &= (7/2)(5/2)^2(1/2)(1/2) > 0, \end{aligned}$$

$$\begin{aligned} g(3) &= (3+2)(3+1)^2(3-1)(2-3) \\ &= (5)(4)^2(2)(-1) < 0. \end{aligned}$$



Figure 4.10.6: Phase Line for Example [4.10.5](#)

Notice that we can give a qualitative description of the solutions using the phase line. For initial data greater than 2, solutions will decrease towards

$x = 2$. For initial data between 1 and 2, solutions will increase toward $x = 2$. Initial data between -1 and 1 launch solutions that decrease to $x = -1$, while initial data between -2 and -1 launch solutions that decrease toward $x = -2$. Finally, solutions with initial data less than -2 increase toward $x = -2$. ■

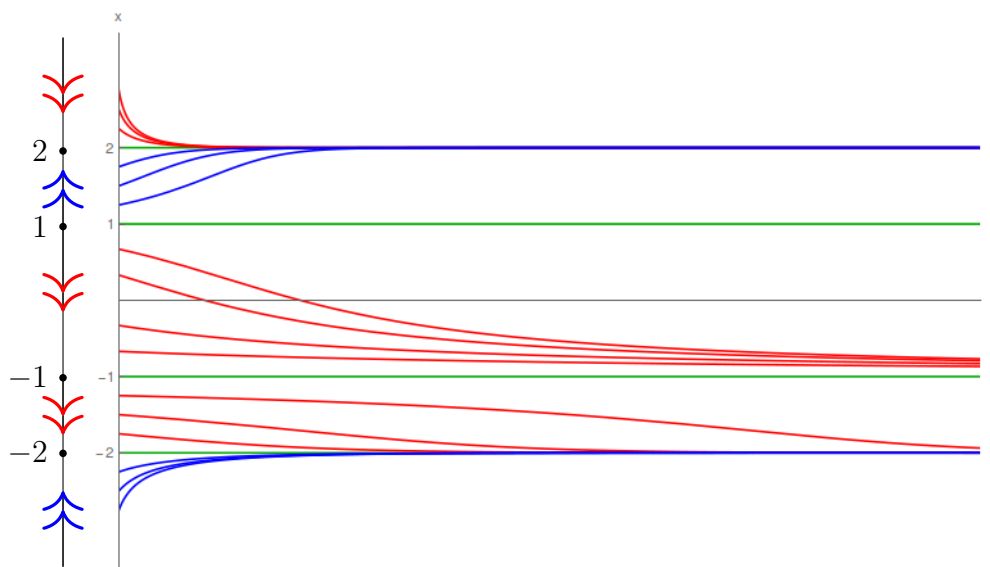


Figure 4.10.7: Phase Line and Typical Solutions for Example 4.10.5

Notice in the last example that the sign of $g(x)$ need not change across an equilibrium. Do not assume that the arrows in the phase line will simply alternate!

The phase line allows us to classify equilibria for autonomous differential equations into one of three types based on their stability. Notice that the definitions given below mirror the definitions given for discrete dynamical systems.⁴ Throughout the following discussion, we assume that g and g' are continuous functions so that the existence and uniqueness theorem stated above is applicable!

⁴Equilibria that are stable but not asymptotically stable do not occur in single autonomous differential equations (at least for the case of continuous $g(x)$ with isolated zeros). Such equilibria are possible for *systems* of autonomous differential equations which we discuss in Chapter 5.

An equilibrium of

$$\frac{dx}{dt} = g(x)$$

is **asymptotically stable** (also called a **sink**) if there is some open interval containing the equilibrium such that any initial datum in the interval different from the equilibrium itself launches a solution that approaches the equilibrium value for all future times.

An equilibrium is **unstable** (also called a **source**) if there is some open interval containing the equilibrium such that any initial datum in the interval different from the equilibrium itself launches a solution that diverges from the equilibrium value for all future times.

An equilibrium is **semi-stable** (also called a **node**) if there is some open interval containing the equilibrium such that initial data on one side of the equilibrium launch solutions that approach the equilibrium value. On the other side of the equilibrium, initial data launch solutions that diverge from the equilibrium value.

Equilibria are easily classified directly from the phase line.⁵

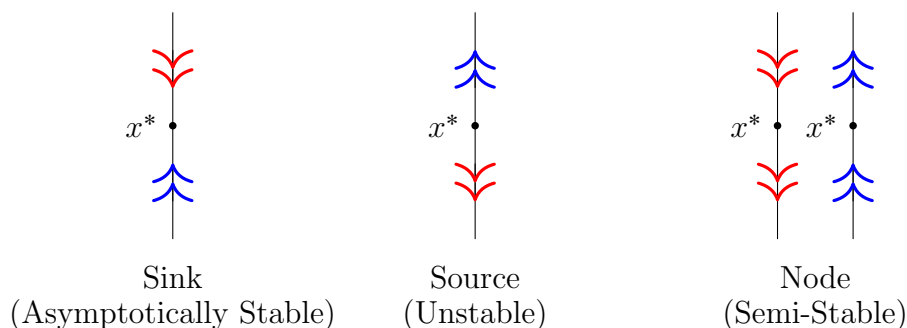


Figure 4.10.8: Stability and Phase Lines

Example 4.10.6. Classify the equilibria for the autonomous differential equation in Example 4.10.1:

$$\frac{dx}{dt} = 2x^3 - x^2 - x.$$

⁵It is possible for a non-zero differentiable function $g(x)$ to have an *entire interval* of zeros. In this case, the zeros in the interior of the interval would be stable (but not asymptotically stable) while the ones on the boundary could be stable or unstable. Such functions rarely (if ever) come up in this context!

Answer: Since we have already sketched the phase line for this differential equation, we only need to classify the equilibria

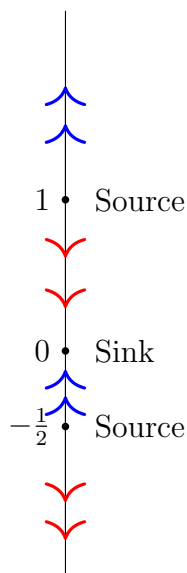


Figure 4.10.9: Phase Line and Stability for Example 4.10.1

So, $x^* = -1/2$ and $x^* = 1$ are unstable while $x^* = 0$ is asymptotically stable.

■

Example 4.10.7. Classify the equilibria for the autonomous differential equation in Example 4.10.5:

$$\frac{dx}{dt} = (x + 2)(x + 1)^2(x - 1)(2 - x).$$

Answer:

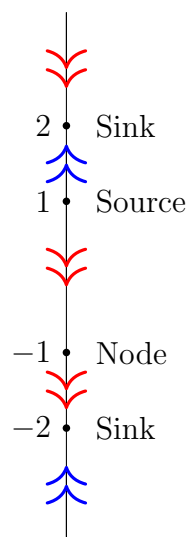


Figure 4.10.10: Phase Line for Example 4.10.5

So, $x^* = -2$ and $x^* = 2$ are both asymptotically stable, $x^* = -1$ is semi-stable, and $x^* = 1$ is unstable. ■

4.10.3 Equilibria and Stability for Important Autonomous Equations

In this section, we will revisit the most important examples of autonomous differential equations. Our primary focus will be on the equilibria of these models together with stability results.

Theorem 4.10.3 (Equilibrium for Newton's Law of Cooling). *The only equilibrium for Newton's Law of Cooling*

$$\frac{dT}{dt} = k(T_e - T)$$

is the environmental temperature $T^ = T_e$. This equilibrium is asymptotically stable.*

Proof: The equilibrium is found by the equation

$$k(T_e - T) = 0.$$

Since the cooling rate k is assumed to be positive, the only equilibrium is $T = T_e$. As we mentioned in the analysis of this differential equation, when $T < T_e$, the right-hand side will be negative; on the other hand, it will be positive when $T > T_e$. This gives us the phase line shown below. ■

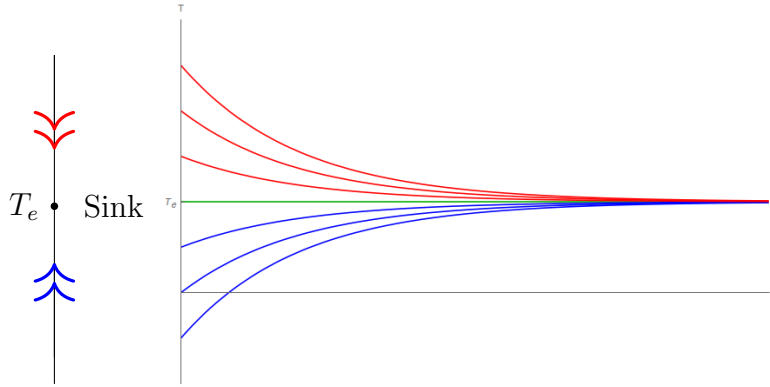


Figure 4.10.11: Phase Line and Typical Solutions for Newton's Law of Cooling

Theorem 4.10.4 (Equilibrium for Selective Diffusion). *The only equilibrium for the selective diffusion model*

$$\frac{dC}{dt} = \alpha\Gamma - \beta C$$

is the concentration

$$C^* = \frac{\alpha}{\beta}\Gamma.$$

This equilibrium is asymptotically stable.

Proof: This system is identical to Newton's Law of Cooling. ■

Theorem 4.10.5 (Equilibria for Exponential Growth and Decay). *The only equilibrium for either exponential growth or decay*

$$\frac{dP}{dt} = kP \quad \text{or} \quad \frac{dA}{dt} = -kA,$$

is 0. For exponential growth, this equilibrium is unstable, while for exponential decay it is asymptotically stable.

Proof: Since the growth rate k (or for decay, the activity k) is assumed to be positive, the only equilibrium is $P^* = 0$ (or $A^* = 0$). For exponential growth, the right-hand side of the differential equation will be positive when $P > 0$ (and negative when $P < 0$). This shows us that $P^* = 0$ is unstable. For exponential decay, the negative sign on the right-hand side of the equation reverses the arrows giving an asymptotically stable equilibrium. ■

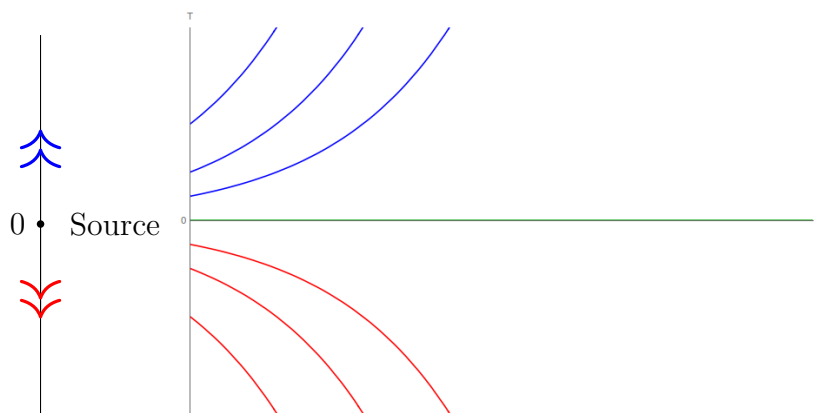


Figure 4.10.12: Phase Line and Typical Solutions for Exponential Growth

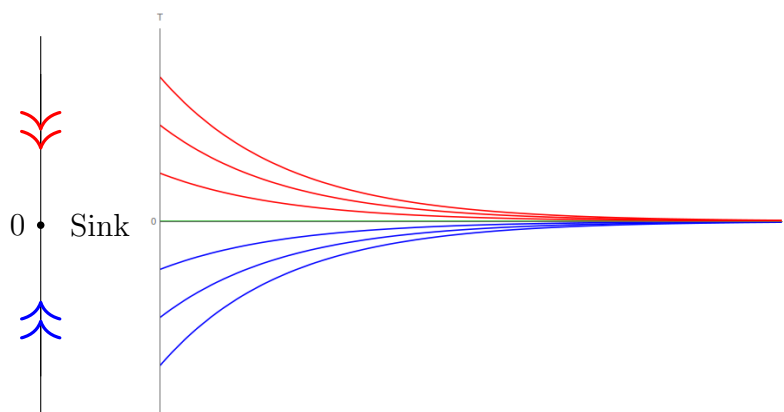


Figure 4.10.13: Phase Line and Typical Solutions for Exponential Decay

Theorem 4.10.6 (Equilibria for Logistic Growth). *The equilibria for the*

logistic growth model

$$\frac{dP}{dt} = kP \left(1 - \frac{P}{N} \right)$$

are $P^* = 0$ which is unstable and the carrying capacity $P^* = N$ which is asymptotically stable.

Proof: The equilibria are given by

$$kP \left(1 - \frac{P}{N} \right) = 0.$$

Since the growth rate k is positive, the only solutions are $P^* = 0$ and $P^* = N$. When $P < 0$ (which is not meaningful for populations of organisms),

$$kP \left(1 - \frac{P}{N} \right) < 0.$$

For populations in the range $0 < P < N$, we can see that

$$kP \left(1 - \frac{P}{N} \right) > 0$$

since k and P are certainly positive and $P/N < 1$. This shows us that $P^* = 0$ is unstable. For $P > N$,

$$kP \left(1 - \frac{P}{N} \right) < 0$$

which shows us that $P^* = N$ is asymptotically stable. ■

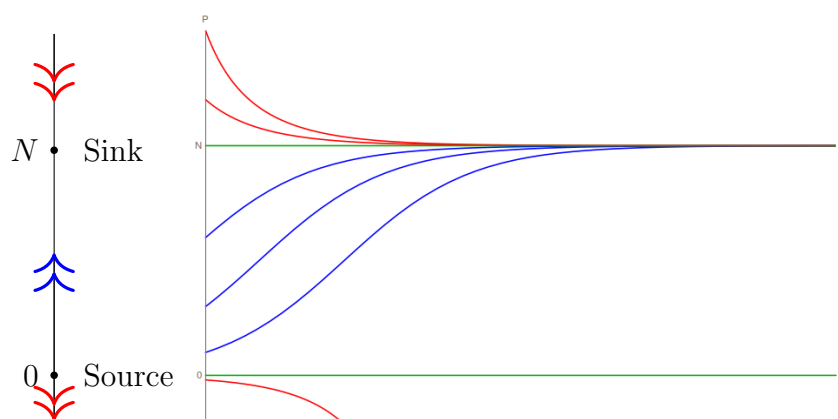


Figure 4.10.14: Phase Line and Typical Solutions for Logistic Growth

4.11 Problems

For problems 1.) – 4.), identify the independent and dependent variables along with any parameters for the given differential equation. Then verify the given function is a solution to the initial value problem.

1.)

$$\begin{cases} \frac{dv}{dt} = 10 - 3v \\ v(0) = 0 \end{cases}$$
$$v(t) = \frac{10}{3} - \frac{10}{3}e^{-3t}$$

2.)

$$\begin{cases} \frac{dv}{dt} = g - bv \\ v(0) = 1 \end{cases}$$
$$v(t) = \frac{g}{b} + \left(1 - \frac{g}{b}\right)e^{-bt}$$

3.)

$$\begin{cases} t \frac{dx}{dt} + x = t^3 \\ x(1) = 2 \end{cases}$$
$$x(t) = \frac{t^4 + 7}{4t}$$

4.)

$$\begin{cases} t \frac{dx}{dt} + \alpha x = t \\ x(1) = 0 \end{cases}$$
$$x(t) = \frac{1}{1 + \alpha} \left(t - \frac{1}{t^\alpha} \right)$$

For problems 5.) – 12.), solve the given initial value problem.

5.)

$$\begin{cases} \frac{dx}{dt} = t^2 x^2 \\ x(1) = 1 \end{cases}$$

6.)

$$\begin{cases} \frac{dx}{dt} = t^2 x \\ x(0) = 5 \end{cases}$$

7.)

$$\begin{cases} \frac{dx}{dt} = \frac{t}{x} \\ x(0) = 5 \end{cases}$$

8.)

$$\begin{cases} \frac{dx}{dt} = t \sin(t)x \\ x(0) = 1 \end{cases}$$

9.)

$$\begin{cases} \frac{dx}{dt} = x(1 - x) \\ x(0) = 6 \end{cases}$$

10.)

$$\begin{cases} \frac{dx}{dt} + 2x = e^{3t} \\ x(0) = 2 \end{cases}$$

11.)

$$\begin{cases} \frac{dx}{dt} + 2tx = t \\ x(0) = 1 \end{cases}$$

12.)

$$\begin{cases} \frac{dx}{dt} + \frac{3}{t}x = \frac{\sin(t)}{t^2} \\ x(\pi) = 0 \end{cases}$$

For problems 13.) – 18.), determine the equilibria for the given autonomous differential equation. Sketch a phase line and classify the equilibria that you find. Also, sketch representative solution curves.

13.)

$$\frac{dx}{dt} = -x^2 + x + 6$$

14.)

$$\frac{dx}{dt} = -x^3 + 8x^2 - 15x$$

15.)

$$\frac{dx}{dt} = (x + 5)(2x + 3)^2(x - 1)(4 - x)$$

16.)

$$\frac{dx}{dt} = -(1 + x^2)$$

17.)

$$\frac{dx}{dt} = \sin(x)$$

18.)

$$\frac{dx}{dt} = \frac{5x^2}{1 + x^2} - 2x$$

Solve the following problems related to the standard differential equations discussed in the chapter.

- 19.) A culture of bacteria is undergoing exponential growth. If initially there are 500 organisms, and there 1500 after 8 hours, how long until the population reaches 10,000 individuals?
- 20.) If a culture of bacteria grows from 1200 to 1600 individuals in 10 hours, what is the doubling time for the population?
- 21.) A certain radioactive element has a half-life of 12 minutes. If there are 100 μg initially, how long until only 20 μg remain?
- 22.) A lump of clay is molded into a vase. Once it has been dried at room temperature (25°C), the vase is placed in a kiln at 1100°C . After 1 hour, the vase has reached 500°C . How long until the vase reaches 1000°C ?

- 23.) A steel rod initially at 50°C will cool to 46°C in 5 seconds if placed into an oil bath held at 30°C . The steel is quenched by first heating it to 900°C and then plunging it into the 30°C oil bath. How long does it take the rod to cool to 100°C ?
- 24.) The concentration of a certain drug in the blood stream is 200 ng/dL (nanograms per deciliter). The drug diffuses into a red blood cell 4 times faster than it leaves the cell. After 10 minutes, the concentration of the drug inside the cell has risen from 0 ng/dL to 300 ng/dL. How long until the concentration in the cell reaches 500 ng/dL? What is the limiting concentration of the drug inside the cell?
- 25.) A population is undergoing logistic growth according to the differential equation

$$\frac{dP}{dt} = \frac{1}{3}P(1000 - P).$$

Identify the growth rate and the carrying capacity for the population. If there are 300 individuals initially, how long until the population reaches 600?

- 26.) A population is living in an area that has a carrying capacity of 50,000 individuals. If initially there were 15,000 individuals, and the population has increased to 20,000 after 5 years, how long until the population reaches 30,000?
- 27.) A tank initially holds 30 liters of a sugar solution at a concentration of 35 grams per liter. Pure water flows into the top of the tank at a rate of 10 liters per minute, and the well-mixed solution flows out of the tank at a matching rate. What is the concentration of sugar in the tank after 20 minutes?
- 28.) A tank initially holds 20 liters of a saline solution at a concentration of 50 grams per liter. A saline solution of concentration of 30 grams per liter flows into the tank at a rate of 5 liters per minute. The well-mixed solution flows out of the tank at the same rate. When will the concentration of salt in the tank fall to 40 grams per liter?

- 29.) A large tank capable of holding 1000 gallons of fluid is initially filled with 200 gallons of pure water. An acid solution at concentration one-quarter pound per gallon flows into the tank at a rate of 3 gallons per minute. The well-mixed solution flows out of the tank at a rate of 1 gallon per minute. What is the concentration of acid in the tank at the point when the tank begins to overflow?
- 30.) **Challenge:** A population is increasing logistically. Suppose that $P(0) = 1$, $P(1) = 9/2$, and $P(2) = 9$. Find the carrying capacity N and the growth rate k .

Chapter 5

Systems of Autonomous Differential Equations

All of the differential equations we have studied so far have been for a *single* unknown function. That is, there was a single dependent variable (e.g. $x = x(t)$), and we were given information about its derivative (and perhaps an appropriate initial condition). In many applications, however, there are several dependent variables which depend on one another (as well as depending on the independent variable t). This gives rise to **systems of differential equations**.

As a particular example, suppose we have two species: one a predator and the other its prey. If there were no predators, then the prey species might increase exponentially (or maybe logistically). Certainly the presence of a predator would change the growth rate! Likewise, the predators would likely die off if there were no prey animals (assuming there were no other sources of food). On the other hand, if the prey animals were plentiful enough, the predator population would likely increase dramatically.

In a simple predator–prey model, there are two populations of animals: $R = R(t)$ is the population of prey while $F = F(t)$ is the population of predators.¹ The generic form of the resulting system of differential equations

¹In one of the early studies of predator–prey models, “ R ” was used for rabbits and “ F ” was used for foxes. These letters have become standard in discussing these models.

is

$$\begin{aligned}\frac{dR}{dt} &= f(R, F) \\ \frac{dF}{dt} &= g(R, F).\end{aligned}$$

In words, the rate of change of the prey population is some function of both the prey population and the predator population. The rate of change of the predator population is a different function of the same two quantities. Notice that there is no explicit t dependence in the model. This means we have a **system of autonomous differential equations**.

While we could consider more general cases than this, systems of autonomous differential equations are already difficult to solve in any but the most simple of cases. In this chapter, we will be content with a more qualitative approach similar to the discussion of equilibria for single autonomous equations in the previous chapter. We will primarily discuss *two-dimensional* autonomous systems in which there are only two dependent variables (like the predator–prey example above). While we discuss a few higher dimensional systems in a very limited way, the qualitative approach we develop becomes extremely difficult once there are more than two dependent variables. If you are comfortable with matrix techniques (see Chapter 8 for the appropriate material), you can find a rigorous treatment of these results in Chapter 9.

5.1 Equilibria and Nullclines

The equilibria of the two-dimensional autonomous system

$$\begin{aligned}\frac{dx}{dt} &= f(x, y) \\ \frac{dy}{dt} &= g(x, y)\end{aligned}$$

are *pairs* of values (x^*, y^*) such that

$$f(x^*, y^*) = 0 \quad \text{AND} \quad g(x^*, y^*) = 0.$$

This means that the equilibria are the *simultaneous solutions* to the equations

$$\begin{cases} f(x, y) = 0 \\ g(x, y) = 0 \end{cases}.$$

As was the case for a single autonomous equation, a system of autonomous equations can have any number of equilibria – from none at all to infinitely many! If the functions f and g are fairly complicated, the equilibria can be difficult to identify. Fortunately, we will deal with relatively simple cases.

Just as with single autonomous equations, the equilibria give us the constant solutions to the autonomous system.

Theorem 5.1.1. *Suppose that (x^*, y^*) is an equilibrium of the autonomous system*

$$\begin{aligned}\frac{dx}{dt} &= f(x, y) \\ \frac{dy}{dt} &= g(x, y).\end{aligned}$$

Then the constant functions $x(t) = x^$ and $y(t) = y^*$ are solutions to the system. Conversely, if $x(t) = x^*$ and $y(t) = y^*$ are constant solutions to the system of autonomous equations, then (x^*, y^*) must be an equilibrium of the system.*

Proof: Suppose that (x^*, y^*) is an equilibrium. Then $f(x^*, y^*) = 0$ and $g(x^*, y^*) = 0$. Checking the differential equation for the functions $x(t) = x^*$ and $y(t) = y^*$ gives

$$\begin{aligned}0 &= x'(t) = f(x(t), y(t)) = f(x^*, y^*) = 0, \\ 0 &= y'(t) = g(x(t), y(t)) = g(x^*, y^*) = 0.\end{aligned}$$

So, these constant functions are a solution. For the converse, since $x'(t) = 0$ and $y'(t) = 0$, we must have

$$\begin{aligned}f(x(t), y(t)) &= f(x^*, y^*) = 0, \\ g(x(t), y(t)) &= g(x^*, y^*) = 0,\end{aligned}$$

for these functions to be a solution. Therefore, (x^*, y^*) must be an equilibrium. ■

We begin with a relatively simple example.

Example 5.1.1. Find all equilibria for the autonomous system

$$\begin{aligned}\frac{dx}{dt} &= x + y - 1 \\ \frac{dy}{dt} &= x - y - 3.\end{aligned}$$

Answer: As per usual, the independent variable is t . This system has two dependent variables, $x = x(t)$ and $y = y(t)$, and is autonomous since there is no explicit dependence on t in either right-hand sides of the differential equations. To find equilibria, we have to solve the system of equations

$$\begin{cases} x + y - 1 = 0 \\ x - y - 3 = 0 \end{cases}.$$

Since these are linear equations, the solution is fairly simple to get! If we add the two equations, we get $2x - 4 = 0$, and so $x = 2$. Substituting this for x in the first equation (or the second) gives

$$\begin{aligned} 2 + y - 1 &= 0 \\ y + 1 &= 0 \\ y &= -1. \end{aligned}$$

So, the only equilibrium for this system is $(2, -1)$. ■

Example 5.1.2. Find all equilibria for the autonomous system

$$\begin{aligned} \frac{dx}{dt} &= x(x - y) \\ \frac{dy}{dt} &= xy + y - 2. \end{aligned}$$

Answer: To find equilibria, we have to solve the system of equations

$$\begin{cases} x(x - y) = 0 \\ xy + y - 2 = 0 \end{cases}.$$

The first equation gives us two possible cases:

$$\begin{aligned} x(x - y) &= 0 \\ x = 0 \quad \text{OR} \quad x - y &= 0 \\ x = 0 \quad \text{OR} \quad x &= y. \end{aligned}$$

We have to take each of these cases individually, apply them to the second equation, and see what the corresponding equilibria are.

Case 1: $x = 0$

Substituting $x = 0$ into the second equation gives:

$$(0)y + y - 2 = 0$$

$$y - 2 = 0$$

$$y = 2.$$

So, $(0, 2)$ is an equilibrium for the system.

Case 2: $x = y$

Substituting $x = y$ into the second equation gives

$$(y)y + y - 2 = 0$$

$$y^2 + y - 2 = 0$$

$$(y - 1)(y + 2) = 0$$

$$y = -2, 1.$$

Since $x = y$ in this case, the equilibria for Case 2 are $(-2, -2)$ and $(1, 1)$.

So, this system has 3 equilibria altogether: $(0, 2)$, $(-2, -2)$, and $(1, 1)$. ■

In order to examine the stability of equilibria, we need to introduce the **phase plane** for two-dimensional autonomous systems. The idea is that since the right-hand sides of the system

$$\begin{aligned}\frac{dx}{dt} &= f(x, y) \\ \frac{dy}{dt} &= g(x, y),\end{aligned}$$

do not depend on the independent variable t , we can display a lot of information about the system in the xy -plane. In this case, the xy -plane is called the phase plane for the system. While the phase plane will be extraordinarily useful for describing the system, it is important to remember that quite a lot of information about the t -dependence of the solutions $x = x(t)$ and $y = y(t)$ is suppressed in the process.

Graphically, the equations giving us the equilibria (i.e. $f(x, y) = 0$ and $g(x, y) = 0$) are curves in the xy -plane. The equilibria are precisely the intersection points of these two curves! So, for the two-dimensional autonomous

system

$$\begin{aligned}\frac{dx}{dt} &= f(x, y) \\ \frac{dy}{dt} &= g(x, y),\end{aligned}$$

all curves given by $f(x, y) = 0$ in the xy -plane are called **x -nullclines** for the system. All curves given by $g(x, y) = 0$ in the xy -plane are called **y -nullclines** for the system. The equilibria of the system are precisely the points where an x -nullcline intersects a y -nullcline.

Example 5.1.3. Plot the nullclines and the equilibria for the system

$$\begin{aligned}\frac{dx}{dt} &= x + y - 1 \\ \frac{dy}{dt} &= x - y - 3.\end{aligned}$$

from Example 5.1.1 in the phase plane.

Answer: The x -nullcline is the curve $x + y - 1 = 0$ which is just the line $y = -x + 1$. Similarly, the y -nullcline is the line $x - y - 3 = 0$ or $y = x - 3$.

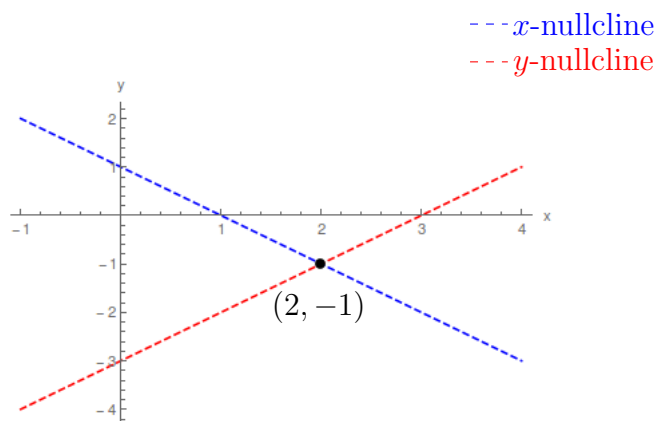


Figure 5.1.1: Nullclines and Equilibrium for Example 5.1.3

Notice that the nullclines intersect in a single point which is the only equilibrium for the system. ■

Example 5.1.4. Plot the nullclines and the equilibria for the system

$$\begin{aligned}\frac{dx}{dt} &= x(x - y) \\ \frac{dy}{dt} &= xy + y - 2.\end{aligned}$$

from Example 5.1.2 in the phase plane.

Answer: The x -nullclines are given by $x(x - y) = 0$ which are the two separate lines $x = 0$ and $x = y$. Similarly, the y -nullcline is the curve $xy + y - 2 = 0$ which is the hyperbola

$$y = \frac{2}{x + 1}.$$

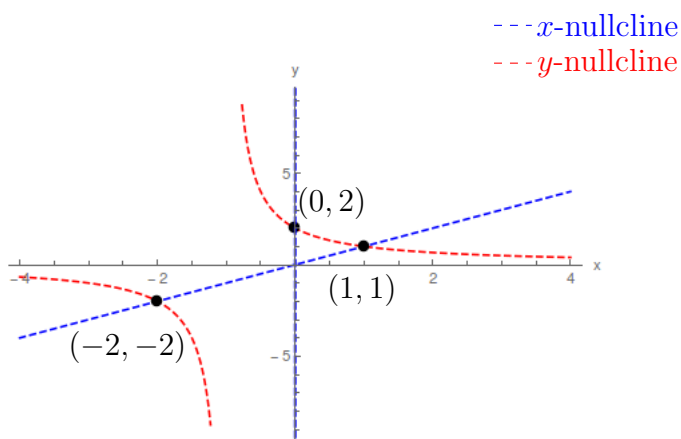


Figure 5.1.2: Nullclines and Equilibrium for Example 5.1.3

Note that the origin is NOT an equilibrium. Even though two nullclines meet there, they are both x -nullclines. ■

The last example brings up a very important point. *Equilibria only occur where an x -nullcline intersects a y -nullcline.* When you make a sketch of the nullclines for a system, it is very important to keep track of which are x - and which are y -nullclines. If you do not have separate colors available, try making one set solid and the other set dashed. Otherwise, the phase plane can get very confusing!

5.2 Stability of Equilibria

Now that we know how to find equilibria for systems of autonomous equations, our next job is to classify them. It turns out there are 6 main types of equilibria.

- 1.) Unstable Nodes²
- 2.) Unstable Spiral Points
- 3.) (Unstable) Saddle Points
- 4.) Asymptotically Stable Nodes
- 5.) Asymptotically Stable Spiral Points
- 6.) Stable Center Points

There are also a few other kinds of equilibria that are intermediate between the ones listed above, but these rarer types arise infrequently and will not concern us here. For the purposes of this chapter, we will mostly be interested in whether an equilibrium is unstable, stable, or asymptotically stable rather than its specific type.

The idea of stability is in spirit no different from what we discussed in Chapter 1. Essentially, an equilibrium is **unstable** if there are initial data points arbitrarily close to the equilibrium that launch trajectories which leave the vicinity of the equilibrium. An equilibrium is **stable** if all initial data sufficiently close to the equilibrium launch trajectories that stay in the vicinity of the equilibrium for all time. If all such trajectories actually limit to the equilibrium rather than just staying near it, we say that the equilibrium is **asymptotically stable**.

It is possible to determine the stability of an equilibrium by examining the signs of the derivatives dx/dt and dy/dt in the various regions determined by the nullclines.

²The use of the term “Node” for systems of autonomous differential equations is quite different than its use for a single autonomous equations. For a single differential equation, a node is always semi-stable. For systems, the term “Node” refers to the shape solution curves make as they approach or diverge from the equilibrium.

Example 5.2.1. Determine whether the equilibrium $(2, -1)$ for the system

$$\begin{aligned}\frac{dx}{dt} &= x + y - 1 \\ \frac{dy}{dt} &= x - y - 3.\end{aligned}$$

from Example 5.1.1 is asymptotically stable, stable, or unstable.

Answer: We will prepare a sign chart that will help us determine how solutions behave in the various regions in the phase plane.

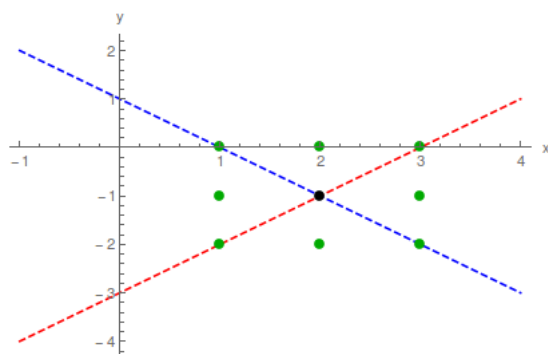


Figure 5.2.1: Phase Plane for Example 5.2.1

Notice that we have selected 8 points in total: 1 for each region between nullclines and 1 for each boundary nullcline between these regions. This is likely many more points than are needed to determine stability, but they will certainly make the result clear!

For each of the 8 points above, we substitute the x - and y -values of these points into the functions

$$\begin{aligned}f(x, y) &= x + y - 1 \\ g(x, y) &= x - y - 3,\end{aligned}$$

and keep track of the signs. We will start with $(1, 0)$ and move clockwise.

Point	$f(x, y)$	$g(x, y)$	Signs
$(1, 0)$	$f(1, 0) = 1 + 0 - 1 = 0$	$g(1, 0) = 1 - 0 - 3 = -2$	$(0, -)$
$(2, 0)$	$f(2, 0) = 2 + 0 - 1 = 1$	$g(2, 0) = 2 - 0 - 3 = -1$	$(+, -)$
$(3, 0)$	$f(3, 0) = 3 + 0 - 1 = 2$	$g(3, 0) = 3 - 0 - 3 = 0$	$(+, 0)$
$(3, -1)$	$f(3, -1) = 3 - 1 - 1 = 1$	$g(3, -1) = 3 + 1 - 3 = 1$	$(+, +)$
$(3, -2)$	$f(3, -2) = 3 - 2 - 1 = 0$	$g(3, -2) = 3 + 2 - 3 = 2$	$(0, +)$
$(2, -2)$	$f(2, -2) = 2 - 2 - 1 = -1$	$g(2, -2) = 2 + 2 - 3 = 1$	$(-, +)$
$(1, -2)$	$f(1, -2) = 1 - 2 - 1 = -2$	$g(1, -2) = 1 + 2 - 3 = 0$	$(-, 0)$
$(1, -1)$	$f(1, -1) = 1 - 1 - 1 = -1$	$g(1, -1) = 1 + 1 - 3 = -1$	$(-, -)$

What exactly do these signs tell us? Consider the sign chart $(+, -)$ for the point $(2, 0)$. The $+$ in the x -component tells us that dx/dt is positive at this point. That means that the solution is trying to move to the right in the phase plane (toward larger x -values). Similarly, the $-$ in the y -component tells us that dy/dt is negative here. Thus, the solution will move downwards in the phase plane (toward smaller y -values). We can indicate this with a small arrow at $(2, 0)$ pointing to the right and downward.

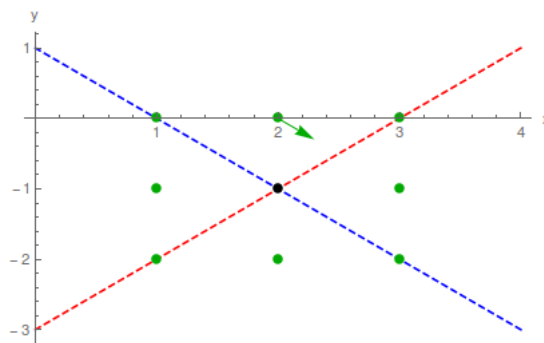


Figure 5.2.2: Phase Plane for Example 5.2.1 with Arrow at $(2, 0)$

If we continue this process for the other 7 points, we get the following picture.

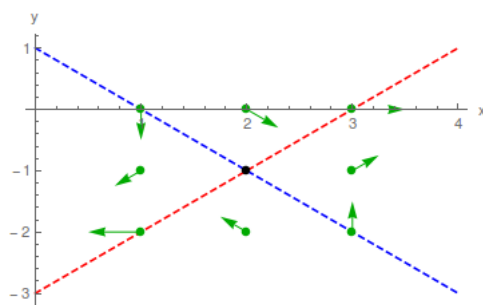


Figure 5.2.3: Phase Plane for Example 5.2.1 with Arrows

Notice that the arrows seem to indicate the equilibrium is **unstable**, no matter how close you start to the equilibrium, you are eventually pushed away from it. In more advanced treatments, we could prove that this equilibrium is a *saddle point* (which is one of the variety of unstable equilibria). See Chapter 9 for more details.

In the figure below, we plot a few solution curves for this system.

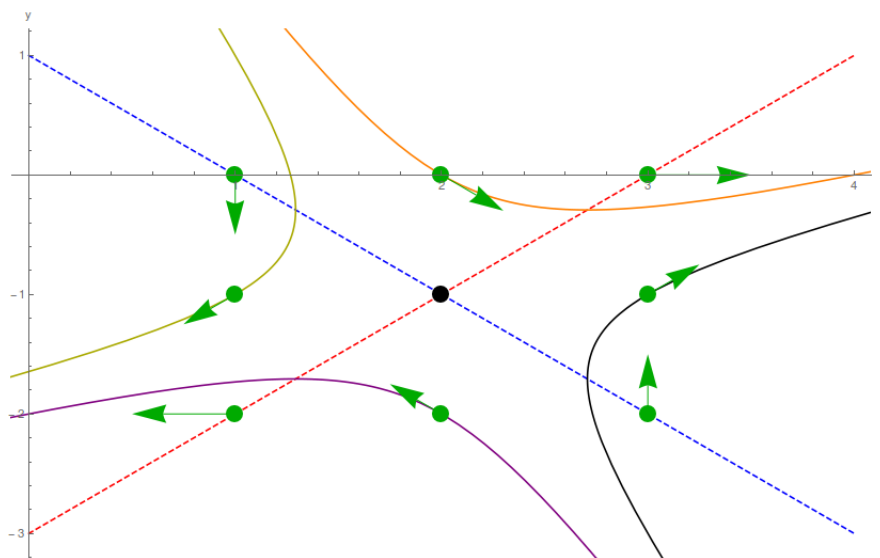


Figure 5.2.4: Phase Plane for Example 5.2.1 with Sample Solution Curves

These curves are called *parametric curves* since they are generated by following the point $(x(t), y(t))$ as t is varied. Each of the four curves shown above represent a solution curve for a particular choice of initial data. For exam-

ple, if we focus on the black curve in the figure above (which goes through $(3, -1)$), we get the following curves for $x = x(t)$ and $y = y(t)$.

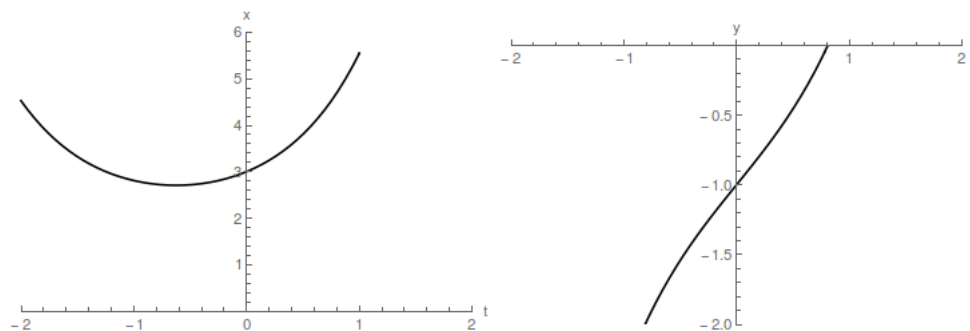


Figure 5.2.5: Solutions through $(3, -1)$ for Example 5.2.1

Even though this will not concern us much, the reason this equilibrium is a *saddle point* is that almost all solution curves move toward the equilibrium for times sufficiently far in the past but after reaching a point of closest approach move away from the equilibrium for all future times. This is also why this type of equilibrium is inherently unstable. In the long run, there are solutions starting relatively close to the equilibrium that eventually end up very far from it. ■

Note that the details of how we move along the parametric curve in the phase plane are lost. All we see is the overall path produced by tracking $(x(t), y(t))$ over time. Despite the loss of information, these parametric curves are more than sufficient to determine the stability of the equilibria. To make detailed predictions about the behavior of the system as time progresses, we would need the actual solutions or at least some way to numerically approximate solutions. Such techniques exist (as evidenced by the solution curves shown above), but they are beyond the scope of this text. ³

Example 5.2.2. Determine whether the equilibria for the system

$$\begin{aligned}\frac{dx}{dt} &= x(x - y) \\ \frac{dy}{dt} &= xy + y - 2.\end{aligned}$$

³There are a number of techniques for numerically approximating solutions to differential equations. For a nice discussion of several of them, see [BDH12, Chap. 7].

from Example 5.1.2 are asymptotically stable, stable, or unstable.

Answer: For this system, we have

$$\begin{aligned} f(x, y) &= x(x - y) \\ g(x, y) &= xy + y - 2. \end{aligned}$$

We will consider the three equilibria separately. We begin with $(0, 2)$. In the figure below, we have used the following sign chart to produce the arrows.

Point	$f(x, y)$	$g(x, y)$	Signs
$(-0.1, 2.5)$	$f(-0.1, 2.5) > 0$	$g(-0.1, 2.5) > 0$	$(+, +)$
$(0, 2.5)$	$f(0, 2.5) = 0$	$g(0, 2.5) > 0$	$(0, +)$
$(0.1, 2.2)$	$f(0.1, 2.2) < 0$	$g(0.1, 2.2) > 0$	$(-, +)$
$(1/10, 20/11)$	$f(1/10, 20/11) < 0$	$g(1/10, 20/11) = 0$	$(-, 0)$
$(0.1, 1.5)$	$f(0.1, 1.5) < 0$	$g(0.1, 1.5) < 0$	$(-, -)$
$(0, 1.5)$	$f(0, 1.5) = 0$	$g(0, 1.5) < 0$	$(0, -)$
$(-0.1, 1.8)$	$f(-0.1, 1.8) > 0$	$g(-0.1, 1.8) < 0$	$(+, -)$
$(-1/10, 20/9)$	$f(-1/10, 20/9) > 0$	$g(-1/10, 20/9) = 0$	$(+, 0)$

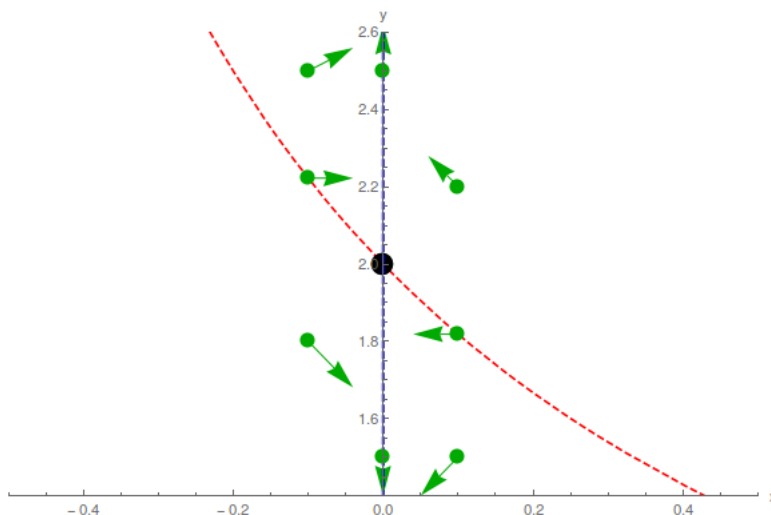
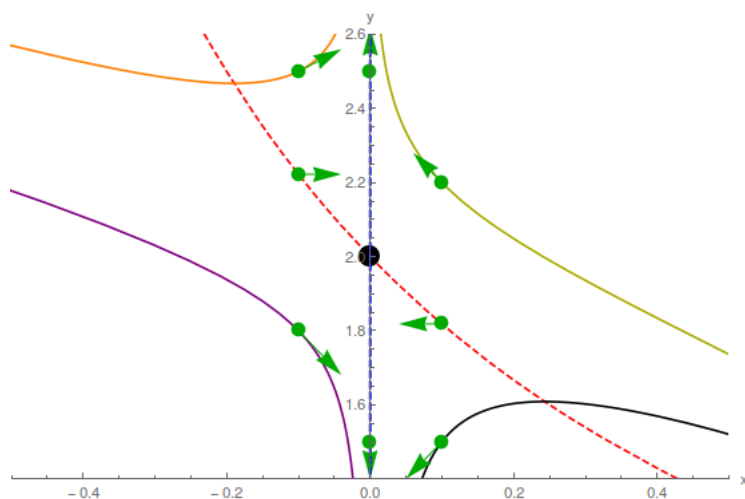
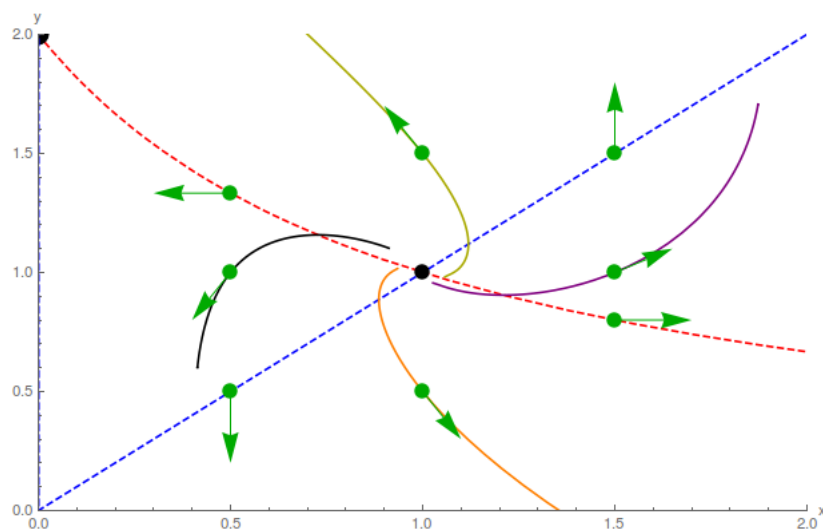


Figure 5.2.6: Behavior Near $(0, 2)$ for Example 5.2.2

Note that this looks similar to the equilibrium from the previous case. As you might guess, $(0, 2)$ is unstable (and is a saddle point).

Figure 5.2.7: Solutions Near $(0, 2)$ for Example 5.2.2

If we repeat the process for $(1, 1)$, we find the following plot.

Figure 5.2.8: Solutions Near $(1, 1)$ for Example 5.2.2

Looking back at our list of possible types of equilibria, you would probably guess that this one is an *unstable spiral point*. Certainly it is unstable since solution curves starting near $(1, 1)$ definitely move away from the equilibrium.

The solutions also seem to spiral outwards giving rise to the name of this type of equilibrium.

Finally, for $(-2, -2)$ we find the following plot.

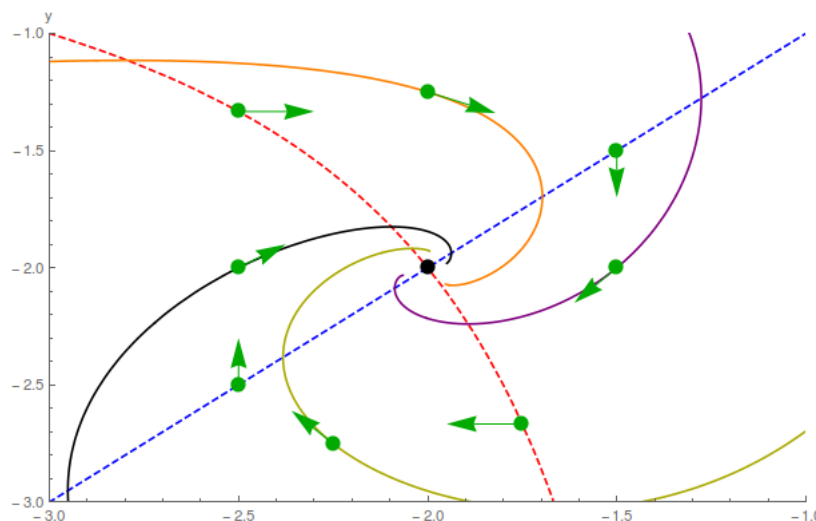


Figure 5.2.9: Solutions Near $(-2, -2)$ for Example 5.2.2

This equilibrium is asymptotically stable. Solution curves starting near $(-2, -2)$ move closer to the equilibrium as time goes on. This point is technically an *asymptotically stable spiral point*. ■

Our plots of direction vectors near an equilibrium are stripped down examples of **vector plots** or **vector field plots**. For a system of differential equations

$$\begin{aligned}\frac{dx}{dt} &= f(x, y) \\ \frac{dy}{dt} &= g(x, y),\end{aligned}$$

we plot arrows at various points having slopes given by

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{g(x, y)}{f(x, y)}.$$

This gives us a very nice idea of how solutions to the system must behave. For example, the system from Example 5.1.2 has the following vector plot.

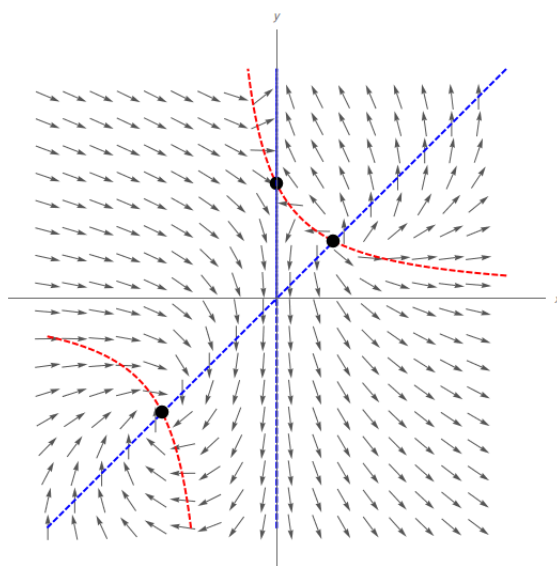


Figure 5.2.10: Vector Plot for Example 5.1.2

In addition to getting the behavior near each of the equilibria, we also have some idea how general solutions will behave. For example, the vector plot above seems to indicate that solutions starting in Quadrant II should travel down to Quadrant III and then be attracted to the asymptotically stable equilibrium at $(-2, -2)$. The plot below shows an approximate solution curve demonstrating this behavior.

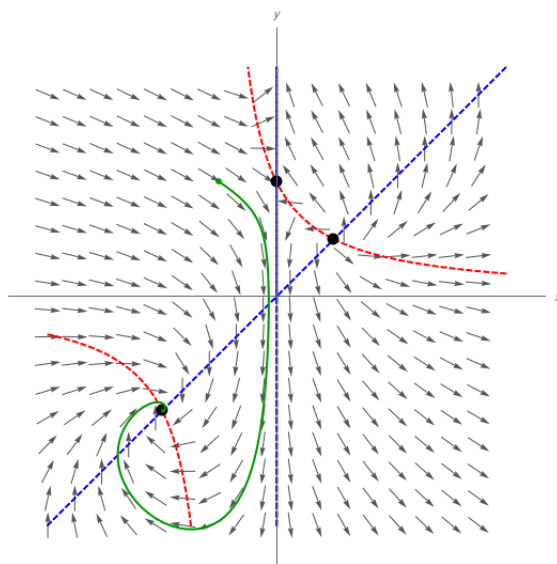


Figure 5.2.11: Vector Plot with Sample Solution Curve for Example 5.1.2

Vector plots are difficult to draw by hand. Computing derivatives at enough points to render a decent plot is time intensive, and drawing small arrows with the appropriate slope is quite hard. For these reasons, it is best to generate vector plots using technology. Appendix E shows how to generate them using the TI-89. It is also possible to generate such plots in most computer algebra systems (e.g. *Mathematica* or Maple).

Even given a decent vector plot, it can still be challenging to determine the exact behavior of solutions near an equilibrium. Fortunately, there is an analytic method for determining stability that gives clear cut answers in most cases. This method is discussed in Chapter 9, but the technique requires a decent knowledge of matrices. A brief introduction to matrix algebra can be found in Chapter 8.

5.3 Predator–Prey Models

As we mentioned in the beginning of this section, one of the more celebrated systems of autonomous differential equations is the predator–prey model, also known as the Lotka–Volterra Model. We will use $R = R(t)$ for the population of prey animals and $F = F(t)$ for the population of predators at

time t . The model has the form

$$\begin{aligned}\frac{dR}{dt} &= \alpha R - \beta RF = R(\alpha - \beta F) \\ \frac{dF}{dt} &= -\gamma F + \delta FR = -F(\gamma - \delta R)\end{aligned}$$

where α, β, γ , and δ are positive parameters.

If you look at the differential equation for dR/dt , you notice that if $F = 0$, the equation looks like $dR/dt = \alpha R$. That is, in the absence of predators, the prey grow exponentially. When there are predators, the RF term measures the likelihood of an interaction between a predator and a prey animal; each such interaction reduces the prey population. For dF/dt , the predator population decays exponentially in the absence of prey (when $R = 0$). Similarly, an interaction between a predator and prey (again with likelihood RF) increases the predator population.

Theorem 5.3.1 (Equilibria for the Lotka-Volterra Model). *The predator-prey model*

$$\begin{aligned}\frac{dR}{dt} &= R(\alpha - \beta F) \\ \frac{dF}{dt} &= -F(\gamma - \delta R)\end{aligned}$$

always has two equilibria: $(0, 0)$ and

$$(R^*, F^*) = \left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta} \right).$$

The equilibrium at $(0, 0)$ is unstable while the other equilibrium is a stable center point.

Proof: From the factored form, the R -nullclines are $R = 0$ and $F = \alpha/\beta$ while the F -nullclines are $F = 0$ and $R = \gamma/\delta$. The typical phase plane diagram for this system is

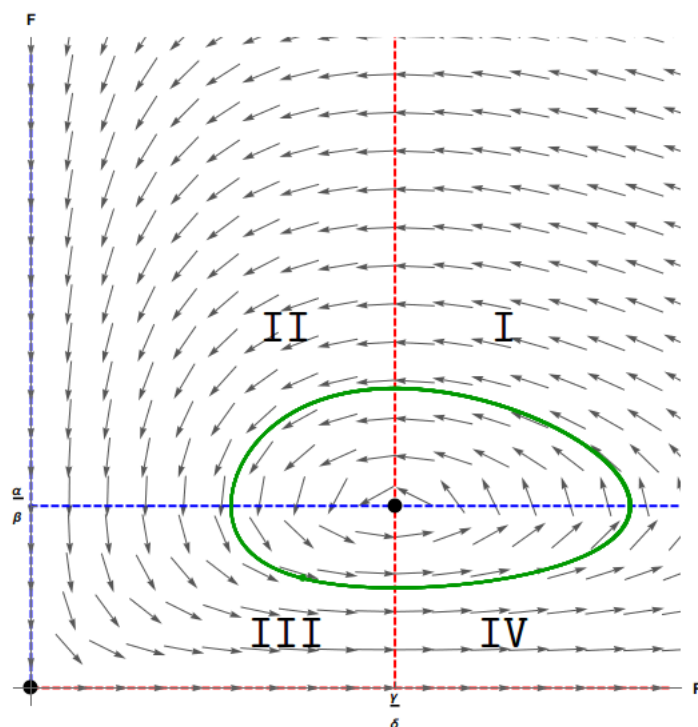


Figure 5.3.1: Phase Plane for the Predator–Prey Model

Notice that solution curves form closed loops around the central equilibrium. As such, solutions starting near this equilibrium point stay close to it for all future times, but the solutions do not necessarily get any closer than their starting points. This is why the equilibrium is said to be *stable* rather than asymptotically stable.

We should also note that it is impossible for solutions to leave the first quadrant of the phase plane. For example, if a solution were to cross from Quadrant I to Quadrant II, it would need to intersect the F -axis. But on the F -axis, dR/dt is zero while dF/dt is negative, and so the solution curve must move directly downwards toward the origin rather than crossing into Quadrant II. In fact, uniqueness of solutions prevents a curve in the interior of Quadrant I from intersecting the F -axis in the first place! Similar statements are true about the R -axis.

Notice that the nullclines divide Quadrant I into four regions (labeled I – IV in the figure above). In region I, there are large populations of both predators and prey (large as compared to the equilibrium values). The population

of prey is large enough that the predator population is growing. However, the population of predators is large enough that the prey population is dying off.

In Region II, both the populations of predators and prey are declining. The predator population is still large enough to kill off the prey at a large rate. But the prey population is no longer large enough to sustain the number of predators.

In both regions III and IV, the population of predators is rather small, allowing the prey population to grow. In region III, the prey population is increasing, but the actual population size is not large enough to sustain growth in the predator population. In region IV, the prey population has increased enough to support an increase in the predator population, but the number of predators is still small enough to allow the prey to continue growing.

If we consider how the populations of predators and prey change over time, we typically find curves like the following.

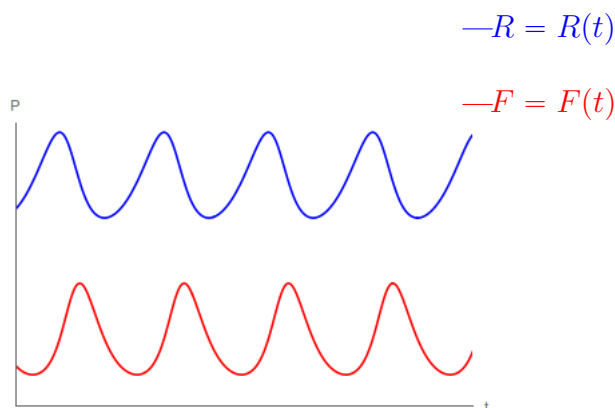


Figure 5.3.2: Typical Solution Curves for the Predator–Prey Model

It is clear from these plots that the solutions to the predator–prey model are periodic functions (though not sines or cosines). It is also apparent that the predator population (shown in red) lags behind the prey population as we discussed above. ■

5.4 Competing Species Models

The next autonomous system that we will investigate is a model describing competing species. In this model, there are two species living in a common area which utilize the same resources for their survival. As a result, interactions between the two species are to the detriment of both.

We will let $a = a(t)$ and $b = b(t)$ be the populations of the two species and assume that either species would grow logistically in the absence of the other. As is usual, we will model the likelihood of interactions between the groups as the product of their populations. These assumptions lead to the following system of differential equations.

$$\begin{aligned}\frac{da}{dt} &= k_a a \left(1 - \frac{a}{N_a}\right) - \alpha ab \\ \frac{db}{dt} &= k_b b \left(1 - \frac{b}{N_b}\right) - \beta ab\end{aligned}$$

The parameters k_a and k_b are the logistic growth rates and N_a and N_b are the carrying capacities for the two species. The parameters α and β are competition parameters measuring the rate at which one species steals resources from the other.

It turns out that there are 3 distinct qualitative behaviors for the competing species model (or maybe 4 depending on how you count). Rather than presenting a theorem with all the results up front, we will explore these three cases separately and then present a theorem summarizing our results.

We first lay some ground work. Our phase plane will be set up so that a is the horizontal axis while b is the vertical (so points will be listed in the order (a, b)). Since we are dealing with populations ($a = a(t) \geq 0$ and $b = b(t) \geq 0$) we are restricted to the first quadrant of the phase plane. Next, we identify the a - and b -nullclines.

- a -nullclines:

$$\begin{aligned}k_a a \left(1 - \frac{a}{N_a}\right) - \alpha ab &= 0 \\ k_a a \left(1 - \frac{1}{N_a}a - \frac{\alpha}{k_a}b\right) &= 0 \\ a = 0 \quad \text{or} \quad 1 - \frac{1}{N_a}a - \frac{\alpha}{k_a}b &= 0\end{aligned}$$

The first is the vertical axis in the phase plane. The second is the oblique line in the phase plane connecting the points

$$(N_a, 0) \quad \text{and} \quad \left(0, \frac{k_a}{\alpha}\right).$$

• *b*-nullclines:

$$\begin{aligned} k_b b \left(1 - \frac{b}{N_b}\right) - \beta a b &= 0 \\ k_b b \left(1 - \frac{1}{N_b} b - \frac{\beta}{k_b} a\right) &= 0 \\ b = 0 \quad \text{or} \quad 1 - \frac{1}{N_b} b - \frac{\beta}{k_b} a &= 0 \end{aligned}$$

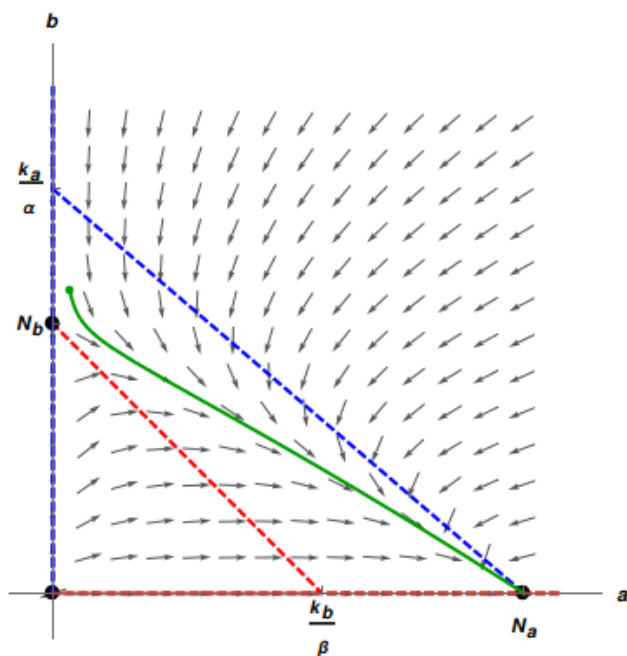
The first is the horizontal axis in the phase plane. The second is the oblique line in the phase plane connecting the points

$$(0, N_b) \quad \text{and} \quad \left(\frac{k_b}{\beta}, 0\right).$$

The different qualitative behaviors for the system depend on the relative positions of the 4 points determining the oblique nullclines.

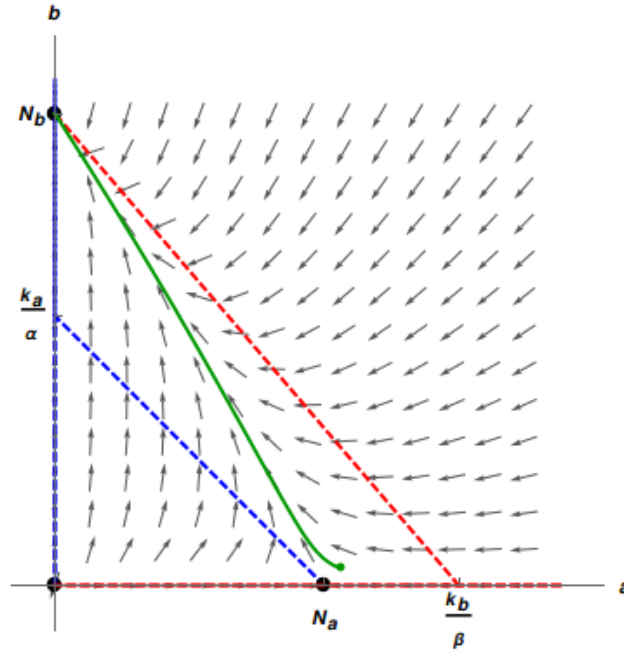
Case 1: Competitive Exclusion

In this case, the two oblique nullclines do not intersect in the first quadrant of the phase plane, and there are only 3 equilibria: $(0, 0)$, $(N_a, 0)$, and $(0, N_b)$. The origin will always be unstable. Depending on the relative sizes of the parameters for the two species, one of the other two equilibria will be asymptotically stable while the remaining one will be unstable. This means that one species will dominate the other. If $N_b < k_a/\alpha$ and $k_b/\beta < N_a$, $(N_a, 0)$ will be asymptotically stable.

Figure 5.4.1: Competing Species Model, Species a Dominant

In this case, species a will definitely out-compete species b (so long as the population of species a is not zero initially). More precisely, species a will approach its carrying capacity, N_a , while species b will tend to die out.

When $k_a/\alpha < N_b$ and $N_a < k_b/\beta$, the behavior is exactly reversed and $(0, N_b)$ will be asymptotically stable.

Figure 5.4.2: Competing Species Model, Species b Dominant

Even though these are technically two different cases, they are qualitatively identical. If we simply interchange the axes in the second phase plane (i.e. switch a and b), the two diagrams would be indistinguishable.

Case 2: Weak Competition (Coexistence)

When $N_b < k_a/\alpha$ and $N_a < k_b/\beta$, the two oblique nullclines intersect in the first quadrant of the phase plane and a fourth equilibrium appears:

$$(a^*, b^*) = \left(\frac{k_b N_a (k_a - \alpha N_b)}{k_a k_b - \alpha \beta N_a N_b}, \frac{k_a N_b (k_b - \beta N_a)}{k_a k_b - \alpha \beta N_a N_b} \right).$$

We will refer to this new equilibrium as the *coexistence point* for the two species. When both of the points representing the carrying capacities of the two species are closer to the origin than the other two points determining the oblique nullclines, the coexistence point will be asymptotically stable while the other three will be unstable. In this case, the species are said to *compete weakly*.

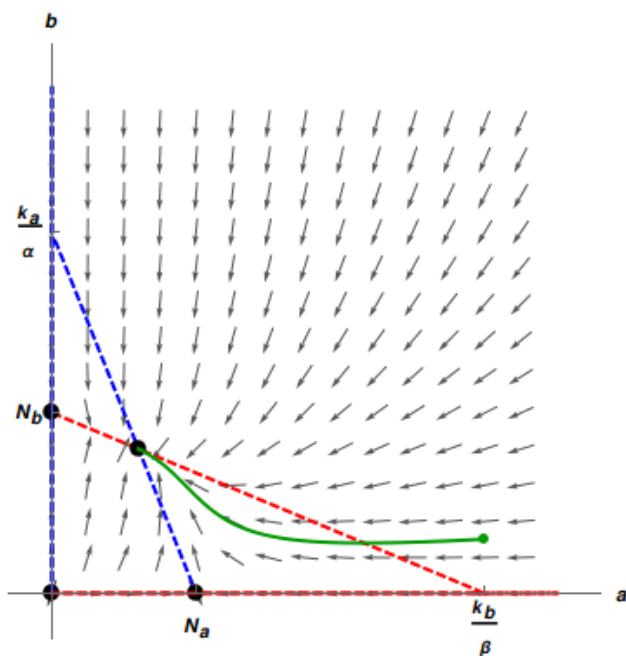


Figure 5.4.3: Competing Species Model, Weak Competition

This is the only case where the two competing species will coexist in the long term.

Case 3: Strong Competition

When $k_a/\alpha < N_b$ and $k_b/\beta < N_a$, the two oblique nullclines still intersect at the coexistence point

$$(a^*, b^*) = \left(\frac{k_b N_a (k_a - \alpha N_b)}{k_a k_b - \alpha \beta N_a N_b}, \frac{k_a N_b (k_b - \beta N_a)}{k_a k_b - \alpha \beta N_a N_b} \right).$$

However, when both carrying capacities are relatively large, the coexistence point becomes unstable (in fact, it becomes an unstable saddle point). The points $(N_a, 0)$ and $(0, N_b)$ will now BOTH be asymptotically stable.

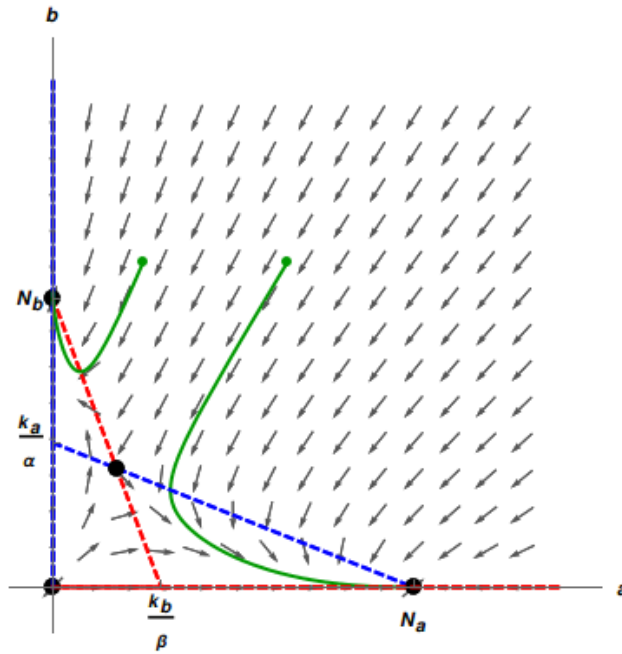


Figure 5.4.4: Competing Species Model, Strong Competition

Notice that in the long run, one species will dominate the other. The difference between this case and the first one we discussed is that either of the two species can dominate. Which species wins out in the long run depends on the initial data for the system – a fact that biologists refer to as the *founder effect*. Since they cannot coexist, the species in this case are said to *compete strongly*.

We summarize all of this information in a final theorem.

Theorem 5.4.1 (Equilibria for the Competing Species Model). *The competing species model*

$$\begin{aligned}\frac{da}{dt} &= k_a a \left(1 - \frac{a}{N_a}\right) - \alpha ab \\ \frac{db}{dt} &= k_b b \left(1 - \frac{b}{N_b}\right) - \beta ab\end{aligned}$$

always has the three equilibria $(0, 0)$, $(N_a, 0)$, and $(0, N_b)$. When $N_b < k_a/\alpha$ and $N_a < k_b/\beta$ or $k_a/\alpha < N_b$ and $k_b/\beta < N_a$, there is a fourth equilibrium at

$$(a^*, b^*) = \left(\frac{k_b N_a (k_a - \alpha N_b)}{k_a k_b - \alpha \beta N_a N_b}, \frac{k_a N_b (k_b - \beta N_a)}{k_a k_b - \alpha \beta N_a N_b} \right).$$

The origin, $(0, 0)$, is always unstable. The other 3 equilibria can be unstable or asymptotically stable depending on the parameters in the model. The asymptotically stable equilibria are summarized in the following table.

	$N_b < k_a/\alpha$	$k_a/\alpha < N_b$
$N_a < k_b/\beta$	Weak Competition (a^*, b^*) Asy. Stable	Competitive Exclusion $(0, N_b)$ Asy. Stable
$k_b/\beta < N_a$	Competitive Exclusion $(N_a, 0)$ Asy. Stable	Strong Competition $(N_a, 0)$ and $(0, N_b)$ Asy. Stable

5.5 Disease Modeling

5.5.1 The SIR Model

Next we examine a well-known disease model called the SIR Model. The acronym SIR stands for the three types of people in the model: Susceptible (people who could become infected), Infected, and Removed or Resistant (people who are naturally immune or have developed an immunity to the disease). We will use these three letters to represent the sizes of these populations. We also assume that the total population is constant (no births or deaths). This means that the total population is a fixed size N and

$$S(t) + I(t) + R(t) = N,$$

for all time. This condition is guaranteed so long as the derivatives of these three functions add up to zero:

$$S'(t) + I'(t) + R'(t) = 0.$$

This condition will be built into all of the models we consider in this section. More complicated models can incorporate births and deaths, but we will be content with this simplifying assumption.

In the SIR model, once a person recovers from the disease they are immune from the disease for all future times. When a susceptible person interacts with an infected person, the susceptible person has some probability of becoming infected. As with the predator-prey model, the likelihood of

an interaction is assumed to be given by the product of these populations: SI . Finally, infected persons recover from the disease at a certain rate. This gives rise to the following differential equations

$$\begin{aligned}\frac{dS}{dt} &= -\alpha SI \\ \frac{dI}{dt} &= \alpha SI - \beta I \\ \frac{dR}{dt} &= \beta I\end{aligned}$$

where α and β are positive parameters. The parameter α is the infection rate while β is the rate that infected people recover (and so become immune). Since $R(t) = N - S(t) - I(t)$, the removed people play a passive role in this model. So, we can focus on the susceptible and infected people.

Theorem 5.5.1 (Equilibria for the SIR Model). *The equilibria for the SIR Model*

$$\begin{aligned}\frac{dS}{dt} &= -\alpha SI \\ \frac{dI}{dt} &= \alpha SI - \beta I = I(\alpha S - \beta).\end{aligned}$$

are every point of the form $(S^, I^*) = (s, 0)$. In other words, every point on the positive S -axis is an equilibrium.*

Proof: The S -nullclines are the lines $S = 0$ and $I = 0$. The I -nullclines are $I = 0$ and $S = \beta/\alpha$. Note that $I = 0$ is a nullcline for both variables. This means that all points $(s, 0)$ are equilibria. The stability of these equilibria is actually a little tricky, but things are easier since we are only concerned with cases for which $S(t) \geq 0$ and $I(t) \geq 0$. This means that the number of susceptible people is always decreasing. The number of infected people will increase when $S > \beta/\alpha$ but decrease when S falls below this level.

A typical phase plane for the SIR Model is shown below. The horizontal line $I = 0$ is displayed in purple since it is a nullcline for both dependent variables (i.e. every point on the S -axis is an equilibrium point). The yellow triangle represents the region in the first quadrant where $S + I \leq N$. Since the total population is always N (recall that we some portion of the population is resistant), solutions must stay in this region for all time. Note that on the oblique boundary, all arrows point towards the interior of the triangle!

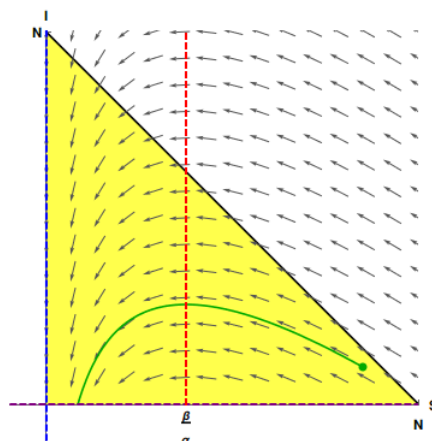


Figure 5.5.1: A Typical Phase Plane for the SIR Model

As we can see, the equilibria $(s, 0)$ for $s < \beta/\alpha$ are asymptotically stable while the others are unstable. The individual solution curves for the typical solution above are shown in the following figure.

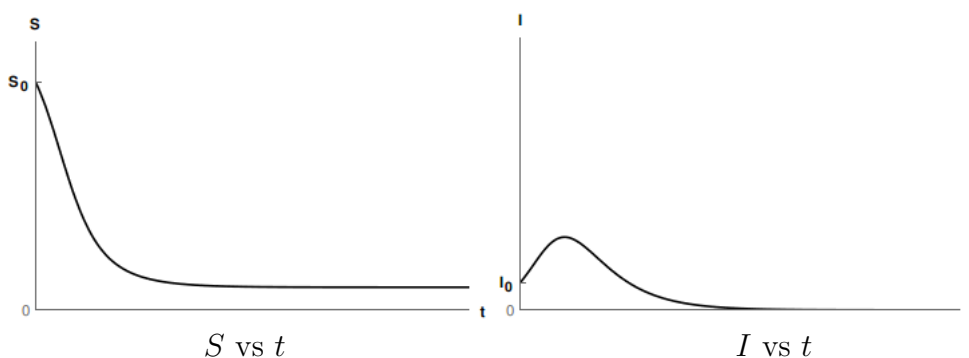


Figure 5.5.2: Typical Solution Curves for the SIR Model

Notice that the susceptible population continually decreases, but approaches a limiting value as time goes on. The number of infected individuals initially is increasing, but once the number of susceptible people falls to a certain threshold, the number of infected decreases and approaches zero.

To summarize, the number of infected individuals in the SIR Model approaches zero as time goes on. The remaining population will be split between the removed (or resistant) people and the individuals who are still

susceptible. The idea is that once the infected population falls to a fairly low level, there will not be enough interactions between susceptible and infected persons to keep the number of infected people from approaching zero. The number of people who remain susceptible once the disease has effectively died out depends on the initial data. If the ratio β/α is close to zero, that means the rate of infection is much larger than the rate of recovery. In such a case, the limiting population that remains susceptible is fairly small indicating that most of the population becomes infected and recovers. On the other hand, if the rate of infection is much smaller than the rate of recovery, the disease can effectively die out before much of the population becomes infected.

5.5.2 The SIRS Model

The SIR Model only works well for relatively short-lived diseases (as it assumes the population is fixed in size) for which recovery grants complete immunity from future infection. While we do not want to give up the assumption that the population remains constant, we can alter our model to take into account the fact that people who recover can sometimes lose their immunity from the disease. This gives rise to the SIRS Model; the final “S” in the name is meant to emphasize that recovered people can again become susceptible over time. This model is given by the following differential equations.

$$\begin{aligned}\frac{dS}{dt} &= -\alpha SI + \gamma R \\ \frac{dI}{dt} &= \alpha SI - \beta I \\ \frac{dR}{dt} &= \beta I - \gamma R\end{aligned}$$

The positive parameters α and β play the same role as before: α is the infection rate and β is the rate that infected people recover. The positive parameter γ represents the rate at which people who recover from the disease lose their immunity.

Using the fact that the total population is fixed at size N , we can replace

R in the differential equation for S with $R = N - S - I$.

$$\begin{aligned}\frac{dS}{dt} &= -\alpha SI + \gamma(N - S - I) \\ \frac{dI}{dt} &= I(\alpha S - \beta)\end{aligned}$$

The behavior of this model can be far more complicated than the SIR Model. In particular, it is possible for the number of infected people to approach a non-zero limiting value. In such a case, we say that the disease is *endemic* to the population.

Theorem 5.5.2 (Equilibria for the SIRS Model). *The SIRS Model*

$$\begin{aligned}\frac{dS}{dt} &= -\alpha SI + \gamma(N - S - I) \\ \frac{dI}{dt} &= I(\alpha S - \beta)\end{aligned}$$

either has 1 or 2 equilibria. The point $(N, 0)$ is always an equilibrium. The point

$$(S^*, I^*) = \left(\frac{\beta}{\alpha}, \frac{\gamma(\alpha N - \beta)}{\alpha(\beta + \gamma)} \right)$$

is a second equilibrium provided $\beta/\alpha < N$. This second equilibrium is asymptotically stable when it is realistic for the model. $(N, 0)$ is asymptotically stable when it is the only sensible equilibrium and unstable otherwise.

Proof: The S -nullcline is the curve

$$I = \frac{\gamma(N - S)}{\gamma + \alpha S}.$$

The I -nullclines are the same as in the SIR Model, $I = 0$ and $S = \beta/\alpha$. For the equilibria, we first consider the case when $I = 0$. Substituting this into the S -nullcline gives $S = N$ which is the first equilibrium. For the second case, we substitute $S = \beta/\alpha$ into the S -nullcline to get

$$I = \frac{\gamma(\alpha N - \beta)}{\alpha(\beta + \gamma)}.$$

If $\beta/\alpha > N$, then the second equilibrium cannot be reached by any realistic solution (as N is the total population size and populations must be

positive). If $\beta/\alpha = N$, the two equilibria are identical. Whenever $(N, 0)$ is the only equilibrium, it will be asymptotically stable (this can be shown by more advanced techniques).

Assume that $\beta/\alpha < N$. It turns out that in this case

$$\frac{\beta}{\alpha} + \frac{\gamma(\alpha N - \beta)}{\alpha(\beta + \gamma)} < N$$

which makes the second equilibrium realistic for our model. When there are two equilibria, $(N, 0)$ will be unstable while the second one will be asymptotically stable. ■

When $\beta/\alpha \geq N$, the phase plane for the SIRS Model generically looks like the following plot.

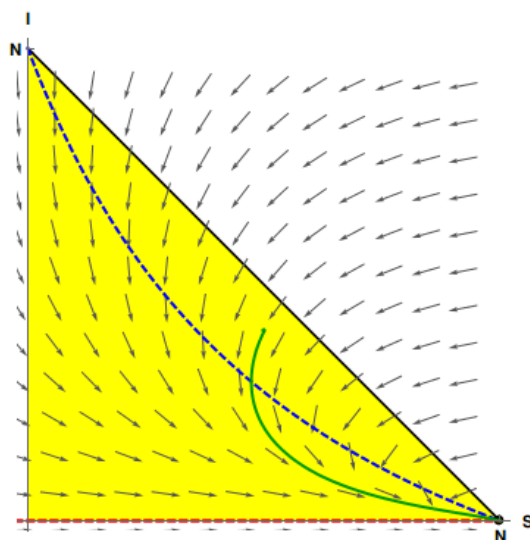
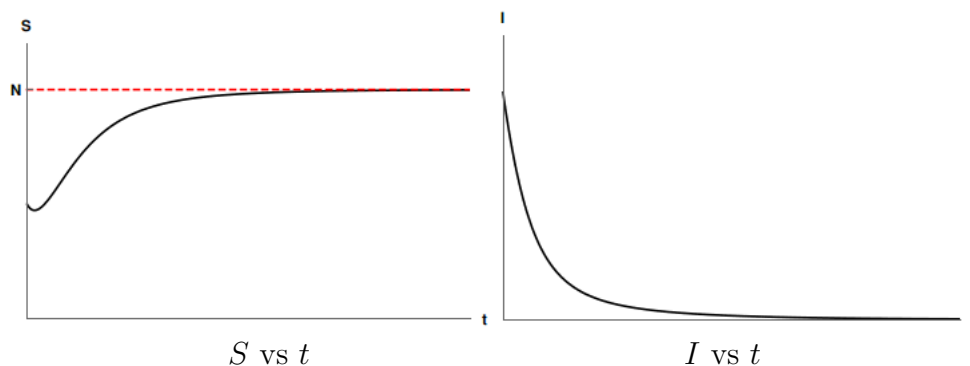


Figure 5.5.3: A Typical Phase Plane for the SIRS Model, $\beta/\alpha \geq N$

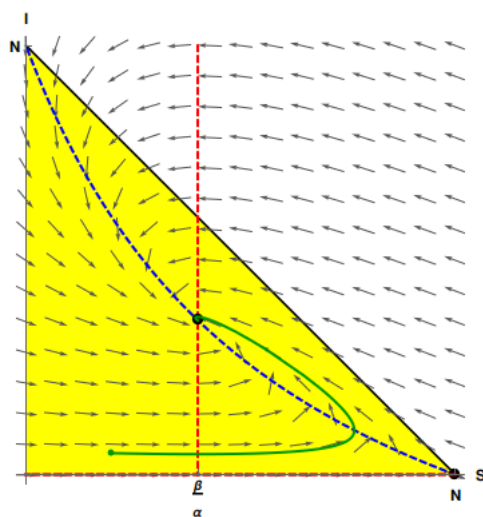
As with the SIR Model, the shaded triangle represents the region in the first quadrant where $S + I \leq N$. Again, note that on the oblique boundary all arrows point towards the interior of the triangle.

As we can see, there is a single equilibrium at $(N, 0)$ and all realistic solution curves are attracted toward it (hence it is asymptotically stable). The parametric curve shown in green above represents the solutions $S = S(t)$ and $I = I(t)$ shown below.

Figure 5.5.4: Typical Solution Curves for the SIRS Model, $\beta/\alpha \geq N$

Notice how the population of infected people fairly rapidly approaches zero. The susceptible population gradually rises to the total population N as the recovered individuals slowly lose their immunity.

For the case $\beta/\alpha < N$, the second equilibrium becomes realistic and asymptotically stable.

Figure 5.5.5: A Typical Phase Plane for the SIRS Model, $\beta/\alpha < N$

Note that $(N, 0)$ has become unstable in this case. Looking once again at the parametric curve in green gives us the following solution curves.

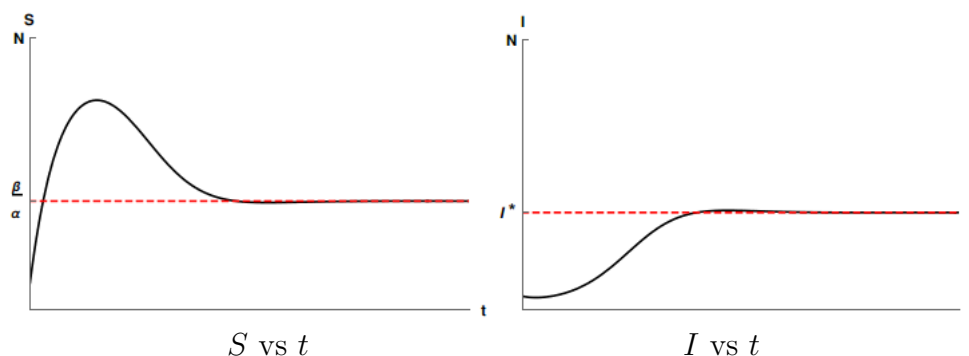


Figure 5.5.6: Typical Solution Curves for the SIRS Model, $\beta/\alpha < N$

In this case, both $S = S(t)$ and $I = I(t)$ oscillate (slightly) around their equilibrium values as they approach these limiting population sizes. This is because the equilibrium they are attracted to is an asymptotically stable spiral point. While the second equilibrium (S^*, I^*) is always asymptotically stable, it is not always a spiral point (it can also be an asymptotically stable node). The exact type of this equilibrium is difficult to classify, but in any event solutions limit to it!

5.5.3 The SEIR Model, Covid-19, and Social Distancing

A slightly more sophisticated model of disease proliferation is the SEIR Model. For this model, there are four types of individuals in the population.

- 1) **S**usceptible people who could contract the illness but have not yet been exposed to it (population given by $S(t)$)
- 2) **E**xposed people who have come into contact with the disease but are not actively ill or contagious (population given by $E(t)$)
- 3) **I**nfectious people who are capable of spreading the illness (population given by $I(t)$)
- 4) **R**ecovered people who are no longer susceptible (populations given by $R(t)$)

As with the other models, the total population is assumed to stay constant, but *equal* birth and death rates are allowed. To make incorporating the birth and death rates easier, we assume the total population size is 1. So, we can naturally think of the populations S, E, I , and R as being *percentages* of the total population in this model. Note that the death rate is the normal rate in the overall population in the absence of the disease. As with the other two models, there are no deaths due to the disease itself (which was very sadly not the case for the Covid-19 outbreak).

The system of differential equations for the SEIR Model have striking similarities to the previous two models.

$$\begin{aligned}\frac{dS}{dt} &= \mu - \alpha SI - \mu S \\ \frac{dE}{dt} &= \alpha SI - (\mu + \gamma)E \\ \frac{dI}{dt} &= \gamma E - (\mu + \beta)I \\ \frac{dR}{dt} &= \beta I - \mu R\end{aligned}$$

The positive parameters in the model are as follows.

- μ : the natural birth and death rate in the population
- α : the rate of transmission of the virus from infected people to susceptible people
- γ : the rate of infection amongst the exposed population
- β : the rate of recovery from the disease

Note that the model assumes that all newborns are susceptible to the disease (if we had wanted a population of size N , the first term in the differential equation for dS/dt should have been μN). Notice that each population has a matching death rate ($-\mu S$, $-\mu E$, $-\mu I$, and $-\mu R$ in each respective equation). Notice that for transmission to occur in the model, a susceptible person must interact with an infected person – not an exposed one. Since the total population size is fixed at 1, we can safely ignore the differential equation for

R and focus on the remaining three.

$$\begin{aligned}\frac{dS}{dt} &= \mu - \alpha SI - \mu S \\ \frac{dE}{dt} &= \alpha SI - (\mu + \gamma)E \\ \frac{dI}{dt} &= \gamma E - (\mu + \beta)I\end{aligned}$$

We can still find equilibria for this system just as we did before (the only difference is that we have three equations to solve instead of two). Unfortunately, our phase plane techniques do not apply since we would need three dimensions to display everything (the correct term here would be *phase space*).

Theorem 5.5.3 (Equilibria for the SEIR Model). *The SEIR Model either has 1 or 2 equilibria. The point $S = 1, E = 0, I = 0$, and $R = 0$ is always an equilibrium regardless of the parameters. The other possible equilibrium is given by the values*

$$\begin{aligned}S^* &= \frac{(\beta + \mu)(\gamma + \mu)}{\alpha\gamma} \\ E^* &= \mu \left(\frac{1}{\gamma + \mu} - \frac{\beta + \mu}{\alpha\gamma} \right) \\ I^* &= \mu \left(\frac{\gamma}{(\beta + \mu)(\gamma + \mu)} - \frac{1}{\alpha} \right) \\ R^* &= \beta \left(\frac{\gamma}{(\beta + \mu)(\gamma + \mu)} - \frac{1}{\alpha} \right)\end{aligned}$$

which is sensible for the model precisely when

$$S^* = \frac{(\beta + \mu)(\gamma + \mu)}{\alpha\gamma} < 1.$$

If the birth/death rate μ is considerably smaller than the other parameters, then

$$\begin{aligned}S^* &\approx \frac{\beta}{\alpha} \\ E^* &\approx 0 \\ I^* &\approx 0 \\ R^* &\approx 1 - \frac{\beta}{\alpha}.\end{aligned}$$

Proof: The algebra to find these equilibria is straightforward (if messy). Showing that the second equilibrium exists when $S^* < 1$ requires some work but is omitted for space. ■

The stability of equilibria in higher dimensions is a delicate subject, and classification becomes far more difficult. It can be shown that if $(S = 1, E = 0, I = 0, R = 0)$ is the only equilibrium for the system, then it is asymptotically stable.⁴ If the second equilibrium (S^*, E^*, I^*, R^*) is present, then $(S = 1, E = 0, I = 0, R = 0)$ becomes an unstable saddle point. The stability of (S^*, E^*, I^*, R^*) is far more difficult to work out, but it can be shown to be asymptotically stable (when it is sensible for the model).

During the Covid-19 outbreak, much attention was paid to the idea that *social distancing* could reduce the maximum number of people who were infected (and likely needing medical attention) at any given time. If too many people require hospitalization over a relatively short period of time, then hospital resources will quickly become overwhelmed, and the overall mortality rate will spike due to the inability of patients to receive adequate care.

While all the parameters in the model are more or less sensitive to the population's response to the disease, social distancing acts to decrease the transmission rate α . In the graph below, we plot only the susceptible and infected populations to keep the graphs from being cluttered. For demonstration purposes only, we take $\mu = 0.001$, $\beta = 0.05$, and $\gamma = 0.3$. The initial population sizes are taken to be $S(0) = 0.999$, $E(0) = 0.001$, $I(0) = 0$, and $R(0) = 0$ (i.e. 0.1% of the population has been exposed to the disease while the remaining 99.9% of the population is susceptible). If we start with a relatively large α ($\alpha = 0.5$ here), we see that there is a large spike in infectious people before the populations approach their equilibrium values (shown as dashed lines), and hospitals are likely to be overwhelmed.

⁴This requires techniques we develop in Section 9.5.

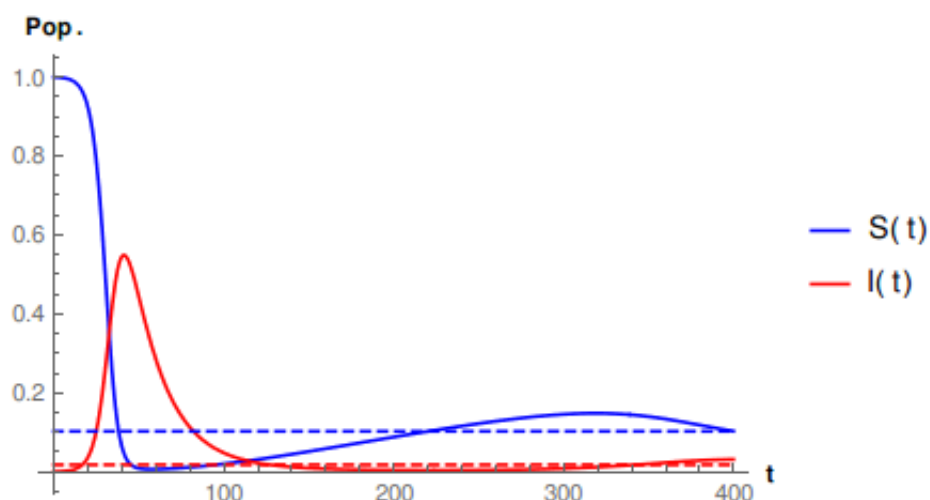


Figure 5.5.7: SEIR Model with Large Transmission Rate

The intended effect of social distancing is a reduction of the transmission rate α . Below is a plot of the susceptible and infected populations with the same model parameters as above except α has been reduced by a factor of 5 (so $\alpha = 0.1$ in this particular model.)

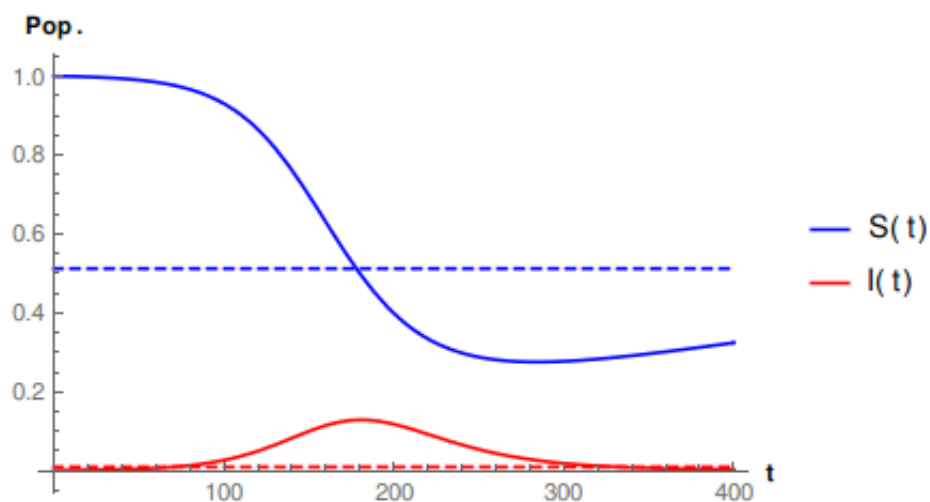


Figure 5.5.8: SEIR Model with Reduced Transmission Rate

Notice that the peak of the infected population has shifted a bit later but is

significantly smaller.

Since we are only looking at numerical solutions, we can also allow the transmission rate to change with time, $\alpha = \alpha(t)$. This allows us to model relaxation of social distancing and its impact on the total number of infected people. For this example, we model the relaxation of social distancing by a function $\alpha(t)$ that increases linearly from $\alpha = 0.1$ to $\alpha = 0.5$ between $t = 180$ and $t = 190$ (the time around the peak infectious cases for the model using $\alpha = 0.1$ for all time).

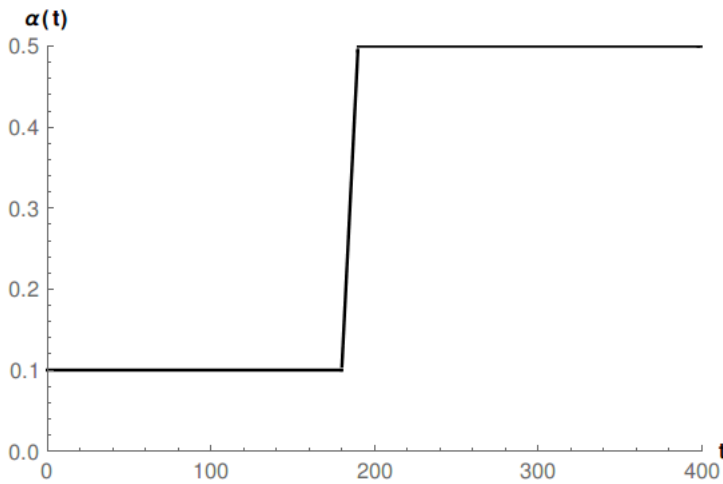


Figure 5.5.9: Variable Transmission Rate for Relaxation of Social Distancing

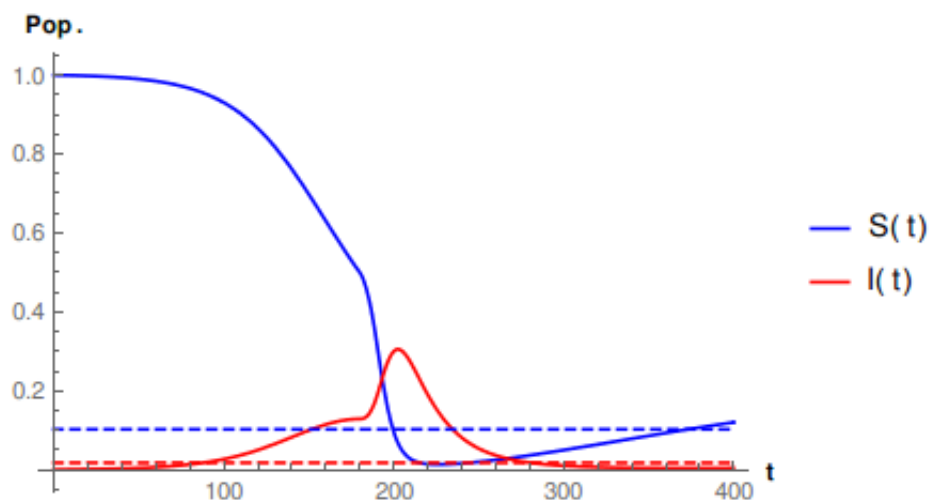


Figure 5.5.10: Early Relaxation of Social Distancing

Notice that while the new peak of the infected population is still smaller than the peak when there was no social distancing at all, relaxing the social distancing guidelines too soon resulted in a tremendous spike in the number of infected persons (more than doubling the number of infections in a relatively short amount of time in our particular model).

On the other hand, if we relax social distancing after the initial peak has passed results are much better. The next graphs show the results of ramping $\alpha = 0.1$ to $\alpha = 0.5$ between $t = 300$ and $t = 310$ (well after the peak of infectious cases when $\alpha = 0.1$ for all time).

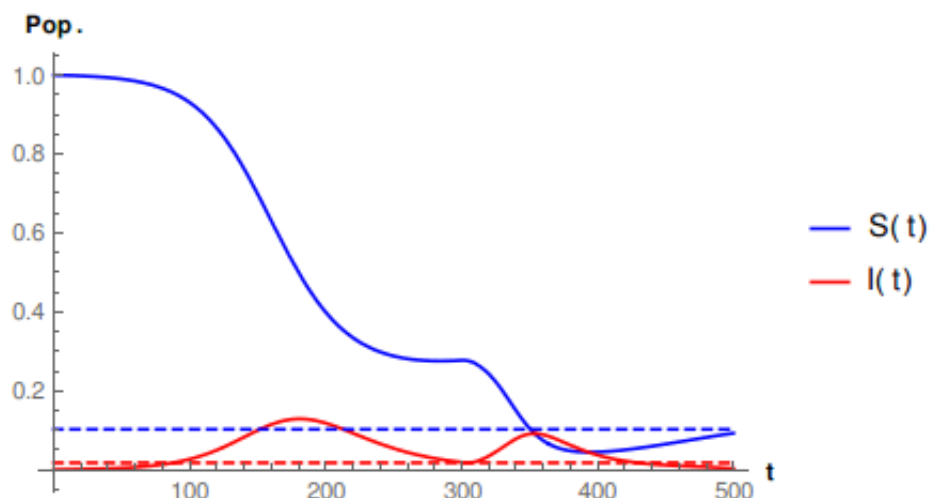


Figure 5.5.11: Later Relaxation of Social Distancing

While there is a second peak in the infected population, both peaks are dramatically smaller than the case of no social distancing measures.

While the numbers used in this particular example are fictitious, they serve to illustrate the main points that epidemiologists stressed during the Covid-19 pandemic.

- 1) Implementing social distancing early in an outbreak can drastically reduce the burden on the healthcare system (and so save lives).
- 2) Relaxing social distancing measures too soon during the outbreak can result in a dramatic surge in the infected population and result in hospitals being overwhelmed with patients (potentially costing lives in the process).

5.6 Problems

For problems 1.) – 5.), find all equilibria for the given autonomous system. Sketch a plot of the nullclines in a phase plane and try to determine whether each equilibrium is asymptotically stable or unstable.

1.)

$$\begin{aligned}\frac{dx}{dt} &= (1-x)(y+1) \\ \frac{dy}{dt} &= x-y\end{aligned}$$

2.)

$$\begin{aligned}\frac{dx}{dt} &= xy - x - y \\ \frac{dy}{dt} &= (x-2)(y-3)\end{aligned}$$

3.)

$$\begin{aligned}\frac{dx}{dt} &= x(1-2y) \\ \frac{dy}{dt} &= -y(3-x)\end{aligned}$$

4.)

$$\begin{aligned}\frac{dx}{dt} &= x\left(1 - \frac{x}{3}\right) - \frac{1}{6}xy \\ \frac{dy}{dt} &= y\left(1 - \frac{y}{4}\right) - \frac{1}{8}xy\end{aligned}$$

5.)

$$\begin{aligned}\frac{dx}{dt} &= -\frac{1}{3}xy \\ \frac{dy}{dt} &= \frac{1}{3}xy - 2y\end{aligned}$$

Solve the following problems related to the standard autonomous systems discussed in the chapter.

- 6.) A population of mice are being preyed upon by a number of owls. In the absence of owls, the population of mice would grow exponentially with a rate $\alpha = 2$. The predation rate is $\beta = 1/15$. If there were no mice,

the owls would die out exponentially fast at a rate $\gamma = 5$. Preying on the mice gives an increase to owl reproduction with a rate $\delta = 1/100$. What is the stable equilibrium point for these two species? If initially there are 600 mice and 20 owls in the region, will each population start out increasing or decreasing?

- 7.) A population of grasshoppers are being preyed upon by a number of frogs. In the absence of frogs, the population of grasshoppers would grow exponentially with a rate $\alpha = 3$. The predation rate is $\beta = 1/20$. If there were no grasshoppers, the frogs would die out exponentially fast at a rate $\gamma = 6$. Preying on the grasshoppers gives an increase to frog reproduction with a rate $\delta = 1/200$. What is the stable equilibrium point for these two species? If initially there are 1300 grasshoppers and 70 frogs in the region, will each population start out increasing or decreasing?
- 8.) Suppose in the SIR Model that the rate of infection is $\alpha = 1/1000$ while the recovery rate is $\beta = 1/2$. What is the largest number of susceptible people that the model predicts is stable?
- 9.) Suppose in the SIRS Model that the rate of infection is $\alpha = 1/1000$, the recovery rate is $\beta = 1/2$, and the rate at which people lose immunity is $\gamma = 1/100$. If the total population is $N = 10,000$, what populations of susceptible and infected persons are asymptotically stable? Is the disease endemic to the population in this case?
- 10.) Suppose in the SIRS Model that the rate of infection is $\alpha = 1/2000$, the recovery rate is $\beta = 2$, and the rate at which people lose immunity is $\gamma = 1/10$. If the total population is $N = 3,000$, what populations of susceptible and infected persons are asymptotically stable? Is the disease endemic to the population in this case?
- 11.) There are two species competing for the same resources. For species a , the logistic growth rate is $k_a = 1/2$ while the carrying capacity is $N_a = 250$. Interspecies competition reduces the rate of reproduction at a rate $\alpha = 1/500$. For species b , the logistic growth rate is $k_b = 1/3$ while the carrying capacity is $N_b = 150$. Interspecies competition reduces the rate of reproduction at a rate $\beta = 1/600$. What will be the size of these populations in the long-term?

- 12.) There are two species competing for the same resources. For species a , the logistic growth rate is $k_a = 1/2$ while the carrying capacity is $N_a = 100$. Interspecies competition reduces the rate of reproduction at a rate $\alpha = 1/500$. For species b , the logistic growth rate is $k_b = 1/3$ while the carrying capacity is $N_b = 150$. Interspecies competition reduces the rate of reproduction at a rate $\beta = 1/600$. What will be the size of these populations in the long-term?
- 13.) There are two species competing for the same resources. For species a , the logistic growth rate is $k_a = 1/2$ while the carrying capacity is $N_a = 100$. Interspecies competition reduces the rate of reproduction at a rate $\alpha = 1/500$. For species b , the logistic growth rate is $k_b = 1/3$ while the carrying capacity is $N_b = 300$. Interspecies competition reduces the rate of reproduction at a rate $\beta = 1/600$. What will be the size of these populations in the long-term?
- 14.) Suppose in the previous problem that the growth rate of species b drops to $k_b = 1/8$. Can you say for certain what will happen to the two populations in the long term?
- 15.) **Challenge:** Consider the following modified predator–prey model.

$$\begin{aligned}\frac{dR}{dt} &= \alpha R \left(1 - \frac{R}{N_R}\right) - \beta RF \\ \frac{dF}{dt} &= -\gamma F + \delta RF\end{aligned}$$

The difference between this model and the traditional Lotka-Volterra Model is that the prey population is assumed to grow *logistically* in the absence of predators. $\alpha > 0$ is now the logistic growth rate and N_R is the carrying capacity for the prey. The other coefficients have the same interpretation as the standard predator–prey model.

Identify all equilibria for the model and determine their stability. HINT: This will depend on the relative sizes of N_R and the quantity γ/δ .

Chapter 6

Discrete Probability

6.1 Introduction and Basic Terminology

Quite often, the final outcome of some process cannot be predicted with complete confidence. We could imagine several reasons why this might be the case, but there are two possibilities which are often cited.

- 1.) The outcome is completely determined once the process is set up, but the system being studied is so complex that exact predictions are next to impossible.
- 2.) The outcome is not completely determined by the initial setup of the process. That is, there is inherent randomness in the process.

There has been a long running discussion in the scientific and philosophical communities about these possibilities and the nature of probability. We bypass all such issues here and focus strictly on the mathematical theory of probability. For an introduction to the various interpretations of probability, see [\[Hj12\]](#).

As always, we begin with some basic terminology. When discussing probability, an **experiment** is any process whose final outcome cannot be predicted with complete confidence in advance. As examples, we could imagine tossing a fair coin, rolling a six-sided die, or flicking a spinner on a game board.

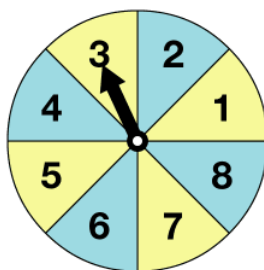


Figure 6.1.1: A Typical Spinner

The set of all possible outcomes for a particular experiment is said to be the **sample space** for the experiment. For instance, there are two possibilities when throwing a coin – heads and tails. So, we can think of the sample space as the set

$$S = \{H, T\}.$$

For the case of a six-sided die, the sample space is the set

$$S = \{1, 2, 3, 4, 5, 6\}.$$

The sample space for the spinner is a little more complicated. Ideally, the outcome of the spinner could be any angle between 0° and 360° . Thus, the sample space would be

$$S = \{\theta | 0^\circ \leq \theta < 360^\circ\} = [0^\circ, 360^\circ).$$

This last example of a sample space brings up an important distinction. An experiment is said to be **discrete** if it has a finite number of outcomes or if its outcomes can be labeled by the positive integers 1, 2, 3, 4, ... If an experiment has an entire range of real numbers for its sample space, it is said to be **continuous**.

An **event** for an experiment is simply a subset of its sample space. In symbols, an event is a set of outcomes E which satisfies $E \subset S$. If our experiment is rolling a six-sided die, then an event might be rolling an even number.

$$E = \{2, 4, 6\} \subset S = \{1, 2, 3, 4, 5, 6\}$$

Often in probability, we describe events in words rather than giving an explicit list of outcomes. Consider the spinner ($S = [0^\circ, 360^\circ)$). We might

want to consider the event E where the spinner lands in the first quadrant. Translating this into set notation gives

$$E = (0^\circ, 90^\circ) \subset S = [0^\circ, 360^\circ).$$

Since sample spaces and events are sets, we can apply basic set operations to them. The most important such operations are listed below. Let S be the sample space for some experiment, and let E_1 and E_2 be two events in S .

- 1.) **Intersection:** The event “ E_1 AND E_2 ” is the event that is the intersection of the two separate events:

$$E_1 \cap E_2 = E_1 E_2 = \{x | x \in E_1 \text{ and } x \in E_2\}.$$

(NOTE: Writing the intersection of two sets as $E_1 E_2$ is a peculiar habit in probability, but it is common enough to be aware of.)

- 2.) **Union:** The event “ E_1 OR E_2 ” is the event that is the union of the two separate events:

$$E_1 \cup E_2 = \{x | x \in E_1 \text{ or } x \in E_2\}.$$

- 3.) **Complement:** The event “NOT E ” is the event that is the set complement of E (relative to S):

$$E^c = \{x | x \in S \text{ and } x \notin E\}.$$

Two events are said to be **mutually exclusive** if

$$E_1 \cap E_2 = \emptyset$$

where $\emptyset$ is the *empty set* (i.e. the set containing no elements). In other words, two events are mutually exclusive if they share no possible outcomes in common.

A **probability** on a sample space S is a function *on the events of S* which returns a number between zero and one. Specifically, a probability must have the following properties.

- For any event $E \subset S$,

$$0 \leq \mathbb{P}(E) \leq 1,$$

- $\mathbb{P}(S) = 1$,
- If E_1 and E_2 are mutually exclusive events ($E_1 \cap E_2 = \emptyset$),

$$\mathbb{P}(E_1 \cup E_2) = \mathbb{P}(E_1) + \mathbb{P}(E_2).$$

A few comments are in order. We already mentioned that a probability is a real number between zero and one (which is the content of the first property above). The second property, $\mathbb{P}(S) = 1$, simply reinforces the requirement that the sample space S contains all of the possible outcomes.

The third property above is the most critical one. It asserts that the probability of non-overlapping events is just the sum of the individual probabilities. In fact, this property instantly scales up to any finite number of mutually exclusive events. That is, if $E_1, E_2, \dots, E_N$ are any finite number of mutually exclusive events (so $E_i \cap E_j = \emptyset$ for any pair of distinct indices i and j), then

$$\mathbb{P}(E_1 \cup E_2 \cup \dots \cup E_N) = \mathbb{P}(E_1) + \mathbb{P}(E_2) + \dots + \mathbb{P}(E_N).$$

In more advanced treatments of probability, it is important to extend this property to *countably infinite* mutually exclusive events:

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(E_i)$$

provided $E_i \cap E_j = \emptyset$ for any pair of distinct indices i and j . Since we will never deal with countably many events at one time, this more general version will not be terribly important for our purposes.

There are a number of important consequences of this definition. We list the two most crucial ones below.

Lemma 6.1.1. *Let $\mathbb{P}$ be a probability on the events in a sample space S .*

- *If E is any event in S , we have*

$$\mathbb{P}(E^c) = 1 - \mathbb{P}(E).$$

- *For any two events E and F in S ,*

$$\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F).$$

Proof: For the first equality, we note that E and E^c are mutually exclusive and $E \cup E^c = S$ by definition. Invoking the second and third properties of probabilities, we have

$$\begin{aligned}\mathbb{P}(S) &= \mathbb{P}(E \cup E^c) \\ 1 &= \mathbb{P}(E) + \mathbb{P}(E^c).\end{aligned}$$

Rearranging this equality gives the result.

The second equality is intuitively obvious from the following Venn diagram.

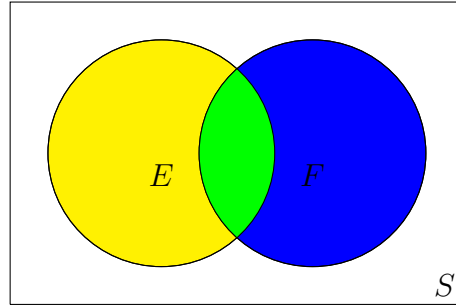


Figure 6.1.2: Venn Diagram of Two Events E and F

Here, the probability of an event is represented by its area in the diagram (and so the total area of the rectangle representing S is equal to 1). The event $E \cup F$ is the totality of all three shaded regions shown. If we add the probabilities of E and F together, then the green region (which is $E \cap F$) is counted twice. Hence,

$$\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F). \blacksquare$$

Finally, a **random variable** is any function defined on the sample space S . As an example, consider the experiment where we roll two, fair, six-sided dice. The sample space S is all possible pairs of rolls:

$$S = \{(i, j) | i, j = 1, 2, 3, 4, 5, 6\} = \{(1, 1); (1, 2); (1, 3); \cdots (6, 4); (6, 5); (6, 6)\}.$$

The random variable most often considered in such a case is the *sum of the two dice*: $R(i, j) = i + j$. It turns out that we can think of the random variable R as a new experiment whose sample space is

$$S' = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\},$$

and the probabilities of the outcomes for R can be computed knowing the probabilities for the original experiment (rolling two dice).

We will often use the terms “random variable” and “experiment” interchangeably in what follows. Given any experiment, we can define the random variable which is just the identity function on the sample space! In other words, the concept of random variable generalizes the notion of experiment.

6.2 Discrete Random Variables

As mentioned above, a **discrete random variable** is any experiment where the outcomes can be labeled by a subset of the natural numbers. This means the sample space looks like

$$S = \{d_1, d_2, d_3, \dots, d_N\}$$

if the sample space is finite. For a discrete random variable, we only need to specify the probabilities of the individual outcomes d_i .

Lemma 6.2.1. *A probability on a finite sample space*

$$S = \{d_1, d_2, d_3, \dots, d_N\}$$

is a specification of probabilities for the individual outcomes

$$\mathbb{P}(\{d_i\}) = p_i$$

where the probabilities p_i satisfy

$$1.) \ 0 \leq p_i \leq 1,$$

$$2.) \ \sum_{i=1}^N p_i = 1.$$

Proof: Since $\{d_i\}$ and $\{d_j\}$ are disjoint subsets of S as long as $i \neq j$, the probability of any event E in S is simply the sum of the probabilities of the outcomes in E . The axiom that the probability of S must equal 1 forces the sum of all the individual probabilities to be 1. This shows that a probability on a discrete sample space must have the properties above. Reversing these arguments shows that the two properties above give us a probability on a discrete space. ■

Example 6.2.1 (A Fair Coin). A fair coin is the random variable with sample space $S = \{H, T\}$ and probabilities

$$\mathbb{P}(\{H\}) = \mathbb{P}(\{T\}) = \frac{1}{2}. \blacksquare$$

Example 6.2.2 (A Biased Coin). A biased coin is any random variable with sample space $S = \{H, T\}$ and probabilities

$$\begin{aligned}\mathbb{P}(\{H\}) &= p \\ \mathbb{P}(\{T\}) &= 1 - p\end{aligned}$$

where $0 \leq p \leq 1$. Notice that the biased coin contains the fair coin as the special case $p = 1/2$. $\blacksquare$

Example 6.2.3 (A Fair Six-Sided Die). A fair six-sided die is the random variable with sample space $S = \{1, 2, 3, 4, 5, 6\}$ and probabilities

$$\mathbb{P}(\{1\}) = \mathbb{P}(\{2\}) = \cdots = \mathbb{P}(\{6\}) = \frac{1}{6}.$$

The term “fair” usually indicates that all possible outcomes are equally likely while “biased” indicates that some outcomes are more likely than others.

Example 6.2.4 (A Biased Six-Sided Die). Suppose we have a six-sided die where the outcomes 2, 3, 4, and 5 all have probability $1/10$ and the probabilities of 1 and 6 are equal. What are these two probabilities?

Answer: We know that the sum of the probabilities must equal 1.

$$1 = \mathbb{P}(\{1\}) + \mathbb{P}(\{2\}) + \mathbb{P}(\{3\}) + \mathbb{P}(\{4\}) + \mathbb{P}(\{5\}) + \mathbb{P}(\{6\})$$

$$1 = \mathbb{P}(\{1\}) + \frac{1}{10} + \frac{1}{10} + \frac{1}{10} + \frac{1}{10} + \mathbb{P}(\{6\})$$

$$1 = \mathbb{P}(\{1\}) + \mathbb{P}(\{6\}) + \frac{2}{5}$$

$$\frac{3}{5} = \mathbb{P}(\{1\}) + \mathbb{P}(\{6\}).$$

Using $\mathbb{P}(\{1\}) = \mathbb{P}(\{6\})$, we have

$$\frac{3}{5} = 2\mathbb{P}(\{1\})$$

$$\mathbb{P}(\{1\}) = \frac{3}{10}.$$

This gives

$$\mathbb{P}(\{1\}) = \mathbb{P}(\{6\}) = \frac{3}{10}. \blacksquare$$

Example 6.2.5 (Rolling Two Fair Six-Sided Dice). If we roll two fair six-sided dice, the sample space consists of 36 pairs of outcomes

$$S = \{(i, j) | i, j = 1, 2, 3, 4, 5, 6\} = \{(1, 1); (1, 2); (1, 3); \cdots (6, 4); (6, 5); (6, 6)\}.$$

Since the dice are fair, each pair is equally likely:

$$\mathbb{P}(\{(i, j)\}) = \frac{1}{36}.$$

We will have more to say about this and similar situations in later sections.

Suppose now that we want to consider the random variable $R = R(i, j) = i + j$ which is the sum of the outcomes for the two dice. What are the possible outcomes for R and their respective probabilities?

Answer: The sum of two dice can be any integer between 2 and 12. This means that the sample space for R is the set

$$S = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}.$$

In order to work out the probabilities for R , we only have to count the outcomes from the original sample space which give the particular value of

R we are considering.

$$\begin{aligned}
\mathbb{P}(R = 2) &= \mathbb{P}(\{(1, 1)\}) = \frac{1}{36}, \\
\mathbb{P}(R = 3) &= \mathbb{P}(\{(1, 2); (2, 1)\}) = \frac{2}{36} = \frac{1}{18}, \\
\mathbb{P}(R = 4) &= \mathbb{P}(\{(1, 3); (2, 2); (3, 1)\}) = \frac{3}{36} = \frac{1}{12}, \\
\mathbb{P}(R = 5) &= \mathbb{P}(\{(1, 4); (2, 3); (3, 2); (4, 1)\}) = \frac{4}{36} = \frac{1}{9}, \\
\mathbb{P}(R = 6) &= \mathbb{P}(\{(1, 5); (2, 4); (3, 3); (4, 2); (5, 1)\}) = \frac{5}{36}, \\
\mathbb{P}(R = 7) &= \mathbb{P}(\{(1, 6); (2, 5); (3, 4); (4, 3); (5, 2); (6, 1)\}) = \frac{6}{36} = \frac{1}{6}, \\
\mathbb{P}(R = 8) &= \mathbb{P}(\{(2, 6); (3, 5); (4, 4); (5, 3); (6, 2)\}) = \frac{5}{36}, \\
\mathbb{P}(R = 9) &= \mathbb{P}(\{(3, 6); (4, 5); (5, 4); (6, 3)\}) = \frac{4}{36} = \frac{1}{9}, \\
\mathbb{P}(R = 10) &= \mathbb{P}(\{(4, 6); (5, 5); (6, 4)\}) = \frac{3}{36} = \frac{1}{12}, \\
\mathbb{P}(R = 11) &= \mathbb{P}(\{(5, 6); (6, 5)\}) = \frac{2}{36} = \frac{1}{18}, \\
\mathbb{P}(R = 12) &= \mathbb{P}(\{(6, 6)\}) = \frac{1}{36}. \blacksquare
\end{aligned}$$

This last example shows a couple of notational peculiarities common in the theory of probability. First, probabilists rarely display the function arguments for random variables. We see this in the example above where we wrote R instead of $R(i, j)$ for the sum of the two dice. The second is that we often leave off the set braces when describing an event. Above, we used the short-hand $R = 3$ to stand for the event

$$\{(i, j) | R(i, j) = 3\} = \{(1, 2); (2, 1)\}.$$

Typically, these abuses of notation do not cause much confusion. In this text, we will typically reserve set braces for listing individual outcomes.

6.3 Independent Events

One of the most important concepts in probability is the idea of **independence**. Two events E and F from a sample space S are said to be independent if

$$\mathbb{P}(E \cap F) = \mathbb{P}(E) \cdot \mathbb{P}(F).$$

The definition extends to any number of events $E_1, E_2, \dots, E_N$ by requiring the probability of *any intersection* of these events to be the product of the probabilities of the events involved in the intersection.

Example 6.3.1. Consider the experiment where we roll two fair six-sided dice. We will consider three random variables related to this experiment:

- 1.) $D_1 = D_1(i, j) = i = \text{outcome of first die,}$
- 2.) $D_2 = D_2(i, j) = j = \text{outcome of second die,}$
- 3.) $R = R(i, j) = i + j = \text{sum of the two dice.}$

Determine whether the following events are independent.

- a.) $D_1 = 4$ and $D_2 = 3$
- b.) $R = 7$ and $D_1 = 3$
- c.) $R = 6$ and $D_2 = 4$

Answer: To determine whether the events are independent, we compare the probability of the intersection of the two events with the product of their individual probabilities.

- a.) The event $D_1 = 4$ consists of the following separate outcomes:

$$\{D_1 = 4\} = \{(4, 1); (4, 2); (4, 3); (4, 4); (4, 5); (4, 6)\}.$$

So, the probability of this event is $\mathbb{P}(D_1 = 4) = 1/6$. Similarly,

$$\{D_2 = 3\} = \{(1, 3); (2, 3); (3, 3); (4, 3); (5, 3); (6, 3)\}$$

giving a probability $\mathbb{P}(D_2 = 3) = 1/6$.

Next, the intersection of these two events is

$$\{D_1 = 4\} \cap \{D_2 = 3\} = \{(4, 3)\},$$

and so the probability of the intersection is

$$\mathbb{P}(\{D_1 = 4\} \cap \{D_2 = 3\}) = \frac{1}{36} = \mathbb{P}(D_1 = 4) \cdot \mathbb{P}(D_2 = 3).$$

That means these two events are independent.

Note there is nothing special about the choices for these two random variables. The events $D_1 = i$ and $D_2 = j$ are independent of one another for any choice of i and j . This means that the outcome of the first roll is independent of the outcome of the second roll, and vice versa.

- b.) We computed the probability that $R = 7$ in Example 6.2.5 above:
 $\mathbb{P}(R = 7) = 1/6$. As in the last problem

$$\{D_1 = 3\} = \{(3, 1); (3, 2); (3, 3); (3, 4); (3, 5); (3, 6)\}$$

giving $\mathbb{P}(D_1 = 3) = 1/6$. Notice that intersection of these two events is

$$\{R = 7\} \cap \{D_1 = 3\} = \{(3, 4)\},$$

and so the probability of the intersection is

$$\mathbb{P}(\{R = 7\} \cap \{D_1 = 3\}) = \frac{1}{36} = \mathbb{P}(R = 7) \cdot \mathbb{P}(D_1 = 3).$$

So, these two events are independent.

As above, there is nothing special about the event $D_1 = 3$. In fact, the event $R = 7$ is independent of any events of the form $D_1 = i$ or $D_2 = j$!

- c.) The probability that $R = 6$ was computed to be $\mathbb{P}(R = 6) = 5/36$. For $D_2 = 4$, we have $\mathbb{P}(D_2 = 4) = 1/6$. For the intersection, we have

$$\mathbb{P}(\{R = 6\} \cap \{D_2 = 4\}) = \mathbb{P}(\{(2, 4)\}) = \frac{1}{36} \neq \mathbb{P}(R = 6) \cdot \mathbb{P}(D_2 = 4) = \frac{5}{216}.$$

So, these two events are not independent. ■

Instead of directly checking whether events are independent, we often assume independence to make the computation of certain probabilities easier. The idea is that if an experiment is comprised of a number of independent sub-experiments, then we simply have to multiply the probabilities of the component events to get the probability of the overall outcome. The following example makes that clear.

Example 6.3.2. Suppose we independently toss 3 fair coins. Determine the sample space and the probabilities of all possible outcomes.

Answer: We can list each possible outcome as a string consisting of H 's and T 's of length 3.

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

Let C_1, C_2 , and C_3 be the outcomes of coins 1, 2, or 3 respectively. To compute probabilities, we use the assumption that the coin tosses are independent. For instance,

$$\begin{aligned}\mathbb{P}(\{HHH\}) &= \mathbb{P}(C_1 = H \text{ and } C_2 = H \text{ and } C_3 = H) \\ &= \mathbb{P}(C_1 = H) \cdot \mathbb{P}(C_2 = H) \cdot \mathbb{P}(C_3 = H) \\ &= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.\end{aligned}$$

Since the coins are fair, the computation will be the same for any of the 8 outcomes.

$$\mathbb{P}(\{HHH\}) = \mathbb{P}(\{HHT\}) = \cdots = \mathbb{P}(\{TTT\}) = \frac{1}{8}. \blacksquare$$

This method is exactly how we came by the probabilities for rolling two fair six-sided dice. We can also compute the probabilities of fairly complicated experiments so long as they are composed of independent parts. Consider the next simple example.

Example 6.3.3. Suppose we perform an experiment where we toss a fair coin and independently roll a fair six-sided die. List all possible outcomes and find their probabilities.

Answer: We can list the outcomes as an H or a T followed by a number between 1 and 6.

$$S = \{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}.$$

Let C be the outcome of the coin toss and D be the outcome of the die roll. Using the assumption of independence, we find

$$\begin{aligned}\mathbb{P}(\{H1\}) &= \mathbb{P}(C = H \text{ and } D = 1) \\ &= \mathbb{P}(C = H) \cdot \mathbb{P}(D = 1) \\ &= \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}.\end{aligned}$$

Since both outcomes of the coin toss have probability $1/2$ and all six outcomes of the die roll have probability $1/6$, each outcome of this experiment has probability $1/12$. ■

As a special note, be careful not to confuse the notions of independent events and mutually exclusive events. If two events E and F are mutually exclusive, then $\mathbb{P}(E \cap F) = 0$ since $E \cap F$ is the empty set. If the events were also independent, we would need

$$0 = \mathbb{P}(E \cap F) = \mathbb{P}(E) \cdot \mathbb{P}(F).$$

This can only happen if $\mathbb{P}(E) = 0$ or $\mathbb{P}(F) = 0$. So, two mutually exclusive events can only be independent if at least one of them has probability zero! Generically speaking, mutually exclusive events are not independent, and vice versa.

6.4 Basic Combinatorics and the Computation of Probabilities

Quite often, the computation of probabilities for discrete random variables comes down to counting the number of outcomes matching some property.

Example 6.4.1. We (independently) toss three fair coins. What is the probability that we have at least two heads in our outcome?

Answer: We listed the sample space earlier:

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

Let E be the event that there are two or more heads in the outcome. Looking through the outcomes, we see

$$E = \{HHH, HHT, HTH, THH\}.$$

Since each outcome has probability $1/8$, we find

$$\mathbb{P}(E) = \frac{4}{8} = \frac{1}{2}. \blacksquare$$

In the last example, we simply listed all possible outcomes and determined which were in our particular event. But what if there had been 10 independent coin tosses? How many outcomes would there be? How many would have 2 or more heads? Once the number of outcomes becomes rather large, listing all of them becomes impractical. Fortunately, there are mathematical techniques which can help us count outcomes matching some pattern. The branch of mathematics that studies counting techniques is called *combinatorics*.

Theorem 6.4.1 (The Fundamental Principle of Counting). *Suppose that a task consists of selecting one element from each of N sets: $S_1, S_2, \dots, S_N$. Let $|S_i|$ represent the number of elements in the set S_i . Then, the number of ways of accomplishing the task is given by the product of the number of elements in each set:*

$$|S_1| \cdot |S_2| \cdots |S_N|.$$

Proof: If the task consists of picking an element from a single set S , then clearly there are $|S|$ ways to accomplish the task. If the task consists of picking one element from S_1 and one element from S_2 , then after we make a choice from S_1 there are exactly $|S_2|$ choices we can make. Since there were $|S_1|$ initial possible choices, the total number of choices must be

$$|S_1| \cdot |S_2|.$$

The general form above follows by induction on the number of sets. $\blacksquare$

Example 6.4.2. How many outcomes are possible if we toss 10 coins?

Answer: We can think of this as a task where we have to pick one item from 10 sets:

$$S_1 = \{H, T\}, S_2 = \{H, T\}, \dots, S_{10} = \{H, T\}.$$

Since each set has 2 elements, we have

$$\underbrace{2 \cdot 2 \cdots 2}_{10 \text{ times}} = 2^{10} = 1024. \blacksquare$$

Example 6.4.3. How many outcomes are possible if we roll 3 six-sided dice?

Answer: We have to pick an element from 3 sets

$$S_1 = S_2 = S_3 = \{1, 2, 3, 4, 5, 6\}.$$

Since each set has 6 elements, we have $6 \cdot 6 \cdot 6 = 6^3 = 216$ possible outcomes.

■

Lemma 6.4.2 (Permutations). *Suppose a task consists of choosing k objects (without replacement) in a particular order from a set of N distinguishable objects ($N \geq k$). Then the number of ways to accomplish the task is given by*

$$P_k^N = \frac{N!}{(N-k)!} = N(N-1)(N-2) \cdots (N-k+1).$$

Proof: We have N ways to make the first choice. Once we remove the first item, there are $N-1$ one items remaining. Thus, there are $N(N-1)$ ways to make the first two choices. Continuing in this manner proves the general formula above. ■

Example 6.4.4. A committee consists of 20 people. From these 20 people, a chairperson, secretary, and treasurer must be selected. How many different slates of officers are possible?

Answer: We have to select three people (in order) from a group of 20. This means there are

$$P_3^{20} = 20 \cdot 19 \cdot 18 = 6840 \text{ slates of officers. } \blacksquare$$

Lemma 6.4.3 (Combinations). *Suppose a task consists of choosing k objects (without replacement) from a set of N distinguishable objects without regard to order. Then the number of ways to accomplish the task is given by*

$$\binom{N}{k} = \frac{N!}{k!(N-k)!}.$$

Proof: If we choose the k objects in particular order, then there are

$$P_k^N = \frac{N!}{(N-k)!}$$

possible outcomes. This over counts the number of actual outcomes since we want do not care about different orderings of the same set of k objects. Since there are $P_k^k = k!$ different orderings of k objects, we have

$$\binom{N}{k} = \frac{N!}{k!(N-k)!} \text{ possible outcomes. } \blacksquare$$

The coefficients $\binom{N}{k}$ are known as *binomial coefficients* and are often read as “ N choose k .”

Example 6.4.5. A committee consists of 20 people. If a subcommittee of 6 people needs to be formed, how many different subcommittees are possible?

Answer: We need to select 6 people out of the original 20, but the order in which they are selected is unimportant. So, the number of subcommittees is

$$\binom{20}{6} = \frac{20!}{6!14!} = \frac{20 \cdot 19 \cdot 18 \cdot 17 \cdot 16 \cdot 15}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 38,760. \blacksquare$$

Example 6.4.6. A standard deck consists of 52 distinguishable cards. A poker hand consists of 5 cards in no particular order. How many different poker hands are possible?

Answer: We need to choose 5 cards from the deck of 52. This means there are

$$\binom{52}{5} = \frac{52!}{5!47!} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 2,598,960$$

different hands of poker. $\blacksquare$

Quite often, we have to combine these two methods of counting.

Example 6.4.7. A committee consists of 20 people. A subcommittee consists of a chairperson, a secretary, and 4 regular members. How many different subcommittees are possible?

Answer: The task of selecting a subcommittee consists of two parts. First, we have to pick a chairperson and a secretary for the subcommittee. Since there are 20 people to choose from, the number of way of choosing the officers for the subcommittee is

$$P_2^{20} = 20 \cdot 19 = 380.$$

After selecting these two positions, we have to choose the 4 regular members from the remaining 18 people. This means there are

$$\binom{18}{4} = \frac{18!}{4!14!} = 3,060$$

ways to choose the remaining members of the subcommittee. Combining this with the number of ways of selecting the officers gives

$$380 \cdot 3,060 = 1,162,800$$

possible subcommittees.

Notice that this answer is independent of how we order the two different tasks. If we choose the 4 regular members first, then there are

$$\binom{20}{4} = \frac{20!}{4!16!} = 4,845$$

ways to select this component of the subcommittee. From the remaining 16 people, we have to choose a chairperson and a secretary:

$$P_2^{16} = 16 \cdot 15 = 240.$$

Combining these two gives

$$4,845 \cdot 240 = 1,162,800. \blacksquare$$

Example 6.4.8. Suppose we flip a coin 10 times. Of the total number of outcomes, how many have exactly 2 heads?

Answer: We can label the different tosses with the numbers 1 through 10. If we think of these different tosses as “slots,” our job is to pick the two slots where the heads will go. The remaining slots must necessarily be tails. Since the order we choose the slots in is unimportant, we have

$$\binom{10}{2} = \frac{10!}{2!8!} = \frac{10 \cdot 9}{2} = 45 \text{ tosses with exactly 2 heads. } \blacksquare$$

These basic counting tools allow us to answer some interesting probability questions.

Example 6.4.9. Suppose we (independently) flip a fair coin 10 times. What is the probability of getting exactly 3 heads?

Answer: We know there are $2^{10} = 1024$ possible outcomes. Since the flips are independent, we know that

$$\mathbb{P}(\{\text{any single outcome}\}) = \frac{1}{1024}.$$

We only need to count the number of outcomes with exactly 3 heads. Repeating our analysis from above, we find

$$\binom{10}{3} = \frac{10!}{3!7!} = 120$$

possible outcomes with exactly 3 heads. Since each has the same probability, we find

$$\mathbb{P}(\{\text{exactly 3 heads}\}) = \frac{120}{1024} = \frac{15}{128}. \blacksquare$$

Example 6.4.10. We (independently) flip a fair coin 6 times. What is the probability that more than half of the outcomes are heads?

Answer: There are $2^6 = 64$ possible outcomes. We want to know how many outcomes have 4, 5, or 6 heads. Notice that these are mutually exclusive outcomes. So, we need to count the number of outcomes for each type separately and sum them up to get the total.

$$\binom{6}{4} = 15 \text{ outcomes with 4 heads}$$

$$\binom{6}{5} = 6 \text{ outcomes with 5 heads}$$

$$\binom{6}{6} = 1 \text{ outcome with 6 heads}$$

$$15 + 6 + 1 = 22 \text{ outcomes with 4, 5, or 6 heads}$$

So, the probability of having more than half of the flips turn up heads is

$$\mathbb{P}(\{\text{more than 3 heads}\}) = \frac{22}{64} = \frac{11}{32}. \blacksquare$$

Example 6.4.11. Suppose we have an urn containing 3 red balls, 3 green balls, and 4 yellow balls, and we randomly select 3 balls from the urn (without replacement). What is the probability that exactly two of the balls are yellow?

Answer: Since we choose three balls at random from the urn, the number of possible outcomes from the experiment is given by

$$\binom{10}{3} = 120.$$

Any of these outcomes is equally likely, so the probability of any particular one is $1/120$.

We need to count the number of outcomes that have exactly 2 yellow balls. So, we need to choose 2 from the 4 yellow balls and then choose any one of the 6 non-yellow balls. The number of ways of doing this is

$$\binom{4}{2} \cdot 6 = 6 \cdot 6 = 36.$$

So the probability of having exactly 2 yellow balls is

$$\mathbb{P}(\{\text{exactly 2 yellows}\}) = \frac{36}{120} = \frac{3}{10}.$$

Alternatively, we could compute the probability by pretending the balls are distinguishable and that we take the balls out in some order. We just have to be sure to count all possible outcomes that correspond to having 2 yellow balls. Let us pretend the balls are numbered 1 through 10. Balls 1, 2, and 3 are red; balls 4, 5, and 6 are green; balls 7 – 10 are yellow. Since we are pulling three balls out of the urn in order, the number of outcomes is $P_3^{10} = 10 \cdot 9 \cdot 8 = 720$. Each outcome is equally likely, and so the probability of an particular outcome is $1/720$.

Since we want to have exactly two yellows, there are three possible types of outcomes we need to count:

$$\begin{array}{ccc} \underline{Y} & \underline{Y} & \underline{\text{not } Y} \\ \underline{Y} & \underline{\text{not } Y} & \underline{Y} \\ \underline{\text{not } Y} & \underline{Y} & \underline{Y} \end{array}$$

The slots labeled “Y” (for yellow) have to be chosen from balls 7 – 10 while the slot labeled “not Y” (for not yellow) has to be chosen from balls 1 – 6. This leads to the following counts:

$$\begin{array}{l} \underline{Y} \quad \underline{Y} \quad \underline{\text{not } Y} = 4 \cdot 3 \cdot 6 = 72, \\ \underline{Y} \quad \underline{\text{not } Y} \quad \underline{Y} = 4 \cdot 6 \cdot 3 = 72, \\ \underline{\text{not } Y} \quad \underline{Y} \quad \underline{Y} = 6 \cdot 4 \cdot 3 = 72. \end{array}$$

Since these individual types are mutually exclusive, there are $72 + 72 + 72 = 216$ outcomes that have exactly two yellow balls. Computing the probability gives

$$\mathbb{P}(\{\text{exactly 2 yellows}\}) = \frac{216}{720} = \frac{3}{10}. \blacksquare$$

Example 6.4.12. Consider the urn from the previous problem (containing 3 red, 3 green, and 4 yellow balls). If we now select 4 balls from the urn (without replacement), what is the probability that there are 2 or more yellow balls in our selection?

Answer: There are

$$\binom{10}{4} = 210$$

different outcomes for this experiment. Hence, the probability of any particular outcome is $1/210$.

For this particular problem, it is easier to find the probability of the opposite event:

$$\begin{aligned} \mathbb{P}(\{\text{at least two yellow balls}\}) \\ = 1 - \mathbb{P}(\{0 \text{ yellow balls or } 1 \text{ yellow ball}\}). \end{aligned}$$

Notice that

$$\begin{aligned} \mathbb{P}(\{0 \text{ yellow balls or } 1 \text{ yellow ball}\}) \\ = \mathbb{P}(\{0 \text{ yellow balls}\}) + \mathbb{P}(\{1 \text{ yellow ball}\}) \end{aligned}$$

since these two events are mutually exclusive.

If there are no yellow balls in our group of 4, then we selected 4 balls from the 6 non-yellow balls:

$$\binom{6}{4} = 15 \text{ outcomes with no yellow balls.}$$

This gives

$$\mathbb{P}(\{0 \text{ yellow balls}\}) = \frac{15}{210} = \frac{1}{14}.$$

If there is exactly 1 yellow ball, then we have 4 ways to pick a yellow ball and the remaining 3 balls must be selected from the non-yellow ones. This means that there are

$$4 \cdot \binom{6}{3} = 4 \cdot 20 = 80 \text{ outcomes with exactly 1 yellow ball.}$$

So,

$$\mathbb{P}(\{1 \text{ yellow ball}\}) = \frac{80}{210} = \frac{8}{21}.$$

Putting these two together gives

$$\begin{aligned} \mathbb{P}(\{\text{at least two yellow balls}\}) &= 1 - \frac{1}{14} - \frac{8}{21} \\ &= \frac{23}{42}. \quad \blacksquare \end{aligned}$$

Example 6.4.13 (The Birthday Problem). Out of a group of N people ($N \geq 2$), what is the probability that at least two people share a common birthday? How big should N be so that there is at least a 50% chance that two or more people share a common birthday?

Answer: To make things simple, we will assume there are 365 different birthdays (so we ignore leap years). We will also assume that all birthdays are equally likely and independent. Both of these assumptions are suspect! It is well known that birth rates vary somewhat over the year, and if there are twins in the group, then birthdays are certainly not independent! However, none of these issues make a significant difference to the outcome.

First note that $N \geq 2$ for the problem to make any sense at all. The other obvious fact is that once $N \geq 366$ there *must be* a pair of people sharing the same birthday (since there are only 365 different possible dates to choose from). So, we assume $2 \leq N \leq 365$ in what follows.

Instead of arguing directly, we first find the probability that no two people share a birthday. To compute this probability, we imagine that we line up the people in the group in some order (any order will do) and ask for their birthdays one at a time. Each time a person tells us their birthday, we place a mark on that date in a calendar. The first person can have any particular birthday at all which means there are 365 possible choices. If no two people are going to share a birthday, then the second person's birthday can be any one of the remaining $365 - 1 = 364$ days. The third person's birthday can be any of the remaining $365 - 2 = 363$ days; the fourth person's birthday can be any of the remaining $365 - 3 = 362$ days, etc. For the last person in the group, $N - 1$ dates have already been marked on the calendar, so $365 - (N - 1) = 365 - N + 1$ days are left for their birthday (again assuming no two people share a birthday). This means there are

$$(365)(364)(363) \cdots (365 - N + 1) = P_N^{365} = \frac{365!}{(365 - N)!}$$

ways that no two people in the group can share a birthday. Since there are 365^N total arrangements of birthdays, we find

$$\mathbb{P}(\{\text{no two share a common birthday}\}) = \frac{365!}{365^N(365 - N)!}.$$

Since the complementary event to at least two people share a common birthday is precisely the event that no two people do, we find

$$\mathbb{P}(\{\text{at least two share a common birthday}\}) = 1 - \frac{365!}{365^N(365 - N)!}$$

for $2 \leq N \leq 365$. We were asked to find N such that this probability is $1/2$ or larger. Unfortunately, the form of the answer makes this difficult to compute directly. Instead, we present a partial chart of these probabilities.

N	$\mathbb{P}(\{\text{at least two share a birthday}\})$
21	0.4437
22	0.4757
23	0.5073
24	0.5383
25	0.5687

So for a group of 23 people, there is just over a 50% chance that two people share a common birthday. ■

The answer above surprises most people since 23 people does not seem like a very large group. However, there are

$$\binom{23}{2} = 253$$

different pairs of people in a group of 23. The fact that the total number of distinct pairs grows rapidly with N ,

$$\binom{N}{2} = \frac{N(N-1)}{2},$$

is why the likelihood of a common birthday increases so quickly. It turns out that once the group has more than 46 people, the probability that at least two people share a birthday is over 95%!

6.5 Conditional Probability and the Law of Total Probability

Frequently, we have partial information about the outcome of an experiment, and we want to update our assessment of probabilities based on this extra knowledge. For instance, suppose we (independently) flip 3 fair coins. Since there are $2^3 = 8$ equally likely outcomes, any particular outcome has probability $1/8$. Imagine a game where someone flips the 3 coins but does not reveal to us the actual outcome. Without any information about the outcome, we would rightly claim that the probability of getting all heads (i.e. HHH) is $1/8$. However, if the person who flipped the coins tells us there is at least one tails in the outcome, we would update our assessment of the probability – the probability of HHH becomes zero!

The set of possible outcomes for this particular experiment is

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

The information we have been given is that the outcome actually lies in the more restricted event

$$E = \{\text{at least one } T\} = \{HHT, HTH, HTT, THH, THT, TTH, TTT\},$$

which has 7 outcomes. Since these outcomes were equally likely before the additional information was given to us, we have no reason to suspect that the probabilities are not still equally likely. So, the probability of any outcome in E should be updated to $1/7$.

The idea above was that the partial information essentially restricted us to a new sample space – from S to the event E . We then updated the probabilities so that E had total probability 1. Generalizing this idea leads us to the notion of conditional probability.

Let S be a sample space with some probability defined on it. Let E and F be events in S , and suppose that $\mathbb{P}(E) > 0$. The **conditional probability** of F given E is defined to be

$$\mathbb{P}(F|E) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)}.$$

The quantity $\mathbb{P}(F|E)$ is supposed to represent our updated assessment of the likelihood that F has happened given that we are certain the outcome lies in the event E . Graphically, the situation is shown below.

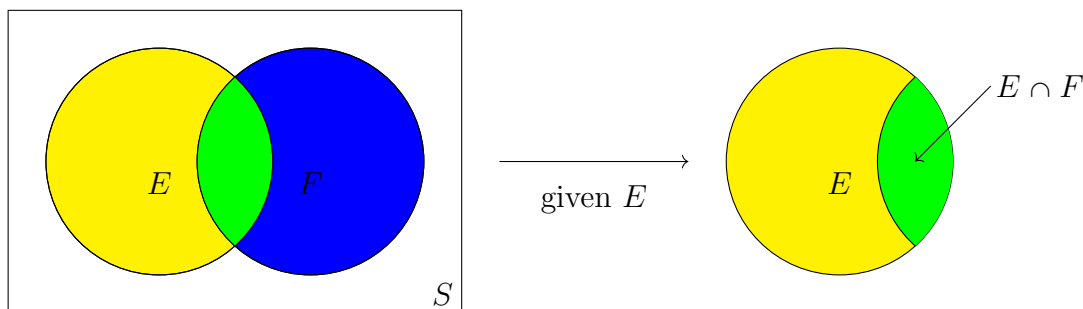


Figure 6.5.1: Diagram Showing Conditional Probability

The information that the outcome is definitely in E restricts the sample space from all of S to just E . If we want to know the probability that F also occurs, we must be looking at the outcomes in $E \cap F$. The final thing to do is to rescale the probabilities so that the event E has probability 1. This is accomplished in the definition by dividing all probabilities by $\mathbb{P}(E)$ (which is why E cannot be allowed to have probability zero).

Example 6.5.1. Suppose we (independently) flip three fair coins. Given that there is at least one tails in the outcome, what is the probability that we have two heads?

Answer: As above, let E be the event that the outcome has at least one tails:

$$E = \{HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

Let F be the event that the outcome has at least 2 heads:

$$F = \{HHH, HHT, HTH, THH\}.$$

The intersection of these two events is

$$E \cap F = \{HHT, HTH, THH\}.$$

We can now use the definition of conditional probability to compute the answer. Remember that the probabilities in the formula are the ones associated to the full sample space S ! In this case, all individual outcomes have

probability $1/8$.

$$\begin{aligned}\mathbb{P}(F|E) &= \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)} \\ &= \frac{\mathbb{P}(\{HHT, HTH, THH\})}{\mathbb{P}(\{HHT, HTH, HTT, THH, THT, TTH, TTT\})} \\ &= \frac{3/8}{7/8} = \frac{3}{7}. \blacksquare\end{aligned}$$

Example 6.5.2. Suppose we (independently) roll 2 fair six-sided dice. Given that the sum of the two dice is 5, what is the probability that one of the dice rolled a 3?

Answer: Let E be the event that the sum of the two dice is 5:

$$E = \{(1, 4); (2, 3); (3, 2); (4, 1)\}.$$

Let F be the event that at least one of the dice shows a 3:

$$F = \{(3, 1); (3, 2); (3, 3); (3, 4); (3, 5); (3, 6); (1, 3); (2, 3); (4, 3); (5, 3); (6, 3)\}.$$

Since the intersection of these two events is

$$E \cap F = \{(3, 2); (2, 3)\},$$

we see that

$$\begin{aligned}\mathbb{P}(F|E) &= \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)} \\ &= \frac{\mathbb{P}(\{(3, 2); (2, 3)\})}{\mathbb{P}(\{(1, 4); (2, 3); (3, 2); (4, 1)\})} \\ &= \frac{2/36}{4/26} = \frac{1}{2}. \blacksquare\end{aligned}$$

Before giving more interesting examples, we stop to give an alternate characterization of independent events.

Lemma 6.5.1 (Independent Events and Conditional Probability). *Suppose E and F are two events in a sample space S and $\mathbb{P}(E) > 0$. Then E and F are independent if and only if*

$$\mathbb{P}(F|E) = \mathbb{P}(F).$$

Proof: If E and F are independent, then $\mathbb{P}(E \cap F) = \mathbb{P}(E) \cdot \mathbb{P}(F)$. So,

$$\mathbb{P}(F|E) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)} = \frac{\mathbb{P}(E) \cdot \mathbb{P}(F)}{\mathbb{P}(E)} = \mathbb{P}(F).$$

On the other hand, if $\mathbb{P}(F|E) = \mathbb{P}(F)$, then

$$\begin{aligned} \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)} &= \mathbb{P}(F) \\ \mathbb{P}(E \cap F) &= \mathbb{P}(E) \cdot \mathbb{P}(F), \end{aligned}$$

and so the events are independent. ■

This lemma gives us the crucial insight into the name “independent.” That is, two events are independent if knowing that one of the events has occurred does not alter our assessment of how likely the other event is. This also makes clear that, in general, mutually exclusive events are definitely not independent. Consider the experiment where we (independently) roll 2 fair six-sided die. Let E be the event that the sum of the two dice is 6 or less, and let F be the event that at least one of the two dice rolls a 6. These events are mutually exclusive since rolling a 6 on one of the two dice guarantees that the sum of the two is at least 7. So, if we are told that the sum of the two dice is not greater than 6, we know that the probability of having rolled a 6 on one of the dice is zero! This is certainly different from the unconditional probability of rolling a 6:

$$\mathbb{P}(\{\text{roll at least one 6}\}) = \frac{11}{36}.$$

Example 6.5.3. Consider an urn filled with 3 red, 3 green, and 4 yellow balls. We perform an experiment where 3 balls are drawn from the urn without replacement. Given that at least one of the balls is red, what is the probability that we have one ball of each color?

Answer: We will compute the probability in two ways. We begin by pretending all of the balls are distinguishable: balls 1 – 3 are red, 4 – 6 green, and 7 – 10 yellow. We will also assume that the order in which the balls are pulled from the urn matters. Since there are 10 balls, that means there are

$$P_3^{10} = 10 \cdot 9 \cdot 8 = 720 \text{ outcomes.}$$

Let E be the event that at least one ball is red, and let F be the event that we have one ball of each color. To compute the probability of E , we need to know how many outcomes are in this event. We have to be very careful here. There can be exactly 1 red ball, exactly 2 red balls, or exactly 3 red balls in our selection. We should count these outcomes separately.

Case 1: Exactly 1 Red Ball

Here, there are three types of outcomes:

$$\begin{aligned}\underline{R} \quad \underline{\text{not } R} \quad \underline{\text{not } R} &= 3 \cdot 7 \cdot 6 = 126, \\ \underline{\text{not } R} \quad \underline{R} \quad \underline{\text{not } R} &= 7 \cdot 3 \cdot 6 = 126, \\ \underline{\text{not } R} \quad \underline{\text{not } R} \quad \underline{R} &= 7 \cdot 6 \cdot 3 = 126.\end{aligned}$$

Thus, there are 378 outcomes with exactly one red ball.

Case 2: Exactly 2 Red Balls

Here again, there are three types of outcomes:

$$\begin{aligned}\underline{R} \quad \underline{R} \quad \underline{\text{not } R} &= 3 \cdot 2 \cdot 7 = 42, \\ \underline{R} \quad \underline{\text{not } R} \quad \underline{R} &= 3 \cdot 7 \cdot 2 = 42, \\ \underline{\text{not } R} \quad \underline{R} \quad \underline{R} &= 7 \cdot 3 \cdot 2 = 42.\end{aligned}$$

So, there are 126 outcomes with exactly two red balls.

Case 3: Exactly 3 Red Balls

$$\underline{R} \quad \underline{R} \quad \underline{R} = 3 \cdot 2 \cdot 1 = 6$$

So, there are a total of 510 outcomes with at least one red ball, and the probability of this event is

$$\mathbb{P}(E) = \frac{510}{720} = \frac{17}{24}.$$

Next, we need to consider the event $E \cap F$ which in words is the event that there is at least one red ball and a ball of each color. Clearly, if there is a ball of each color then there will be exactly one red ball! In this case $E \cap F = F$.

One Ball of Each Color

There are six types of outcomes:

$$\begin{aligned}\frac{R}{\quad} \frac{G}{\quad} \frac{Y}{\quad} &= 3 \cdot 3 \cdot 4 = 36, \\ \frac{R}{\quad} \frac{Y}{\quad} \frac{G}{\quad} &= 3 \cdot 4 \cdot 3 = 36, \\ \frac{G}{\quad} \frac{R}{\quad} \frac{Y}{\quad} &= 3 \cdot 3 \cdot 4 = 36, \\ \frac{G}{\quad} \frac{Y}{\quad} \frac{R}{\quad} &= 3 \cdot 4 \cdot 3 = 36, \\ \frac{Y}{\quad} \frac{R}{\quad} \frac{G}{\quad} &= 4 \cdot 3 \cdot 3 = 36, \\ \frac{Y}{\quad} \frac{G}{\quad} \frac{R}{\quad} &= 4 \cdot 3 \cdot 3 = 36.\end{aligned}$$

This gives 216 outcomes with exactly one ball of each color, which means

$$\mathbb{P}(F) = \mathbb{P}(E \cap F) = \frac{216}{720} = \frac{3}{10}.$$

Finally, the conditional probability we are asked for is

$$\mathbb{P}(F|E) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)} = \frac{3/10}{17/24} = \frac{36}{85}.$$

The method above was certainly the lengthy way of obtaining the answer! The first simplification is to forget about the order in which the balls are removed from the urn. This means there are

$$\binom{10}{3} = 120 \text{ outcomes.}$$

Next,

$$\mathbb{P}(\{\text{at least one red}\}) = 1 - \mathbb{P}(\{\text{no reds}\}).$$

Since there are 7 non-red balls, there are

$$\binom{7}{3} = 35 \text{ outcomes with no red balls.}$$

This gives

$$\mathbb{P}(\{\text{at least one red}\}) = 1 - \mathbb{P}(\{\text{no reds}\}) = 1 - \frac{35}{120} = \frac{17}{24}.$$

As before, $E \cap F = F$. The number of outcomes with one ball of each color is

$$\binom{3}{1} \cdot \binom{3}{1} \cdot \binom{4}{1} = 3 \cdot 3 \cdot 4 = 36.$$

As before,

$$\mathbb{P}(F) = \mathbb{P}(E \cap F) = \frac{36}{120} = \frac{3}{10}.$$

Once again, we have

$$\mathbb{P}(F|E) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)} = \frac{3/10}{17/24} = \frac{36}{85}. \blacksquare$$

If we rearrange the formula for conditional probability, we find an alternate one:

$$\mathbb{P}(E \cap F) = \mathbb{P}(F|E)\mathbb{P}(E).$$

This simple observation leads us to the following very useful theorem.

Theorem 6.5.2 (The Law of Total Probability). *Let E and F be two events of a sample space S , and assume that $0 < \mathbb{P}(E) < 1$. Then*

$$\mathbb{P}(F) = \mathbb{P}(F|E)\mathbb{P}(E) + \mathbb{P}(F|E^c)\mathbb{P}(E^c).$$

Proof: Using the rearranged form of conditional probability above, we have

$$\mathbb{P}(F|E)\mathbb{P}(E) + \mathbb{P}(F|E^c)\mathbb{P}(E^c) = \mathbb{P}(F \cap E) + \mathbb{P}(F \cap E^c).$$

Since E and E^c are mutually exclusive events, so are $F \cap E$ and $F \cap E^c$. This means that the sum of the probabilities of these two events is just the probability of their union:

$$\begin{aligned} \mathbb{P}(F|E)\mathbb{P}(E) + \mathbb{P}(F|E^c)\mathbb{P}(E^c) &= \mathbb{P}(F \cap E) + \mathbb{P}(F \cap E^c) \\ &= \mathbb{P}((F \cap E) \cup (F \cap E^c)). \end{aligned}$$

Basic results about set operations give

$$\begin{aligned} (F \cap E) \cup (F \cap E^c) &= F \cap (E \cup E^c) \\ &= F \cap S \\ &= F. \end{aligned}$$

Hence,

$$\mathbb{P}(F|E)\mathbb{P}(E) + \mathbb{P}(F|E^c)\mathbb{P}(E^c) = \mathbb{P}(F). \blacksquare$$

The reason that this formula is useful is that it allows us to compute the probability of the event F by conditioning on whether the separate event E has happened or not. The idea is to pick E so that knowing whether E happens makes the computation of F easier. For this reason, the law of total probability is often referred to as *conditioning*.

Example 6.5.4. Consider an experiment that consists of two stages. First, we flip a fair coin. If the coin comes up heads, we then roll a fair six-sided die. If the coin comes up tails, we roll a fair 12-sided die. What is the probability that the second stage of the experiment gives us a 4?

Answer: Let $F = \{\text{outcome from die is 4}\}$. While we could list all the possibilities (there are 18) and compute their probabilities using independence, we will proceed by conditioning. It would be very helpful to know whether the coin toss resulted in a heads or tails.

Let $E = \{\text{outcome of coin toss is } H\}$ (which means E^c is the event that the coin toss is a tails). By the law of total probability

$$\mathbb{P}(F) = \mathbb{P}(F|E)\mathbb{P}(E) + \mathbb{P}(F|E^c)\mathbb{P}(E^c).$$

Since the coin toss is fair, we know

$$\mathbb{P}(E) = \mathbb{P}(E^c) = \frac{1}{2}.$$

For the conditional probabilities, we only have to think about the situations they imply. For instance, $\mathbb{P}(F|E)$ is asking us for the probability of rolling a 4 given that the coin toss resulted in a heads. If the coin toss is heads, we use a fair 6-sided die. Therefore,

$$\mathbb{P}(F|E) = \frac{1}{6}.$$

Similarly, if the coin toss results in tails, we use a fair 12-sided die. Thus,

$$\mathbb{P}(F|E^c) = \frac{1}{12}.$$

Putting these numbers together gives

$$\begin{aligned} \mathbb{P}(F) &= \mathbb{P}(F|E)\mathbb{P}(E) + \mathbb{P}(F|E^c)\mathbb{P}(E^c) \\ &= \frac{1}{6} \cdot \frac{1}{2} + \frac{1}{12} \cdot \frac{1}{2} \\ &= \frac{3}{24} = \frac{1}{8}. \quad \blacksquare \end{aligned}$$

Example 6.5.5. *Dichromatism* is a genetic disorder resulting in a kind of color blindness. The disorder affects 5% of men and 1% of women. If a population is 52% female, what is the probability that a randomly sampled person will have dichromatism?

Answer: Let D be the event that a person has dichromatism and F be the event that a person is female (so F^c is the event that a person is male). Our information gives us

$$\mathbb{P}(F) = 0.52 \quad \text{and} \quad \mathbb{P}(F^c) = 1 - 0.52 = 0.48.$$

The information about the rates of dichromatism are actually conditional statements:

$$\mathbb{P}(D|F) = 0.01 \quad \text{and} \quad \mathbb{P}(D|F^c) = 0.05.$$

Using the law of total probability, we have

$$\begin{aligned} \mathbb{P}(D) &= \mathbb{P}(D|F)\mathbb{P}(F) + \mathbb{P}(D|F^c)\mathbb{P}(F^c) \\ &= 0.01(0.52) + 0.05(0.48) \\ &= 0.0292. \end{aligned}$$

So, there is a 2.92% chance that a randomly selected person in the population will have dichromatism. ■

The law of total probability is not limited to two events!

Corollary 6.5.2.1 (The Generalized Law of Total Probability). *Let $E_1, E_2, \dots, E_N$ be a collection of mutually exclusive events with positive probability that exhaust the sample space S . In other words,*

- $E_i \cap E_j = \emptyset$ whenever $i \neq j$,
- $E_1 \cup E_2 \cup \dots \cup E_N = S$,
- $\mathbb{P}(E_i) > 0$ for all i .

Then for any event F in S ,

$$\mathbb{P}(F) = \mathbb{P}(F|E_1)\mathbb{P}(E_1) + \mathbb{P}(F|E_2)\mathbb{P}(E_2) + \dots + \mathbb{P}(F|E_N)\mathbb{P}(E_N).$$

Proof: This result follows from the law of total probability by induction on the number of events E_i . ■

Example 6.5.6. Craps is a game where we roll 2 fair six-sided dice. If we roll a sum of 2, 3, or 12, we automatically lose. If we roll a sum of 7 or 11, we automatically win. If we roll any of the other possible sums (4, 5, 6, 8, 9, or 10), we have to continue rolling the dice until either we roll our original sum (in which case we win) or we roll a 7 (in which case we lose). What is the probability that we win the game on the first roll? What is the probability that the game will continue for more than one roll?

Answer: Let R be the sum of the two dice on the first roll. Note that the 11 events

$$R = 2, R = 3, \dots, R = 12$$

are mutually exclusive and exhaust the sample space (that is, they account for all of the possibilities). As such, we can use them for conditioning! Remember that we computed the probabilities of the random variable R in Example 6.2.5.

Let W be the event that we win the game on the first roll. Then using the law of total probability

$$\begin{aligned} \mathbb{P}(W) &= \mathbb{P}(W|R = 2)\mathbb{P}(R = 2) + \mathbb{P}(W|R = 3)\mathbb{P}(R = 3) \\ &\quad + \mathbb{P}(W|R = 4)\mathbb{P}(R = 4) + \dots + \mathbb{P}(W|R = 12)\mathbb{P}(R = 12). \end{aligned}$$

These conditional probabilities make the problem simple! According to the rules,

$$\mathbb{P}(W|R = 7) = 1 \quad \text{and} \quad \mathbb{P}(W|R = 11) = 1$$

while all of the other conditional probabilities are zero (the only way to win on the first roll is to get a 7 or 11)! That means

$$\begin{aligned} \mathbb{P}(W) &= \mathbb{P}(W|R = 7)\mathbb{P}(R = 7) + \mathbb{P}(W|R = 11)\mathbb{P}(R = 11) \\ &= \mathbb{P}(R = 7) + \mathbb{P}(R = 11) \\ &= \frac{1}{6} + \frac{1}{18} = \frac{2}{9}. \end{aligned}$$

Of course, we could have obtained this by the simpler observation that we need to roll a 7 or 11 to win outright. But this serves to illustrate the basic idea of conditioning. Likewise, if C is the event that the game continues past the first roll, then

$$\begin{aligned} \mathbb{P}(C) &= \mathbb{P}(C|R = 2)\mathbb{P}(R = 2) + \mathbb{P}(C|R = 3)\mathbb{P}(R = 3) \\ &\quad + \mathbb{P}(C|R = 4)\mathbb{P}(R = 4) + \dots + \mathbb{P}(C|R = 12)\mathbb{P}(R = 12). \end{aligned}$$

Now,

$$\mathbb{P}(C|R = 2) = \mathbb{P}(C|R = 3) = \mathbb{P}(C|R = 12) = 0$$

since we automatically lose the game if those are our first roll, and

$$\mathbb{P}(C|R = 7) = \mathbb{P}(C|R = 11) = 0$$

since we win on the first roll with those two outcomes. All of the remaining conditional probabilities are 1. Hence,

$$\begin{aligned}\mathbb{P}(C) &= \mathbb{P}(C|R=4)\mathbb{P}(R=4) + \mathbb{P}(C|R=5)\mathbb{P}(R=5) + \mathbb{P}(C|R=6)\mathbb{P}(R=6) \\ &\quad + \mathbb{P}(C|R=8)\mathbb{P}(R=8) + \mathbb{P}(C|R=9)\mathbb{P}(R=9) + \mathbb{P}(C|R=10)\mathbb{P}(R=10) \\ &= \mathbb{P}(R=4) + \mathbb{P}(R=5) + \mathbb{P}(R=6) + \mathbb{P}(R=8) + \mathbb{P}(R=9) + \mathbb{P}(R=10) \\ &= \frac{1}{12} + \frac{1}{9} + \frac{5}{36} + \frac{5}{36} + \frac{1}{9} + \frac{1}{12} = \frac{2}{3}.\end{aligned}$$

Even though we were not asked, we can also calculate that the probability of losing on the first roll is $1/9$. ■

With slightly more advanced techniques, it can be shown that the overall probability of winning at craps is

$$\frac{244}{495} \approx 0.4929.$$

The difficulty in computing this probability is that in theory the game can go on for arbitrarily large number of rolls; if you roll a 4, 5, 6, 8, 9, or 10, you continue rolling until you get a 7 or the first roll again. Even though the game could go on forever, it is also possible to show that games of craps typically last between 1 and 9 rolls with an average length of about 3 rolls.

6.6 Application: The Hardy–Weinberg Principle

Consider the case of a trait whose expression is controlled by a single gene with two different alleles: A and a (the dominant and recessive versions of the gene, respectively). Assuming that each individual has two copies of the gene (one from each parent), there are three different genotypes: AA , Aa , and aa . Since A is the dominant version of the gene, both AA and Aa individuals will have the same phenotype (physical expression of the trait) while aa individuals will have a different expression of the trait.

Suppose that in a given population, p percentage of the genes are dominant while $q = 1 - p$ is the percentage of the genes that are recessive. This means that a randomly chosen copy of the gene has a probability p of being A and a probability q of being a . The first question to address is how

these probabilities relate to the overall frequency of the three genotypes in the population.

Let P, Q , and R be the probabilities that a randomly selected individual is AA, Aa , or aa , respectively (with $P + Q + R = 1$). To find a relationship between these quantities, we can compute the probability that a randomly selected copy of the gene is of type A by conditioning on the type of individual we sample the gene from.

$$\mathbb{P}(A) = \mathbb{P}(A|AA)\mathbb{P}(AA) + \mathbb{P}(A|Aa)\mathbb{P}(Aa) + \mathbb{P}(A|aa)\mathbb{P}(aa)$$

The quantity $\mathbb{P}(A|AA)$ is clearly 1 since the probability of selecting an A gene from an AA individual is 1. Similarly, $\mathbb{P}(A|aa) = 0$ since you cannot select an A gene from an aa individual. The remaining conditional probability is $1/2$ since there is a 50-50 chance of selecting an A copy of the gene from an Aa individual. Putting this together with the fact that $\mathbb{P}(A) = p$, $\mathbb{P}(AA) = P$, and $\mathbb{P}(Aa) = Q$, gives

$$p = P + \frac{1}{2}Q.$$

Similarly,

$$\begin{aligned} \mathbb{P}(a) &= \mathbb{P}(a|AA)\mathbb{P}(AA) + \mathbb{P}(a|Aa)\mathbb{P}(Aa) + \mathbb{P}(a|aa)\mathbb{P}(aa) \\ q &= \frac{1}{2}Q + R. \end{aligned}$$

If we think of p and q as the known quantities and try to solve for P, Q , and R in terms of them, we see that there is not enough information to compute them. This is because there are an infinite number of ways that the individual genes could be sorted into the three groups AA, Aa , and aa and still retain the overall frequencies p and q for the individual alleles. In one extreme case, the genes could be completely stratified into the two homozygous genotypes AA and aa (in which case $P = p$, $Q = 0$, and $R = q$). In another extreme case, if $p > q$ then there could be no homozygous recessive individuals (with $P = p - q$, $Q = 2q$, and $R = 0$). Of course, both of these scenarios seem unlikely in general.

To make progress, we need to make some assumptions about mating in our population. The following work was first done independently by the mathematician G.H. Hardy and the physician Wilhelm Weinberg in 1908. The primary assumptions in their modeling are as follows.

- 1) Mating is totally random among the individuals in the population.

- 2) The number of offspring capable of successful reproduction for any mating pair is roughly equal (i.e. $AA \times AA$ is not likely to produce more successful offspring than $AA \times Aa$, etc.).
- 3) No mutations or migrations occur (so the overall frequencies p and q of the two alleles stays constant over generations).

The frequency of genotypes for offspring of each mating pair are easily found by Punnett Square. In our case, the Punnett Square is a diagram with two rows and two columns. Each row corresponds to a single allele from one parent, and each column corresponds to a single allele from the other parent. The possible genotypes of the offspring are then easily read off from the resulting table. For example, a mating of one heterozygous parent with a homozygous recessive parent would yield the following Punnett Square.

	A	a
a	Aa	aa
a	Aa	aa

As we can see, the offspring would have a 50% chance of being homozygous recessive (aa) and a 50% chance of being heterozygous (Aa). It is impossible for any offspring to be homozygous dominant (AA).

Running through the Punnett Squares for all possible mating pairs yields the frequency table below.

Mating Pair	Frequency of Offspring		
	AA	Aa	aa
$AA \times AA$	1	0	0
$AA \times Aa$	$\frac{1}{2}$	$\frac{1}{2}$	0
$AA \times aa$	0	1	0
$Aa \times Aa$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$
$Aa \times aa$	0	$\frac{1}{2}$	$\frac{1}{2}$
$aa \times aa$	0	0	1

Suppose that the percentages of AA , Aa , and aa individuals in the first generation are P_1 , Q_1 , and R_1 , respectively (with their sum being 1). What will the percentages be in the subsequent generation (call them P_2 , Q_2 , and R_2). To compute these probabilities, we use our assumption of random mating to compute the probabilities of the six different mating pairs.

Mating Pair	Frequency
$AA \times AA$	P_1^2
$AA \times Aa$	$2P_1Q_1$
$AA \times aa$	$2P_1R_1$
$Aa \times Aa$	Q_1^2
$Aa \times aa$	$2Q_1R_1$
$aa \times aa$	R_1^2

The factors of 2 above come from the fact that there are two possible ways such a mating could occur (e.g. with $AA \times Aa$, the father could be AA and the mother Aa or vice versa). Note that

$$P_1^2 + 2P_1Q_1 + 2P_1R_1 + Q_1^2 + 2Q_1R_1 + R_1^2 = (P_1 + Q_1 + R_1)^2 = 1.$$

To compute the frequencies P_2 , Q_2 , and R_2 in the next generation, we can condition on the parentage of the offspring.

$$\begin{aligned}
P_2 &= \mathbb{P}(\text{child is } AA) \\
&= \mathbb{P}(\text{child is } AA | \text{parents are } AA \times AA) \mathbb{P}(\text{parents are } AA \times AA) \\
&\quad + \mathbb{P}(\text{child is } AA | \text{parents are } AA \times Aa) \mathbb{P}(\text{parents are } AA \times Aa) \\
&\quad + \mathbb{P}(\text{child is } AA | \text{parents are } AA \times aa) \mathbb{P}(\text{parents are } AA \times aa) \\
&\quad + \mathbb{P}(\text{child is } AA | \text{parents are } Aa \times Aa) \mathbb{P}(\text{parents are } Aa \times Aa) \\
&\quad + \mathbb{P}(\text{child is } AA | \text{parents are } Aa \times aa) \mathbb{P}(\text{parents are } Aa \times aa) \\
&\quad + \mathbb{P}(\text{child is } AA | \text{parents are } aa \times aa) \mathbb{P}(\text{parents are } aa \times aa) \\
&= 1 \cdot P_1^2 + \frac{1}{2} \cdot 2P_1Q_1 + 0 + \frac{1}{4} \cdot Q_1^2 + 0 + 0 \\
&= P_1^2 + P_1Q_1 + \frac{1}{4}Q_1^2 \\
&= \left(P_1 + \frac{1}{2}Q_1\right)^2 = p^2
\end{aligned}$$

where in the last step we have used the fact computed above that in any generation $p = P + Q/2$.

For the heterozygous offspring, we find

$$\begin{aligned}
 Q_2 &= \mathbb{P}(\text{child is } Aa) \\
 &= \mathbb{P}(\text{child is } Aa | \text{parents are } AA \times AA) \mathbb{P}(\text{parents are } AA \times AA) \\
 &\quad + \mathbb{P}(\text{child is } Aa | \text{parents are } AA \times Aa) \mathbb{P}(\text{parents are } AA \times Aa) \\
 &\quad + \mathbb{P}(\text{child is } Aa | \text{parents are } AA \times aa) \mathbb{P}(\text{parents are } AA \times aa) \\
 &\quad + \mathbb{P}(\text{child is } Aa | \text{parents are } Aa \times Aa) \mathbb{P}(\text{parents are } Aa \times Aa) \\
 &\quad + \mathbb{P}(\text{child is } Aa | \text{parents are } Aa \times aa) \mathbb{P}(\text{parents are } Aa \times aa) \\
 &\quad + \mathbb{P}(\text{child is } Aa | \text{parents are } aa \times aa) \mathbb{P}(\text{parents are } aa \times aa) \\
 &= 0 + \frac{1}{2} \cdot 2P_1Q_1 + 1 \cdot 2P_1R_1 + \frac{1}{2} \cdot Q_1^2 + \frac{1}{2} 2Q_1R_1 + 0 \\
 &= P_1Q_1 + 2P_1R_1 + \frac{1}{2}Q_1^2 + Q_1R_1 \\
 &= 2 \left(P_1 + \frac{1}{2}Q_1 \right) \left(\frac{1}{2}Q_1 + R_1 \right) = 2pq.
 \end{aligned}$$

The homozygous recessive offspring frequency can be computed equivalently with the result

$$R_2 = \mathbb{P}(\text{child is } aa) = \left(\frac{1}{2}Q_1 + R_1 \right)^2 = q^2$$

What these computations show is that regardless of the initial distribution of the three genotypes, random mating (along with the other assumptions listed above) will produce a distribution of genotypes given solely by the underlying frequency of the alleles:

$$\begin{aligned}
 \mathbb{P}(AA) &= p^2, \\
 \mathbb{P}(Aa) &= 2pq, \\
 \mathbb{P}(aa) &= q^2.
 \end{aligned}$$

Moreover, these frequencies will remain constant in all subsequent generations so long as the assumptions in the model remain true. This fact leads us to call the special frequencies above the *Hardy–Weinberg Equilibrium* of the population, and the constancy of this frequency over time is known as the *Hardy–Weinberg Principle*.

Example 6.6.1. The dominant coloration of pea plants is actually yellow (the fact that green peas are the ones we commonly consume is due to very

selective farming practices). The coloration of pea plants is controlled by a dominant allele Y and a recessive allele y . YY and Yy plants are yellow while yy plants are green. Suppose that we have an initial population of pea plants where one-eighth have genotype YY , one-quarter Yy , and five-eighths yy . Under the assumptions of the Hardy–Weinberg principle, what should the proportions of the genotypes be in subsequent generations? What percentage of the plants should be yellow in the future?

Answer: We have that $P_1 = \frac{1}{8}$, $Q_1 = \frac{1}{4}$, and $R_1 = \frac{5}{8}$. These allow us to compute the underlying frequency of the individual alleles.

$$p = \mathbb{P}(Y) = P_1 + \frac{1}{2}Q_1 = \frac{1}{4}$$

$$q = \mathbb{P}(y) = \frac{1}{2}Q_1 + R_1 = \frac{3}{4}$$

Using the Hardy–Weinberg Equilibrium, we expect that for future generations of the plants

$$\mathbb{P}(YY) = p^2 = \frac{1}{16},$$

$$\mathbb{P}(Yy) = 2pq = \frac{3}{8},$$

$$\mathbb{P}(yy) = q^2 = \frac{9}{16}.$$

Since only the yy plants are green, we anticipate that

$$\mathbb{P}(\text{yellow pea plant}) = \frac{7}{16}. \blacksquare$$

Example 6.6.2. Suppose that the length of rabbit ears is controlled by a single gene with S being the dominant allele giving short ears and s being the recessive allele giving long, floppy ears. In a particular population of 1000 rabbits, only 90 have long, floppy ears. Assuming the population is in Hardy–Weinberg Equilibrium, find the overall frequency of the two alleles in the population.

Answer: If the population is in equilibrium, then

$$\mathbb{P}(SS) = p^2,$$

$$\mathbb{P}(Ss) = 2pq,$$

$$\mathbb{P}(ss) = q^2.$$

Since only the ss individuals have long ears, we must have

$$q^2 = \mathbb{P}(ss) = \frac{90}{1000} = 0.09.$$

This gives us

$$q = \sqrt{0.09} = 0.3.$$

This means that $p = 1 - q = 0.7$. Hence, 70% of all copies of the gene are the dominant type, and only 30% are recessive. ■

6.7 Bayes' Formula and Medical Testing

It frequently happens that we are given the conditional probability $\mathbb{P}(F|E)$ (or at least this probability is easy to compute), but we actually need $\mathbb{P}(E|F)$. Fortunately, there is a formula connecting these two numbers.

Theorem 6.7.1 (Bayes' Formula). *Suppose that E and F are two events in a sample space S and that $\mathbb{P}(E) > 0$ and $\mathbb{P}(F) > 0$. Then,*

$$\mathbb{P}(E|F) = \frac{\mathbb{P}(F|E)\mathbb{P}(E)}{\mathbb{P}(F)}.$$

Proof: Using the alternate form of conditional probability we noted just before the law of total probability,

$$\begin{aligned} \mathbb{P}(E|F) &= \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)} \\ &= \frac{\mathbb{P}(F|E)\mathbb{P}(E)}{\mathbb{P}(F)}. \quad \blacksquare \end{aligned}$$

We will begin with an example where we can explicitly compute both conditional probabilities and then compare the results (verifying Bayes' Formula). We then examine a couple of problems which are more typical examples of Bayesian reasoning.

Example 6.7.1. Suppose we independently flip 10 fair coins. Let E be the event that there are exactly 4 heads, and let F be the event that the first coin toss is a heads. Compute $\mathbb{P}(E|F)$, $\mathbb{P}(F|E)$, and verify that they satisfy Bayes' Formula.

Answer: We know that there are $2^{10} = 1024$ outcomes. We first compute the probabilities of the events themselves. For E , there must be 4 heads and 6 tails. The idea for counting the outcomes here is that once we place the 4 heads, the remaining positions have to be filled with tails. Since there are 10 positions and we have to choose 4 of them to be heads, there are

$$\binom{10}{4} = 210 \text{ outcomes.}$$

This means

$$\mathbb{P}(E) = \frac{210}{1024} = \frac{105}{512}.$$

For F , since the first coin flip is independent of the remaining 9 flips,

$$\mathbb{P}(F) = \frac{1}{2}.$$

For the conditional probabilities, we need the probability of $E \cap F$. Outcomes in this event all start with a heads and have exactly 4 heads in total. So among the 9 coin flips after the first one, there are 3 heads in total:

$$\binom{9}{3} = 84.$$

This gives

$$\mathbb{P}(E \cap F) = \frac{84}{1024} = \frac{21}{256}.$$

For the conditional probabilities,

$$\begin{aligned} \mathbb{P}(E|F) &= \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)} \\ &= \frac{21/256}{1/2} \\ &= \frac{21}{128}, \\ \mathbb{P}(F|E) &= \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)} \\ &= \frac{21/256}{105/512} \\ &= \frac{42}{105} = \frac{2}{5}. \end{aligned}$$

Notice that we could have computed these conditional probabilities more directly. For instance, $\mathbb{P}(E|F)$ is asking for the probability that there are 4 heads in total given that the first coin flip results in a heads. This is the same as the probability that there are exactly 3 heads in the remaining 9 flips.

$$\mathbb{P}(E|F) = \frac{\binom{9}{3}}{2^9} = \frac{84}{512} = \frac{21}{128}.$$

The second conditional probability, $\mathbb{P}(F|E)$, is a little harder to compute directly but can be done. In this case, we know there are exactly 4 heads in the outcome and are being asked for the probability that the first coin toss was one of them. There are $\binom{10}{4}$ possible outcomes (all equally likely) but only $\binom{9}{3}$ of them definitely have a heads for the first coin flip. So,

$$\mathbb{P}(F|E) = \frac{\binom{9}{3}}{\binom{10}{4}} = \frac{84}{210} = \frac{2}{5}.$$

Finally, we verify Bayes' Formula:

$$\frac{21}{128} = \mathbb{P}(E|F) = \frac{\mathbb{P}(F|E)\mathbb{P}(E)}{\mathbb{P}(F)} = \frac{(2/5)(105/512)}{1/2} = \frac{21}{128}. \blacksquare$$

Example 6.7.2. A teacher gives a true/false test. If a student does not know the correct answer for a particular question, he or she will guess at random. The teacher estimates that a student has a 75% chance of knowing the correct answer for any given question. Given that a student got the correct answer, what is the probability that they knew the right answer?

Answer: Let C be the event that the student gets the correct answer, and K be the event that the student knew the correct answer. We are told $\mathbb{P}(K) = 0.75$, and so $\mathbb{P}(K^c) = 1 - 0.75 = 0.25$.

The rest of the information we know is all conditional. If a student knows the correct answer, then they will certainly choose the right answer:

$$\mathbb{P}(C|K) = 1 \quad \text{and} \quad \mathbb{P}(C^c|K) = 0.$$

If a student does not know the correct answer, then they will guess the correct answer with probability 1/2:

$$\mathbb{P}(C|K^c) = \frac{1}{2} \quad \text{and} \quad \mathbb{P}(C^c|K^c) = \frac{1}{2}.$$

We are being asked for $\mathbb{P}(K|C)$. Using Bayes' Formula, we have

$$\mathbb{P}(K|C) = \frac{\mathbb{P}(C|K)\mathbb{P}(K)}{\mathbb{P}(C)}.$$

We have the probabilities needed for the numerator, but the probability in the denominator is the overall probability that a student gets the correct answer to a question. To compute this, we use conditioning.

$$\begin{aligned}\mathbb{P}(C) &= \mathbb{P}(C|K)\mathbb{P}(K) + \mathbb{P}(C|K^c)\mathbb{P}(K^c) \\ &= 1(0.75) + (0.5)(0.25) = 0.875.\end{aligned}$$

Notice that the first term in this sum is exactly the numerator from Bayes' Formula.

$$\mathbb{P}(K|C) = \frac{\mathbb{P}(C|K)\mathbb{P}(K)}{\mathbb{P}(C)} = \frac{1(0.75)}{0.875} \approx 0.8571.$$

So, there is over an 85% chance that a student who selects the correct answer to the question actually knew the right answer. ■

Example 6.7.3. A pharmaceutical company is developing a new test for bubonic plague. For a sample known to be infected, the test returns a false negative with probability of 0.02. For a sample known to be disease-free, the test returns a false positive with probability 0.001. Suppose the test is being administered in a region where bubonic plague is estimated to infect 1 out of every 10,000 people. What is the probability that a person who tests positive for the plague is actually disease-free? Is the company's test reliable in this case?

Answer: Let I be the event that a given person is infected with the disease (so I^c is the event that the person is disease-free). Let P be the event that the test returns a positive result (i.e. the test indicates the person has the disease). We are told

$$\mathbb{P}(I) = \frac{1}{10000} \quad \text{and} \quad \mathbb{P}(I^c) = 1 - \frac{1}{10000} = \frac{9999}{10000}.$$

When a pharmaceutical company tries to determine how effective their test is, they run the test on samples already known to be infected or disease-free (presumably by running a costlier and/or more time consuming test). So, the rates reported by the company are *conditional probabilities*. A false

positive in this context would be for the test to return a positive on a sample known to be disease-free:

$$\underbrace{\mathbb{P}(P|I^c) = 0.001}_{\text{false positive}} \quad \text{and} \quad \mathbb{P}(P^c|I^c) = 1 - 0.001 = 0.999.$$

A false negative is when the test returns a negative result on a sample known to be infected:

$$\mathbb{P}(P|I) = 1 - 0.02 = 0.98 \quad \text{and} \quad \underbrace{\mathbb{P}(P^c|I) = 0.02}_{\text{false negative}}.$$

We are being asked for the probability that a person who tests positive for the disease is actually disease-free: $\mathbb{P}(I^c|P)$. Fortunately, Bayes' Formula comes to the rescue:

$$\mathbb{P}(I^c|P) = \frac{\mathbb{P}(P|I^c)\mathbb{P}(I^c)}{\mathbb{P}(P)}.$$

We know the two probabilities in the numerator. The probability in the denominator is the overall probability that a test returns a positive. For that, we use conditioning!

$$\begin{aligned} \mathbb{P}(P) &= \mathbb{P}(P|I)\mathbb{P}(I) + \mathbb{P}(P|I^c)\mathbb{P}(I^c) \\ &= 0.98(0.0001) + 0.001(0.9999) \\ &= 0.0010979. \end{aligned}$$

Notice that the second term is precisely the numerator in Bayes' Formula!

$$\begin{aligned} \mathbb{P}(I^c|P) &= \frac{\mathbb{P}(P|I^c)\mathbb{P}(I^c)}{\mathbb{P}(P)} \\ &= \frac{0.001(0.9999)}{0.0010979} \approx 0.9107. \end{aligned}$$

This means that there is over a 91% chance that a person who receives a positive test result is actually healthy! In other words, this test is not very reliable in determining who is actually infected with the disease.

Even though we were not asked, we can determine the complimentary probability $\mathbb{P}(I|P^c)$, the probability that a person who receives a negative test result is actually infected.

$$\mathbb{P}(I|P^c) = \frac{\mathbb{P}(P^c|I)\mathbb{P}(I)}{\mathbb{P}(P^c)}.$$

We can use conditioning for the denominator

$$\begin{aligned}\mathbb{P}(P^c) &= \mathbb{P}(P^c|I)\mathbb{P}(I) + \mathbb{P}(P^c|I^c)\mathbb{P}(I^c) \\ &= 0.02(0.0001) + 0.999(0.9999) \\ &= 0.9989021,\end{aligned}$$

or we can realize

$$\mathbb{P}(P^c) = 1 - \mathbb{P}(P) = 1 - 0.0010979 = 0.9989021.$$

Completing the computation gives

$$\begin{aligned}\mathbb{P}(I|P^c) &= \frac{\mathbb{P}(P^c|I)\mathbb{P}(I)}{\mathbb{P}(P^c)} \\ &= \frac{0.02(0.0001)}{0.9989021} \approx 2 \times 10^{-6}.\end{aligned}$$

Despite the very poor reliability of a positive test result, a negative test result is pretty significant. If a person tests negative, then they are very likely to be healthy. If a person receives a positive test result, they are still likely to be healthy but should be retested or sent off for a costlier, more accurate test. ■

The example above, while certainly contrived, demonstrates some very important facts to know about medical tests. The *sensitivity* of a test is the probability that a person who has the disease tests positive for it while the *specificity* of the test is the probability that a disease-free individual tests negative. In terms of conditional probabilities, these are given by

$$\begin{aligned}\text{sensitivity} &= \mathbb{P}(\text{test is positive} \mid \text{individual has disease}), \\ \text{specificity} &= \mathbb{P}(\text{test is negative} \mid \text{individual is disease-free}).\end{aligned}$$

These quantities are related to the rate of false positives and negatives in the obvious way:

$$\begin{aligned}\mathbb{P}(\text{false positive}) &= \mathbb{P}(\text{test is positive} \mid \text{individual is disease-free}) \\ &= 1 - \mathbb{P}(\text{test is negative} \mid \text{individual is disease-free}) \\ &= 1 - \text{specificity}, \\ \mathbb{P}(\text{false negative}) &= \mathbb{P}(\text{test is negative} \mid \text{individual has disease}) \\ &= 1 - \mathbb{P}(\text{test is positive} \mid \text{individual has disease}) \\ &= 1 - \text{sensitivity}.\end{aligned}$$

The sensitivity and specificity of a test are often confused for their reverse conditional statements. The *positive predictive value* (PPV) of a test is the probability that an individual actually has the disease if they have tested positive for it. Similarly, the *negative predictive value* (NPV) is the probability that a person is actually disease-free if they test negative.

$$\text{PPV} = \mathbb{P}(\text{individual has disease} \mid \text{test is positive})$$

$$\text{NPV} = \mathbb{P}(\text{individual is disease-free} \mid \text{test is negative})$$

Bayes' Formula gives the connection between these quantities. As in the example above, let P be the event that an individual tests positive for the disease and I the event that they actually have the disease.

$$\begin{aligned} \text{PPV} &= \mathbb{P}(I|P) \\ &= \frac{\mathbb{P}(P|I)\mathbb{P}(I)}{\mathbb{P}(P)} \\ &= \frac{\mathbb{P}(P|I)\mathbb{P}(I)}{\mathbb{P}(P|I)\mathbb{P}(I) + \mathbb{P}(P|I^c)\mathbb{P}(I^c)} \\ &= \frac{\mathbb{P}(P|I)\mathbb{P}(I)}{\mathbb{P}(P|I)\mathbb{P}(I) + (1 - \mathbb{P}(P^c|I^c))\mathbb{P}(I^c)} \end{aligned}$$

In the next to last step, we conditioned on whether a person has the disease to compute the overall probability of receiving a positive test. For the final step, we used the complementary event to work in the specificity of the test, $\mathbb{P}(P^c|I^c)$. Notice that the positive predictive value depends on both the sensitivity and specificity of the test as well as the overall prevalence of the disease in the population, $\mathbb{P}(I)$. Using similar computations, we have

$$\text{NPV} = \frac{\mathbb{P}(P^c|I^c)\mathbb{P}(I^c)}{(1 - \mathbb{P}(P|I))\mathbb{P}(I) + \mathbb{P}(P^c|I^c)\mathbb{P}(I^c)}.$$

In the previous example, our test for bubonic plague had a sensitivity of 98% and a specificity of 99.9%. Despite these very favorable numbers, we saw that a person who tested positive had a fairly high probability of being disease-free. In fact, the PPV for the test was only about 9%. While the test has very good sensitivity and specificity, the number of people actually infected with the disease is so low that the number of false positives is significantly larger than the number of true positives. What we take away from this

is that even a remarkably accurate test can have low positive predictive value if the disease is fairly rare. Similarly, a reasonably good test can have poor negative predictive value if the disease is very common in the population.

Example 6.7.4. The Centers for Disease Control and Prevention (CDC) sets minimum standards that a rapid diagnostic test for the flu must meet. In particular, they require that a test have a sensitivity of at least 80% and a specificity of at least 95%. If a test meets the minimum requirements and the flu is suspected to infect 15% of the population, what are the positive and negative predictive values of the test?

Answer: We have

$$\begin{aligned} \text{sensitivity} &= \mathbb{P}(P|I) = 0.8, \\ \text{specificity} &= \mathbb{P}(P^c|I^c) = 0.95, \\ \text{prevalence} &= \mathbb{P}(I) = 0.15. \end{aligned}$$

Using the formulas developed above:

$$\begin{aligned} \text{PPV} &= \frac{\mathbb{P}(P|I)\mathbb{P}(I)}{\mathbb{P}(P|I)\mathbb{P}(I) + (1 - \mathbb{P}(P^c|I^c))\mathbb{P}(I^c)} \\ &= \frac{(0.8)(0.15)}{(0.8)(0.15) + (0.05)(0.85)} \approx 0.7385, \end{aligned}$$

$$\begin{aligned} \text{NPV} &= \frac{\mathbb{P}(P^c|I^c)\mathbb{P}(I^c)}{(1 - \mathbb{P}(P|I))\mathbb{P}(I) + \mathbb{P}(P^c|I^c)\mathbb{P}(I^c)} \\ &= \frac{(0.95)(0.85)}{(0.2)(0.15) + (0.95)(0.85)} \approx 0.9642. \end{aligned}$$

This means that a person who tests positive has slightly over a 1 in 4 chance of not having the flu and should probably be given a more accurate test to confirm the results. On the other hand, a person who tests negative for the flu can be fairly confident they are not infected. ■

Example 6.7.5. An antibody test for a disease checks for immunological factors that indicate whether a person has had the disease in the past. So, a positive test hopefully indicates that a person has recovered from the disease and has some immunity to it. During the Covid-19 outbreak, many antibody

tests were produced with doubtful efficacy. One such test had a sensitivity of around 89% and a specificity of about 91%. If the novel coronavirus infected 5% of the total population at the time the test was administered, what are the positive and negative predictive values of this test? If the specificity is 95%, what is the largest PPV we can hope for?

Answer:

$$\begin{aligned}\text{PPV} &= \frac{\mathbb{P}(P|I)\mathbb{P}(I)}{\mathbb{P}(P|I)\mathbb{P}(I) + (1 - \mathbb{P}(P^c|I^c))\mathbb{P}(I^c)} \\ &= \frac{(0.89)(0.05)}{(0.89)(0.05) + (0.09)(0.95)} \approx 0.3423,\end{aligned}$$

$$\begin{aligned}\text{NPV} &= \frac{\mathbb{P}(P^c|I^c)\mathbb{P}(I^c)}{(1 - \mathbb{P}(P|I))\mathbb{P}(I) + \mathbb{P}(P^c|I^c)\mathbb{P}(I^c)} \\ &= \frac{(0.91)(0.95)}{(0.11)(0.05) + (0.91)(0.95)} \approx 0.9937.\end{aligned}$$

So, a person receiving a negative test result has very likely not had the disease. On the other hand, a person receiving a positive test only has about a 34% chance of actually having the antibodies for the coronavirus! As such, relying on this test to determine who has been exposed to the disease is highly dubious.

Now suppose the specificity is 95%. To get the best PPV we assume the test is 100% sensitive.¹

$$\begin{aligned}\text{PPV} &= \frac{\mathbb{P}(P|I)\mathbb{P}(I)}{\mathbb{P}(P|I)\mathbb{P}(I) + (1 - \mathbb{P}(P^c|I^c))\mathbb{P}(I^c)} \\ &= \frac{(1)(0.05)}{(1)(0.05) + (0.05)(0.95)} \approx 0.5128.\end{aligned}$$

So even with 100% sensitivity, there is little better than a 50-50 chance that a person who tests positive for antibodies actually has immunity to the virus!

■

¹While this may seem obvious, letting the sensitivity be x gives

$$\text{PPV} = \frac{x}{x + 0.95}$$

which is a strictly increasing function over the range $0 \leq x \leq 1$.

6.8 Expected Value, Variance, and Standard Deviation

Now that we have discussed the basics of probability, we want to explore some quantities associated to a random variable which attempt to give us some intuition about “typical outcomes” and “generic behaviors” that we can expect from experiments modeled by the random variable. There are too many such quantities for us to cover here. Instead, we focus on the three which are arguably the most important: *expected value*, *variance*, and *standard deviation*.

Let X be a discrete random variable with sample space

$$S = \{d_1, d_2, \dots, d_N\}$$

where the outcomes are assumed to be *real numbers*. Let the probabilities of the individual outcomes be given by

$$\mathbb{P}(\{d_i\}) = p_i.$$

Then the **expected value** of the random variable X (denoted $\mathbb{E}[X]$ or $\bar{X}$) is given by

$$\mathbb{E}[X] = p_1 \cdot d_1 + p_2 \cdot d_2 + \dots + p_N \cdot d_N.$$

In other words, the expected value of any discrete random variable is just the sum of “outcome $\times$ probability” over all possible outcomes. As the name suggests, the expected value of a random variable X is the value we would expect to see on average. That is, if we do a large number of independent trials of the experiment modeled by X and average all of the outcomes, the result should be roughly $\mathbb{E}[X]$.² For this reason, the expected value of X is also called the *average value* or the *mean value* of X .

Example 6.8.1. Let R be the sum of two fair six-sided dice. What is the expected value of R ?

Answer: The sample space for R is $S = \{2, 3, 4, \dots, 12\}$ and the probabilities of these outcomes were given in Example 6.2.5. For the expected

²In more advanced treatments of probability, this fact is known as the **law of large numbers**, but the proof of this important theorem is beyond the scope of this text.

value, all we need to do is sum up “outcome $\times$ probability” for all 11 outcomes.

$$\begin{aligned}\mathbb{E}[R] &= 2 \cdot \frac{1}{36} + 3 \cdot \frac{1}{18} + 4 \cdot \frac{1}{12} + 5 \cdot \frac{1}{9} + 6 \cdot \frac{5}{36} + 7 \cdot \frac{1}{6} \\ &\quad + 8 \cdot \frac{5}{36} + 9 \cdot \frac{1}{9} + 10 \cdot \frac{1}{12} + 11 \cdot \frac{1}{18} + 12 \cdot \frac{1}{36} \\ &= 7.\end{aligned}$$

So, if we roll our dice a large number of times and average the results, we would expect to see 7 on average. This hardly seems surprising since 7 is in the middle of the outcomes, and the probabilities are exactly symmetric for outcomes around 7. ■

In the last example, the expected value also happened to be the outcome with the highest probability.³ This is certainly not the case in general. In fact, the expected value of a random variable may not be one of the possible outcomes!

Example 6.8.2. A weighted six-sided die has the property that the even numbers are twice as likely as the odd numbers. What is the expected value for this die?

Answer: Let D be the outcome of rolling this die. Clearly the sample space is $S = \{1, 2, 3, 4, 5, 6\}$. If

$$\mathbb{P}(D = 1) = \mathbb{P}(D = 3) = \mathbb{P}(D = 5) = p$$

then

$$\mathbb{P}(D = 2) = \mathbb{P}(D = 4) = \mathbb{P}(D = 6) = 2p$$

since the even numbers must be twice as likely as the odd numbers. Since the probabilities must add up to 1, we have

$$\begin{aligned}1 &= \mathbb{P}(D = 1) + \mathbb{P}(D = 2) + \mathbb{P}(D = 3) + \mathbb{P}(D = 4) + \mathbb{P}(D = 5) + \mathbb{P}(D = 6) \\ &= p + 2p + p + 2p + p + 2p \\ &= 9p.\end{aligned}$$

³The outcome (or outcomes) with the highest probability is the **mode** of the discrete random variable.

So $p = 1/9$, the odds have probability $1/9$, and the evens have probability $2/9$. For the expected value, we have

$$\begin{aligned}\mathbb{E}[D] &= 1 \cdot \frac{1}{9} + 2 \cdot \frac{2}{9} + 3 \cdot \frac{1}{9} + 4 \cdot \frac{2}{9} + 5 \cdot \frac{1}{9} + 6 \cdot \frac{2}{9} \\ &= \frac{11}{3} = 3.\bar{6}. \blacksquare\end{aligned}$$

The expected value gives us a measure of the size of a typical outcome, but a particular outcome may be quite far from this value. It would be beneficial to have some quantity which would tell us how tightly grouped outcomes tend to be around the expected value. While there are several candidates, the most important is the standard deviation.⁴ The standard deviation is obtained by first computing a related value – the variance.

Let X be a discrete random variable with sample space

$$S = \{d_1, d_2, \dots, d_N\}$$

where the outcomes are assumed to be *real numbers*. Let the probabilities of the individual outcomes be given by

$$\mathbb{P}(\{d_i\}) = p_i.$$

Then the **variance** of X (denoted $\text{Var}(X)$) is given by

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= p_1 \cdot (d_1 - \mathbb{E}[X])^2 + p_2 \cdot (d_2 - \mathbb{E}[X])^2 + \dots + p_N \cdot (d_N - \mathbb{E}[X])^2.\end{aligned}$$

The **standard deviation** of X (denoted $\sigma(X)$) is simply the square root of the variance:

$$\sigma(X) = \sqrt{\text{Var}(X)}.$$

Lemma 6.8.1.

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p_1 \cdot d_1^2 + p_2 \cdot d_2^2 + \dots + p_N \cdot d_N^2 - (\mathbb{E}[X])^2.$$

Proof: Using the identity

$$(x - y)^2 = x^2 - 2xy + y^2$$

⁴Another popular group of quantities are **percentiles** – which include the **quartiles** and **median**.

on each of the terms, we get

$$\begin{aligned}
 \text{Var}(X) &= p_1 \cdot (d_1 - \mathbb{E}[X])^2 + p_2 \cdot (d_2 - \mathbb{E}[X])^2 + \cdots + p_N \cdot (d_N - \mathbb{E}[X])^2 \\
 &= p_1 \cdot (d_1^2 - 2d_1\mathbb{E}[X] + \mathbb{E}[X]^2) + p_2 \cdot (d_2^2 - 2d_2\mathbb{E}[X] + \mathbb{E}[X]^2) \\
 &\quad + \cdots + p_N \cdot (d_N^2 - 2d_N\mathbb{E}[X] + \mathbb{E}[X]^2) \\
 &= p_1 \cdot d_1^2 + p_2 \cdot d_2^2 + \cdots + p_N \cdot d_N^2 \\
 &\quad - 2(p_1 \cdot d_1 + p_2 \cdot d_2 + \cdots + p_N \cdot d_N) \mathbb{E}[X] \\
 &\quad + (p_1 + p_2 + \cdots + p_N) \mathbb{E}[X]^2 \\
 &= p_1 \cdot d_1^2 + p_2 \cdot d_2^2 + \cdots + p_N \cdot d_N^2 - 2\mathbb{E}[X]^2 + 1\mathbb{E}[X]^2 \\
 &= p_1 \cdot d_1^2 + p_2 \cdot d_2^2 + \cdots + p_N \cdot d_N^2 - \mathbb{E}[X]^2. \blacksquare
 \end{aligned}$$

So to compute the variance for a discrete random variable, we sum up “outcome² $\times$ probability” over all possibilities *and then subtract the square of the expected value*. Intuitively, you should think of the expected value as the center of the possible outcomes while the standard deviation gives us a range around this central point where we would typically expect to see most of the outcomes.

Example 6.8.3. Find the expected value, variance, and standard deviation for the sum of two fair six-sided dice.

Answer: As before, we let R be the sum of the two dice. We have the probabilities for all outcomes listed in Example 6.2.5 and the expected value was already computed to be $\mathbb{E}[R] = 7$ in Example 6.8.1. The variance is

$$\begin{aligned}
 \text{Var}(R) &= 2^2 \cdot \frac{1}{36} + 3^2 \cdot \frac{1}{18} + 4^2 \cdot \frac{1}{12} + 5^2 \cdot \frac{1}{9} + 6^2 \cdot \frac{5}{36} + 7^2 \cdot \frac{1}{6} \\
 &\quad + 8^2 \cdot \frac{5}{36} + 9^2 \cdot \frac{1}{9} + 10^2 \cdot \frac{1}{12} + 11^2 \cdot \frac{1}{18} + 12^2 \cdot \frac{1}{36} - (7)^2 \\
 &= \frac{35}{6}.
 \end{aligned}$$

The standard deviation of R is

$$\sigma(R) = \sqrt{\frac{35}{6}} \approx 2.415.$$

Taken together, these quantities tell us that if we roll two dice and look at their sum, then the outcome will be 7 on average, and most outcomes will lie between $\mathbb{E}[R] - \sigma(R) \approx 4.585$ and $\mathbb{E}[R] + \sigma(R) \approx 9.415$. That is, most rolls will be 5, 6, 7, 8, or 9 (which accounts for 66.6% of all possible outcomes). $\blacksquare$

Example 6.8.4. We flip a fair coin 6 times. Let H be the number of heads in the outcome of the experiment. Find the expected value, variance, and standard deviation of H .

Answer: Since H is the number of heads in 6 coin flips, the sample space must be

$$S = \{0, 1, 2, 3, 4, 5, 6\}.$$

We need to compute the probabilities of all these outcomes. Note that there are $2^6 = 64$ possible outcomes.

$$\mathbb{P}(H = 0) = \mathbb{P}(\{\text{all tails}\}) = \frac{1}{64}$$

$$\mathbb{P}(H = 1) = \frac{6}{64} = \frac{3}{32} \quad (\text{pick the position for heads, all rest tails})$$

$$\mathbb{P}(H = 2) = \frac{\binom{6}{2}}{64} = \frac{15}{64} \quad (\text{pick two positions for heads, all rest tails})$$

$$\mathbb{P}(H = 3) = \frac{\binom{6}{3}}{64} = \frac{20}{64} = \frac{5}{16} \quad (\text{pick three positions for heads, all rest tails})$$

$$\mathbb{P}(H = 4) = \frac{\binom{6}{4}}{64} = \frac{15}{64} \quad (\text{pick four positions for heads, all rest tails})$$

$$\mathbb{P}(H = 5) = \frac{\binom{6}{5}}{64} = \frac{6}{64} = \frac{3}{32} \quad (\text{pick five positions for heads, all rest tails})$$

$$\mathbb{P}(H = 6) = \mathbb{P}(\{\text{all heads}\}) = \frac{1}{64}$$

So, the expected value of H is

$$\begin{aligned} \mathbb{E}[H] &= 0 \cdot \frac{1}{64} + 1 \cdot \frac{3}{32} + 2 \cdot \frac{15}{64} + 3 \cdot \frac{5}{16} + 4 \cdot \frac{15}{64} + 5 \cdot \frac{3}{32} + 6 \cdot \frac{1}{64} \\ &= 3 \end{aligned}$$

which is thoroughly unsurprising. For the variance, we have

$$\begin{aligned} \text{Var}(H) &= 0^2 \cdot \frac{1}{64} + 1^2 \cdot \frac{3}{32} + 2^2 \cdot \frac{15}{64} + 3^2 \cdot \frac{5}{16} + 4^2 \cdot \frac{15}{64} + 5^2 \cdot \frac{3}{32} + 6^2 \cdot \frac{1}{64} - (3)^2 \\ &= \frac{3}{2}, \end{aligned}$$

and so the standard deviation is

$$\sigma(H) = \sqrt{\frac{3}{2}} \approx 1.225.$$

This means we expect 3 heads on average, with most results being 2, 3, or 4 heads (which accounts for just over 78% of the total probability). ■

Next, we note two basic results about expected value and variance that are crucial in applications. The proofs of these results are beyond the scope of this book, but see [Kh07] and [Rs03] for all relevant details.

Theorem 6.8.2 (Linearity of Expected Value). *Let X_1 and X_2 be two random variables and a and b be real numbers. Then*

$$\mathbb{E}[aX_1 + bX_2] = a\mathbb{E}[X_1] + b\mathbb{E}[X_2].$$

Theorem 6.8.3 (Variance of Independent Random Variables). *Let X_1 and X_2 be independent random variables ⁵ and a, b and c be real numbers. Then*

$$\text{Var}(aX_1 + bX_2 + c) = a^2 \text{Var}(X_1) + b^2 \text{Var}(X_2).$$

Example 6.8.5. Consider the experiment where we roll 2 fair six-sided dice. Let R be the sum of the two dice. Use the theorems above to compute the expected value and variance of R .

Answer: We can think of R as the sum of two independent random variables: D_1 is the outcome of the first die, and D_2 is the outcome of the second. You should verify that

$$\begin{aligned}\mathbb{E}[D_1] &= \mathbb{E}[D_2] = 3.5, \\ \text{Var}(D_1) &= \text{Var}(D_2) = \frac{35}{12}.\end{aligned}$$

Using the two results above:

$$\mathbb{E}[R] = \mathbb{E}[D_1 + D_2] = \mathbb{E}[D_1] + \mathbb{E}[D_2] = 3.5 + 3.5 = 7,$$

$$\text{Var}(R) = \text{Var}(D_1 + D_2) = \text{Var}(D_1) + \text{Var}(D_2) = \frac{35}{12} + \frac{35}{12} = \frac{35}{6}.$$

Notice that these results match Example 6.8.3.

⁵This theorem is actually true for *uncorrelated* random variables. This notion is explored in more advanced treatments of probability.

6.9 A Catalog of Important Discrete Random Variables

In this section, we list a number of common discrete random variables and try to give an indication of typical situations they model. We also provide the expected value and standard deviation. While it is not important to memorize every last detail about these random variables, it is useful to know the sorts of experiments they are meant to describe.

6.9.1 The Bernoulli Random Variable

The Bernoulli random variable is the most basic of all discrete random variables. It models any situation where there are two outcomes – a “success” and a “failure”. The random variable has a single parameter p (with $0 \leq p \leq 1$) which is the probability of a success.

To make things definite, we take a **Bernoulli random variable** with probability of success p to be a random variable X whose sample space is $S = \{0, 1\}$ having probabilities

$$\begin{aligned}\mathbb{P}(X = 0) &= 1 - p, \\ \mathbb{P}(X = 1) &= p.\end{aligned}$$

We often write $X \sim \text{Bernoulli}(p)$ to indicate that X is a Bernoulli random variable with probability of success p .

Notice that we use the number 0 to denote a “failure” and 1 to denote a “success.” We have made extensive use of one kind of Bernoulli random variable already; namely, a fair coin is simply $\text{Bernoulli}(1/2)$. The only apparent difference is that for a fair coin we used H and T for the outcomes, but 0 and 1 work just as well!

In many treatments, the quantity $1 - p$ is given its own name: $q = 1 - p$. In terms of this new quantity, we have

$$\begin{aligned}\mathbb{P}(X = 0) &= q, \\ \mathbb{P}(X = 1) &= p,\end{aligned}$$

where $p + q = 1$. There is absolutely no new content in this version of the Bernoulli random variable, but the quantity $1 - p$ shows up often enough in results that having a name for it can be useful.

Considering how basic this random variable is, there is not too much more to say about it. Its importance lies in the fact that several of the other random variables we discuss are defined in terms of it. The only additional thing we mention is the expected value, variance, and standard deviation of these random variables.

Lemma 6.9.1 (Expected Value, Variance, and Standard Deviation for Bernoulli(p)). *Let $X \sim \text{Bernoulli}(p)$. Then*

$$\begin{aligned}\mathbb{E}[X] &= p, \\ \text{Var}(X) &= p(1 - p) = pq, \\ \sigma(X) &= \sqrt{p(1 - p)} = \sqrt{pq}.\end{aligned}$$

Proof:

$$\begin{aligned}\mathbb{E}[X] &= 0(1 - p) + 1(p) = p, \\ \text{Var}(X) &= 0^2(1 - p) + 1^2(p) - (p)^2 = p - p^2 = p(1 - p). \blacksquare\end{aligned}$$

6.9.2 The Binomial Random Variable

Imagine an experiment where we run n independent trials of a Bernoulli(p) random variable. In other words, we run n independent experiments where the outcome of each separate experiment is a success with probability p (and a failure with probability $q = 1 - p$). The binomial random variable is the random variable which counts the *total number of successes* out of the n trials.

So, a **binomial random variable** with n trials and a probability of success p is a random variable X with sample space

$$S = \{0, 1, 2, \dots, n - 1, n\},$$

and probabilities

$$\mathbb{P}(X = i) = \binom{n}{i} p^i (1 - p)^{n-i} = \binom{n}{i} p^i q^{n-i}.$$

We write $X \sim \text{binomial}(n, p)$ to indicate that X is a binomial random variable with n trials and a probability of success p .

The formula for the probabilities makes sense from basic counting considerations. We have n trials and we want to know the probability that i of them

are successes. From a counting perspective, we have to choose i positions out of the n trials to place the successes – which explains the appearance of the binomial coefficient! The remaining part is simply the probability of any single outcome with i successes and $n - i$ failures.

Example 6.9.1. Suppose that we flip a fair coin 20 times. What is the probability of getting exactly 7 heads? What is the most likely outcome?

Answer: Let H be the random variable that counts the number of heads in our 20 coin tosses. Since the probability of getting heads on a fair coin is $1/2$, it follows that

$$H \sim \text{binomial}\left(20, \frac{1}{2}\right).$$

Using the formula above

$$\mathbb{P}(H = 7) = \binom{20}{7} \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^{13} = \frac{77520}{2^{20}} = \frac{4845}{65536} \approx 0.0739.$$

As we have seen in similar cases, the most likely outcome is exactly in the middle $H = 10$ (i.e. 10 heads). Below, we just list a few probabilities around $H = 10$.

$$\begin{aligned} \mathbb{P}(H = 8) &= \binom{20}{8} \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^{12} = \frac{125970}{2^{20}} \approx 0.1201, \\ \mathbb{P}(H = 9) &= \binom{20}{9} \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^{11} = \frac{167960}{2^{20}} \approx 0.1602, \\ \mathbb{P}(H = 10) &= \binom{20}{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^{10} = \frac{184756}{2^{20}} \approx 0.1762, \\ \mathbb{P}(H = 11) &= \binom{20}{11} \left(\frac{1}{2}\right)^{11} \left(\frac{1}{2}\right)^9 = \frac{167960}{2^{20}} \approx 0.1602, \\ \mathbb{P}(H = 12) &= \binom{20}{12} \left(\frac{1}{2}\right)^{12} \left(\frac{1}{2}\right)^8 = \frac{125970}{2^{20}} \approx 0.1201. \blacksquare \end{aligned}$$

Before giving another example, we compute the expected value, variance, and standard deviation for all binomial random variables.

Lemma 6.9.2 (Expected Value, Variance, and Standard Deviation for binomial(n, p)).
Let $X \sim \text{binomial}(n, p)$. Then

$$\begin{aligned}\mathbb{E}[X] &= np, \\ \text{Var}(X) &= np(1 - p) = npq, \\ \sigma(X) &= \sqrt{np(1 - p)} = \sqrt{npq}.\end{aligned}$$

Proof: Let $X \sim \text{binomial}(n, p)$. Then we can think of X as

$$X = X_1 + X_2 + \cdots + X_n$$

where the random variables X_i are all Bernoulli(p) and independent of one another. Using Theorems 6.8.2 and 6.8.3, we have

$$\begin{aligned}\mathbb{E}[X] &= \mathbb{E}[X_1 + X_2 + \cdots + X_n] \\ &= \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_n] \\ &= p + p + \cdots + p \\ &= np, \\ \text{Var}(X) &= \text{Var}(X_1 + X_2 + \cdots + X_n) \\ &= \text{Var}(X_1) + \text{Var}(X_2) + \cdots + \text{Var}(X_n) \\ &= p(1 - p) + p(1 - p) + \cdots + p(1 - p) \\ &= np(1 - p). \blacksquare\end{aligned}$$

Example 6.9.2. Let H be the total number of heads in 20 tosses of a fair coin. What are the expected value and standard deviation for H ?

Answer: We have $H \sim \text{binomial}(20, 1/2)$. So,

$$\begin{aligned}\mathbb{E}(H) &= 20 \left(\frac{1}{2} \right) = 10, \\ \text{Var}(H) &= 20 \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) = 5, \\ \sigma(H) &= \sqrt{5} \approx 2.2361.\end{aligned}$$

So, we expect to see 10 heads out of 20 coin flips on average, with most results being 8, 9, 10, 11, or 12 heads total. Looking back at the probabilities we computed for this case, we see that these 5 outcomes account for about 73.7% of the probability. $\blacksquare$

Example 6.9.3. Dichromatism is a genetic condition that affects 5% of men and causes a form of color blindness. Suppose that we pick a sample of 1000 men at random. How many men in this sample should we expect to have dichromatism? What is the probability that we have 5 or fewer men with the condition?

Answer: Let X be the total number of men in the sample who have dichromatism. Then

$$X \sim \text{binomial}(1000, 0.05)$$

since we can think of each person as an independent Bernoulli trial with “success” meaning the person has the condition. So,

$$\begin{aligned}\mathbb{E}(X) &= 1000(0.05) = 50, \\ \text{Var}(X) &= 1000(0.05)(0.95) = 47.5, \\ \sigma(X) &= \sqrt{47.5} \approx 6.892.\end{aligned}$$

So we would expect anywhere between 44 and 56 men in the sample to have dichromatism with an expected value of 50.

The probability that we have 5 or fewer men with the condition is

$$\begin{aligned}\mathbb{P}(0 \leq X \leq 5) &= \mathbb{P}(X = 0) + \mathbb{P}(X = 1) + \mathbb{P}(X = 2) + \mathbb{P}(X = 3) \\ &\quad + \mathbb{P}(X = 4) + \mathbb{P}(X = 5) \\ &= \binom{1000}{0}(0.05)^0(0.95)^{1000} + \binom{1000}{1}(0.05)^1(0.95)^{999} \\ &\quad + \binom{1000}{2}(0.05)^2(0.95)^{998} + \binom{1000}{3}(0.05)^3(0.95)^{997} \\ &\quad + \binom{1000}{4}(0.05)^4(0.95)^{996} + \binom{1000}{5}(0.05)^5(0.95)^{995} \\ &\approx 1.9 \times 10^{-16}. \blacksquare\end{aligned}$$

The probability computation in this last example is VERY tedious due to the fact that n is rather large. See Appendix E for information on how to compute these probabilities using the TI-83/84 and TI-89.

It turns out that for cases when the number of trials n is quite large and the probability of success p is quite small, $X \sim \text{binomial}(n, p)$ is well approximated by a slightly simpler random variable – the Poisson random

variable. In other words, the Poisson random variable is a very good approximation to a binomial random variable when we are doing a rather large number of independent trials where the probability of success in each trial is rather small (so successes are *rare events*). We will discuss the Poisson random variable in Section 6.9.4.

6.9.3 The Geometric Random Variable

Consider the setup for the binomial random variable; we perform n independent Bernoulli(p) trials and count the number of successes. Suppose instead that we keep performing the independent trials *until the first success occurs* in which case we stop. Let T be the total number of trials performed until the first success (which also counts the final trial which ends in success). This sort of random variable is called a geometric random variable.

A **geometric random variable** with probability of success p is a random variable T with sample space

$$S = \{1, 2, 3, 4, \dots\}$$

and probabilities given by

$$\mathbb{P}(T = i) = (1 - p)^{i-1}p = q^{i-1}p.$$

We write $T \sim \text{geometric}(p)$ to indicate that T is a geometric random variable with a probability of success p .

There are a few things to note about this random variable. First, the sample space consists of *all positive integers*! This is simply because we could have an arbitrarily large number of failures until the first success occurs. The probabilities are easy to understand since if $T = i$, we had $i - 1$ failures (each with probability $1 - p$) and then a final success (with probability p). We will discuss the proof that these probabilities sum to one later.

Example 6.9.4. Suppose we flip a fair coin until we get the first tails. What is the probability that the first tails appears in the first three tosses? What is the probability that we have to flip the coin 5 or more times?

Let T be the number of flips until the first tails appears. Then $T \sim$

geometric($1/2$). We are being asked for the probability that $1 \leq T \leq 3$. So,

$$\begin{aligned}\mathbb{P}(1 \leq T \leq 3) &= \mathbb{P}(T = 1) + \mathbb{P}(T = 2) + \mathbb{P}(T = 3) \\ &= \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \\ &= \frac{7}{8}. \blacksquare\end{aligned}$$

For the second question, we are being asked $\mathbb{P}(T \geq 5)$. Attempting this directly would require us to add up an infinite number of terms (which turns out to be easily doable in this case). However, it is probably easier at this point to look at the opposite event:

$$\begin{aligned}\mathbb{P}(T \geq 5) &= 1 - \mathbb{P}(T < 5) \\ &= 1 - \mathbb{P}(T = 1) - \mathbb{P}(T = 2) - \mathbb{P}(T = 3) - \mathbb{P}(T = 4) \\ &= 1 - \frac{1}{2} - \frac{1}{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \\ &= \frac{1}{16}. \blacksquare\end{aligned}$$

Lemma 6.9.3 (Expected Value, Variance, and Standard Deviation for geometric(p)).
Let $T \sim \text{geometric}(p)$. Then

$$\begin{aligned}\mathbb{E}[T] &= \frac{1}{p}, \\ \text{Var}(T) &= \frac{1-p}{p^2} = \frac{q}{p^2}, \\ \sigma(T) &= \frac{\sqrt{1-p}}{p} = \frac{\sqrt{q}}{p}.\end{aligned}$$

Proof: The computation of these quantities is a little cumbersome (and rely on results about power series). C.f. [Rs03][Chapter 2] for most of the details. However, we will discuss the proof that the probabilities add up to 1.

Using summation notation, we have

$$\sum_{i=1}^{\infty} \mathbb{P}(T = i) = \sum_{i=1}^{\infty} q^{i-1} p = p \sum_{i=1}^{\infty} q^{i-1},$$

where q is a real number in the range $0 < q < 1$. We just need to focus on the summation, which we call S for convenience. Notice that

$$S = \sum_{i=1}^{\infty} q^{i-1} = 1 + q + q^2 + q^3 + q^4 + \cdots$$

If we multiply S by q , we see an interesting pattern

$$\begin{aligned} S &= \sum_{i=1}^{\infty} q^{i-1} = 1 + q + q^2 + q^3 + q^4 + \cdots \\ qS &= \sum_{i=1}^{\infty} q^i = q + q^2 + q^3 + q^4 + \cdots \end{aligned}$$

If we subtract these two quantities, all of the terms involving q cancel out, leaving a 1. That means,

$$S - qS = (1 - q)S = 1.$$

Solving for S gives

$$S = \frac{1}{1 - q} = \frac{1}{p}.$$

The last equality follows from the fact that $p + q = 1$. Putting everything together, we have

$$\sum_{i=1}^{\infty} \mathbb{P}(T = i) = p \sum_{i=1}^{\infty} q^{i-1} = \frac{p}{p} = 1. \quad \blacksquare$$

Example 6.9.5. Consider again dichromatism in men (which has a probability of 0.05). How many men should we expect to sample at random until we find someone with the condition? What is the standard deviation for this number?

Answer: Let N be the number of men sampled until we find one with dichromatism. Then $N \sim \text{geometric}(0.05)$. According to the formulas above

$$\begin{aligned} \mathbb{E}[N] &= \frac{1}{0.05} = 20, \\ \text{Var}(N) &= \frac{0.95}{(0.05)^2} = 380, \\ \sigma(N) &= \sqrt{380} \approx 19.49. \end{aligned}$$

So we would expect to sample 20 men before finding one with dichromatism (i.e. 19 men without the condition followed by 1 with it), but the typical range is anywhere between 1 and 39 men. Using a calculator (see Appendix E) or computer, we can compute that

$$\begin{aligned}\mathbb{P}(1 \leq N \leq 39) &= \mathbb{P}(N = 1) + \mathbb{P}(N = 2) + \cdots + \mathbb{P}(N = 39) \\ &\approx 0.8647. \blacksquare\end{aligned}$$

6.9.4 The Poisson Random Variable

A **Poisson random variable** with positive parameter λ is a random variable X with sample space

$$S = \{0, 1, 2, 3, 4, \dots\}$$

and probabilities given by

$$\mathbb{P}(X = i) = e^{-\lambda} \frac{\lambda^i}{i!}.$$

The proof that these numbers sum to one is a lot more involved than for the geometric random variable, but power series techniques prove that they do. We write $X \sim \text{Poisson}(\lambda)$ to indicate that X is a Poisson random variable with parameter $\lambda > 0$.

Lemma 6.9.4 (Expected Value, Variance, and Standard Deviation for $\text{Poisson}(\lambda)$). *Let $X \sim \text{Poisson}(\lambda)$. Then*

$$\begin{aligned}\mathbb{E}[X] &= \lambda, \\ \text{Var}(X) &= \lambda, \\ \sigma(X) &= \sqrt{\lambda}.\end{aligned}$$

Proof: Again, the computations here are a little tricky. C.f. [Rs03][Chapter 2] for most of the details. $\blacksquare$

Example 6.9.6. Suppose that the number of typos on any given page of a textbook has a Poisson distribution with $\lambda = 2$. How many typos per page would we expect on average? What is the standard deviation for this expected value? What is the probability that any given page has at least one typo?

Answer: Let T be the number of typos on a page. Then we are told $T \sim \text{Poisson}(2)$. From the formulas above,

$$\begin{aligned}\mathbb{E}[T] &= 2, \\ \text{Var}(T) &= 2, \\ \sigma(T) &= \sqrt{2} \approx 1.414.\end{aligned}$$

So, we would expect about 2 typos per page with typical pages having 1, 2, or 3 typos. The probability of these three outcomes is

$$\begin{aligned}\mathbb{P}(T = 1, 2, \text{ or } 3) &= \mathbb{P}(T = 1) + \mathbb{P}(T = 2) + \mathbb{P}(T = 3) \\ &= e^{-2} \frac{2^1}{1!} + e^{-2} \frac{2^2}{2!} + e^{-2} \frac{2^3}{3!} \\ &= e^{-2} \left(\frac{16}{3} \right) \approx 0.7218.\end{aligned}$$

The second question is asking for the probability that $T \geq 1$. Rather than summing up an infinite number of terms, it is easier to look at the complementary event:

$$\begin{aligned}\mathbb{P}(T \geq 1) &= 1 - \mathbb{P}(T = 0) \\ &= 1 - e^{-2} \frac{2^0}{0!} = 1 - e^{-2} \approx 0.8647. \blacksquare\end{aligned}$$

As we mentioned above, the Poisson random variable is a decent approximation of a binomial random variable in the case when the number of trials n is quite large, and the probability of success p is quite small. In such a case, we can take $\lambda = np$ (so that the expected values match) and use the Poisson random variable instead of the binomial since probabilities for Poisson random variables are typically easier to compute. Intuitively,

$$\text{binomial}(n, p) \approx \text{Poisson}(np) \text{ when } n \text{ is very large and } p \text{ is very small.}$$

In practical terms, this means that a Poisson random variable is a *decent approximation* to the number of successes in a very large number of trials where the occurrence of a success is relatively rare.

Example 6.9.7. Consider dichromatism in men (which occurs with a probability of 0.05). Suppose that we pick a sample of 1000 men at random. How many men in this sample should we expect to have dichromatism? What is the probability that we have 5 or fewer men with the condition?

Answer: Let X be the total number of men in the sample who have dichromatism. Then $X \sim \text{binomial}(1000, 0.05)$. As we saw in Example 6.9.3, binomial probabilities can be difficult to compute. Since $np = 1000(0.05) = 50$, we take $X \sim \text{Poisson}(50)$ instead.

$$\begin{aligned}\mathbb{E}(X) &= 50, \\ \text{Var}(X) &= 50, \\ \sigma(X) &= \sqrt{50} \approx 7.071.\end{aligned}$$

Notice that these values are fairly close to values obtained in Example 6.9.3; we would expect anywhere between 43 and 57 men in the sample to have dichromatism with an expected value of 50.

The probability that we have 5 or fewer men with the condition is

$$\begin{aligned}\mathbb{P}(0 \leq X \leq 5) &= \mathbb{P}(X = 0) + \mathbb{P}(X = 1) + \mathbb{P}(X = 2) + \mathbb{P}(X = 3) \\ &\quad + \mathbb{P}(X = 4) + \mathbb{P}(X = 5) \\ &= e^{-50} \left(\frac{50^0}{0!} + \frac{50^1}{1!} + \frac{50^2}{2!} + \frac{50^3}{3!} + \frac{50^4}{4!} + \frac{50^5}{5!} \right) \\ &\approx 5.6 \times 10^{-16}.\end{aligned}$$

This is roughly 3 times the actual probability predicted by the binomial random variable! However, the results are much more encouraging for events that are more likely. For instance

$$\mathbb{P}(43 \leq X \leq 57) \approx 0.7116$$

using $X \sim \text{Poisson}(50)$. If we use the correct distribution, $X \sim \text{binomial}(1000, 0.05)$, we get

$$\mathbb{P}(43 \leq X \leq 57) \approx 0.7239.$$

For this range of (more likely) outcomes, the Poisson random variable approximates the actual probability fairly well. ■

Example 6.9.8. Suppose you enter 30 different lotteries, and in each of them your probability of winning is $1/1000$. What is the (approximate) probability that you win exactly once? What is the (approximate) probability that you win at least once?

Answer: Let W be the number of lotteries that you win. If we were computing the probabilities exactly, we would have

$$W \sim \text{binomial}\left(30, \frac{1}{1000}\right).$$

Since the probability of winning is fairly low, we can approximate this with the Poisson random variable instead. Since $np = 30/1000 = 0.03$, we use

$$W \sim \text{Poisson}(0.03).$$

As far as the expected value and standard deviation go,

$$\begin{aligned}\mathbb{E}(W) &= 0.03, \\ \text{Var}(W) &= 0.03, \\ \sigma(W) &= \sqrt{0.03} \approx 0.1732.\end{aligned}$$

Given the expected value and standard deviation, we should not expect to win any of the lotteries! We are first asked to compute the probability of winning exactly once:

$$\mathbb{P}(W = 1) = e^{-0.03} \frac{0.03^1}{1!} = 0.03e^{-0.03} \approx 0.0291.$$

So, there is almost a 3% chance we will win exactly one of the lotteries. The second question asks for the probability that we will win 1 or more lotteries:

$$\mathbb{P}(W \geq 1) = 1 - \mathbb{P}(W = 0) = 1 - e^{-0.03} \approx 0.0296.$$

Notice from the last computation that the probability we lose all 30 lotteries is

$$\mathbb{P}(W = 0) = e^{-0.03} \approx 0.9704.$$

So, the probability of winning one or more of the lotteries is also around 3%. This means that if we win at all, we are only likely to win once. We can make this last assertion more precise by computing the conditional probability that we win exactly one lottery given that we win at least one. Below, we use $\{W = 1\} \cap \{W \geq 1\} = \{W = 1\}$ which is true because any event where

we win exactly one of the lotteries is clearly an event where we win at least one lottery!

$$\begin{aligned}\mathbb{P}(W = 1|W \geq 1) &= \frac{\mathbb{P}(\{W = 1\} \cap \{W \geq 1\})}{\mathbb{P}(W \geq 1)} \\ &= \frac{\mathbb{P}(W = 1)}{\mathbb{P}(W \geq 1)} \\ &= \frac{0.03e^{-0.03}}{1 - e^{-0.03}} \approx 0.9851.\end{aligned}$$

This means that if we manage to win at least one of the lotteries, there is over a 98% chance that we win only one of them. ■

6.10 Problems

- 1.) The random variable X has sample space

$$S = \{0, 1, 2, 3, 4, 5\}.$$

The probabilities for the first five outcomes are

$$\begin{aligned}\mathbb{P}(X = 0) &= \frac{1}{9} = \mathbb{P}(X = 1), \\ \mathbb{P}(X = 2) &= \frac{2}{9}, \\ \mathbb{P}(X = 3) &= \frac{1}{9} = \mathbb{P}(X = 4).\end{aligned}$$

Compute $\mathbb{P}(X = 5)$, $\mathbb{E}[X]$, and $\sigma(X)$.

- 2.) The random variable Y has sample space

$$S = \{1, 2, 3, 4, 5, 6, 7, 8\}.$$

The probability of all the odd outcomes are the same, and the probability of all of the even outcomes are the same. However, the probability of an even outcome is one-half the probability of an odd outcome. Find the probability of each outcome and compute the expected value and standard deviation of this random variable.

- 3.) A random variable Z has sample space

$$S = \{1, 2, 3, 4, 5\}.$$

The probability that $X = 1$ is double the probability that $X = 2$. The probability that $X = 2$ is double the probability that $X = 3$, and so on. Find the probabilities for all five outcomes and compute the expected value, variance, and standard deviation for Z .

- 4.) We perform an experiment where we first toss a biased coin which has a probability of $3/5$ to come up heads. If the coin toss comes up heads, we then roll a fair six-sided die. If the coin toss comes up tails, we then pick a random number between 1 and 10 out of a hat (all 10 numbers are equally likely). What is the sample space for this experiment, and what are the probabilities of the individual outcomes? Compute the probability that the second stage of the experiment gives us the number 3 (either on the die or from the hat).
- 5.) Suppose we perform an experiment where we independently roll three fair six-sided dice. Let R be the sum of the three dice. What is the sample space for R ? Compute $\mathbb{P}(R = 5)$.
- 6.) We have an urn containing 2 red balls, 3 yellow balls, and 5 green balls. We select two balls from the urn (without replacement). What is the probability that both balls pulled from the urn are green? What is the probability that one of the balls is yellow and the other is green?
- 7.) Consider the urn from Problem 6.). Now suppose we pull two balls from the urn *but with replacement*. In other words, we pull a ball from the urn, note its color, then place it back in the urn. After re-randomizing the urn (i.e. shaking it up), we select a second ball and note its color. What is the probability that both balls pulled from the urn are green? What is the probability that one of the balls is yellow and the other is green?
- 8.) We roll a fair six-sided die 5 times. Let T be the total number of twos in the outcome of the rolls. Compute the probability of each possible outcome for T .
- 9.) We roll two fair six-sided dice. Let R be their sum. Given that at least one of the two dice shows a 1, what is the probability that $R = 5$? Is

the event that $R = 5$ independent of the event that at least one of the two dice shows a 1?

- 10.) Consider the experiment described in Problem 6.) (without replacement). What is the probability that both balls are green given that at least one of the balls is green?
- 11.) We have an urn with balls 10 balls numbered 1 through 10. Balls 1 – 4 are green, Ball 5 is yellow, and Balls 6 – 10 are red. We select two balls from the urn (without replacement). Given that both balls are odd numbered, what is the probability that the yellow ball was selected? Is the event that the yellow ball was selected independent of the event that both balls are odd?
- 12.) Consider the experiment in Problem 11.). Given that one of the two balls is yellow, what is the probability that one of the balls is labeled with an even number?
- 13.) A teacher is giving a multiple choice test, and each question has 5 choices. From past experience, the teacher estimates that a student has a 70% of knowing the correct answer for any particular question. If the student does not know the correct answer, the teacher assumes he or she guesses randomly. Given that the student answered the question correctly, what is the probability that he or she knew the right answer?
- 14.) A pharmaceutical company has developed a new test for dengue fever. The test returns a false positive with a probability of 1% and a false negative with a probability of 0.5%. If the test is being administered in a region where dengue fever affects one in 5,000 people, what are the positive predictive value (PPV) and negative predictive value (NPV) for this test?
- 15.) Consider the diagnostic test from Problem 14.). However, we are administering the test in a region where we are not sure of the actual prevalence of dengue fever. Given that 2.5% of the population test positive, what is the actual prevalence of dengue fever among the population?
- 16.) Consider the diagnostic test from Problem 14.). However, we are administering the test in a region where we are not sure of the actual

prevalence of dengue fever. If a person who tests positive for the disease only has a 50% chance of actually having the disease, what is the actual prevalence of dengue fever among the population?

- 17.) We have an urn with 24 red balls and 1 green ball. We select 10 balls from the urn *with replacement*. Let G be the number of times the green ball is selected from the urn. Which family of random variables does G belong to? Use this to compute the probability that we select the green ball at least once.
- 18.) We toss a fair coin 15 times. Let T be the total number of tails in the outcome. What is the expected value and standard deviation of T ? What is the probability that the number of tails is within one standard deviation on either side of the expected value.

- 19.) We toss a fair coin 20 times. Given that there are at least 8 heads in the outcome, what is the most likely value for T , the total number of tails in the outcome?
- 20.) There is roughly a 2% chance that a person has green eyes. If we select 100 people at random, what is the probability that we have 3 or more people with green eyes?
- 21.) Suppose we have a biased coin that comes up heads 55% of the time. We perform an experiment where we toss the coin until the first tails appears. Let T be the number of tosses until the first tails. What is the expected value and standard deviation for T ? What is the probability that it takes 5 or more tosses before the first tails appears?
- 22.) A bag contains 4 identical white socks and 12 identical black socks. Suppose that the first sock I pull out of the bag is black. How many socks should I expect to have to pull from the bag (with replacement) until I get a matching black sock. What is the probability that I pull 3 white socks out of the bag before the black sock that I need?
- 23.) Suppose that the number of accidents on any given day for a particular stretch of highway is a Poisson random variable with an expected value of 3. What is the probability that there are no accidents on the highway tomorrow? Exactly one accident? Given that there were at least 4 accidents today, find the probability that there were exactly 5 accidents.
- 24.) Klinefelter syndrome is a relatively common genetic disorder among men which occurs when a male has 2 (or more) X chromosomes (also known as XXY syndrome). The actual rate at which this disorder occurs in men is not exactly known, but is somewhere close to 1 out of every 750 live male births. Suppose we pull a random sample of 4500 men. Let N be the number of men in the sample who have Klinefelter syndrome. Using a Poisson random variable to approximate the distribution of N , how many men in our sample should we expect to have the syndrome? What is the probability that there are 4 or fewer men in the group with the syndrome?
- 25.) In sheep, the color of wool is controlled by a single gene with two alleles: W is dominant and gives white wool while w is recessive and gives black wool. In a group of 5000 sheep, there are 1800 individuals with black

wool. Assuming the population is in Hardy–Weinberg Equilibrium, what are the overall frequency of the genes W and w in the population? What proportion of sheep are heterozygous for this gene (i.e. are Ww)?

- 26.) Handedness in humans is controlled by a single gene for which there are two alleles: H is dominant and codes for right-handedness while h is recessive and codes for left-handedness. Assuming that the world population is in Hardy–Weinberg Equilibrium (at least for the handedness gene) and that 10% of the population is left-handed, what are the overall frequencies of the alleles H and h in the overall population? What percentage of the population is homozygous dominant (i.e. are HH)?
- 27.) A fish breeder sells angelfish to pet stores. One gene responsible for the coloration of these fish has two alleles: D is dominant and gives dark colored fish while d is recessive and gives gold colored fish. Suppose that the breeder's initial population of fish is 10% homozygous dominant (DD individuals), 50% heterozygous (Dd), and 40% homozygous recessive (dd). Assuming random mating and the other conditions of the Hardy–Weinberg Principle, what proportion of fish are expected to be gold colored in future generations?
- 28.) In fruit flies, wing length is controlled by a gene with two possible alleles: W gives long wings and is dominant while w gives short vestigial wings and is recessive. Fruit flies with short vestigial wings cannot fly. Suppose that an initial population of flies has 70% homozygous dominant individuals (WW) and 30% heterozygous individuals (Ww). Assuming that the Hardy–Weinberg Principle applies, what proportion of future generations will be incapable of flying?

29.) **Challenge: Hardy–Weinberg Equilibrium for Tetraploidy**

Polyploidy is when an organism has more than two copies of a particular gene. While this is a rare occurrence in animals, polyploidy is rather common in plants. Suppose that there is a particular plant gene with two alleles, A dominant and a recessive, but each plant has four copies of the gene (two from each parent). Note that there are 5 types of individuals in this population: $AAAA$, $AAAa$, $AAaa$, $Aaaa$, and $aaaa$. Derive the Hardy–Weinberg Equilibrium frequencies of these genotypes

in terms of the underlying frequencies of the individual alleles (i.e. $\mathbb{P}(A) = p$ and $\mathbb{P}(a) = q$).

30.) **Challenge: The Negative Binomial Distribution**

Suppose we perform independent trials of a Bernoulli(p) random variable until r successes are obtained. Let X be the total number of trials needed until the r -th success. Determine the sample space for X and find a formula for $\mathbb{P}(X = n)$. A random variable distributed in this fashion is listed as $X \sim \text{nb}(r, p)$.

HINT: Clearly $n \geq r$ since you must do at least r trials to get r successes. How many successes must there be among the first $n - 1$ trials?

Chapter 7

Continuous Random Variables

We now turn to the study of **continuous random variables**. Recall that a continuous random variable is one whose sample space consists of an entire range of real numbers. A classic example is the spinner whose sample space is the set of angles $S = [0, 2\pi)$ if we choose to use radians or $S = [0^\circ, 360^\circ)$ in degrees.

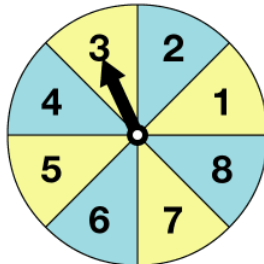


Figure 7.0.1: A Typical Spinner

As another example, suppose we select a random adult and measured their height (in inches). Since heights are non-negative real numbers, we could say that this experiment has sample space $S = [0, \infty)$. Of course, the vast majority of measurements should land in the range $[12, 120]$! For a third example, we might select a light bulb and ask how long the bulb will work (in hours). The sample space for this is certainly $S = [0, \infty)$.

Recall that a probability on a sample space S is a set function defined on the events in S which enjoys the following properties.

- For any event $E \subset S$,

$$0 \leq \mathbb{P}(E) \leq 1,$$

- $\mathbb{P}(S) = 1$,

- If E_1 and E_2 are mutually exclusive events ($E_1 \cap E_2 = \emptyset$),

$$\mathbb{P}(E_1 \cup E_2) = \mathbb{P}(E_1) + \mathbb{P}(E_2).$$

While probabilities for discrete random variables relied heavily on counting techniques (i.e. the branch of mathematics called combinatorics), computing probabilities for continuous random variables relies heavily on integration. The key insight here comes from the final requirement in the definition: **probabilities of mutually exclusive events simply add together**. This is a lot like area! If we have two completely disjoint regions of the plane, the total area of the region is simply the sum of the two individual areas. Combining this with the fact that integrals were designed to compute areas gives us a clue that integration should be useful for continuous random variables!

7.1 Probability Density Functions

Associated to every continuous random variable X is a special function f called the **probability density function** (or pdf) of X . Suppose that X is a continuous random variable with sample space $S = [a, b]$ (we can also allow infinite sample spaces). The probability density function associated to X is a function f with the following properties.

- $f(x) \geq 0$ over the sample space $S = [a, b]$

- $\int_a^b f(x) dx = 1$

The pdf allows us to compute probabilities associated to X . In particular, if we have an event $[c, d] \subseteq [a, b]$ in the sample space, then the probability that the outcome lies in this interval is

$$\mathbb{P}(c \leq X \leq d) = \int_c^d f(x) dx.$$

In other words, probabilities associated to X are computed by integrating f over an appropriate range of outcomes. The pdf is an essential part of the specification for any continuous random variable. If you do not know the pdf, then you do not know the random variable!

Example 7.1.1. Suppose that X is a random variable with sample space $S = [0, 2]$ and pdf

$$f(x) = \frac{3}{2}x - \frac{3}{4}x^2 = \frac{3}{4}x(2 - x).$$

Verify that f is a valid pdf, and use it to compute the probability that the outcome of X lies in the interval $[0, 1/2]$.

Answer: In order for f to be a pdf, it has to be non-negative over the sample space. Note that the graph of f is a parabola which opens downward and has zeros at $x = 0$ and $x = 2$.

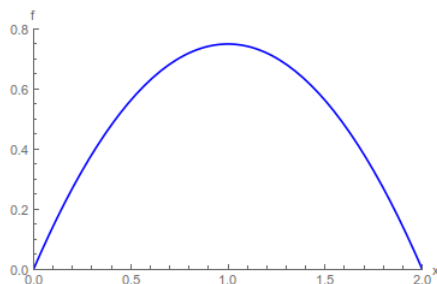


Figure 7.1.1: Graph of the pdf for Example 7.1.1

Clearly this function is non-negative over the sample space $0 \leq x \leq 2$, and the fact that this function is negative outside of the sample space is not a concern! The second requirement is that the pdf must integrate to 1 over the sample space.

$$\begin{aligned} \int_0^2 f(x) dx &= \int_0^2 \left(\frac{3}{2}x - \frac{3}{4}x^2 \right) dx \\ &= \left(\frac{3}{4}x^2 - \frac{1}{4}x^3 \right) \Big|_0^2 \\ &= \frac{3}{4}(4) - \frac{1}{4}(8) \\ &= 1. \end{aligned}$$

So f is a valid pdf for the sample space. In order to compute the probability that the outcome of the experiment is in the range $[0, 1/2]$, we simply integrate the pdf over this interval.

$$\begin{aligned}\mathbb{P}\left(0 \leq X \leq \frac{1}{2}\right) &= \int_0^{1/2} \left(\frac{3}{2}x - \frac{3}{4}x^2\right) dx \\ &= \left(\frac{3}{4}x^2 - \frac{1}{4}x^3\right)\Big|_0^{1/2} \\ &= \frac{3}{4}\left(\frac{1}{4}\right) - \frac{1}{4}\left(\frac{1}{8}\right) \\ &= \frac{5}{32}. \blacksquare\end{aligned}$$

Quite often, we are not given the pdf in its entirety. Instead, we are given a function which is non-negative over the sample space and asked to **normalize** the function. That is, we are tasked with finding a coefficient which ensures that the function integrates to 1 over the sample space.

Example 7.1.2. A continuous random variable Y has sample space $S = [1, 9]$ and a pdf of the form

$$f(y) = Ay^2.$$

Find that value of A which normalizes the pdf, and then compute the probability that Y will lie in the interval $[2, 4]$.

Answer: So long as A is positive, the function $f(y) = Ay^2$ will be positive over the sample space. To find the appropriate value of A , we force the pdf to integrate to 1 over this range.

$$\begin{aligned}\int_1^9 Ay^2 dy &= A \int_1^9 y^2 dy \\ &= A \left.\frac{1}{3}y^3\right|_1^9 \\ &= \frac{1}{3}A(9^3 - 1^3) \\ &= \frac{728}{3}A = 1.\end{aligned}$$

So, we should take $A = 3/728$ giving a pdf of

$$f(y) = \frac{3}{728}y^2$$

over the sample space. We can now compute probabilities for Y . In particular,

$$\begin{aligned}
 \mathbb{P}(2 \leq Y \leq 4) &= \int_2^4 \frac{3}{728} y^2 dy \\
 &= \frac{3}{728} \int_2^4 y^2 dy \\
 &= \frac{3}{728} \left(\frac{1}{3} y^3 \right) \Big|_2^4 \\
 &= \frac{1}{728} (4^3 - 2^3) \\
 &= \frac{56}{728} = \frac{1}{13}. \blacksquare
 \end{aligned}$$

The convention in most probability texts is that a random variable is given a capital letter (like X or Y) while the corresponding lower case letter (x or y , respectively) is used for the argument of the pdf. We will stick to this convention throughout this text.

Example 7.1.3. The random variable T is the time (in years) until a car stereo fails. The sample space for T is the set of non-negative real numbers, $[0, \infty)$, and the pdf has the form

$$f(t) = Ae^{-t/10}.$$

Compute the probability that stereo fails in the first 5 years you own the car. What is the probability the radio lasts 12 or more years?

Answer: The first thing we have to do is normalize the pdf.

$$\begin{aligned}
 \int_0^\infty Ae^{-t/10} dt &= A \lim_{R \rightarrow \infty} \int_0^R e^{-t/10} dt \\
 &= A \lim_{R \rightarrow \infty} -10 e^{-t/10} \Big|_0^R \\
 &= -10A \lim_{R \rightarrow \infty} (e^{-R/10} - 1) \\
 &= 10A = 1.
 \end{aligned}$$

So, the coefficient must be $A = 1/10$.

The first probability we are being asked for is the probability that T is in the range $[0, 5]$.

$$\begin{aligned}\mathbb{P}(0 \leq T \leq 5) &= \int_0^5 \frac{1}{10} e^{-t/10} dt \\ &= -e^{-t/10} \Big|_0^5 \\ &= -e^{-5/10} - (-1) \\ &= 1 - e^{-1/2} \approx 0.3935.\end{aligned}$$

So, there is almost a 40% chance the radio will fail in the first five years!

The second probability we are being asked to compute is the probability that T lies in the range $[12, \infty)$. We could compute this probability directly, but we choose to proceed in a different way.

$$\begin{aligned}\mathbb{P}(T \geq 12) &= 1 - \mathbb{P}(0 \leq T \leq 12) \\ &= 1 - \int_0^{12} \frac{1}{10} e^{-t/10} dt \\ &= 1 - \left(-e^{-t/10} \Big|_0^{12} \right) \\ &= 1 - \left(-e^{-12/10} - (-1) \right) \\ &= 1 - (1 - e^{-6/5}) \\ &= e^{-6/5} \approx 0.3012.\end{aligned}$$

So, there is just over a 30% chance the radio will last at least 12 years. ■

The attentive reader will have noticed something strange in the last example. When we computed the probability that T is 12 or larger, we used

$$\mathbb{P}(T \geq 12) = 1 - \mathbb{P}(0 \leq T \leq 12).$$

Since we want to include the probability that $T = 12$ in our final answer, we really should have used

$$\mathbb{P}(T \geq 12) = 1 - \mathbb{P}(0 \leq T < 12).$$

The curious thing about continuous random variables is that these two probabilities are exactly the same! To drive the point home, we note that

$$\mathbb{P}(0 \leq T \leq 12) = \mathbb{P}(0 \leq T < 12) + \mathbb{P}(T = 12).$$

However,

$$\begin{aligned}\mathbb{P}(T = 12) &= \mathbb{P}(12 \leq T \leq 12) \\ &= \int_{12}^{12} f(t) dt \\ &= 0.\end{aligned}$$

This strange fact is true in general! If X is any continuous random variable and c is any particular value in its sample space, then

$$\mathbb{P}(X = c) = \int_c^c f(x) dx = 0.$$

So, the probability that a continuous random variable takes on any particular value is always zero! This rather counter-intuitive fact tells us that *the only sensible question we can ask about a continuous random variable is whether the outcome lies in some range of values*. In practice, this strange state of affairs is completely unimportant. In any real world measurement, there is always some associated error, and so all we can ever determine is that the outcome of an experiment lies within some range.

Example 7.1.4. Let X be a random variable with sample space $S = (-\infty, \infty)$ and whose pdf is given by

$$f(x) = \begin{cases} 0, & x < 0 \\ \frac{3}{2}x - \frac{3}{4}x^2, & 0 \leq x \leq 2 \\ 0, & x > 2 \end{cases}.$$

Verify that f is a pdf and find the probability that X is greater than $3/2$.

Answer: Even though the sample space is listed as $S = (-\infty, \infty)$, the fact that f is zero outside $[0, 2]$ means all integrals are limited to this range.

$$\begin{aligned}\int_{-\infty}^{\infty} f(x) dx &= \int_0^2 \left(\frac{3}{2}x - \frac{3}{4}x^2 \right) dx \\ &= \left(\frac{3}{4}x^2 - \frac{1}{4}x^3 \right) \Big|_0^2 \\ &= 3 - 2 = 1.\end{aligned}$$

So, f is a pdf over the sample space for X .

$$\begin{aligned}
 \mathbb{P}\left(X > \frac{3}{2}\right) &= \int_{3/2}^{\infty} f(x) \, dx \\
 &= \int_{3/2}^2 \left(\frac{3}{2}x - \frac{3}{4}x^2\right) \, dx \\
 &= \left(\frac{3}{4}x^2 - \frac{1}{4}x^3\right) \Big|_{3/2}^2 \\
 &= (3 - 2) - \left(\frac{27}{16} - \frac{27}{32}\right) \\
 &= 1 - \frac{27}{32} \\
 &= \frac{5}{32}. \quad \blacksquare
 \end{aligned}$$

Comparing the random variable from Example 7.1.1 with the one above, we see that the sample space from the former is $S = [0, 2]$ while the latter is all real numbers. However, the probability that an outcome for the random variable above will lie in the range $[0, 2]$ is 1. Hence, these two random variables are essentially identical.

The point of this discussion is that we can suppress mentioning the sample space of a continuous random variable altogether. Since the outcome must lie in the range $(-\infty, \infty)$ (by definition), we can simply list the pdf for the random variable. There will automatically be zero probability of an outcome occurring in any interval of the real line where the pdf is identically zero.

Example 7.1.5. Suppose that W is a random variable with pdf

$$f(w) = \begin{cases} Ae^{2w}, & w \leq 0 \\ 0, & w > 0 \end{cases}$$

Normalize f , and find a so that the interval $[a, 0]$ has probability $1/2$.

Answer: Notice that the effective sample space for W is $(-\infty, 0]$. To

normalize the pdf, we force the total probability to be 1.

$$\begin{aligned}
 1 &= \int_{-\infty}^{\infty} f(w) \, dw \\
 &= \lim_{R \rightarrow -\infty} \int_R^0 A e^{2w} \, dw \\
 &= \frac{A}{2} \lim_{R \rightarrow -\infty} e^{2w} \Big|_R^0 \\
 &= \frac{A}{2} \lim_{R \rightarrow -\infty} (1 - e^R) \\
 &= \frac{A}{2}.
 \end{aligned}$$

So, we need to take $A = 2$ to normalize the pdf. To find the value a so that $[a, 0]$ has probability $1/2$, we need to solve

$$\begin{aligned}
 \frac{1}{2} &= \mathbb{P}(a \leq W \leq 0) \\
 &= \int_a^0 2e^{2w} \, dw \\
 &= e^{2w} \Big|_a^0 \\
 &= 1 - e^{2a}.
 \end{aligned}$$

This gives the equation

$$\begin{aligned}
 \frac{1}{2} &= 1 - e^{2a} \\
 e^{2a} &= \frac{1}{2} \\
 2a &= \ln\left(\frac{1}{2}\right) = -\ln(2) \\
 a &= -\frac{\ln(2)}{2}. \quad \blacksquare
 \end{aligned}$$

Another useful tool for studying continuous random variables is the **cumulative distribution function** or cdf. Suppose that X is a continuous random variable with pdf f . The cumulative distribution function of X is given by

$$F(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f(t) \, dt.$$

Notice that the convention is that a lower case letter (like f or g) is used for a pdf while the corresponding upper case letter is used for the corresponding cdf.

Knowing the cdf makes the computation of probabilities very easy!

$$\begin{aligned}\mathbb{P}(a \leq X \leq b) &= \int_a^b f(x) \, dx \\ &= \int_{-\infty}^b f(x) \, dx - \int_{-\infty}^a f(x) \, dx \\ &= \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a) \\ &= F(b) - F(a).\end{aligned}$$

Computing the probability of any interval simply comes down to subtracting the values of F at the endpoints of the interval!

Example 7.1.6. Find the cdf for the random variable X in Example 7.1.4.

Answer: We need to compute

$$F(x) = \mathbb{P}(X \leq x).$$

The first thing to note is that $F(x) = 0$ for $x \leq 0$. This is because X has zero probability of lying outside the interval $[0, 2]$. For any x in the range $0 \leq x \leq 2$, we have

$$\begin{aligned}F(x) &= \mathbb{P}(X \leq x) \\ &= \int_0^x \left(\frac{3}{2}t - \frac{3}{4}t^2 \right) \, dt \\ &= \frac{3}{4}x^2 - \frac{1}{4}x^3.\end{aligned}$$

Notice that we used t for the integration variable since it is bad form (and potentially confusing) to use the same variable in both the limits of integration and inside the integral itself.

For values of x that are larger than 2, we have

$$\begin{aligned}F(x) &= \mathbb{P}(X \leq x) \\ &= \int_0^2 \left(\frac{3}{2}t - \frac{3}{4}t^2 \right) \, dt \\ &= 1\end{aligned}$$

since the pdf is zero outside the interval $[0, 2]$. Hence, the cdf for X is

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{3}{4}x^2 - \frac{1}{4}x^3, & 0 \leq x \leq 2 \\ 1, & x > 2 \end{cases} \quad \blacksquare$$

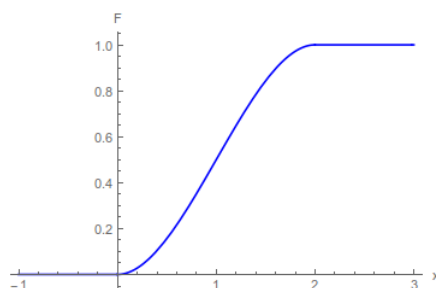


Figure 7.1.2: Graph of the cdf for Example 7.1.6

It turns out that all cdfs share certain properties.

Lemma 7.1.1. *Let X be a continuous random variable with pdf f and corresponding cdf F . Then F has the following properties.*

- F is continuous.
- F is non-decreasing.
- $\lim_{x \rightarrow -\infty} F(x) = 0$
- $\lim_{x \rightarrow \infty} F(x) = 1$

Proof: Since F is the integral of f , it must be continuous. Moreover, since f is never negative, F can never decrease. The final two properties come from the fact that all outcomes of X must be in the range $(-\infty, \infty)$. $\blacksquare$

Example 7.1.7. Find the cdf for the random variable T in Example 7.1.3. Use the cdf to find the probability that the car stereo will last between 10 and 20 years.

Answer: Since the sample space for T is $[0, \infty)$, the cdf for T will be zero for any $t < 0$. For $t > 0$, we have

$$\begin{aligned} F(t) &= \int_0^t f(x) \, dx \\ &= \int_0^t \frac{1}{10} e^{-x/10} \, dx \\ &= -e^{-x/10} \Big|_0^t \\ &= 1 - e^{-t/10}. \end{aligned}$$

This means that the cdf is the function

$$F(t) = \begin{cases} 0, & t < 0 \\ 1 - e^{-t/10}, & t \geq 0 \end{cases}.$$

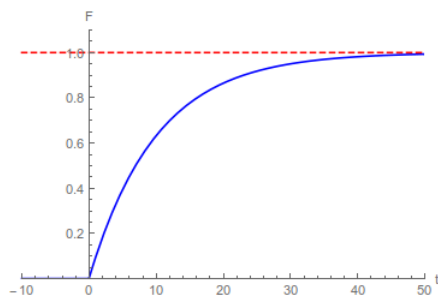


Figure 7.1.3: Graph of the cdf for Example 7.1.3

Notice that the graph of F approaches 1 as t tends to infinity. To find the probability that T is in the range $10 \leq T \leq 20$, we simply subtract the values of the cdf at the endpoints.

$$\begin{aligned} \mathbb{P}(10 \leq T \leq 20) &= F(20) - F(10) \\ &= (1 - e^{-20/10}) - (1 - e^{-10/10}) \\ &= e^{-1} - e^{-2} \approx 0.2325. \blacksquare \end{aligned}$$

7.2 Expected Value, Variance, and Standard Deviation

Just as with discrete random variables, the most important descriptive quantities associated to a continuous random variable are the expected value, variance, and standard deviation. We begin with expected value.

Let X be a continuous random variable with sample space $S = [a, b]$ and pdf f . The **expected value** of X is the quantity

$$\mathbb{E}[X] = \int_a^b x f(x) dx,$$

whenever this integral is finite.

The definition of expected value for continuous random variables is formally similar to the definition in the discrete case. The product of x with $f(x) dx$ is essentially “outcome $\times$ probability”, and the integral acts to “sum up” all of these contributions. Just as with discrete random variables, the expected value is the outcome we would expect to see on average.

Example 7.2.1. Suppose Y is a random variable with pdf of the form

$$f(y) = \begin{cases} 0, & y < 1 \\ \frac{A}{y^4}, & y \geq 1 \end{cases}.$$

Find the expected value of Y .

Answer: The first thing to do is normalize the pdf for Y .

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} f(y) dy \\ &= \int_1^{\infty} \frac{A}{y^4} dy \\ &= A \lim_{R \rightarrow \infty} \int_1^R \frac{1}{y^4} dy \\ &= A \lim_{R \rightarrow \infty} \left. -\frac{1}{3y^3} \right|_1^R \\ &= \frac{A}{3} \lim_{R \rightarrow \infty} \left(1 - \frac{1}{R^3} \right) \\ &= \frac{A}{3}. \end{aligned}$$

So, A must equal 3. To find the expected value, we compute the following integral.

$$\begin{aligned}
 \mathbb{E}[Y] &= \int_{-\infty}^{\infty} y f(y) dy \\
 &= \int_1^{\infty} y \left(\frac{3}{y^4} \right) dy \\
 &= \lim_{R \rightarrow \infty} \int_1^R \frac{3}{y^3} dy \\
 &= \lim_{R \rightarrow \infty} -\frac{3}{2y^2} \Big|_1^R \\
 &= \frac{3}{2} \lim_{R \rightarrow \infty} \left(1 - \frac{1}{R} \right) \\
 &= \frac{3}{2}. \blacksquare
 \end{aligned}$$

Not every continuous random variable has an expected value. Consider the following example.

Example 7.2.2. Suppose Z is a random variable with pdf of the form

$$f(z) = \begin{cases} 0, & z < 1 \\ \frac{A}{z^2}, & z \geq 1 \end{cases}.$$

What happens when you try to find the expected value of Z ?

Answer: As always, we first need to normalize the pdf.

$$\begin{aligned}
 1 &= \int_{-\infty}^{\infty} f(z) dz \\
 &= \int_1^{\infty} \frac{A}{z^2} dz \\
 &= A \lim_{R \rightarrow \infty} \int_1^R \frac{1}{z^2} dz \\
 &= A \lim_{R \rightarrow \infty} -\frac{1}{z} \Big|_1^R \\
 &= A \lim_{R \rightarrow \infty} \left(1 - \frac{1}{R} \right) \\
 &= A.
 \end{aligned}$$

So, A is just 1. If we try to find the expected value, we find

$$\begin{aligned}
 \mathbb{E}[Z] &= \int_{-\infty}^{\infty} z f(z) dz \\
 &= \int_1^{\infty} z \left(\frac{1}{z^2} \right) dz \\
 &= \lim_{R \rightarrow \infty} \int_1^R \frac{1}{z} dz \\
 &= \lim_{R \rightarrow \infty} \ln(z) \Big|_1^R \\
 &= \lim_{R \rightarrow \infty} (\ln(R)) \\
 &= \infty.
 \end{aligned}$$

Since the result of this integration is not a real number, Z does not have an expected value. ■

Again, the expected value gives us a measure of the size of a typical outcome. But, the result of any particular experiment might be quite far from this value. The **standard deviation** of a random variable is a measure of how far we can expect outcomes to vary around the expected value.

Let X be a continuous random variable with sample space $S = [a, b]$ and pdf f . Then the **variance** of X (denoted $\text{Var}(X)$) is given by

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

whenever this integral is finite. The **standard deviation** of X (denoted $\sigma(X)$) is simply the square root of the variance:

$$\sigma(X) = \sqrt{\text{Var}(X)}.$$

Just as with discrete random variables, there is a simpler formula for computing the variance of a continuous random variable.

Lemma 7.2.1.

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \int_a^b x^2 f(x) dx - (\mathbb{E}[X])^2.$$

Proof: The proof is analogous to the discrete case. Remember that $\mathbb{E}[X]$ is

simply a number (and not involved in the integrals below).

$$\begin{aligned}
 \text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\
 &= \int_a^b (x - \mathbb{E}[X])^2 f(x) dx \\
 &= \int_a^b (x^2 - 2x\mathbb{E}[X] + (\mathbb{E}[X])^2) f(x) dx \\
 &= \int_a^b x^2 f(x) dx - 2\mathbb{E}[X] \int_a^b x f(x) dx + (\mathbb{E}[X])^2 \int_a^b f(x) dx \\
 &= \int_a^b x^2 f(x) dx - 2(\mathbb{E}[X])^2 + (\mathbb{E}[X])^2 \\
 &= \int_a^b x^2 f(x) dx - (\mathbb{E}[X])^2. \blacksquare
 \end{aligned}$$

Example 7.2.3. Compute the expected value and standard deviation for the random variable X in Example 7.1.1. What is the probability that an outcome for X will be within one standard deviation of the expected value?

Answer: The expected value of X is

$$\begin{aligned}
 \mathbb{E}[X] &= \int_0^2 x f(x) dx \\
 &= \int_0^2 x \left(\frac{3}{2}x - \frac{3}{4}x^2 \right) dx \\
 &= \int_0^2 \left(\frac{3}{2}x^2 - \frac{3}{4}x^3 \right) dx \\
 &= \left(\frac{1}{2}x^3 - \frac{3}{16}x^4 \right) \Big|_0^2 \\
 &= \frac{1}{2}(2)^3 - \frac{3}{16}(2)^4 = 1.
 \end{aligned}$$

So, the expected value of X is 1. To find the standard deviation, we first

have to find the variance

$$\begin{aligned}
 \text{Var}(X) &= \int_0^2 x^2 f(x) dx - (\mathbb{E}[X])^2 \\
 &= \int_0^2 x^2 \left(\frac{3}{2}x - \frac{3}{4}x^2 \right) dx - 1^2 \\
 &= \int_0^2 \left(\frac{3}{2}x^3 - \frac{3}{4}x^4 \right) dx - 1 \\
 &= \left(\frac{3}{8}x^4 - \frac{3}{20}x^5 \right) \Big|_0^2 - 1 \\
 &= \frac{3}{8}(2)^4 - \frac{3}{20}(2)^5 - 1 = \frac{1}{5}.
 \end{aligned}$$

Once we have the variance, the standard deviation is simple.

$$\sigma(X) = \sqrt{\text{Var}(X)} = \sqrt{\frac{1}{5}} = \frac{\sqrt{5}}{5}.$$

To find the probability that an outcome for X will lie within one standard deviation of its expected value, we need to compute

$$\mathbb{P}\left(1 - \frac{\sqrt{5}}{5} \leq X \leq 1 + \frac{\sqrt{5}}{5}\right).$$

To compute this probability, we can use the cdf for X that we computed in Example 7.1.6.

$$\begin{aligned}
 \mathbb{P}\left(1 - \frac{\sqrt{5}}{5} \leq X \leq 1 + \frac{\sqrt{5}}{5}\right) &= F\left(1 + \frac{\sqrt{5}}{5}\right) - F\left(1 - \frac{\sqrt{5}}{5}\right) \\
 &= \left[\frac{3}{4} \left(1 + \frac{\sqrt{5}}{5}\right)^2 - \frac{1}{4} \left(1 + \frac{\sqrt{5}}{5}\right)^3 \right] - \left[\frac{3}{4} \left(1 - \frac{\sqrt{5}}{5}\right)^2 - \frac{1}{4} \left(1 - \frac{\sqrt{5}}{5}\right)^3 \right] \\
 &= \frac{7\sqrt{5}}{25} \approx 0.6261.
 \end{aligned}$$

So, there is almost a 63% chance that an outcome for X will lie within one standard deviation of its expected value. ■

Example 7.2.4. Compute the expected value and standard deviation for the random variable, T , in Example 7.1.3. What is the probability that an outcome for T will be within one-half of a standard deviation of the expected value?

Answer: The expected value of T is

$$\begin{aligned}
 \mathbb{E}[T] &= \int_0^{\infty} t f(t) dt \\
 &= \int_0^{\infty} t \left(\frac{1}{10} e^{-t/10} \right) dt \\
 &= \lim_{R \rightarrow \infty} \int_0^R \frac{1}{10} t e^{-t/10} dt \\
 &= \lim_{R \rightarrow \infty} \left(-(10 + t) e^{-t/10} \right) \Big|_0^R \\
 &= \lim_{R \rightarrow \infty} (10 - (10 + R) e^{-R/10}) \\
 &= 10.
 \end{aligned}$$

Notice that the integral was evaluated by integration by parts. The resulting limit can be computed via L'Hôpital's Rule. This result says that we expect the car stereo to last 10 years.

We will also need to use integration by parts (twice) to compute the variance.

$$\begin{aligned}
 \text{Var}(T) &= \int_0^{\infty} t^2 f(t) dt - 10^2 \\
 &= \lim_{R \rightarrow \infty} \int_0^R \frac{1}{10} t^2 e^{-t/10} dt - 100 \\
 &= \lim_{R \rightarrow \infty} \left(-t^2 e^{-t/10} \Big|_0^R + \int_0^R 2t e^{-t/10} dt \right) - 100 \\
 &= \lim_{R \rightarrow \infty} \left(-t^2 e^{-t/10} - 20(10 + t) e^{-t/10} \right) \Big|_0^R - 100 \\
 &= \lim_{R \rightarrow \infty} (200 - R^2 e^{-R/10} - 20(10 + R) e^{-R/10}) - 100 \\
 &= 100.
 \end{aligned}$$

So, the standard deviation of T is

$$\sigma(T) = \sqrt{\text{Var}(T)} = \sqrt{100} = 10.$$

As in the last example, we can use the cdf computed in Example 7.1.7 to compute the probability that T is within one-half of a standard deviation of

the expected value.

$$\begin{aligned}
 \mathbb{P}(10 - 5 \leq T \leq 10 + 5) &= \mathbb{P}(5 \leq T \leq 15) \\
 &= F(15) - F(5) \\
 &= (1 - e^{-15/10}) - (1 - e^{-5/10}) \\
 &= e^{-1/2} - e^{-3/2} \approx 0.3834.
 \end{aligned}$$

So, there is just over a 38% chance that the car stereo will last within one-half of a standard deviation of its expected lifetime. ■

If a random variable does not have an expected value (as in Example 7.2.2), then it cannot have a variance or standard deviation. However, a random variable can have an expected value but no variance.

Example 7.2.5. Suppose that X is a random variable with sample space $S = [1, \infty]$ and pdf of the form

$$f(x) = \frac{A}{x^3}.$$

Find the expected value of X . What happens when you attempt to compute the variance?

Answer: We first normalize the pdf.

$$\begin{aligned}
 1 &= \int_1^\infty \frac{A}{x^3} dx \\
 &= A \lim_{R \rightarrow \infty} \int_1^R \frac{1}{x^3} dx \\
 &= A \lim_{R \rightarrow \infty} \left(-\frac{1}{2x^2} \right) \Big|_1^R \\
 &= \frac{A}{2} \lim_{R \rightarrow \infty} \left(1 - \frac{1}{R^2} \right) \\
 &= \frac{A}{2}.
 \end{aligned}$$

So, $A = 2$. The expected value of X is

$$\begin{aligned}
 \mathbb{E}[X] &= \int_1^\infty x \left(\frac{2}{x^3} \right) dx \\
 &= 2 \lim_{R \rightarrow \infty} \int_1^R \frac{1}{x^2} dx \\
 &= 2 \lim_{R \rightarrow \infty} \left(-\frac{1}{x} \right) \Big|_0^R \\
 &= 2 \lim_{R \rightarrow \infty} \left(1 - \frac{1}{R} \right) \\
 &= 2.
 \end{aligned}$$

However, when we try to compute the variance we find

$$\begin{aligned}
 \text{Var}(X) &= \int_1^\infty x^2 \left(\frac{2}{x^3} \right) dx - 2^2 \\
 &= 2 \lim_{R \rightarrow \infty} \int_1^R \frac{1}{x} dx - 4 \\
 &= 2 \lim_{R \rightarrow \infty} \ln(x) \Big|_0^R - 4 \\
 &= 2 \lim_{R \rightarrow \infty} \ln(R) - 4 \\
 &= \infty.
 \end{aligned}$$

Since the variance of X is infinite, so is the standard deviation. ■

7.3 A Catalog of Important Continuous Random Variables

As with discrete random variables, there are a number of families of continuous random variables that come up often in applications. For each family, we will compute all of the relevant descriptive quantities as well as provide typical applications of those random variables.

7.3.1 Uniform Random Variables

A continuous random variable X is said to be a uniform random variable over the *finite* interval $S = [a, b]$ if the pdf for X is given by

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}.$$

We write $X \sim \text{Unif}(a, b)$ to indicate that the random variable X is distributed as a uniform random variable over $S = [a, b]$.

The uniform random variable tries to generalize the situation for discrete random variables where all outcomes are equally likely. Of course, the probability of any particular outcome for a continuous random variable is always zero! For the uniform random variable, any two intervals in the sample space that have the same length will have the same probability.

Theorem 7.3.1. *Suppose $X \sim \text{Unif}(a, b)$. Then, the cdf for X is given by*

$$F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x > b \end{cases}.$$

The expected value, variance, and standard deviation of X are

$$\begin{aligned} \mathbb{E}[X] &= \frac{a+b}{2}, \\ \text{Var}(X) &= \frac{(b-a)^2}{12}, \\ \sigma(X) &= \frac{b-a}{2\sqrt{3}}. \end{aligned}$$

Proof: For the cdf, clearly $\mathbb{P}(X \leq x)$ will be zero when $x < a$ (since these values of x are smaller than the smallest possible outcome). Likewise, $\mathbb{P}(X \leq x)$ will be one when $x > b$ (since these values of x are larger than the

largest possible outcome). For values between these extremes

$$\begin{aligned}
 F(x) &= \mathbb{P}(X \leq x) \\
 &= \int_a^x f(t) dt \\
 &= \int_a^x \frac{1}{b-a} dt \\
 &= \left. \frac{t}{b-a} \right|_a^x \\
 &= \frac{x}{b-a} - \frac{a}{b-a} = \frac{x-a}{b-a}.
 \end{aligned}$$

We now compute the expected value, variance, and standard deviation.

$$\begin{aligned}
 \mathbb{E}[X] &= \int_a^b \frac{x}{b-a} dx \\
 &= \left. \frac{x^2}{2(b-a)} \right|_a^b \\
 &= \frac{b^2}{2(b-a)} - \frac{a^2}{2(b-a)} = \frac{b^2 - a^2}{2(b-a)} \\
 &= \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2}.
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(X) &= \int_a^b \frac{x^2}{b-a} dx - \left(\frac{a+b}{2} \right)^2 \\
 &= \left. \frac{x^3}{3(b-a)} \right|_a^b - \frac{a^2 + 2ab + b^2}{4} \\
 &= \frac{b^3 - a^3}{3(b-a)} - \frac{a^2 + 2ab + b^2}{4} \\
 &= \frac{(b-a)(b^2 + ab + a^2)}{3(b-a)} - \frac{a^2 + 2ab + b^2}{4} \\
 &= \frac{b^2 + ab + a^2}{3} - \frac{a^2 + 2ab + b^2}{4} \\
 &= \frac{b^2 - 2ab + a^2}{12} = \frac{(b-a)^2}{12}.
 \end{aligned}$$

As always, the standard deviation is just the square root of the variance. ■

Calculating the probability of an interval is particularly easy for a uniform random variable.

Lemma 7.3.2. *Suppose $X \sim \text{Unif}(a, b)$. If $[c, d] \subseteq [a, b]$ (i.e. $a \leq c \leq d \leq b$), then*

$$\mathbb{P}(c \leq X \leq d) = \frac{d - c}{b - a}.$$

In words, the probability of a sub-interval of $[a, b]$ is just the length of the sub-interval divided by the total length of the sample space.

Proof:

$$\begin{aligned} \mathbb{P}(c \leq X \leq d) &= F(d) - F(c) \\ &= \frac{d}{b - a} - \frac{c}{b - a} \\ &= \frac{d - c}{b - a}. \quad \blacksquare \end{aligned}$$

Example 7.3.1. A standard spinner on a game board is a uniform random variable. If we use radians, then we can model the spinner as

$$\Theta \sim \text{Unif}(0, 2\pi).$$

Compute the expected value, variance, and standard deviation for Θ . What is the probability that the spinner will land in Quadrant III?

Answer: Since $\Theta \sim \text{Unif}(0, 2\pi)$, we can simply use the results above to read off the expected value and standard deviation.

$$\begin{aligned} \mathbb{E}[\Theta] &= \frac{0 + 2\pi}{2} = \pi, \\ \sigma(\Theta) &= \frac{2\pi - 0}{2\sqrt{3}} = \frac{\pi}{\sqrt{3}}. \end{aligned}$$

Since Quadrant III consists of the angles $\pi < \theta < \frac{3\pi}{2}$, we have

$$\begin{aligned} \mathbb{P}\left(\pi < \Theta < \frac{3\pi}{2}\right) &= \frac{3\pi/2 - \pi}{2\pi - 0} \\ &= \frac{\pi/2}{2\pi} \\ &= \frac{1}{4}. \end{aligned}$$

So, the spinner has a 25% chance of landing in Quadrant III. Of course, this is the same probability that the spinner will land in any of the other quadrants! ■

Example 7.3.2. A particular subway train runs every 15 minutes. If we do not know the exact train schedule, then we can think of the time T that we will have to wait for the train as a uniform random variable over the interval $S = [0, 15]$. How much time should we expect to wait on average? What is the probability that our waiting time is within one standard deviation of the expected value?

Answer: Since $T \sim \text{Unif}(0, 15)$, we know that

$$\begin{aligned}\mathbb{E}[T] &= \frac{0 + 15}{2} = \frac{15}{2} = 7.5, \\ \sigma(T) &= \frac{15 - 0}{2\sqrt{3}} = \frac{5\sqrt{3}}{2} \approx 4.33.\end{aligned}$$

So, we should expect to wait about seven and a half minutes on average. The probability that our wait time is within one standard deviation of the expected value is

$$\begin{aligned}\mathbb{P}\left(\frac{15}{2} - \frac{5\sqrt{3}}{2} \leq T \leq \frac{15}{2} + \frac{5\sqrt{3}}{2}\right) &= \left[\frac{15}{2} + \frac{5\sqrt{3}}{2} - \left(\frac{15}{2} - \frac{5\sqrt{3}}{2}\right)\right] \left(\frac{1}{15 - 0}\right) \\ &= \frac{5\sqrt{3}}{15} = \frac{\sqrt{3}}{3} \approx 0.5774.\end{aligned}$$

This means that there is almost a 58% chance that we will be waiting between about 3.2 and 11.8 minutes. ■

7.3.2 Exponential Random Variables

An exponential random variable with positive parameter λ is any random variable T with sample space $S = [0, \infty)$ and pdf

$$f(t) = \begin{cases} \lambda e^{-\lambda t}, & t \geq 0 \\ 0, & \text{otherwise} \end{cases}.$$

We write $T \sim \text{Exp}(\lambda)$ to indicate that the random variable T is distributed as an exponential random variable with parameter $\lambda > 0$. For a variety of theoretical reasons, exponential random variables are excellent models for waiting times – especially waiting times for relatively rare events.

Theorem 7.3.3. *Suppose $T \sim \text{Exp}(\lambda)$. Then, the cdf for T is given by*

$$F(t) = \begin{cases} 0, & t < 0 \\ 1 - e^{-\lambda t}, & t \geq 0 \end{cases}.$$

The expected value, variance, and standard deviation of X are

$$\begin{aligned}\mathbb{E}[T] &= \frac{1}{\lambda}, \\ \text{Var}(T) &= \frac{1}{\lambda^2}, \\ \sigma(T) &= \frac{1}{\lambda}.\end{aligned}$$

Proof: We first verify the cdf. Notice that $F(t) = 0$ for $t < 0$ since that t -value is smaller than the smallest possible outcome for T . For non-negative values of t ,

$$\begin{aligned}F(t) &= \mathbb{P}(T \leq t) \\ &= \int_0^t \lambda e^{-\lambda x} dx \\ &= -e^{\lambda x} \Big|_0^t \\ &= 1 - e^{-\lambda t}.\end{aligned}$$

For the expected value, we must use integration by parts.

$$\begin{aligned}\mathbb{E}[T] &= \int_0^\infty \lambda t e^{-\lambda t} dt \\ &= \lim_{R \rightarrow \infty} \int_0^R \lambda t e^{-\lambda t} dt \\ &= \lim_{R \rightarrow \infty} \left(-t e^{-\lambda t} - \frac{1}{\lambda} e^{-\lambda t} \right) \Big|_0^R \\ &= \lim_{R \rightarrow \infty} \left(\frac{1}{\lambda} - R e^{-\lambda R} - \frac{1}{\lambda} e^{-\lambda R} \right) \\ &= \frac{1}{\lambda}.\end{aligned}$$

The variance can be found similarly.

$$\begin{aligned}
 \text{Var}(T) &= \int_0^\infty \lambda t^2 e^{-\lambda t} dt - \left(\frac{1}{\lambda}\right)^2 \\
 &= \lim_{R \rightarrow \infty} \int_0^R \lambda t^2 e^{-\lambda t} dt - \frac{1}{\lambda^2} \\
 &= \lim_{R \rightarrow \infty} \left(-t^2 e^{-\lambda t} - \frac{1}{\lambda} 2t e^{-\lambda t} - \frac{1}{\lambda^2} 2e^{-\lambda t} \right) \Big|_0^R - \frac{1}{\lambda^2} \\
 &= \lim_{R \rightarrow \infty} \left(\frac{2}{\lambda^2} - R^2 e^{-\lambda R} - \frac{1}{\lambda} 2R e^{-\lambda R} - \frac{1}{\lambda^2} 2e^{-\lambda R} \right) - \frac{1}{\lambda^2} \\
 &= \frac{1}{\lambda^2}. \blacksquare
 \end{aligned}$$

As with the uniform random variable, there is a nice formula for the probability of an interval.

Lemma 7.3.4. Suppose $T \sim \text{Exp}(\lambda)$. If $[c, d] \subseteq [0, \infty)$ (i.e. $0 \leq c \leq d$), then

$$\mathbb{P}(c \leq T \leq d) = e^{-\lambda c} - e^{-\lambda d}.$$

Proof: Using the cdf of an exponential random variable, we find

$$\begin{aligned}
 \mathbb{P}(c \leq T \leq d) &= F(d) - F(c) \\
 &= (1 - e^{-\lambda d}) - (1 - e^{-\lambda c}) \\
 &= e^{-\lambda c} - e^{-\lambda d}. \blacksquare
 \end{aligned}$$

Example 7.3.3. The waiting time between customers in a checkout queue is an exponential random variable with expected value of 2 minutes. What is the probability that the waiting time is within one standard deviation of the expected value? What is the probability that the queue goes longer than 12 minutes without a customer?

Answer: Since the expected value is 2, we need to take $\lambda = 1/2$. So, the waiting time is given by

$$T \sim \text{Exp}\left(\frac{1}{2}\right)$$

with pdf

$$f(t) = \begin{cases} \frac{1}{2}e^{-t/2}, & t \geq 0 \\ 0, & \text{otherwise} \end{cases}.$$

Notice that the expected value and standard deviation are both equal to 2 for this random variable. The probability that the waiting time is within one standard deviation of the expected value is

$$\begin{aligned}\mathbb{P}(2 - 2 \leq T \leq 2 + 2) &= \mathbb{P}(0 \leq T \leq 4) \\ &= e^{-0/2} - e^{-4/2} \\ &= 1 - e^{-2} \approx 0.8647.\end{aligned}$$

For the probability that the waiting time is longer than 12 minutes, we find

$$\begin{aligned}\mathbb{P}(T > 12) &= 1 - \mathbb{P}(0 \leq T \leq 12) \\ &= 1 - (e^{-0/2} - e^{-12/2}) \\ &= e^{-6} \approx 0.0025.\end{aligned}$$

So, there is only about a quarter of a percent chance that the queue will go more than 12 minutes without a customer. ■

The value we found for the probability of being within one standard deviation of the expected value is the same for all exponential random variables.

Lemma 7.3.5. *The probability that an exponential random variable is within one standard deviation of its expected value is always*

$$1 - e^{-2} \approx 0.8647.$$

Proof:

$$\begin{aligned}\mathbb{P}\left(\frac{1}{\lambda} - \frac{1}{\lambda} \leq T \leq \frac{1}{\lambda} + \frac{1}{\lambda}\right) &= \mathbb{P}\left(0 \leq T \leq \frac{2}{\lambda}\right) \\ &= 1 - e^{-\lambda(2/\lambda)} \\ &= 1 - e^{-2}. \quad \blacksquare\end{aligned}$$

Example 7.3.4. Suppose the time T until a particular car part fails is an exponential random variable. Data has shown that 95% of the parts will fail within 25 years. What is the expected lifetime of the part?

Answer: We know that $\mathbb{P}(0 \leq T \leq 25) = 0.95$. Using the formula for

the cdf of an exponential random variable, we find

$$\begin{aligned}
 0.95 &= \mathbb{P}(0 \leq T \leq 25) \\
 0.95 &= 1 - e^{-25\lambda} \\
 e^{-25\lambda} &= 1 - 0.95 = \frac{1}{20} \\
 -25\lambda &= \ln\left(\frac{1}{20}\right) = -\ln(20) \\
 \lambda &= \frac{\ln(20)}{25}.
 \end{aligned}$$

Now that we have the parameter λ , we know

$$\begin{aligned}
 \mathbb{E}[T] &= \frac{1}{\lambda} \\
 &= \frac{25}{\ln(20)} \approx 8.35.
 \end{aligned}$$

So, the expected lifetime of the part is 8.35 years. ■

7.3.3 Beta Random Variables

A beta random variable with positive shape parameters α and β is any random variable X with sample space $S = [0, 1]$ and pdf

$$f(x) = \begin{cases} \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}.$$

The number $1/B(\alpha, \beta)$ is the appropriate normalizing factor for the random variable. We write $X \sim \text{Beta}(\alpha, \beta)$ to indicate that the random variable X is distributed as a beta random variable with shape parameters α and β .

The formula for $B(\alpha, \beta)$ is complicated in general,¹ but when α and β are natural numbers, there is a closed form.

Lemma 7.3.6. *When the shape parameters are positive integers m and n , the normalizing factor takes the form*

$$B(m, n) = \frac{(m-1)!(n-1)!}{(m+n-1)!}.$$

¹The general form for $B(\alpha, \beta)$ involves the *gamma function* which extends the notion of factorials to most real numbers.

Proof: The form for $B(m, n)$ can be determined by induction. We will only verify it for the cases when $n = 1$ and $n = 2$. When $n = 1$, we have

$$f(x) = \frac{x^{m-1}}{B(m, 1)}$$

over $0 \leq x \leq 1$. This means we need

$$\begin{aligned} 1 &= \frac{1}{B(m, 1)} \int_0^1 x^{m-1} dx \\ &= \frac{1}{B(m, 1)} \left. \frac{x^m}{m} \right|_0^1 \\ &= \frac{1}{B(m, 1)} \left(\frac{1}{m} \right). \end{aligned}$$

So, we need

$$B(m, 1) = \frac{1}{m}.$$

In the form given above

$$B(m, 1) = \frac{(m-1)!(0)!}{m!} = \frac{1}{m}.$$

When $n = 2$,

$$f(x) = \frac{x^{m-1}(1-x)}{B(m, 2)}$$

over $0 \leq x \leq 1$. Repeating our computation above.

$$\begin{aligned} 1 &= \frac{1}{B(m, 2)} \int_0^1 x^{m-1}(1-x) dx \\ &= \frac{1}{B(m, 2)} \left(\frac{x^m}{m} - \frac{x^{m+1}}{m+1} \right) \Big|_0^1 \\ &= \frac{1}{B(m, 2)} \left(\frac{1}{m} - \frac{1}{m+1} \right) \\ &= \frac{1}{B(m, 2)} \left(\frac{1}{(m+1)(m)} \right). \end{aligned}$$

So, we need

$$B(m, 2) = \frac{1}{(m+1)(m)}.$$

In the form given above

$$B(m, 2) = \frac{(m-1)!(1)!}{(m+1)!} = \frac{1}{(m+1)(m)}. \blacksquare$$

Lemma 7.3.7. *Suppose that $X \sim \text{Beta}(\alpha, \beta)$. Then*

$$\begin{aligned}\mathbb{E}[X] &= \frac{\alpha}{\alpha + \beta}, \\ \text{Var}(X) &= \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}, \\ \sigma(X) &= \sqrt{\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}}.\end{aligned}$$

The cdf for X is quite complicated in general.

Proof: Again, we will only verify this information for $\alpha = m$ (a positive integer), $\beta = 1$, and $\beta = 2$. In the case $X \sim \text{Beta}(m, 1)$ we find

$$\begin{aligned}\mathbb{E}[X] &= \int_0^1 x \left(\frac{x^{m-1}}{B(m, 1)} \right) dx \\ &= \frac{1}{B(m, 1)} \int_0^1 x^m dx \\ &= \frac{1}{B(m, 1)} \frac{x^{m+1}}{m+1} \Big|_0^1 \\ &= \frac{1}{B(m, 1)} \left(\frac{1}{m+1} \right) = \frac{m}{m+1}.\end{aligned}$$

$$\begin{aligned}
\text{Var}(X) &= \int_0^1 x^2 \left(\frac{x^{m-1}}{B(m, 1)} \right) dx - \left(\frac{m}{m+1} \right)^2 \\
&= \frac{1}{B(m, 1)} \int_0^1 x^{m+1} dx - \frac{m^2}{(m+1)^2} \\
&= \frac{1}{B(m, 1)} \frac{x^{m+2}}{m+2} \Big|_0^1 - \frac{m^2}{(m+1)^2} \\
&= \frac{1}{B(m, 1)} \left(\frac{1}{m+2} \right) - \frac{m^2}{(m+1)^2} \\
&= \frac{m}{m+2} - \frac{m^2}{(m+1)^2} \\
&= \frac{m(m+1)^2 - m^2(m+2)}{(m+1)^2(m+2)} \\
&= \frac{m(m^2 + 2m + 1) - m^3 - 2m^2}{(m+1)^2(m+2)} \\
&= \frac{m}{(m+1)^2(m+2)}.
\end{aligned}$$

Both of these agree with the general forms given above.

For $X \sim \text{Beta}(m, 2)$ we find

$$\begin{aligned}
\mathbb{E}[X] &= \int_0^1 x \left(\frac{x^{m-1}(1-x)}{B(m, 2)} \right) dx \\
&= \frac{1}{B(m, 2)} \int_0^1 x^m - x^{m+1} dx \\
&= \frac{1}{B(m, 2)} \left(\frac{x^{m+1}}{m+1} - \frac{x^{m+2}}{m+2} \right) \Big|_0^1 \\
&= \frac{1}{B(m, 2)} \left(\frac{1}{m+1} - \frac{1}{m+2} \right) \\
&= \frac{(m+1)m}{(m+2)(m+1)} = \frac{m}{m+2}.
\end{aligned}$$

$$\begin{aligned}
\text{Var}(X) &= \int_0^1 x^2 \left(\frac{x^{m-1}(1-x)}{B(m, 2)} \right) dx - \left(\frac{m}{m+2} \right)^2 \\
&= \frac{1}{B(m, 2)} \int_0^1 x^{m+1} - x^{m+2} dx - \frac{m^2}{(m+2)^2} \\
&= \frac{1}{B(m, 2)} \left(\frac{x^{m+2}}{m+2} - \frac{x^{m+3}}{m+3} \right) \Big|_0^1 - \frac{m^2}{(m+2)^2} \\
&= \frac{1}{B(m, 2)} \left(\frac{1}{m+2} - \frac{1}{m+3} \right) - \frac{m^2}{(m+2)^2} \\
&= \frac{(m+1)m}{(m+3)(m+2)} - \frac{m^2}{(m+2)^2} \\
&= \frac{m(m+1)(m+2) - m^2(m+3)}{(m+3)(m+2)^2} \\
&= \frac{m(m^2 + 3m + 2) - m^3 - 3m^2}{(m+2)^2(m+3)} \\
&= \frac{2m}{(m+2)^2(m+3)}.
\end{aligned}$$

Once again, these agree with the general forms given above. ■

The beta distribution is often used to model the frequency that some particular property occurs in a population. For example, the percentage of a population that carries a particular allele (as in the Hardy-Weinberg Equilibrium in Section 6.6) is often modeled as a beta random variable. To understand why this is the case, we use Bayesian inference. Suppose that we are considering an allele whose frequency P in the population is unknown. Initially, we will assume that $P \sim \text{Unif}(0, 1)$. This is called a *prior distribution* since we assume this before having any data. Now suppose that we test a random sample of individuals to determine whether they have the allele. The results are that m individuals carry the allele while n do not.

The question we would like to explore is how the probabilities we associate to P should change given the data we have collected. In the considerations below, we will write “Data” to stand for the results of our experiment. Using Bayes’ Formula, we find

$$\mathbb{P}(P = p|\text{Data}) = \frac{\mathbb{P}(\text{Data}|P = p)\mathbb{P}(P = p)}{\mathbb{P}(\text{Data})}$$

There are a number of issues with what we are trying to do. First, $\mathbb{P}(P = p|\text{Data})$ and $\mathbb{P}(P = p)$ are technically zero since P is a continuous

random variable. Even worse, $\mathbb{P}(\text{Data}|P = p)$ is ill-defined since we cannot condition on an event that has probability zero! These technical issues can be resolved by considering pdfs rather than actual probabilities, but we will simply pretend as though what is written above is sensible. A major problem is that we have no idea how to compute $\mathbb{P}(\text{Data})$. Nevertheless, we can find the functional form for the pdf of P .

If we know that the allele occurs with frequency p in the population (i.e. we know $P = p$), then the probability that m randomly selected individuals have the allele while n do not is a binomial random variable with $m + n$ trials and a probability of success p . This means

$$\mathbb{P}(\text{Data}|P = p) = \binom{m+n}{m} p^m (1-p)^n.$$

As a result,

$$\begin{aligned} \mathbb{P}(P = p|\text{Data}) &= \frac{\mathbb{P}(\text{Data}|P = p)\mathbb{P}(P = p)}{\mathbb{P}(\text{Data})} \\ &= \frac{\binom{m+n}{m} p^m (1-p)^n}{\mathbb{P}(\text{Data})} \\ &= C p^m (1-p)^n. \end{aligned}$$

Notice that we replaced $\mathbb{P}(P = p)$ by 1 since the pdf for P is 1 according to our assumption that P is distributed uniformly over $[0, 1]$. Even though it has not been normalized, we recognize this as the functional form of a beta random variable. In particular,

$$P \sim \text{Beta}(m+1, n+1).$$

So, the data updates our prior uniform distribution for P to a beta random variable. Oddly, the expected value of P due to these considerations is

$$\mathbb{E}[P] = \frac{m+1}{m+n+2}$$

rather than the more sensible $m/(m+n)$. By reconsidering the choice of prior distribution, it is possible to end up with $P \sim \text{Beta}(m, n)$.² These technical

²An intuitive argument for this would be that we need at least a minimum amount of data to come up with our prior distribution. In order to assume that P is some value between 0 and 1, we at least need to know that one individual has the allele and another one does not. This means we have already used a portion of our data to construct the prior. This leaves $m-1$ positive results and $n-1$ negatives. Using this data in our Bayesian analysis gives $P \sim \text{Beta}(m, n)$.

considerations are beyond the scope of this text. The thing to take away from the discussion is that the beta distribution is quite often a reasonable model for the unknown frequency of some characteristic of a population when we have data about the characteristic from a sample group.

Example 7.3.5. We want to estimate the probability that paramecia will carry a particular allele related to cilia formation. We examine 10 individuals and determine that 7 of them have the allele in question.³ Determine the expected value and standard deviation of the frequency P that the allele occurs among paramecia. What is the probability that P will be within one standard deviation of its expected value?

Answer: We can take $P \sim \text{Beta}(7, 3)$ since there were 7 samples with the allele and 3 without. We then know

$$\begin{aligned}\mathbb{E}[P] &= \frac{7}{10}, \\ \text{Var}(P) &= \frac{21}{(10)^2(11)} = \frac{21}{1100}, \\ \sigma(P) &= \frac{1}{10} \sqrt{\frac{21}{11}} \approx 0.1382.\end{aligned}$$

We also know the pdf is given by

$$f(p) = \frac{p^6(1-p)^2}{B(7, 3)}$$

over $S = [0, 1]$ where

$$B(7, 3) = \frac{6!2!}{9!} = \frac{1}{252}.$$

³This is an absurdly small sample size, but for larger values the integrals would need to be done by computer.

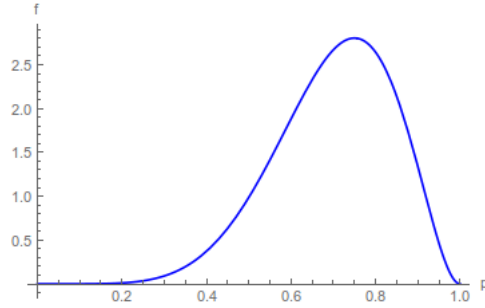


Figure 7.3.1: Graph of the pdf for Beta(7, 3)

In order to find the probability that P is within one standard deviation of its expected value, we first find its cdf. We know that $F(p)$ is zero for $p < 0$ and 1 for $p > 1$. For values of p between these two values, we have

$$\begin{aligned}
 F(p) &= \mathbb{P}(0 \leq P \leq p) \\
 &= \int_0^p 252x^6(1-x)^2 dx \\
 &= 252 \int_0^p x^6(1-2x+x^2) dx \\
 &= 252 \int_0^p (x^6 - 2x^7 + x^8) dx \\
 &= 252 \left(\frac{x^7}{7} - \frac{x^8}{4} + \frac{x^9}{9} \right) \Big|_0^p \\
 &= 252 \left(\frac{p^7}{7} - \frac{p^8}{4} + \frac{p^9}{9} \right) \\
 &= 36p^7 - 63p^8 + 28p^9.
 \end{aligned}$$

So, the probability that P is within one standard deviation of its expected value is

$$\begin{aligned}
 &\mathbb{P}(\mathbb{E}[P] - \sigma(X) \leq P \leq \mathbb{E}[P] + \sigma(X)) \\
 &= F\left(\frac{7}{10} + \frac{1}{10}\sqrt{\frac{21}{11}}\right) - F\left(\frac{7}{10} - \frac{1}{10}\sqrt{\frac{21}{11}}\right) \\
 &= \frac{55141509717}{114382812500} \sqrt{\frac{21}{11}} \approx 0.6661.
 \end{aligned}$$

This means that there is almost a 67% chance that the frequency of the allele in the population is between roughly 0.56 and 0.84. ■

Notice that even for relatively small values of α and β , the computations for beta random variables can be onerous. For reasonable amounts of data, there is not much choice but to use a calculator or computer mathematics package to work out predictions from the beta distribution.

Example 7.3.6. Suppose that our data on the paramecia had been that 76 out of 100 individuals carry the allele. What is the expected value and standard deviation for the frequency P ? What is the probability that P is within one standard deviation of its expected value?

Answer: We take $P \sim \text{Beta}(76, 24)$. This means that

$$f(p) = \frac{p^{75}(1-p)^{23}}{B(76, 24)} \quad \text{on } 0 \leq p \leq 1,$$

$$\mathbb{E}[P] = \frac{76}{100} = 0.76,$$

$$\text{Var}(P) = \frac{76(24)}{(100)^2(101)} = \frac{114}{63125},$$

$$\sigma(P) = \sqrt{\frac{114}{63125}} \approx 0.0425.$$

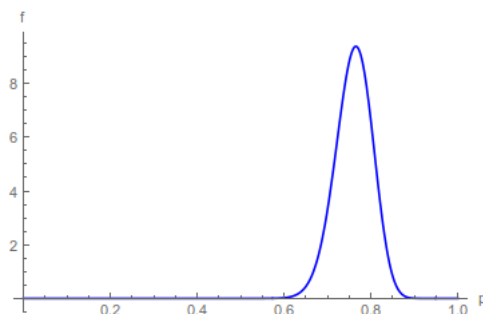


Figure 7.3.2: Graph of the pdf for Beta(76, 24)

Notice that the larger amount of data gives us a much smaller standard deviation. This is a consequence of the fact that the pdf for Beta(76, 24) is more strongly concentrated around its expected value.

To compute the probability that P lies within one standard deviation of its expected value, we must compute the following integral. Given the exponents involved, the only choice we have is to use a computer or calculator to compute the probability.

$$\begin{aligned}\mathbb{P}(0.7175 \leq P \leq 0.8025) &= \int_{0.7175}^{0.8025} \frac{p^{75}(1-p)^{23}}{B(76, 24)} dp \\ &\approx 0.6815.\end{aligned}$$

So, there is just over a 68% chance that the allele occurs in between 71.75% and 80.25% of all paramecia. ■

7.3.4 Normal Random Variables

Arguably the most important random variable, continuous or discrete, is the normal random variable. A random variable X is said to be normal with expected value μ and standard deviation $\sigma > 0$ if its sample space is $S = (-\infty, \infty)$ and its pdf is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

We write $X \sim N(\mu, \sigma^2)$ to indicate that the random variable X is distributed as a normal random variable with expected value μ and variance σ^2 . The graph of this pdf is often referred to as the “bell curve” or Gaussian curve.

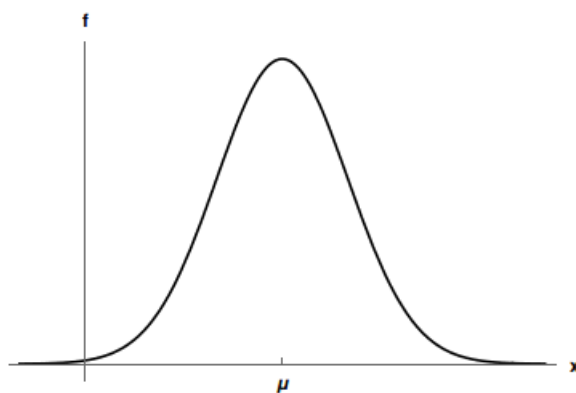


Figure 7.3.3: Typical Plot of a Normal pdf

Lemma 7.3.8. For $X \sim N(\mu, \sigma^2)$ we have

$$\begin{aligned}\mathbb{E}[X] &= \mu, \\ \text{Var}(X) &= \sigma^2, \\ \sigma(X) &= \sigma.\end{aligned}$$

Proof: These facts can be verified by integration which we omit here. Notice that the convention is to list the variance σ^2 rather than the standard deviation. Unfortunately, the cdf for a normal random variable is not a simple combination of the usual functions we are familiar with (that is, not a combination of polynomial, rational, exponential, logarithmic, or trigonometric functions). ■

There is one very special normal random variable. The **standard normal random variable** is $Z \sim N(0, 1)$ with sample space $S = (-\infty, \infty)$ and pdf

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}.$$

The use of the letter Z for the standard normal random variable is nearly universal in the literature.

Considering the importance of the standard normal, its cdf (conventionally written as Φ) has been extensively studied.

$$\Phi(z) = \mathbb{P}(Z \leq z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

Before the advent of modern computing devices, textbooks used tables of numerical values for Φ like those found in Appendix D.

Example 7.3.7. What is the probability that a standard normal random variable, Z , will take on a value between 0.5 and 0.75? What is the probability that Z will be within one-half of a standard deviation of its expected value?

Answer: The first question is asking for

$$\mathbb{P}(0.5 \leq Z \leq 0.75) = \Phi(0.75) - \Phi(0.5).$$

To find the value $\Phi(0.75)$, we go to the second page of the table. We find the row labeled “0.7” and then move over to the column labeled “.05.” This

gives us $\Phi(0.75) = 0.7734$. For $\Phi(0.5)$, we find the row labeled “0.5” and then move over to the first column (labeled “.00”). We find $\Phi(0.5) = 0.6915$. So,

$$\mathbb{P}(0.5 \leq Z \leq 0.75) = \Phi(0.75) - \Phi(0.5) = 0.7734 - 0.6915 = 0.0819.$$

Since the expected value of Z is 0 and the standard deviation is 1, the second question is asking for

$$\mathbb{P}(-0.5 \leq Z \leq 0.5) = \Phi(0.5) - \Phi(-0.5) = 0.6915 - 0.3085 = 0.3830. \blacksquare$$

It turns out that the cdf for the standard normal random variable can be used to compute probabilities for any normal random variable.

Theorem 7.3.9 (Z-Scores). *Suppose that $X \sim N(\mu, \sigma^2)$. Then the random variable*

$$Z = \frac{X - \mu}{\sigma}$$

is distributed as the standard normal random variable.

Proof: Suppose that $X \sim N(\mu, \sigma^2)$, then for the definition of Z given above we have

$$\begin{aligned} \mathbb{P}(a \leq Z \leq b) &= \mathbb{P}\left(a \leq \frac{X - \mu}{\sigma} \leq b\right) \\ &= \mathbb{P}(\mu + a\sigma \leq X \leq \mu + b\sigma) \\ &= \int_{\mu+a\sigma}^{\mu+b\sigma} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx. \end{aligned}$$

If we make the change of variables $z = (x - \mu)/\sigma$ in the integral above, we find

$$\mathbb{P}(a \leq Z \leq b) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz.$$

But this is just the pdf of the standard normal random variable. $\blacksquare$

The proof shows us how to use the standard normal to find probabilities for any $X \sim N(\mu, \sigma^2)$.

$$\begin{aligned} \mathbb{P}(a \leq X \leq b) &= \mathbb{P}\left(\frac{a - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{b - \mu}{\sigma}\right) \\ &= \mathbb{P}\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right) \\ &= \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right). \end{aligned}$$

This process is called finding the **z-scores** for X .

Example 7.3.8. Suppose that $X \sim N(2, 16)$. Find the probability that an outcome for X will lie between 3 and 5.

Answer: Since $\mu = 2$ and $\sigma = 4$, we find

$$\begin{aligned}\mathbb{P}(3 \leq X \leq 5) &= \mathbb{P}\left(\frac{3-2}{4} \leq \frac{X-2}{4} \leq \frac{5-2}{4}\right) \\ &= \mathbb{P}(0.25 \leq Z \leq 0.75) \\ &= \Phi(0.75) - \Phi(0.25) \\ &= 0.7734 - 0.5987 = 0.1747. \blacksquare\end{aligned}$$

Example 7.3.9. Suppose that $Y \sim N(10, 25)$. Find the probability that an outcome for Y will lie within two standard deviations of its expected value. What is the probability that an outcome for Y will be greater than 20?

Answer: Since $\mu = 10$ and $\sigma = 5$, we find

$$\begin{aligned}\mathbb{P}(10 - 10 \leq Y \leq 10 + 10) &= \mathbb{P}\left(\frac{0-10}{5} \leq \frac{Y-10}{5} \leq \frac{20-10}{5}\right) \\ &= \mathbb{P}(-2 \leq Z \leq 2) \\ &= \Phi(2) - \Phi(-2) \\ &= 0.9772 - 0.0228 = 0.9544.\end{aligned}$$

For the second question, we rewrite the probability so that it directly involves the cdf for standard normal random variable.

$$\begin{aligned}\mathbb{P}(Y > 20) &= 1 - \mathbb{P}(Y \leq 20) \\ &= 1 - \mathbb{P}\left(\frac{Y-10}{5} \leq \frac{20-10}{5}\right) \\ &= 1 - \mathbb{P}(Z \leq 2) \\ &= 1 - \Phi(2) \\ &= 1 - 0.9772 = 0.0228. \blacksquare\end{aligned}$$

The reason that normal random variables are so important is due to a theoretical result known as the **central limit theorem**. Suppose that $X_1, X_2, \dots, X_n$ are n identically distributed and independent⁴ real-valued

⁴The requirement that the X_i be independent can be relaxed. In fact, the X_i can be *weakly dependent*, where weak dependence has a very technical definition.

random variables. Suppose that $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$ (since the X_i are all distributed the same way, they must have the same expected value and variance). If we take the average of these n values, we get

$$S_n = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

Since the values of all of the X_i are random, so is their average value, S_n . Since S_n is a new random variable, we can ask how it is distributed. The central limit theorem says that for large n (i.e. a large sample size), the distribution of S_n is very well approximated by $N(\mu, \sigma^2/n)$. Notice that the averages limit to a normal random variable *regardless of the underlying probability distribution of the individual X_i* . Also notice that the standard deviation of S_n , $\sigma(S_n) = \sigma/\sqrt{n}$, decreases as we average more and more samples.

Example 7.3.10. Suppose that the height, H , of a woman in the United States is a normal random variable with expected value of 64 inches and a standard deviation of 3 inches. What is the probability that a woman in the US is within 5 inches of the average height? What percentage of women in the US are taller than 6 feet (72 inches)?

Answer: The first question is asking for the probability that H is between 59 and 69 inches. So,

$$\begin{aligned} \mathbb{P}(59 \leq H \leq 69) &= \mathbb{P}\left(\frac{59 - 64}{3} \leq \frac{H - 64}{3} \leq \frac{69 - 64}{3}\right) \\ &= \mathbb{P}\left(-\frac{5}{3} \leq Z \leq \frac{5}{3}\right) \\ &= \Phi(1.\bar{6}) - \Phi(-1.\bar{6}). \end{aligned}$$

The problem is that $5/3 = 1.\bar{6}$ is not in our table! We will round this number to 1.67 and use this to get an approximation.

$$\begin{aligned} \mathbb{P}(59 \leq H \leq 69) &\approx \Phi(1.67) - \Phi(-1.67) \\ &\approx 0.9525 - 0.0475 = 0.9050. \end{aligned}$$

So, about 90.5% of women in the US are between 59 and 69 inches in height. For the second question, we need to determine the probability that H is

greater than 72.

$$\begin{aligned}
 \mathbb{P}(H > 72) &= 1 - \mathbb{P}(H \leq 72) \\
 &= 1 - \mathbb{P}\left(\frac{H - 64}{3} \leq \frac{72 - 64}{3}\right) \\
 &= 1 - \mathbb{P}\left(Z \leq \frac{8}{3}\right) \\
 &= 1 - \Phi(2.\bar{6}) \approx 1 - \Phi(2.67) \\
 &\approx 1 - 0.9962 = 0.0038.
 \end{aligned}$$

So, only about 0.4% of women in the US are 6 feet or taller. ■

There are several things we should mention about the last example. First, the sample space of a normal random variable is always $(-\infty, \infty)$. This seems strange considering that the height of a person cannot possibly be a negative number! However

$$\mathbb{P}(H \leq 0) = \mathbb{P}\left(\frac{H - 64}{3} \leq -\frac{64}{3}\right) = \mathbb{P}(Z \leq -21.\bar{3}) \approx 3 \times 10^{-101}.$$

In other words, the probability that H is less than zero is so incredibly small that it might as well be zero. This means that the normal random variable $N(64, 9)$ is *approximately* restricted to $[0, \infty)$. The second issue is that we rounded our z-scores so that we could use the table. This undoubtedly introduced some amount of error. In fact, the table entries themselves are just approximate answers.

Before computational devices became portable, Z-Tables (and similar tables for other random variables) were the best one could hope for. However, most modern graphing calculators come with pre-programmed functions that compute probabilities for normal random variables. You can find information on how to use your TI-83, 84, or 89 to compute probabilities associated to normal random variables (and others) in Appendix E.

Example 7.3.11. A machine designed to fill bottles of water is supposed to deliver 500 mL to each bottle. However, the actual volume of water delivered is a normal random variable with expected value 500 mL and a standard deviation of 2 mL. How often will the machine fill a bottle with less than 495 mL? If the maximum volume the bottle can hold is 507 mL, what is the probability that the machine causes a bottle to overflow?

Answer: Letting V stand for the amount of water delivered by the machine, we are told that $V \sim N(500, 4)$ (remember, by convention we list variance rather than standard deviation). The first question is asking for $\mathbb{P}(V \leq 495)$. Using the TI-89, we find

$$\mathbb{P}(V \leq 495) \approx 0.0062.$$

So, there is about a 0.62% chance that a bottle will be filled with less than 495 mL. The second question is asking for the probability that $V > 507$. This gives

$$\mathbb{P}(V > 507) = 1 - \mathbb{P}(V \leq 507) \approx 1 - 0.9998 \approx 0.0002.$$

This means that there is only a 0.02% chance that a bottle will overflow. ■

The final thing we need to mention about the normal random variable is a nice interpretation of the standard deviation σ . Consider the following graph of the standard normal random variable.

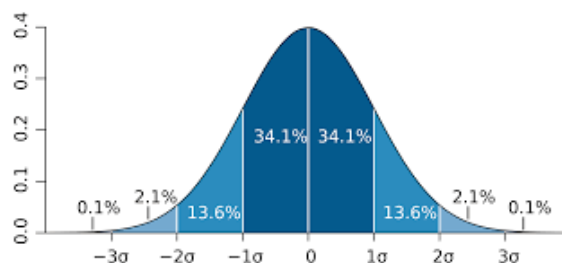


Figure 7.3.4: The Standard Normal Random Variable and Standard Deviation

We see that an outcome from the standard normal Z has a probability of about 68.2% of lying within one standard deviation of its expected value. By taking Z scores, we see that this is true for any normal random variable! To be exact, if $X \sim N(\mu, \sigma^2)$ then

$$\mathbb{P}(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.6827.$$

Similarly, the probability that the outcome from a normal random variable is within two standard deviations of its expected value is

$$\mathbb{P}(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.9545.$$

That means there is over a 95% chance that the outcome of a normal random variable will be within 2 standard deviations of its expected value. If we go out to three standard deviations, we find

$$\mathbb{P}(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.9973.$$

In many disciplines, a result that is more than 3 standard deviations away from the expected value is considered an extreme outlier. Physicists often use a 5σ standard! Using a calculator, we can find that

$$\mathbb{P}(X < -5\sigma) + \mathbb{P}(X > 5\sigma) \approx 5.742 \times 10^{-7}.$$

So, an outcome that is more than 5 standard deviations away from the expected value would almost certainly not occur by chance!

7.4 Problems

- 1.) Consider the random variable X whose pdf has the form

$$f(x) = \begin{cases} \frac{A}{x^4}, & x \geq 1 \\ 0, & \text{otherwise} \end{cases}.$$

What is the effective sample space for X ? Normalize X , and find the probability that an outcome for X will lie between 2 and 4.

- 2.) A random variable Y has pdf of the form

$$f(y) = \begin{cases} Ay(5 - y), & \text{if } 0 \leq y \leq 5 \\ 0, & \text{otherwise} \end{cases}.$$

What is the effective sample space for Y ? Normalize Y , and find the probability that an outcome for Y lies between 1 and 2.

- 3.) A random variable R has pdf of the form

$$f(r) = \begin{cases} \frac{A}{\sqrt{r}}, & 0 < r \leq 1 \\ 0, & \text{otherwise} \end{cases}.$$

What is the effective sample space for R ? Normalize R , and find the probability that an outcome for R lies between $1/2$ and 1.

- 4.) Find the cdf, expected value, variance, and standard deviation for the random variable X in Problem 1.
- 5.) Find the cdf, expected value, variance, and standard deviation for the random variable Y in Problem 2.
- 6.) Find the cdf, expected value, variance, and standard deviation for the random variable R in Problem 3.
- 7.) Let X be a random variable with pdf of the form

$$f(x) = \begin{cases} kx^3(1-x), & \text{if } 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}.$$

- a.) Find the correct value of k which normalizes f .
 - b.) Compute the probability that X lies between 0 and $1/4$.
 - c.) Find the cdf for X .
 - d.) Compute the expected value of X .
 - e.) Compute the variance of X .
- 8.) Suppose we have a random variable S with pdf given by

$$f(s) = \begin{cases} As \sin(s), & \text{if } 0 \leq s \leq \pi \\ 0, & \text{otherwise} \end{cases}.$$

- a.) Find the correct value of A which normalizes f .
 - b.) What is the probability that S lies between 0 and $\pi/2$?
 - c.) What is the cdf for S ?
 - d.) Find the expected value of S . (NOTE: We can also compute the variance of this random variable, but it is a little messy!)
- 9.) Suppose that X is a random variable with pdf of the form

$$f(x) = \begin{cases} \frac{A}{1+x^2}, & \text{if } x \geq 0 \\ 0, & \text{otherwise} \end{cases}.$$

- a.) Find the correct value of A which normalizes f .
- b.) Find the probability that X lies between 0 and 1.

- c.) What goes wrong when you try to find the expected value or variance of X ?
- 10.) The **median value** of a random variable, X , is the value, m , in the sample space such that

$$\mathbb{P}(X \leq m) = \frac{1}{2}.$$

If F is the cdf for X , then the median value satisfies $F(m) = \frac{1}{2}$. Find the medians of the random variables in Problems 1, 3, and 9. Note that you MUST normalize these random variables before finding the median!

- 11.) Another special continuous random variable is the *Chi-Squared Distribution* χ_k^2 which has outcomes in $[0, +\infty)$. The parameter k is a positive integer and is referred to as the degrees of freedom of the distribution. For $k = 4$, χ_4^2 (the chi-squared distribution with 4 degrees of freedom) has pdf

$$f(x) = \begin{cases} \frac{1}{4}xe^{-x/2}, & \text{if } x \geq 0 \\ 0, & \text{otherwise} \end{cases}.$$

Compute the expected value, variance, and standard deviation for this random variable.

- 12.) Consider a random variable $X \sim \text{Unif}(2, 10)$. What is the expected value and standard deviation of X . What is the probability that X lies between 3 and 5?
- 13.) Suppose $X \sim \text{Unif}(0, 1)$. From X , we create a new random variable: $Y = aX + b$ where a and b are fixed real numbers.
- a.) What is the sample space for Y ?
- b.) Compute the cdf for Y . That is, find

$$F(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(aX + b \leq y).$$

- c.) Taking the derivative of the result in b.) (over the sample space for Y) gives the pdf for Y .
- d.) Based on c.), what kind of random variable is Y ?

- 14.) Suppose that $T \sim \text{Exp}(3)$. What is the expected value, variance, and standard deviation of T ? What is the probability that T is within 2 standard deviations of its expected value? (HINT: For the last question, pay close attention to the sample space for T .)
- 15.) A restaurant owner finds that the time customers wait between sitting down and their server arriving is a random variable distributed exponentially with an average of 2.5 minutes. What is the probability that a customer will wait for their server for more than 4 minutes?
- 16.) Consider the situation in the previous problem. The owner wishes to post a sign saying that customers who have waited longer than M minutes will receive a free meal. How long should M be so that only 0.1 % of customers will get a free meal?
- 17.) If a random variable P is distributed as $\text{Beta}(3, 2)$, what is the probability that P will lie within one standard deviation of its expected value?
- 18.) The **skewness** of a random variable X is a measure of how asymmetric its pdf is about the mean. If X has pdf f , expected value $\mathbb{E}[X] = \mu$ and standard deviation $\sigma(X) = \sigma$, then the skewness of X is defined to be

$$\gamma(X) = \frac{\mathbb{E}[X^3] - 3\mu\sigma^2 - \mu^3}{\sigma^3}$$

where

$$\mathbb{E}[X^3] = \int_{-\infty}^{\infty} x^3 f(x) dx.$$

Compute the skewness of a random variable $X \sim \text{Beta}(2, 3)$.

- 19.) Suppose that the scores on an exam are normally distributed with an average of 73 and a standard deviation of 8. What percentage of students made a score of 59 or below?
- 20.) One study found that the amount of paper discarded by a household during any given week is a normal random variable with average 10 lbs and a standard deviation of 4 lbs. What percentage of households discard more than 13 pounds of paper in a given week?

- 21.) **Challenge:** The **moment generating function** (mgf) of a random variable X with pdf f is

$$M_X(t) = \mathbb{E}[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f(x) dx.$$

Find the moment generating function for the standard normal random variable $Z \sim N(0, 1)$. Can you find the mgf for a generic normal random variable $X \sim N(\mu, \sigma^2)$?

Chapter 8

A Primer on Matrices and Vectors

In this chapter, we will explore some basic results about matrices and vectors. Many of the results below will be presented without proof, but the interested reader can consult any textbook on Linear Algebra for more details (e.g. [\[FIS03\]](#)). Our goal is simply to become familiar with the small portion of this vast subject that will be of use to us in the remainder of this book.

This chapter begins with a discussion of systems of linear equations using matrix methods. This will naturally lead us to a general discussion of matrices – especially square matrices. In particular, we will see that a square matrix can be interpreted as a linear function which transforms a vector of the right size into another vector of the same size.

The most important concept that we develop in this chapter is the idea of eigenvalues and eigenvectors. These are sometimes called “characteristic values” and “characteristic vectors,” respectively. And indeed, we will see that they give us crucial information about how a square matrix transforms vectors. There is a nice geometric interpretation of eigenvectors in the case that the eigenvalues are real. However, eigenvalues are often complex numbers. While the geometric interpretation can be extended in a natural way to the complex case, the visualization becomes problematic. If you need to review basic results about complex numbers, see [Appendix A](#).

8.1 Systems of Linear Equations

A system of linear equations is a set of m equations ($m \geq 2$) in the variables $x_1, x_2, \dots, x_n$ of the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m. \end{aligned} \tag{8.1}$$

The coefficients a_{ij} and the numbers b_i are assumed to be constants (real or complex makes no difference). We describe the size of a linear system with m equations and n unknowns as an $m \times n$ system of linear equations. Note that for systems with 2 or 3 unknowns, x, y , and z are often used for variables instead of x_1, x_2 , and x_3 .

The simplest set of such equations are 2×2 linear systems. For this case, the tried and true method of solving one equation for one of the variables and substituting into the other is probably the most efficient.

Example 8.1.1. Solve the following system:

$$\begin{aligned} 2x + 3y &= 8 \\ x - 5y &= -9. \end{aligned}$$

Answer: Solving the second equation for x gives $x = 5y - 9$. Substituting this into the first equation gives

$$\begin{aligned} 2(5y - 9) + 3y &= 8 \\ 10y - 18 + 3y &= 8 \\ 13y &= 26 \\ y &= 2. \end{aligned}$$

This also gives $x = 5(2) - 9 = 1$. Hence, the solution to the system is the single point $x = 1$ and $y = 2$. ■

Unfortunately, this method is not always the best way to attack problems with more than 2 variables. In the sections below, we will develop the matrix technique known as *row reduction* which is a far easier and quicker method of solving such problems.

8.1.1 Basic Results for Systems of Linear Equations

We first give a number of basic results about systems of linear equations that are important to know.

Theorem 8.1.1. *For an $m \times n$ linear system of the form 8.1, there are three possibilities for the set of solutions.*

- *There are **no solutions**.*
- *There is **exactly one solution**.*
- *There are an **infinite number of solutions**.*

*A linear system with no solutions is often said to be **inconsistent**.*

Proof: Clearly no solutions, and exactly one solution are possibilities. Now suppose we have two solutions to 8.1: $(a_1, a_2, \dots, a_n)$ and $(b_1, b_2, \dots, b_n)$. Let α and β be any two real numbers that add up to 1 (i.e. $\alpha + \beta = 1$ or $\beta = 1 - \alpha$). Then the list of numbers

$$(\alpha a_1 + \beta b_1, \alpha a_2 + \beta b_2, \dots, \alpha a_n + \beta b_n)$$

is also a solution. In order to see this, we substitute these new numbers into each of the equations in 8.1 and verify that we have a true statement. Looking at the i -th equation, we see

$$\begin{aligned} a_{i1}(\alpha a_1 + \beta b_1) + a_{i2}(\alpha a_2 + \beta b_2) + \dots + a_{in}(\alpha a_n + \beta b_n) \\ &= \alpha(a_{i1}a_1 + a_{i2}a_2 + \dots + a_{in}a_n) + \beta(a_{i1}b_1 + a_{i2}b_2 + \dots + a_{in}b_n) \\ &= \alpha b_i + \beta b_i \\ &= (\alpha + \beta)b_i \\ &= b_i. \end{aligned}$$

Hence, any two solutions to 8.1 automatically give rise to an infinite family of solutions. ■

If we look carefully at the proof above, the condition that $\alpha + \beta = 1$ is not necessary if ALL of the numbers b_i are zero. This leads us to the following definition.

A system of linear equations is said to be **homogeneous** if all of the numbers b_i appearing on the right hand side of 8.1 are zero. That is, an $m \times n$ linear system is homogeneous if it is of the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0. \end{aligned} \tag{8.2}$$

Otherwise, the system is said to be *inhomogeneous*.

There are a few other terms which are sometimes used to describe systems of linear equations. If a system has more equations than unknowns (i.e. $m > n$ in our notation), then the system is said to be *over-determined*. Over-determined systems tend to have no solutions (though this is by no means always true). If a system has fewer equations than unknowns (i.e. $n > m$), then the system is said to be *under-determined*. If a system is under-determined, it is likely to have an infinite number of solutions (though again, this is not true in all cases). Most of our work will be with systems where the number of equations exactly equals the number of unknowns ($m = n$). Such systems are sometimes referred to as *square systems* (for reasons that will become apparent later).

The following theorem allows us to develop an efficient method for solving linear systems regardless of the number of equations or variables.

Theorem 8.1.2. *Any $m \times n$ system of linear equations (whether homogeneous or inhomogeneous) can be solved by repeated application of three simple operations:*

- E1) Interchange any two equations.*
- E2) Replace any equation by a **non-zero** multiple of itself.*
- E3) Replace any equation by an equation obtained by adding a multiple of another equation to it.*

We will not prove this theorem here, but a thorough discussion can be found in any textbook on linear algebra (for example, [FIS03, Ch. 3]). We will simply look at several (small) examples to see how to use these operations to solve systems of equations. First, a few observations. Operation E1 allows

us to completely reorder the equations in any way we wish. We can also combine operations E2 and E3 to give the auxiliary operation:

Add a non-zero multiple of one equation to a non-zero multiple of another equation replacing exactly one of the equations used.

Example 8.1.2. Solve the system

$$\begin{aligned} 2x - 7y &= -8 \\ 3x + y &= 11 \end{aligned}$$

Answer: If we multiply the second equation by 7, we have

$$\begin{aligned} 2x - 7y &= -8 \\ 21x + 7y &= 77. \end{aligned}$$

Adding the two equations together and replacing the second gives

$$\begin{aligned} 2x - 7y &= -8 \\ 23x &= 69. \end{aligned}$$

Dividing the second equation by 23 gives

$$\begin{aligned} 2x - 7y &= -8 \\ x &= 3. \end{aligned}$$

If we multiply the second equation by -2 and add it to the first (keeping the second) gives

$$\begin{aligned} -7y &= -14 \\ x &= 3. \end{aligned}$$

Dividing the first equation by -7 , and interchanging the two equations gives

$$\begin{aligned} x &= 3 \\ y &= 2. \quad \blacksquare \end{aligned}$$

Notice in the example above that we never substituted the value of a variable into an equation. We will develop a strategy to streamline this process in the next section. For now, we consider two examples showing the cases of no solutions and infinite solutions, respectively.

Example 8.1.3. Solve the following system of linear equations.

$$\begin{aligned}x + 2y + 3z &= 5 \\2x - 5y + z &= 2 \\-4x + 19y + 3z &= 7.\end{aligned}$$

Answer: As we will see, a sound strategy when a variable has a coefficient of 1 (as x does in the first equation) is to use this term to eliminate the variable from the other equations. So, if we subtract 2 copies of the first equation from the second and add 4 copies of the first equation to the third, we have

$$\begin{aligned}x + 2y + 3z &= 5 \\-9y - 5z &= -8 \\27y + 15z &= 27.\end{aligned}$$

If we divide the third equation by 3, we find

$$\begin{aligned}x + 2y + 3z &= 5 \\-9y - 5z &= -8 \\9y + 5z &= 9.\end{aligned}$$

Adding the second and third equations (replacing the third) gives

$$\begin{aligned}x + 2y + 3z &= 5 \\-9y - 5z &= -8 \\0 &= 1.\end{aligned}$$

The third equation above is clearly false. Thus, we conclude that there are *no solutions* to the original system of equations. ■

Example 8.1.4. Solve the following system of equations.

$$\begin{aligned}x - 3y + 7z &= 7 \\2x + y - 7z &= -14 \\8x - 3y - 7z &= -28.\end{aligned}$$

Answer: As before, we use the coefficient of 1 multiplying x in the first equation to eliminate the variable x from the other two equations. So, we

subtract 2 copies of the first equation from the second and 8 copies of the first equation from the third.

$$\begin{aligned}x - 3y + 7z &= 7 \\7y - 21z &= -28 \\21y - 63z &= -84.\end{aligned}$$

if we divide the second equation by 7 and third by 21, we have

$$\begin{aligned}x - 3y + 7z &= 7 \\y - 3z &= -4 \\y - 3z &= -4.\end{aligned}$$

If we subtract the second equation from the third, we are left with

$$\begin{aligned}x - 3y + 7z &= 7 \\y - 3z &= -4 \\0 &= 0.\end{aligned}$$

The last equation is obviously true, and this is the signal that there are an infinite number of solutions.¹ To easily specify the solutions, it will be beneficial to eliminate y from the first equation. So, if we add 3 copies of equation 2 to equation 1, we get

$$\begin{aligned}x - 2z &= -5 \\y - 3z &= -4 \\0 &= 0.\end{aligned}$$

Since the third equation has been reduced to the true statement $0 = 0$, one of the variables is free to be any real number. It is customary to choose the final variable z to be free in this situation. If we then solve for the other two variables in terms of z , we can specify the infinite family of solutions by

$$\begin{aligned}x &= 2z - 5 \\y &= 3z - 4 \\z &\text{ is free. } \blacksquare\end{aligned}$$

The process we have been using to solve systems of linear equations is usually referred to as Gaussian Elimination. In the next section, we will discuss this method in more detail and see how to streamline it.

¹If the process yields both $0 = 0$ and $0 = 1$, then there are still no solutions.

8.1.2 Augmented Matrices and Row Reduction

Our first observation in the examples above is that the variables actually played a relatively passive role. Essentially, we only worked with the coefficients, and the variables served as placeholders. As such, we can dispense with the variables altogether and collect the coefficients in a matrix.

Given a system of equations of the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m, \end{aligned}$$

we can form the associated **augmented matrix**

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right].$$

Each row corresponds to one of the equations while all but the last column contain the coefficients for a particular variable. The last column contains the numbers on the right-hand sides of the equations. The line separating the final column is optional (sometimes displayed with a dashed line or left out altogether), but it reinforces the fact that the numbers in this column play a different role in the system than the coefficients in the columns on the left.

For example, the augmented matrix for the system in Example 8.1.2 is given by

$$\left[\begin{array}{cc|c} 2 & -7 & -8 \\ 3 & 1 & 11 \end{array} \right].$$

Notice that the first column contains the coefficients for x while the second coefficient contains the coefficients for y . Notice that after we solved the system, we were left with a system of equations which has augmented matrix

$$\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 2 \end{array} \right].$$

In this form, we can easily read off the solutions.

In order to transform the augmented matrix corresponding to the original system to one where the solutions are obvious, we first need to rewrite the operations from Theorem 8.1.2 in terms of operations on the augmented matrix. Since all of the operations refer to the individual equations of the system, the operations on the augmented matrix will refer to the individual rows.

The basic operations in Theorem 8.1.2 correspond to the following **Elementary Row Operations** for the augmented matrix.

RO1: Interchange any two rows.

RO2: Multiply a row by a **non-zero** constant.

RO3: Add a multiple of one row to another.

Before we illustrate the use of these row operations (a process known as *row reduction*) we need to discuss notation. First however is a word of caution. Notice that all of the operations above are applied to rows. NEVER apply these operations to individual *columns*. For example, interchanging two columns would be equivalent to interchanging the coefficients of two independent variables. That will certainly change the final answer!

To describe a series of row operations, we will use the following notation. First, the i -th row of an augmented matrix will be denoted R_i . If we use RO1 to interchange two rows, we will denote the operation by $R_i \leftrightarrow R_j$. Just as with equations, we can reorder the rows of an augmented matrix however we wish (often without explicitly noting the changes). If we use RO2 to multiply row i by a non-zero number m , we will write $m \cdot R_i \rightarrow R_i$. For RO3, we would write $m \cdot R_i + R_j \rightarrow R_j$ to indicate that we added m copies of row i to row j (and placed the result in row j). As with equations, we can combine RO2 and RO3 to get an operation of the form $m \cdot R_i + n \cdot R_j \rightarrow R_i$. Notice that we may replace either of the two rows involved in this operation (but not any other row).

Example 8.1.5. Solve the following system of linear equations using row reduction.

$$\begin{aligned}x - 3y - 5z &= 7 \\2x + y + 3z &= 8 \\-x + 4y + z &= -2\end{aligned}$$

Answer: The augmented matrix corresponding to this system is

$$\left[\begin{array}{ccc|c} 1 & -3 & -5 & 7 \\ 2 & 1 & 3 & 8 \\ -1 & 4 & 1 & -2 \end{array} \right]$$

We will discuss the overall goal of row reduction in the next section, but if there is a unique solution to this system, we should be able to reduce the matrix to one of the form

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & A \\ 0 & 1 & 0 & B \\ 0 & 0 & 1 & C \end{array} \right].$$

In this form, we can easily read off the solution: $x = A$, $y = B$, and $z = C$. Essentially, the goal of row reduction is to reduce an augmented matrix as close as possible to this form. To that end, can use the 1 in the first row and first column to clear out the coefficients below it.

So, we begin by subtracting 2 copies of row one from row two (replacing row two) ($-2 \cdot R1 + R2 \rightarrow R2$)

$$\left[\begin{array}{ccc|c} 1 & -3 & -5 & 7 \\ 2 & 1 & 3 & 8 \\ -1 & 4 & 1 & -2 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & -3 & -5 & 7 \\ 0 & 7 & 13 & -6 \\ -1 & 4 & 1 & -2 \end{array} \right]$$

Notice that we use $\sim$ to indicate that a matrix has been obtained from another by an elementary row operation. From now on, we will simply list

the row operations in the notation mentioned above.

$$\begin{aligned}
 & \left[\begin{array}{ccc|c} 1 & -3 & -5 & 7 \\ 2 & 1 & 3 & 8 \\ -1 & 4 & 1 & -2 \end{array} \right] & -2 \cdot R1 + R2 \rightarrow R2 \\
 & \sim \left[\begin{array}{ccc|c} 1 & -3 & -5 & 7 \\ 0 & 7 & 13 & -6 \\ -1 & 4 & 1 & -2 \end{array} \right] & R1 + R3 \rightarrow R3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & -3 & -5 & 7 \\ 0 & 7 & 13 & -6 \\ 0 & 1 & -4 & 5 \end{array} \right] & R2 \leftrightarrow R3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & -3 & -5 & 7 \\ 0 & 1 & -4 & 5 \\ 0 & 7 & 13 & -6 \end{array} \right] & 3 \cdot R2 + R1 \rightarrow R1 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 0 & -17 & 22 \\ 0 & 1 & -4 & 5 \\ 0 & 7 & 13 & -6 \end{array} \right] & -7 \cdot R2 + R3 \rightarrow R3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 0 & -17 & 22 \\ 0 & 1 & -4 & 5 \\ 0 & 0 & 41 & -41 \end{array} \right] & \frac{1}{41} \cdot R3 \rightarrow R3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 0 & -17 & 22 \\ 0 & 1 & -4 & 5 \\ 0 & 0 & 1 & -1 \end{array} \right] & 4 \cdot R3 + R2 \rightarrow R2 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 0 & -17 & 22 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array} \right] & 17 \cdot R3 + R1 \rightarrow R1 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array} \right]
 \end{aligned}$$

Hence, the solution is $x = 5$, $y = 1$, and $z = -1$. ■

We can perform multiple row operations in one step. For example, we could have combined the first two steps above to eliminate the coefficients below the 1 in the first row and column. Be very cautious with this, however. It is probably not wise to perform two row operations simultaneously which both involve replacing the same row.

Example 8.1.6. Solve the following system of equations by row reduction.

$$\begin{aligned}x + y + z &= 9 \\x + 2y + 4z &= 13 \\2x + 3y + 5z &= 22\end{aligned}$$

Answer:

$$\begin{aligned}& \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 1 & 2 & 4 & 13 \\ 2 & 3 & 5 & 22 \end{array} \right] & -R1 + R2 \rightarrow R2 \quad \text{and} \quad -2 \cdot R1 + R3 \rightarrow R3 \\& \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & 1 & 3 & 4 \\ 0 & 1 & 3 & 4 \end{array} \right] & R1 - R2 \rightarrow R1 \quad \text{and} \quad R3 - R2 \rightarrow R3 \\& \sim \left[\begin{array}{ccc|c} 1 & 0 & -2 & 5 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right]\end{aligned}$$

Notice that it is not possible to row reduce the matrix any further due to the row of all zeros. Since this row represents the true statement $0 = 0$, we know that there are an infinite number of solutions. The remaining equations state

$$\begin{aligned}x - 2z &= 5 \\y + 3z &= 4.\end{aligned}$$

Solving these equations in terms of z gives

$$\begin{aligned}x &= 5 + 2z \\y &= 4 - 3z \\z &\text{ is free. } \blacksquare\end{aligned}$$

Example 8.1.7. Solve the following system of equations by row reduction.

$$\begin{aligned}2x + 3y - z &= 2 \\3x + 2z &= 4 \\5x + 3y + z &= 5\end{aligned}$$

Answer: The augmented matrix for the system is

$$\left[\begin{array}{ccc|c} 2 & 3 & -1 & 2 \\ 3 & 0 & 2 & 4 \\ 5 & 3 & 1 & 5 \end{array} \right]$$

We would like to have a 1 for the entry in the first row and first column. We could divide the first row by 2, but there is a better way to proceed. If we subtract row 1 from row 2, we will be left with a 1 for the first entry of row 2. Interchanging rows 1 and 2 will then give us a 1 in the upper left corner of the matrix.

$$\begin{aligned} & \left[\begin{array}{ccc|c} 2 & 3 & -1 & 2 \\ 3 & 0 & 2 & 4 \\ 5 & 3 & 1 & 5 \end{array} \right] & R2 - R1 \rightarrow R2 \\ \sim & \left[\begin{array}{ccc|c} 2 & 3 & -1 & 2 \\ 1 & -3 & 3 & 2 \\ 5 & 3 & 1 & 5 \end{array} \right] & R1 \leftrightarrow R2 \\ \sim & \left[\begin{array}{ccc|c} 1 & -3 & 3 & 2 \\ 2 & 3 & -1 & 2 \\ 5 & 3 & 1 & 5 \end{array} \right] & R2 - 2 \cdot R1 \rightarrow R2 \text{ and } R3 - 5 \cdot R1 \rightarrow R3 \\ \sim & \left[\begin{array}{ccc|c} 1 & -3 & 3 & 2 \\ 0 & 9 & -7 & -2 \\ 0 & 18 & -14 & -5 \end{array} \right] & R3 - 2 \cdot R2 \rightarrow R3 \\ \sim & \left[\begin{array}{ccc|c} 1 & -3 & 3 & 2 \\ 0 & 9 & -7 & -2 \\ 0 & 0 & 0 & -1 \end{array} \right] & -1 \cdot R3 \rightarrow R3 \\ \sim & \left[\begin{array}{ccc|c} 1 & -3 & 3 & 2 \\ 0 & 9 & -7 & -2 \\ 0 & 0 & 0 & 1 \end{array} \right] \end{aligned}$$

In the next section, we will discuss the ultimate goal for row reduction – reduced row echelon form. The final matrix above is not quite in this form, but the third row gives us the false statement $0 = 1$. Hence, we know there are no solutions. ■

8.1.3 Reduced Row Echelon Form

We have seen several examples of row reduction, but so far we have been blindly applying row reductions in order to simplify the augmented matrix to a form where the solutions are fairly obvious. It turns out there is a way to describe the goal we should aim for in the row reduction process.

An augmented matrix is said to be in **reduced row echelon form** (RREF) if the following conditions are satisfied.

- 1) All rows of all zeros are at the bottom of the matrix.
- 2) The first non-zero entry in any row (known as the **leading coefficient**) must be a 1 and must be to the right of the leading coefficient in any row above it.
- 3) Any leading coefficient must be the only non-zero entry in its column.

Theorem 8.1.3. *The reduced row echelon form of an augmented matrix is unique.*

This theorem guarantees that if two people start with the same augmented matrix and perform completely different sets of row operations, they will nonetheless arrive at the same final form (if they take the row reduction all the way to reduced row echelon form). In certain cases, it is not necessary to take an augmented matrix all the way down to RREF to read off the solutions, but it is a decent goal to aim for. We will not prove this result here, but any book on linear algebra will give a thorough discussion of this theorem (c.f. [FIS03, p. 191]).

While our examples below will be 3×3 systems, the technique will work for linear systems of any size.

Example 8.1.8. Solve the following system of linear equations.

$$\begin{aligned}x + 5z &= 3 \\2x + 7y + 3z &= -8 \\2x + 5y + 5z &= -4\end{aligned}$$

Answer: The augmented matrix associated to this system is

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 3 \\ 2 & 7 & 3 & -8 \\ 2 & 5 & 5 & -4 \end{array} \right].$$

We already have a 1 in the upper left corner of the matrix, so we will begin by eliminating the coefficients below it.

$$\begin{aligned}
 & \left[\begin{array}{ccc|c} 1 & 0 & 5 & 3 \\ 2 & 7 & 3 & -8 \\ 2 & 5 & 5 & -4 \end{array} \right] & R2 - 2 \cdot R1 \rightarrow R2 \text{ and } R3 - 2 \cdot R1 \rightarrow R3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 0 & 5 & 3 \\ 0 & 7 & -7 & -14 \\ 0 & 5 & -5 & -10 \end{array} \right] & \frac{1}{7} \cdot R2 \rightarrow R2 \text{ and } \frac{1}{5} \cdot R3 \rightarrow R3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 0 & 5 & 3 \\ 0 & 1 & -1 & -2 \\ 0 & 1 & -1 & -2 \end{array} \right] & R3 - R2 \rightarrow R3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 0 & 5 & 3 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 \end{array} \right]
 \end{aligned}$$

This final matrix is in RREF. Note that the row of all zeros tells that there are an infinite number of solutions, and that one of the variables is free to be any real number. As usual, we will let z be free and solve for the remaining variables in terms of it. Rewriting the non-trivial rows as equations gives

$$\begin{aligned}
 x + 5z &= 3 \\
 y - z &= -2.
 \end{aligned}$$

Hence, the solutions to our system of equations can be expressed as

$$\begin{aligned}
 x &= 3 - 5z \\
 y &= -2 + z \\
 z &\text{ is free. } \blacksquare
 \end{aligned}$$

Example 8.1.9. Solve the following system of equations.

$$\begin{aligned}
 x + y + z &= 4 \\
 2x - 3y - 13z &= 18 \\
 3x - 2y - 12z &= 23
 \end{aligned}$$

Answer: The augmented matrix associated to the system is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & -3 & -13 & 18 \\ 3 & -2 & -12 & 23 \end{array} \right].$$

We begin by eliminating the coefficients under the 1 in the upper left corner of the matrix.

$$\begin{aligned}
 & \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & -3 & -13 & 18 \\ 3 & -2 & -12 & 23 \end{array} \right] & R2 - 2 \cdot R1 \rightarrow R2 \text{ and } R3 - 3 \cdot R1 \rightarrow R3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -5 & -15 & 10 \\ 0 & -5 & -15 & 11 \end{array} \right] & R3 - R2 \rightarrow R3 \text{ and } -\frac{1}{5} \cdot R2 \rightarrow R2 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 1 \end{array} \right]
 \end{aligned}$$

Notice that the last row translates to $0 = 1$, and so the system has no solutions. If this is our only goal, then we can simply stop here. However, the matrix is not yet in RREF.

$$\begin{aligned}
 & \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 1 \end{array} \right] & R1 - R2 \rightarrow R1 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 0 & -2 & 6 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 1 \end{array} \right] & R1 - 6 \cdot R3 \rightarrow R1 \text{ and } R2 + 2 \cdot R3 \rightarrow R2 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]
 \end{aligned}$$

We have now reduced the matrix to RREF. Of course, there are still no solutions to the original system of equations. ■

Example 8.1.10. Solve the following system of linear equations.

$$\begin{aligned}
 2x + y - z &= -2 \\
 3x - 2y + z &= -5 \\
 5x + 4y + 2z &= 7
 \end{aligned}$$

Answer: The augmented matrix associated to this system is

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & -2 \\ 3 & -2 & 1 & -5 \\ 5 & 4 & 2 & 7 \end{array} \right].$$

As we have said several times, it would be nice to have a 1 in the upper left corner. We could divide the first row by 2, but a slightly nicer way to begin is by subtracting row 1 from row 2.

$$\begin{aligned}
 & \left[\begin{array}{ccc|c} 2 & 1 & -1 & -2 \\ 3 & -2 & 1 & -5 \\ 5 & 4 & 2 & 7 \end{array} \right] & R2 - R1 \rightarrow R2 \\
 & \sim \left[\begin{array}{ccc|c} 2 & 1 & -1 & -2 \\ 1 & -3 & 2 & -3 \\ 5 & 4 & 2 & 7 \end{array} \right] & R2 \leftrightarrow R1 \\
 & \sim \left[\begin{array}{ccc|c} 1 & -3 & 2 & -3 \\ 2 & 1 & -1 & -2 \\ 5 & 4 & 2 & 7 \end{array} \right] & R2 - 2 \cdot R1 \rightarrow R2 \text{ and } R3 - 5 \cdot R1 \rightarrow R3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & -3 & 2 & -3 \\ 0 & 7 & -5 & 4 \\ 0 & 19 & -8 & 22 \end{array} \right]
 \end{aligned}$$

Now, we would like a 1 for the entry in the second row and second column, but there does not seem to be an obvious way to make that happen.² However, if we note that $3 \cdot 19 = 57$ and $8 \cdot 7 = 56$, we can get a 1 by $3 \cdot R3 - 8 \cdot R2 \rightarrow R2$.

$$\begin{aligned}
 & \left[\begin{array}{ccc|c} 1 & -3 & 2 & -3 \\ 0 & 7 & -5 & 4 \\ 0 & 19 & -8 & 22 \end{array} \right] & 3 \cdot R3 - 8 \cdot R2 \rightarrow R2 \\
 & \sim \left[\begin{array}{ccc|c} 1 & -3 & 2 & -3 \\ 0 & 1 & 16 & 34 \\ 0 & 19 & -8 & 22 \end{array} \right] & 3 \cdot R2 + R1 \rightarrow R1 \text{ and } R3 - 19 \cdot R2 \rightarrow R3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 0 & 50 & 99 \\ 0 & 1 & 16 & 34 \\ 0 & 0 & -312 & -624 \end{array} \right] & -\frac{1}{312} \cdot R3 \rightarrow R3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 0 & 50 & 99 \\ 0 & 1 & 16 & 34 \\ 0 & 0 & 1 & 2 \end{array} \right] & R1 - 50 \cdot R3 \rightarrow R1 \text{ and } R2 - 16 \cdot R3 \rightarrow R3
 \end{aligned}$$

²It might be tempting to use the -3 in the second entry of the first row to get a 1 in the second entry of the third row ($6 \cdot R1 + R3 \rightarrow R3$). However, this would make the first entry of the resulting row non-zero!

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

We are finally in RREF. So, our system of equations has the unique solution $x = -1, y = 2$, and $z = 2$. ■

Example 8.1.11. Solve the following system of linear equations.

$$\begin{aligned} x - 2y - 5z &= 2 \\ -3x + 6y + 15z &= -6 \\ 2x - 4y - 10z &= 4 \end{aligned}$$

Answer: Writing the system as an augmented matrix and performing the row operations $R2 + 3 \cdot R1 \rightarrow R2$ and $R3 - 2 \cdot R1 \rightarrow R3$ gives

$$\left[\begin{array}{ccc|c} 1 & -2 & -5 & 2 \\ -3 & 6 & 15 & -6 \\ 2 & -4 & -10 & 4 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & -2 & -5 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Note that we have 2 rows of all zeros! This indicates that we have an infinite number of solutions, and two of the variables are free to be any real number. We take y and z to be free and solve the remaining equation for x . Hence, the solutions are given by

$$\begin{aligned} x &= 2 + 2y + 5z \\ y &\text{ is free} \\ z &\text{ is free. } \blacksquare \end{aligned}$$

8.2 Vectors

Often in mathematics, we can learn more about an object by looking at it from a slightly different point of view. This is definitely true of matrices. Before we can tackle this material, however, we need to discuss vectors.

The word vector is used in mathematics for various kinds of objects, but for our purposes, we will mean a very particular type of matrix. An

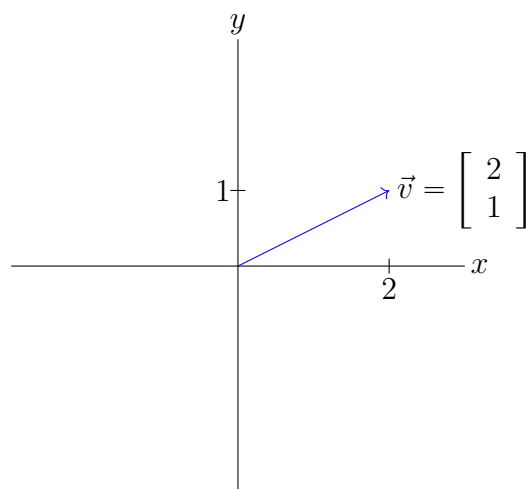


Figure 8.2.1: Geometric Interpretation of Vectors

n -dimensional vector is an $n \times 1$ matrix

$$\vec{v} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

where the x_i are numbers (referred to as the **components** of the vector). If all of the x_i are real numbers, we have a real vector. Otherwise, we have a complex vector.

Notice that we will place a vector sign over quantities that are vectors. It is also possible to use $1 \times n$ matrices to represent vectors, $[x_1 \ x_2 \ \cdots \ x_n]$, but we will not develop the idea further. One-dimensional vectors (i.e. vectors of the form $[x_1]$) are not essentially different than numbers, so we will not make a distinction.

Real vectors have a nice geometric interpretation. If we think of the list of numbers in the vector as the coordinate of a point in n -dimensional space, we can think of the vector as an “arrow” pointing from the origin to the point given by the vector. For example, the vector $\vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ can be envisioned as an arrow pointing from the origin to the point $(2, 1)$. Of course, this interpretation becomes difficult to visualize for vectors of dimension 4 and

higher. There is also a similar interpretation for complex vectors, but to draw a 2×1 complex vector

$$\vec{v} = \begin{bmatrix} a + bi \\ c + di \end{bmatrix}$$

would require 4 dimensions (since a, b, c , and d are all independent quantities). Fortunately, we will only use the geometric interpretation of vectors to motivate some of the later definitions, and two-dimensional vectors will more than suffice.

There are two major operations we can perform on vectors: scalar multiplication and vector addition. In what follows, let α be a scalar (i.e. a real or complex number), and let $\vec{v}_1$ and $\vec{v}_2$ be two vectors of the same size.

- **Scalar multiplication** of a vector by a number is given by

$$\alpha \vec{v}_1 = \alpha \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{bmatrix}.$$

That is, we multiply each component of the vector $\vec{v}_1$ by the number α .

- The **vector sum** of $\vec{v}_1$ and $\vec{v}_2$ is given by

$$\vec{v}_1 + \vec{v}_2 = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}.$$

So, to add two vectors of the same size, we simply add together like components.

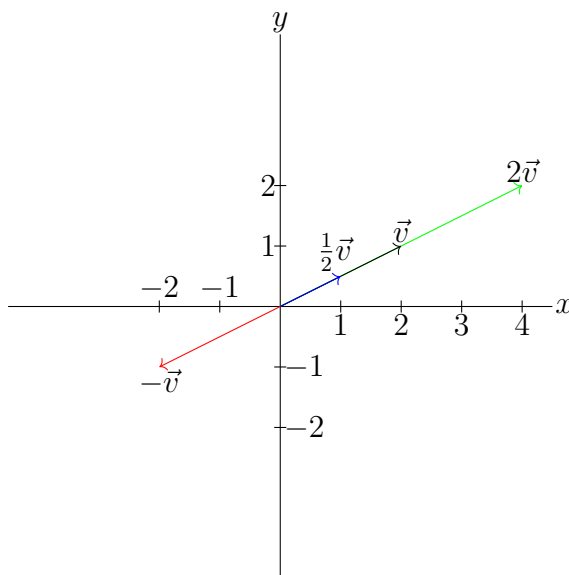
Both of these operations have nice geometric interpretations.

Example 8.2.1. Let $\vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Compute $2\vec{v}$, $\frac{1}{2}\vec{v}$, and $-\vec{v}$, and plot the resulting vectors.

Answer: The actual computations are easy.

$$\begin{aligned} 2\vec{v} &= 2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \cdot 2 \\ 2 \cdot 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \\ \frac{1}{2}\vec{v} &= \frac{1}{2} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \cdot 2 \\ \frac{1}{2} \cdot 1 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix} \\ -\vec{v} &= - \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \cdot 2 \\ -1 \cdot 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \end{bmatrix} \end{aligned}$$

If we plot these vectors, we notice an interesting fact.



Notice that the positive multiples of $\vec{v}$ are simply stretched or shrunk in the same direction as $\vec{v}$. Multiplying by a negative number rotates the vector by 180° (and stretches or shrinks the vector in the process). This is a general fact. *Multiplying a vector by a real number results in a vector that is stretched or shrunk on the same line determined by the original vector. A negative multiple flips the vector by 180° .* ■

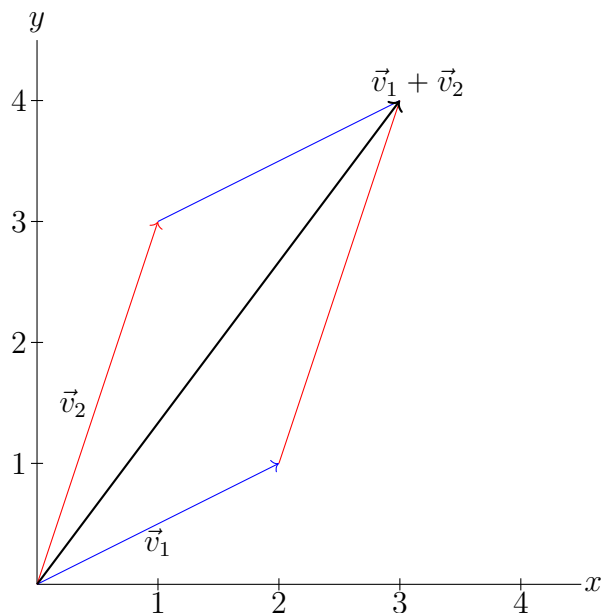
Example 8.2.2. Find the sum of the vectors

$$\vec{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \text{and} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

Answer: Computing the sum is easy.

$$\vec{v}_1 + \vec{v}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2+1 \\ 1+3 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

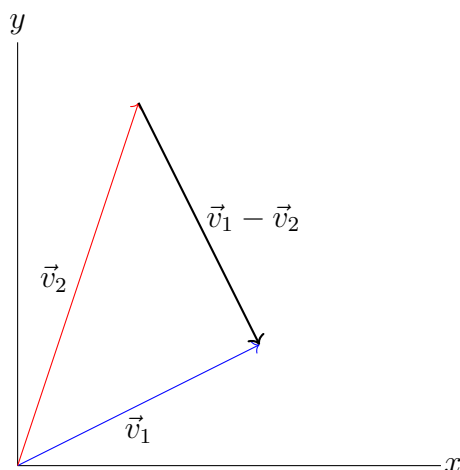
If we plot these vectors, we see something interesting.



Notice that $\vec{v}_1 + \vec{v}_2$ is the result of parallel transporting the tail of $\vec{v}_2$ to the head of $\vec{v}_1$ and connecting the tail of $\vec{v}_1$ (at the origin) to the head of $\vec{v}_2$. As the diagram above shows, you get the same thing by transporting $\vec{v}_1$ to the head of $\vec{v}_2$. This result is known as the *parallelogram law of vector addition*. ■

What this result indicates is that vectors do not have to be thought of as permanently attached to the origin. We should instead think of the vector $\vec{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ as an instruction to move 2 units right and 1 unit up from whatever base point we choose.

A particularly nice use of this idea is the interpretation of vector subtraction. If $\vec{v} = \vec{v}_1 - \vec{v}_2$, then $\vec{v}_1 = \vec{v}_2 + \vec{v}$. So, using the geometric interpretation, $\vec{v} = \vec{v}_1 - \vec{v}_2$ points from the tail of $\vec{v}_2$ to the head of $\vec{v}_1$.



There is one more concept we will need. The **magnitude** (or size) of a real vector is the overall length of the vector and is given by

$$\vec{v} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

is given by

$$|\vec{v}| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

Example 8.2.3. Compute the magnitude of

$$\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Answer:

$$|\vec{v}| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5. \blacksquare$$

The magnitude has several nice properties.

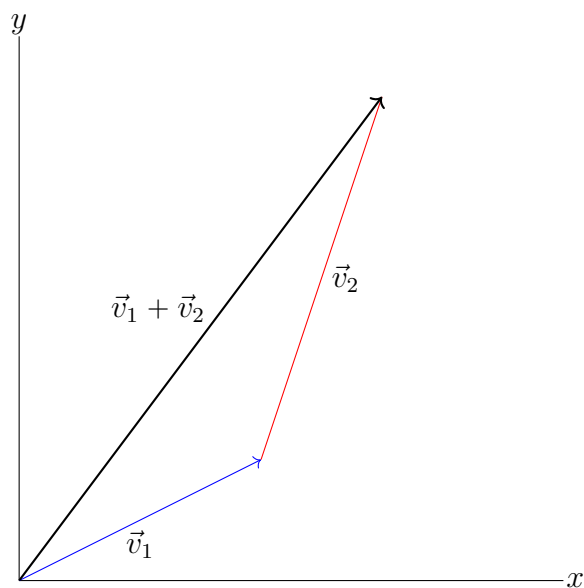
Theorem 8.2.1. *Let α be a scalar and let $\vec{v}_1$ and $\vec{v}_2$ be vectors of the same size, then we have*

- $|\alpha \vec{v}_1| = |\alpha| |\vec{v}_1|$
- $|\vec{v}_1 + \vec{v}_2| \leq |\vec{v}_1| + |\vec{v}_2|$

Partial Proof: The first property is easy to verify.

$$\begin{aligned}
 |\alpha \vec{v}_1| &= \left| \alpha \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \right| \\
 &= \left| \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{bmatrix} \right| \\
 &= \sqrt{(\alpha x_1)^2 + (\alpha x_2)^2 + \cdots + (\alpha x_n)^2} \\
 &= \sqrt{\alpha^2(x_1^2 + x_2^2 + \cdots + x_n^2)} \\
 &= \sqrt{\alpha^2} \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2} \\
 &= |\alpha| |\vec{v}|.
 \end{aligned}$$

The second property is known as the **triangle inequality**. The geometric reason for this fact is evident from the diagram we drew for vector addition.



Notice that $\vec{v}_1$, $\vec{v}_2$, and $\vec{v}_1 + \vec{v}_2$ form three sides of a triangle. Since following the single line segment given by $\vec{v}_1 + \vec{v}_2$ gets us to the same point as first

following $\vec{v}_1$ and then $\vec{v}_2$, it must be that the length of $\vec{v}_1 + \vec{v}_2$ is shorter than the sum of the lengths of the other two vectors. Even though we drew the picture in two-dimensions, the reasoning is sound for two vectors in any number of dimensions. ■

We can extend the definition of magnitude to include complex vectors

$$\vec{v} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}.$$

Instead of simply squaring the entries, we have to multiply them by their conjugates:

$$|\vec{v}| = \sqrt{z_1 \cdot \bar{z}_1 + z_2 \cdot \bar{z}_2 + \cdots + z_n \cdot \bar{z}_n}.$$

Notice that this reduces to our original definition for real vectors ($x \cdot \bar{x} = x^2$ for real numbers). This extended version of magnitude has the same nice properties as the real case. However, we will not need to use this fact in this book.

8.3 Linear Independence

A set of vectors $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ is said to **linearly dependent** if there are scalars $c_1, c_2, \dots, c_k$ *not all of which are zero* so that

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + \cdots + c_k \vec{v}_k = \vec{0}.$$

Notice that if the zero vector itself is one of the vectors in the set, say $\vec{v}_i$, the set of vectors is automatically linearly dependent:

$$1 \cdot \vec{v}_i = 1 \cdot \vec{0} = \vec{0}.$$

Since having the zero vector makes a set trivially dependent, we will assume that none of the vectors in our set is the zero vector from here on out.

Suppose that c_1 is not zero in the equation above. Then we can solve for $\vec{v}_1$ in terms of the other vectors in the set:

$$\vec{v}_1 = -\frac{c_2}{c_1} \vec{v}_2 - \frac{c_3}{c_1} \vec{v}_3 - \cdots - \frac{c_k}{c_1} \vec{v}_k.$$

So, a set of non-zero vectors is linearly dependent if and only if it is possible to write one of the vectors in the set as a combination of the others.

As an example, the set of vectors

$$\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \end{bmatrix} \right\}$$

is linearly dependent because

$$2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 1 \cdot \begin{bmatrix} 1 \\ -2 \end{bmatrix} - 1 \cdot \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

As a result, we can solve for any one of these vectors in terms of the other two. In particular,

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 1 \cdot \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

If a set of non-zero vectors is not linearly dependent, then we say the set is **linearly independent**. In words, a set of non-zero vectors is linearly independent if it is impossible to write any one of the vectors as a combination of the other vectors in the set.

There is a lot to say about linear independence in general, but we will stick the facts that are most important for us moving forward.

Theorem 8.3.1. A pair of non-zero vectors $\{\vec{v}_1, \vec{v}_2\}$ is linearly dependent if and only if one is a multiple of the other.

Proof: Suppose that $\vec{v}_1 = c \cdot \vec{v}_2$ (note that $c \neq 0$ since $\vec{v}_1$ is not the zero vector and so $\vec{v}_2 = (1/c) \cdot \vec{v}_1$). Then clearly we have

$$\vec{v}_1 - c \cdot \vec{v}_2 = \vec{0}$$

and the vectors are linearly dependent. Conversely, if the two vectors are linearly dependent, then

$$c_1 \cdot \vec{v}_1 + c_2 \cdot \vec{v}_2 = \vec{0}$$

for some pair of scalars c_1 and c_2 at least one of which is non-zero. In fact, since neither of our vectors is the zero vector, neither of these scalars can be zero! So,

$$\vec{v}_1 = -\frac{c_2}{c_1} \cdot \vec{v}_2$$

and $\vec{v}_1$ is a multiple of $\vec{v}_2$. Of course, $\vec{v}_2$ is also a multiple of $\vec{v}_1$! ■

Example 8.3.1. Are the vectors

$$\left\{ \begin{bmatrix} 1 \\ -2 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -4 \\ 4 \\ 3 \end{bmatrix} \right\}$$

linearly dependent?

Answer: These vectors are clearly not multiples of one another. If we try to make their first arguments the same, we find

$$2 \cdot \begin{bmatrix} 1 \\ -2 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 4 \\ 2 \end{bmatrix} \neq \begin{bmatrix} 2 \\ -4 \\ 4 \\ 3 \end{bmatrix}.$$

So, these two vectors are linearly independent. ■

Theorem 8.3.2. To determine if a set $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ of 2 or more non-zero vectors is linearly dependent, form the augmented matrix

$$\left[\vec{v}_1 \ \vec{v}_2 \ \cdots \ \vec{v}_k \mid \vec{0} \right]$$

having the vectors $\vec{v}_i$ as its columns on the left and the zero vector on the right. After row reduction, if the system has only a single solution then the set of vectors is linearly independent. If the system has an infinite number of solutions, then the set of vectors is linearly dependent.

Proof: To see if a set of vectors is linearly dependent, we need to find coefficients $c_1, c_2, \dots, c_k$ so that

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + \cdots + c_k \vec{v}_k = \vec{0}$$

We think about this problem as a system of linear equations where the variables are the c_i (this is where the augmented matrix above comes from). Notice that setting each coefficient equal to 0 definitely gives a solution (so the system cannot be inconsistent). If this is the only solution, then the vectors are linearly independent. If there is a solution other than this trivial one, then the set of vectors is linearly dependent. In this case, there will be an infinite number of possible solutions. ■

Example 8.3.2. Is the set of vectors

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

linearly dependent or independent?

Answer: We first form the augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right].$$

Row reducing this matrix gives

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] &\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]. \end{aligned}$$

Since this system has only one solution (the trivial one where all coefficients equal zero), this set of vectors is linearly independent. ■

Example 8.3.3. Is the set of vectors

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} \right\}$$

linearly dependent or independent?

Answer:

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & 1 & 4 & 0 \\ 0 & -1 & 2 & 0 \end{array} \right] &\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 2 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & 0 & 3 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]. \end{aligned}$$

The row of all zeros clearly indicates that there is an infinite number of solutions (remember that these systems can never be inconsistent). As a result, the set of vectors above is linearly dependent.

In fact, we can find a linear combination of these vectors that add up to the zero vector. From the reduced matrix, we must have $c_1 = -3c_3$ and $c_2 = 2c_3$ while c_3 is free to be any value. If we take $c_1 = 1$, we find

$$-3 \cdot \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + 2 \cdot \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + 1 \cdot \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

As a result, can see that

$$\begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} = 3 \cdot \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - 2 \cdot \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}. \quad \blacksquare$$

8.4 Matrices as Linear Transformations of Vectors

So far, matrices have merely served as a kind of shorthand for systems of linear equations. Of course, this short hand has been extremely useful since it led us to a very efficient method for solving such systems. But matrices are so much more than this! In this section, we will explore the notion that a matrix is a very special kind of function – one that accepts a vector as an input and returns another vector as output.

An $m \times n$ matrix (i.e. a matrix with m rows and n columns) acts on n -dimensional vectors in the following way:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n \end{bmatrix}.$$

Notice that the output is the m -dimensional vector obtained by applying the rows of the matrix as coefficients multiplying the entries of the vector.

If we think of an n -dimensional vector as an $n \times 1$ matrix, the input and output sizes can be remembered in the following schematic way:

$$[m \times n][n \times 1] = [m \times 1].$$

In words, the number of columns in the matrix on the left has to match the number of rows in the vector on the right. This will turn out to be a special case of a more general pattern we will develop in this section.

Example 8.4.1. Compute

$$\begin{bmatrix} 3 & -2 & 4 \\ 5 & -1 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 7 \\ 1 \end{bmatrix}.$$

Answer: Since we have a 2×3 matrix, we have a function that accepts 3-dimensional vectors and outputs 2 dimensional vectors. For the particular 3-vector we are given, we find

$$\begin{aligned} \begin{bmatrix} 3 & -2 & 4 \\ 5 & -1 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 7 \\ 1 \end{bmatrix} &= \begin{bmatrix} 3(2) + (-2)(7) + 4(1) \\ 5(2) + (-1)(7) + 6(1) \end{bmatrix} \\ &= \begin{bmatrix} -4 \\ 9 \end{bmatrix}. \end{aligned}$$

Example 8.4.2. Compute

$$\begin{bmatrix} 1 & 2 \\ 4 & -3 \\ 0 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Answer: The matrix we were given is 4×2 , so it accepts 2-dimensional vectors and outputs 4-dimensional ones.

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 4 & -3 \\ 0 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} &= \begin{bmatrix} 1(3) + 2(2) \\ 4(3) + (-3)(2) \\ 0(3) + 3(2) \\ -1(3) + (-2)(2) \end{bmatrix} \\ &= \begin{bmatrix} 7 \\ 6 \\ 6 \\ -7 \end{bmatrix}. \end{aligned}$$

If a matrix has the same number of rows as it does columns, then we say that the matrix is a **square matrix**. A square matrix of size $n \times n$ accepts n -dimensional vectors and outputs vectors of the same size.

Example 8.4.3. Compute

$$\begin{bmatrix} 2 & -5 & 3 \\ 1 & 0 & 4 \\ -2 & 2 & -7 \end{bmatrix} \begin{bmatrix} 1 \\ -3 \\ 6 \end{bmatrix}.$$

Answer: Since we have a 3×3 matrix, it accepts 3-dimensional vectors and returns 3-dimensional vectors.

$$\begin{aligned} \begin{bmatrix} 2 & -5 & 3 \\ 1 & 0 & 4 \\ -2 & 2 & -7 \end{bmatrix} \begin{bmatrix} 1 \\ -3 \\ 6 \end{bmatrix} &= \begin{bmatrix} 2(1) + (-5)(-3) + 3(6) \\ 1(1) + 0(-3) + 4(6) \\ -2(1) + 2(-3) + (-7)(6) \end{bmatrix} \\ &= \begin{bmatrix} 35 \\ 25 \\ -50 \end{bmatrix}. \end{aligned}$$

As functions acting on vectors, matrices have many nice properties. The most important is that a matrix represents a **linear transformation** of vectors.

Theorem 8.4.1 (Matrices are Linear Transformations). *Let M be a $m \times n$ matrix, α a scalar, and $\vec{v}_1$ and $\vec{v}_2$ n -dimensional vectors. Then the following properties hold.*

- 1) $M(\alpha\vec{v}_1) = \alpha(M\vec{v}_1)$
- 2) $M(\vec{v}_1 + \vec{v}_2) = M\vec{v}_1 + M\vec{v}_2$

Partial Proof: We will not provide the general proof of these facts (which can be found in [FIS03, Ch. 2] or any other textbook on linear algebra). We will simply show these facts are true for 2×2 matrices. The general proof is essentially the same (just with fancier notation).

So, let

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \vec{v}_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \quad \text{and} \quad \vec{v}_2 = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}.$$

First,

$$\begin{aligned} M(\alpha\vec{v}_1) &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \left(\alpha \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \right) \\ &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \alpha x_1 \\ \alpha y_1 \end{bmatrix} \\ &= \begin{bmatrix} a(\alpha x_1) + b(\alpha y_1) \\ c(\alpha x_1) + d(\alpha y_1) \end{bmatrix} \\ &= \begin{bmatrix} \alpha(ax_1 + by_1) \\ \alpha(cx_1 + dy_1) \end{bmatrix} \\ &= \alpha \begin{bmatrix} ax_1 + by_1 \\ cx_1 + dy_1 \end{bmatrix} \\ &= \alpha \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \right) \\ &= \alpha(M\vec{v}_1). \end{aligned}$$

For the second property, we have

$$\begin{aligned}
 M(\vec{v}_1 + \vec{v}_2) &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \left(\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \right) \\
 &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \end{bmatrix} \\
 &= \begin{bmatrix} a(x_1 + x_2) + b(y_1 + y_2) \\ c(x_1 + x_2) + d(y_1 + y_2) \end{bmatrix} \\
 &= \begin{bmatrix} (ax_1 + by_1) + (ax_2 + by_2) \\ (cx_1 + dy_1) + (cx_2 + dy_2) \end{bmatrix} \\
 &= \begin{bmatrix} ax_1 + by_1 \\ cx_1 + dy_1 \end{bmatrix} + \begin{bmatrix} ax_2 + by_2 \\ cx_2 + dy_2 \end{bmatrix} \\
 &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \\
 &= M\vec{v}_1 + M\vec{v}_2. \blacksquare
 \end{aligned}$$

Any transformation having properties 1) and 2) above is called a **linear transformation**.

Theorem 8.4.2. *Let M be a linear transformation of vectors, α and β be scalars, and $\vec{v}_1$ and $\vec{v}_2$ be vectors of the appropriate size for M . For definiteness, we will assume that M takes as inputs n -dimensional vectors and outputs m -dimensional ones (i.e. M can be given by a $m \times n$ matrix). Then,*

$$1) \quad M(\alpha\vec{v}_1 + \beta\vec{v}_2) = \alpha M\vec{v}_1 + \beta M\vec{v}_2$$

$$2) \quad M\vec{0}_n = \vec{0}_m$$

where $\vec{0}_n$ represents an n -dimensional vector of all zeros.

Proof: The first property is a direct consequence of the previous theorem.

$$\begin{aligned}
 M(\alpha\vec{v}_1 + \beta\vec{v}_2) &= M(\alpha\vec{v}_1) + M(\beta\vec{v}_2) \\
 &= \alpha M\vec{v}_1 + \beta M\vec{v}_2.
 \end{aligned}$$

In fact, this single result is equivalent to both of the conditions in the previous theorem.

For the second part of the theorem, note that $\vec{0}_n = 0 \cdot \vec{0}_n$ since all of the components of $\vec{0}_n$ are zero.

$$\begin{aligned} M\vec{0}_n &= M(0 \cdot \vec{0}_n) \\ &= 0 \cdot M\vec{0}_n \\ &= \vec{0}_m. \end{aligned}$$

Note that the change from $\vec{0}_n$ to $\vec{0}_m$ in the final step occurs simply because M is assumed to output m -dimensional vectors. ■

Thinking of matrices as linear transformations allows us to define a number of operations on them. In particular, we can multiply a matrix by a scalar, add two matrices of the same size, and multiply two matrices together (if their sizes are compatible). We will give definitions for each of these operations below and supply motivations for them as we go. To keep the notation simple, we will examine the motivation for 2×2 matrices only.

- **Scalar Multiplication:** Let M be an $m \times n$ matrix and α a scalar. Then the matrix αM is the matrix of the same size as M whose entries are the entries of M multiplied by α .

In symbols,

$$\alpha \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} \alpha a_{11} & \alpha a_{12} & \cdots & \alpha a_{1n} \\ \alpha a_{21} & \alpha a_{22} & \cdots & \alpha a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ \alpha a_{m1} & \alpha a_{m2} & \cdots & \alpha a_{mn} \end{bmatrix}.$$

Motivation: Since αM is supposed to be a matrix of the same size as M , we need to ask what action it should have on a vector $\vec{v}$ (of the appropriate size). The property we would like to have is $(\alpha M)\vec{v} = \alpha(M\vec{v})$. Note that this is a choice we are making! We would like our new operation (multiplying a matrix by a scalar) to be compatible with an operation we already understand (multiplying a vector by a scalar). It is this natural choice which leads to the definition above.

For a 2×2 matrix M and $\vec{v}$ a 2-dimensional vector, we find

$$\begin{aligned}\alpha(M\vec{v}) &= \alpha\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}\right) \\ &= \alpha \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix} \\ &= \begin{bmatrix} \alpha(ax + by) \\ \alpha(cx + dy) \end{bmatrix} \\ &= \begin{bmatrix} (\alpha a)x + (\alpha b)y \\ (\alpha c)x + (\alpha d)y \end{bmatrix} \\ &= \begin{bmatrix} \alpha a & \alpha b \\ \alpha c & \alpha d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.\end{aligned}$$

So, our requirement that $(\alpha M)\vec{v} = \alpha(M\vec{v})$ leads us to the conclusion (at least in the 2×2 case) that

$$\alpha \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \alpha a & \alpha b \\ \alpha c & \alpha d \end{bmatrix}.$$

That is, multiplying a matrix by a scalar simply multiplies each of the entries in the matrix by that scalar. ■

- **Matrix Addition:** Let A and B be matrices of the same size. Then the matrix $A + B$ is the matrix of the same size as A and B obtained by adding together corresponding entries.

In symbols, if

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix}$$

are matrices of the same size, then

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{bmatrix}.$$

Motivation: The natural thing to require is the distributive property:

$$(A + B)\vec{v} = A\vec{v} + B\vec{v}.$$

Again, this a choice we make, but it is certainly an obvious choice! In the 2×2 case, if

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} r & s \\ t & u \end{bmatrix}, \quad \text{and} \quad \vec{v} = \begin{bmatrix} x \\ y \end{bmatrix},$$

then

$$\begin{aligned} A\vec{v} + B\vec{v} &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} r & s \\ t & u \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\ &= \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix} + \begin{bmatrix} rx + sy \\ tx + uy \end{bmatrix} \\ &= \begin{bmatrix} (ax + by) + (rx + sy) \\ (cx + dy) + (tx + uy) \end{bmatrix} \\ &= \begin{bmatrix} (a + r)x + (b + s)y \\ (c + t)x + (d + u)y \end{bmatrix} \\ &= \begin{bmatrix} a + r & b + s \\ c + t & d + u \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}. \end{aligned}$$

So, we must have

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} r & s \\ t & u \end{bmatrix} = \begin{bmatrix} a + r & b + s \\ c + t & d + u \end{bmatrix}$$

as promised in the definition. ■

Before we discuss matrix multiplication, we should emphasize a nice bit of notation we used in the test for linear independence. Consider an $m \times n$ matrix

$$M = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

We can think of this matrix as a list of n m -dimensional vectors

$$M = [\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n]$$

where

$$\vec{v}_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \quad \dots, \quad \vec{v}_n = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}.$$

In other words, we can think of each column of M as a vector in its own right.

- **Matrix Multiplication:** Let A be a $p \times m$ matrix and B a $m \times n$ matrix. Let $B = [\vec{v}_1 \ \vec{v}_2 \ \dots \ \vec{v}_n]$ be thought of a list of n m -dimensional vectors. Then the product of these two matrices, AB , is the $p \times n$ matrix with columns given by

$$AB = [A\vec{v}_1 \ A\vec{v}_2 \ \dots \ A\vec{v}_n].$$

Notice that AB being $p \times n$ is a consequence of this definition. We still have n vectors (the columns of AB), but since A is $p \times m$, it transforms the m -dimensional vectors from B into p -dimensional vectors. Schematically, the sizes of matrices that can be multiplied are

$$[p \times m] \cdot [m \times n] = [p \times n].$$

Motivation: If A and B are matrices, then the product AB should be a new matrix. Hence, it should act on vectors of a certain size. As always, we need to ask what we want this product to mean. The natural thing to require is that the product AB should represent the composition of these two transformations:

$$(AB)\vec{v} = A(B\vec{v}).$$

That is, we first let B act on $\vec{v}$, and the output from this transformation is then taken as the input for A . If B is an $m \times n$ matrix, then $\vec{v}$ must be n -dimensional and so $B\vec{v}$ will be m -dimensional. So, the matrix A must accept m -dimensional vectors as input. Hence, A must be $p \times m$ for some p .

To see how this requirement leads to the definition given above, we will look at the 2×2 case in detail. So, if

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} r & s \\ t & u \end{bmatrix}, \quad \text{and} \quad \vec{v} = \begin{bmatrix} x \\ y \end{bmatrix},$$

then

$$\begin{aligned} B\vec{v} &= \begin{bmatrix} r & s \\ t & u \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\ &= \begin{bmatrix} rx + sy \\ tx + uy \end{bmatrix}. \end{aligned}$$

We now input this vector into A .

$$\begin{aligned} A(B\vec{v}) &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} rx + sy \\ tx + uy \end{bmatrix} \\ &= \begin{bmatrix} a(rx + sy) + b(tx + uy) \\ c(rx + sy) + d(tx + uy) \end{bmatrix} \\ &= \begin{bmatrix} (ar + bt)x + (as + bu)y \\ (cr + dt)x + (cs + du)y \end{bmatrix} \\ &= \begin{bmatrix} ar + bt & as + bu \\ cr + dt & cs + du \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}. \end{aligned}$$

So we must have

$$\begin{aligned} AB &= \begin{bmatrix} ar + bt & as + bu \\ cr + dt & cs + du \end{bmatrix} \\ &= \left[A \begin{bmatrix} r \\ t \end{bmatrix} \quad A \begin{bmatrix} s \\ u \end{bmatrix} \right]. \blacksquare \end{aligned}$$

There is an easier way to compute matrix products. Look closely at the product in the 2×2 case.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} r & s \\ t & u \end{bmatrix} = \begin{bmatrix} ar + bt & as + bu \\ cr + dt & cs + du \end{bmatrix}$$

If we look carefully at the $(1,1)$ -entry of the product (in the upper left corner), it looks as though it comes from applying the first row of A to the first column of B (in the usual way that matrices act on vectors). Similarly, the $(1,2)$ -entry (in the upper right corner), looks like the application of the first row of A on the second column of B , and so forth. Most people remember matrix multiplication by this basic idea:

“Multiply row by column.”

Example 8.4.4. Find the product AB where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & 4 \end{bmatrix}$$

Answer: Since A is 3×2 and B is 2×3 , we know that AB will have size 3×3 .

$$AB = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}$$

We get c_{11} by multiplying the first row of A by the first column of B .

$$c_{11} = 1(2) + 2(1) = 4$$

Similarly, c_{12} and c_{13} are obtained by multiplying the first row of A with the second and third columns of B , respectively.

$$c_{12} = 1(0) + 2(-1) = -2$$

$$c_{13} = 1(-3) + 2(4) = 5$$

For c_{21} , we multiply the second row of A by the first column of B .

$$c_{21} = 3(2) + 4(1) = 10$$

The remaining entries are as follows.

$$c_{22} = 3(0) + 4(-1) = -4$$

$$c_{23} = 3(-3) + 4(4) = 7$$

$$c_{31} = 5(2) + 6(1) = 16$$

$$c_{32} = 5(0) + 6(-1) = -6$$

$$c_{33} = 5(-3) + 6(4) = 9$$

So, we have

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & 4 \end{bmatrix} = \begin{bmatrix} 4 & -2 & 5 \\ 10 & -4 & 7 \\ 16 & -6 & 9 \end{bmatrix}. \quad \blacksquare$$

An important fact about matrix multiplication is that it is *not commutative* in general. What this means is that generally $AB \neq BA$. This fact is not too surprising if you remember that matrix multiplication is essentially function composition (and typically $f \circ g \neq g \circ f$).

Example 8.4.5. Compute BA using the matrices given in the previous example.

Answer: Since B is 2×3 and A is 3×2 , the product BA will have size 2×2 . Note that this is already enough to see that AB and BA need not be the same. In this example, they are not even the same size!

$$BA = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$$

where

$$\begin{aligned} c_{11} &= 2(1) + 0(3) - 3(5) = -13, \\ c_{12} &= 2(2) + 0(4) - 3(6) = -14, \\ c_{21} &= 1(1) - 1(3) + 4(5) = 18, \\ c_{22} &= 1(2) - 1(4) + 4(6) = 22. \end{aligned}$$

This gives us

$$BA = \begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} -13 & -14 \\ 18 & 22 \end{bmatrix}. \blacksquare$$

In some instances, not only will $AB \neq BA$, but BA may not even be defined! For example, if A is 2×3 and B is 3×4 , then the product AB is defined and will have size 2×4 . However, the product BA does not make sense since the dimensions of the matrices do not line up correctly. Even if AB and BA both exist and are the same size, these two products need not be the same!

Example 8.4.6. Given the matrices A and B below, compute both AB and BA .

$$A = \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 3 \\ 4 & -5 \end{bmatrix}$$

Answer:

$$\begin{aligned} AB &= \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 4 & -5 \end{bmatrix} = \begin{bmatrix} -7 & 13 \\ 14 & -9 \end{bmatrix} \\ BA &= \begin{bmatrix} 1 & 3 \\ 4 & -5 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 7 & 7 \\ -6 & -23 \end{bmatrix} \end{aligned}$$

Notice that even though AB and BA are both defined and have the same size, they are definitely not equal! ■

On occasion, we need a specific entry from the product of two matrices. Fortunately, there is an easy formula for exactly this purpose.

Lemma 8.4.3. *Suppose that A is an $m \times n$ matrix with entries a_{ij} and B is an $n \times p$ matrix with entries b_{jk} . Then the $m \times p$ matrix AB has entries $c_{\ell m}$ given by the formula*

$$c_{\ell m} = \sum_{j=1}^n a_{\ell j} b_{jm}.$$

Proof: The entry $c_{\ell m}$ of AB is computed by taking the ℓ -th row of A and multiplying its entries with the m -th column of B (and summing the results). But the entries of $a_{\ell j}$ are precisely the elements of the ℓ -th row of A when we let j range from 1 to n . Likewise, the elements b_{jm} give the elements of the m -th column of B when j ranges from 1 to n . ■

Example 8.4.7. Suppose that

$$A = \begin{bmatrix} 1 & 2 & -6 \\ 3 & -4 & 7 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & -1 \\ 4 & 5 \\ -2 & 7 \end{bmatrix}.$$

Find the entry in the second row and first column of AB .

Answer: Notice that AB will be a 2×2 matrix. We will use the notation $[AB]_{21}$ to stand for the entry in the second row and first column of AB .

$$\begin{aligned} [AB]_{21} &= \sum_{j=1}^3 a_{2j} b_{j1} \\ &= a_{21} b_{11} + a_{22} b_{21} + a_{23} b_{31} \\ &= 3(1) + (-4)(4) + 7(-2) \\ &= -27 \quad \blacksquare \end{aligned}$$

Before we leave this section, there is a special square matrix we need to discuss. First, the entries of

$$M = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

of the form a_{ii} (i.e. the entries $a_{11}, a_{22}, \dots, a_{nn}$) are called the **main diagonal**.

The $n \times n$ **identity matrix** is the matrix with 1's on the main diagonal and 0's everywhere else.

$$I_n = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

The most important fact about the identity matrix is that if A is an $n \times n$ matrix

$$AI_n = I_n A = A.$$

In other words, I_n plays an analogous role to the number 1 for multiplication of square matrices.

8.5 Inverse Matrices and Determinants

Let us reconsider the problem that led us to matrices in the first place: finding solutions to linear systems of equations. We will restrict ourselves to $n \times n$ systems moving forward.

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \cdots a_{nn}x_n &= b_n \end{aligned}$$

If we define an $n \times n$ matrix A by

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix},$$

and n -dimensional vectors $\vec{x}$ and $\vec{b}$ by

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{and} \quad \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix},$$

then the system of equations above is equivalent to the matrix equation

$$A\vec{x} = \vec{b}.$$

If A were a non-zero number, we could simply divide by it to solve for $\vec{x}$. The corresponding notion for a square matrix A is called the **inverse matrix** of A .

The inverse matrix of a square matrix A (if it exists) is a square matrix denoted by A^{-1} which has the same size as A and satisfies

$$AA^{-1} = A^{-1}A = I_n.$$

Suppose that A^{-1} exists for our particular matrix A . Using this matrix, the solution to our system of equations is easy to write down.

$$\begin{aligned} A\vec{x} &= \vec{b} \\ A^{-1}A\vec{x} &= A^{-1}\vec{b} \\ I_n\vec{x} &= A^{-1}\vec{b} \\ \vec{x} &= A^{-1}\vec{b}. \end{aligned}$$

Unfortunately, not every square matrix has an inverse. In this section, we will discuss how to tell whether a matrix is invertible, and we will develop a fairly efficient algorithm for computing inverses. We will only scratch the surface of everything that could be said about this topic. For a more in depth discussion, see [FIS03, §3.2 and Ch. 4].

We will begin with the simplest possible case: 2×2 matrices. Since the existence of an inverse matrix is related to solving systems of linear equations, that is where we will begin. Specifically, we will try to solve for the variables x and y in the generic 2×2 system

$$\begin{aligned} ax + by &= \alpha \\ cx + dy &= \beta. \end{aligned}$$

Think of a, b, c, d, α , and β as fixed (but unspecified) numbers, and our job is to solve for x and y in terms of these quantities. We could attempt to reduce the associated augmented matrix to RREF, but for 2×2 systems it is just as simple to solve and substitute. For simplicity, we will assume that none of the coefficients a, b, c , or d are zero. It will end up making no difference in the final answer.

If we solve the first equation for y , we find

$$y = \frac{\alpha - ax}{b}.$$

We then substitute this into the second equation to find

$$\begin{aligned} cx + dy &= \beta \\ cx + d\left(\frac{\alpha - ax}{b}\right) &= \beta \\ bcx + d\alpha - adx &= b\beta \\ (bc - ad)x &= b\beta - d\alpha \\ (ad - bc)x &= d\alpha - b\beta \\ x &= \frac{d\alpha - b\beta}{ad - bc}, \end{aligned}$$

provided that $ad - bc \neq 0$. A little algebra shows that the corresponding value for y is

$$y = \frac{-c\alpha + a\beta}{ad - bc}.$$

Translating these results to matrix form shows that the solution to the matrix equation

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

is given by

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

provided the quantity $ad - bc$ is non-zero. This quantity is known as the **determinant** of the matrix since it determines whether or not the inverse of the matrix exists. The above calculations prove the following theorem.

Theorem 8.5.1 (Inverse of a 2×2 Matrix). A 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

has an inverse if and only if its determinant

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

is non-zero. In which case,

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}.$$

Example 8.5.1. Determine whether the following matrix has an inverse. If it does, find it.

$$A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}.$$

Answer: The determinant of the matrix is

$$\det(A) = \begin{vmatrix} 3 & 4 \\ 1 & 2 \end{vmatrix} = 3(2) - 4(1) = 2 \neq 0.$$

Since the determinant is non-zero, the matrix has an inverse. The formula from the theorem gives us

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 2 & -4 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -\frac{1}{2} & \frac{3}{2} \end{bmatrix}. \blacksquare$$

Example 8.5.2. Determine whether the following matrix has an inverse. If it does, find it.

$$B = \begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix}.$$

Answer: The determinant of the matrix is

$$\det(B) = \begin{vmatrix} 2 & 1 \\ 6 & 3 \end{vmatrix} = 2(3) - 1(6) = 0.$$

Since the determinant is zero, the matrix B has no inverse. $\blacksquare$

Example 8.5.3. Solve the following system of equations by rewriting the system in matrix form and finding an inverse matrix.

$$\begin{aligned} 5x + 3y &= 4 \\ -x + 2y &= 7 \end{aligned}$$

Answer: In matrix form, the system above is given by

$$\begin{bmatrix} 5 & 3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}.$$

Since

$$\begin{vmatrix} 5 & 3 \\ -1 & 2 \end{vmatrix} = 10 - (-1)(3) = 13 \neq 0,$$

the matrix has an inverse. Specifically,

$$\begin{bmatrix} 5 & 3 \\ -1 & 2 \end{bmatrix}^{-1} = \frac{1}{13} \begin{bmatrix} 2 & -3 \\ 1 & 5 \end{bmatrix}.$$

So, the solution to the matrix problem above is

$$\begin{aligned} \begin{bmatrix} x \\ y \end{bmatrix} &= \frac{1}{13} \begin{bmatrix} 2 & -3 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 4 \\ 7 \end{bmatrix} \\ &= \frac{1}{13} \begin{bmatrix} 2(4) - 3(7) \\ 1(4) + 5(7) \end{bmatrix} \\ &= \frac{1}{13} \begin{bmatrix} -13 \\ 39 \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ 3 \end{bmatrix}. \end{aligned}$$

So, the solution to our system of equations is $x = -1$ and $y = 3$. ■

For square matrices of size 3×3 and larger, the story is similar but more complicated. For an $n \times n$ matrix A , the **minor** associated to element a_{ij} is the $(n - 1) \times (n - 1)$ square matrix A_{ij} obtained from A by deleting row i and column j (the row and column containing a_{ij}).

Example 8.5.4. Find the minors associated to the elements in the first row of the matrix

$$A = \begin{bmatrix} 2 & 5 & -1 \\ 3 & -2 & 4 \\ 1 & 1 & -2 \end{bmatrix}.$$

Answer: The minor associated to $a_{11} = 2$ is the matrix

$$A_{11} = \begin{bmatrix} -2 & 4 \\ 1 & -2 \end{bmatrix}.$$

The minor associated to $a_{12} = 5$ is the matrix

$$A_{12} = \begin{bmatrix} 3 & 4 \\ 1 & -2 \end{bmatrix}.$$

The minor associated to $a_{13} = -1$ is the matrix

$$A_{13} = \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix}.$$

As with 2×2 matrices, there is a quantity called the determinant which indicates whether or not an $n \times n$ matrix has an inverse. While there are many ways to define determinants for larger square matrices, we will present a recursive definition.

For a square matrix A of size $n \times n$ ($n \geq 3$), the determinant of A (still symbolized as $\det(A)$) is defined recursively using determinants of minors. If

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix},$$

then

$$\det(A) = a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + a_{13} \det(A_{13}) - \cdots + (-1)^{n+1} \det(A_{1n}).$$

Each of the determinants on the right-hand-side above are determinants of strictly smaller matrices. Hence, they can be expanded in terms of determinants of smaller matrices themselves. This process continues until we reach a series of 2×2 determinants (whose definition is given above).

Notice the alternating signs on the determinants. They can be easily remembered by writing a series of plus and minus signs above the matrix as

Example 8.5.6. Compute the determinant of the matrix

$$B = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 2 \\ 1 & 1 & 1 & 2 \end{bmatrix}.$$

Answer: Computing a 4×4 determinant involves first expanding the determinant into $4 \cdot 3 \times 3$ determinants. Each of those 3×3 determinants is then expanded into $3 \cdot 2 \times 2$ determinants. This means that we eventually have to compute $4 \cdot 3 = 12 \cdot 2 \times 2$ determinants! Fortunately, B has a large number of 0 entries!

$$\begin{array}{cccc} + & - & + & - \\ \left| \begin{array}{cccc} 1 & 0 & 0 & -2 \\ 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 2 \\ 1 & 1 & 1 & 2 \end{array} \right| \end{array}$$

The first expansion gives

$$\begin{aligned} \det(B) &= 1 \cdot \begin{vmatrix} 0 & 1 & 0 \\ 2 & 0 & 2 \\ 1 & 1 & 2 \end{vmatrix} - 0 \cdot \begin{vmatrix} 1 & 1 & 0 \\ 0 & 0 & 2 \\ 1 & 1 & 2 \end{vmatrix} + 0 \cdot \begin{vmatrix} 1 & 0 & 0 \\ 0 & 2 & 2 \\ 1 & 1 & 2 \end{vmatrix} - (-2) \cdot \begin{vmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 1 \end{vmatrix} \\ &= 1 \cdot \begin{vmatrix} 0 & 1 & 0 \\ 2 & 0 & 2 \\ 1 & 1 & 2 \end{vmatrix} + 2 \cdot \begin{vmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 1 \end{vmatrix}. \end{aligned}$$

Expanding the 3×3 determinants gives

$$\begin{aligned}
 \begin{vmatrix} 0 & 1 & 0 \\ 2 & 0 & 2 \\ 1 & 1 & 2 \end{vmatrix} &= -1 \cdot \begin{vmatrix} 2 & 2 \\ 1 & 2 \end{vmatrix} \\
 &= -1(2(2) - 2(1)) \\
 &= -2, \\
 \begin{vmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 1 \end{vmatrix} &= 1 \cdot \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 0 & 2 \\ 1 & 1 \end{vmatrix} \\
 &= 1(2(1) - 0(1)) + 1(0(1) - 2(1)) \\
 &= 2 + (-2) = 0.
 \end{aligned}$$

Finally,

$$\det(B) = 1(-2) + 2(0) = -2. \blacksquare$$

As you can see, computing determinants is a labor intensive process. If we needed to compute the determinant of a 5×5 matrix, we would end up computing $5 \cdot 4 \cdot 3 = 60$ 2×2 determinants! Fortunately, 3×3 and 4×4 will be the largest matrices we will deal with by hand. For larger matrices, calculators and computer algebra systems are more than capable of taking determinants for us. Even with relatively efficient algorithms, computing the determinant of a large matrix (for example, 100×100) will take a significant amount of time. Before we return to inverses, we have the following important theorem.

Theorem 8.5.2 (Properties of the Determinant). *Let A and B be $n \times n$ matrices and c a scalar.*

- 1) $\det(I_n) = 1$
- 2) $\det(cA) = c^n \det(A)$
- 3) $\det(AB) = \det(A) \det(B)$
- 4) $\det(A^{-1}) = 1/\det(A)$
- 5) *An $n \times n$ matrix has an inverse if and only if its determinant is non-zero.*

Proof: Omitted. See [FIS03, §4.3] for details.

For most of what we will be studying, the determinant is the most important notion. However, there is a very efficient algorithm for finding the inverse of a square matrix. In fact, we can use this algorithm without checking the determinant first! If the determinant is zero, the method will clearly show that. Otherwise, the method will produce the inverse matrix.

To see why the method works, we look at part of the defining property:

$$AA^{-1} = I_n.$$

If we write

$$A^{-1} = [\vec{v}_1 \ \vec{v}_2 \ \cdots \ \vec{v}_n],$$

(that is, $\vec{v}_1$ is the first column of A^{-1} , $\vec{v}_2$ is the second column of A^{-1} , etc.) then

$$AA^{-1} = [A\vec{v}_1 \ A\vec{v}_2 \ \cdots \ A\vec{v}_n] = I_n.$$

What this tells us is that

$$\begin{aligned} A\vec{v}_1 &= \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \\ A\vec{v}_2 &= \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \\ &\vdots \\ A\vec{v}_n &= \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}, \end{aligned}$$

which we can solve by row reduction! Instead of performing n different row reductions, we can combine them into one single process. The discussion above establishes the following result.

Theorem 8.5.3 (Algorithm for Finding Inverse Matrices). *If A is an $n \times n$ matrix, form the augmented matrix with A on the left half of the matrix and the $n \times n$ identity matrix on the right:*

$$\left[A \mid I_n \right].$$

Row reduce this matrix until it is in RREF. If the matrix is invertible, the final form will be

$$\left[I_n \mid A^{-1} \right].$$

*If A is not invertible, then the left hand side of the reduced matrix **will not be** the identity matrix I_n .*

Example 8.5.7. Determine whether the matrix

$$M = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 0 & 2 \\ 1 & 1 & 2 \end{bmatrix}$$

is invertible. If it is, find M^{-1} .

Answer: We actually know that the determinant of this matrix is -2 since it appeared as one of the sub-determinants in the 4×4 determinant above. However, we do not need to know the determinant before starting the algorithm. Our job is to reduce

$$\left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 1 & 0 & 0 \\ 2 & 0 & 2 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array} \right]$$

to RREF. The easiest way to begin is to rearrange the matrix so that row 3 is on top and the other two rows have been moved down one row.

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 2 & 0 & 2 & 0 & 1 & 0 \end{array} \right]$$

Next, we use the 1 in the upper left corner to eliminate the two in the final row.

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & -2 & -2 & 0 & 1 & -2 \end{array} \right]$$

Now, we can use the $(2, 2)$ -entry to eliminate the entries above and below it.

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 2 & 1 & -2 \end{array} \right]$$

Dividing the third row by -2 gives

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1/2 & 1 \end{array} \right].$$

Finally, we eliminate the 2 in the $(1, 3)$ -entry.

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & -1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1/2 & 1 \end{array} \right].$$

So, the algorithm shows that M is invertible (which we already knew), and

$$M^{-1} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 0 & 0 \\ -1 & -1/2 & 1 \end{bmatrix}. \blacksquare$$

Example 8.5.8. Determine whether the matrix

$$M = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

is invertible. If it is, find M^{-1} .

Answer: We need to row reduce

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array} \right].$$

We begin by dividing row 2 by 2, and then subtracting row 1 from row 3.

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1/2 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 \end{array} \right]$$

Subtracting row 2 from row 3 gives us

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & -1 & -1/2 & 1 \end{array} \right].$$

Even though this final matrix is not quite in RREF (the leading coefficient in row 3 is not a 1 and is not the only non-zero entry in its row), we can see that the left half of this matrix will not be the identity matrix. Hence, M is not invertible. ■

Example 8.5.9. Determine whether the matrix

$$M = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 2 \\ 1 & 1 & 1 & 2 \end{bmatrix}$$

is invertible. If it is, find M^{-1} .

Answer: If we row reduce

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & -2 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 2 & 0 & 0 & 0 & 1 \end{array} \right]$$

we find

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & -2 & -1 & 2 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 & 3 & 1 & -2 \\ 0 & 0 & 0 & 1 & 0 & -1 & -1/2 & 1 \end{array} \right].$$

Since we have the identity matrix on the left, we know M is invertible, and

$$M^{-1} = \begin{bmatrix} 1 & -2 & -1 & 2 \\ 0 & 1 & 1 & -1 \\ -1 & 3 & 1 & -2 \\ 0 & -1 & -1/2 & 1 \end{bmatrix}. \quad \blacksquare$$

The inverse has several important properties as an operation. The following theorem collects the ones most relevant for our purposes.

Theorem 8.5.4 (Properties of the Matrix Inverse). *Let A and B be invertible matrices and c a non-zero scalar. Then the following properties hold.*

- 1) $(A^{-1})^{-1} = A$
- 2) $(cA)^{-1} = \frac{1}{c}A^{-1}$
- 3) $(AB)^{-1} = B^{-1}A^{-1}$

Proof: Since

$$A^{-1}A = AA^{-1} = I_n,$$

it follows that the inverse of A^{-1} is A . Hence $(A^{-1})^{-1} = A$.

Since

$$\frac{1}{c}A^{-1} \cdot cA = \frac{c}{c}A^{-1}A = I_n$$

and similarly for the other order, we see that $(cA)^{-1} = \frac{1}{c}A^{-1}$.

For the final property

$$\begin{aligned} (AB)B^{-1}A^{-1} &= A(BB^{-1})A^{-1} \\ &= AI_nA^{-1} \\ &= AA^{-1} = I_n, \\ B^{-1}A^{-1}(AB) &= B^{-1}(A^{-1}A)B \\ &= B^{-1}I_nB \\ &= B^{-1}B = I_n. \end{aligned}$$

So, $B^{-1}A^{-1}$ must be the inverse of AB . ■

8.6 Eigenvalues and Eigenvectors

We now explore a topic of central importance for material we wish to discuss in later sections. If M is an $n \times n$ matrix, then an **eigenvector** for M is any *non-zero* n -dimensional vector $\vec{v}$ such that there is a scalar λ satisfying

$$M\vec{v} = \lambda\vec{v}.$$

The scalar λ is the **eigenvalue** associated to the eigenvector $\vec{v}$.

Note that only *non-zero* vectors are allowed to be eigenvectors. Recall that $M\vec{0}_n = \vec{0}_n$, and so every possible scalar λ would be an eigenvalue if

the zero vector is allowed to be an eigenvector. We will see that this is a highly undesirable state of affairs. On the other hand, 0 is allowed to be an *eigenvalue*.

Lemma 8.6.1. *Let M be a square matrix.*

- 1) *If $\vec{v}$ is an eigenvector of M , then there is a unique eigenvalue of M associated to $\vec{v}$.*
- 2) *If $\vec{v}_1$ and $\vec{v}_2$ are eigenvectors corresponding to the same eigenvalue λ , then $\alpha\vec{v}_1 + \beta\vec{v}_2$ is an eigenvector of M corresponding to the same eigenvalue λ for any choice of scalars α and β (not both zero).*

Proof: For 1), if $\vec{v}$ is an eigenvector of M , then there is a scalar eigenvalue λ so that $M\vec{v} = \lambda\vec{v}$. If there was another eigenvalue λ' satisfying this equation, we would have

$$\begin{aligned} M\vec{v} &= \lambda\vec{v} = \lambda'\vec{v} \\ (\lambda - \lambda')\vec{v} &= \vec{0}. \end{aligned}$$

Since $\vec{v}$ is not the zero vector (by the definition of eigenvector), we must have $\lambda = \lambda'$.

For 2), we compute

$$\begin{aligned} M(\alpha\vec{v}_1 + \beta\vec{v}_2) &= \alpha M\vec{v}_1 + \beta M\vec{v}_2 \\ &= \alpha(\lambda\vec{v}_1) + \beta(\lambda\vec{v}_2) \\ &= \lambda(\alpha\vec{v}_1 + \beta\vec{v}_2). \blacksquare \end{aligned}$$

Corollary 8.6.1.1. *Let M be a square matrix with eigenvalue λ . Then there is an infinite family of eigenvectors associated to λ .*

Proof: Since λ is an eigenvalue, it must have a non-zero eigenvector $\vec{v}$ associated to it. Let c be any non-zero scalar, then item 2) in the lemma above (with $\alpha = c$, $\beta = 0$, and $\vec{v}_2 = \vec{0}$) shows that $c\vec{v}$ is also an eigenvector. Since c is arbitrary (other than being non-zero), we have an infinite family of eigenvectors associated to λ . ■

For a matrix with real entries, eigenvectors corresponding to real eigenvalues have a particularly nice geometric interpretation. Recall that multiplying a vector by a real scalar stretches or shrinks the vector (and possibly rotates it by 180°). So an eigenvector corresponding to a real eigenvalue lies along the same line determined by the original vector.

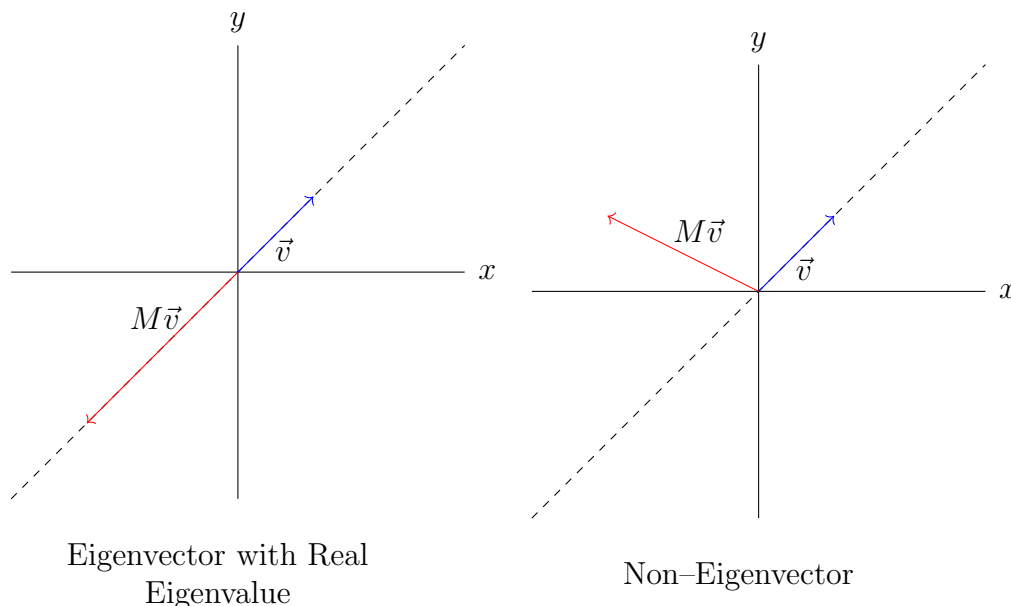


Figure 8.6.1: Geometric Interpretation of Real Eigenvalues

Example 8.6.1. The matrix

$$M_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

rotates vectors through the angle θ . If $0 \leq \theta < 2\pi$, and $\theta \neq 0$ or π , why does M_θ have no real eigenvalues?

Answer: First if $\theta = 0$,

$$M_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

which is the identity matrix. So, every vector is an eigenvector with eigenvalue 1: $M_0 \vec{v} = \vec{v}$. If $\theta = \pi$,

$$M_\pi = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

So, every vector is rotated by 180° : $M_\pi \vec{v} = -\vec{v}$. This means every vector is an eigenvector with eigenvalue -1 .

If θ is neither of these special values, then every vector $\vec{v}$ is rotated off of the line determined by $\vec{v}$. Hence, no vector can be an eigenvector with real eigenvalue. In fact, the eigenvalues of M_θ are precisely $e^{\pm i\theta}$. When the eigenvalues are complex, the corresponding eigenvector will be complex as well. ■

Example 8.6.2. Verify that

$$\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

is an eigenvector of the matrix

$$A = \begin{bmatrix} 2 & 0 & 2 \\ -1 & 3 & 3 \\ 0 & 0 & 4 \end{bmatrix}.$$

Answer:

$$\begin{aligned} \begin{bmatrix} 2 & 0 & 2 \\ -1 & 3 & 3 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} &= \begin{bmatrix} 2(1) + 0(2) + 2(1) \\ -1(1) + 3(2) + 3(1) \\ 0(1) + 0(2) + 4(1) \end{bmatrix} \\ &= \begin{bmatrix} 4 \\ 8 \\ 4 \end{bmatrix} \\ &= 4 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \end{aligned}$$

So, $\vec{v}$ is an eigenvector of A with eigenvalue $\lambda = 4$. ■

Verifying that a given vector is an eigenvector of a particular matrix is relatively easy. The question we need to answer is how to find the eigenvectors for a given matrix. It turns out that finding the eigenvalues is the first step.

Theorem 8.6.2 (Eigenvalues and the Characteristic Polynomial of a Square Matrix). *If M is an $n \times n$ matrix, the eigenvalues are given by the equation*

$$\det(M - \lambda I_n) = 0.$$

The determinant in this equation yields a polynomial of degree n in the variable λ . Hence, a matrix has n eigenvalues (counting multiplicities) which are possibly complex.

Partial Proof: Suppose that $\vec{v}$ is an eigenvector for matrix M with eigenvalue λ . Then by definition, we have

$$M\vec{v} - \lambda\vec{v} = \vec{0}_n.$$

We cannot simply “factor out” $\vec{v}$ since M is a matrix and λ is just a number. We can replace $\vec{v}$ by $I_n\vec{v}$, however.

$$\begin{aligned} M\vec{v} - \lambda\vec{v} &= \vec{0}_n \\ M\vec{v} - \lambda I_n\vec{v} &= \vec{0}_n \\ (M - \lambda I_n)\vec{v} &= \vec{0}_n \end{aligned}$$

The matrix $M - \lambda I_n$ is an $n \times n$ matrix that sends $\vec{v}$ to $\vec{0}_n$. Since this matrix *must* send $\vec{0}_n$ to $\vec{0}_n$, and since $\vec{v} \neq \vec{0}_n$ (eigenvectors are not allowed to be the zero vector), we must have two distinct vectors that are mapped to $\vec{0}_n$. This means $M - \lambda I_n$ cannot be invertible. The only way this can happen is if

$$\det(M - \lambda I_n) = 0.$$

The fact that $\det(M - \lambda I_n)$ is a polynomial will become apparent as we go through several examples. Ultimately, the proof comes down to the fact that taking determinants only involves products and sums of the entries of a matrix. However, we will omit the details here (c.f. [FIS03, Ch. 5]). ■

Corollary 8.6.2.1. *If M is a matrix with real entries, then complex eigenvalues must come in complex conjugate pairs*

$$\lambda = \alpha \pm i\beta$$

where α and β are real numbers.

Proof: If the entries of M are real numbers, then the polynomial $\det(M - \lambda I_n)$ will have real coefficients. It is a standard fact that complex roots of real polynomials must come in complex conjugate pairs. ■

Example 8.6.3. Find the eigenvalues of the matrix

$$M = \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}.$$

Answer:

$$\begin{aligned} M - \lambda I_2 &= \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \\ &= \begin{bmatrix} 1 - \lambda & 4 \\ 3 & 2 - \lambda \end{bmatrix} \end{aligned}$$

Notice that $M - \lambda I_n$ is simply the matrix M with λ subtracted from all entries on the main diagonal. Taking the determinant of this matrix gives

$$\begin{aligned} \det(M - \lambda I_2) &= \begin{vmatrix} 1 - \lambda & 4 \\ 3 & 2 - \lambda \end{vmatrix} \\ &= (1 - \lambda)(2 - \lambda) - 4(3) \\ &= \lambda^2 - 2\lambda - \lambda + 2 - 12 \\ &= \lambda^2 - 3\lambda - 10. \end{aligned}$$

So, the characteristic polynomial of M is $\lambda^2 - 3\lambda - 10$ (a polynomial of degree 2 as promised by the theorem). Setting this polynomial equal to zero gives us the eigenvalues of M .

$$\begin{aligned} \lambda^2 - 3\lambda - 10 &= 0 \\ (\lambda + 2)(\lambda - 5) &= 0 \\ \lambda &= -2 \text{ or } 5. \end{aligned}$$

So, the eigenvalues of M are precisely $\lambda = -2$ and $\lambda = 5$. ■

Example 8.6.4. Find the eigenvalues of the matrix

$$A = \begin{bmatrix} -1 & 5 \\ -2 & -3 \end{bmatrix}.$$

Answer:

$$\begin{aligned} \det(A - \lambda I_2) &= \begin{vmatrix} -1 - \lambda & 5 \\ -2 & -3 - \lambda \end{vmatrix} \\ &= (-1 - \lambda)(-3 - \lambda) - (5)(-2) \\ &= \lambda^2 + 3\lambda + \lambda + 3 + 10 \\ &= \lambda^2 + 4\lambda + 13 \end{aligned}$$

Setting the characteristic polynomial of A equal to zero gives us

$$\lambda^2 + 4\lambda + 13 = 0.$$

Since this does not factor easily, we complete the square (or use the quadratic formula).

$$\begin{aligned}\lambda^2 + 4\lambda &= -13 \\ \lambda^2 + 4\lambda + 4 &= -13 + 4 \\ (\lambda + 2)^2 &= -9 \\ \lambda + 2 &= \pm 3i \\ \lambda &= -2 \pm 3i. \blacksquare\end{aligned}$$

Example 8.6.5. Find the eigenvalues of the matrix

$$B = \begin{bmatrix} -4 & 4 & -2 \\ -9 & 9 & -5 \\ -6 & 6 & -4 \end{bmatrix}.$$

Answer:

$$\begin{aligned}\det(B - \lambda I_3) &= \begin{vmatrix} -4 - \lambda & 4 & -2 \\ -9 & 9 - \lambda & -5 \\ -6 & 6 & -4 - \lambda \end{vmatrix} \\ &= (-4 - \lambda) \cdot \begin{vmatrix} 9 - \lambda & -5 \\ 6 & -4 - \lambda \end{vmatrix} - 4 \cdot \begin{vmatrix} -9 & -5 \\ -6 & -4 - \lambda \end{vmatrix} \\ &\quad + (-2) \cdot \begin{vmatrix} -9 & 9 - \lambda \\ -6 & 6 \end{vmatrix} \\ &= (-4 - \lambda) \left((9 - \lambda)(-4 - \lambda) - (-5)(6) \right) \\ &\quad - 4 \left((-9)(-4 - \lambda) - (-5)(-6) \right) \\ &\quad - 2 \left((-9)(6) - (9 - \lambda)(-6) \right) \\ &= (-4 - \lambda)(\lambda^2 + 4\lambda - 9\lambda - 36 + 30) - 4(9\lambda + 36 - 30) \\ &\quad - 2(-54 + 54 - 6\lambda) \\ &= (-4 - \lambda)(\lambda^2 - 5\lambda - 6) - 4(9\lambda + 6) + 12\lambda\end{aligned}$$

$$\begin{aligned}
&= -\lambda^3 + 5\lambda^2 + 6\lambda - 4\lambda^2 + 20\lambda + 24 - 36\lambda - 24 + 12\lambda \\
&= -\lambda^3 + \lambda^2 + 2\lambda
\end{aligned}$$

Setting the characteristic polynomial equal to zero gives

$$\begin{aligned}
-\lambda^3 + \lambda^2 + 2\lambda &= 0 \\
-\lambda(\lambda^2 - \lambda - 2) &= 0 \\
-\lambda(\lambda - 2)(\lambda + 1) &= 0.
\end{aligned}$$

So, the eigenvalues of B are $\lambda = -1, 0$, and 2 . Remember, $\vec{0}_n$ is not allowed to be an eigenvector, but 0 is allowed to be an eigenvalue. ■

Now that we know how to find eigenvalues, we have to figure out how to find the corresponding eigenvectors. We begin with the following theorem.

Theorem 8.6.3. *Let M be a square matrix with eigenvalue λ_0 .*

- 1) *Suppose M has real entries. If λ_0 is a real eigenvalue then it has an associated eigenvector that is purely real.*
- 2) *Suppose M has real entries, and λ_0 is a complex eigenvalue with associated eigenvector $\vec{v}_0$. Then $\overline{\lambda_0}$ is an eigenvalue of M with associated eigenvector $\overline{\vec{v}_0}$.*

Proof: We begin with the following observation. If M is a real matrix, then $\overline{M}$ (the matrix whose entries are the complex conjugates of the entries in M) is equal to M . If λ_0 has an associated eigenvector $\vec{v}_0$, then $M\vec{v}_0 = \lambda_0\vec{v}_0$ implies that $\overline{M\vec{v}_0} = \overline{\lambda_0\vec{v}_0}$. But,

$$\begin{aligned}
\overline{M\vec{v}_0} &= \overline{M} \overline{\vec{v}_0} = M \overline{\vec{v}_0}, \\
\overline{\lambda_0\vec{v}_0} &= \overline{\lambda_0} \overline{\vec{v}_0}
\end{aligned}$$

(recall that $\overline{\vec{v}}$ is the vector whose components are the complex conjugate of the components in $\vec{v}$). So, we have shown that

$$M \overline{\vec{v}_0} = \overline{\lambda_0} \overline{\vec{v}_0}.$$

This is enough to prove 2). We already know that for real matrices complex eigenvalues must come in conjugate pairs. The computation above shows that if $\vec{v}_0$ is an eigenvector associated to a complex eigenvalue λ_0 , then $\overline{\vec{v}_0}$ is an eigenvector associated to $\overline{\lambda_0}$. Incidentally, this also shows that eigenvectors

of real matrices associated to complex eigenvalues must be complex-valued. Otherwise, $\overline{\vec{v}_0} = \vec{v}_0$, and $\vec{v}_0$ would be associated to two different eigenvalues.

For 1), if λ_0 is real, then we have

$$M \overline{\vec{v}_0} = \lambda_0 \overline{\vec{v}_0}$$

which shows that $\vec{v}_0$ and $\overline{\vec{v}_0}$ correspond to the same eigenvalue. In that case,

$$\begin{aligned} M(\vec{v}_0 + \overline{\vec{v}_0}) &= M\vec{v}_0 + M\overline{\vec{v}_0} \\ &= \lambda_0 \vec{v}_0 + \lambda_0 \overline{\vec{v}_0} \\ &= \lambda_0(\vec{v}_0 + \overline{\vec{v}_0}). \end{aligned}$$

So, $\vec{v}_0 + \overline{\vec{v}_0}$ is an eigenvector with the same eigenvalue λ_0 . But $\vec{v}_0 + \overline{\vec{v}_0}$ is a real vector. ■

With the preliminaries out of the way, we now present the method of finding the family of eigenvectors associated to an eigenvalue.

Theorem 8.6.4. *Let M be an $n \times n$ matrix with eigenvalue λ_0 . Then the associated family of eigenvectors can be found by row reducing the augmented matrix*

$$\left[M - \lambda_0 I_n \mid \vec{0}_n \right].$$

Proof: If $\vec{v}$ is an eigenvector associated to λ_0 , then we already know it satisfies the matrix equation

$$(M - \lambda_0 I_n) \vec{v} = \vec{0}_n.$$

But this is equivalent to an $n \times n$ system of equations, and so we can find the solutions by row reducing the augmented matrix above. ■

Remember, we are expecting an *infinite family* of eigenvectors. So, there should be an infinite number of solutions to the associated system of linear equations. This means we should get at least one row of all zeros during the row reduction process. If a row of all zeros does not appear, then either the eigenvalue is incorrect, or there was an error during the row reduction process.

Example 8.6.6. Find the families of eigenvectors associated to each of the eigenvalues of the matrix in Example 8.6.3.

Answer: We already know that the eigenvalues of M are $\lambda = -2, 5$. We begin with $\lambda = -2$.

$$\begin{aligned} \left[M - (-2)I_2 \mid \vec{0}_2 \right] &= \left[\begin{array}{cc|c} 1 - (-2) & 4 & 0 \\ 3 & 2 - (-2) & 0 \end{array} \right] \\ &= \left[\begin{array}{cc|c} 3 & 4 & 0 \\ 3 & 4 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cc|c} 3 & 4 & 0 \\ 0 & 0 & 0 \end{array} \right] \end{aligned}$$

While this matrix is not in RREF, it is reduced enough to easily find the family of eigenvectors. If we let $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$, then we must have

$$3v_1 + 4v_2 = 0.$$

Solving this for v_1 gives $v_1 = -(4/3)v_2$ with v_2 free. So, we can take

$$\vec{v} = \begin{bmatrix} -(4/3)v_2 \\ v_2 \end{bmatrix} = v_2 \begin{bmatrix} -(4/3) \\ 1 \end{bmatrix}$$

as the family of eigenvectors associated to $\lambda = -2$. However, there is a better way to write this family. If we take *any* basic solution to the equation for v_1 and v_2 , then any multiple of that basic one will also be a solution. That means, any solution we pick (other than the trivial $v_1 = 0 = v_2$) will generate the family of eigenvectors. So, we could have picked $v_1 = 4$ and $v_2 = -3$. This gives us the equivalent form

$$\vec{v} = c \begin{bmatrix} 4 \\ -3 \end{bmatrix}$$

for the family of eigenvectors associated to $\lambda = -2$.

Repeating the process for $\lambda = 5$ gives us

$$\begin{aligned} \left[M - 5I_2 \mid \vec{0}_2 \right] &= \left[\begin{array}{cc|c} 1 - 5 & 4 & 0 \\ 3 & 2 - 5 & 0 \end{array} \right] \\ &= \left[\begin{array}{cc|c} -4 & 4 & 0 \\ 3 & -3 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 1 & -1 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array} \right]. \end{aligned}$$

So, we need $v_1 - v_2 = 0$ for these eigenvectors. Clearly, the simplest eigenvector we can choose is $v_1 = v_2 = 1$. So,

$$\vec{v} = c \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

is the family of eigenvectors associated to $\lambda = 5$. ■

Example 8.6.7. Find the families of eigenvectors associated to each of the eigenvalues of the matrix in Example 8.6.5.

Answer: We already know that the eigenvalues for B are $\lambda = -1, 0$, and 2 . Beginning with $\lambda = -1$, we have

$$\begin{aligned} \left[\begin{array}{ccc|c} -4 - (-1) & 4 & -2 & 0 \\ -9 & 9 - (-1) & -5 & 0 \\ -6 & 6 & -4 - (-1) & 0 \end{array} \right] &= \left[\begin{array}{ccc|c} -3 & 4 & -2 & 0 \\ -9 & 10 & -5 & 0 \\ -6 & 6 & -3 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 3 & -4 & 2 & 0 \\ -9 & 10 & -5 & 0 \\ 2 & -2 & 1 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ -9 & 10 & -5 & 0 \\ 2 & -2 & 1 & 0 \end{array} \right] \end{aligned}$$

where in the first row reduction we performed $-R1 \rightarrow R1$ and $-(1/3)R3 \rightarrow R3$. In the second row reduction, we did $R1 - R3 \rightarrow R1$. Next, we use the 1 in the upper left corner to eliminate the entries below it: $R2 + 9R1 \rightarrow R2$ and $R3 - 2R1 \rightarrow R3$.

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ -9 & 10 & -5 & 0 \\ 2 & -2 & 1 & 0 \end{array} \right] &\sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & -8 & 4 & 0 \\ 0 & 2 & -1 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 2 & -1 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]. \end{aligned}$$

The final steps above were $-(1/4)R2 \rightarrow R2$, $R3 - R2 \rightarrow R3$, and $R1 + R2 \rightarrow R1$. Even though this is not in RREF, we can easily read off the family of

eigenvectors from here. Letting $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$, the matrix above states

$$\begin{aligned} v_1 &= 0 \\ 2v_2 - v_3 &= 0 \\ v_3 &\text{ is free.} \end{aligned}$$

The simplest solution to this equation is $v_1 = 0$, $v_2 = 1$, and $v_3 = 2$. This gives us that

$$\vec{v} = c \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

is the family of eigenvectors associated to $\lambda = -1$.

For $\lambda = 0$, we find

$$\begin{aligned} \left[\begin{array}{ccc|c} -4-0 & 4 & -2 & 0 \\ -9 & 9-0 & -5 & 0 \\ -6 & 6 & -4-0 & 0 \end{array} \right] &= \left[\begin{array}{ccc|c} -4 & 4 & -2 & 0 \\ -9 & 9 & -5 & 0 \\ -6 & 6 & -4 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

(try the row reduction for yourself). So, we need

$$\begin{aligned} v_1 - v_2 &= 0 \\ v_3 &= 0 \\ v_2 &\text{ is free.} \end{aligned}$$

Notice that v_3 cannot be free in this system since it must be equal to 0. The simplest solution to this set of equations is $v_1 = 1 = v_2$ and $v_3 = 0$. This means

$$\vec{v} = c \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

is the family of eigenvectors associated to $\lambda = 0$.

For $\lambda = 2$, we have

$$\left[\begin{array}{ccc|c} -4-2 & 4 & -2 & 0 \\ -9 & 9-2 & -5 & 0 \\ -6 & 6 & -4-2 & 0 \end{array} \right] = \left[\begin{array}{ccc|c} -6 & 4 & -2 & 0 \\ -9 & 7 & -5 & 0 \\ -6 & 6 & -6 & 0 \end{array} \right] \\ \sim \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

This tells us

$$\begin{aligned} v_1 &= v_3 \\ v_2 &= 2v_3 \\ v_3 &\text{ is free.} \end{aligned}$$

The simplest solution comes from choosing $v_3 = 1$. Hence,

$$\vec{v} = c \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

is the family of eigenvectors associated to $\lambda = 2$. ■

The process is no different for complex eigenvalues. Remember that for real matrices, complex eigenvalues and eigenvectors come in conjugate pairs.

Example 8.6.8. Find the families of eigenvectors associated to each of the eigenvalues of the matrix in Example 8.6.4.

Answer: We have already calculated that A has eigenvalues $\lambda = -2 \pm 3i$. We only need to go through the row reduction process for one of these eigenvalues. So, for $\lambda = -2 + 3i$ we have

$$\left[\begin{array}{cc|c} -1 - (-2 + 3i) & 5 & 0 \\ -2 & -3 - (-2 + 3i) & 0 \end{array} \right] = \left[\begin{array}{cc|c} 1 - 3i & 5 & 0 \\ -2 & -1 - 3i & 0 \end{array} \right].$$

A step that often works in the case of complex eigenvalues is to multiply the top row by the conjugate of the complex entry. In this case, we try

$$(1 + 3i)R1 \rightarrow R1.$$

$$\begin{aligned} \left[\begin{array}{cc|c} 1-3i & 5 & 0 \\ -2 & -1-3i & 0 \end{array} \right] &\sim \left[\begin{array}{cc|c} (1-3i)(1+3i) & 5(1+3i) & 0(1+3i) \\ -2 & -1-3i & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cc|c} 10 & 5(1+3i) & 0 \\ 2 & 1+3i & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cc|c} 2 & 1+3i & 0 \\ 2 & 1+3i & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cc|c} 2 & 1+3i & 0 \\ 0 & 0 & 0 \end{array} \right]. \end{aligned}$$

So, we have

$$\begin{aligned} 2v_1 &= -(1+3i)v_2 \\ v_2 &\text{ is free.} \end{aligned}$$

There are many choices we could make, but $v_1 = 1 + 3i$ and $v_2 = -2$ is a particularly simple choice. So,

$$\vec{v} = c \begin{bmatrix} 1+3i \\ -2 \end{bmatrix}$$

is the family of eigenvectors associated to $\lambda = -2 + 3i$.

We do not need to repeat these calculations for the other eigenvalue. Taking conjugates, we have

$$\vec{v} = c \begin{bmatrix} 1-3i \\ -2 \end{bmatrix}$$

is the family of eigenvectors associated to $\lambda = -2 - 3i$. ■

So far, we have only considered the case when all of the eigenvalues are distinct. The **algebraic multiplicity** of an eigenvalue of matrix M is its multiplicity as a root of the characteristic polynomial of the matrix M .

Example 8.6.9. Find the eigenvalues and eigenvectors for

$$M = \begin{bmatrix} 3 & 0 & -1 \\ -1 & 4 & -1 \\ -1 & 0 & 3 \end{bmatrix}.$$

Answer: We begin by finding eigenvalues.

$$\begin{aligned}
 \begin{vmatrix} 3-\lambda & 0 & -1 \\ -1 & 4-\lambda & -1 \\ -1 & 0 & 3-\lambda \end{vmatrix} &= (3-\lambda) \cdot \begin{vmatrix} 4-\lambda & -1 \\ 0 & 3-\lambda \end{vmatrix} + (-1) \begin{vmatrix} -1 & 4-\lambda \\ -1 & 0 \end{vmatrix} \\
 &= (3-\lambda)(4-\lambda)(3-\lambda) - (4-\lambda) \\
 &= (4-\lambda)((3-\lambda)^2 - 1) \\
 &= (4-\lambda)(\lambda^2 - 6\lambda + 8) \\
 &= -(\lambda - 2)(\lambda - 4)^2
 \end{aligned}$$

So, the eigenvalues of M are $\lambda = 2, 4$, and 4 . So, 2 is an eigenvalue of algebraic multiplicity 1 and 4 is an eigenvalue of algebraic multiplicity 2 .

For $\lambda = 2$, the procedure is no different than before.

$$\begin{aligned}
 \left[\begin{array}{ccc|c} 3-2 & 0 & -1 & 0 \\ -1 & 4-2 & -1 & 0 \\ -1 & 0 & 3-2 & 0 \end{array} \right] &= \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & 0 & 1 & 0 \end{array} \right] \\
 &\sim \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]
 \end{aligned}$$

So,

$$\vec{v} = c \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

is the family of eigenvectors associated to $\lambda = 2$.

For $\lambda = 4$, we find

$$\begin{aligned}
 \left[\begin{array}{ccc|c} 3-4 & 0 & -1 & 0 \\ -1 & 4-4 & -1 & 0 \\ -1 & 0 & 3-4 & 0 \end{array} \right] &= \left[\begin{array}{ccc|c} -1 & 0 & -1 & 0 \\ -1 & 0 & -1 & 0 \\ -1 & 0 & -1 & 0 \end{array} \right] \\
 &\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].
 \end{aligned}$$

The two rows of zeros indicate that there are two free variables.

$$v_1 = -v_3$$

v_2 is free

v_3 is free.

This indicates that there are two linearly independent families of eigenvectors. The easiest way to proceed here is to choose one of the free variables to be 1 and the other to be 0. Then choose the variables the other way. Choosing $v_2 = 1$ and $v_3 = 0$ gives the family

$$c_1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

The other choice, $v_2 = 0$ and $v_3 = 1$ gives

$$c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

This means there is a “doubly infinite” family of eigenvectors for $\lambda = 4$:

$$\vec{v} = c_1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Notice that these two vectors are *linearly independent* (since neither is a multiple of the other). ■

To make rigorous sense of what we just did, we need to introduce the notion of a **basis**. Essentially, a set of non-zero vectors forms a basis for a family of vectors if the set is linearly independent and every vector in the family can be expressed as a combination of the vectors in the family.³ So in the previous example, the set of vectors

$$\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

is a basis for the family of eigenvectors associated to the eigenvalue $\lambda = 4$. Clearly these vectors are linearly independent, and our work showed that every eigenvector in the family is a combination of these two.

³Technically, we should be using the more rigorous term *vector space* instead of “family.” The exact definition of a vector space will not concern us much here. Also note that the number of vectors in a basis is the *dimension* of the vector space.

In the previous example, $\lambda = 4$ had algebraic multiplicity 2 (i.e. 4 was two of the roots of the characteristic polynomial) and we found two linearly independent eigenvectors corresponding to it. Unfortunately, this does not always happen. All we are guaranteed is the following theorem (whose proof can be found in [FIS03, p. 264]).

Theorem 8.6.5. *If a square matrix M has an eigenvalue λ of multiplicity m , then M has between 1 and m linearly independent families of eigenvectors corresponding to λ .*

*The number of linearly independent families of eigenvectors associated to an eigenvalue is called the **geometric multiplicity** of the eigenvalue. In terms of the two different kinds of multiplicities, we have*

$$1 \leq \text{geometric multiplicity of } \lambda \leq \text{algebraic multiplicity of } \lambda.$$

So, we are only guaranteed a single family of eigenvectors for any given eigenvalue (no matter how large its multiplicity). At the other extreme, there can be no more independent families than the algebraic multiplicity of the eigenvalue.

Example 8.6.10. Find the eigenvalues and eigenvectors for

$$A = \begin{bmatrix} 3 & 0 & -1 \\ 1 & 4 & -3 \\ -1 & 0 & 3 \end{bmatrix}.$$

Answer: It turns out that this matrix has the same set of eigenvalues as the previous matrix.

$$\begin{aligned} \begin{vmatrix} 3-\lambda & 0 & -1 \\ 1 & 4-\lambda & -3 \\ -1 & 0 & 3-\lambda \end{vmatrix} &= (3-\lambda) \cdot \begin{vmatrix} 4-\lambda & -3 \\ 0 & 3-\lambda \end{vmatrix} + (-1) \cdot \begin{vmatrix} 1 & 4-\lambda \\ -1 & 0 \end{vmatrix} \\ &= -(\lambda-2)(\lambda-4)^2 \end{aligned}$$

Again, $\lambda = 2$ has algebraic multiplicity 1 while $\lambda = 4$ has algebraic multiplicity 2. We know that there can be only a single eigenvector family associated to $\lambda = 2$ (i.e. it must also have geometric multiplicity 1). Row reduction once again gives

$$\vec{v} = c \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

is the family of eigenvectors associated to $\lambda = 2$.

When we attempt to find the eigenvectors associated to $\lambda = 4$, we find

$$\left[\begin{array}{ccc|c} 3-4 & 0 & -1 & 0 \\ 1 & 4-4 & -3 & 0 \\ -1 & 0 & 3-4 & 0 \end{array} \right] = \left[\begin{array}{ccc|c} -1 & 0 & -1 & 0 \\ 1 & 0 & -3 & 0 \\ -1 & 0 & -1 & 0 \end{array} \right] \\ \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

This gives us $v_1 = 0, v_3 = 0$, and v_2 is free, and we only get a single infinite family of eigenvectors associated to $\lambda = 4$. So,

$$\vec{v} = c \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

is the entire family of eigenvectors associated to $\lambda = 4$. This means that the geometric multiplicity of the eigenvalue $\lambda = 4$ is 1 even though its algebraic multiplicity is 2. ■

If the geometric multiplicity of an eigenvalue is strictly smaller than its algebraic multiplicity, we say that the eigenvalue has a **deficiency**. In other words, an eigenvalue has a deficiency if it has fewer linearly independent families of eigenvectors than the number of times it appears as a root of the characteristic polynomial.

8.7 Transposition and Eigenvalues

The **transpose** of a matrix M is the matrix M^T whose entries are given by

$$[M^T]_{ij} = [M]_{ji}.$$

In words, the matrix M^T is obtained from M by interchanging rows and columns.

Example 8.7.1. Find the transpose of the following matrices.

$$\text{a) } M_1 = \begin{bmatrix} 1 & 2 \\ 0 & 3 \\ 1 & 5 \end{bmatrix}$$

$$\text{b) } M_2 = \begin{bmatrix} 2 & 14 \\ -7 & -3 \end{bmatrix}$$

$$\text{c) } M_3 = [1 \quad 2 \quad 3 \quad 4]$$

Answer:

$$\text{a) } M_1^T = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 5 \end{bmatrix}$$

$$\text{b) } M_2^T = \begin{bmatrix} 2 & -7 \\ 14 & -3 \end{bmatrix}$$

$$\text{c) } M_3^T = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

Notice that if M is a $m \times n$ matrix, M^T is a $n \times m$ matrix. In particular, the transpose of a square matrix is another square matrix of the same size. ■

The transpose has many nice properties and uses that we do not explore in this book. For later chapters, we are most interested in how the eigenvalues of a square matrix relate to the eigenvalues of its transpose. Before tackling that question, we need some basic properties of transposition. The proof of the lemma below is omitted, but see any text on linear algebra for details (for instance, [FIS03]).

Lemma 8.7.1. *Let M and N be two matrices whose sizes are appropriate for the operations below, let c be a scalar, and let n be a positive integer. Then the transpose has the following properties.*

$$\text{a) } (M^T)^T = M$$

$$\text{b) } (cM)^T = cM^T$$

$$\text{c) } (M + N)^T = M^T + N^T$$

$$\text{d) } (MN)^T = N^T M^T$$

$$\text{e) } (M^n)^T = (M^T)^n$$

$$\text{f) } \det(M^T) = \det(M)$$

Using these properties, we can prove the only major result we need about transposes.

Theorem 8.7.2. *Let M be a square matrix. Then the eigenvalues of M^T are the same as the eigenvalues of M .*

Proof: For the $n \times n$ identity matrix, we clearly have $I_n^T = I_n$. Looking at the characteristic polynomial for M^T and using the properties of the transpose given above, we find

$$\begin{aligned}\det(M^T - \lambda I_n) &= \det(M^T - (\lambda I_n)^T) \\ &= \det((M - \lambda I_n)^T) \\ &= \det(M - \lambda I_n).\end{aligned}$$

So, the characteristic polynomial of M^T is exactly the same as the characteristic polynomial of M . As a result, the eigenvalues of M^T must be the same as those of M . Note that the eigenvectors for M and M^T corresponding to the same eigenvalue are typically different! ■

8.8 Diagonalization

It turns out that every square matrix is “closely related” to one of a collection of relatively simple matrices. We will only pursue the simplest possible case of *diagonalizable matrices*. Before we can explore this concept, we have to specify what “closely related” means. Two $n \times n$ matrices, A and B , are said to be **similar** if there is an invertible $n \times n$ matrix M such that

$$A = MBM^{-1}.$$

Similarity has several nice properties. We will begin with the following theorem.

Theorem 8.8.1. *Similarity is an equivalence relation. This means it has the following three properties.*

- **Similarity is reflexive:** A is similar to itself.

- **Similarity is symmetric:** If A is similar to B , then B is similar to A .
- **Similarity is transitive:** If A is similar to B , and B is similar to C , then A is similar to C .

Proof: Let A , B , and C be $n \times n$ matrices. For reflexivity, we note that I_n is invertible and $(I_n)^{-1} = I_n$. Since

$$A = I_n A I_n = I_n A (I_n)^{-1},$$

we see that A is similar to itself.

For symmetry, assume A is similar to B . So, $A = MBM^{-1}$ for some invertible matrix M . But then

$$M^{-1}AM = M^{-1}MBM^{-1}M = I_n B I_n = B.$$

Since $(M^{-1})^{-1} = M$, this means $B = M^{-1}A(M^{-1})^{-1}$ and so B is similar to A .

For transitivity, assume A is similar to be B (say, $A = M_1 B M_1^{-1}$), and B is similar to C (say, $B = M_2 C M_2^{-1}$). This means

$$\begin{aligned} A &= M_1 B M_1^{-1} \\ &= M_1 M_2 C M_2^{-1} M_1^{-1} \\ &= (M_1 M_2) C (M_1 M_2)^{-1}. \end{aligned}$$

This means A is similar to C . ■

Our less-than-precise statement above now becomes: every matrix is similar to one of a family of relatively simple matrices. This is still not terribly rigorous since we have not specified the family of matrices. The general description is outside the scope of this book, so we will make do with the simplest type of matrix in this special class.

An $n \times n$ matrix D is a **diagonal matrix** if its only non-zero entries lie on the main diagonal.

$$D = \begin{bmatrix} d_1 & 0 & 0 & \cdots & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ 0 & 0 & d_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & d_n \end{bmatrix}$$

Diagonal matrices have many nice properties. The following theorem (whose proof is quite simple and left to the reader) collects the most important ones.

Theorem 8.8.2. *Let D be a diagonal matrix.*

- 1) *If all of the diagonal entries of D are non-zero, then D is invertible and*

$$D^{-1} = \begin{bmatrix} \frac{1}{d_1} & 0 & 0 & \cdots & 0 \\ 0 & \frac{1}{d_2} & 0 & \cdots & 0 \\ 0 & 0 & \frac{1}{d_3} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{1}{d_n} \end{bmatrix}.$$

- 2) *The determinant of D is the product of the diagonal entries:*

$$\det(D) = d_1 \cdot d_2 \cdots d_n.$$

- 3) *The eigenvalues of D are precisely the diagonal entries of D . The geometric multiplicity of each eigenvalue exactly equals its algebraic multiplicity. So, no eigenvalue has a deficiency.*

- 4) *If m is a positive integer, then*

$$D^m = \begin{bmatrix} d_1^m & 0 & 0 & \cdots & 0 \\ 0 & d_2^m & 0 & \cdots & 0 \\ 0 & 0 & d_3^m & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & d_n^m \end{bmatrix}$$

With the preliminaries out of the way, we are ready for the main definition and theorem of this section. An $n \times n$ matrix is said to be **diagonalizable** if it is similar to a diagonal matrix.

Theorem 8.8.3. *A square matrix is diagonalizable if and only if the algebraic multiplicity of each eigenvalue equals its geometric multiplicity. If A is a diagonalizable matrix with eigenvalues λ_i and associated eigenvectors $\vec{v}_i$, then*

$$A = MDM^{-1}$$

where

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{bmatrix},$$

and $M = [\vec{v}_1 \ \vec{v}_2 \ \vec{v}_3 \ \cdots \ \vec{v}_n]$ is the matrix whose columns are eigenvectors listed in the same order as their corresponding eigenvalues in D .

We will not prove this theorem here, but [FIS03, p. 268] has all of the details.

Corollary 8.8.3.1. *If a square matrix A has distinct eigenvalues (i.e. every eigenvalue has algebraic multiplicity 1), then A is diagonalizable.*

Proof: The geometric multiplicity of an eigenvalue must be at least one, and the algebraic multiplicity is assumed to be equal to one. Then, the algebraic and geometric multiplicities must be equal for each eigenvalue (both have to be 1). By the theorem above, the matrix must be diagonalizable. ■

Example 8.8.1. Determine if the matrix in Example 8.6.3 is diagonalizable. If so, find an invertible matrix B and a diagonal matrix D such that the given matrix is equal to BDB^{-1} .

Answer: We have already shown that

$$M = \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$$

has eigenvalues $\lambda_1 = -2$ and $\lambda_2 = 5$ with corresponding eigenvectors

$$\vec{v}_1 = \begin{bmatrix} 4 \\ -3 \end{bmatrix} \quad \text{and} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

This means we can take

$$B = \begin{bmatrix} 4 & 1 \\ -3 & 1 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} -2 & 0 \\ 0 & 5 \end{bmatrix}.$$

So,

$$\begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} -2 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ -3 & 1 \end{bmatrix}^{-1}. \quad \blacksquare$$

Of course, the big question is why we should care about such a decomposition. There are many useful applications, but the following result is an important one.

Theorem 8.8.4. *If a matrix M can be decomposed as*

$$M = BDB^{-1},$$

then for any positive integer n

$$M^n = BD^nB^{-1}.$$

Proof: For M^2 , we have

$$\begin{aligned} M^2 &= (BDB^{-1})(BDB^{-1}) \\ &= BD(B^{-1}B)DB^{-1} \\ &= BDI_nDB^{-1} \\ &= BD^2B^{-1}. \end{aligned}$$

Now suppose we know the result is true for $n-1$ (i.e. $M^{n-1} = BD^{n-1}B^{-1}$), then

$$\begin{aligned} M^n &= MM^{n-1} \\ &= (BDB^{-1})(BD^{n-1}B^{-1}) \\ &= BD(B^{-1}B)D^{n-1}B^{-1} \\ &= BDI_nD^{n-1}B^{-1} \\ &= BD^nB^{-1}. \blacksquare \end{aligned}$$

If we just need M^2 , then this method is probably not very efficient. If you need M^{100} , however, then having a diagonalization will probably save you quite a lot of computing time.

Example 8.8.2. Determine if the matrix in Example 8.6.5 is diagonalizable. If so, find an invertible matrix M and a diagonal matrix D such that the given matrix is equal to MDM^{-1} . Use this decomposition to find B^5 .

Proof: We already know that the eigenvalues for B are $\lambda_1 = -1$, $\lambda_2 = 0$, and $\lambda_3 = 2$ with corresponding eigenvectors

$$\vec{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \vec{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}.$$

If we take

$$M = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix},$$

then we have

$$\begin{bmatrix} -4 & 4 & -2 \\ -9 & 9 & -5 \\ -6 & 6 & -4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix}^{-1}.$$

Per the theorem, we know that

$$\begin{aligned} B^5 &= \begin{bmatrix} -4 & 4 & -2 \\ -9 & 9 & -5 \\ -6 & 6 & -4 \end{bmatrix}^5 \\ &= \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}^5 \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} (-1)^5 & 0 & 0 \\ 0 & 0^5 & 0 \\ 0 & 0 & 2^5 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 32 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} -64 & 64 & -32 \\ -129 & 129 & -65 \\ -66 & 66 & -34 \end{bmatrix}. \quad \blacksquare \end{aligned}$$

In the previous example, we omitted the computation of the inverse matrix which is another computational cost of this method. So whether B^5 can be computed faster by diagonalization is not completely clear. However, at some point the diagonalization procedure will become faster. This is especially true if we need to compute more than one power of a particular matrix.

Finally, note that the matrix in Example 8.6.10 is not diagonalizable. This is because the eigenvalue $\lambda = 4$ has algebraic multiplicity 2 but geometric

multiplicity 1. Since the matrix has a deficiency, it cannot be diagonalized. It turns out that there is a rather nice form that every matrix is similar to (known as the Jordan Canonical Form), but this form is not a diagonal matrix in general.

8.9 Problems

8.9.1 Systems of Linear Equations

For the following systems of linear equations, write each system as an augmented matrix, reduce the matrix to RREF, and use the reduced matrix to give the solutions to the system.

1.)

$$\begin{aligned}2x - y &= 7 \\ 3x + 2y &= 0\end{aligned}$$

2.)

$$\begin{aligned}x + 2y + z &= 16 \\ x + 3y - z &= 22 \\ 2x + 5y &= 2\end{aligned}$$

3.)

$$\begin{aligned}x + y + 3z &= 10 \\ x + 2y + z &= 16 \\ 2x + 3y + 4z &= 26\end{aligned}$$

4.)

$$\begin{aligned}x + y + z &= 0 \\ 2x - 2y + z &= 11 \\ 3x + y - 2z &= 1\end{aligned}$$

5.)

$$x - 2y + 3z = 4$$

$$2x + y + z = 8$$

$$3x + 4y - 5z = 8$$

6.)

$$2x + 3y - z = 7$$

$$3x + y + 2z = -7$$

$$5x + 4y + 2z = -4$$

7.)

$$x + y + 5w = 3$$

$$x + 2y - z + 8w = 4$$

$$2x + 3y - z + 13w = 7$$

$$x + 2y - z + 8w = 4$$

8.9.2 Basics of Vectors and Matrices

8.) Compute the following quantities for the vectors

$$\vec{v}_1 = \begin{bmatrix} 3 \\ -2 \end{bmatrix} \quad \text{and} \quad \vec{v}_2 = \begin{bmatrix} 5 \\ 4 \end{bmatrix}.$$

a) $2\vec{v}_1$

b) $-3\vec{v}_2$

c) $2\vec{v}_1 + 3\vec{v}_2$

d) $3\vec{v}_1 - 2\vec{v}_2$

e) $|\vec{v}_1|$ and $|\vec{v}_2|$

9.) Compute the following quantities for the vectors

$$\vec{v}_1 = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} \quad \text{and} \quad \vec{v}_2 = \begin{bmatrix} 4 \\ 2 \\ -7 \end{bmatrix}.$$

- a) $3\vec{v}_1$
- b) $-\frac{1}{2}\vec{v}_2$
- c) $3\vec{v}_1 + 2\vec{v}_2$
- d) $\vec{v}_2 - 4\vec{v}_1$
- e) $|\vec{v}_1|$ and $|\vec{v}_2|$

- 10.) Determine if the following set of vectors is linearly dependent or independent. If the vectors are linearly dependent, find a combination of them that sums to the zero vector.

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} \right\}$$

- 11.) Determine if the following set of vectors is linearly dependent or independent. If the vectors are linearly dependent, find a combination of them that sums to the zero vector.

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 18 \\ -5 \end{bmatrix} \right\}$$

Problems 12.) and 13.) refer to the following matrices.

$$A = \begin{bmatrix} 2 & 2 & 3 \\ 1 & -3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 5 & 7 \end{bmatrix}, \quad C = \begin{bmatrix} -2 & 0 & 1 \\ 5 & 2 & -3 \\ -1 & -1 & -1 \end{bmatrix},$$

$$D = \begin{bmatrix} 1 & -4 \\ -2 & 5 \end{bmatrix}$$

If the given quantity exists, compute it. If it does not, give a reason why not.

- 12.)
- a) $2A$
 - b) $-3C$
 - c) $C + 2D$
 - d) $2A - 3B$

- 13.) a) AB
b) AC
c) DA
d) D^2
e) C^2

8.9.3 Inverse Matrices and Determinants

- 14.) Find the determinants of the following matrices.

$$\begin{array}{ll} \text{a) } \begin{bmatrix} 1 & -3 \\ 2 & 5 \end{bmatrix} & \text{b) } \begin{bmatrix} 1 & -2 & -3 \\ 4 & -1 & 0 \\ -3 & -1 & 1 \end{bmatrix} \\ \text{c) } \begin{bmatrix} 1 & 2 & -3 \\ 2 & 1 & 1 \\ 1 & -1 & 4 \end{bmatrix} & \text{d) } \begin{bmatrix} 1 & -2 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 1 & 1 & 1 & 2 \\ 1 & -3 & 2 & -1 \end{bmatrix} \end{array}$$

- 15.) Determine whether the following matrices are invertible. If so, find the inverse matrix.

$$\begin{array}{ll} \text{a) } \begin{bmatrix} 2 & 4 \\ -1 & 6 \end{bmatrix} & \text{b) } \begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & 0 \\ 1 & 1 & -1 \end{bmatrix} \\ \text{c) } \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 4 & 0 & -1 \end{bmatrix} & \text{d) } \begin{bmatrix} 1 & 0 & 1 & 2 \\ 1 & 1 & 2 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & -1 & 5 & 3 \end{bmatrix} \end{array}$$

- 16.) Solve the following systems of linear equations by translating them into matrix form and finding the inverse of the resulting square matrix.

a)

$$\begin{aligned} 2x + y &= 1 \\ x - 3y &= 11 \end{aligned}$$

b)

$$\begin{aligned}x + 2y + z &= 3 \\2x + 3y + z &= 2 \\-x + 5y - 2z &= 1\end{aligned}$$

8.9.4 Eigenvalues and Eigenvectors

17.) – 20.) Find all eigenvalues and eigenvectors for the following 2×2 matrices.

$$17.) \begin{bmatrix} 5 & -6 \\ 3 & -4 \end{bmatrix}$$

$$18.) \begin{bmatrix} -1 & 1 \\ 3 & 1 \end{bmatrix}$$

$$19.) \begin{bmatrix} 4 & -2 \\ 1 & 2 \end{bmatrix}$$

$$20.) \begin{bmatrix} 7 & -13 \\ 1 & 1 \end{bmatrix}$$

21.) – 24.) Find all eigenvalues and eigenvectors for the following 3×3 matrices.

$$21.) \begin{bmatrix} 3 & -2 & -3 \\ 5 & -4 & -5 \\ 2 & -2 & -2 \end{bmatrix}$$

$$22.) \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$23.) \begin{bmatrix} 1 & -1 & -2 \\ 0 & 2 & 0 \\ 1 & -4 & -1 \end{bmatrix}$$

$$24.) \begin{bmatrix} -10 & -6 & 28 \\ 6 & 5 & -19 \\ -2 & -1 & 5 \end{bmatrix}$$

25.), 26.) Find all eigenvalues and eigenvectors for the following 4×4 matrices.

$$25.) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

$$26.) \begin{bmatrix} -1 & 4 & -2 & -4 \\ -2 & 2 & 2 & 1 \\ -2 & -1 & -1 & 1 \\ 0 & 3 & 0 & 0 \end{bmatrix}$$

8.9.5 Diagonalization

For the matrices in problems 17.) – 26.), determine which are diagonalizable. For the ones that are, find an invertible matrix M and a diagonal matrix D such that the given matrix is equal to MDM^{-1} . Use this decomposition to find the fifth power of these matrices.

Chapter 9

Analysis of Linear Systems

In Chapter 5, we studied systems of autonomous differential equations from a purely qualitative point of view. While we found the equilibria exactly, the determination of whether a particular equilibrium was stable or unstable was handled in a rather haphazard way. Now that we have matrix techniques at our disposal, we can study these systems more precisely. The key to understanding the behavior of equilibria in systems of autonomous differential equations is to first understand the solutions to *linear* systems of differential equations.

9.1 Linear Systems of Differential Equations

A **linear system of first-order differential equations** is a set of differential equations for n unknown functions $x_1(t), x_2(t), \dots, x_n(t)$ of the form

$$\begin{aligned}x'_1(t) &= m_{11}x_1(t) + m_{12}x_2(t) + m_{13}x_3(t) + \cdots + m_{1n}x_n(t) + b_1(t) \\x'_2(t) &= m_{21}x_1(t) + m_{22}x_2(t) + m_{23}x_3(t) + \cdots + m_{2n}x_n(t) + b_2(t) \\x'_3(t) &= m_{31}x_1(t) + m_{32}x_2(t) + m_{33}x_3(t) + \cdots + m_{3n}x_n(t) + b_3(t) \\&\vdots \\x'_n(t) &= m_{n1}x_1(t) + m_{n2}x_2(t) + m_{n3}x_3(t) + \cdots + m_{nn}x_n(t) + b_n(t)\end{aligned}$$

where the quantities m_{ij} and b_i are given. If the functions $b_i(t)$ are all zero, then the system is said to be *homogeneous*; otherwise, it is *inhomogeneous*. Throughout this chapter, we will assume that the coefficients m_{ij} are real

numbers.¹ As is readily apparent from the setup of the problem, we can re-express this system using matrices.

We begin by collecting the unknown function $x_i(t)$ into a single vector-valued function $\vec{x}(t)$:

$$\vec{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ \vdots \\ x_n(t) \end{bmatrix} \quad \text{and} \quad \vec{x}'(t) = \begin{bmatrix} x'_1(t) \\ x'_2(t) \\ x'_3(t) \\ \vdots \\ x'_n(t) \end{bmatrix}.$$

The inhomogeneous terms $b_i(t)$ can be collected into a similar vector function:

$$\vec{b}(t) = \begin{bmatrix} b_1(t) \\ b_2(t) \\ b_3(t) \\ \vdots \\ b_n(t) \end{bmatrix}.$$

Finally, the coefficients m_{ij} can be collected into a square matrix M :

$$M = \begin{bmatrix} m_{11} & m_{12} & m_{13} & \cdots & m_{1n} \\ m_{21} & m_{22} & m_{23} & \cdots & m_{2n} \\ m_{31} & m_{32} & m_{33} & \cdots & m_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ m_{n1} & m_{n2} & m_{n3} & \cdots & m_{nn} \end{bmatrix}.$$

This allows us to write the system of differential equations in the far more compact *vector form* or *matrix form*:

$$\vec{x}'(t) = M\vec{x}(t) + \vec{b}(t).$$

As we will see, solving these systems relies heavily on computing the eigenvalues and eigenvectors of M .

Example 9.1.1. Rewrite the system of differential equations below in matrix form.

$$\begin{aligned} x'(t) &= x(t) - 2y(t) + 2t + 1 \\ y'(t) &= -2x(t) + y(t) + t^2 \end{aligned}$$

¹Allowing the coefficients to be functions of t (i.e. $m_{ij} = m_{ij}(t)$) would still give us a linear system, but such systems are far more difficult to solve in general.

Answer: Notice that this is a 2×2 inhomogeneous system. We have

$$\vec{x}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}, \vec{b}(t) = \begin{bmatrix} 2t+1 \\ t^2 \end{bmatrix}, \text{ and } M = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}.$$

This allows us to write

$$\begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} + \begin{bmatrix} 2t+1 \\ t^2 \end{bmatrix}. \blacksquare$$

Example 9.1.2. Rewrite the system of differential equations below in matrix form.

$$\begin{aligned} x' &= 3x - z \\ y' &= x + 2y - z \\ z' &= -x + 3z \end{aligned}$$

Answer: Notice that this is a 3×3 homogeneous system. We have

$$\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } M = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 2 & -1 \\ -1 & 0 & 3 \end{bmatrix}.$$

This allows us to write

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 2 & -1 \\ -1 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}. \blacksquare$$

As with single first-order differential equations, an initial condition is also required to specify a unique answer. Since we have n unknown functions, we need initial data for each of the functions. Since such data is often given at $t = 0$, a typical initial condition would be of the form

$$\vec{x}(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \\ \vdots \\ x_n(0) \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{bmatrix}$$

where the a_i are given real numbers.

The *solution* to a linear system of differential equations is any particular set of functions $x_i(t)$ such that when substituted into the correct places in each of the differential equations yields a true statement.

Example 9.1.3. Verify that the functions $x(t) = -e^{-t} + e^{3t} + 1$ and $y(t) = -e^{-t} - e^{3t} + 2$ give a solution to the initial value problem

$$\begin{aligned}x' &= x - 2y + 3 \\y' &= -2x + y \\x(0) &= 1 \\y(0) &= 0.\end{aligned}$$

Answer: Clearly $x(0) = -1 + 1 + 1 = 1$ and $y(0) = -1 - 1 + 2 = 0$. For the first differential equation, we find

$$\begin{aligned}x'(t) &= e^{-t} + 3e^{3t}, \\x(t) - 2y(t) + 3 &= (-e^{-t} + e^{3t} + 1) - 2(-e^{-t} - e^{3t} + 2) + 3 \\&= e^{-t} + 3e^{3t}.\end{aligned}$$

Since both sides yield the same quantity, the two given functions satisfy the first differential equation. For the second,

$$\begin{aligned}y'(t) &= e^{-t} - 3e^{3t}, \\-2x(t) + y(t) &= -2(-e^{-t} + e^{3t} + 1) + (-e^{-t} - e^{3t} + 2) \\&= e^{-t} - 3e^{3t}.\end{aligned}$$

Since the given functions satisfy both differential equations and the initial conditions, they are a solution to the given initial value problem. ■

Of course, the biggest question we need to answer is whether or not a given system of differential equations even has a solution and if it does whether that solution is unique. The following result addresses these issues.

Theorem 9.1.1 (Existence and Uniqueness). *The initial value problem*

$$\begin{aligned}\vec{x}'(t) &= M\vec{x}(t) + \vec{b}(t) \\ \vec{x}(t_0) &= \vec{x}_0\end{aligned}$$

is guaranteed to have a unique solution for some interval containing the initial time t_0 provided that the functions $b_1(t), \dots, b_n(t)$ comprising $\vec{b}(t)$ are continuous in some open interval containing the initial time t_0 .

The proof of this result is well beyond the scope of this text, but most any text on dynamical systems should discuss it (c.f. [Ro95][Chpts. 4 and 5]). In fact, the requirement that $\vec{b}(t)$ be continuous in some interval containing t_0 is a bit stronger than what is strictly necessary. However, this result will more than suffice for the examples we consider.

9.1.1 Homogeneous Systems with Real Eigenvalues

In this section, we consider linear systems of the form $\vec{x}'(t) = M\vec{x}(t)$ where M is a matrix of constant coefficients. The first result we consider is the following lemma.

Lemma 9.1.2. *Suppose that the matrix M has eigenvalue λ with corresponding eigenvector $\vec{v}$. Then the function $\vec{x}(t) = ce^{\lambda t}\vec{v}$ is a solution to the system of differential equations*

$$\vec{x}'(t) = M\vec{x}(t)$$

for any constant scalar c .

Proof: Recall that $\vec{v}$ being an eigenvector of matrix M with eigenvalue λ means $M\vec{v} = \lambda\vec{v}$. On the one hand, we have

$$\begin{aligned}\vec{x}'(t) &= \frac{d}{dt}(ce^{\lambda t}\vec{v}) \\ &= \lambda ce^{\lambda t}\vec{v}\end{aligned}$$

since c and $\vec{v}$ are constant. On the other hand,

$$\begin{aligned}M\vec{x}(t) &= M(ce^{\lambda t}\vec{v}) \\ &= ce^{\lambda t}(M\vec{v}) \\ &= ce^{\lambda t}\lambda\vec{v} \\ &= \lambda ce^{\lambda t}\vec{v}\end{aligned}$$

since $ce^{\lambda t}$ is a scalar quantity (and matrices only act on vectors). ■

This lemma immediately leads us to the following theorem.

Theorem 9.1.3 (General Solution to Diagonalizable Homogeneous Systems). *Suppose that no eigenvalue of the $n \times n$ matrix M is deficient. Then the general solution to $\vec{x}'(t) = M\vec{x}(t)$ is of the form*

$$\vec{x}(t) = c_1 e^{\lambda_1 t} \vec{v}_1 + c_2 e^{\lambda_2 t} \vec{v}_2 + \cdots c_n e^{\lambda_n t} \vec{v}_n$$

where λ_i are the eigenvalues of M , $\vec{v}_i$ the corresponding eigenvectors, and the c_i are arbitrary coefficients.

Recall that an eigenvalue is said to have a deficiency if its algebraic multiplicity is strictly larger than its geometric multiplicity. In other words, an eigenvalue is deficient if it has fewer linearly independent eigenvectors than its multiplicity as a root of the characteristic polynomial. Also remember that every eigenvalue must have at least one eigenvector associated to it!

Example 9.1.4. Find the general solution to the following system of differential equations.

$$\begin{aligned} x' &= x - 2y \\ y' &= -2x + y \end{aligned}$$

Answer: If we define $\vec{x}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$, the system is given by

$$\vec{x}'(t) = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \vec{x}(t)$$

in matrix form. The characteristic polynomial for this matrix is

$$\begin{vmatrix} 1 - \lambda & -2 \\ -2 & 1 - \lambda \end{vmatrix} = (1 - \lambda)^2 - 4$$

which gives eigenvalues

$$\begin{aligned} (1 - \lambda)^2 - 4 &= 0 \\ (1 - \lambda)^2 &= 4 \\ 1 - \lambda &= \pm 2 \\ \lambda &= 1 \pm 2 = -1, 3. \end{aligned}$$

For $\lambda = -1$, we find by row reduction

$$\left[\begin{array}{cc|c} 1 - (-1) & -2 & 0 \\ -2 & 1 - (-1) & 0 \end{array} \right] = \left[\begin{array}{cc|c} 2 & -2 & 0 \\ -2 & 2 & 0 \end{array} \right] \\ \sim \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array} \right].$$

If we represent the eigenvector as $\vec{v}_1 = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$, then the row reduced matrix above tells us that we must have $v_1 = v_2$. So, we can take $\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ as a basis eigenvector for $\lambda = -1$.

For $\lambda = 3$, we find

$$\left[\begin{array}{cc|c} 1 - (3) & -2 & 0 \\ -2 & 1 - (3) & 0 \end{array} \right] = \left[\begin{array}{cc|c} -2 & -2 & 0 \\ -2 & -2 & 0 \end{array} \right] \\ \sim \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right].$$

Taking v_1 and v_2 for the components of the second eigenvector, we must have $v_1 = -v_2$. So, we can take $\vec{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ as a basis eigenvector for $\lambda = 3$.

Putting these result together gives

$$\vec{x}(t) = c_1 e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Since the original problem was stated in component form, we ought to express the final answer the same way. In our definition of $\vec{x}(t)$, the top component of the vector gives $x(t)$ while the bottom component is $y(t)$. Reading these from the vector form above gives

$$\begin{aligned} x(t) &= c_1 e^{-t} + c_2 e^{3t} \\ y(t) &= c_1 e^{-t} - c_2 e^{3t}. \quad \blacksquare \end{aligned}$$

A word of caution: When we studied single differential equations in Chapter 4, we often absorbed numerical coefficients into the unknown coefficient in the general solution (i.e. replacing $2C$ by C or $-C$ by C). This was perfectly acceptable in that setting so long as the unknown coefficient

appeared only once in the solution. For systems of differential equations, the coefficients c_i appear *in multiple components* of the solution. As such, you should not make changes to any single instance of an arbitrary coefficient without making the exact same change to all other instances of it. As an example, you should not absorb the negative sign on c_2 in the right-hand-side of $y(t)$. If you do replace $-c_2$ by c_2 in $y(t)$, you would need to make the exact same change in $x(t)$ (and so a negative sign would appear there).

Example 9.1.5. Find the general solution to the following system of linear differential equations.

$$\vec{x}' = \begin{bmatrix} 5 & 1 & -2 \\ 1 & 5 & -2 \\ -1 & -1 & 6 \end{bmatrix} \vec{x}.$$

Answer: For eigenvalues, we have

$$\begin{vmatrix} 5 - \lambda & 1 & -2 \\ 1 & 5 - \lambda & -2 \\ -1 & -1 & 6 - \lambda \end{vmatrix} = -\lambda^3 + 16\lambda^2 - 80\lambda + 128 \\ = -(\lambda - 4)^2(\lambda - 8) = 0.$$

We see that $\lambda_1 = 4$ has algebraic multiplicity 2. For eigenvectors, we find

$$\left[\begin{array}{ccc|c} 5-4 & 1 & -2 & 0 \\ 1 & 5-4 & -2 & 0 \\ -1 & -1 & 6-4 & 0 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 1 & 1 & -2 & 0 \\ -1 & -1 & 2 & 0 \end{array} \right] \\ \sim \left[\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

The two rows of zeros tell us that there are two linearly independent families of eigenvectors, and the geometric multiplicity of $\lambda_1 = 4$ is also 2. Since we need $v_1 + v_2 - 2v_3 = 0$, we can take

$$\vec{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \text{ and } \vec{v}_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

as our independent basis eigenvectors. The other eigenvalue $\lambda_2 = 8$ has multiplicity 1, and we leave it to the reader to find

$$\vec{v}_3 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

This gives us the general solution

$$\vec{x}(t) = c_1 e^{4t} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + c_2 e^{4t} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + c_3 e^{8t} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}. \blacksquare$$

9.1.2 Homogeneous Systems with Complex Eigenvalues

Recall that if a real matrix has a complex eigenvalue $a + bi$ with eigenvector $\vec{v}$, then the complex conjugate $a - bi$ must also be an eigenvalue with corresponding eigenvector $\vec{\bar{v}}$ (the vector whose components are the conjugate of the components of $\vec{v}$). Just as with real eigenvalues, this means that

$$c_1 e^{(a+bi)t} \vec{v} + c_2 e^{(a-bi)t} \vec{\bar{v}}$$

is part of the general solution. The problem with this form of the solution is that we expect *real-values* solutions if our initial data consists of all real numbers. Fortunately, we can convert this complex form of the solution into an equivalent real form. The key to accomplishing this transformation is using **Euler's Formula**:

$$e^{x+iy} = e^x e^{iy} = e^x \cos(y) + i e^x \sin(y).$$

Rather than giving a generic formula, we demonstrate the process with a simple example.

Example 9.1.6. Find the general solution to

$$\begin{aligned} x' &= x - 3y \\ y' &= 6x - 5y. \end{aligned}$$

Answer: In matrix form, the system is given by

$$\frac{d\vec{x}}{dt} = \begin{bmatrix} 1 & -3 \\ 6 & -5 \end{bmatrix} \vec{x}.$$

The eigenvalues for this matrix are $\lambda = -2 \pm 3i$. For $\lambda = -2 + 3i$, we find

$$\begin{aligned} \left[\begin{array}{cc|c} 1 - (-2 + 3i) & -3 & 0 \\ 6 & -5 - (-2 + 3i) & 0 \end{array} \right] &= \left[\begin{array}{cc|c} 3 - 3i & -3 & 0 \\ 6 & -3 - 3i & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cc|c} 1 - i & -1 & 0 \\ 2 & -1 - i & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cc|c} (1 - i)(1 + i) & -1(1 + i) & 0 \\ 2 & -(1 + i) & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cc|c} 2 & -(1 + i) & 0 \\ 0 & 0 & 0 \end{array} \right]. \end{aligned}$$

We must have $2v_1 = (1 + i)v_2$ which means we can take

$$\vec{v} = \begin{bmatrix} 1 + i \\ 2 \end{bmatrix}$$

as our basis eigenvector. Using basic facts about conjugates, the general solution to our system must be

$$\vec{x}(t) = d_1 e^{(-2+3i)t} \begin{bmatrix} 1 + i \\ 2 \end{bmatrix} + d_2 e^{(-2-3i)t} \begin{bmatrix} 1 - i \\ 2 \end{bmatrix}.$$

The reason for using the coefficients d_1 and d_2 (instead of c_1 and c_2) will become apparent below.

If we examine the first part of this solution, we find the following.

$$\begin{aligned} e^{(-2+3i)t} \begin{bmatrix} 1 + i \\ 2 \end{bmatrix} &= e^{-2t} e^{3it} \begin{bmatrix} 1 + i \\ 2 \end{bmatrix} \\ &= e^{-2t} (\cos(3t) + i \sin(3t)) \begin{bmatrix} 1 + i \\ 2 \end{bmatrix} \\ &= e^{-2t} \begin{bmatrix} (1 + i)(\cos(3t) + i \sin(3t)) \\ 2(\cos(3t) + i \sin(3t)) \end{bmatrix} \\ &= e^{-2t} \begin{bmatrix} \cos(3t) + i \sin(3t) + i \cos(3t) - \sin(3t) \\ 2 \cos(3t) + 2i \sin(3t) \end{bmatrix} \\ &= e^{-2t} \begin{bmatrix} \cos(3t) - \sin(3t) \\ 2 \cos(3t) \end{bmatrix} + i e^{-2t} \begin{bmatrix} \sin(3t) + \cos(3t) \\ 2 \sin(3t) \end{bmatrix} \end{aligned}$$

In the last step above, we separated the vector into real and imaginary parts. Since the second part of the solution is the conjugate of the first part, we have

$$e^{(-2-3i)t} \begin{bmatrix} 1-i \\ 2 \end{bmatrix} = e^{-2t} \begin{bmatrix} \cos(3t) - \sin(3t) \\ 2 \cos(3t) \end{bmatrix} - ie^{-2t} \begin{bmatrix} \sin(3t) + \cos(3t) \\ 2 \sin(3t) \end{bmatrix}.$$

Combining these results, we have

$$\begin{aligned} \vec{x}(t) &= d_1 \left(e^{-2t} \begin{bmatrix} \cos(3t) - \sin(3t) \\ 2 \cos(3t) \end{bmatrix} + ie^{-2t} \begin{bmatrix} \sin(3t) + \cos(3t) \\ 2 \sin(3t) \end{bmatrix} \right) \\ &\quad + d_2 \left(e^{-2t} \begin{bmatrix} \cos(3t) - \sin(3t) \\ 2 \cos(3t) \end{bmatrix} - ie^{-2t} \begin{bmatrix} \sin(3t) + \cos(3t) \\ 2 \sin(3t) \end{bmatrix} \right) \\ &= (d_1 + d_2)e^{-2t} \begin{bmatrix} \cos(3t) - \sin(3t) \\ 2 \cos(3t) \end{bmatrix} + (id_1 - id_2)e^{-2t} \begin{bmatrix} \sin(3t) + \cos(3t) \\ 2 \sin(3t) \end{bmatrix}. \end{aligned}$$

If we rename the coefficients as

$$\begin{aligned} c_1 &= d_1 + d_2 \\ c_2 &= id_1 - id_2, \end{aligned}$$

we have the *real form* of the solution

$$\vec{x}(t) = c_1 e^{-2t} \begin{bmatrix} \cos(3t) - \sin(3t) \\ 2 \cos(3t) \end{bmatrix} + c_2 e^{-2t} \begin{bmatrix} \sin(3t) + \cos(3t) \\ 2 \sin(3t) \end{bmatrix}$$

where c_1 and c_2 are completely independent quantities. Note that despite their appearance, the coefficients c_1 and c_2 can both be real quantities for certain choices of d_1 and d_2 . In fact, c_1 and c_2 can be any pair of quantities we wish! The proof that these new coefficients are independent of one another is simply that d_1 and d_2 can be written uniquely in terms of c_1 and c_2 :

$$\begin{aligned} d_1 &= \frac{c_1 - ic_2}{2} \\ d_2 &= \frac{c_1 + ic_2}{2}. \end{aligned}$$

If we write our answer in component form, we have

$$\begin{aligned} x(t) &= c_1 e^{-2t} (\cos(3t) - \sin(3t)) + c_2 e^{-2t} (\sin(3t) + \cos(3t)) \\ y(t) &= 2c_1 e^{-2t} \cos(3t) + 2c_2 e^{-2t} \sin(3t). \quad \blacksquare \end{aligned}$$

What we learn from this example is the following fact.

Lemma 9.1.4 (Real Form for Complex Eigenvalues). *If $\lambda = a \pm bi$ is a conjugate pair of eigenvalues for a real $n \times n$ matrix M with basis eigenvectors $\vec{v}, \bar{\vec{v}}$, then the portion of the general solution to $\vec{x}' = M\vec{x}$ corresponding to these eigenvalues is of the form*

$$c_1 e^{at} \begin{bmatrix} r_1 \cos(bt) + s_1 \sin(bt) \\ r_2 \cos(bt) + s_2 \sin(bt) \\ \vdots \\ r_n \cos(bt) + s_n \sin(bt) \end{bmatrix} + c_2 e^{at} \begin{bmatrix} u_1 \cos(bt) + v_1 \sin(bt) \\ u_2 \cos(bt) + v_2 \sin(bt) \\ \vdots \\ u_n \cos(bt) + v_n \sin(bt) \end{bmatrix}$$

where the two independent solutions are the real and imaginary parts of $e^{(a+bi)t}\vec{v}$:

$$e^{(a+bi)t}\vec{v} = e^{at} \begin{bmatrix} r_1 \cos(bt) + s_1 \sin(bt) \\ r_2 \cos(bt) + s_2 \sin(bt) \\ \vdots \\ r_n \cos(bt) + s_n \sin(bt) \end{bmatrix} + ie^{at} \begin{bmatrix} u_1 \cos(bt) + v_1 \sin(bt) \\ u_2 \cos(bt) + v_2 \sin(bt) \\ \vdots \\ u_n \cos(bt) + v_n \sin(bt) \end{bmatrix}.$$

Example 9.1.7. Solve the following initial value problem.

$$\begin{aligned} x' &= 6x - 13y \\ y' &= 4x - 6y \\ x(0) &= 2 \\ y(0) &= -4 \end{aligned}$$

Answer: The eigenvalues for this matrix are given by

$$\begin{vmatrix} 6 - \lambda & -13 \\ 4 & -6 - \lambda \end{vmatrix} = (6 - \lambda)(-6 - \lambda) + 52 \\ = \lambda^2 + 16 = 0.$$

So, the eigenvalues are $\lambda = \pm 4i$. We only need to find an eigenvector for one of the conjugate pair. Choosing $\lambda = 4i$, we find

$$\begin{aligned} \left[\begin{array}{cc|c} 6 - 4i & -13 & 0 \\ 4 & -6 - 4i & 0 \end{array} \right] &\sim \left[\begin{array}{cc|c} (6 - 4i)(6 + 4i) & -13(6 + 4i) & 0 \\ 2 & -(3 + 2i) & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cc|c} 52 & -26(3 + 2i) & 0 \\ 2 & -(3 + 2i) & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cc|c} 2 & -(3 + 2i) & 0 \\ 0 & 0 & 0 \end{array} \right]. \end{aligned}$$

So, we need $2v_1 = (3 + 2i)v_2$ and can take

$$\vec{v} = \begin{bmatrix} 3 + 2i \\ 2 \end{bmatrix}$$

as a basis eigenvector. Next,

$$\begin{aligned} e^{4it} \begin{bmatrix} 3 + 2i \\ 2 \end{bmatrix} &= (\cos(4t) + i \sin(4t)) \begin{bmatrix} 3 + 2i \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 3 \cos(4t) + 3i \sin(4t) + 2i \cos(4t) - 2 \sin(4t) \\ 2 \cos(4t) + 2i \sin(4t) \end{bmatrix} \\ &= \begin{bmatrix} 3 \cos(4t) - 2 \sin(4t) \\ 2 \cos(4t) \end{bmatrix} + i \begin{bmatrix} 3 \sin(4t) + 2 \cos(4t) \\ 2 \sin(4t) \end{bmatrix}. \end{aligned}$$

This form directly gives us the general solution to our system of differential equations:

$$\vec{x}(t) = c_1 \begin{bmatrix} 3 \cos(4t) - 2 \sin(4t) \\ 2 \cos(4t) \end{bmatrix} + c_2 \begin{bmatrix} 3 \sin(4t) + 2 \cos(4t) \\ 2 \sin(4t) \end{bmatrix}.$$

For the initial data, we need

$$\vec{x}(0) = c_1 \begin{bmatrix} 3 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}.$$

This is equivalent to the equations

$$\begin{aligned} 3c_1 + 2c_2 &= 2 \\ 2c_1 &= -4 \end{aligned}$$

which have solution $c_1 = -2$ and $c_2 = 4$. So, the solution to our initial value problem is

$$\vec{x}(t) = -2 \begin{bmatrix} 3 \cos(4t) - 2 \sin(4t) \\ 2 \cos(4t) \end{bmatrix} + 4 \begin{bmatrix} 3 \sin(4t) + 2 \cos(4t) \\ 2 \sin(4t) \end{bmatrix}.$$

If we write this in component form and simplify, we have

$$\begin{aligned} x(t) &= 2 \cos(4t) + 16 \sin(4t) \\ y(t) &= -4 \cos(4t) + 8 \sin(4t). \quad \blacksquare \end{aligned}$$

Example 9.1.8. Find the general solution to

$$\frac{d\vec{x}}{dt} = \begin{bmatrix} 5 & -16 & 10 \\ 2 & -3 & 0 \\ 0 & 4 & -5 \end{bmatrix} \vec{x}.$$

Answer: The characteristic polynomial for this matrix is

$$\begin{vmatrix} 5 - \lambda & -16 & 10 \\ 2 & -3 - \lambda & 0 \\ 0 & 4 & -5 - \lambda \end{vmatrix} = -\lambda^3 - 3\lambda^2 - 7\lambda - 5 = -(\lambda + 1)(\lambda^2 + 2\lambda + 5).$$

This gives us eigenvalues of $\lambda = -1, -1 \pm 2i$. For $\lambda_1 = -1$, a basis eigenvector is

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

We leave the details to the reader.

For the two complex eigenvalues, we only need find an eigenvector for one of them. For $\lambda_2 = -1 + 2i$, we find

$$\begin{aligned} & \left[\begin{array}{ccc|c} 5 - (-1 + 2i) & -16 & 10 & 0 \\ 2 & -3 - (-1 + 2i) & 0 & 0 \\ 0 & 4 & -5 - (-1 + 2i) & 0 \end{array} \right] \\ &= \left[\begin{array}{ccc|c} 6 - 2i & -16 & 10 & 0 \\ 2 & -2 - 2i & 0 & 0 \\ 0 & 4 & -4 - 2i & 0 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 2 & 0 & -(1 + 3i) & 0 \\ 0 & 2 & -(2 + i) & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]. \end{aligned}$$

For an eigenvector, we need

$$2v_1 = (1 + 3i)v_3$$

$$2v_2 = (2 + i)v_3.$$

This means we can take

$$\vec{v}_2 = \begin{bmatrix} 1 + 3i \\ 2 + i \\ 2 \end{bmatrix}.$$

Next, we find the real and imaginary parts of the solution corresponding to this eigenvalue/eigenvector pair.

$$\begin{aligned}
 e^{(-1+2i)t} \begin{bmatrix} 1+3i \\ 2+i \\ 2 \end{bmatrix} &= e^{-t} e^{2it} \begin{bmatrix} 1+3i \\ 2+i \\ 2 \end{bmatrix} \\
 &= e^{-t} (\cos(2t) + i \sin(2t)) \begin{bmatrix} 1+3i \\ 2+i \\ 2 \end{bmatrix} \\
 &= e^{-t} \begin{bmatrix} \cos(2t) + i \sin(2t) + 3i \cos(2t) - 3 \sin(2t) \\ 2 \cos(2t) + 2i \sin(2t) + i \cos(2t) - \sin(2t) \\ 2 \cos(2t) + 2i \sin(2t) \end{bmatrix} \\
 &= e^{-t} \begin{bmatrix} \cos(2t) - 3 \sin(2t) \\ 2 \cos(2t) - \sin(2t) \\ 2 \cos(2t) \end{bmatrix} + i e^{-t} \begin{bmatrix} \sin(2t) + 3 \cos(2t) \\ 2 \sin(2t) + \cos(2t) \\ 2 \sin(2t) \end{bmatrix}.
 \end{aligned}$$

Taking these two solutions together with the one from $\lambda_1 = -1$ gives us the general solution:

$$\vec{x}(t) = c_1 e^{-t} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} \cos(2t) - 3 \sin(2t) \\ 2 \cos(2t) - \sin(2t) \\ 2 \cos(2t) \end{bmatrix} + c_3 e^{-t} \begin{bmatrix} \sin(2t) + 3 \cos(2t) \\ 2 \sin(2t) + \cos(2t) \\ 2 \sin(2t) \end{bmatrix}. \blacksquare$$

9.1.3 Solutions to Inhomogeneous Systems

The key to solving inhomogeneous systems is the following simple result.

Theorem 9.1.5. *If $\vec{x}_1(t)$ and $\vec{x}_2(t)$ are two solutions to the inhomogeneous system*

$$\vec{x}'(t) = M\vec{x}(t) + \vec{b}(t),$$

then their difference $\vec{x}_1(t) - \vec{x}_2(t)$ is a solution of the associated homogeneous system

$$\vec{x}'(t) = M\vec{x}(t).$$

Proof: Let $\vec{x}_d(t) = \vec{x}_1(t) - \vec{x}_2(t)$. Since $\vec{x}_1(t)$ and $\vec{x}_2(t)$ satisfy the same differential equation, we have

$$\begin{aligned}\vec{x}_d'(t) &= \vec{x}_1'(t) - \vec{x}_2'(t) \\ &= M\vec{x}_1(t) + \vec{b}(t) - (M\vec{x}_2(t) + \vec{b}(t)) \\ &= M\vec{x}_1(t) - M\vec{x}_2(t) \\ &= M(\vec{x}_1(t) - \vec{x}_2(t)) \\ &= M\vec{x}_d(t). \blacksquare\end{aligned}$$

As an immediate consequence, we have the following.

Corollary 9.1.5.1 (General Solutions of Inhomogeneous Systems). *Suppose that $\vec{x}_p(t)$ is **any single solution** to the inhomogeneous system*

$$\vec{x}'(t) = M\vec{x}(t) + \vec{b}(t),$$

*and $\vec{x}_h(t)$ is the **general solution** to the associated homogeneous system*

$$\vec{x}'(t) = M\vec{x}(t).$$

Then the general solution to the inhomogeneous system above is of the form

$$\vec{x}(t) = \vec{x}_p(t) + \vec{x}_h(t).$$

The single solution to the inhomogeneous system, $\vec{x}_p(t)$ is usually referred to as the **particular solution** to the inhomogeneous system. Since we already know how to find the general solution to homogeneous systems (at least for diagonalizable systems), this corollary reduces the problem of solving inhomogeneous systems to finding any *single* solution to the system of differential equations.

Of course, this leaves us the question of how to find a particular solution! The method we will develop is usually called the *Method of Undetermined Coefficients*. The primary advantage of this method is that it is relatively simple to understand. Its major drawback is that it only works for very special kinds of inhomogeneous terms, $\vec{b}(t)$. A more general method, *Variation of Parameters*, will be developed in the exercises at the end of this chapter.

We begin with a simple example.

Example 9.1.9. Find the general solution to the following system of differential equations.

$$\begin{aligned}x' &= x - 2y + 9t \\ y' &= -2x + y\end{aligned}$$

Answer: In matrix form, this system is

$$\vec{x}' = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \vec{x} + \begin{bmatrix} 9t \\ 0 \end{bmatrix}.$$

First, we have to find the general solution to the associated homogeneous problem

$$\vec{x}' = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \vec{x}.$$

Since this is the same system as in Example 9.1.4, we know

$$\vec{x}_h(t) = c_1 e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

To finish the problem, we need to find any particular solution to

$$\vec{x}_p' = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \vec{x}_p + \begin{bmatrix} 9t \\ 0 \end{bmatrix}.$$

Since the inhomogeneous term is linear in t , we can make a reasonable guess about the form of the particular solution:

$$\vec{x}_p(t) = \begin{bmatrix} At + B \\ Ct + D \end{bmatrix}.$$

In other words, we guess that *both components* of the particular solution are linear functions of t . Our job is to figure out the coefficients A, B, C , and D that will make this work. This is the origin of the name “Method of Undetermined Coefficients.”

First,

$$\vec{x}_p'(t) = \begin{bmatrix} A \\ C \end{bmatrix}.$$

The right-hand-side of the differential equation gives

$$\begin{aligned}
 \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \vec{x}_p(t) + \begin{bmatrix} 9t \\ 0 \end{bmatrix} &= \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} At + B \\ Ct + D \end{bmatrix} + \begin{bmatrix} 9t \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} 1(At + B) - 2(Ct + D) \\ -2(At + B) + 1(Ct + D) \end{bmatrix} + \begin{bmatrix} 9t \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} (A - 2C)t + (B - 2D) \\ (-2A + C)t + (-2B + D) \end{bmatrix} + \begin{bmatrix} 9t \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} (A - 2C + 9)t + (B - 2D) \\ (-2A + C)t + (-2B + D) \end{bmatrix}.
 \end{aligned}$$

So, the differential equation requires

$$\begin{aligned}
 \vec{x}_p' &= \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \vec{x}_p + \begin{bmatrix} 9t \\ 0 \end{bmatrix} \\
 \begin{bmatrix} A \\ C \end{bmatrix} &= \begin{bmatrix} (A - 2C + 9)t + (B - 2D) \\ (-2A + C)t + (-2B + D) \end{bmatrix}.
 \end{aligned}$$

To make this work, we need the following to be true:

$$\begin{aligned}
 0 &= A - 2C + 9 \\
 0 &= -2A + C \\
 A &= B - 2D \\
 C &= -2B + D.
 \end{aligned}$$

The first two equations are due to the fact that the left-hand-side of the differential equation has no t -terms whatsoever (and so the t -terms on the right need coefficients of 0). The last two equations force the constant terms on the left and right sides to agree. Solving the first two gives $A = 3$ and $C = 6$. Substituting these into the last two equations gives

$$\begin{aligned}
 3 &= B - 2D \\
 6 &= -2B + D
 \end{aligned}$$

which has solution $B = -5$ and $D = -4$. So, one particular solution is given by

$$\vec{x}_p(t) = \begin{bmatrix} 3t - 5 \\ 6t - 4 \end{bmatrix}.$$

So, the general solution to our original problem is given by

$$\vec{x}(t) = \vec{x}_p(t) + \vec{x}_h(t) = c_1 e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 3t - 5 \\ 6t - 4 \end{bmatrix}.$$

Writing this in component form gives

$$\begin{aligned} x(t) &= c_1 e^{-t} + c_2 e^{3t} + 3t - 5 \\ y(t) &= c_1 e^{-t} - c_2 e^{3t} + 6t - 4. \blacksquare \end{aligned}$$

Essentially, the Method of Undetermined Coefficients is a viable option whenever the inhomogeneous term is composed of the same type of functions that typically show up as solutions of homogeneous problems.

Method of Undetermined Coefficients²

Table 1: Inhomogeneous Term Unrelated to Homogeneous Solution	
Component of Inhomogeneous Term	Add to EACH component of $\vec{x}_p$
$a_n t^n + \cdots + a_1 t + a_0$	$A_n t^n + \cdots + A_1 t + A_0$
$a e^{bt}$	$A e^{bt}$
$a \sin(bt)$ or $c \cos(bt)$	$A \sin(bt) + B \cos(bt)$

Table 2: Inhomogeneous Term Related to Homogeneous Solution	
Component of Inhomogeneous Term	Add to EACH component of $\vec{x}_p$
$a_n t^n + \cdots + a_1 t + a_0$	$A_{n+1} t^{n+1} + \cdots + A_1 t + A_0$
$a e^{bt}$	$(A_1 t + A_0) e^{bt}$
$a \sin(bt)$ or $c \cos(bt)$	$(A_1 t + A_0) \sin(bt) + (B_1 t + B_0) \cos(bt)$

What this table tells us is that the appropriate guess *for each component of the particular solution* is the most general function like the given inhomogeneous term. If the inhomogeneous term is part of the homogeneous solution, we may have to add an extra power of t to have a particular solution that will work correctly.

Example 9.1.10. Find the solution to the following initial value problem. Find the general solution to

$$\begin{aligned} x' &= x - 3y + 3e^{-2t} \\ y' &= 6x - 5y - 6e^{-2t} \\ x(0) &= 0 \\ y(0) &= 0. \end{aligned}$$

²We are still assuming that the system is diagonalizable. If an eigenvalue is deficient, then Table 2 would need to be altered.

Answer: In matrix form, we have

$$\vec{x}' = \begin{bmatrix} 1 & -3 \\ 6 & -5 \end{bmatrix} \vec{x} + \begin{bmatrix} 3e^{-2t} \\ -6e^{-2t} \end{bmatrix}.$$

So we first need to consider the homogeneous system

$$\vec{x}' = \begin{bmatrix} 1 & -3 \\ 6 & -5 \end{bmatrix} \vec{x}.$$

Since this is identical to the system in Example 9.1.6, we have

$$\vec{x}_h(t) = c_1 e^{-2t} \begin{bmatrix} \cos(3t) - \sin(3t) \\ 2 \cos(3t) \end{bmatrix} + c_2 e^{-2t} \begin{bmatrix} \sin(3t) + \cos(3t) \\ 2 \sin(3t) \end{bmatrix}.$$

For our particular solution, notice that the homogeneous solution involves functions like $e^{-2t} \cos(3t)$ and $e^{-2t} \sin(3t)$. Since our inhomogeneous term only involves e^{-2t} , we can simply take

$$\vec{x}_p(t) = \begin{bmatrix} Ae^{-2t} \\ Be^{-2t} \end{bmatrix}.$$

Since

$$\begin{aligned} \vec{x}_p'(t) &= \begin{bmatrix} -2Ae^{-2t} \\ -2Be^{-2t} \end{bmatrix} \\ \begin{bmatrix} 1 & -3 \\ 6 & -5 \end{bmatrix} \vec{x}_p + \begin{bmatrix} 3e^{-2t} \\ -6e^{-2t} \end{bmatrix} &= \begin{bmatrix} 1 & -3 \\ 6 & -5 \end{bmatrix} \begin{bmatrix} Ae^{-2t} \\ Be^{-2t} \end{bmatrix} + \begin{bmatrix} 3e^{-2t} \\ -6e^{-2t} \end{bmatrix} \\ &= \begin{bmatrix} Ae^{-2t} - 3Be^{-2t} \\ 6Ae^{-2t} - 5Be^{-2t} \end{bmatrix} + \begin{bmatrix} 3e^{-2t} \\ -6e^{-2t} \end{bmatrix} \\ &= \begin{bmatrix} (A - 3B + 3)e^{-2t} \\ (6A - 5B - 6)e^{-2t} \end{bmatrix}, \end{aligned}$$

we need

$$\begin{bmatrix} -2Ae^{-2t} \\ -2Be^{-2t} \end{bmatrix} = \begin{bmatrix} (A - 3B + 3)e^{-2t} \\ (6A - 5B - 6)e^{-2t} \end{bmatrix}.$$

This gives us the equations

$$\begin{aligned} -2A &= A - 3B + 3 \\ -2B &= 6A - 5B - 6 \end{aligned}$$

which have solution $A = 3$ and $B = 4$. This means the general solution to our system of differential equations is

$$\begin{aligned}\vec{x}(t) &= \vec{x}_h(t) + \vec{x}_p(t) \\ &= c_1 e^{-2t} \begin{bmatrix} \cos(3t) - \sin(3t) \\ 2 \cos(3t) \end{bmatrix} + c_2 e^{-2t} \begin{bmatrix} \sin(3t) + \cos(3t) \\ 2 \sin(3t) \end{bmatrix} + \begin{bmatrix} 3e^{-2t} \\ 4e^{-2t} \end{bmatrix}.\end{aligned}$$

Now that we have the general solution, we can find the solution to the initial value problem. INITIAL DATA SHOULD ONLY BE CONSIDERED AFTER YOU HAVE THE GENERAL SOLUTION.

$$\vec{x}(0) = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

So, we need

$$\begin{aligned}c_1 + c_2 + 3 &= 0 \\ 2c_1 + 4 &= 0\end{aligned}$$

which has solution $c_1 = -2$ and $c_2 = -1$. So, the solution to our initial value problem is

$$\vec{x}(t) = -2e^{-2t} \begin{bmatrix} \cos(3t) - \sin(3t) \\ 2 \cos(3t) \end{bmatrix} - e^{-2t} \begin{bmatrix} \sin(3t) + \cos(3t) \\ 2 \sin(3t) \end{bmatrix} + \begin{bmatrix} 3e^{-2t} \\ 4e^{-2t} \end{bmatrix}$$

or in component form

$$\begin{aligned}x(t) &= -3e^{-2t} \cos(3t) + e^{-2t} \sin(3t) + 3e^{-2t} \\ y(t) &= -4e^{-2t} \cos(3t) - 2e^{-2t} \sin(3t) + 4e^{-2t}. \blacksquare\end{aligned}$$

Example 9.1.11. Find the general solution to

$$\vec{x}' = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 24e^{3t} \end{bmatrix}.$$

Answer: Since the homogeneous system is again like Example 9.1.4, we have

$$\vec{x}_h(t) = c_1 e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Since e^{3t} is part of the homogeneous solution and 3 has multiplicity 1 for the matrix, we have to guess

$$\vec{x}_p(t) = \begin{bmatrix} (At + B)e^{3t} \\ (Ct + D)e^{3t} \end{bmatrix}.$$

$$\vec{x}_p'(t) = \begin{bmatrix} Ae^{3t} + (3At + 3B)e^{3t} \\ Ce^{3t} + (3Ct + 3D)e^{3t} \end{bmatrix} = \begin{bmatrix} 3Ate^{3t} + (A + 3B)e^{3t} \\ 3Cte^{3t} + (C + 3D)e^{3t} \end{bmatrix}$$

$$\begin{aligned} \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 24e^{3t} \end{bmatrix} &= \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} (At + B)e^{3t} \\ (Ct + D)e^{3t} \end{bmatrix} + \begin{bmatrix} 0 \\ 24e^{3t} \end{bmatrix} \\ &= \begin{bmatrix} (At + B)e^{3t} - 2(Ct + D)e^{3t} \\ -2(At + B)e^{3t} + (Ct + D)e^{3t} + 24e^{3t} \end{bmatrix} \\ &= \begin{bmatrix} (A - 2C)te^{3t} + (B - 2D)e^{3t} \\ (-2A + C)te^{3t} + (-2B + D + 24)e^{3t} \end{bmatrix} \end{aligned}$$

Equating like coefficients gives

$$\begin{aligned} 3A &= A - 2C \\ 3C &= -2A + C \\ A + 3B &= B - 2D \\ C + 3D &= -2B + D + 24. \end{aligned}$$

Notice that *both* of the first two equations tell us that $A = -C$. This makes it seem as though there is some freedom in the choice of A (or C). However, using this information in the last two equations gives us

$$\begin{aligned} A + 3B &= B - 2D \\ -A + 3D &= -2B + D + 24 \end{aligned}$$

which simplify to

$$\begin{aligned} 2B + 2D &= -A \\ 2B + 2D &= A + 24 \end{aligned}$$

So, we must have $-A = A + 24$ which means $A = -12$ (and so $C = 12$). Finally, we have to choose B and D to satisfy $2B + 2D = 12$. We genuinely do

have freedom to pick B or D anyway we wish,³ and so we choose $B = D = 3$. ($B = 0$ and $D = 6$ or any other combination satisfying $B + D = 6$ would do just as well). So, the general solution to our problem is

$$\vec{x}(t) = c_1 e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} (-12t + 3)e^{3t} \\ (12t + 3)e^{3t} \end{bmatrix}.$$

The general solution shows us why there is some freedom in the choice of particular solution. Note that the second term of the homogeneous solution allows us to add any term of the form

$$ce^{3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

to the particular solution without penalty. No matter what multiple we add, we end up with

$$\vec{x}_p(t) = \begin{bmatrix} (-12t + 3 + c)e^{3t} \\ (12t + 3 - c)e^{3t} \end{bmatrix},$$

and the new B and D coefficients still sum up to 6:

$$(3 + c) + (3 - c) = 3 + 3 = 6. \blacksquare$$

9.2 Application: Multiple Compartment Mixing Problems

In this section, we examine mixing problems similar to those we studied in Section 4.9. Instead of a single tank, we will now consider several tanks connected together. Just as with a single tank, the key idea is to consider the total amount of solute moving into and out of each tank per unit of time.

³A more systematic way to see this is to row reduce the augmented matrix corresponding to this system of equations.

$$\left[\begin{array}{cccc|c} 2 & 0 & 2 & 0 & 0 \\ 2 & 0 & 2 & 0 & 0 \\ 1 & 2 & 0 & 2 & 0 \\ 0 & 2 & 1 & 2 & 24 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -12 \\ 0 & 1 & 0 & 1 & 6 \\ 0 & 0 & 1 & 0 & 12 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

The row of all zeros tells us there is a free variable!

Throughout this section, we will assume that the solutions in the various tanks remain well-mixed.

When we had a single tank, we could easily handle cases where the volume of fluid in the tank was increasing or decreasing. That is no longer the case when there are multiple tanks. While the resulting system would still be linear, the coefficients of the matrix describing the system would be functions of t , and solving such systems is substantially more difficult than systems with constant coefficients.

Example 9.2.1. Predict the concentration of salt as a function of t in the following tank system. The tank on the left starts with 50 L at a concentration of 20 g/L. The tank on the right starts with 50 L at a concentration of 40 g/L.

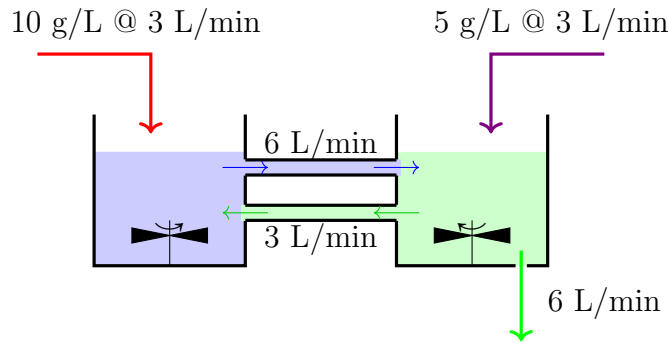


Figure 9.2.1: Tank System for Example 9.2.1

Answer: Let $C_\ell(t)$ be the concentration in the tank on the left and $C_r(t)$ be the concentration of salt in the tank on the right. The corresponding amounts of salt in each of the tanks are $A_\ell(t)$ and $A_r(t)$. Since for both tanks the total volume of fluid in and out balances, both tanks remain at their original 50 L for all time. This means

$$C_\ell(t) = \frac{A_\ell(t)}{50},$$

$$C_r(t) = \frac{A_r(t)}{50}.$$

Analyzing the flows in and out of the tank on the left gives

$$\frac{dA_\ell}{dt} = 30 - 6 \left(\frac{A_\ell}{50} \right) + 3 \left(\frac{A_r}{50} \right).$$

Similarly for the tank on the right,

$$\frac{dA_r}{dt} = 15 + 6 \left(\frac{A_\ell}{50} \right) - 9 \left(\frac{A_r}{50} \right).$$

Notice that the two streams flowing out of the right tank are combined in the differential equation for A_r . As far as the tank on the right is concerned, it is losing 9 L/min of its contents. The fact that 3 L/min is flowing into the tank on the left is handled in the differential equation for A_ℓ . For initial data, we have

$$\begin{aligned} A_\ell(0) &= 1000, \\ A_r(0) &= 2000. \end{aligned}$$

This gives us the following inhomogeneous system.

$$\begin{aligned} \begin{bmatrix} A'_\ell \\ A'_r \end{bmatrix} &= \begin{bmatrix} -3/25 & 3/50 \\ 3/25 & -9/50 \end{bmatrix} \begin{bmatrix} A_\ell \\ A_r \end{bmatrix} + \begin{bmatrix} 30 \\ 15 \end{bmatrix} \\ \begin{bmatrix} A_\ell(0) \\ A_r(0) \end{bmatrix} &= \begin{bmatrix} 1000 \\ 2000 \end{bmatrix} \end{aligned}$$

The homogeneous part of the differential equation has eigenvalue/eigenvector pairs

$$\begin{aligned} \lambda_1 &= -\frac{6}{25}, & \vec{v}_1 &= \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \\ \lambda_2 &= -\frac{3}{50}, & \vec{v}_2 &= \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \end{aligned}$$

So, the homogeneous solution is

$$\vec{A}_h(t) = c_1 e^{-6t/25} \begin{bmatrix} 1 \\ -2 \end{bmatrix} + c_2 e^{-3t/50} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

For the particular solution, we can take

$$\vec{A}_p(t) = \begin{bmatrix} a \\ b \end{bmatrix}.$$

Substituting this into the differential equation gives us

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -3/25 & 3/50 \\ 3/25 & -9/50 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} 30 \\ 15 \end{bmatrix}.$$

This gives us the system

$$-6a + 3b + 1500 = 0$$

$$6a - 9b + 750 = 0$$

which has solution $a = 437.5$ and $b = 375$. This means the general solution to our system is

$$\begin{bmatrix} A_\ell(t) \\ A_r(t) \end{bmatrix} = c_1 e^{-6t/25} \begin{bmatrix} 1 \\ -2 \end{bmatrix} + c_2 e^{-3t/50} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 437.5 \\ 375 \end{bmatrix}.$$

For the initial data, we need

$$\begin{bmatrix} 1000 \\ 2000 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 437.5 \\ 375 \end{bmatrix}.$$

This system has solution

$$c_1 = -\frac{2125}{6} \quad \text{and} \quad c_2 = \frac{2750}{3}.$$

This gives

$$\begin{aligned} A_\ell(t) &= -\frac{2125}{6} e^{-6t/25} + \frac{2750}{3} e^{-3t/50} + 437.5, \\ A_r(t) &= \frac{2125}{3} e^{-6t/25} + \frac{2750}{3} e^{-3t/50} + 375. \end{aligned}$$

For the concentrations in each tank, we find

$$\begin{aligned} C_\ell(t) &= \frac{A_\ell(t)}{50} = -\frac{85}{12} e^{-6t/25} + \frac{55}{3} e^{-3t/50} + \frac{35}{4}, \\ C_r(t) &= \frac{A_r(t)}{50} = \frac{85}{6} e^{-6t/25} + \frac{55}{3} e^{-3t/50} + \frac{15}{2}. \end{aligned}$$

Notice that the limiting concentration in the left tank is

$$\lim_{t \rightarrow \infty} C_\ell(t) = \frac{35}{4} \text{ g/L},$$

while in the right tank

$$\lim_{t \rightarrow \infty} C_r(t) = \frac{15}{2} \text{ g/L.}$$

So, after this mixing process has gone on for a relatively long period of time, the concentration of salt in the left tank should be close to 8.75 grams per liter. In the right tank, we expect the concentration to be roughly 7.5 grams per liter. ■

Example 9.2.2. Consider the system of tanks shown below. The outer two tanks (Tanks 1 and 3) both have 35 gallons of a sugar solution while the tank in the middle (Tank 2) contains 40 gallons. All flow rates below are in gallons per minute. Do the concentrations in the tanks limit to a steady value? If so, what will the concentrations limit to in the long term?

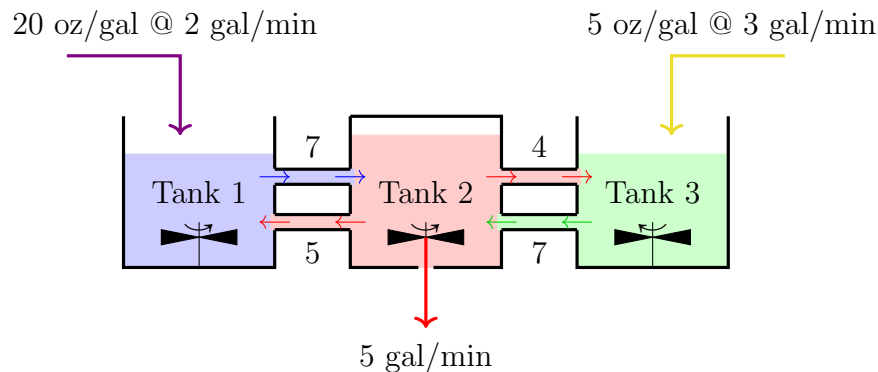


Figure 9.2.2: Tank System for Example 9.2.2

Answer: The concentrations will be labeled as $C_1(t)$, $C_2(t)$, and $C_3(t)$ for Tanks 1, 2 and 3, respectively. Since the volume of fluid flowing into and out of each tank is perfectly balanced, the total volume of fluid in each tank is constant. This means the total amounts of sugar in each tank ($A_1(t)$, $A_2(t)$,

and $A_3(t)$ respectively) are related to the concentrations by

$$\begin{aligned}C_1(t) &= \frac{A_1(t)}{35}, \\C_2(t) &= \frac{A_2(t)}{40}, \\C_3(t) &= \frac{A_3(t)}{35}.\end{aligned}$$

The system of differential equations describing this system is

$$\begin{bmatrix} A_1' \\ A_2' \\ A_3' \end{bmatrix} = \begin{bmatrix} -1/5 & 1/8 & 0 \\ 1/5 & -7/20 & 1/5 \\ 0 & 1/10 & -1/5 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix} + \begin{bmatrix} 40 \\ 0 \\ 15 \end{bmatrix}.$$

The characteristic polynomial for this matrix is (after much work)

$$-\lambda^3 - \frac{3\lambda^2}{4} - \frac{27\lambda}{200} - \frac{1}{200} = -\frac{1}{200}(2\lambda + 1)(5\lambda + 1)(20\lambda + 1).$$

This means the eigenvalues are all negative. As such, the homogeneous part of the solution decays to zero as $t \rightarrow \infty$.

Since we only want to know the long-term behavior of the system, we only need to find the particular solution of the form

$$\vec{A}_p(t) = \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

The particular solution must satisfy

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/5 & 1/8 & 0 \\ 1/5 & -7/20 & 1/5 \\ 0 & 1/10 & -1/5 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} + \begin{bmatrix} 40 \\ 0 \\ 15 \end{bmatrix}.$$

This system of equations has solution $a = 475$, $b = 440$, and $c = 295$. This means that the concentrations in the tank limit to

$$\begin{aligned}\lim_{t \rightarrow \infty} C_1(t) &= \frac{475}{35} = \frac{95}{7} \approx 13.571 \text{ oz/gal}, \\ \lim_{t \rightarrow \infty} C_2(t) &= \frac{440}{40} = 11 \text{ oz/gal}, \\ \lim_{t \rightarrow \infty} C_3(t) &= \frac{295}{35} = \frac{59}{7} \approx 8.429 \text{ oz/gal.} \blacksquare\end{aligned}$$

9.3 Application: Spring–Mass Systems

Many phenomena in nature can be described by higher order differential equations – differential equations involving derivatives of the unknown function beyond the first derivative. For example, Newton’s Second Law tells us that the acceleration of an object is proportional to the total force acting on the object. Since acceleration is the second derivative of position, we naturally get a second order differential equation for the position of the object.

Consider a block of mass m connected to a spring sliding horizontally on a surface. We set up a coordinate system where $x = x(t)$ measures how far the center of the block has moved from its equilibrium position – the position where the mass would be able to sit at rest for all time. According to Hooke’s Law, the force exerted by a spring is proportional to how much it has been stretched⁴: $F_{\text{spring}} = -kx$. The positive parameter k is called the spring constant and is determined by the particulars of how the spring was made. The negative sign is crucial as it indicates that the force from a spring is opposite to the direction it has been stretched (or compressed).

The other major force acting on the block is friction with the surface. Friction is a complex phenomenon, and there are many ways to model it that are useful in various situations. We will use the simplest model; the force due to friction is proportional to and opposite the velocity of the object: $F_{\text{friction}} = -bx'$. The positive coefficient of friction, b , depends on the materials the block and table are made of. Putting these two forces together gives us $mx'' = -kx - bx'$.

It is very easy to turn a higher order differential equation into a system of first order differential equations. The key idea is to give names to all but the highest derivative appearing in the equation. Since our spring–mass system is second order, we need to give a name to the first derivative: $x'(t) = v(t)$. Using v as a second unknown function gives us a 2×2 linear system:

$$\begin{aligned}x' &= v \\v' &= x'' = -\frac{k}{m}x - \frac{b}{m}v.\end{aligned}$$

Notice that we replaced x' by v in the force due to friction! If we write this

⁴Hooke’s Law is typically a reasonable model for the force exerted by a spring so long as the spring is not overly stretched or compressed.

in vector form, we find

$$\begin{bmatrix} x' \\ v' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}.$$

This allows us to use the techniques from the previous sections.

Example 9.3.1. A block of mass $m = 2$ kg is attached to a spring of strength $k = 3$ N/m and allowed to slide horizontally on a table. The coefficient of linear friction between the table and the mass is $b = 7$ N·s/m. If the mass starts from rest half a meter to the right of equilibrium, predict the motion of the mass for all time (measured in seconds).

Answer: If we let $x = x(t)$ be the position of the mass (measured from equilibrium), we have the differential equation

$$2x'' = -3x - 7x'.$$

For initial data, we have

$$\begin{aligned} x(0) &= \frac{1}{2} \\ v(0) &= 0. \end{aligned}$$

Using $v(t) = x'(t)$, we have the system

$$\begin{aligned} x' &= v \\ v' &= -\frac{3}{2}x - \frac{7}{2}v \end{aligned}$$

The eigenvalues of this matrix are give by

$$\begin{aligned} \begin{vmatrix} -\lambda & 1 \\ -\frac{3}{2} & -\frac{7}{2} - \lambda \end{vmatrix} &= -\lambda \left(-\frac{7}{2} - \lambda \right) + \frac{3}{2} \\ &= \lambda^2 + \frac{7}{2}\lambda + \frac{3}{2} \\ &= \left(\lambda + \frac{1}{2} \right) (\lambda + 3) = 0. \end{aligned}$$

So, the eigenvalues are $\lambda = -\frac{1}{2}, -3$.

The eigenvectors for $\lambda_1 = -\frac{1}{2}$ are given by

$$\left[\begin{array}{cc|c} \frac{1}{2} & 1 & 0 \\ -\frac{3}{2} & -3 & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right].$$

So we can take

$$\vec{v}_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

as a basis eigenvector for the first eigenvalue. For $\lambda_2 = -3$, we find

$$\left[\begin{array}{cc|c} 3 & 1 & 0 \\ -\frac{3}{2} & -\frac{1}{2} & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 3 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right].$$

So we can take

$$\vec{v}_2 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

as a basis eigenvector for the second eigenvalue. This means the general solution to our differential equation is

$$\begin{bmatrix} x(t) \\ v(t) \end{bmatrix} = c_1 e^{-t/2} \begin{bmatrix} 2 \\ -1 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} 1 \\ -3 \end{bmatrix}.$$

Next, we fit the initial data.

$$\begin{bmatrix} x(0) \\ v(0) \end{bmatrix} = c_1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -3 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}$$

This gives us the system

$$\begin{aligned} 2c_1 + c_2 &= \frac{1}{2} \\ -c_1 - 3c_2 &= 0 \end{aligned}$$

which has solution $c_1 = 3/10$ and $c_2 = -1/10$. Substituting this into the general solution gives us

$$\begin{aligned} x(t) &= \frac{3}{5} e^{-t/2} - \frac{1}{10} e^{-3t} \\ v(t) &= -\frac{3}{10} e^{-t/2} + \frac{3}{10} e^{-3t}. \end{aligned}$$

Notice that $v(t)$ is just the derivative of $x(t)$ as it must be! ■

If there is an external force, $F_{ext}(t)$, separate from the spring and friction acting on the mass, then our second order differential equation becomes

$$mx'' = -kx - bx' + F_{ext}.$$

This gives us an inhomogeneous system:

$$\begin{bmatrix} x'(t) \\ v'(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \begin{bmatrix} x(t) \\ v(t) \end{bmatrix} + \begin{bmatrix} 0 \\ F_{ext}(t) \end{bmatrix}.$$

Example 9.3.2. A mass of weight 32 lbs is suspended vertically from a spring of strength $k = 13$ lbs/ft. Suppose that the coefficient of linear friction for the system is $b = 4$ lbs · s / ft. Find the general solution to the associated differential equation.

Answer: The first thing to note is that the pound (lb) is **not** a measurement of mass! In British Imperial units, the unit of mass is called the *slug*. The relation between the weight, w , of an object and its mass, m , is

$$w = mg$$

where $g = 32$ ft/s is the acceleration due to gravity. Since the weight is suspended vertically, gravity is acting on the object with a constant force:

$$F_{ext} = -mg = -32 \text{ lbs.}$$

The negative sign is due to the fact that force of gravity acts downwards. So, our weight has mass $m = 1$ slug. So, the motion of the weight satisfies

$$x'' = -13x - 4x' - 32.$$

Notice that $x = x(t)$ measures the displacement of the weight from the equilibrium position of the system *in the absence of any external force*. In particular $x = 0$ for our system would be the position that the weight would naturally sit if it were set up horizontally (and gravity was not a factor). The vector form of this differential equation is

$$\begin{bmatrix} x'(t) \\ v'(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -13 & -4 \end{bmatrix} \begin{bmatrix} x(t) \\ v(t) \end{bmatrix} + \begin{bmatrix} 0 \\ -32 \end{bmatrix}.$$

The matrix has eigenvalues $\lambda = -2 \pm 3i$. A basis eigenvector associated with $\lambda = -2 + 3i$ is

$$\vec{v} = \begin{bmatrix} -2 - 3i \\ 13 \end{bmatrix}.$$

Using the methods from the previous section, we find

$$\begin{aligned} e^{(-2+3i)t} \begin{bmatrix} -2 - 3i \\ 13 \end{bmatrix} \\ = e^{-2t} \begin{bmatrix} -2 \cos(3t) + 3 \sin(3t) \\ 13 \cos(3t) \end{bmatrix} + ie^{-2t} \begin{bmatrix} -2 \sin(3t) - 3 \cos(3t) \\ 13 \sin(3t) \end{bmatrix}. \end{aligned}$$

So, the homogeneous solution to our system is

$$\vec{x}_h(t) = c_1 e^{-2t} \begin{bmatrix} -2 \cos(3t) + 3 \sin(3t) \\ 13 \cos(3t) \end{bmatrix} + c_2 e^{-2t} \begin{bmatrix} -2 \sin(3t) - 3 \cos(3t) \\ 13 \sin(3t) \end{bmatrix}.$$

For a particular solution, we can try

$$\vec{x}_p(t) = \begin{bmatrix} A \\ B \end{bmatrix}.$$

Substituting this into the differential equation gives

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -13 & -4 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} + \begin{bmatrix} 0 \\ -32 \end{bmatrix}.$$

This gives us $A = -32/13$ and $B = 0$. So, the general solution to our differential equation is

$$\begin{aligned} \begin{bmatrix} x(t) \\ v(t) \end{bmatrix} &= c_1 e^{-2t} \begin{bmatrix} -2 \cos(3t) + 3 \sin(3t) \\ 13 \cos(3t) \end{bmatrix} \\ &+ c_2 e^{-2t} \begin{bmatrix} -2 \sin(3t) - 3 \cos(3t) \\ 13 \sin(3t) \end{bmatrix} + \begin{bmatrix} -32/13 \\ 0 \end{bmatrix}. \end{aligned}$$

If we examine the position, we find

$$x(t) = c_1 e^{-2t} (-2 \cos(3t) + 3 \sin(3t)) + c_2 e^{-2t} (-2 \sin(3t) - 3 \cos(3t)) - \frac{32}{13}.$$

We can rearrange this into the form

$$x(t) = d_1 e^{-2t} \cos(3t) + d_2 e^{-2t} \sin(3t) - \frac{32}{13}$$

where d_1 and d_2 are combinations of c_1 and c_2 . Note that the only effect gravity has is to shift the natural equilibrium point of the spring–mass system downward 32/13 feet. What this means is that there is no essential difference between the motion of a mass hanging vertically from a spring and the motion of a mass connected to a spring that is sliding horizontally. After we shift the equilibrium downward, the motion is completely identical!⁵ ■

If we compute the characteristic polynomial for

$$\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix}$$

then the eigenvalues are given by the equation

$$\lambda^2 + \frac{b}{m}\lambda + \frac{k}{m} = 0.$$

If we multiply this equation by m , we get the equation

$$m\lambda^2 + b\lambda + k = 0$$

which has eigenvalues

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4mk}}{2m}.$$

The quantity under the square root (known as the *discriminant*) controls the type of solutions.

If $b^2 - 4mk > 0$, then both eigenvalues are real and negative. In this case, the system is said to be **over-damped**, and solutions have the form

$$x(t) = d_1 e^{-r_1 t} + d_2 e^{-r_2 t}.$$

Notice that an over-damped spring–mass system does not oscillate. The damping parameter is large enough that the mass creeps back toward equilibrium without ever crossing it.

⁵While this is mathematically accurate, there may be practical differences. Remember that Hooke's Law will stop being a good model for the spring force when the spring is stretched or compressed by a large amount. Since the force of gravity stretches the spring by some amount, this might be a concern. Also, air resistance when the mass is suspended vertically is likely to be very different than sliding friction when the mass is moving horizontally!

If $b^2 - 4mk < 0$, then the eigenvalues are complex with negative real part:

$$\lambda = -\frac{b}{2m} \pm \mu i$$

where $\mu = \sqrt{4mk - b^2}/2m$. In this case, the spring-mass system is said to be **under-damped**, and the solutions are of the form

$$x(t) = d_1 e^{-bt/2m} \cos(\mu t) + d_2 e^{-bt/2m} \sin(\mu t).$$

For an under-damped spring-mass system, the damping is small enough that the mass can oscillate while it tries to return the equilibrium.

In the event that $b^2 - 4mk = 0$, we have a repeated, negative eigenvalue. In this unlikely case, we say the system is **critically damped**. It turns out that solutions in this case are of the form

$$x(t) = d_1 e^{-bt/2m} + d_2 t e^{-bt/2m},$$

and the spring-mass system does not oscillate. In all three cases, the mass tries to return to equilibrium exponentially fast. Only in the under-damped case does the system exhibit any oscillatory behavior.

9.4 Analysis of 2×2 Homogeneous Systems

If we are only interested in long term behavior, we do not need to compute the full solution to gain a qualitative understanding of solutions to linear homogeneous systems. In fact, the eigenvalues alone can tell us quite a lot about the nature of the solutions. For instance, suppose a 2×2 linear system has eigenvalues $\lambda_1 = -2$ and $\lambda_2 = -1$. The general solution would then look like

$$\vec{x}(t) = c_1 e^{-2t} \vec{v}_1 + c_2 e^{-t} \vec{v}_2$$

where $\vec{v}_1$ and $\vec{v}_2$ are the corresponding eigenvectors. The solution in component form would be

$$\begin{aligned} x(t) &= A e^{-2t} + B e^{-t} \\ y(t) &= C e^{-2t} + D e^{-t} \end{aligned}$$

for some coefficients A, B, C , and D which would be determined by the initial data. Regardless of what these coefficients are (i.e. regardless of the initial data), we necessarily have

$$\lim_{t \rightarrow \infty} x(t) = \lim_{t \rightarrow \infty} y(t) = 0.$$

To make progress in our understanding of 2×2 homogeneous systems, we need to recall some ideas from Chapter 5. For a linear system of the form

$$\begin{aligned} \frac{dx}{dt} &= ax + by \\ \frac{dy}{dt} &= cx + dy, \end{aligned}$$

the x - and y -nullclines are the lines $ax + by = 0$ and $cx + dy = 0$, respectively, both of which pass through the origin. Assuming these two lines are distinct, the origin will be the only equilibrium for the system. In Chapter 5, we classified equilibria based on vector plots. Without technology, vector plots are difficult to draw well, and even with a machine-drawn vector plot it can often be difficult to determine the behavior. Fortunately, we can easily predict the long term behavior of solutions to 2×2 systems from the eigenvalues (as long as they are distinct). For 3×3 and larger systems, the classification becomes a lot more intricate (as we will see).

Theorem 9.4.1. *Suppose that a 2×2 system $\vec{x}' = M\vec{x}$ has distinct eigenvalues λ_1 and λ_2 .*

1. *If **both eigenvalues are negative**, then solutions approach $(0, 0)$ as $t \rightarrow \infty$, and we say the origin is an **asymptotically stable node**.*

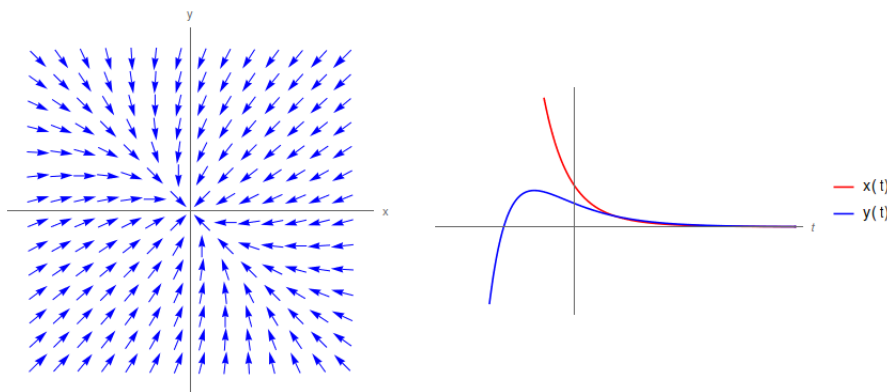


Figure 9.4.1: Asymptotically Stable Node

2. If **both eigenvalues are positive**, then solutions diverge to $\pm\infty$ as $t \rightarrow \infty$, and we say the origin is an **unstable node**.

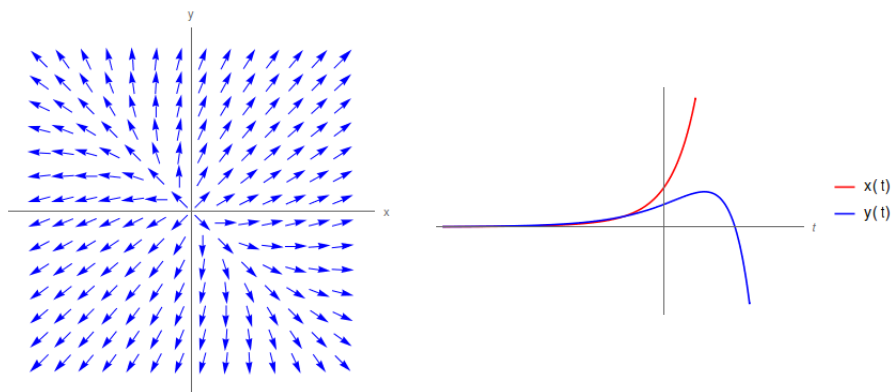


Figure 9.4.2: Unstable Node

3. If **one eigenvalue is positive and the other negative**, then almost all solutions diverge to $\pm\infty$ as $t \rightarrow \pm\infty$, and we say the origin is an **unstable saddle point**.

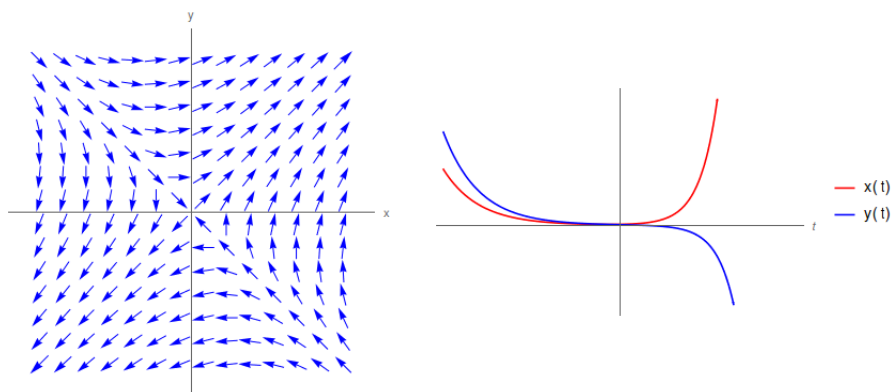


Figure 9.4.3: Saddle Point

4. If the *eigenvalues are complex with negative real part*, then solutions approach $(0,0)$ as $t \rightarrow \infty$, and we say the origin is an **asymptotically stable spiral point**.

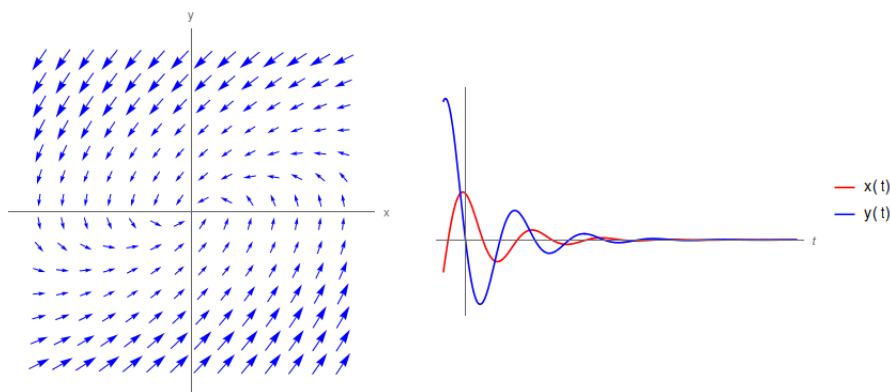


Figure 9.4.4: Asymptotically Stable Spiral

5. If the *eigenvalues are purely imaginary*, then solutions starting near $(0,0)$ will stay near the origin for all time, and we say the origin is a **stable center**.

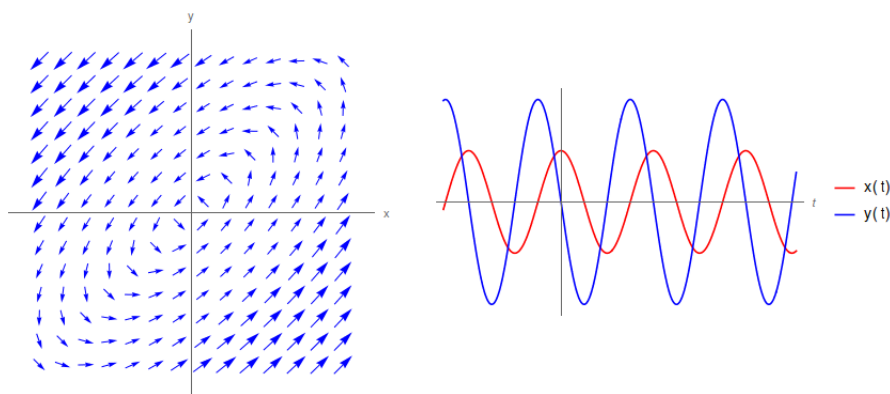


Figure 9.4.5: Stable Center

6. If the eigenvalues are complex with positive real part, then solutions diverge to $\pm\infty$ as $t \rightarrow \infty$ and we say the origin is an **unstable spiral point**.

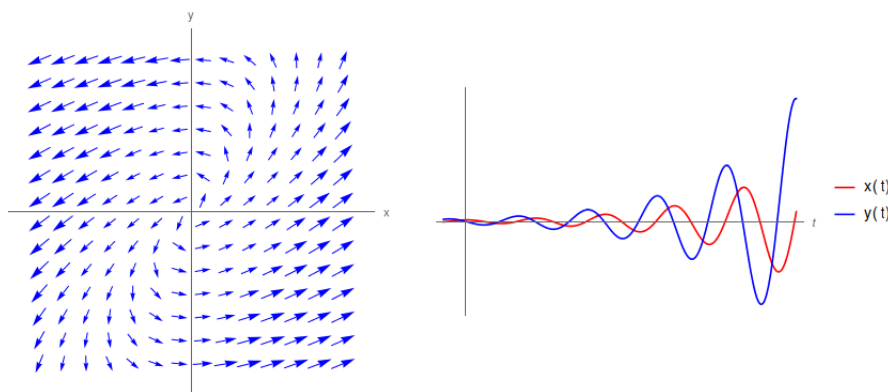


Figure 9.4.6: Unstable Spiral

Proof: The proof follows from the form of solutions given the eigenvalues. If the eigenvalues are real, then

$$\begin{aligned}x(t) &= Ae^{\lambda_1 t} + Be^{\lambda_2 t} \\y(t) &= Ce^{\lambda_1 t} + De^{\lambda_2 t},\end{aligned}$$

for some coefficients $A - D$ depending on the initial data. This readily gives us the first three behaviors in the theorem. If the eigenvalues are complex, then

$$\begin{aligned}x(t) &= Ae^{\lambda t} \cos(\mu t) + Be^{\lambda t} \sin(\mu t) \\y(t) &= Ce^{\lambda t} \cos(\mu t) + De^{\lambda t} \sin(\mu t),\end{aligned}$$

if the eigenvalues are $\lambda \pm i\mu$. This gives us the last three behaviors in the theorem. ■

Before we move on, there are some things to note about the theorem above. First, the term *asymptotically stable equilibrium* indicates that solutions starting relatively close to the equilibrium will move closer to it as

time increases. This is clearly the case for the point $(x, y) = (0, 0)$ in cases 1 and 4 above. A *stable* equilibrium is one where starting relatively close to it guarantees that solutions will stay close to the equilibrium for all future times. In case 5 above, solutions starting near $(0, 0)$ will stay near that point for all time. However, they do not necessarily approach the origin as time goes on.

In the theorem, we have listed *saddle points* as being unstable. Generically speaking, this is certainly true. Typically, solutions will achieve some point of closest approach to a saddle point before departing from the vicinity for all future times. However, there are special initial conditions that will generate solutions that approach the saddle point as t tends to infinity. If you look at the vector field for the saddle point case in the theorem, it looks as though an initial condition on one particular line in quadrants II and III leads to solutions that will limit to the origin. In fact, this line is precisely the direction indicated by the eigenvector corresponding to the negative eigenvalue! Since a randomly chosen initial condition is unlikely to lie perfectly on this line, we do not typically expect solutions to limit to a saddle point. This is the reason for listing a saddle point as unstable.

Given the theorem above, it would be nice to have some way of determining which case we are in directly from the matrix describing the system. If we try computing the eigenvalues for a generic 2×2 matrix

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

we find

$$\begin{aligned} \det(M - \lambda I_2) &= \begin{vmatrix} a - \lambda & b \\ c & d - \lambda \end{vmatrix} \\ &= (a - \lambda)(d - \lambda) - bc \\ &= \lambda^2 - (a + d)\lambda + (ad - bc) = 0. \end{aligned}$$

The constant term is clearly $\det(M)$. The quantity appearing as a coefficient of λ (without the minus sign) is known as the **trace** of the matrix M and is denoted $\text{tr}(M)$. Specifically,

$$\text{tr} \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = a + d$$

is simply the sum of the entries on the main diagonal.

So, the eigenvalues for a 2×2 matrix M are controlled by $\text{tr}(M)$ and $\det(M)$:

$$\lambda^2 - \text{tr}(M)\lambda + \det(M) = 0.$$

Hence, the eigenvalues are given by

$$\lambda = \frac{\text{tr}(M) \pm \sqrt{(\text{tr}(M))^2 - 4\det(M)}}{2}.$$

This simple observation leads us to the following classification scheme.

Theorem 9.4.2 (Classification of 2×2 Linear Systems). *For a 2×2 matrix M , we can determine the behavior of solutions as follows.*

- If $\det(M) < \frac{(\text{tr}(M))^2}{4}$, both eigenvalues are real. Moreover,
 1. If $\det(M) > 0$ and $\text{tr}(M) < 0$, the origin is an asymptotically stable node.
 2. If $\det(M) > 0$ and $\text{tr}(M) > 0$, the origin is an unstable node.
 3. If $\det(M) < 0$, the origin is a saddle point.
- If $\det(M) > \frac{(\text{tr}(M))^2}{4}$, the eigenvalues are complex (or imaginary). Moreover,
 4. If $\text{tr}(M) < 0$, the origin is an asymptotically stable spiral point.
 5. If $\text{tr}(M) = 0$, the origin is a stable center point.
 6. If $\text{tr}(M) > 0$, the origin is an unstable spiral point.
- If $\det(M) = \frac{(\text{tr}(M))^2}{4}$, M has a real eigenvalue of algebraic multiplicity 2.
 2. If $\text{tr}(M) < 0$, the origin will be asymptotically stable, and it will be unstable if $\text{tr}(M) > 0$.
- If $\det(M) = 0$, at least one of the eigenvalues is zero (and the classification fails).

Proof: The discriminant $(\text{tr}(M))^2 - 4\det(M)$ appearing under the radical in the expression for the eigenvalues

$$\lambda = \frac{\text{tr}(M) \pm \sqrt{(\text{tr}(M))^2 - 4\det(M)}}{2}$$

controls whether the eigenvalues are real or complex: $(\text{tr}(M))^2 - 4\det(M) > 0$ indicates two, distinct real solutions; $(\text{tr}(M))^2 - 4\det(M) = 0$ indicates a repeated solution; $(\text{tr}(M))^2 - 4\det(M) < 0$ indicates complex (or imaginary) solutions. This is the main classification in the system.

If the eigenvalues are complex, then the long term behavior of solutions is controlled by the real part of the eigenvalues (since this part appears in the exponential part of the solution). Since $\text{Re}(\lambda) = \text{tr}(M)/2$, we get classifications 4 – 6 in the theorem.

If the eigenvalues are real and $\det(M) < 0$, then

$$\sqrt{(\text{tr}(M))^2 - 4\det(M)} > |\text{tr}(M)|$$

since we have added a positive quantity to $(\text{tr}(M))^2$ under the radical. This means $\text{tr}(M) + \sqrt{(\text{tr}(M))^2 - 4\det(M)}$ is definitely positive while $\text{tr}(M) - \sqrt{(\text{tr}(M))^2 - 4\det(M)}$ is definitely negative. Hence, we have a saddle point at the origin which is classification 3 above.

If the eigenvalues are real and $\det(M) > 0$, then

$$\sqrt{(\text{tr}(M))^2 - 4\det(M)} < |\text{tr}(M)|$$

since we have subtracted a positive quantity from $(\text{tr}(M))^2$. Notice that $\text{tr}(M) \neq 0$ in this case since $0 < \det(M) < \frac{(\text{tr}(M))^2}{4}$. If $\text{tr}(M) > 0$, then both eigenvalues will be positive (since at worst we subtract a quantity smaller than $\text{tr}(M)$). Likewise, if $\text{tr}(M) < 0$, then both eigenvalues will be negative. This gives classifications 1 and 2 in the theorem.

Finally, if one of the eigenvalues is 0 (and so the other eigenvalue must be real), the solutions look will look something like

$$\begin{aligned}x(t) &= A + Be^{\lambda t} \\ y(t) &= C + De^{\lambda t},\end{aligned}$$

and so $(0, 0)$ is no longer the only equilibrium! In fact, a zero eigenvalue would mean that the two nullclines are in fact the same line giving us an infinity of equilibria! ■

Before we look at some examples, there is a convenient way of picturing the information in the theorem above – the *Trace–Determinant Plane*.

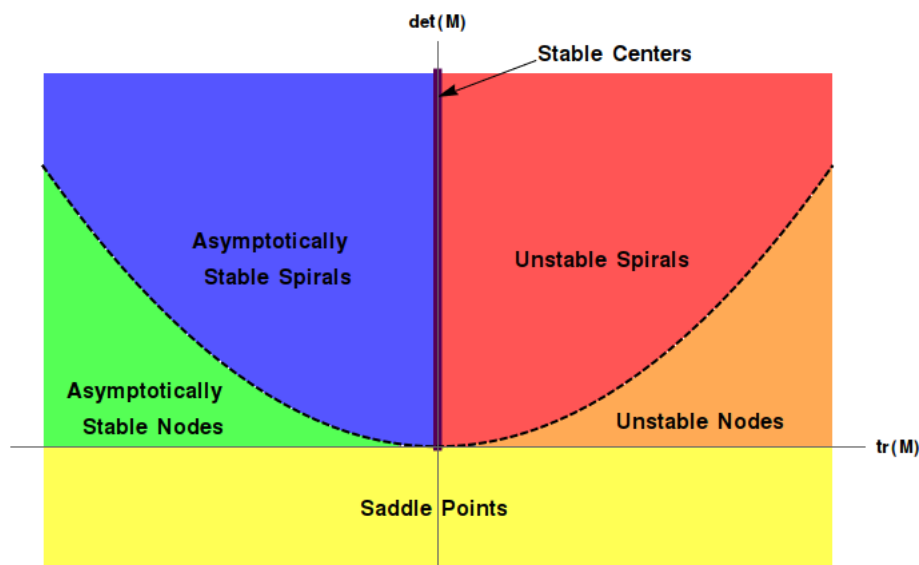


Figure 9.4.7: The Trace–Determinant Plane (Not to Scale)

The dashed parabola in Quadrants I and II dividing Nodes and Spirals is $\det(M) = \frac{(\operatorname{tr}(M))^2}{4}$. Notice that the positive $\det(M)$ axis dividing Asymptotically Stable Spirals from Unstable Spirals is the *only* place where Stable Center Points occur! The trace axis (points with $\det(M) = 0$) is where our classification theorem fails. We will illustrate the use of the trace–determinant plane in several examples below.

Example 9.4.1. Classify the behavior of solutions for the system

$$\vec{x}' = \begin{bmatrix} -2 & -2 \\ 2 & -7 \end{bmatrix} \vec{x}.$$

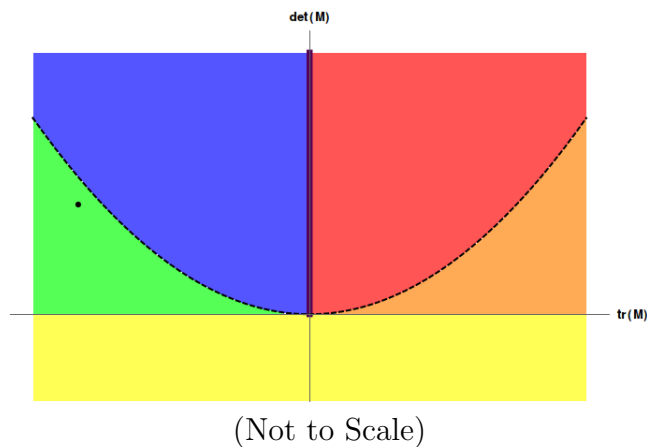
Answer: Let

$$M = \begin{bmatrix} -2 & -2 \\ 2 & -7 \end{bmatrix}.$$

Then,

$$\begin{aligned} \operatorname{tr}(M) &= -2 + -7 = -9, \\ \det(M) &= (-2)(-7) - (-2)(2) = 18, \\ \det(M) &= 18 < \frac{(\operatorname{tr}(M))^2}{4} = \frac{81}{4} = 20.25. \end{aligned}$$

Since the determinant is smaller than $(\text{tr}(M))^2/4$, we must be *under* the parabola. Since the trace is negative and determinant positive, we are in the green region corresponding to asymptotically stable nodes.



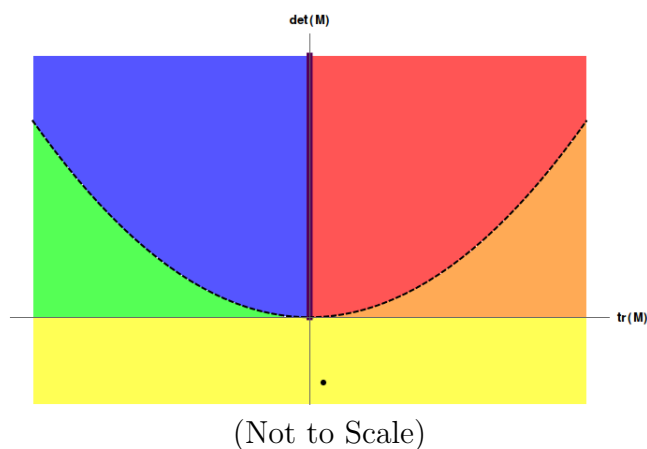
So, we know that there are two negative eigenvalues and that all solutions will limit to zero as $t \rightarrow \infty$. ■

Example 9.4.2. Classify the behavior of solutions for the system

$$\vec{x}' = \begin{bmatrix} -6 & 6 \\ -6 & 9 \end{bmatrix} \vec{x}.$$

Answer: We have

$$\begin{aligned} \text{tr}(M) &= -6 + 9 = 3, \\ \det(M) &= (-6)(9) - (6)(-6) = -18. \end{aligned}$$



Since the determinant is negative, we are clearly in the saddle point region. This means we have one negative and one positive eigenvalue and solutions tend to $\pm\infty$ both for future times and in the past for almost all initial data. As a result, the origin is an unstable equilibrium point. ■

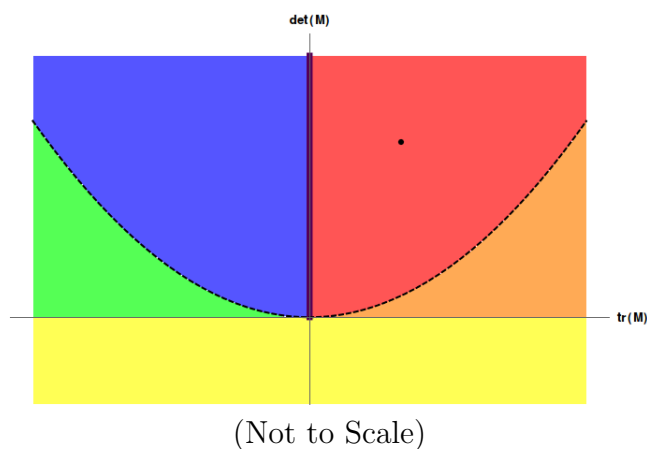
Example 9.4.3. Classify the behavior of solutions for the system

$$\vec{x}' = \begin{bmatrix} 3 & -5 \\ 1 & 1 \end{bmatrix} \vec{x}.$$

Answer: We have

$$\begin{aligned} \text{tr}(M) &= 3 + 1 = 4, \\ \det(M) &= (3)(1) - (-5)(1) = 8, \\ \det(M) &= 8 > \frac{(\text{tr}(M))^2}{4} = \frac{16}{4} = 4. \end{aligned}$$

Since the determinant is greater than $(\text{tr}(M))^2/4$, we must be *above* the parabola. Since the trace and determinant are both positive, we are in the red region corresponding to unstable spiral points.



So, we know that there are two complex eigenvalues with positive real part. All solutions will tend to $\pm\infty$ as $t \rightarrow \infty$ and that the origin is unstable. ■

9.5 Partial Derivatives and Linearization of Non-linear Systems

In Chapter 5, we looked at the stability of equilibria of 2×2 *non-linear* systems from a graphical point of view. Now that we understand the solutions to linear systems, it turns out we can classify the behavior of solutions near equilibrium points exactly! Before we can do that, we need to discuss *partial derivatives*.

9.5.1 Partial Derivatives

Suppose that we have a function f which has two or more independent variables. For concreteness, we will take x and y to be our independent variables (and so, $f = f(x, y)$). Since f depends on two quantities, there are two different derivatives we can consider.

The **partial derivative of $f = f(x, y)$ with respect to the variable x** is given the symbol $f_x(x, y)$ (or sometimes $\frac{\partial f}{\partial x}(x, y)$) and is defined by

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}.$$

Similarly, the **partial derivative of $f = f(x, y)$ with respect to the variable y** is given the symbol $f_y(x, y)$ (or sometimes $\frac{\partial f}{\partial y}(x, y)$) and is defined by

$$f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y + h) - f(x, y)}{h}.$$

As you can see, these definitions look extremely similar to the definition of derivative for a function of a single variable. In fact, you should think of f_x as the quantity you obtain by taking the derivative of f with respect to x *treating y as a constant*. Similarly, f_y is the quantity you obtain by taking the derivative of f with respect to y *treating x as a constant*. All derivative rules from your first course in calculus apply to partial derivatives!

Example 9.5.1. Find the x - and y -partial derivatives for

$$f(x, y) = x^2y^2 + 3x^3y^2 + 4x^2 + 2y + 3.$$

Answer: For the x -partial, we find

$$f_x(x, y) = 2xy^2 + 9x^2y^2 + 8x.$$

Notice that the last two terms ($2y + 3$) do not involve x at all! So, their x -partial is 0. For the y -partial, we have

$$f_y(x, y) = 2x^2y + 6x^3y + 2.$$

Again, notice that the terms $4x^2$ and 3 do not involve y ! ■

Example 9.5.2. Find the x - and y -partial derivatives for

$$g(x, y) = x^2 \sin(5x - 3y).$$

Answer: For the x -partial, we have to use the product rule (and then the chain rule to take the x -partial of the sine term).

$$f_x(x, y) = 2x \sin(5x - 3y) + 5x^2 \cos(5x - 3y)$$

Since the x^2 is a constant as far as the y -partial is concerned, we only need the chain rule to find f_y .

$$f_y(x, y) = -3x^2 \cos(5x - 3y) \quad \blacksquare$$

Example 9.5.3. Find the x - and y -partial derivatives for

$$w(x, y) = xy^3e^{x^2-2xy-y^2}.$$

Answer: For both the x - and y -partial derivatives, we need to use both the product and chain rules.

$$\begin{aligned} w_x(x, y) &= y^3e^{x^2-2xy-y^2} + xy^3(2x - 2y)e^{x^2-2xy-y^2} \\ w_y(x, y) &= 3xy^2e^{x^2-2xy-y^2} + xy^3(-2x - 2y)e^{x^2-2xy-y^2} \quad \blacksquare \end{aligned}$$

The idea of partial derivatives can be extended to a function of any number of independent variables!

Example 9.5.4. Find all three partial derivatives for the function

$$f(x, y, z) = x^3y^4z^2 + ze^{x^2+y^2}.$$

Answer:

$$\begin{aligned} f_x(x, y, z) &= 3x^2y^4z^2 + 2xz e^{x^2+y^2} \\ f_y(x, y, z) &= 4x^3y^3z^2 + 2yz e^{x^2+y^2} \\ f_z(x, y, z) &= 2x^3y^4z + e^{x^2+y^2} \quad \blacksquare \end{aligned}$$

9.5.2 Linearization

We can now state the main theorem of this chapter. Recall that the equilibrium points of a 2×2 system

$$\begin{aligned} x' &= f(x, y) \\ y' &= g(x, y) \end{aligned}$$

are all simultaneous solutions to the equations $f(x, y) = 0$ and $g(x, y) = 0$.⁶

⁶As with single differential equations, there is an *existence and uniqueness* theorem for this type of system. While we will not concern ourselves with the details too much, certainly if f and g are continuous in some region surrounding an equilibrium point, there will be a unique solution to the system for some range of t for any initial condition close to the equilibrium.

Theorem 9.5.1 (Linearization Near an Equilibrium Point). *Suppose that (x^*, y^*) is an equilibrium point for*

$$\begin{aligned}x' &= f(x, y) \\ y' &= g(x, y).\end{aligned}$$

Then the behavior of solutions near the equilibrium point is the same as the behavior of solutions near the origin in the corresponding linear system

$$\begin{aligned}u' &= f_x(x^*, y^*)u + f_y(x^*, y^*)v \\ v' &= g_x(x^*, y^*)u + g_y(x^*, y^*)v\end{aligned}$$

so long as this system has no eigenvalue with zero real part. Here, $u = x - x^$ and $v = y - y^*$ are simply new coordinates centered on the equilibrium (x^*, y^*) . This system is called the linearization of our original non-linear system near the equilibrium (x^*, y^*) .*

Proof: The proof of this theorem is beyond the scope of this text, but see [HK91, Chpt. 9] for a discussion.

For our purposes, the theorem above gives us the following useful lemma.

Lemma 9.5.2. *The stability of an equilibrium (x^*, y^*) for the system*

$$\begin{aligned}x' &= f(x, y) \\ y' &= g(x, y)\end{aligned}$$

is controlled by the eigenvalues of the matrix

$$\begin{bmatrix} f_x(x^*, y^*) & f_y(x^*, y^*) \\ g_x(x^*, y^*) & g_y(x^*, y^*) \end{bmatrix}$$

so long as the eigenvalues have non-zero real part.

In particular, the trace-determinant test we developed in the last section can help us determine whether an equilibrium is asymptotically stable or unstable! A word of caution, however. We already know that cases where $\det(M) = 0$ are problematic (as this indicates a zero eigenvalue). Our linearization theorem also fails to classify equilibria if the eigenvalues are purely imaginary. In other words, our theorem cannot directly classify an equilibrium as a stable center!

Example 9.5.5. Find and classify all equilibria for the system

$$\begin{aligned}\frac{dx}{dt} &= y - x \\ \frac{dy}{dt} &= y^2 - x.\end{aligned}$$

Answer: First, we find the equilibria by solving

$$\begin{aligned}y - x &= 0 \\ y^2 - x &= 0.\end{aligned}$$

The first equation clearly requires $y = x$. Substituting this into the second equation gives

$$\begin{aligned}x^2 - x &= 0 \\ x(x - 1) &= 0 \\ x &= 0, 1.\end{aligned}$$

So, there are two equilibria: $(0, 0)$ and $(1, 1)$.

To classify these equilibria, we need the partial derivatives of the functions

$$\begin{aligned}f(x, y) &= y - x, \\ g(x, y) &= y^2 - x.\end{aligned}$$

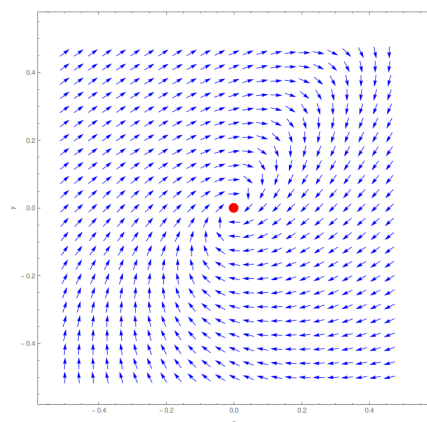
These are given as follows.

$$\begin{aligned}f_x(x, y) &= -1 & f_y(x, y) &= 1 \\ g_x(x, y) &= -1 & g_y(x, y) &= 2y\end{aligned}$$

For $(0, 0)$, we have

$$M = \begin{bmatrix} f_x(0, 0) & f_y(0, 0) \\ g_x(0, 0) & g_y(0, 0) \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}.$$

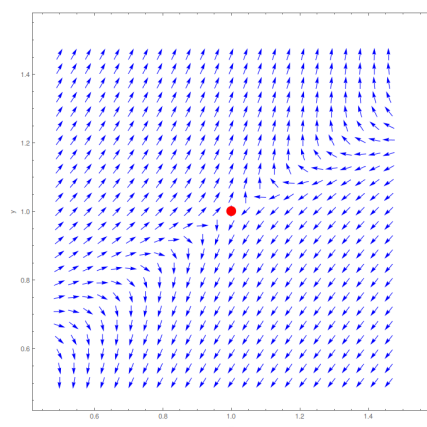
So, $\text{tr}(M) = -1$, $\det(M) = 1$, and $1 = \det(M) > (\text{tr}(M))^2/4 = 1/4$. This means that $(0, 0)$ is an asymptotically stable spiral point.

Figure 9.5.1: Vector Field near $(0, 0)$ for Example 9.5.5

For $(1, 1)$, we have

$$M = \begin{bmatrix} f_x(1, 1) & f_y(1, 1) \\ g_x(1, 1) & g_y(1, 1) \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & 2 \end{bmatrix}.$$

This gives us $\text{tr}(M) = 1$ and $\det(M) = -1$. Since the determinant is negative, $(1, 1)$ must be an unstable saddle point. ■

Figure 9.5.2: Vector Field near $(1, 1)$ for Example 9.5.5

Example 9.5.6. Find and classify all equilibria for the system

$$\begin{aligned}\frac{dx}{dt} &= (y - x^2)(x^2 - 9) \\ \frac{dy}{dt} &= x^2 + y - 8.\end{aligned}$$

Answer: For the equilibria, we need

$$\begin{aligned}(y - x^2)(x^2 - 9) &= 0 \\ x^2 + y - 8 &= 0.\end{aligned}$$

The first equation clearly gives us two cases.

- Case 1: $y - x^2 = 0$
- Case 2: $x^2 - 9 = 0$

In Case 1, we must have $y = x^2$. Substituting this into the second equation gives

$$\begin{aligned}x^2 + x^2 - 8 &= 0 \\ 2x^2 &= 8 \\ x &= \pm 2.\end{aligned}$$

So, $(-2, 4)$ and $(2, 4)$ are equilibrium points for the system.

For Case 2, $x = \pm 3$. For $x = -3$, we find

$$\begin{aligned}(-3)^2 + y - 8 &= 0 \\ 1 + y &= 0 \\ y &= -1,\end{aligned}$$

which gives us an equilibrium at $(-3, -1)$. Similarly, when $x = 3$ we have

$$\begin{aligned}(3)^2 + y - 8 &= 0 \\ 1 + y &= 0 \\ y &= -1.\end{aligned}$$

So, there is a fourth equilibrium at $(3, -1)$.

In order to classify these equilibria, we need the partial derivatives of the functions

$$\begin{aligned}f(x, y) &= (y - x^2)(x^2 - 9), \\g(x, y) &= x^2 + y - 8.\end{aligned}$$

These are given as follows.

$$\begin{aligned}f_x(x, y) &= -2x(x^2 - 9) + (y - x^2)(2x) & f_y(x, y) &= x^2 - 9 \\g_x(x, y) &= 2x & g_y(x, y) &= 1\end{aligned}$$

If we begin with $(-2, 4)$, the linearization matrix near this equilibrium is

$$M = \begin{bmatrix} f_x(-2, 4) & f_y(-2, 4) \\ g_x(-2, 4) & g_y(-2, 4) \end{bmatrix} = \begin{bmatrix} -20 & -5 \\ -4 & 1 \end{bmatrix}.$$

Since $\text{tr}(M) = -19$ and $\det(M) = -40$, we know that $(-2, 4)$ must be a saddle point. So, this equilibrium is unstable for almost all initial data.

For $(2, 4)$, we find

$$M = \begin{bmatrix} f_x(2, 4) & f_y(2, 4) \\ g_x(2, 4) & g_y(2, 4) \end{bmatrix} = \begin{bmatrix} 20 & -5 \\ 4 & 1 \end{bmatrix}.$$

Since $\text{tr}(M) = 21$, $\det(M) = 40$, and $\det(M) = 40 < (\text{tr}(M))^2/4 = 110.25$, $(2, 4)$ is an unstable node.

For $(-3, -1)$, we find

$$M = \begin{bmatrix} f_x(-3, -1) & f_y(-3, -1) \\ g_x(-3, -1) & g_y(-3, -1) \end{bmatrix} = \begin{bmatrix} 60 & 0 \\ -6 & 1 \end{bmatrix}.$$

Since $\text{tr}(M) = 61$, $\det(M) = 60$, and $\det(M) = 60 < (\text{tr}(M))^2/4 = 930.25$, $(-3, -1)$ is also an unstable node.

Finally, for the linearization near $(3, -1)$ we have

$$M = \begin{bmatrix} f_x(3, -1) & f_y(3, -1) \\ g_x(3, -1) & g_y(3, -1) \end{bmatrix} = \begin{bmatrix} -60 & 0 \\ 6 & 1 \end{bmatrix}.$$

Since $\text{tr}(M) = -59$ and $\det(M) = -60$, $(3, -1)$ must be an unstable saddle point.

The following figure shows plots of the (unscaled) vector field for this system near all four equilibria. The unstable nodes in the plots above are

very clearly unstable. The saddle point at $(3, -1)$ is a prime example of this class of equilibria. Notice that there seems to be a pair of directions where you could limit to $(3, -1)$ (i.e. the horizontal directions in the center of the plot). However, starting even very slightly off of the center line will cause the solution to limit to infinity as $t \rightarrow \infty$. Hence, this equilibrium is unstable. The behavior at $(-2, 4)$ is similar, though the arrows are slightly more difficult to read in this instance. ■

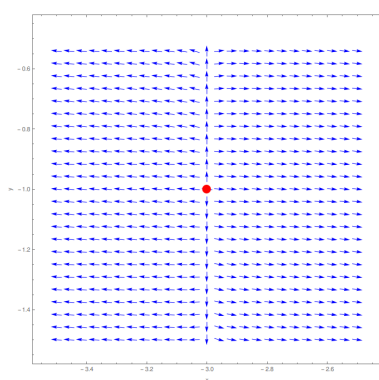
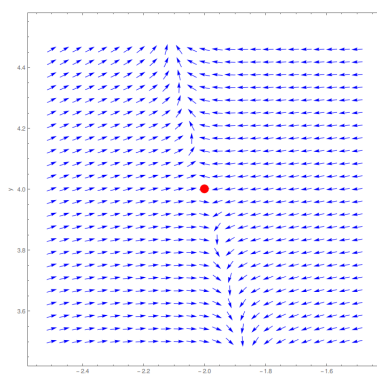
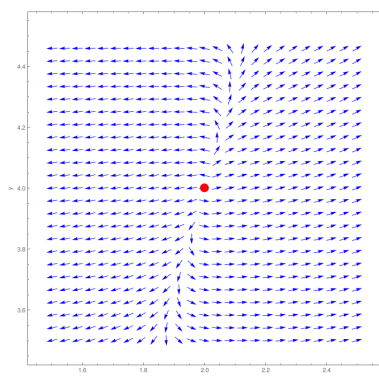
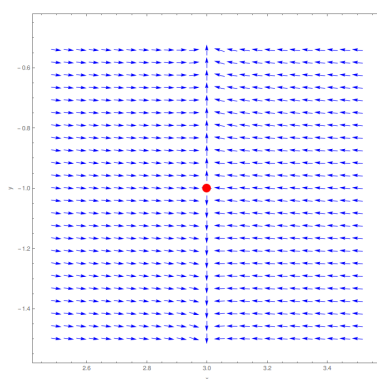
a) $(-3, -1)$: Unstable Nodeb) $(-2, 4)$: Saddle Pointc) $(2, 4)$: Unstable Noded) $(3, -1)$: Saddle Point

Figure 9.5.3: Vector Fields near Equilibria for Example 9.5.6

9.6 Analysis of Particular Models

Linearization allows us to rigorously classify the behavior of solutions to the major systems we studied in Chapter 5. In this section, we will revisit the Predator–Prey, Competing Species, and two of our disease models to rigorously verify the claims made earlier. We will also consider another interesting system, the chemostat.

9.6.1 The Predator–Prey Model

Recall that the Predator–Prey Model (or Lotka–Volterra Model) is a system of differential equations designed to model two species: a prey species with population $R = R(t)$ and a predator species with population $F = F(t)$. The system is given by

$$\begin{aligned}\frac{dR}{dt} &= \alpha R - \beta RF = R(\alpha - \beta F) \\ \frac{dF}{dt} &= -\gamma F + \delta FR = -F(\gamma - \delta R)\end{aligned}$$

where α, β, γ , and δ are positive parameters. The parameter α plays the role of a growth rate for the prey species. In the absence of any predators ($F = 0$), the prey population grows exponentially. Similarly, γ plays the role of a death rate for the predators, and in the absence of any prey ($R = 0$), the predators die off exponentially fast. The other two parameters, β and δ , control the interaction between predator and prey. For both species, the product RF serves as a measure of how frequently the two species interact. Since an interaction is detrimental to the prey, their population is reduced by βRF . Likewise, an interaction increases the predator population at the rate δRF .

The Lotka–Volterra Model always has two equilibria: $(0, 0)$ and $\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$. Recall that in Chapter 5 we said that $(0, 0)$ was unstable while $\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$ was a stable center. We can now study these facts in a more rigorous fashion.

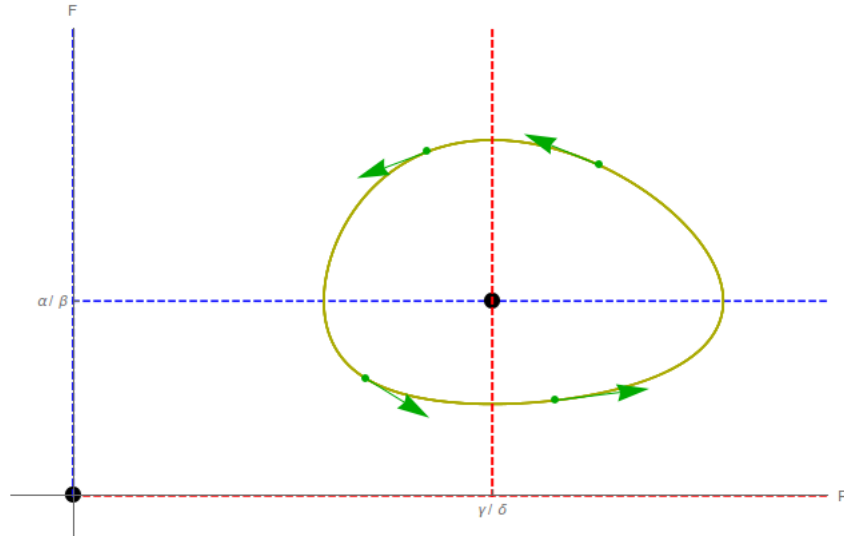


Figure 9.6.1: Typical Solution to the Predator-Prey Model

First, solutions that start in Quadrant I must stay in Quadrant I for all time. Any solution that starts on the positive R axis stays on the axis for all t . This is because when $F = 0$, $dF/dt = 0$ and F stays zero forever. Similarly, any solution that starts on the positive F axis stays on the axis for all time. Hence, any solution that starts in the interior of Quadrant I cannot intersect the axes. Therefore, solutions starting in Quadrant I must stay there forever.

To study the stability of the equilibria, we need the partial derivatives of

$$\begin{aligned} f(R, F) &= \alpha R - \beta RF \\ g(R, F) &= -\gamma F + \delta FR. \end{aligned}$$

These are

$$\begin{aligned} f_R(R, F) &= \alpha - \beta F, & f_F(R, F) &= -\beta R, \\ g_R(R, F) &= \delta F, & g_F(R, F) &= -\gamma + \delta R. \end{aligned}$$

For $(0, 0)$, we find

$$M = \begin{bmatrix} f_R(0, 0) & f_F(0, 0) \\ g_R(0, 0) & g_F(0, 0) \end{bmatrix} = \begin{bmatrix} \alpha & 0 \\ 0 & -\gamma \end{bmatrix}.$$

Since $\det(M) = -\alpha\gamma < 0$, we see that $(0, 0)$ is an unstable saddle point. In fact, the eigenvalues are exactly α and $-\gamma$.

For $\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$, we have

$$M = \begin{bmatrix} f_R\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right) & f_F\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right) \\ g_R\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right) & g_F\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right) \end{bmatrix} = \begin{bmatrix} 0 & -\frac{\beta\gamma}{\delta} \\ \frac{\alpha\delta}{\beta} & 0 \end{bmatrix}.$$

Since $\text{tr}(M) = 0$ and $\det(M) = \alpha\gamma > 0$, the linearized system has a stable center. However, Theorem 9.5.1 does not directly apply in this case!

It is easy to see that solutions must spiral around the equilibrium point in counter-clockwise fashion. The linearization is just not sensitive enough to determine whether $\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$ is unstable, stable, or asymptotically stable. A more detailed analysis (c.f. [HS74, pp. 258 – 265]) can show that the solutions are in fact closed orbits. So for the Predator–Prey Model, the linearization happens to classify both equilibria correctly. But remember that Theorem 9.5.1 cannot be used to establish that an equilibrium is a stable center!

9.6.2 The Competing Species Model

The system of differential equations

$$\begin{aligned} \frac{da}{dt} &= k_a a \left(1 - \frac{a}{N_a}\right) - \alpha ab \\ \frac{db}{dt} &= k_b b \left(1 - \frac{b}{N_b}\right) - \beta ab \end{aligned}$$

models the populations of two species, $a = a(t)$ and $b = b(t)$, that compete for the same resources. The parameters k_a and k_b are logistic growth rates and N_a and N_b are carrying capacities for the two species. The parameters α and β are competition parameters measuring the rate at which one species takes resources away from the other. The nullclines for this system are lines:

- a -nullclines: $a = 0$ and $1 - \frac{1}{N_a}a - \frac{\alpha}{k_a}b = 0$
- b -nullclines: $b = 0$ and $1 - \frac{1}{N_b}b - \frac{\beta}{k_b}a = 0$

Since $a = 0$ and $b = 0$ are nullclines, solutions that begin in Quadrant I are definitely trapped there for all time (just as with the Predator–Prey Model).

The points $(0, 0)$, $(N_a, 0)$, and $(0, N_b)$ are always equilibria. There is a fourth equilibrium if the two oblique nullclines intersect in the first quadrant. Recall that in Chapter 5 we saw that there were three distinct behaviors.

- Competitive Exclusion (3 equilibria)
- Weak Competition (4 equilibria)
- Strong Competition (4 equilibria)

To analyze stability, we need the following partial derivatives.

$$\begin{aligned} f_a(a, b) &= k_a - \frac{2k_a}{N_a}a - \alpha b, & f_b(a, b) &= -\alpha a \\ g_a(a, b) &= -\beta b, & g_b(a, b) &= k_b - \frac{2k_b}{N_b}b - \beta a \end{aligned}$$

At $(0, 0)$, we have

$$M = \begin{bmatrix} f_a(0, 0) & f_b(0, 0) \\ g_a(0, 0) & g_b(0, 0) \end{bmatrix} = \begin{bmatrix} k_a & 0 \\ 0 & k_b \end{bmatrix}.$$

We could compute trace and determinant, but the eigenvalues of this matrix are k_a and k_b – both of which are positive. So, $(0, 0)$ is unstable in all cases!

Case 1: Competitive Exclusion In this case, the two oblique nullclines do not intersect in the first quadrant.

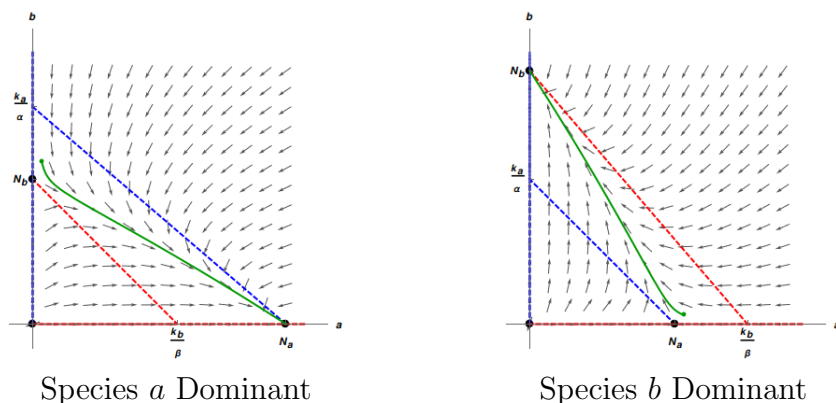


Figure 9.6.2: Typical Solutions for Competing Species: Competitive Exclusion

As we can see from the figure above, this happens in one of two cases:

$$\begin{aligned} \text{case a)} \quad N_b < \frac{k_a}{\alpha} \quad \text{and} \quad \frac{k_b}{\beta} < N_a, \\ \text{or} \\ \text{case b)} \quad \frac{k_a}{\alpha} < N_b \quad \text{and} \quad N_a < \frac{k_b}{\beta}. \end{aligned}$$

Since we have already examined $(0, 0)$, we need to consider the equilibria $(N_a, 0)$ and $(0, N_b)$. For the first of these

$$M_1 = \begin{bmatrix} f_a(N_a, 0) & f_b(N_a, 0) \\ g_a(N_a, 0) & g_b(N_a, 0) \end{bmatrix} = \begin{bmatrix} -k_a & -\alpha N_a \\ 0 & k_b - \beta N_a \end{bmatrix}.$$

The eigenvalues for this matrix are clearly $-k_a$ and $k_b - \beta N_a$. Similarly,

$$M_2 = \begin{bmatrix} f_a(0, N_b) & f_b(0, N_b) \\ g_a(0, N_b) & g_b(0, N_b) \end{bmatrix} = \begin{bmatrix} k_a - \alpha N_b & 0 \\ -\beta N_b & -k_b \end{bmatrix}$$

with eigenvalues $k_a - \alpha N_b$ and $-k_b$.

In case a), we have $k_a > \alpha N_b$ and $k_b < \beta N_a$. This means the linearization matrix M_1 for $(N_a, 0)$ has two negative eigenvalues which makes this equilibrium an asymptotically stable node. Likewise, $(0, N_b)$ has one positive and one negative eigenvalue which makes it an unstable saddle point. So in case a), species a out-competes species b, and species b will die out.

Case b) is exactly the reverse. $(N_a, 0)$ has one positive and one negative eigenvalue making it an unstable saddle point. $(0, N_b)$ has two negative eigenvalues which makes it an asymptotically stable node. So in case b), species b out-competes species a.

Case 2: Weak Competition In this case,

$$N_a < \frac{k_b}{\beta} \quad \text{and} \quad N_b < \frac{k_a}{\alpha}.$$

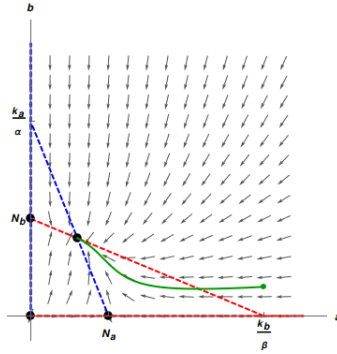


Figure 9.6.3: Typical Solutions for Competing Species: Weak Competition

We've already seen that $(0, 0)$ is unstable. The linearization matrices for $(N_a, 0)$ and $(0, N_b)$ are exactly the same as before. In the case of weak competition however, both of these equilibria are unstable saddle points (as both have one negative and one positive eigenvalue).

The fourth equilibrium point is a bit more complicated in form:

$$\left(\frac{k_b N_a (k_a - \alpha N_b)}{k_a k_b - \alpha \beta N_a N_b}, \frac{k_a N_b (k_b - \beta N_a)}{k_a k_b - \alpha \beta N_a N_b} \right).$$

Since $k_a > \alpha N_b$ and $k_b > \beta N_a$ in this case, both the numerator and denominator of each coordinate are positive numbers – verifying that this equilibrium point is in Quadrant I. Instead of displaying the linearization matrix M_3 for this equilibrium, we simply give the trace and determinant.

$$\begin{aligned} \text{tr}(M_3) &= -\frac{k_a k_b (k_a - \alpha N_b + k_b - \beta N_a)}{k_a k_b - \alpha \beta N_a N_b} \\ \det(M_3) &= \frac{k_a k_b (k_a - \alpha N_b)(k_b - \beta N_a)}{k_a k_b - \alpha \beta N_a N_b} \end{aligned}$$

Our assumption that $k_a > \alpha N_b$ and $k_b > \beta N_a$ guarantees that $\text{tr}(M_3) < 0$ while $\det(M_3) > 0$. This means that the fourth equilibrium is asymptotically stable (whether it is a node or spiral point depends on the specific parameters). So, weak competition allows the two species to coexist long term.

Case 3: Strong Competition In this case,

$$N_a > \frac{k_b}{\beta} \quad \text{and} \quad N_b > \frac{k_a}{\alpha}.$$

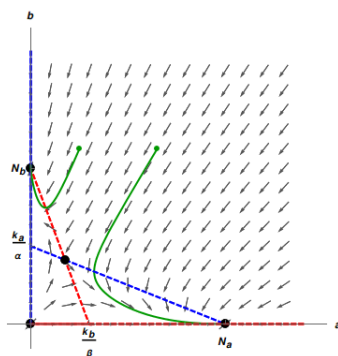


Figure 9.6.4: Typical Solutions for Competing Species: Strong Competition

Since we have already studied the linearization matrices for all four equilibria (or at least the trace and determinant for the fourth equilibrium), we only need to summarize their stability in this case.

- $(0, 0)$: Unstable Node
- $(N_a, 0)$: Asymptotically Stable Node
- $(0, N_b)$: Asymptotically Stable Node
- $\left(\frac{k_b N_a (k_a - \alpha N_b)}{k_a k_b - \alpha \beta N_a N_b}, \frac{k_a N_b (k_b - \beta N_a)}{k_a k_b - \alpha \beta N_a N_b} \right)$: Unstable Saddle Point

So in the case of strong competition, the species cannot coexist long term. One of the two species will dominate while the other dies out. However, which species survives depends on the initial population sizes.

9.6.3 The SIR Model

The SIR Disease Model assumes that there are three categories of people.

- Susceptible people (with population $S = S(t)$)
- Infected people (with population $I = I(t)$)
- Recovered people (with population $R = R(t)$)

To make the analysis simpler, we assume that the total population is a constant size N : $S + I + R = N$. This allows us to ignore the recovered population and focus on the susceptible and infected populations. We also assume that recovered people are immune from the disease forever.

$$\begin{aligned}\frac{dS}{dt} &= -\alpha SI \\ \frac{dI}{dt} &= \alpha SI - \beta I\end{aligned}$$

As with the Predator–Prey Model, interactions between susceptible and infected people are modeled by the product SI ; susceptible people contract the disease at a rate αSI and so become infected. Infected people recover at the rate βI . As usual, the parameters α and β are positive. Recall that in Chapter 5 we proved that any point of the form $(S, I) = (s, 0)$ is an equilibrium point. In words, the equilibria are points where the disease dies out; all infected people recover leaving some number of susceptible people who never contract the illness. Just like the Predator–Prey Model, solutions that start in Quadrant I must stay in Quadrant I for all time. In fact, solutions must stay in the region $0 \leq S + I \leq N$ for all t (since the population is constant).

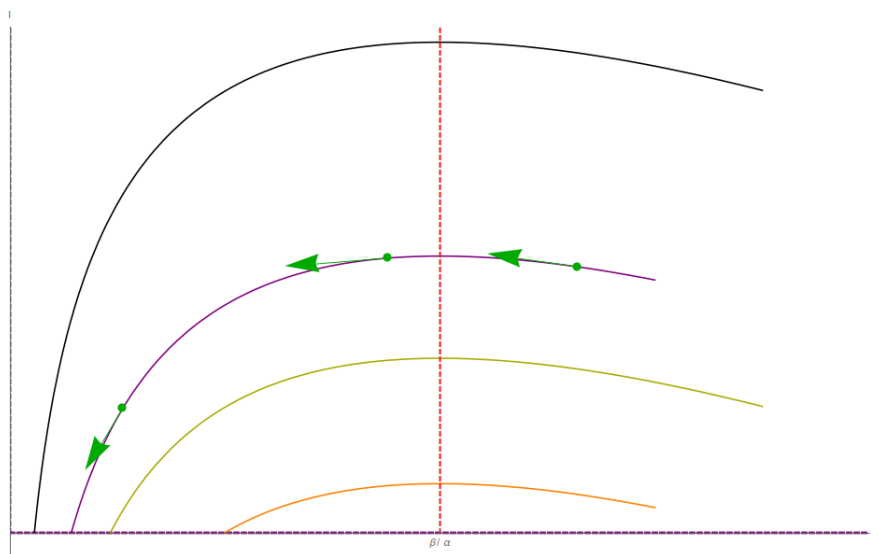


Figure 9.6.5: Typical Solutions to the SIR Model

To explore the stability of these equilibria, we need to find the first partial derivatives of

$$\begin{aligned}f(S, I) &= -\alpha SI \\g(S, I) &= \alpha SI - \beta I.\end{aligned}$$

These are given by

$$\begin{aligned}f_S(S, I) &= -\alpha I, & f_I(S, I) &= -\alpha S, \\g_S(S, I) &= \alpha I, & g_I(S, I) &= \alpha S - \beta.\end{aligned}$$

For any equilibrium point $(s, 0)$, we have

$$M = \begin{bmatrix} f_S(s, 0) & f_I(s, 0) \\ g_S(s, 0) & g_I(s, 0) \end{bmatrix} = \begin{bmatrix} 0 & -\alpha s \\ 0 & \alpha s - \beta \end{bmatrix}.$$

This gives us $\det(M) = 0$ and $\text{tr}(M) = \alpha s - \beta$.

The determinant being zero means that one of the eigenvalues of the linearization matrix is zero. The zero eigenvalue stems from the fact that every point on the S -axis is an equilibrium. Even though Theorem 9.5.1 does not directly apply, a slight extension of it can establish stability of these equilibria for solutions starting in Quadrant I. In particular, if $\text{tr}(M) > 0$, then the non-zero eigenvalue must be positive, and the equilibrium is unstable. Likewise, $\text{tr}(M) < 0$ would indicate that the second eigenvalue is negative, and the equilibrium is asymptotically stable (again, for solutions starting in Quadrant I). So if $s > \beta/\alpha$, the equilibrium $(s, 0)$ is unstable; if $s < \beta/\alpha$, the equilibrium $(s, 0)$ is asymptotically stable. The case $s = \beta/\alpha$ is still ambiguous as *both* eigenvalues are 0.

9.6.4 The SIRS Model

If we change the SIR Model by assuming recovered individuals lose their immunity to the disease over time, we have the SIRS Disease Model. Note that we still assume the population is a fixed size.

$$\begin{aligned}\frac{dS}{dt} &= -\alpha SI + \gamma R \\ \frac{dI}{dt} &= \alpha SI - \beta I \\ \frac{dR}{dt} &= \beta I - \gamma R\end{aligned}$$

The positive parameters α and β play the same role as before: α is the infection rate and β is the rate that infected people recover. The positive parameter γ represents the rate at which people who recover from the disease lose their immunity and become susceptible to the disease again.

We can ignore the population $R = R(t)$ since $R = N - S - I$ where N is the fixed population size. This reduces the system of differential equations to

$$\begin{aligned}\frac{dS}{dt} &= -\alpha SI + \gamma(N - S - I) \\ \frac{dI}{dt} &= \alpha SI - \beta I.\end{aligned}$$

As shown in Chapter 5, there are two equilibria for this system:

$$(S, I) = (N, 0) \text{ and } \left(\frac{\beta}{\alpha}, \frac{\gamma(\alpha N - \beta)}{\alpha(\beta + \gamma)} \right).$$

The second equilibrium only makes sense for the model when $\beta/\alpha \leq N$ (as N is the total population), and when $\beta/\alpha = N$, the second equilibrium reduces to the first.

Since the total population is fixed at size N , clearly we need $0 \leq S + I \leq N$ for solutions to be sensible. This is the region shown in yellow below. Looking at the vector field for the system, we can see solutions that start in this triangular region cannot leave it.

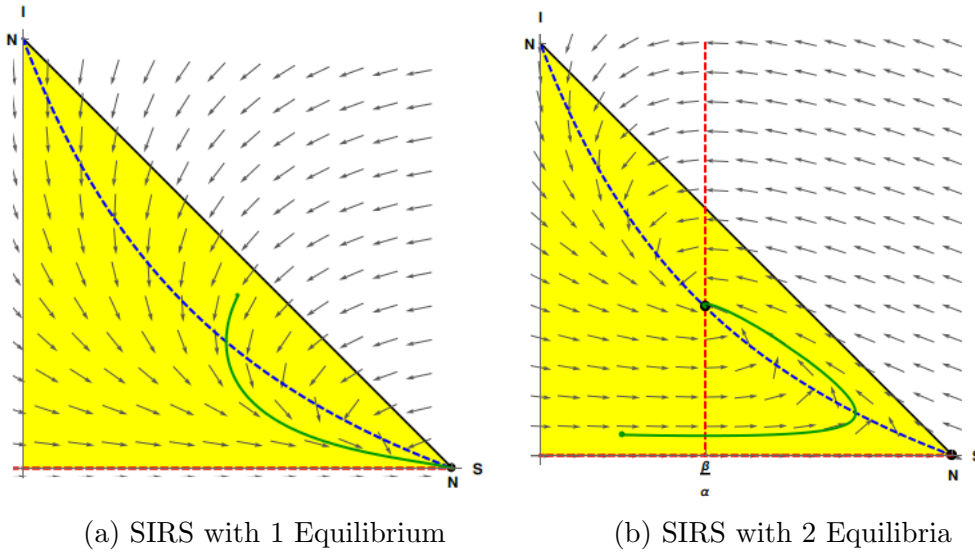


Figure 9.6.6: Typical Vector Plots for the SIRS Model

Note that solutions on the bottom edge of the triangular region flow along it toward $S = N$. In these cases, the number of infected people is zero and any recovered people eventually lose their immunity. On the other two edges, the direction arrows all point toward the interior of the region. Hence, solutions starting in $0 \leq S + I \leq N$ will stay there for all time.

To determine stability, we need to following partial derivatives.

$$\begin{aligned} f_S(S, I) &= -\alpha I - \gamma & f_I(S, I) &= -\alpha S - \gamma \\ g_S(S, I) &= \alpha I & g_I(S, I) &= \alpha S - \beta \end{aligned}$$

For $(N, 0)$, we find

$$M = \begin{bmatrix} f_S(N, 0) & f_I(N, 0) \\ g_S(N, 0) & g_I(N, 0) \end{bmatrix} = \begin{bmatrix} -\gamma & -\alpha N - \gamma \\ 0 & \alpha N - \beta \end{bmatrix}.$$

This gives us $\det(M) = -\gamma(\alpha N - \beta)$ and $\text{tr}(M) = \alpha N - \beta - \gamma$. When $\alpha N - \beta > 0$ (which means $\beta/\alpha < N$), $\det(M) < 0$ and $(N, 0)$ is an unstable saddle point. Similarly, $\det(M) > 0$ when $\alpha N - \beta < 0$. In this case $(N, 0)$ is either an asymptotically stable node or an asymptotically stable spiral point since $\text{tr}(M) < 0$. So, when $(N, 0)$ is the only sensible equilibrium, it is asymptotically stable. When the second equilibrium is sensible for the

system, $(N, 0)$ is unstable. Notice that our analysis is inconclusive when $\alpha N - \beta = 0$.

For the second equilibrium, a little algebra shows us that the linearization matrix is

$$M = \begin{bmatrix} -\frac{(\alpha N - \beta)\gamma}{\beta + \gamma} - \gamma & -(\beta + \gamma) \\ \frac{(\alpha N - \beta)\gamma}{\beta + \gamma} & 0 \end{bmatrix}.$$

The trace and determinant of this matrix are

$$\begin{aligned} \text{tr}(M) &= -\left(\gamma + \frac{\gamma(\alpha N - \beta)}{\beta + \gamma}\right), \\ \det(M) &= \gamma(\alpha N - \beta). \end{aligned}$$

Since this equilibrium is only meaningful when $\beta/\alpha < N$ (and so $\alpha N - \beta > 0$), we know that $\text{tr}(M) < 0$ and $\det(M) > 0$. Hence, whenever

$$\left(\frac{\beta}{\alpha}, \frac{\gamma(\alpha N - \beta)}{\alpha(\beta + \gamma)}\right)$$

is a sensible equilibrium, it is asymptotically stable. Whether the equilibrium is an asymptotically stable node or an asymptotically stable spiral point depends on the specific parameters.

9.6.5 Chemostats

A chemostat is a chamber that is used to continually grow microbes. We supply the microbes in the chamber with a sterile supply of some nutrient solution (for example, a sugar solution) at a fixed rate (f liters per hour) and concentration (c_0 grams per liter). The mixture of nutrient solution and microbes in the chamber is continuously agitated to make sure the material is well-mixed. The microbe/nutrient mixture is collected from the chemostat at the same rate as the sterile nutrient solution enters the chamber.

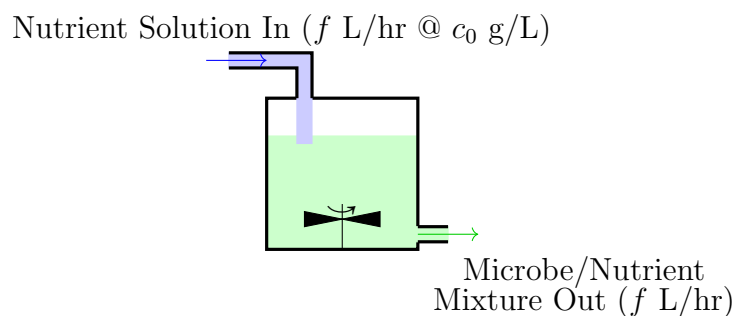


Figure 9.6.7: A Chemostat

Since the flow rates in and out are both f liters per hour, the total volume of fluid in the chamber is constant (V liters). The quantities we are interested in modeling are the concentration of the nutrient in the tank, $C_N(t)$, and the concentration of microbes in the tank, $C_M(t)$, at time t (both measured in grams per liter). These are related to the total amounts (in grams) of nutrient, $N(t)$, and microbes, $M(t)$, in the tank at time t by

$$C_N(t) = \frac{N(t)}{V} \quad \text{and} \quad C_M(t) = \frac{M(t)}{V}.$$

The microbes consume the nutrient in the chamber, and the nutrient provides energy for the microbes to reproduce. What we need to discuss is how the growth rate of the microbes, μ , depends on the concentration of nutrient in the chemostat, C_N . We are assuming that the microbes are totally dependent on the nutrient being supplied to the chamber. So, if the concentration of nutrient falls to zero, the microbes will not be able to reproduce. On the other hand, there is some fixed amount of time required for the microbes to reproduce even in an over-abundance of nutrients. The model typically used for chemostats is known as the Monod Equation:⁷

$$\mu(C_N) = \mu_{max} \left(\frac{C_N}{K_M + C_N} \right) = \mu_{max} \left(\frac{N}{K_M V + N} \right).$$

The parameter μ_{max} is the ideal growth rate (in units 1/hr) that the microbes would reproduce at given an unlimited amount of nutrients. The parameter

⁷This empirical model for the growth of microbes is identical in form to the Michealis-Menten kinetics which models the rate of a chemical reaction in the presence of a enzyme that facilitates the reaction.

K_M is the concentration of nutrient at which the growth rate drops to half of μ_{max} . Taking into account the growth of the microbes in the chamber along with the flow of microbes out of the chemostat, we have

$$\frac{dM}{dt} = \mu_{max} \left(\frac{N}{K_M V + N} \right) M - \frac{f}{V} M = \left(\frac{\mu_{max} N}{K_M V + N} - \frac{f}{V} \right) M.$$

For the concentration of nutrient in the tank, we clearly have nutrient flowing into the tank at a constant rate and concentration. And we have nutrient flowing out of the tank at a matching rate. In addition, the microbes are consuming the nutrient as they reproduce. We model the consumption of nutrients as being proportional to the growth rate of the microbes:

$$\text{rate of consumption} = \frac{1}{\gamma} \left(\frac{\mu_{max} N}{K_M V + N} \right) M.$$

The parameter γ is called the *yield constant* and is in the range $0 < \gamma \leq 1$. So,

$$\frac{dN}{dt} = f c_0 - \frac{1}{\gamma} \left(\frac{\mu_{max} N}{K_M V + N} \right) M - f \left(\frac{N}{V} \right).$$

To summarize, our chemostat is modeled by the system of non-linear equations:

$$\begin{aligned} \frac{dM}{dt} &= \left(\frac{\mu_{max} N}{K_M V + N} - \frac{f}{V} \right) M, \\ \frac{dN}{dt} &= f c_0 - \frac{1}{\gamma} \left(\frac{\mu_{max} N}{K_M V + N} \right) M - f \left(\frac{N}{V} \right). \end{aligned}$$

This system has two equilibria:

$$\begin{aligned} (M_1, N_1) &= (0, c_0 V), \\ (M_2, N_2) &= (\gamma(c_0 V - \bar{N}), \bar{N}), \end{aligned}$$

where

$$\bar{N} = \frac{f K_M V}{\mu_{max} V - f}.$$

If we express these in terms of concentrations, then the two equilibria are

$$(0, c_0) \text{ and } (\gamma(c_0 - \bar{c}), \bar{c})$$

where the concentration appearing in the second equilibrium is given by

$$\bar{c} = \frac{fK_M}{\mu_{max}V - f}.$$

Notice that this equilibrium is physically meaningful only when $\mu_{max}V > f$.

The linearization matrix for these two equilibria is a little complicated, so we will simply give the stability results. Details can be found in [F18, pp. 15 – 18].

- **Case 1:** $f < \mu_{max}V \left(\frac{c_0}{c_0 + K_M} \right)$

In this case, the equilibrium $(0, c_0)$ is an unstable saddle point. The other equilibrium, $(\gamma(c_0 - \bar{c}), \bar{c})$, is an asymptotically stable node. This means that in the long-term, the concentration of microbes in the tank limits to

$$\lim_{t \rightarrow \infty} C_M(t) = \gamma(c_0 - \bar{c}).$$

- **Case 2:** $\mu_{max}V \left(\frac{c_0}{c_0 + K_M} \right) < f < \mu_{max}V$

The equilibrium $(0, c_0)$ becomes an asymptotically stable node in this case while $(\gamma(c_0 - \bar{c}), \bar{c})$ becomes an unstable saddle point. In this case, the microbes are completely washed out of the chemostat and

$$\lim_{t \rightarrow \infty} C_M(t) = 0.$$

- **Case 3:** $f > \mu_{max}V$

In this case, the only physically meaningful equilibrium is $(0, c_0)$ which is asymptotically stable. So once again,

$$\lim_{t \rightarrow \infty} C_M(t) = 0,$$

and the microbes are completely washed out of the chemostat.

What we have learned from our study is that for a given microbial species, nutrient concentration c_0 , and volume V , there is a critical flow rate

$$f_{crit} = \mu_{max}V \left(\frac{c_0}{c_0 + K_M} \right)$$

below which the chemostat will achieve a steady-state concentration of the microbes. Above this critical flow rate, the microbes will be completely washed out of the system.

Another question we might want to answer is how to maximize the rate at which microbes are collected from the chemostat (and so we must be in Case 1 above). The parameters γ and K_M are particular to the microbial species and type of nutrient being used. The volume V is limited by the vessel being used for the chemostat. The concentration of the nutrient supply, c_0 , is somewhat under our control, but once the solution is saturated with the nutrient it will be impossible to increase the concentration. The aspect of the chemostat most directly under our control is the flow rate f .

The limiting microbial concentration is given by

$$\gamma c_0 \left[1 - \frac{f K_M}{c_0(\mu_{max} V - f)} \right]$$

which means that the *rate* at which the microbes leave the chemostat is

$$\gamma c_0 f \left[1 - \frac{f K_M}{c_0(\mu_{max} V - f)} \right]$$

This rate will be maximized as a function of f if we maximize the function

$$g(f) = f \left[1 - \frac{f K_M}{c_0(\mu_{max} V - f)} \right].$$

After some lengthy computations, it can be shown that the flow rate

$$f^* = \mu_{max} V \left(1 - \sqrt{\frac{K_M}{c_0 + K_M}} \right)$$

maximizes the microbial harvest.

9.7 A Word on Chaotic Systems

Systems that are 3×3 or larger can have far more complex behavior than 2×2 systems. To demonstrate this, we will look at a particularly famous

system of differential equations known as the Lorenz System.

$$\begin{aligned}\frac{dx}{dt} &= \sigma(y - x) \\ \frac{dy}{dt} &= x(\rho - z) - y \\ \frac{dz}{dt} &= xy - \beta z\end{aligned}$$

This system was first developed in 1963 by Edward Lorenz to study convection in the atmosphere. The specifics of what the variables $x = x(t)$, $y = y(t)$, and $z = z(t)$ represent are not important for our discussion here. For demonstration purposes, we will take $\sigma = 10$, $\rho = 28$, and $\beta = 3$.

As in the 2×2 case, we begin by finding equilibria.

$$\begin{aligned}10(y - x) &= 0 \\ x(28 - z) - y &= 0 \\ xy - 3z &= 0\end{aligned}$$

The first equation requires $x = y$. Combining this with the third equation gives

$$z = \frac{1}{3}x^2.$$

Combining these facts with the second equation gives

$$\begin{aligned}x \left(28 - \frac{1}{3}x^2 \right) - x &= 0 \\ x \left(27 - \frac{1}{3}x^2 \right) &= 0 \\ x &= -9, 0, 9.\end{aligned}$$

So, our version of the Lorenz System has three equilibria:

$$(-9, -9, 27), (0, 0, 0), \text{ and } (9, 9, 27).$$

To study the behavior near these equilibria, we need the first partial derivatives of the three functions appearing in the system.

$$\begin{array}{lll}f_x(x, y, z) = -10 & f_y(x, y, z) = 10 & f_z(x, y, z) = 0 \\ g_x(x, y, z) = 28 - z & g_y(x, y, z) = -1 & g_z(x, y, z) = -x \\ h_x(x, y, z) = y & h_y(x, y, z) = x & h_z(x, y, z) = -3\end{array}$$

Unfortunately, the trace and determinant of the linearization matrix are no longer sufficient to determine stability. Instead, we have to look directly at the eigenvalues.

For $(0, 0, 0)$, we find

$$M_1 = \begin{bmatrix} f_x(0, 0, 0) & f_y(0, 0, 0) & f_z(0, 0, 0) \\ g_x(0, 0, 0) & g_y(0, 0, 0) & g_z(0, 0, 0) \\ h_x(0, 0, 0) & h_y(0, 0, 0) & h_z(0, 0, 0) \end{bmatrix} = \begin{bmatrix} -10 & 10 & 0 \\ 28 & -1 & 0 \\ 0 & 0 & -3 \end{bmatrix}.$$

This matrix has eigenvalues

$$\begin{aligned} \lambda_1 &= \frac{1}{2} \left(-\sqrt{1201} - 11 \right) \approx -22.83, \\ \lambda_2 &= -3, \\ \lambda_3 &= \frac{1}{2} \left(\sqrt{1201} - 11 \right) \approx 11.83. \end{aligned}$$

The presence of one positive eigenvalue makes $(0, 0, 0)$ makes this equilibrium unstable (in fact, a saddle point).

For $(-9, -9, 27)$, we find

$$M_2 = \begin{bmatrix} f_x(-9, -9, 27) & f_y(-9, -9, 27) & f_z(-9, -9, 27) \\ g_x(-9, -9, 27) & g_y(-9, -9, 27) & g_z(-9, -9, 27) \\ h_x(-9, -9, 27) & h_y(-9, -9, 27) & h_z(-9, -9, 27) \end{bmatrix} = \begin{bmatrix} -10 & 10 & 0 \\ 1 & -1 & 9 \\ -9 & -9 & -3 \end{bmatrix}.$$

The eigenvalues for this matrix are far more difficult to compute, but they can be estimated numerically.

$$\begin{aligned} \lambda_1 &\approx -14.08 \\ \lambda_2 &\approx 0.04 + 10.73i \\ \lambda_3 &\approx 0.04 - 10.73i \end{aligned}$$

It turns out that $(9, 9, 27)$ has the same eigenvalues.

$$M_3 = \begin{bmatrix} f_x(9, 9, 27) & f_y(9, 9, 27) & f_z(9, 9, 27) \\ g_x(9, 9, 27) & g_y(9, 9, 27) & g_z(9, 9, 27) \\ h_x(9, 9, 27) & h_y(9, 9, 27) & h_z(9, 9, 27) \end{bmatrix} = \begin{bmatrix} -10 & 10 & 0 \\ 1 & -1 & -9 \\ 9 & 9 & -3 \end{bmatrix}.$$

$$\begin{aligned} \lambda_1 &\approx -14.08 \\ \lambda_2 &\approx 0.04 + 10.73i \\ \lambda_3 &\approx 0.04 - 10.73i \end{aligned}$$

The behavior near these equilibria is not analogous to any of the equilibria we studied for 2×2 systems. Solutions can be attracted to each of these equilibria from the direction of the eigenvector associated to the negative eigenvalue. However, solutions slowly spiral out from the equilibrium in the plane perpendicular to this eigenvector.

If we look at the trajectory of solutions we see some really interesting behavior.

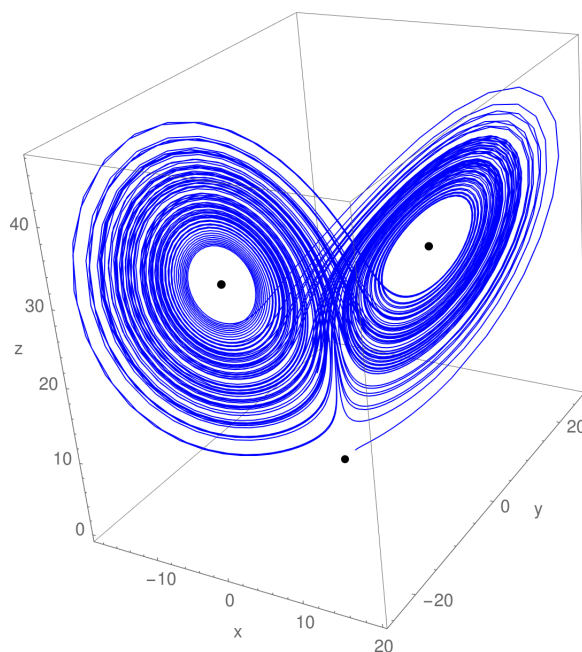


Figure 9.7.1: Lorenz Attractor: $x(0) = 1, y(0) = 1$, and $z(0) = 1$

The butterfly shape eventually traced out by the solutions is generic and is known as the *Lorenz Attractor*. The plots below show trajectories for the same parameters $\sigma = 10, \rho = 28$, and $\beta = 3$ but with different initial data.

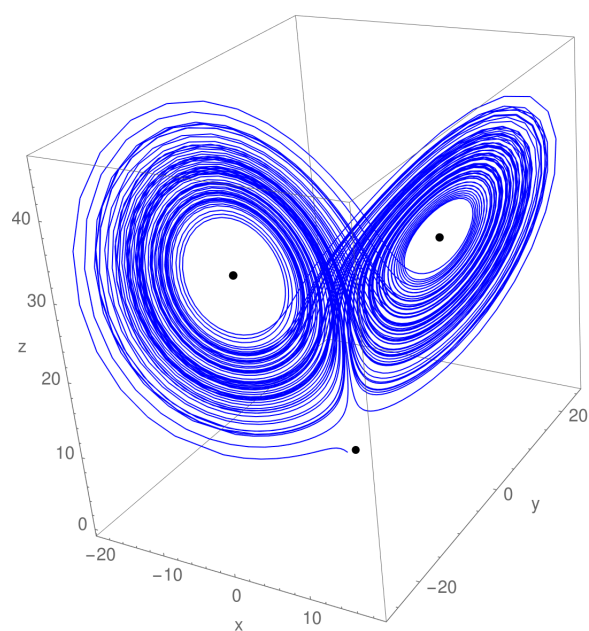


Figure 9.7.2: Lorenz Attractor: $x(0) = 2$, $y(0) = -5$, and $z(0) = 3$

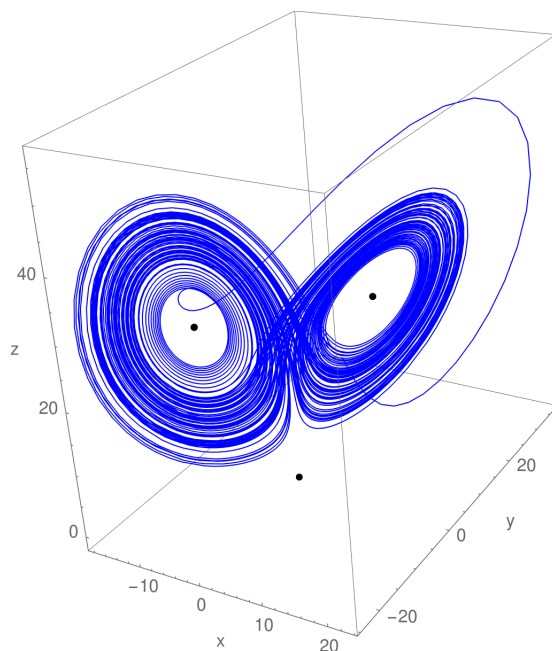


Figure 9.7.3: Lorenz Attractor: $x(0) = -10$, $y(0) = -30$, and $z(0) = 0$

In addition to the Lorenz Attractor, the other interesting fact about the Lorenz System is that it was one of the earliest systems of differential equations known to exhibit *chaotic behavior*. Chaos is a general term used to indicate that the long term behavior of a system can depend quite sensitively on the specific choice of initial data. More precisely, a chaotic system is one in which two sets of initial data which are relatively close together can lead to solutions which are eventually quite different.

To see this phenomenon, we consider trajectories launched by two sets of initial data that are relatively close together.

- 1) $x(0) = 1$, $y(0) = 1$, and $z(0) = 1$
- 2) $x(0) = 1.1$, $y(0) = 0.9$, and $z(0) = 1$

In the plot below, the trajectory launched by initial data 1) is shown in blue while initial data 2) is shown in red.

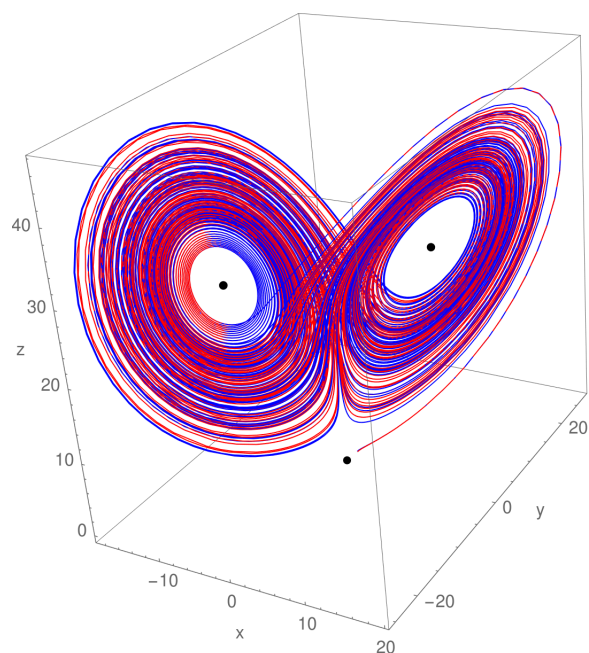
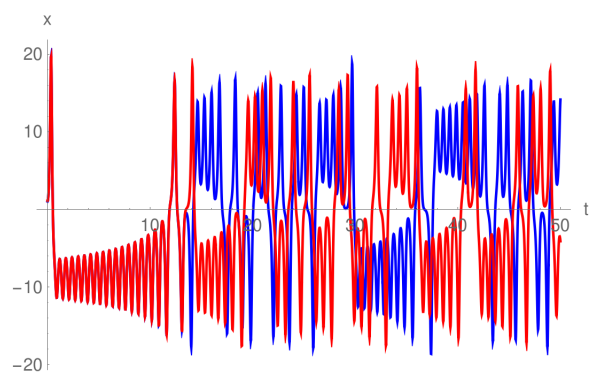
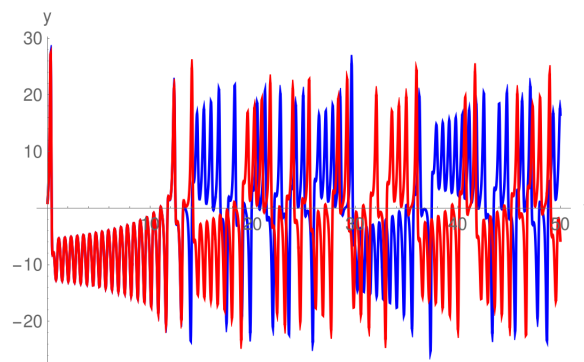
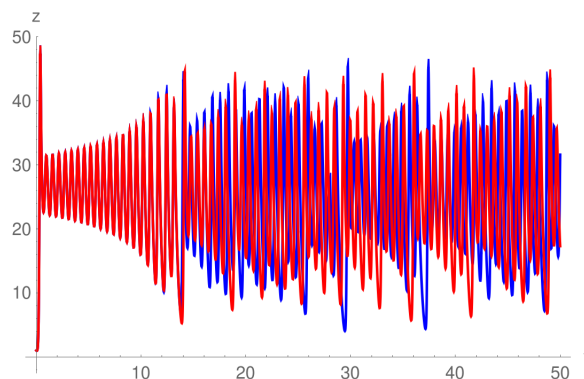


Figure 9.7.4: Lorenz System for Similar Initial Data

While both trajectories trace out the Lorenz Attractor, it looks as though the two solutions cease to be close together after some amount of time. This becomes very clear when we plot the individual functions.

Figure 9.7.5: Lorenz System for Similar Initial Data: $x(t)$

Figure 9.7.6: Lorenz System for Similar Initial Data: $y(t)$ Figure 9.7.7: Lorenz System for Similar Initial Data: $z(t)$

Despite being very similar near $t = 0$ (where the red and blue curves practically overlap one another), the solutions eventually diverge from one another after a finite amount of time.

From a practical point of view, this means that chaotic systems are very difficult to predict in the long term. Since in any real-world system knowledge of the initial data will come with some amount of error, we will eventually lose the ability to make accurate predictions after some amount of time.

The interesting thing about chaotic systems is that they are deterministic (solutions are determined completely by the initial data) but they are inherently unpredictable in any realistic situation – especially over long time periods. Many systems in the natural sciences exhibit some degree of chaos making long term predictions problematic.

9.8 Problems

- 1.) Find the general solution to the following system of differential equations.

$$\begin{cases} x' = -2x + y \\ y' = 4x + y. \end{cases}$$

- 2.) Solve the following initial value problem.

$$\begin{cases} x' = -x + 3y \\ y' = 4x + 3y \\ x(0) = 1 \\ y(0) = -2 \end{cases}$$

- 3.) Solve the following initial value problem.

$$\begin{cases} x' = 2x - \frac{7}{2}y \\ y' = \frac{5}{2}x - 4y \\ x(0) = 1 \\ y(0) = -3 \end{cases}$$

- 4.) Find the general solution to the following system of differential equations.

$$\begin{cases} x' = -4x + 3y \\ y' = -6x + 2y \end{cases}$$

- 5.) Solve the following initial value problem.

$$\begin{cases} x' = 2x - 5y \\ y' = 5x + 2y \\ x(0) = 2 \\ y(0) = 3 \end{cases}$$

- 6.) Solve the following initial value problem.

$$\begin{cases} x' = 2x - 13y \\ y' = 2x - 8y \\ x(0) = 1 \\ y(0) = 0 \end{cases}$$

- 7.) Find the general solution to the following system of differential equations.

$$\begin{cases} x' = 6x + 3y - 3z \\ y' = 4x - 4z \\ z' = 10x + 3y - 7z \end{cases}$$

- 8.) Solve the following initial value problem.

$$\begin{cases} x' = 12x + 3y - 6z \\ y' = -4x + y + 4z \\ z' = 13x + 5y - 7z \\ x(0) = 1 \\ y(0) = 3 \\ z(0) = 2 \end{cases}$$

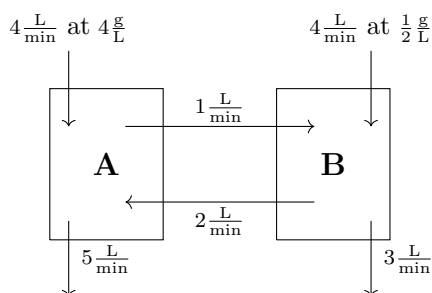
- 9.) Solve the following initial value problem.

$$\begin{cases} x' = 4x + 3y - 5z \\ y' = -4x + y + 5z \\ z' = 4x + y - 3z \\ x(0) = 0 \\ y(0) = 0 \\ z(0) = 2 \end{cases}$$

- 10.) Find the general solution to the following system of differential equations.

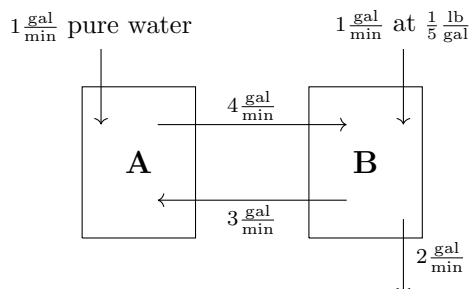
$$\begin{cases} x' = -x - 2z \\ y' = 4x + y + 4z \\ z' = -6x - 4y - z \end{cases}$$

- 11.) Consider the following system of two tanks. Tank A initially contains 50 L of pure water; tank B initially contains 50 L of a salt solution at concentration 5 g/L. The rate at which salt solution is pumped into each tank, the rate of fluid collection out of each tank, and the various mixing rates between the tanks are shown below.



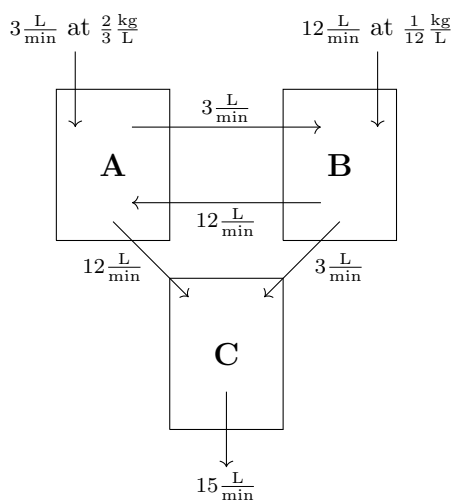
Assuming that the solutions in both tanks remain well-mixed throughout the process, find $C_A(t)$ and $C_B(t)$, the concentrations of salt in tanks A and B respectively. What concentrations do the tanks limit to in the long-run?

- 12.) Consider the following system of two tanks. Tank A initially contains 25 gallons of a sugar solution at $2/25$ lb/gal; tank B initially contains 25 gallons of a pure water. The rate at which sugar solution is pumped into each tank, the rate of fluid collection out of each tank, and the various mixing rates between the tanks are shown below.



Assuming that the solutions in both tanks remain well-mixed throughout the process, find $C_A(t)$ and $C_B(t)$, the concentrations of salt in tanks A and B respectively. What concentrations do the tanks limit to in the long-run?

- 13.) Consider the following system of three tanks. Tanks A and B contain 300 L of fluid while Tank C contains 600 L. The rate at which salt solution is pumped into each tank, the rate of fluid collection out of each tank, and the various mixing rates between the tanks are shown below.



Assuming that the salt solutions in all three tanks remain well-mixed throughout the process, what do the concentrations, C_A , C_B , and C_C , in each tank limit to in the long-run?

- 14.) Find the general solution to the following second order linear differential equation by converting it into a system of first order equations.

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 5x = 13\cos(2t)$$

- 15.) A block of mass $m = 1$ kg is attached to a spring of strength $k = 21$ N/m and is allowed to slide horizontally on table. An external force of $F(t) = 25\sin(t)$ acts on the block. Assume the coefficient of linear resistance for the system is $b = 10$ Ns/m. If the block starts from equilibrium moving to the right at 1 m/s, solve for the position of the block as a function of time.

- 16.) Find the general solution to the following third order linear differential equation by converting it into a system of first order equations.

$$\frac{d^3x}{dt^3} - 4\frac{d^2x}{dt^2} + \frac{dx}{dt} - 4x = -15e^{-2t}$$

- 17.) Find and classify all equilibria for the following nonlinear system of differential equations.

$$\begin{cases} x' = -(x-y)(1-x-y) \\ y' = x(y+2) \end{cases}$$

- 18.) Find and classify all equilibria for the following nonlinear system of differential equations.

$$\begin{cases} x' = xy - 9x \\ y' = (x - 2)(y^2 + 3y + 2) \end{cases}$$

- 19.) Find and classify all equilibria for the following nonlinear system of differential equations.

$$\begin{cases} x' = x(y^2 + 2y - 3) \\ y' = -y(x + 5) \end{cases}$$

- 20.) Consider the modified predator–prey model with logistic growth for the prey species.

$$\begin{aligned} \frac{dR}{dt} &= \alpha R \left(1 - \frac{R}{N_R}\right) - \beta RF \\ \frac{dF}{dt} &= -\gamma F + \delta RF \end{aligned}$$

- a.) Show that the equilibria are $(R, F) = (0, 0)$ and $(N_R, 0)$ when $N_R < \frac{\gamma}{\delta}$. In the case $N_R > \frac{\gamma}{\delta}$, show that there are three equilibria: $(0, 0)$, $(N_R, 0)$, and the coexistence equilibrium

$$\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta} \left(1 - \frac{\gamma}{\delta N_R}\right)\right).$$

- b.) Show that $(0, 0)$ is always an unstable saddle point.
 c.) Show that $(N_R, 0)$ is an asymptotically stable node when $N_R < \frac{\gamma}{\delta}$ and an unstable saddle point when the reverse inequality holds.
 d.) In the special case that $\alpha = \frac{1}{6}, \beta = \frac{1}{3}, \gamma = \frac{1}{12}, \delta = \frac{1}{24}$, and $N_R = 4$, show that the coexistence equilibrium is an asymptotically stable spiral point.

- 21.) Recall the SEIR Disease Model discussed in Section 5.5.3.

$$\begin{aligned} \frac{dS}{dt} &= \mu - \alpha SI - \mu S \\ \frac{dE}{dt} &= \alpha SI - (\mu + \gamma)E \\ \frac{dI}{dt} &= \gamma E - (\mu + \beta)I \end{aligned}$$

- a.) Verify the equilibria for this system are those given in Theorem 5.5.3.
- b.) For the case when $\alpha = 1/20, \beta = 1/5, \gamma = 1/5$, and $\mu = 1/100$, show that the trivial equilibrium $(S, E, I, R) = (1, 0, 0, 0)$ is asymptotically stable by showing (numerically) that all eigenvalues are negative.
- c.) For the case when $\alpha = 1/2, \beta = 1/20, \gamma = 1/5$, and $\mu = 1/100$, show that the trivial equilibrium $(S, E, I, R) = (1, 0, 0, 0)$ is unstable by showing (numerically) that there is a positive eigenvalue. (It can be shown that the other equilibrium in this case is asymptotically stable).
- 22.) A 2×2 matrix with a repeated eigenvalue is almost certainly going to have a deficiency in the number of independent eigenvector families.⁸ If a 2×2 matrix M has a repeated eigenvalue λ and a single family of eigenvectors with basis vector $\vec{v}$, then the general solution to

$$\vec{x}' = M\vec{x}$$

is of the form

$$\vec{x}(t) = c_1 \vec{v} e^{\lambda t} + c_2 (t\vec{v} + \vec{\eta}) e^{\lambda t}$$

where the vector $\vec{\eta}$ is any *generalized eigenvector* satisfying the equation

$$(M - \lambda I_2)\vec{\eta} = \vec{v}$$

(which is equivalent to $M\vec{\eta} = \lambda\vec{\eta} + \vec{v}$). Verify that this is indeed a solution to the differential equation.

- 23.) Using the results discussed in the previous problem, find the general solution to the following systems of differential equations.

a.)

$$\vec{x}' = \begin{bmatrix} 1 & 9 \\ -1 & -5 \end{bmatrix} \vec{x}$$

b.)

$$\vec{x}' = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} \vec{x}$$

⁸If a 2×2 matrix has a repeated eigenvalue and two independent families of eigenvectors, then the matrix must be a multiple of the 2×2 identity matrix.

- 24.) a.) For any square matrix M , it is possible to define the matrix exponential, e^M . If a matrix is diagonalizable, then

$$e^M = Be^DB^{-1}$$

where $M = BDB^{-1}$, D is the diagonal matrix of eigenvalues similar to M , and

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{bmatrix} \longrightarrow e^D = \begin{bmatrix} e^{\lambda_1} & 0 & 0 & \cdots & 0 \\ 0 & e^{\lambda_2} & 0 & \cdots & 0 \\ 0 & 0 & e^{\lambda_3} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & e^{\lambda_n} \end{bmatrix}.$$

Find the matrix exponential for the matrices in problems 15.) and 16.) in Section 8.9.4.

- b.) For an inhomogeneous system of differential equations

$$\vec{x}' = M\vec{x} + \vec{b}(t)$$

where M is a matrix of *constant coefficients*, a particular solution is given by

$$\vec{x}_p(t) = e^{tM} \int_0^t e^{-\tau M} \vec{b}(\tau) d\tau.$$

Verify that this is indeed a solution to the inhomogeneous differential equation above using the following facts about the matrix exponential.

$$e^{tM} e^{-tM} = e^{-tM} e^{tM} = I_n$$

$$\frac{d}{dt} [e^{tM}] = e^{tM} M = M e^{tM}$$

- 25.) Using the results discussed in the previous problem, find a particular solution to the following systems of differential equations.

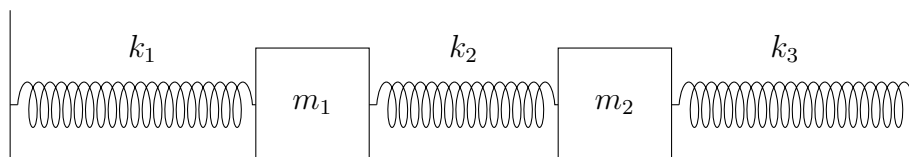
- a.)

$$\vec{x}' = \begin{bmatrix} 5 & -6 \\ 3 & -4 \end{bmatrix} \vec{x} + \begin{bmatrix} 3e^{2t} \\ 0 \end{bmatrix}$$

b.)

$$\vec{x}' = \begin{bmatrix} -1 & 1 \\ 3 & 1 \end{bmatrix} \vec{x} + \begin{bmatrix} 4e^{-2t} \\ 4e^{2t} \end{bmatrix}$$

- 26.) **Challenge: Coupled Spring–Mass Systems** Two blocks are connected by three springs as shown below.



The masses of the blocks are m_1 and m_2 , respectively; the strengths of the three springs are given by k_1, k_2 , and k_3 . If $x_1 = x_1(t)$ and $x_2 = x_2(t)$ are the horizontal positions of the blocks measured from equilibrium, then the motion of the two springs is governed by the following equations (assuming friction is negligible and the oscillations of the blocks are not too large).⁹

$$\begin{aligned} m_1 x_1'' &= -k_1 x_1 + k_2 (x_2 - x_1) \\ m_2 x_2'' &= -k_2 (x_2 - x_1) - k_3 x_2 \end{aligned}$$

- a.) Write this second order system of differential equations as a system of four first order differential equations.
- b.) Find the general solution to this system of differential equations when $m_1 = m_2 = 1$, $k_1 = k_3 = 2$, and $k_2 = 1$.

⁹If the oscillations become so large that the blocks contact each other or the wall, then certainly our model will not accurately predict the positions of the blocks. Even if the oscillations are not that big, Hooke's Law (the force supplied by a spring is proportional to how much the string has been stretched) is only valid for a modest amount of stretching.

Chapter 10

Discrete-Time Stochastic Processes

Often in applications, we have a sequence of random events which happen at different times rather than a single random variable. These random events may be completely independent of one another, or more typically, the next event in the sequence may depend on the current and previous events in some way (while still being random).

For example, suppose that we are allowed to walk along a straight line. We start out on the point labeled “0,” and every few seconds we take either a step forward or a step backwards. The decision of whether to move forward or backwards is random – perhaps determined by the flip of a fair coin. Let X_i be our position after the i -th step, and note that $X_0 = 0$ since we started from the origin. A coin flip of heads (H) will mean we take one step forward, while tails (T) will mean we take one step backwards. The start of one possible sequence of events is the following:

Step	Old Position	Coin Flip	New Position
1	$X_0 = 0$	H	$X_1 = X_0 + 1 = 1$
2	$X_1 = 1$	H	$X_2 = X_1 + 1 = 2$
3	$X_2 = 2$	T	$X_3 = X_2 - 1 = 1$
4	$X_3 = 1$	H	$X_4 = X_3 + 1 = 2$
5	$X_4 = 2$	T	$X_5 = X_4 - 1 = 1$
6	$X_5 = 1$	T	$X_6 = X_5 - 1 = 0$
7	$X_6 = 0$	T	$X_7 = X_6 - 1 = -1$
$\vdots$	$\vdots$	$\vdots$	$\vdots$

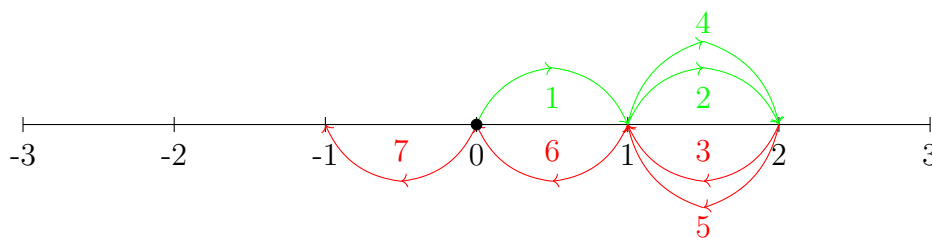


Figure 10.0.1: The Beginning of a Random Walk

This sequence of random variables $(X_i) = (X_0, X_1, X_2, X_3, \dots)$ is an example of a **discrete-time stochastic process**: a dynamical system with some inherent randomness in each step and where time passes in whole blocks marked by integers (typically $n = 0, 1, 2, 3, \dots$). Notice that these are very similar to the discrete dynamical systems we discussed in Chapter 1. The difference is that the discrete dynamical systems we discussed before were *deterministic*. If we knew $q_0, q_1, q_2, \dots, q_n$, then the updating function for the system completely determined q_{n+1} :

$$q_{n+1} = f(n, q_n, q_{n-1}, q_{n-2}, \dots, q_{n-m}).$$

With a stochastic process, however, knowing $X_0, X_1, X_2, \dots, X_n$ only gives us partial information about X_{n+1} because there is an element of randomness in each step. In an extreme case, X_{n+1} may be completely independent of all the values that came before! In this chapter, our goal is to explore some of the more common discrete-time stochastic processes. In addition to time being discrete, all of the random variable (X_i) in the stochastic processes we study are assumed to have the same *discrete* sample space (usually labeled by the integers). Since we will only deal with discrete-time stochastic processes with discrete sample spaces, we will often just call them stochastic processes in the following sections.¹ This chapter follows closely the presentation in Chapter 4 of [Rs03], but there will be a number of results whose proofs will be omitted. Appropriate references will be cited for the interested reader.

¹Many of our results apply just as well to discrete-time stochastic processes with *continuous sample spaces*. This is especially true for Markov Chains. We can also allow stochastic processes with time measured continuously, though the mathematical tools required are significantly more sophisticated than what we have at our disposal.

10.1 Markov Chains

Despite already being rather specialized, the universe of discrete-time stochastic processes is still too broad a subject for our purposes. We first make the assumption that our random variables, X_n , have values in the integers:

$$X_n = 0, \pm 1, \pm 2, \pm 3, \dots$$

These integers are the possible *states* of the process. Suppose after n units of time, the system has the following history:

$$X_0 = i_0, X_1 = i_1, X_2 = i_2, \dots, X_n = i_n.$$

That is, i_0 is some integer (a state of the system), and the system is initially in this state. After one time step ($n = 1$), the system transitions from state i_0 to the state i_1 (another integer representing a possible state of the system), and so on.

In order to make predictions, we need to know the probability distribution for X_{n+1} given this past history. In more precise language, we need to know the following conditional probabilities:

$$\mathbb{P}(X_{n+1} = i \mid X_0 = i_0, X_1 = i_1, X_2 = i_2, \dots, X_n = i_n).$$

These conditional probabilities are known as *transition probabilities* since they tell us the probability that the system will move to state i given all of its previous history.

In general, these probabilities can depend on the entire history up to time $n + 1$ (and perhaps on the time n itself). However, we will make the simplifying assumption that

$$\begin{aligned} \mathbb{P}(X_{n+1} = i \mid X_0 = i_0, \dots, X_n = i_n) &= \mathbb{P}(X_{n+1} = i \mid X_n = i_n) \\ &= \mathbb{P}(X_1 = i \mid X_0 = i_n). \end{aligned}$$

In other words, the probabilities for the next time step (time $n + 1$) only depend on the *current state* of the system (the state at time n), and these transition probabilities do not change over time. Such a stochastic process is called a **Markov Chain**. Essentially, a Markov Chain is memoryless; it not only forgets what has occurred before the present time, it is also unaware of how much time has passed!

Since for a Markov Chain the transition probabilities depend only on the current state (which are integers), we can refer to these probabilities with labels:

$$\mathbb{P}(X_{n+1} = i \mid X_n = j) = \mathbb{P}(X_1 = i \mid X_0 = j) = P_{ij}.$$

You should think of P_{ij} as “the probability that a Markov Chain currently in state j will transition to state i in the next time step.” The labeling here may seem a little strange (i.e. P_{ij} indicates a change *to* state i *from* state j rather than the other way around), but it will make later formulas seem more natural.

Since the P_{ij} are supposed to be probabilities, they must satisfy certain basic requirements. First,

$$P_{ij} \geq 0 \text{ for all pairs } i \text{ and } j.$$

This is clear since probabilities are supposed to be non-negative! Second, if the system is currently in state j , all possible transition probabilities must add up to be 1. That is, we require

$$\sum_i P_{ij} = 1 \text{ for each } j.$$

If we make the simplifying assumption that there are only a *finite* number of possible states, then we can display the transition probabilities in a square matrix. We suppose that the possible states of the system are $\{0, 1, 2, \dots, m\}$. This means that for each time step i , X_i must be equal to one of $0, 1, 2, \dots, m$. We display the probabilities as

$$P = \begin{bmatrix} P_{00} & P_{01} & P_{02} & \cdots & P_{0m} \\ P_{10} & P_{11} & P_{12} & \cdots & P_{1m} \\ P_{20} & P_{21} & P_{22} & \cdots & P_{2m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ P_{m0} & P_{m1} & P_{m2} & \cdots & P_{mm} \end{bmatrix}.$$

This matrix is called a **(left) stochastic matrix**² or a **transition probability matrix**. Note that in a transition probability matrix, the probabilities in each column must add up to 1.

²The “left” in “left stochastic matrix” indicates that each column must sum to 1. A right stochastic matrix is a matrix where each row sums to 1. A doubly stochastic matrix is one where each row and each column sum to 1.

Example 10.1.1. Consider the following overly simple weather model. Each day, the sky is either clear, partly cloudy, or overcast. If today is clear, then tomorrow will be clear with probability 0.5, partly cloudy with probability 0.3, or overcast with probability 0.2. If today is partly cloudy, then tomorrow will be clear with probability 0.3, partly cloudy with probability 0.4, or overcast with probability 0.3. If today is overcast, then tomorrow will be clear with probability 0.2, partly cloudy with probability 0.4, or overcast with probability 0.4. Write this process as a Markov Chain, and determine the matrix of transition probabilities.

Answer: We begin by arbitrarily labeling the possible states with integers: let 0 stand for the state that the sky is clear, 1 for partly cloudy, and 2 for overcast. Let $(X_i) = (X_0, X_1, X_2, \dots)$ be the sequence of weather states. For the transition probabilities, we find

$$\begin{aligned} P_{00} &= \text{"probability that it will be clear tomorrow given it is clear today"} = 0.5, \\ P_{01} &= \text{"probability that it will be partly cloudy tomorrow given that is clear today"} = 0.3, \\ P_{02} &= \text{"probability that it will be overcast tomorrow given that is clear today"} = 0.2, \\ P_{10} &= \text{"probability that it will be clear tomorrow given it is partly cloudy today"} = 0.3, \\ P_{11} &= \text{"probability that it will be partly cloudy tomorrow given it is partly cloudy today"} = 0.4, \\ P_{12} &= \text{"probability that it will be overcast tomorrow given it is partly cloudy today"} = 0.3, \\ P_{20} &= \text{"probability that it will be clear tomorrow given it is overcast today"} = 0.2, \\ P_{21} &= \text{"probability that it will be partly cloudy tomorrow given it is overcast today"} = 0.4, \\ P_{22} &= \text{"probability that it will be overcast tomorrow given it is overcast today"} = 0.4. \end{aligned}$$

Putting this into a 3×3 matrix gives

$$P = \begin{bmatrix} P_{00} & P_{01} & P_{02} \\ P_{10} & P_{11} & P_{12} \\ P_{20} & P_{21} & P_{22} \end{bmatrix} = \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.4 & 0.4 \end{bmatrix}.$$

Notice that the requirement that

$$\sum_i P_{ij} = 1 \text{ for each } j$$

translates into the sum of the entries in any given column must equal 1. ■

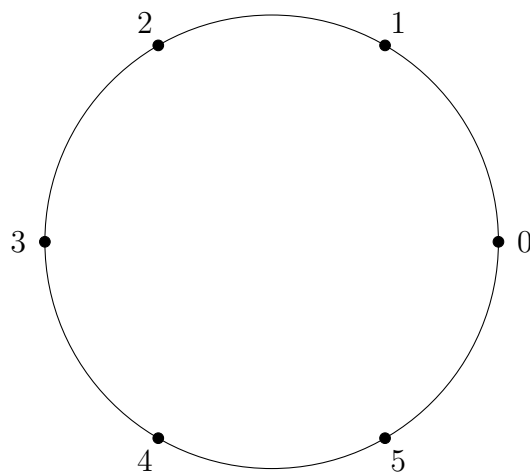


Figure 10.1.1: A Circular Walk

Example 10.1.2. Suppose we take a random walk on the circular path shown in Figure 10.1.1. At each time step, we will flip a fair coin to determine our next position. If the coin flip is heads, we will move one unit counterclockwise; for tails, we will move one unit clockwise. Write this process as a Markov Chain, and determine the matrix of transition probabilities.

Answer: Clearly, the states are the positions on the circular path (which are already labeled with integers). Let $(X_i) = (X_0, X_1, X_2, \dots)$ be the sequence of nodes we take on our random walk. We need only determine the transition probabilities. Suppose that we are currently standing on the node labeled 0. The only possible moves that can happen are a transition to node 1 or a transition to node 5. Hence, we have the following probabilities:

$$\begin{aligned} P_{00} &= \text{"probability of moving from node 0 to node 0"} = 0, \\ P_{01} &= \text{"probability of moving from node 1 to node 0"} = 0.5, \\ P_{02} &= \text{"probability of moving from node 2 to node 0"} = 0, \\ P_{03} &= \text{"probability of moving from node 3 to node 0"} = 0, \\ P_{04} &= \text{"probability of moving from node 4 to node 0"} = 0, \\ P_{05} &= \text{"probability of moving from node 5 to node 0"} = 0.5. \end{aligned}$$

Likewise, if we are currently standing on a particular node, then there is a 50% chance of moving to the node one space counterclockwise and a 50%

chance of moving to the node one space clockwise. All other transition probabilities must be zero. Putting this into a transition matrix gives

$$P = \begin{bmatrix} P_{00} & P_{01} & P_{02} & P_{03} & P_{04} & P_{05} \\ P_{10} & P_{11} & P_{12} & P_{13} & P_{14} & P_{15} \\ P_{20} & P_{21} & P_{22} & P_{23} & P_{24} & P_{25} \\ P_{30} & P_{31} & P_{32} & P_{33} & P_{34} & P_{35} \\ P_{40} & P_{41} & P_{42} & P_{43} & P_{44} & P_{45} \\ P_{50} & P_{51} & P_{52} & P_{53} & P_{54} & P_{55} \end{bmatrix} = \begin{bmatrix} 0 & 0.5 & 0 & 0 & 0 & 0.5 \\ 0.5 & 0 & 0.5 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0.5 & 0 & 0.5 \\ 0.5 & 0 & 0 & 0 & 0.5 & 0 \end{bmatrix}.$$

Once again, note that each column sums up to 1. Unlike the previous example, the rows of this transition matrix also sum to 1. Such a matrix is known as a *doubly stochastic matrix*. ■

Example 10.1.3. Consider the following simple maze.

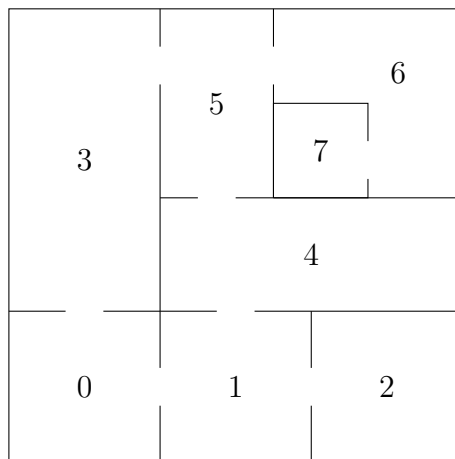


Figure 10.1.2: A Simple Maze

Suppose a mouse is running this maze, and Room 7 is the goal (i.e. once it reaches that room, the mouse gets a reward and the experiment is over). If the mouse is in any other room, we assume that it chooses any of the connecting rooms with equal probability. Write this process as a Markov Chain, and determine the matrix of transition probabilities.

Answer: Let $(X_i) = (X_0, X_1, X_2, \dots)$ be the sequence of rooms that the mouse inhabits. If the mouse is in room 0, then it will move to either room 1 or room 3 each with probability $1/2$. All other transitions are impossible.

Similarly, if the mouse is in room 1, then it will move to either of rooms 0, 2, or 4 each with a probability of $1/3$. We continue this way until we get to room 7. Once the mouse is in this room, the experiment is over. Putting these probabilities in the transition matrix gives

$$P = \begin{bmatrix} P_{00} & P_{01} & P_{02} & P_{03} & P_{04} & P_{05} & P_{06} & P_{07} \\ P_{10} & P_{11} & P_{12} & P_{13} & P_{14} & P_{15} & P_{16} & P_{17} \\ P_{20} & P_{21} & P_{22} & P_{23} & P_{24} & P_{25} & P_{26} & P_{27} \\ P_{30} & P_{31} & P_{32} & P_{33} & P_{34} & P_{35} & P_{36} & P_{37} \\ P_{40} & P_{41} & P_{42} & P_{43} & P_{44} & P_{45} & P_{46} & P_{47} \\ P_{50} & P_{51} & P_{52} & P_{53} & P_{54} & P_{55} & P_{56} & P_{57} \\ P_{60} & P_{61} & P_{62} & P_{63} & P_{64} & P_{65} & P_{66} & P_{67} \\ P_{70} & P_{71} & P_{72} & P_{73} & P_{74} & P_{75} & P_{76} & P_{77} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1/3 & 0 & 1/2 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 1 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 1/3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 0 & 0 & 0 & 1/3 & 0 & 0 \\ 0 & 1/3 & 0 & 0 & 0 & 1/3 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1/2 & 1 \end{bmatrix} \cdot \blacksquare$$

10.2 The Chapman–Kolmogorov Equations

The transition probability matrix gives us the probabilities of states exactly one time step in the future. Our next goal is to find a way of predicting probabilities n time steps in the future. The n -step transition probabilities are defined by

$$P_{ij}^n = \mathbb{P}(X_{n+m} = i | X_m = j) = \mathbb{P}(X_n = i | X_0 = j).$$

The two conditional probabilities are equal because Markov Chains are memoryless. Note that $P_{ij}^1 = P_{ij}$ while P_{ij}^0 equals one when $i = j$ and 0 otherwise. In other words, the matrix giving the values of P_{ij}^0 is simply the identity matrix.

Theorem 10.2.1 (The Chapman–Kolmogorov Equations). *The n -step transition probabilities satisfy the following equations*

$$P_{ij}^{n+m} = \sum_k P_{ik}^n P_{kj}^m.$$

Proof: We begin by conditioning on the state of the system at time m . Note that we have to preserve the original condition as we go!

$$\begin{aligned} P_{ij}^{n+m} &= \mathbb{P}(X_{n+m} = i | X_0 = j) \\ &= \sum_k \mathbb{P}(X_{n+m} = i | X_m = k, X_0 = j) \mathbb{P}(X_m = k | X_0 = j) \\ &= \sum_k \mathbb{P}(X_{n+m} = i | X_m = k) \mathbb{P}(X_m = k | X_0 = j). \end{aligned}$$

The last step above utilizes the fact that Markov Chains are memoryless (and so we only have to keep the latest condition). Given the initial state is j , the probability that the system ends up in state k at time step m is

$$P_{kj}^m = \mathbb{P}(X_m = k | X_0 = j).$$

The other term is given by

$$\mathbb{P}(X_{n+m} = i | X_m = k) = \mathbb{P}(X_n = i | X_0 = k) = P_{ik}^n.$$

As a result,

$$P_{ij}^{n+m} = \sum_k P_{ik}^n P_{kj}^m. \quad \blacksquare$$

Corollary 10.2.1.1. *Let P be the matrix of transition probabilities and $P^{(n)}$ be the matrix of n -step transition probabilities. Then $P^{(n)}$ can be found by raising P to the n -th power. That is*

$$P^{(n)} = P^n.$$

Proof: We already know $P^{(0)} = P^0 = I_n$ and $P^{(1)} = P^1 = P$. If we consider the two step transition probabilities, we find by the Chapman–Kolmogorov equations that

$$P_{ij}^2 = \sum_k P_{ik}^1 P_{kj}^1 = \sum_k P_{ik} P_{kj}.$$

This summation multiplies the entries in a row of the matrix on the left by the entries in a column of the matrix on the right and adds up the total. But that is the definition of matrix multiplication! Hence, we see that $P^{(2)} = P^2$. Assume we have already shown that $P^{(n-1)} = P^{n-1}$ and consider P^n . By the Chapman–Kolmogorov equations

$$P^{(n)} = P^{(n-1)} \cdot P^{(1)} = P^{n-1} \cdot P = P^n.$$

Thus, the corollary follows by mathematical induction. ■

Since the matrix $P^{(n)}$ of n -step transition probabilities is simply the n -th power of the transition matrix P , we drop this notation moving forward and simply use P^n .

Example 10.2.1. Consider the simple weather model in Example 10.1.1. Compute the 2-step transition probabilities. What do these probabilities tell us?

Answer: Recall that state 0 is clear skies, 1 is partly cloudy, and 2 is overcast. We found that the appropriate transition probability matrix is

$$P = \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.3 & 0.4 & 0.4 \\ 0.2 & 0.3 & 0.4 \end{bmatrix}.$$

The 2-step transition probabilities are given by

$$\begin{aligned} \begin{bmatrix} P_{00}^2 & P_{01}^2 & P_{02}^2 \\ P_{10}^2 & P_{11}^2 & P_{12}^2 \\ P_{20}^2 & P_{21}^2 & P_{22}^2 \end{bmatrix} &= P^2 \\ &= \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.3 & 0.4 & 0.4 \\ 0.2 & 0.3 & 0.4 \end{bmatrix} \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.3 & 0.4 & 0.4 \\ 0.2 & 0.3 & 0.4 \end{bmatrix} \\ &= \begin{bmatrix} 0.38 & 0.33 & 0.3 \\ 0.35 & 0.37 & 0.38 \\ 0.27 & 0.3 & 0.32 \end{bmatrix} \end{aligned}$$

This matrix gives us the probability of the weather two days in the future. For instance, if today the sky is overcast, then the probability that it will be partly cloudy two days from now is

$$P_{12}^2 = 0.38. \quad \blacksquare$$

Example 10.2.2. Consider the circular random walk in Example 10.1.2. Compute the 5-step transition probabilities. What do these probabilities tell us?

Answer: Recall that the transition probability matrix is

$$P = \begin{bmatrix} 0 & \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 \end{bmatrix}.$$

So, the 5-step transition probability matrix is given by

$$\begin{aligned} & \begin{bmatrix} P_{00}^5 & P_{01}^5 & P_{02}^5 & P_{03}^5 & P_{04}^5 & P_{05}^5 \\ P_{10}^5 & P_{11}^5 & P_{12}^5 & P_{13}^5 & P_{14}^5 & P_{15}^5 \\ P_{20}^5 & P_{21}^5 & P_{22}^5 & P_{23}^5 & P_{24}^5 & P_{25}^5 \\ P_{30}^5 & P_{31}^5 & P_{32}^5 & P_{33}^5 & P_{34}^5 & P_{35}^5 \\ P_{40}^5 & P_{41}^5 & P_{42}^5 & P_{43}^5 & P_{44}^5 & P_{45}^5 \\ P_{50}^5 & P_{51}^5 & P_{52}^5 & P_{53}^5 & P_{54}^5 & P_{55}^5 \end{bmatrix} = P^{(5)} \\ & = P^5 \\ & = \begin{bmatrix} 0 & \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 \end{bmatrix}^5 \\ & = \begin{bmatrix} 0 & \frac{11}{32} & 0 & \frac{5}{16} & 0 & \frac{11}{32} \\ \frac{11}{32} & 0 & \frac{11}{32} & 0 & \frac{5}{16} & 0 \\ 0 & \frac{11}{32} & 0 & \frac{11}{32} & 0 & \frac{5}{16} \\ \frac{5}{16} & 0 & \frac{11}{32} & 0 & \frac{11}{32} & 0 \\ 0 & \frac{5}{16} & 0 & \frac{11}{32} & 0 & \frac{11}{32} \\ \frac{11}{32} & 0 & \frac{5}{16} & 0 & \frac{11}{32} & 0 \end{bmatrix}. \end{aligned}$$

This matrix of transition probabilities gives us probabilities 5 time steps into the future. For example, if we start at node 0, the probability that we land on node 3 after 5 steps is $P_{30}^5 = 5/16$. The probabilities that we land on nodes 1 or 5 after 5 steps are both $P_{10}^5 = P_{50}^5 = 11/32$. The probabilities of landing on nodes 0, 2, or 4 after 5 steps are all zero. This last fact makes sense since on every step of our random walk we must change from an even node to an odd node or vice versa. So, if we start on node 0 (or any other

even node), one step in the future must put us on an odd node. The second step must put us back on an even node, and so on. ■

The transition probabilities in each $P^{(n)}$ are *conditional probabilities*. In the example above, P_{30}^5 is the probability of landing on node 3 after 5 steps *given that we started on node 0*. To go from conditional probabilities to unconditional probabilities, we have to specify the probability distribution of the initial state of the system. That is, we have to specify the probabilities $\mathbb{P}(X_0 = i)$ for every possible state. We can write these probabilities as a column vector

$$\vec{p}_0 = \begin{bmatrix} \mathbb{P}(X_0 = 0) \\ \mathbb{P}(X_0 = 1) \\ \vdots \\ \mathbb{P}(X_0 = m) \end{bmatrix}.$$

Notice that the entries of this vector must sum up to 1. Once we specify the initial probability distribution, we can compute the unconditional probabilities for the outcome of X_n ,

$$\vec{p}_n = \begin{bmatrix} \mathbb{P}(X_n = 0) \\ \mathbb{P}(X_n = 1) \\ \vdots \\ \mathbb{P}(X_n = m) \end{bmatrix},$$

as follows.

Corollary 10.2.1.2. *For a Markov Chain with transition probability matrix P and initial probability distribution $\vec{p}_0$, the unconditional probability distribution of X_n (the state of the system at time step n) is denoted $\vec{p}_n$ and is given by*

$$\vec{p}_n = P^{(n)}\vec{p}_0 = P^n\vec{p}_0.$$

Proof: By conditioning,

$$\begin{aligned} \mathbb{P}(X_n = i) &= \sum_j \mathbb{P}(X_n = i | X_0 = j) \mathbb{P}(X_0 = j) \\ &= \sum_j P_{ij}^n \mathbb{P}(X_0 = j) \\ &= i\text{-th component of } P^n \vec{p}_0. \quad \blacksquare \end{aligned}$$

Example 10.2.3. Consider the simplified weather model in Example 10.1.1. Suppose that on a given day, the probability that the sky is clear is 50%, the probability of being partly cloudy is 25%, and the probability of being overcast is also 25%. What is the probability of having an overcast day 5 days later?

Answer: The information given to us tells us that

$$\vec{p}_0 = \begin{bmatrix} 0.5 \\ 0.25 \\ 0.25 \end{bmatrix}.$$

So, the probability distribution 5 days later is given by

$$\begin{aligned} \vec{p}_5 &= P^{(5)}\vec{p}_0 \\ &= \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.3 & 0.4 & 0.4 \\ 0.2 & 0.3 & 0.4 \end{bmatrix}^5 \begin{bmatrix} 0.5 \\ 0.25 \\ 0.25 \end{bmatrix} \\ &= \begin{bmatrix} 0.33878 & 0.33789 & 0.33734 \\ 0.36591 & 0.36625 & 0.36646 \\ 0.29531 & 0.29586 & 0.2962 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.25 \\ 0.25 \end{bmatrix} \\ &= \begin{bmatrix} 0.3381975 \\ 0.3661325 \\ 0.29567 \end{bmatrix}. \end{aligned}$$

The probability that the sky will be overcast 5 days later is given by the final component of this vector.

$$\mathbb{P}(X_5 = 2) = 0.29567. \blacksquare$$

Example 10.2.4. Consider the maze from Example 10.1.3. If the mouse starts out in room 0, what is the probability that it has reached room 7 after 4 random moves?

Answer: Since the mouse must start from room 0, the initial probability

distribution is given by the column vector

$$\vec{p}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

So, the probability distribution after 4 moves is given by

$$\begin{aligned} \vec{p}_4 &= P^{(4)}\vec{p}_0 \\ &= \begin{bmatrix} 0 & 1/3 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 1 & 0 & 1/2 & 0 & 0 & 0 & 0 \\ 0 & 1/3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 0 & 0 & 0 & 1/3 & 0 & 0 & 0 \\ 0 & 1/3 & 0 & 0 & 0 & 1/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1/2 & 1 & 0 \end{bmatrix}^4 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} \frac{43}{144} & \frac{1}{36} & \frac{11}{36} & \frac{1}{36} & \frac{13}{72} & \frac{11}{72} & \frac{1}{36} & 0 & 0 \\ \frac{1}{24} & \frac{1}{72} & 0 & \frac{1}{48} & \frac{2}{24} & \frac{1}{36} & \frac{1}{24} & 0 & 0 \\ \frac{1}{72} & 0 & \frac{2}{9} & \frac{1}{36} & \frac{1}{11} & \frac{1}{36} & \frac{1}{36} & 0 & 0 \\ \frac{1}{36} & \frac{13}{72} & \frac{1}{18} & \frac{1}{48} & \frac{1}{72} & \frac{1}{36} & \frac{1}{8} & 0 & 0 \\ \frac{1}{13} & \frac{1}{72} & \frac{1}{5} & \frac{1}{11} & \frac{1}{72} & \frac{1}{36} & \frac{1}{9} & 0 & 0 \\ \frac{1}{72} & \frac{1}{36} & \frac{1}{18} & \frac{1}{72} & \frac{1}{4} & \frac{1}{36} & 0 & 0 & 0 \\ \frac{1}{48} & \frac{1}{36} & \frac{1}{12} & \frac{1}{24} & \frac{1}{24} & \frac{1}{72} & 0 & 0 & 0 \\ \frac{1}{36} & \frac{1}{36} & \frac{1}{18} & \frac{1}{8} & \frac{1}{9} & 0 & \frac{1}{12} & 0 & 0 \\ \frac{1}{24} & \frac{1}{36} & 0 & \frac{1}{12} & \frac{1}{12} & \frac{1}{4} & \frac{1}{12} & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 43/144 \\ 1/24 \\ 11/72 \\ 1/36 \\ 13/72 \\ 11/48 \\ 1/36 \\ 1/24 \end{bmatrix}. \end{aligned}$$

So, the probability that the mouse finds room 7 after 4 random moves is

$$\mathbb{P}(X_4 = 7) = \frac{1}{24}. \blacksquare$$

10.3 Transient and Recurrent States

A state i is **accessible** from state j if it is possible that the Markov Chain is in state i at some time in the future if it starts out in state j . An equivalent definition is that

$$P_{ij}^n > 0 \text{ for some } n \geq 0.$$

Clearly a state is accessible from itself. In the case state i is accessible from state j , we write $j \rightarrow i$.

In our simple weather model (Example 10.1.1), state 1 is directly accessible from state 0: $0 \rightarrow 1$. In words, it is possible that tomorrow will be partly cloudy if it is clear today. In fact, each state in this example is directly accessible from every other state.

In our circular random walk (Example 10.1.2), we see that every state is accessible from every other state (though not after a single step). Consider starting from node 0. After 1 step, we could only be on nodes 1 or 5 (so $0 \rightarrow 1$ and $0 \rightarrow 5$). After two steps, we could be on nodes 0, 2, or 4 ($0 \rightarrow 0$, $0 \rightarrow 2$, and $0 \rightarrow 4$). After three steps, we could be on nodes 1, 3, 5 (giving us the final $0 \rightarrow 3$). Hence, every state is accessible from 0. This same argument can be applied to any of the other nodes as well.

For the maze in Example 10.1.3, the situation is a little more interesting. Starting in rooms 0 through 6, any other room in the maze is clearly accessible. However, in our setup of the maze, if the mouse starts in room 7 it stays in room 7. So, no other room is accessible from room 7.

Two states i and j are said to **communicate** if i is accessible from j **AND** j is accessible from i . If two states communicate with each other, we write $i \leftrightarrow j$. Communication has several nice properties. First, any state communicates with itself: $i \leftrightarrow i$. If a state i communicates with a state j , then j communicates with i . Finally, if i communicates with j and j communicates with k , then i must communicate with k : $i \leftrightarrow j$ and $j \leftrightarrow k$ implies $i \leftrightarrow k$. This means that communication is an *equivalence relation*, and it divides up the states of a Markov Chain into a number of classes. Each class consists of states that communicate with each other, and states in separate classes do not communicate with each other.

In Examples 10.1.1 and 10.1.2, all the states belong to the same class. A Markov Chain in which all states communicate is said to be **irreducible**. The maze in Example 10.1.3 has two classes of states: $\{0, 1, 2, 3, 4, 5, 6\}$ and $\{7\}$. As another example, consider the following variant of our circular random walk.

Example 10.3.1. Considered the connected loops below.

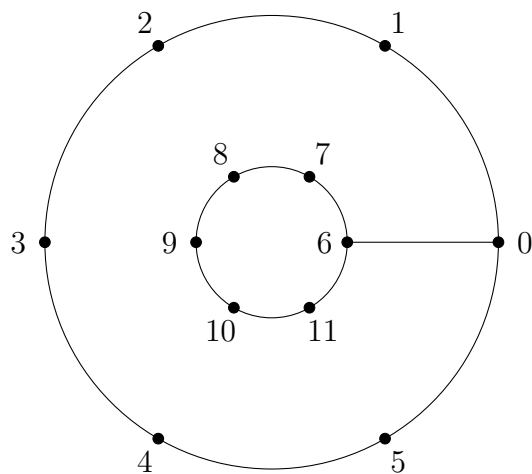


Figure 10.3.1: An Alternate Walk

For nodes 1, 2, 3, 4, 5, 7, 8, 9, 10, and 11, there is a probability of $1/2$ of moving to the node on either side of it. For node 0, there is a probability of $1/3$ of moving to any of the three nodes connected to it. For node 6, there is a probability of $1/2$ of moving to nodes 7 or 11. It is impossible to move from node 6 back to node 0.

There are two classes of states in this example: $\{0, 1, 2, 3, 4, 5\}$ and $\{6, 7, 8, 9, 10, 11\}$. Notice that 0 and 6 do not communicate. Node 6 is accessible from node 0, but node 0 is not accessible from node 6. ■

Let q_i be the probability that the Markov Chain will ever reenter state i if it starts out in state i . If $q_i = 1$, then the system must return to state i at some point in the future. Such states are said to be **recurrent**. If $q_i < 1$, then the system may never return to state i in the future. Such states are called **transient**.

If a state is recurrent, then it follows that the Markov Chain will visit that state *infinitely often* if it ever enters that state. Indeed, if the Markov Chain starts out in a recurrent state, then it will definitely return to that state at some point in the future. But once it returns to that state, the process starts over anew. Hence, it must return to the same state again and again.

On the other hand, if a state is transient, the number of times the system is expected to be in that state is finite. Suppose that state i is transient with $q_i < 1$ being the probability that the system will return to state i if it starts there. The probability that the system visits state i exactly n times must be

$$q_i^{n-1}(1 - q_i)$$

which means that the number of times the system revisits a transient state is a geometric random variable with probability $1 - q_i$. Hence, the expected number of times the system will visit state i is

$$\frac{1}{1 - q_i}.$$

Theorem 10.3.1. *Let (X_i) be a Markov Chain. The following facts about recurrent and transient states are true.*³

³Infinite sums (or series as they are usually called) require care to handle correctly. Fortunately, we are dealing with the simplest possible case when all of the terms are non-negative. See Appendix C for the relevant details.

1.) State i is recurrent if

$$\sum_{n=1}^{\infty} P_{ii}^n = \infty.$$

2.) State i is transient if

$$\sum_{n=1}^{\infty} P_{ii}^n < \infty.$$

3.) If a Markov Chain has a finite number of states, then at least one state must be recurrent.

4.) If state i is recurrent and state i communicates with state j , then state j must also be recurrent. The same statement holds for transient states.

Proof: For any state i in the Markov Chain, let I_n^i be the random variable

$$I_n^i = \begin{cases} 1, & X_n = i \\ 0, & X_n \neq i \end{cases}.$$

In words, I_n^i indicates whether the Markov Chain is in state i at time step n . To determine whether i is recurrent or transient, we need to look at the conditional expectation $\mathbb{E} \left[\sum_{n=0}^{\infty} I_n^i \middle| X_0 = i \right]$ which is precisely the expected total number of times the Markov Chain is in state i given that it started in this state.

$$\begin{aligned} \mathbb{E} \left[\sum_{n=0}^{\infty} I_n^i \middle| X_0 = i \right] &= \sum_{n=0}^{\infty} \mathbb{E} [I_n^i | X_0 = i] \\ &= \sum_{n=0}^{\infty} \mathbb{P}(X_n = i | X_0 = i) \\ &= \sum_{n=0}^{\infty} P_{ii}^n \end{aligned}$$

This computation establishes the first two statements of the theorem since a recurrent state must be revisited an infinite number of times while a transient state is only expected to be visited a finite number of times.

For the third statement, if *all* states in a finite state Markov Chain were transient, then after some (possibly large) number of time steps none of

the states in the chain are expected to be visited again. But this is clearly impossible since the Markov Chain continues for all future time steps!

The fourth statement of the theorem says that recurrence and transience are properties of *communication classes*. Namely, every state in a communication class is either recurrent or transient. We need only prove the theorem for recurrent states since the result for transient states follows logically. Suppose state i communicates with state j and i is recurrent. Since there must be a strictly positive probability of starting in state j and transitioning to state i in the future, there is some integer k such that $P_{ij}^k > 0$. Similarly, there is an integer m such that $P_{ji}^m > 0$ (starting from state i , there must be a positive probability of transitioning to state j at some point in the future). Consider P_{jj}^{k+m+n} which is the probability of returning to state j in $k+m+n$ steps if the process starts in state j . We must have

$$P_{jj}^{k+m+n} \geq P_{ji}^m P_{ii}^n P_{ij}^k$$

since the probability on the left represents the overall probability of returning to state j in $k+m+n$ steps while the product of probabilities on the right represents the probability of going from state j to state i in k steps, returning to state i in n steps, and finally going from state i to state j in m steps (which is only one of many possible kinds of paths from j back to itself). But then

$$\begin{aligned} \sum_{\ell=0}^{\infty} P_{jj}^{\ell} &\geq \sum_{n=0}^{\infty} P_{jj}^{k+m+n} \\ &\geq \sum_{n=0}^{\infty} P_{ji}^m P_{ii}^n P_{ij}^k = P_{ji}^m P_{ij}^k \sum_{n=0}^{\infty} P_{ii}^n. \end{aligned}$$

Since i is recurrent, the sum on the right hand side of the inequality is infinite. As a result, j must also be recurrent by the first statement in the theorem. ■

In the simple weather model, Example 10.1.1, all states must be recurrent since all states communicate. The same is true for the circular random walk in Example 10.1.2. Notice that for any irreducible Markov Chain with a finite number of states, all states must be recurrent!

For the maze in Example 10.1.3, we can argue heuristically that rooms $\{0, 1, 2, 3, 4, 5, 6\}$ must be transient. Suppose that these rooms are recurrent. Then the process would revisit room 6 an infinite number of times. In that case, the probability that the system would *never* enter room 7 is 0. But

once the system enters room 7, it is trapped in room 7 forever. Hence, room 6 cannot be recurrent. That means rooms $\{0, 1, 2, 3, 4, 5, 6\}$ must all be transient. We also know room 7 must be recurrent. A similar argument works for the nested loops in Example 10.3.1. Nodes $\{0, 1, 2, 3, 4, 5\}$ must be transient while nodes $\{6, 7, 8, 9, 10, 11\}$ are recurrent. Of course, the rigorous proof of these statements is through Theorem 10.3.1.

In all of the examples so far, there has been a single recurrent family. That need not be the case in every Markov Chain!

Example 10.3.2. In the diagram below, the nodes 0, 1, 2, 3, 4, and 5 all have equal probability of transitioning to any of the nodes they are connected to. Nodes 6, 7, and 8 can only transition amongst themselves (according to how they are connected). If the process enters node 9, then the system remains on that node for all future times.

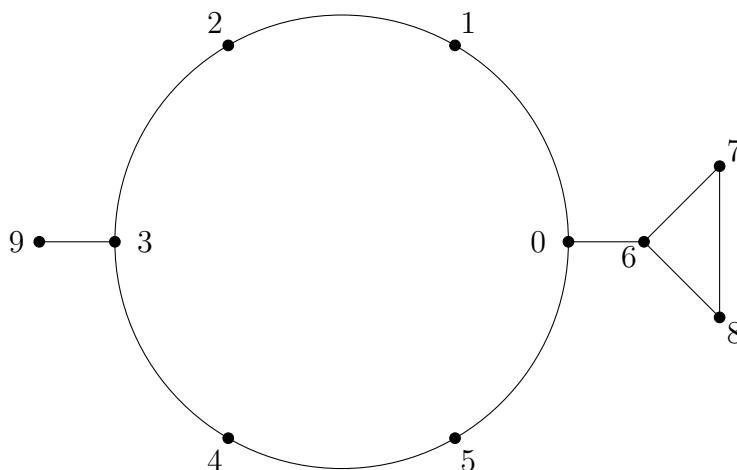


Figure 10.3.2: Random Walk for Example 10.3.2

As before, nodes $\{0, 1, 2, 3, 4, 5\}$ communicate and are transient. The other two families of communicating states, $\{6, 7, 8\}$ and $\{9\}$, are both recurrent. ■

10.4 Application: The Gambler's Ruin Problem

Consider a game where the player has probability p of winning (and so a probability of $q = 1 - p$ of losing). If the player wins, they receive a fixed amount of money. For simplicity, we will assume the player receives \$1 for a win. On the other hand, the player must pay \$1 if they lose the game. Suppose the player starts with \$ M and plays the game repeatedly. The player decides to stop the game if they reach \$ N (with $N \geq M \geq 0$) or if they lose all of their money. What is the probability that the player loses all of their money?

To analyze this situation, we set up a Markov Chain (X_n) with states $\{0, 1, 2, 3, \dots, N\}$ representing the different amounts of money the player can have. The random variable X_n represents the total money the player has after n plays of the game. For the first state, we have $X_0 = M$. Since the player ends the process whenever $X_n = 0$ or N , we must have $P_{00} = 1$, $P_{i0} = 0$ ($i > 0$), $P_{NN} = 1$, and $P_{iN} = 0$ ($i < N$). If $X_n = i$ (i.e. the gambler

has i dollars after n rounds) with $0 < i < N$, then $X_{n+1} = i + 1$ with probability p or $X_{n+1} = i - 1$ with probability q . So

$$\begin{aligned} P_{i+1,i} &= p, \\ P_{i-1,i} &= q, \end{aligned}$$

and all other transition probabilities are 0. The $(N + 1) \times (N + 1)$ transition matrix has the form

$$P = \begin{bmatrix} 1 & q & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & q & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & p & 0 & q & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & p & 0 & q & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & p & 0 & q & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & q & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & p & 1 \end{bmatrix}$$

There are three communication classes: $\{0\}$, $\{1, 2, 3, \dots, N - 1\}$, and $\{N\}$. Clearly 0 and N are recurrent. All of the other states are transient.

Let P_M be the probability that the Markov Chain reaches state N starting from state M ; clearly $P_0 = 0$ (since if the gambler starts with no money they stop the game immediately), and $P_N = 1$ (since the gambler stops the game immediately if they start with N dollars). To compute the other probabilities, we can condition on the outcome of the first play of the game.

$$\begin{aligned} P_M &= \mathbb{P}(X_n = N | X_0 = M) \\ &= \mathbb{P}(X_n = N | X_1 = M + 1, X_0 = M) \mathbb{P}(X_1 = M + 1 | X_0 = M) \\ &\quad + \mathbb{P}(X_n = N | X_1 = M - 1, X_0 = M) \mathbb{P}(X_1 = M - 1 | X_0 = M) \\ &= p \mathbb{P}(X_n = N | X_1 = M + 1, X_0 = M) + q \mathbb{P}(X_n = N | X_1 = M - 1, X_0 = M) \\ &= p P_{M+1} + q P_{M-1} \quad (M = 1, 2, 3, \dots, N - 1) \end{aligned}$$

where in the last step we used the basic fact that transition probabilities do not change from one time step to the next. Using the fact that $p + q = 1$, we

have

$$\begin{aligned}(p+q)P_M &= pP_{M+1} + qP_{M-1} \\ p(P_{M+1} - P_M) &= q(P_M - P_{M-1}) \\ P_{M+1} - P_M &= \frac{q}{p}(P_M - P_{M-1}).\end{aligned}$$

If we look at these terms individually, we see a clear pattern:

$$\begin{aligned}P_2 - P_1 &= \frac{q}{p}(P_1 - P_0) = \frac{q}{p}P_1, \\ P_3 - P_2 &= \frac{q}{p}(P_2 - P_1) = \left(\frac{q}{p}\right)^2 P_1, \\ P_4 - P_3 &= \frac{q}{p}(P_3 - P_2) = \left(\frac{q}{p}\right)^3 P_1, \\ &\vdots \\ P_i - P_{i-1} &= \frac{q}{p}(P_{i-1} - P_{i-2}) = \left(\frac{q}{p}\right)^{i-1} P_1, \\ &\vdots \\ P_N - P_{N-1} &= \frac{q}{p}(P_{N-1} - P_{N-2}) = \left(\frac{q}{p}\right)^{N-1} P_1.\end{aligned}$$

If we add up the terms $(P_2 - P_1) + (P_3 - P_2) + \cdots + (P_i - P_{i-1})$ and notice the cancellations that occur, we find

$$P_i - P_1 = \left[\frac{q}{p} + \left(\frac{q}{p}\right)^2 + \cdots + \left(\frac{q}{p}\right)^{i-1} \right] P_1$$

or after rearranging

$$P_i = \left[1 + \frac{q}{p} + \left(\frac{q}{p}\right)^2 + \cdots + \left(\frac{q}{p}\right)^{i-1} \right] P_1.$$

If $p = 1/2$ (and so $q/p = 1$), we find that $P_i = iP_1$. Otherwise

$$P_i = \frac{1 - (q/p)^i}{1 - (q/p)} P_1.$$

Finally, we use the fact that $P_N = 1$ to compute P_1 . If $p = 1/2$, we have $P_1 = 1/N$. In all other cases, we find

$$P_1 = \frac{1 - (q/p)}{1 - (q/p)^N}.$$

Combining this result with our previous calculations gives

$$P_M = \mathbb{P}(\text{gambler wins } \$N \text{ before going bankrupt} \mid \text{gambler started with } \$M) \\ = \begin{cases} \frac{1 - (q/p)^M}{1 - (q/p)^N}, & p \neq 1/2 \\ \frac{M}{N}, & p = 1/2 \end{cases}$$

Example 10.4.1. Suppose Alice and Bob play the following variant of horse-shoes. Both players throw one of their horseshoes at the stake, and the person whose horseshoe is closest to the stake keeps both horseshoes. Both players start out with 10 horseshoes, and the game is over when one player has won all of the horseshoes. Alice is more skilled at the game and has a probability of 0.6 of winning each round. What is the probability that Alice wins the game?

Answer: This is a variant of the Gambler's Ruin Problem. We want to know that probability that the gambler (here Alice) reaches a total of $N = 20$ starting with $M = 10$ given that her probability of winning is $p = 0.6$ (which means $q = 1 - p = 0.4$). According to our results, Alice's probability of winning the game is

$$P_{10} = \frac{1 - (0.4/0.6)^{10}}{1 - (0.4/0.6)^{20}} \\ = \frac{59049}{60073} \approx 0.983.$$

So, Alice has over a 98% chance of winning the game. ■

In the last example, N was the total amount of winnings that could possibly be earned. If the gambler is playing against a very wealthy opponent (like a casino), then as $N \rightarrow \infty$ we find

$$P_M \rightarrow \begin{cases} 1 - (q/p)^M, & p > 1/2 \\ 0, & p \leq 1/2 \end{cases}$$

since $q/p \geq 1$ when $p \leq 1/2$. So, a player who has a 50% chance or less of winning will eventually go broke playing against an incredibly wealthy opponent. Even if the game is in the player's favor ($p > 0.5$), there is still a positive probability that the player will go broke since their opponent has a virtually unlimited fortune.

Example 10.4.2. Suppose a gambler finds a casino game where their probability of winning is $p = 0.51$. What is the probability that they still go broke playing the game if they start with \$20 (assuming \$1 bets each game play)?

Answer: We can safely let $N \rightarrow \infty$ since the casino surely has far more than \$20 in its coffers! In this case,

$$P_{20} = 1 - \left(\frac{0.49}{0.51} \right)^{20} \approx 0.551.$$

So, there is still roughly a 45% chance that the gambler will go bankrupt even when the game is slightly in their favor. ■

Of course, casino games are set up so that the casino has at least a slight advantage over the player. For example, the probability of winning at craps is about $p = 0.49$.⁴ Even though this is as close to fair as you can hope for in a casino, the slight disadvantage to the player is enough to tilt things dramatically in favor of the house in the long run.

Theorem 10.4.1. *For the Gambler's Ruin Problem, let E_M be the expected number of game plays if the player starts with \$M and has a probability p of winning. Then if $p \neq 1/2$, we have*

$$E_M = \frac{M}{q-p} - \frac{N}{q-p} \left(\frac{1 - (q/p)^M}{1 - (q/p)^N} \right).$$

In the case $p = 1/2$, $E_M = M(N - M)$.

Sketch of Proof: If the gambler has no money, then no games are played. Similarly, if the gambler has all of the money ($X_0 = N$) no games are played. Hence $E_0 = E_N = 0$. The proof follows the same steps as the computation of

⁴This is strictly the probability that the person rolling the dice wins. There are other side bets that take place during game play which have different odds of paying out.

P_M . Namely, we condition on the outcome of the first game. If we let G be the total number of games played, then for $M = 1, 2, 3, \dots, N - 1$ we have

$$\begin{aligned}
 E_M &= \sum_{n=1}^{\infty} n\mathbb{P}(G = n|X_0 = M) \\
 &= \sum_{n=1}^{\infty} n\mathbb{P}(G = n|X_1 = M + 1, X_0 = M)\mathbb{P}(X_1 = M + 1|X_0 = M) \\
 &\quad + \sum_{n=1}^{\infty} n\mathbb{P}(G = n|X_1 = M - 1, X_0 = M)\mathbb{P}(X_1 = M - 1|X_0 = M) \\
 &= p \sum_{n=1}^{\infty} n\mathbb{P}(G = n|X_1 = M + 1, X_0 = M) + q \sum_{n=1}^{\infty} n\mathbb{P}(G = n|X_1 = M - 1, X_0 = M) \\
 &= p \sum_{n=1}^{\infty} n\mathbb{P}(G = n - 1|X_0 = M + 1) + q \sum_{n=1}^{\infty} n\mathbb{P}(G = n - 1|X_0 = M - 1) \\
 &= p \sum_{n=1}^{\infty} (n - 1)\mathbb{P}(G = n - 1|X_0 = M + 1) + p \sum_{n=1}^{\infty} \mathbb{P}(G = n - 1|X_0 = M + 1) \\
 &\quad + q \sum_{n=1}^{\infty} (n - 1)\mathbb{P}(G = n - 1|X_0 = M - 1) + q \sum_{n=1}^{\infty} \mathbb{P}(G = n - 1|X_0 = M - 1) \\
 &= pE_{M+1} + p + qE_{M-1} + q \\
 &= pE_{M+1} + qE_{M-1} + 1.
 \end{aligned}$$

Rearranging as we did for P_M gives the relationship

$$E_{M+1} - E_M = \frac{q}{p}(E_M - E_{M-1}) + \frac{1}{p}.$$

The analysis of this relationship is a bit more involved than for P_M , so we simply quote the final result above. ■

Example 10.4.3. We play the following game. You are given two dice and the game follows the basic rules of craps given in Example 6.5.6. You initially have \$10, and you gain or lose a dollar on each play of the game. The game continues until you double your money or you go bankrupt. What is your probability of winning the game? How many turns is the game expected to take?

Answer: We will take the probability of winning at craps to be $p = 0.49$. Since $N = 20$ and $M = 10$, we have

$$P_{10} = \frac{1 - (0.51/0.49)^{10}}{1 - (0.51/0.49)^{20}} \approx 0.401.$$

So, there is only a 40% probability that you win the game. As for the expected number of turns, the theorem above gives

$$E_{10} = \frac{10}{0.51 - 0.49} - \frac{20}{0.51 - 0.49} \left(\frac{1 - (0.51/0.49)^{10}}{1 - (0.51/0.49)^{20}} \right) \approx 98.7.$$

So, the game is expected to take about 99 turns before finishing. ■

Example 10.4.4. Suppose that we are trying to test whether a new drug therapy for a disease is more effective than an older, well-established therapy. Suppose the new drug therapy has a cure rate of p_1 which we believe is larger than the cure rate for the older therapy, $p_2 = 0.6$. To test whether the new therapy is more effective, we perform the following experiment. All participants in a drug trial are separated into pairs. The first person in each pair receives the new drug and the second person receives the traditional drug. We keep track of the net number of people cured by the new therapy: (total cured by the new drug) – (total cured by the old drug). We set a threshold N , and if the net total reaches N before it reaches $-N$, we decide that the new therapy has the higher cure rate. Otherwise, we decide that the traditional drug has a higher cure rate. We have evidence from a preliminary study that suggests the new drug has a cure rate of $p_1 = 0.68$. What should our threshold N be to ensure that we make the correct decision with a probability of 0.95? How many pairs of people do we expect to need for the trial if we use this value of N ?

Answer: To relate this to the Gambler's Ruin Problem, we first throw out all pairs where both therapies cure the illness or where both therapies fail (since these cases do not contribute to the *net* number cured by therapy 1). Note that for a pair (a, b) , where a receives the new drug and b receives

the older one,

$$\begin{aligned}\mathbb{P}(a \text{ cured but not } b) &= (0.68)(1 - 0.6) = 0.272, \\ \mathbb{P}(\text{both cured}) &= (0.68)(0.6) = 0.408, \\ \mathbb{P}(b \text{ cured but not } a) &= (1 - 0.68)(0.6) = 0.192, \\ \mathbb{P}(\text{neither cured}) &= (1 - 0.68)(1 - 0.6) = 0.128\end{aligned}$$

(where we assume independence of the individual trials). Notice that we will be throwing out pairs where either both are cured or neither are cured. This means that 53.6% of our pairs will not contribute to our test! For the remaining pairs, note that there is a probability $p = 0.272/0.464 \approx 0.586$ that the new drug cures the patient and the older drug does not. Likewise, $q = 0.192/0.464 \approx 0.414$ is the probability that the traditional therapy cures the illness in a pair while the new one fails.

We consider a Gambler's Ruin set up with the player starting with a fortune of N ; for each of the remaining pairs where the new therapy is effective the player wins 1 point while losing a point for each pair where the older therapy is effective. The game stops when the player's fortune reaches $2N$ or zero. Notice that this is equivalent to the *net* number of people cured by the new drug reaching N before reaching $-N$. So, the probability of the player winning is

$$P_N = \frac{1 - (0.414/0.586)^N}{1 - (0.414/0.586)^{2N}}$$

which we want to equal 0.95 or larger. Instead of trying to solve this for N directly, it is easier to look at a table of values.

N	P_N
5	0.850
6	0.889
7	0.919
8	0.942
9	0.958
10	0.970

So it would appear we need to take $N = 9$ to ensure that there is a 95% chance of making the correct decision. Using our results in Theorem [10.4.1](#)

(with $M = 9$, $N = 18$) we have

$$E_9 = \frac{9}{0.464 - 0.586} - \frac{18}{0.464 - 0.586} \left(\frac{1 - (0.414/0.586)^9}{1 - (0.414/0.586)^{18}} \right) \approx 67.6.$$

So we expect to need 68 pairs where one of the therapies cures the illness *but not the other*. Since such pairs only account for 46.4% of all possible pairs, we expect to need

$$68/0.464 \approx 146.6$$

pairs of patients. Hence, we expect that the trial will need at least 147 pairs and so 294 patients in total. ■

10.5 Properties of Transition Matrices

Transition matrices for finite state Markov Chains have a number of nice properties. Recall that the matrix of transition probabilities P has entries P_{ij} which give the probability of transitioning from state j to state i . As a result, each *column* of the transition matrix sums to 1. The following theorem tells us that all eigenvalues of a transition matrix are no bigger than 1 in size (so all real eigenvalues lie in the range $[-1, 1]$), and 1 is *always* an eigenvalue of a transition matrix.

Theorem 10.5.1. *If P is a transition matrix for a finite state Markov Chain, then 1 must be an eigenvalue of P . Moreover, every eigenvalue λ of P satisfies $|\lambda| \leq 1$.*

Proof: Suppose that P is an $n \times n$ transition matrix; then the matrix P^T has *rows* that all sum to one. Let $\vec{v}$ be the column vector consisting of all ones. Then we must have

$$P^T \vec{v} = \vec{v}$$

since each entry in the resulting vector is the sum of a row from P^T . So, P^T has an eigenvalue of one. By the results in Section 8.7, P must also have an eigenvalue of one.

For the second part of the theorem, if λ is an eigenvalue for P , it is also an eigenvalue for P^T . Hence, there is a non-zero vector $\vec{v}$ such that $P^T \vec{v} = \lambda \vec{v}$.

If we let the components of $\vec{v}$ be given by v_i , then we have

$$P^T \vec{v} = \begin{bmatrix} p_{11} & p_{21} & \cdots & p_{n1} \\ p_{12} & p_{22} & \cdots & p_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ p_{1n} & p_{2n} & \cdots & p_{nn} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} \lambda v_1 \\ \lambda v_2 \\ \vdots \\ \lambda v_n \end{bmatrix} = \lambda \vec{v}.$$

Since $\vec{v}$ is not the zero vector, at least one of the v_i must be different from zero, and so one of them must have the largest (non-zero) absolute value. Suppose this happens for component j (which means $|v_i| \leq |v_j|$ for $i = 1, 2, \dots, n$ and $v_j \neq 0$). If we look at row j , we must have

$$p_{1j}v_1 + p_{2j}v_2 + \cdots + p_{nj}v_n = \lambda v_j.$$

Using the triangle inequality ($|x + y| \leq |x| + |y|$), we find

$$\begin{aligned} |\lambda||v_j| &= |p_{1j}v_1 + p_{2j}v_2 + \cdots + p_{nj}v_n| \\ &\leq p_{1j}|v_1| + p_{2j}|v_2| + \cdots + p_{nj}|v_n| \\ &\leq (p_{1j} + p_{2j} + \cdots + p_{nj})|v_j| \\ &\leq |v_j|. \end{aligned}$$

In the third line of the inequality, we used that $|v_j|$ is larger than the other components; in the final line we used the fact the rows of P^T sum to one. Dividing by $|v_j|$ gives the desired result $|\lambda| \leq 1$. ■

To state the next result, we need the following definitions. A matrix is said to be **positive** if all of its entries are strictly positive. A square matrix is said to be **regular** if some power of the matrix is positive. So,

$$M_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

is a positive matrix (and regular) while

$$M_2 = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$$

is not a positive matrix. However,

$$M_2^2 = \begin{bmatrix} 2 & 3 \\ 6 & 11 \end{bmatrix}$$

is a positive matrix, and so M_2 is regular. Notice that if P^n is a positive matrix, then P^{n+1} and every higher power of P must also be a positive matrix.

It turns out that regular transition matrices have a very nice property.

Theorem 10.5.2. *Let P be an $n \times n$ transition matrix for a finite state Markov Chain. If P is a regular matrix and λ is any eigenvalue that is not equal to 1, then $|\lambda| < 1$.*

Proof: We prove that if an eigenvalue has the property $|\lambda| = 1$, then $\lambda = 1$. Suppose first that the transition matrix P is positive. Since λ is also an eigenvalue for P^T , there is a non-zero vector $\vec{v}$ with $P^T \vec{v} = \lambda \vec{v}$. Looking at our proof of the previous theorem, we showed

$$|\lambda||v_j| \leq p_{1j}|v_1| + p_{2j}|v_2| + \cdots + p_{nj}|v_n| \leq |v_j|$$

where v_j was the entry of $\vec{v}$ with the largest magnitude. But if $|\lambda| = 1$, all of the inequalities in the previous proof must actually be equalities! Hence,

$$p_{1j}|v_1| + p_{2j}|v_2| + \cdots + p_{nj}|v_n| = |v_j| = (p_{1j} + p_{2j} + \cdots + p_{nj})|v_j|$$

since each column of P sums to 1. Rearranging the outer equality gives

$$p_{1j}(|v_j| - |v_1|) + p_{2j}(|v_j| - |v_2|) + \cdots + p_{nj}(|v_j| - |v_n|) = 0.$$

Since all of the entries of P are positive and $|v_j| - |v_i| \geq 0$, the only way the sum above can be zero is if $|v_i| = |v_j|$ for all $i = 1, 2, \dots, n$. Since our proof of the theorem above also gives

$$|p_{1j}v_1 + p_{2j}v_2 + \cdots + p_{nj}v_n| = p_{1j}|v_1| + p_{2j}|v_2| + \cdots + p_{nj}|v_n|$$

when $|\lambda| = 1$, it must be the case that all of the v_i have the same sign (since all of the p_{ij} are positive). Hence, one eigenvector for λ must be

$$\vec{v} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

since the other possibility (all entries -1) is just a multiple of the one given. But the eigenvalue for this particular vector is definitely $\lambda = 1$.

If P is regular, then there is an exponent n so that P^n is a positive matrix. Of course, it is still a transition matrix and the previous calculation shows that if P^n has an eigenvalue of magnitude 1, then that eigenvalue must be 1. Now, if λ is an eigenvalue of P , then λ^n is an eigenvalue of P^n with the same eigenvector.⁵ So, if P has an eigenvector satisfying $|\lambda| = 1$, then P^n has an eigenvector λ^n satisfying $|\lambda^n| = |\lambda|^n = 1$. Hence $\lambda^n = 1$. But since P^{n+1} is also a positive matrix, we must have $\lambda^{n+1} = 1$. Therefore,

$$\begin{aligned}\lambda^n &= \lambda^{n+1} \\ \lambda^n(\lambda - 1) &= 0 \\ \lambda &= 1\end{aligned}$$

since $|\lambda| = 1$ (and so $\lambda \neq 0$). ■

Corollary 10.5.2.1. *If P is a regular $n \times n$ transition matrix that is diagonalizable, then 1 is an eigenvalue of P with algebraic multiplicity 1. All other eigenvalues of P must satisfy $|\lambda| < 1$.*

Proof: That 1 is an eigenvalue of P and any eigenvalue of P not equal to 1 must satisfy $|\lambda| < 1$ are the conclusions of the previous two theorems. The new result here is that 1 can only have algebraic multiplicity 1. If P is diagonalizable, then the algebraic multiplicity of an eigenvalue must equal its geometric multiplicity. But the considerations of the previous theorem tell us that when P is regular, the only family corresponding to $\lambda = 1$ for P^T is

$$\vec{v} = c \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

⁵If $P\vec{v} = \lambda\vec{v}$, then

$$\begin{aligned}P^n \vec{v} &= P^{n-1}(P\vec{v}) \\ &= \lambda P^{n-1} \vec{v} \\ &= \lambda P^{n-2}(P\vec{v}) \\ &= \lambda^2 P^{n-2} \vec{v} = \dots = \lambda^n \vec{v}.\end{aligned}$$

Hence, the geometric multiplicity of $\lambda = 1$ is 1 for P^T . But this must be the case for P as well. As a result, the algebraic multiplicity of $\lambda = 1$ must also be 1. ■

We can now state the most important result of this section.

Theorem 10.5.3. *Let P be a regular $n \times n$ transition matrix. Then there is an $n \times n$ matrix L such that*

$$\lim_{k \rightarrow \infty} P^k = L.$$

Each column of L is identical to the same vector $\vec{\pi}$ with entries summing to one.

Proof: We only prove this result in the special case that P is diagonalizable. If P is diagonalizable, then the corollary above tells us that P has 1 as eigenvalue with algebraic multiplicity 1, and all other eigenvalues satisfy $|\lambda| < 1$.⁶ This means that we can decompose P as $P = BDB^{-1}$ where D is a diagonal matrix of the form

$$D = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

Notice that

$$\lim_{k \rightarrow \infty} D^k = \lim_{k \rightarrow \infty} \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2^k & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3^k & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n^k \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

⁶The main difficulty in extending this theorem to the non-diagonalizable case is that 1 may have a deficiency for eigenvalue 1. Handling this correctly requires the *Jordan Canonical Form* for a matrix which is a topic beyond the scope of this text.

since all of the other eigenvalues have size less than 1. This gives us

$$\lim_{k \rightarrow \infty} P^k = \lim_{k \rightarrow \infty} BD^k B^{-1} = B \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} B^{-1}$$

which proves that the limiting matrix exists.

Rather than focusing on the matrix B to determine the form of the limiting matrix L , we will argue in a more indirect way. Recall that one way of thinking about matrix multiplication $M_1 M_2$ is that the resulting matrix has columns which result from the action of M_1 on each column of M_2 . If we look at the product PL , we find

$$PL = P \lim_{k \rightarrow \infty} P^k = \lim_{k \rightarrow \infty} P^{k+1} = L.$$

This implies that each column of L is an eigenvector of P corresponding to the eigenvalue 1. Since 1 has algebraic multiplicity 1, this tells us that each column of L is a multiple of the same eigenvector. Since P is a regular matrix, it must be the case that L is a positive matrix. In fact, each column of L must sum to one. To see this, let $\vec{1}$ be the column vector of all ones. Since the rows of P^T sum to one, we must have $P^T \vec{1} = \vec{1}$. Then

$$\begin{aligned} L^T \vec{1} &= \left(\lim_{k \rightarrow \infty} P^k \right)^T \vec{1} \\ &= \lim_{k \rightarrow \infty} (P^T)^k \vec{1} \\ &= \lim_{k \rightarrow \infty} (P^T)^{k-1} P^T \vec{1} \\ &= \lim_{k \rightarrow \infty} (P^T)^{k-1} \vec{1} \\ &= \vec{1} \end{aligned}$$

by induction. But this means that each row of L^T sums to one. Hence, each column of L must sum to one. Hence, L is a matrix of transition probabilities. Since the algebraic multiplicity is 1, there is a *unique* eigenvector of P corresponding to $\lambda = 1$ with positive entries that sum to one. ■

Example 10.5.1. Consider the Markov Chain with transition matrix

$$P = \begin{bmatrix} 1/3 & 1/2 & 0 \\ 1/3 & 0 & 1/2 \\ 1/3 & 1/2 & 1/2 \end{bmatrix}.$$

Determine if P is a regular matrix. If so, find its eigenvalues and determine the limiting matrix

$$L = \lim_{k \rightarrow \infty} P^k.$$

Solution: Squaring P , we see that

$$P^2 = \begin{bmatrix} 5/18 & 1/6 & 1/4 \\ 5/18 & 5/12 & 1/4 \\ 4/9 & 5/12 & 1/2 \end{bmatrix}.$$

So, P is a regular transition matrix. The characteristic polynomial for P is

$$\det(P - \lambda I) = -\lambda^3 + \frac{5\lambda^2}{6} + \frac{\lambda}{4} - \frac{1}{12} = -\frac{1}{12}(\lambda - 1)(12\lambda^2 + 2\lambda - 1).$$

We already knew that 1 was going to be an eigenvalue (which allowed us to factor the polynomial), the only question is whether 1 has algebraic multiplicity 1. The remaining eigenvalues are

$$\lambda = \frac{1}{12}(-1 \pm \sqrt{13})$$

which are both in the range $-1 < \lambda < 1$. So, the limiting matrix L must be the matrix whose columns are all the unique eigenvector corresponding to $\lambda = 1$ whose entries add up to 1. A basis eigenvector for $\lambda = 1$ is given by

$$\vec{v} = \begin{bmatrix} 1/2 \\ 2/3 \\ 1 \end{bmatrix},$$

but the entries of this vector sum to 13/6! So to get a vector of probabilities, we take for the eigenvector

$$\vec{\pi} = \frac{6}{13} \begin{bmatrix} 1/2 \\ 2/3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3/13 \\ 4/13 \\ 6/13 \end{bmatrix}.$$

This means that the limiting matrix L is

$$L = \lim_{k \rightarrow \infty} P^k = \begin{bmatrix} 3/13 & 3/13 & 3/13 \\ 4/13 & 4/13 & 4/13 \\ 6/13 & 6/13 & 6/13 \end{bmatrix}. \blacksquare$$

We end this section with a useful formula for determining whether certain states are transient or recurrent in the case that the transition matrix is diagonalizable.

Theorem 10.5.4. *Suppose that P is an $(m+1) \times (m+1)$ diagonalizable transition matrix for a Markov Chain. Let $P = BDB^{-1}$ with*

$$D = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_1 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_m \end{bmatrix}$$

the matrix of eigenvalues and B the matrix whose columns consist of the corresponding eigenvectors. Recall that 1 is guaranteed to be one of the eigenvalues (here labeled λ_0). Then,

$$P_{ii}^n = \sum_{j=0}^m \lambda_j^n [B]_{ij} [B^{-1}]_{ji}.$$

Proof: Since $P^n = BD^nB^{-1}$, we have

$$\begin{aligned} P_{ii}^n &= [BD^nB^{-1}]_{ii} \\ &= \sum_{j=0}^m [B]_{ij} [D^nB^{-1}]_{ji} \\ &= \sum_{j=0}^m [B]_{ij} \left(\sum_{k=0}^m [D^n]_{jk} [B^{-1}]_{ki} \right) \\ &= \sum_{j=0}^m \sum_{k=0}^m [B]_{ij} [D^n]_{jk} [B^{-1}]_{ki}. \end{aligned}$$

The matrix D^n is diagonal with $[D^n]_{jj} = \lambda_j^n$ and all other entries equal to zero. As a result, the sum over k will only produce a non-zero term when $k = j$. This gives us

$$\begin{aligned} P_{ii}^n &= \sum_{j=0}^m \sum_{k=0}^m [B]_{ij} [D^n]_{jk} [B^{-1}]_{ki} \\ &= \sum_{j=0}^m [B]_{ij} [D^n]_{jj} [B^{-1}]_{ji} \\ &= \sum_{j=0}^m \lambda_j^n [B]_{ij} [B^{-1}]_{ji}. \blacksquare \end{aligned}$$

As a simple example of how useful this theorem can be, we present the following corollaries which allow us to easily determine whether certain states are transient in special cases.

Corollary 10.5.4.1. *Suppose that $\lambda_0 = 1, \dots, \lambda_k$ are the eigenvalues of P with magnitude 1 while $|\lambda_j| < 1$ for $k+1 \leq j \leq m$. If $[B]_{ij} [B^{-1}]_{ji} = 0$ for $0 \leq j \leq k$, then state i is transient.*

Proof: We break up the sum from the previous theorem into two pieces:

$$P_{ii}^n = \sum_{j=0}^k \lambda_j^n [B]_{ij} [B^{-1}]_{ji} + \sum_{j=k+1}^m \lambda_j^n [B]_{ij} [B^{-1}]_{ji},$$

where the first sum is over the indices where $|\lambda_j| = 1$. According to our assumptions, the first part of the sum is zero leaving

$$P_{ii}^n = \sum_{j=k+1}^m \lambda_j^n [B]_{ij} [B^{-1}]_{ji}.$$

Using the results on geometric series from Appendix C,

$$\begin{aligned}
 \sum_{n=1}^{\infty} P_{ii}^n &= \sum_{n=1}^{\infty} \left(\sum_{j=k+1}^m \lambda_j^n [B]_{ij} [B^{-1}]_{ji} \right) \\
 &= \sum_{j=k+1}^m \left(\sum_{n=1}^{\infty} \lambda_j^n \right) [B]_{ij} [B^{-1}]_{ji} \\
 &= \sum_{j=k+1}^m \left(\frac{\lambda_j}{1 - \lambda_j} \right) [B]_{ij} [B^{-1}]_{ji} \\
 &< \infty. \blacksquare
 \end{aligned}$$

Corollary 10.5.4.2. *Suppose that $\lambda_0 = 1$ is the unique eigenvalue of magnitude 1 for P . Then state i is transient if and only if $[B]_{i0} [B^{-1}]_{0i} = 0$. This means that i is transient precisely when the i -th row of the limiting probability matrix consists of all zeros.*

Proof: By the theorem above,

$$P_{ii}^n = [B]_{i0} [B^{-1}]_{0i} + \sum_{j=1}^m \lambda_j^n [B]_{ij} [B^{-1}]_{ji}.$$

Since $|\lambda_j| < 1$ for all $j > 0$, we have

$$\begin{aligned}
 \sum_{n=1}^{\infty} \sum_{j=1}^m \lambda_j^n [B]_{ij} [B^{-1}]_{ji} &= \sum_{j=1}^m \left(\sum_{n=1}^{\infty} \lambda_j^n \right) [B]_{ij} [B^{-1}]_{ji} \\
 &= \sum_{j=1}^m \left(\frac{\lambda_j}{1 - \lambda_j} \right) [B]_{ij} [B^{-1}]_{ji} \\
 &< \infty.
 \end{aligned}$$

So if $[B]_{i0} [B^{-1}]_{0i} = 0$, state i is definitely transient. On the other hand, if $[B]_{i0} [B^{-1}]_{0i} \neq 0$ the sum $\sum_{n=1}^{\infty} P_{ii}^n$ will definitely diverge.

From the proof of Theorem 10.5.3, we know that the limiting matrix L

is given by

$$L = \lim_{\ell \rightarrow \infty} P^\ell = B \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} B^{-1}.$$

This in turn tells us that

$$[L]_{ij} = [B]_{i0}[B^{-1}]_{0j}.$$

As a result, if $[B]_{i0}[B^{-1}]_{0i} = 0$ then $L_{ii} = 0$. Since we know that the columns of L are identical, the entire i -th row of L must be zero. ■

10.6 Limiting Probabilities

The previous section showed that the transition matrix of a Markov Chain has many nice properties – especially if the matrix is regular. In this section, we will explore some conditions that guarantee that the transition matrix of a Markov Chain is regular. As a result, these special Markov Chains will have a definite limiting behavior thanks to Theorem 10.5.3.

The **set of possible recurrence times** for a state i is

$$\{n | P_{ii}^n > 0\}.$$

In words, this is the set of time steps where it is possible to return to state i if the system starts in state i . If the set of recurrence times happens to be empty (i.e. if the system can never return to state i once it has visited it), then the period is undefined. The *period*, d , of state i is the greatest common divisor of all of the recurrence times:

$$d = \gcd\{n | P_{ii}^n > 0\}.$$

A state is said to be *aperiodic* if its period is 1. Notice that if a Markov Chain can remain in a particular state from one time step to the next, then that state is necessarily aperiodic. However, a state can be aperiodic in other ways. For instance, if a state can recur after 2 steps and after 3 steps, then the gcd of the recurrence times is 1, and the state is aperiodic. We present the following lemma without proof.

Lemma 10.6.1 (Periodicity of Classes). *Periodicity is a property of communication classes. That is, if state i has period d and i and j communicate, then state j must also have period d .*

Example 10.6.1. Consider the circular random walk in Example 10.1.2. The 1-step through 4-step transition probability matrices are

$$\begin{aligned}
 P &= \begin{bmatrix} 0 & 0.5 & 0 & 0 & 0 & 0.5 \\ 0.5 & 0 & 0.5 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0.5 & 0 & 0.5 \\ 0.5 & 0 & 0 & 0 & 0.5 & 0 \end{bmatrix}, \\
 P^2 &= \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{4} & 0 & \frac{1}{4} & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{4} & 0 & \frac{1}{4} \\ \frac{1}{4} & 0 & \frac{1}{2} & 0 & \frac{1}{4} & 0 \\ 0 & \frac{1}{4} & 0 & \frac{1}{2} & 0 & \frac{1}{4} \\ \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{2} \end{bmatrix}, \\
 P^3 &= \begin{bmatrix} 0 & \frac{3}{8} & 0 & \frac{1}{4} & 0 & \frac{3}{8} \\ \frac{3}{8} & 0 & \frac{3}{8} & 0 & \frac{1}{4} & 0 \\ 0 & \frac{3}{8} & 0 & \frac{3}{8} & 0 & \frac{1}{4} \\ \frac{1}{4} & 0 & \frac{3}{8} & 0 & \frac{3}{8} & 0 \\ 0 & \frac{1}{4} & 0 & \frac{3}{8} & 0 & \frac{3}{8} \\ \frac{3}{8} & 0 & \frac{1}{4} & 0 & \frac{3}{8} & 0 \end{bmatrix}, \\
 P^4 &= \begin{bmatrix} \frac{3}{8} & 0 & \frac{5}{16} & 0 & \frac{5}{16} & 0 \\ 0 & \frac{3}{8} & 0 & \frac{5}{16} & 0 & \frac{5}{16} \\ \frac{5}{16} & 0 & \frac{3}{8} & 0 & \frac{5}{16} & 0 \\ 0 & \frac{5}{16} & 0 & \frac{3}{8} & 0 & \frac{5}{16} \\ \frac{5}{16} & 0 & \frac{5}{16} & 0 & \frac{3}{8} & 0 \\ 0 & \frac{5}{16} & 0 & \frac{5}{16} & 0 & \frac{3}{8} \end{bmatrix}.
 \end{aligned}$$

Notice that $P_{ii} = P_{ii}^3 = 0$ (the diagonal elements) for all 6 states. In fact, this pattern continues for all the odd powers of P . On the other hand P_{ii}^2 and P_{ii}^4 are all positive, and this pattern continues for all even powers of P . So, the set of recurrence times for any state is

$$\{2, 4, 6, 8, \dots\}.$$

The largest integer d that divides all of these possible recurrence times is 2. Hence, all states in our circular random walk have period 2. This makes sense since you must leave a particular node on the next step, but after 2 steps, you could come back to the node you started on. ■

Example 10.6.2. Consider the maze in Example 10.1.3. Recall that the rooms come in two separate communication classes: $\{0, 1, 2, 3, 4, 5, 6\}$ and $\{7\}$. If we look at the first few n -step transition probability matrices, we find

$$P = \begin{bmatrix} 0 & 1/3 & 0 & 1/2 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 1 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 1/3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 0 & 0 & 0 & 1/3 & 0 & 0 \\ 0 & 1/3 & 0 & 0 & 0 & 1/3 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1/2 & 1 \end{bmatrix},$$

$$P^2 = \begin{bmatrix} \frac{5}{12} & 0 & \frac{1}{3} & 0 & \frac{1}{6} & \frac{1}{6} & 0 & 0 \\ 0 & \frac{2}{3} & 0 & \frac{1}{4} & 0 & \frac{1}{6} & 0 & 0 \\ \frac{1}{6} & 0 & \frac{1}{3} & 0 & \frac{1}{6} & 0 & 0 & 0 \\ 0 & \frac{1}{6} & 0 & \frac{5}{12} & \frac{1}{6} & 0 & \frac{1}{6} & 0 \\ \frac{1}{6} & 0 & \frac{1}{3} & \frac{1}{6} & \frac{1}{3} & 0 & \frac{1}{6} & 0 \\ \frac{1}{4} & \frac{1}{6} & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{6} & \frac{1}{6} & 0 & \frac{1}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{6} & \frac{1}{2} & 1 \end{bmatrix},$$

$$P^3 = \begin{bmatrix} 0 & \frac{11}{36} & 0 & \frac{7}{24} & \frac{1}{12} & \frac{1}{18} & \frac{1}{12} & 0 \\ \frac{11}{24} & 0 & \frac{2}{3} & \frac{1}{12} & \frac{5}{12} & \frac{1}{12} & \frac{1}{12} & 0 \\ 0 & \frac{2}{9} & 0 & \frac{1}{12} & 0 & \frac{1}{18} & 0 & 0 \\ \frac{7}{24} & \frac{1}{18} & \frac{1}{6} & 0 & \frac{1}{12} & \frac{1}{4} & 0 & 0 \\ \frac{1}{12} & \frac{5}{18} & 0 & \frac{1}{12} & 0 & \frac{2}{9} & 0 & 0 \\ \frac{1}{12} & \frac{1}{12} & \frac{1}{6} & \frac{3}{8} & \frac{1}{3} & 0 & \frac{1}{4} & 0 \\ \frac{1}{12} & \frac{1}{18} & 0 & 0 & 0 & \frac{1}{6} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{12} & \frac{1}{12} & \frac{1}{6} & \frac{7}{12} & 1 \end{bmatrix},$$

$$P^4 = \begin{bmatrix} \frac{43}{144} & \frac{1}{36} & \frac{11}{36} & \frac{1}{36} & \frac{13}{72} & \frac{11}{72} & \frac{1}{36} & 0 \\ \frac{1}{24} & \frac{37}{72} & 0 & \frac{1}{48} & \frac{1}{24} & \frac{36}{7} & \frac{1}{24} & 0 \\ \frac{11}{72} & 0 & \frac{2}{9} & \frac{1}{36} & \frac{5}{11} & \frac{36}{1} & \frac{1}{36} & 0 \\ \frac{1}{72} & \frac{13}{72} & \frac{1}{18} & \frac{1}{48} & \frac{11}{72} & \frac{36}{1} & \frac{1}{36} & 0 \\ \frac{36}{13} & \frac{72}{1} & \frac{5}{18} & \frac{11}{48} & \frac{1}{72} & \frac{36}{1} & \frac{1}{8} & 0 \\ \frac{72}{11} & \frac{36}{7} & \frac{1}{18} & \frac{72}{1} & \frac{4}{1} & \frac{36}{23} & \frac{9}{0} & 0 \\ \frac{48}{1} & \frac{36}{1} & \frac{12}{1} & \frac{24}{1} & \frac{24}{72} & 0 & \frac{1}{12} & 0 \\ \frac{36}{1} & \frac{36}{1} & \frac{18}{1} & \frac{8}{1} & \frac{9}{1} & 0 & \frac{12}{7} & 0 \\ \frac{1}{24} & \frac{1}{36} & 0 & \frac{1}{12} & \frac{1}{12} & \frac{1}{4} & \frac{1}{12} & 1 \end{bmatrix}.$$

At first glance, it would seem that the class $\{0, 1, 2, 3, 4, 5, 6\}$ should have period 2 since the mouse must leave the room it is currently in each time step. However, it is possible to return to Room 1 in 5 steps: $1 \rightarrow 4 \rightarrow 5 \rightarrow 3 \rightarrow 0 \rightarrow 1$ (looking at P^5 would also show that it is possible to return in 5 steps). This means the set of possible recurrence times for Room 1 is $\{2, 4, 5, \dots\}$. Since 2 and 5 are both prime numbers, the gcd is 1. Hence, the entire class of transient states is aperiodic. Room 7 is also aperiodic. In fact, once the mouse enters this room, it can never leave! ■

If a state i is recurrent, it is said to be **positive recurrent** if the expected time to return to state i (after starting in state i) is finite. The following lemma is again presented without proof.

Lemma 10.6.2. *Positive recurrence is a property of communication classes. That is, if state i is positive recurrent and i and j communicate, then state j must also be positive recurrent. Furthermore, all recurrent states in a finite-state Markov Chain must be positive recurrent.*

For the interested reader, proofs of both these lemmas can be found in [Sh96][Chapter VIII].

For a Markov Chain with an *infinite* number of possible states, it is possible for states to be recurrent but not positive recurrent. That is, a state could be recurrent, but the expected time it takes for the system to return to the state is infinite. Recurrent states that are not positive recurrent are said to be **null recurrent**. For a null recurrent state, the system is certain to return to the state at some point in the future, but the expected number of time steps before a recurrence happens is unbounded (i.e. you may be waiting a VERY long time before the system returns to the state). However, the expected number of times the system will revisit a null recurrent state is still infinite! This is one of the bizarre things that can happen with Markov Chains with an infinite number of possible states!

We need two final definitions. A state i in a Markov Chain is said to be **ergodic** if it is aperiodic and positive recurrent. Finally, an **irreducible ergodic Markov Chain** is one where all the states belong to the same communication class, each state is aperiodic, and each state is positive recurrent.

Example 10.6.3. The simple weather model in Example 10.1.1 is an irreducible ergodic Markov Chain. Indeed, we know that all the states belong to a single communication class (which is the definition of irreducible). Since it is possible to go from any state in the chain to any other state (including the state the system is currently in), the possible recurrence times are $\{1, 2, 3, 4, \dots\}$. Hence, each state has period 1 (i.e. each state is aperiodic). Finally, since the states are certainly recurrent, they must be positive recurrent by the lemma above. Hence, the Markov Chain in Example 10.1.1 must be an irreducible ergodic Markov Chain. ■

The following theorem is crucial to what follows.

Theorem 10.6.3 (Limiting Distributions of Irreducible Ergodic Markov Chains). *Let (X_i) be an irreducible ergodic Markov Chain with transition probability matrix P . Then*

$$\lim_{n \rightarrow \infty} P^n$$

exists. Moreover, this limiting matrix has identical columns. If we let

$$\pi_i = \lim_{n \rightarrow \infty} P_{ij}^n$$

be the entries of this limiting column of probabilities, then the π_i are the unique solutions to the equations

$$\begin{aligned} \pi_i &= \sum_j P_{ij} \pi_j, \\ \sum_i \pi_i &= 1. \end{aligned}$$

Sketch of Proof: The basic idea is that the transition matrix for an irreducible ergodic Markov Chain must be a regular transition matrix. Since the Markov Chain is irreducible and ergodic, each state communicates with every other state. This means there is a $k \geq 1$ such that $P_{ij}^k > 0$ for every pair of states. Since the number of recurrence times for each state is infinite (since each state is positive recurrent) and each state is aperiodic, it can

be shown that there is a single positive integer $K \geq 1$ such that $P_{ij}^K > 0$ for *every pair* of indices i and j . Hence, the transition matrix P is regular (since P^K is a positive matrix). By Theorem 10.5.3, there is a limiting matrix $L = \lim_{k \rightarrow \infty} P^k$ whose columns consist precisely of the eigenvector $\vec{\pi}$ corresponding to 1 whose entries sum to 1. But this is precisely the statement of the theorem above! ■

The limiting probabilities π_i guaranteed by the theorem should be thought of as the percentage of time the process spends in state i in the long-term (i.e. after a large amount of time has passed).

Example 10.6.4. Consider again the simple weather model from Example 10.1.1. Since we already know that this Markov Chain is irreducible and ergodic, we know there should be a limiting probability distribution. To find it, we have to solve

$$\begin{aligned}\vec{\pi} &= P\vec{\pi}, \\ \pi_0 + \pi_1 + \pi_2 &= 1.\end{aligned}$$

In matrix form, the first equation is

$$\begin{bmatrix} \pi_0 \\ \pi_1 \\ \pi_2 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.3 & 0.4 & 0.4 \\ 0.2 & 0.3 & 0.4 \end{bmatrix} \begin{bmatrix} \pi_0 \\ \pi_1 \\ \pi_2 \end{bmatrix}.$$

This translates into the equations

$$\begin{aligned}\pi_0 &= \frac{1}{2}\pi_0 + \frac{3}{10}\pi_1 + \frac{1}{5}\pi_2, \\ \pi_1 &= \frac{3}{10}\pi_0 + \frac{2}{5}\pi_1 + \frac{2}{5}\pi_2, \\ \pi_2 &= \frac{1}{5}\pi_0 + \frac{3}{10}\pi_1 + \frac{2}{5}\pi_2.\end{aligned}$$

These equations do not have a unique solution. What they tell us is

$$\begin{aligned}\pi_1 &= \frac{13}{12}\pi_0, \\ \pi_2 &= \frac{7}{8}\pi_0.\end{aligned}$$

Combining this with the requirement that $\pi_0 + \pi_1 + \pi_2 = 1$ gives us

$$\pi_0 = \frac{24}{71} \approx 0.3380, \quad \pi_1 = \frac{26}{71} \approx 0.3662, \quad \text{and} \quad \pi_2 = \frac{21}{71} \approx 0.2958.$$

So, we see that days with clear skies, days that are partly cloudy, and days that are overcast are almost equally likely. Partly cloudy days are slightly more likely in our model however. This also means that

$$\lim_{n \rightarrow \infty} P^n = \begin{bmatrix} 24/71 & 24/71 & 24/71 \\ 26/71 & 26/71 & 26/71 \\ 21/71 & 21/71 & 21/71 \end{bmatrix}. \quad \blacksquare$$

Example 10.6.5. Consider an irreducible ergodic Markov Chain (X_i) with states 0, 1, 2, and 3. The matrix of transition probabilities is given by

$$P = \begin{bmatrix} 1/4 & 1/2 & 1/2 & 1/6 \\ 1/4 & 0 & 0 & 1/3 \\ 1/4 & 0 & 1/2 & 1/3 \\ 1/4 & 1/2 & 0 & 1/6 \end{bmatrix}.$$

Compute the limiting probabilities of this Markov Chain.

Answer: The equation

$$\vec{\pi} = P\vec{\pi}$$

yields the following system of four equations.

$$\begin{aligned} \pi_0 &= \frac{1}{4}\pi_0 + \frac{1}{2}\pi_1 + \frac{1}{2}\pi_2 + \frac{1}{6}\pi_3, \\ \pi_1 &= \frac{1}{4}\pi_0 + 0\pi_1 + 0\pi_2 + \frac{1}{3}\pi_3, \\ \pi_2 &= \frac{1}{4}\pi_0 + 0\pi_1 + \frac{1}{2}\pi_2 + \frac{1}{3}\pi_3, \\ \pi_3 &= \frac{1}{4}\pi_0 + \frac{1}{2}\pi_1 + 0\pi_2 + \frac{1}{6}\pi_3. \end{aligned}$$

Solving these equations leads to the following infinite set of solutions.

$$\pi_1 = \frac{7}{16}\pi_0, \quad \pi_2 = \frac{7}{8}\pi_0, \quad \text{and} \quad \pi_3 = \frac{9}{16}\pi_0.$$

Combining this with the requirement that $\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1$ gives

$$\pi_0 = \frac{8}{23}, \quad \pi_1 = \frac{7}{46}, \quad \pi_2 = \frac{7}{23}, \quad \text{and} \quad \pi_3 = \frac{9}{46}. \quad \blacksquare$$

10.7 Application: Shape of Epithelial Cells

Epithelial cells form tissues that line the outer surfaces of many organs and blood vessels throughout the bodies of most animals. Perhaps the largest example of epithelial tissue in the human body is the outermost layer of the skin – the epidermis. From the earliest studies of these tissues, scientists have observed that these cells often take on roughly polygonal shapes with hexagons being most common. A very elegant Markov Chain model developed by Gibson, Patel, Nagpal, and Perrimon in 2006 explains the distribution of these polygons quite well.[\[GPNP06\]](#)

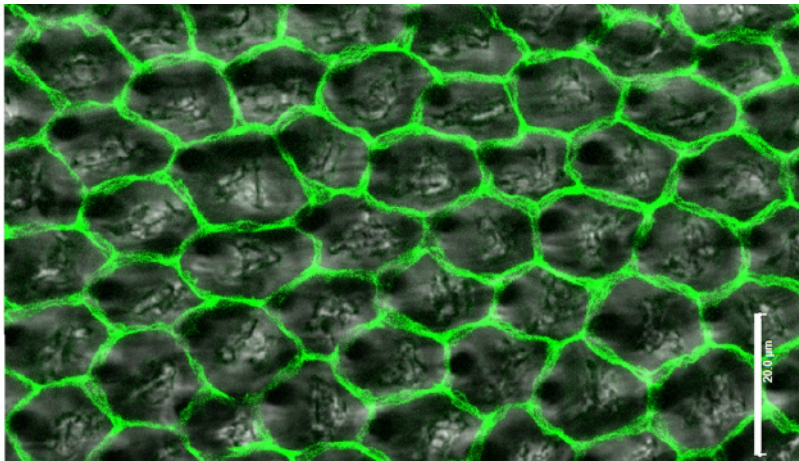


Figure 10.7.1: Epithelial Cells in the Wing of a Fruit Fly
(Iyengar, Balaji (2012) from figshare)

The authors make six basic assumptions about cell division in epithelial cells (all of which are based on observational data).

- 1) Cells are polygons with a minimum of four sides.
- 2) Cells do not sort themselves after division.
- 3) After division, the two daughter cells retain a common side.
- 4) Cells have asynchronous but roughly equal cycle times.
- 5) Cell divisions cut through sides of polygons and not vertices.

- 6) Junctions of three cells are randomly distributed among the daughter cells upon division.

From these assumptions, the authors are able to compute transition probabilities in a two step process. The first step is to consider the mitotic division of a single cell. Vastly simplifying the cell division process, suppose that a new cell wall forms joining two existing sides of a parent cell with j -sides (assumption 5 rules out new walls joining vertices). Since triangular cells are observed exceedingly rarely (hence assumption 1 above), the new cell wall cannot connect two adjacent sides. Through various arguments regarding the nature of mitosis, the authors effectively assume that the probability of the new cell wall connecting a given side A with the one of the other possible sides follows a binomial distribution with number of trials $j - 4$ and probability of success $1/2$ (note that a binomial distribution with $j - 4$ trials has $j - 3$ possible outcomes).

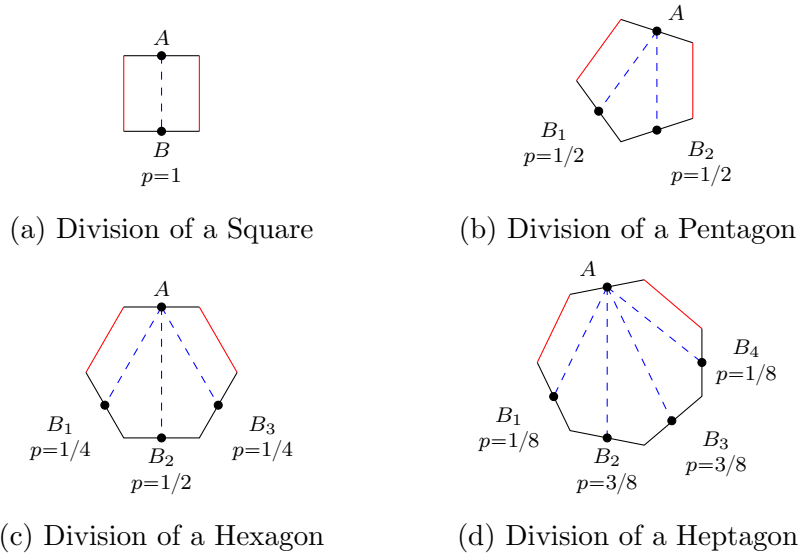


Figure 10.7.2: Division Probabilities for Smaller Polygons

These considerations lead to the following transition probabilities for the first step of the process:

$$\mathbb{P}(\text{daughter cell has } i \text{ sides} | \text{parent cell has } j \text{ sides}) = \binom{j-4}{i-4} \frac{1}{2^{j-4}}$$

where $4 \leq i \leq j$. For i outside this range, the transition probabilities are zero.

The second step in the process takes into account that the division of a cell can add sides to a neighboring cell. In general, the number of sides a cell will gain due to the division of a neighboring cell is difficult to predict. However, general considerations led the authors to the conclusion that each cell picks up one additional side per time step *on average*. If we take one extra side as a given, we finally arrive at our transition probabilities for the entire process:

$$p_{ij} = \binom{j-4}{i-5} \frac{1}{2^{j-4}}$$

for $5 \leq i \leq j + 1$.

At the moment, we have an “infinite matrix” of transition probabilities of the form

$$\begin{array}{c} \begin{matrix} & 4 & 5 & 6 & 7 & 8 & 9 & \dots \\ \begin{matrix} 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ \vdots \end{matrix} & \left[\begin{array}{cccccc} 0 & 0 & 0 & 0 & 0 & 0 & \dots \\ 1 & 1/2 & 1/4 & 1/8 & 1/16 & 1/32 & \dots \\ 0 & 1/2 & 1/2 & 3/8 & 1/4 & 5/32 & \dots \\ 0 & 0 & 1/4 & 3/8 & 3/8 & 5/16 & \dots \\ 0 & 0 & 0 & 1/8 & 1/4 & 5/16 & \dots \\ 0 & 0 & 0 & 0 & 1/16 & 5/32 & \dots \\ 0 & 0 & 0 & 0 & 0 & 1/32 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{array} \right] \end{matrix} \end{array}.$$

To use our methods, we need a genuine matrix. To proceed, we decide to truncate this matrix by only considering side lengths up to size 13. This choice is completely arbitrary (other than the fact that cells with more than

13 sides are exceedingly rare), but it results in the following 10×10 matrix.

$$\begin{array}{c} \begin{matrix} & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 \end{matrix} \\ \begin{matrix} 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ 11 \\ 12 \\ 13 \end{matrix} \left[\begin{array}{cccccccccc} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1/2 & 1/4 & 1/8 & 1/16 & 1/32 & 1/64 & 1/128 & 1/256 & 1/512 & \\ 0 & 1/2 & 1/2 & 3/8 & 1/4 & 5/32 & 3/32 & 7/128 & 1/32 & 9/512 & \\ 0 & 0 & 1/4 & 3/8 & 3/8 & 5/16 & 15/64 & 21/128 & 7/64 & 9/128 & \\ 0 & 0 & 0 & 1/8 & 1/4 & 5/16 & 5/16 & 35/128 & 7/32 & 21/128 & \\ 0 & 0 & 0 & 0 & 1/16 & 5/32 & 15/64 & 35/128 & 35/128 & 63/256 & \\ 0 & 0 & 0 & 0 & 0 & 1/32 & 3/32 & 21/128 & 7/32 & 63/256 & \\ 0 & 0 & 0 & 0 & 0 & 0 & 1/64 & 7/128 & 7/64 & 21/128 & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/128 & 1/32 & 9/128 & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/256 & 9/512 & \end{array} \right] \end{array}$$

The problem with this last matrix is that it is not a stochastic matrix! Since we truncated at 13, the final column is missing its final non-zero transition probability (which is $1/512$). To fix this, we make the ridiculous assumption that the cell of size 14 which we have removed from the transition probabilities collapses to a cell of size 4! So, we take as our transition matrix for this process

$$P = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/512 \\ 1 & 1/2 & 1/4 & 1/8 & 1/16 & 1/32 & 1/64 & 1/128 & 1/256 & 1/512 \\ 0 & 1/2 & 1/2 & 3/8 & 1/4 & 5/32 & 3/32 & 7/128 & 1/32 & 9/512 \\ 0 & 0 & 1/4 & 3/8 & 3/8 & 5/16 & 15/64 & 21/128 & 7/64 & 9/128 \\ 0 & 0 & 0 & 1/8 & 1/4 & 5/16 & 5/16 & 35/128 & 7/32 & 21/128 \\ 0 & 0 & 0 & 0 & 1/16 & 5/32 & 15/64 & 35/128 & 35/128 & 63/256 \\ 0 & 0 & 0 & 0 & 0 & 1/32 & 3/32 & 21/128 & 7/32 & 63/256 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1/64 & 7/128 & 7/64 & 21/128 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/128 & 1/32 & 9/128 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/256 & 9/512 \end{bmatrix}.$$

This slightly silly process is certainly irreducible and ergodic. Hence, there must be a set of limiting probabilities by our previous theorem. Solving the associated systems of equations

$$\begin{aligned} \vec{\pi} &= P\vec{\pi} \\ \sum_{i=4}^{13} \pi_i &= 1 \end{aligned}$$

is not something anyone would want to do by hand. Thankfully, a computer can easily deal with the problem. The limiting set of probabilities turns out to be approximately

$$\vec{\pi} = \begin{matrix} 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ 11 \\ 12 \\ 13 \end{matrix} \begin{bmatrix} 1 \times 10^{-13} \\ 0.2888 \\ 0.4640 \\ 0.2085 \\ 0.0359 \\ 0.0027 \\ 9 \times 10^{-5} \\ 2 \times 10^{-6} \\ 1 \times 10^{-8} \\ 5 \times 10^{-11} \end{bmatrix}.$$

As you can see, pentagons, hexagons, and heptagons make up over 96% of the shapes of epithelial cells in the model with almost half being hexagons.⁷ The authors of this study went on to do observational studies of epithelial tissues in various animals. Their reported findings matched reasonably well with the probabilities listed above.

10.8 Random Walks

The very first example we discussed at the beginning of this chapter was a one-dimensional **random walk**. Recall that we considered a process where the possible states are the integers, and at each time step we are allowed to move either one step forwards or one step backwards from our current position. This process can be represented as a Markov Chain – though one with an infinite number of states. As such, not all of our results from the study of finite state Markov Chains will apply.

In this section, we want to present the details of random walks not just in one-dimension but also in two, three, and higher dimensions. Many of the results we give will be presented without proof. As always, a reference will be given for the curious reader. By the way, a random walk is sometimes referred to as a “drunkard’s walk” or a “drunken walk” in older literature. This name

⁷Truncating the matrix at a larger number of sides does not significantly change the results for these three polygons.

is often the source of some funny descriptions of these stochastic processes, but we will generally stick to the name random walk in the following.

10.8.1 Random Walks in One Dimension

Consider a Markov Chain (X_n) where the possible states are the integers: $\{0, \pm 1, \pm 2, \pm 3, \dots\}$. If our current state is the integer j (i.e. $X_n = j$), then the next state can only be $j + 1$ or $j - 1$. So, only a single step forward or backwards is allowed at each time step, and the process *must* change states each time step. So, the only non-zero transition probabilities are

$$\begin{aligned} P_{j+1,j} &= p, \\ P_{j-1,j} &= 1 - p = q. \end{aligned}$$

A few comments are in order. First, we have employed a slightly different notation for the transition probabilities: $P_{j+1,j}$. The comma that has been inserted is purely to make the indices easier to read. The order of the coefficients is still consistent with our previous notation; $P_{j+1,j}$ is the probability of transitioning from state j to state $j + 1$.

Second, notice that we have allowed for *biased* (or *asymmetric*) random walks. That is, the probability of a step forward (which is p) could be different than the probability of a step backwards ($q = 1 - p$).

Third, notice that the probabilities p and q for forward and backward transitions are independent of the state j . In other words, the transition $10 \rightarrow 11$ has the same probability as the transition $-1000 \rightarrow -999$ (both are equal to p). This requirement could be relaxed, but the analysis becomes much more difficult.

It is easy to see that every state in our random walk communicates with every other state. Indeed, starting at 0, the first step will take us to either 1 or -1 . The second step will take us to one of $-2, 0$, or 2. The third step will take us to $-3, -1, 1$, or 3. Continuing this line of thought shows us that it is possible to travel from any given state to any other state given enough time steps. Hence, our random walk is an irreducible Markov Chain.

Since our Markov Chain is forced to change state every time step (we are not allowed to stay on the same integer during the next step), our states cannot be aperiodic. In fact, all states have period 2. This is easy to see if we remember that period is a common property of a communication class. Since we can only return to 0 at times $\{2, 4, 6, 8, \dots\}$, we see that 0 must have period 2. Hence, all states have this property.

Since all the states belong to the same class, either all the states are recurrent or all are transient. The way to decide is to use Theorem 10.3.1. In order to use this theorem, we need to compute the probabilities P_{ii}^n for some state i in the communication class. We might as well consider state 0. Recall that P_{00}^n is the probability of starting from state 0 and returning to state 0 in exactly n steps. We already know that we cannot return to 0 in an odd number of steps (each step switches between even and odd states). So, all recurrence probabilities for an odd number of steps must be zero:

$$P_{00}^{2n+1} = 0 \text{ for all } n.$$

This means we only need to consider

$$P_{00}^{2n} = \text{probability of returning to 0 in } 2n \text{ steps starting from 0.}$$

If we start from 0 and then return to 0, we must have taken just as many steps forward (which adds 1 to the state) as we have backwards (which adds -1 to the state). This is equivalent to flipping a (biased) coin $2n$ times and getting heads exactly n of those times. But this is just a binomial random variable (with probability of success p). Therefore,

$$P_{00}^{2n} = \binom{2n}{n} p^n (1-p)^n = \frac{(2n)!}{n!n!} (p(1-p))^n.$$

Our question becomes whether the summation

$$\sum_{n=1}^{\infty} P_{00}^{2n} = \sum_{n=1}^{\infty} \frac{(2n)!}{n!n!} (p(1-p))^n$$

gives a finite quantity or infinity. This question can be decided using the well-known result

$$n! \approx \sqrt{2\pi} \frac{n^{n+1/2}}{e^n} \text{ for large } n,$$

which is usually referred to as *Stirling's Approximation* for factorials. Combining this with basic facts about infinite series gives the following result (see [Ro95, pp. 194, 195] for the details).

Theorem 10.8.1 (Recurrence of 1-D Random Walks). *For the random walk (X_n) with states labeled by the integers $\{0, \pm 1, \pm 2, \dots\}$ and transition probabilities*

$$\begin{aligned} P_{j+1,j} &= p, \\ P_{j-1,j} &= 1 - p = q, \end{aligned}$$

all states are recurrent if $p = q = 1/2$. If $p \neq q$ (i.e. neither is $1/2$), then all states are transient.

When $p = q = 1/2$, there is an equal probability of taking a step forward or backwards. Such walks are said to be *unbiased* or *symmetric* random walks. If $p \neq q$, then the random walk is biased. This means that either forward steps are more likely than backward steps or vice versa. Our theorem tell us that all states in an unbiased one-dimensional random walk are recurrent (i.e. they are expected to recur an infinite number of times). On the other hand, all states in a biased one-dimensional random walk are transient (and so, any particular state is expected to be visited at most a finite number of times).

Despite the fact that all states in an unbiased one-dimensional random walk are recurrent, none of the states are positive recurrent!

Theorem 10.8.2. *For the random walk (X_n) with states labeled by the integers $\{0, \pm 1, \pm 2, \dots\}$ and transition probabilities*

$$P_{j+1,j} = \frac{1}{2},$$

$$P_{j-1,j} = \frac{1}{2},$$

the expected number of steps before a state recurs is infinite. Hence, all states are null recurrent.

See [D96, §3.1] for a discussion of this fact. In other words, we know that each state in a symmetric random walk has to occur an infinite number of times. However, the expected number of time steps until a state is repeated can be arbitrarily large!

To clarify this rather unintuitive result, consider the first 25 steps of the following symmetric random walk.

$n:$	0	1	2	3	4	5	6	7	8	9	10	11	12
$X_n:$	0	-1	0	1	2	1	2	1	0	-1	0	1	0
$n:$	13	14	15	16	17	18	19	20	21	22	23	24	25
$X_n:$	1	2	3	2	1	2	1	0	-1	0	1	2	3

Notice that the initial state ($X_0 = 0$) is repeated quite often. The fact that the *expected* time until a state recurs is infinite does not mean that a

state cannot be repeated in a finite amount of time! It only means that the probabilities of very large waiting times between recurrences is significant enough that we cannot say how long we should expect to wait for a state to be repeated. For instance, we could have started out making 10 consecutive moves to the right (i.e. $X_{10} = 10$). While this is not a highly likely state of affairs since its probability is

$$\mathbb{P}(X_{10} = 10 | X_0 = 0) = \left(\frac{1}{2}\right)^{10} = \frac{1}{1024}$$

(roughly one chance in a thousand), it is far from an impossibility. Once we are in state $X_{10} = 10$, the chances of returning to 0 in a short amount of time are relatively slim!

Compare the previous symmetric random walk with the following biased random walk. In this example,

$$P_{j+1,j} = \frac{4}{5},$$

$$P_{j-1,j} = \frac{1}{5},$$

(so a step forward is four times more likely than a step backwards).

$n:$	0	1	2	3	4	5	6	7	8	9	10	11	12
$X_n:$	0	1	0	1	2	3	4	3	2	3	4	5	6
$n:$	13	14	15	16	17	18	19	20	21	22	23	24	25
$X_n:$	7	8	9	10	9	10	9	10	11	10	9	10	11

Since this is a biased random walk, we know that all states are transient, and we can clearly see this from the overall pattern. While we occasionally take one or two steps backwards, there is a clear tendency toward bigger and bigger integers. So, we expect that any particular state will only be visited a finite number of times (if at all), and that the process will more or less stream off to positive infinity.

Another question of interest is how far from the starting location do we expect to be after n transitions. In order to tackle such questions, it is better to rethink how we describe our Markov Chain. Let $X_0 = 0$ be the initial state of the chain. At each time step, we transition either one step forward or one step backwards. We can think about this as

$$X_1 = X_0 + T_1$$

where the random variable T_1 is either -1 or 1 with probabilities

$$\begin{aligned}\mathbb{P}(T_1 = 1) &= p, \\ \mathbb{P}(T_1 = -1) &= q = 1 - p.\end{aligned}$$

(i.e. T_1 is a slight variant of the usual Bernoulli random variable). Similarly, we can think of X_2 as

$$X_2 = X_1 + T_2 = X_0 + T_1 + T_2$$

where T_2 is identical to but independent from T_1 . Going on in this fashion, we can write

$$X_n = X_0 + T_1 + T_2 + \cdots + T_n$$

where all of the random variables T_i are identical to and independent from T_1 . This allows us to easily compute expected values and standard deviations for the states X_n .

Theorem 10.8.3. *For the random walk (X_n) with states labeled by the integers $\{0, \pm 1, \pm 2, \dots\}$, $X_0 = 0$, and transition probabilities*

$$\begin{aligned}P_{j+1,j} &= p, \\ P_{j-1,j} &= q,\end{aligned}$$

we have

$$\begin{aligned}\mathbb{E}[X_n] &= n(p - q), \\ \sigma(X_n) &= \sqrt{n(1 - (p - q)^2)}.\end{aligned}$$

Proof: We note that for each of the random variables T_i , we have

$$\begin{aligned}\mathbb{E}[T_i] &= (1)p + (-1)q = p - q, \\ \text{Var}(T_i) &= (1)^2p + (-1)^2q - (p - q)^2 = p + q - (p - q)^2 = 1 - (p - q)^2.\end{aligned}$$

So, using the basic properties of expectation and variance for independent random variables we find

$$\begin{aligned}\mathbb{E}(X_n) &= \mathbb{E}[X_0] + \mathbb{E}[T_1] + \mathbb{E}[T_2] + \cdots + \mathbb{E}[T_n] = 0 + n(p - q), \\ \text{Var}(X_n) &= \text{Var}(X_0) + \text{Var}(T_1) + \text{Var}(T_2) + \cdots + \text{Var}(T_n) = 0 + n(1 - (p - q)^2).\end{aligned}$$

Since we forced $X_0 = 0$ in the statement of the problem, the expected value and variance of X_0 are both 0. ■

The standard deviation is particularly interesting. If we consider the symmetric case ($p = q = 1/2$), then the expected position is always $\mathbb{E}(X_n) = 0$. The standard deviation *grows* with n , however. This means that as time goes on, we are less certain of the state the system should be in. A more precise result gives us an even better indication of the long-term behavior.

Lemma 10.8.4. *For the symmetric random walk in one-dimension,*

$$\lim_{n \rightarrow \infty} \frac{\mathbb{E}(|X_n|)}{\sqrt{n}} = \sqrt{\frac{2}{\pi}}.$$

The first proof of this result was due to Grünbaum in 1960.⁸ What this result says is that for sufficiently large n , we expect to be a distance of

$$\sqrt{\frac{2n}{\pi}}$$

from our starting point after n time steps. This is yet another indication that while every state in the one-dimensional random walk is recurrent, the states cannot be positive recurrent.

10.8.2 Random Walks in Two Dimensions

When we go from random walks in one-dimension to ones in two and higher dimensions, one of the difficult changes is the labeling of the states. For two-dimensional random walks, we will assume that the walker must stay on a rectangular grid labeled by *pairs of integers*, (i, j) . You should think of the label (i, j) as directions for how to get to the node in question. The first coordinate, i , tells you how many steps left or right to take from the origin. The second coordinate, j , tells you how many steps forward or backwards to take. In the figure below, we have only labeled a few nodes around $(0, 0)$.

So, each state in our Markov Chain is given by a pair of integers, $X_n = (i, j)$, and we will start our process at the origin $X_0 = (0, 0)$. At each step in the process, the state is only allowed to change to one of the four nodes directly connected to it.

⁸Grünbaum, B. “Projection Constants.” *Trans. Amer. Math. Soc.* **95**, 451–465, 1960.

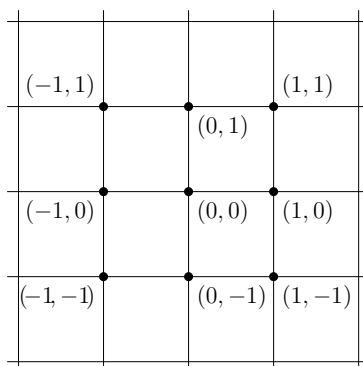


Figure 10.8.1: Standard Rectangular Grid for 2-D Random Walks

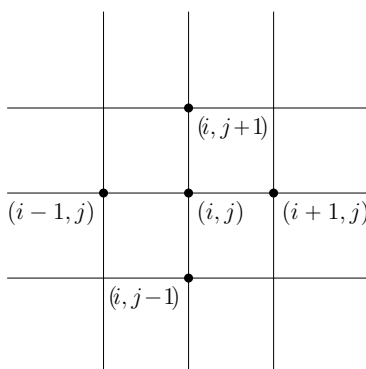


Figure 10.8.2: Allowed Transitions for 2-D Random Walks

So, the only states directly accessible from (i, j) are the adjacent nodes $(i+1, j)$, $(i, j+1)$, $(i-1, j)$, and $(i, j-1)$. We will denote the probabilities of these transitions as

$$P_{(i+1,j),(i,j)} = r,$$

$$P_{(i,j+1),(i,j)} = f,$$

$$P_{(i-1,j),(i,j)} = l,$$

$$P_{(i,j-1),(i,j)} = b,$$

where $r + f + l + b = 1$. As before, if $r = f = l = b = 1/4$ then the walk is totally symmetric (or unbiased). Otherwise the walk is biased. The choice of letters is designed to make these probabilities easier to remember. The probability of moving right is r , of moving forward is f , of moving left is l , and of moving backward is b .

Just as with the one-dimensional version, the generic two-dimensional random walk is irreducible – each state communicates with every other state.⁹ The period of each state requires a little more thought. If we are currently at node (i, j) , then we cannot stay there on the next time step. Once we move to one of the adjacent nodes, then we could certainly return to (i, j) on the next time step (assuming that none of the transition probabilities is zero). So, 2 is a possible recurrence time. In fact, we can make an argument that all recurrence times must be even numbers (just as in the one-dimensional random walk).

The way to think about this fact is to consider how we could possibly start from (i, j) and then return there. Since the x -coordinate begins and ends at i , the total number of steps to the right must be balanced by the total number of steps to the left (both could be zero). Likewise, the total number of steps forward must be balanced by the total number backwards. So, BOTH the number of left–right steps and the number of forward–backward steps must be even. Therefore, we can only return to a state after an even number of steps. Since 2 is a possible recurrence time, we find that the period of every state is 2.

The figures below show simulations of random walks in two-dimensions which start from $(0,0)$. The first two figures show symmetric random walks allowed to run for 10,000 steps. The second figure shows 10,000 steps of a biased random walk with probabilities

$$\begin{aligned} P_{(i+1,j),(i,j)} &= 1/3, \\ P_{(i,j+1),(i,j)} &= 1/3, \\ P_{(i-1,j),(i,j)} &= 1/6, \\ P_{(i,j-1),(i,j)} &= 1/6. \end{aligned}$$

What this means is that the system is twice as likely to move forward or right as it is to move backward or left.

We should note that the scales of these three figures (which are not shown to keep the plots from being cluttered) are very different. The last plot is especially zoomed out from the others. As you can see, the symmetric random walks meander around in no preferred direction. However, our biased

⁹This is not quite true if some of r, f, l , or b are zero. In that case, there are some states which will be inaccessible from a given state. In an extreme example, if $f = b = 0$, then the only states accessible from $(0, 0)$ are states of the form $(i, 0)$ - i.e. states to the left or right of the origin.

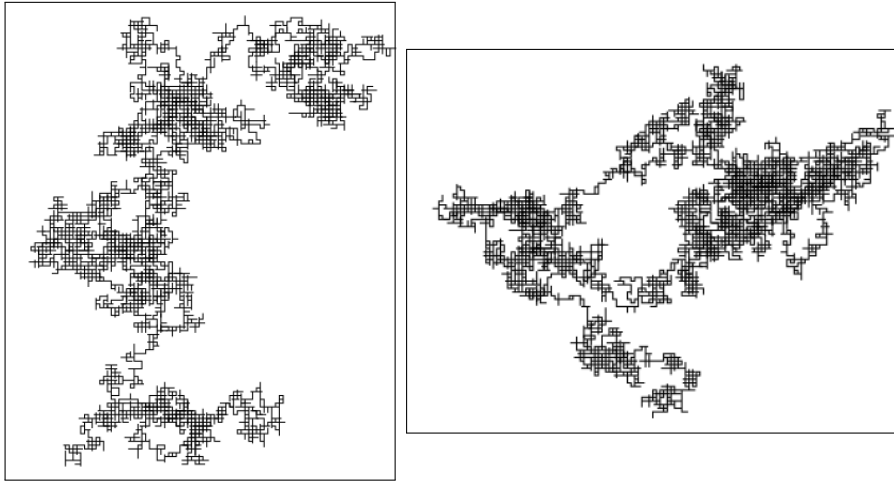


Figure 10.8.3: Sample 2-D Symmetric Random Walks: 10,000 Steps

example has a clear tendency to move up and to the right. The inset panel shows a tight zoom on the beginning of the biased walk. Clearly, moves down and to the left are possible, but they are infrequent enough to be overwhelmed.

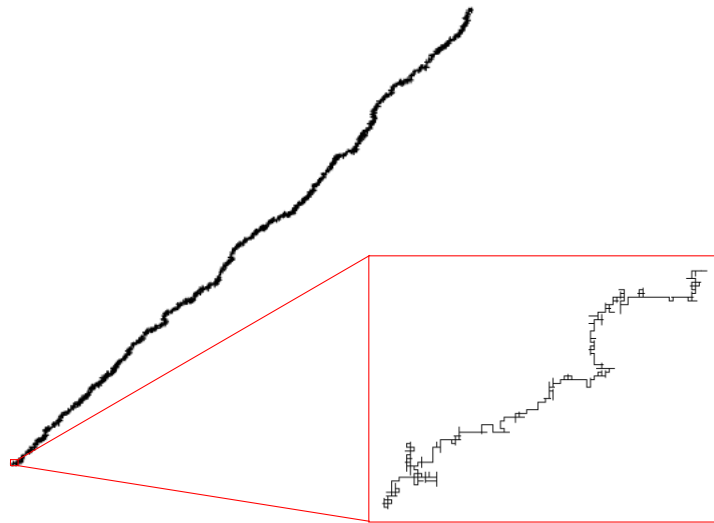


Figure 10.8.4: Sample 2-D Biased Random Walk: 10,000 Steps

So far, our examples seem to indicate that the same result that was

true in one-dimension is also true in two-dimensions. Namely, states in the symmetric random walk are recurrent, while they are transient in biased random walks. However, the situation is a bit more complicated than that. We will say that a two-dimensional random walk is *directionally unbiased* if the probability of moving forwards is equal to the probability of moving backwards AND the probability of moving right is equal to the probability of moving left. That is,

$$\begin{aligned} P_{(i+1,j),(i,j)} &= r, \\ P_{(i,j+1),(i,j)} &= f, \\ P_{(i-1,j),(i,j)} &= r, \\ P_{(i,j-1),(i,j)} &= f, \end{aligned}$$

where $r + f + r + f = 1$ or equivalently, $r + f = 1/2$. We will refer to the special case where

$$r = f = \frac{1}{4}$$

as the *totally symmetric* random walk in two-dimensions.

Theorem 10.8.5 (Recurrence of 2-D Random Walks). *For the two-dimensional random walk (X_n) with states labeled by pairs of integers (i, j) and transition probabilities*

$$\begin{aligned} P_{(i+1,j),(i,j)} &= r, \\ P_{(i,j+1),(i,j)} &= f, \\ P_{(i-1,j),(i,j)} &= l, \\ P_{(i,j-1),(i,j)} &= b, \end{aligned}$$

all states are recurrent if the walk is directionally unbiased. Otherwise, all states are transient.

A partial proof of this result can be found in [D96, §3.2]. We will only give an outline of why this is true. As with the one-dimensional random walk, we appeal to Theorem 10.3.1 and the fact that the process is irreducible. For the recurrence times $P_{(0,0),(0,0)}^n$, we have already observed that a state cannot recur in an odd number of time steps:

$$P_{(0,0),(0,0)}^{2n+1} = 0 \text{ for all } n \geq 0.$$

This means we only need to consider $P_{(0,0),(0,0)}^{2n}$. In order to start at $(0,0)$ and return there, we have to take an equal number of steps to the left and right as well as an equal number of steps forward and backward. If we take i steps to the right (and so i steps to the left), we must have taken $n - i$ steps forward (and $n - i$ steps backward) to get a total of $2n$ steps. Here, i could be any integer from 0 to n . Since these steps can be in any different order, we need to count the number of ways the steps can be arranged. This is done with a *multinomial* coefficient:

$$\binom{2n}{i, i, n-i, n-i} = \frac{(2n)!}{i!i!(n-i)!(n-i)!}.$$

The exact definition of multinomial coefficients is not important here, but the idea is simple enough. We have $2n$ steps to take, and they can be reordered in $(2n)!$ different ways – *if all steps were distinguishable*. However, the i steps to the right are essentially the same move. Since there are $i!$ ways of rearranging these steps to the right, we divide by $i!$ so we do not over-count the number of steps. The same is done for steps to the left, forward, and backward. Using this idea, we can compute the probability of returning to the origin in $2n$ steps:

$$P_{(0,0),(0,0)}^{2n} = \sum_{i=0}^n \frac{(2n)!}{i!i!(n-i)!(n-i)!} f^i b^i r^{n-i} l^{n-i}.$$

So, the determination of whether the states are transient or recurrent depends on whether the sum

$$\sum_{n=1}^{\infty} P_{(0,0),(0,0)}^{2n} = \sum_{n=1}^{\infty} \sum_{i=0}^n \frac{(2n)!}{i!i!(n-i)!(n-i)!} f^i b^i r^{n-i} l^{n-i}$$

is finite or infinite, respectively.

When $f = r = b = l = 1/4$, this sum becomes

$$P_{(0,0),(0,0)}^{2n} = \sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^{2n} \sum_{i=0}^n \frac{(2n)!}{i!i!(n-i)!(n-i)!} \approx \mathcal{C} \sum_{n=1}^{\infty} \frac{1}{n} = \infty,$$

($\mathcal{C}$ is a positive constant whose exact form does not have any bearing on the conclusion). So, the totally symmetric random walk in two-dimensions is

recurrent. For the directionally unbiased case (i.e. $f = b$ and $r = l$), it is possible to show that

$$\begin{aligned} P_{(0,0),(0,0)}^{2n} &= \sum_{i=0}^n \frac{(2n)!}{i!i!(n-i)!(n-i)!} (f^i)^2 (r^{n-i})^2 \\ &\geq \sum_{i=0}^n \frac{(2n)!}{i!i!(n-i)!(n-i)!} \left(\frac{1}{4}\right)^{2i} \left(\frac{1}{4}\right)^{2(n-i)}. \end{aligned}$$

But this smaller term is exactly the transition probability for the totally symmetric case! Since the recurrence probabilities are bigger in the directionally unbiased random walk, the states must also be recurrent in this case. Analyzing the biased case is far more challenging.

As before, we would like to know our expected position and the standard deviation of the distance for a two-dimensional random walk. We assume our starting state is always $X_0 = (0, 0)$. Each of our independent transition random variables T_i are distributed as

$$\begin{aligned} \mathbb{P}(T_i = (1, 0)) &= r, \\ \mathbb{P}(T_i = (0, 1)) &= f, \\ \mathbb{P}(T_i = (-1, 0)) &= l, \\ \mathbb{P}(T_i = (0, -1)) &= b. \end{aligned}$$

Just as in the one-dimensional case, we can write

$$X_n = X_0 + T_1 + T_2 + \cdots + T_n$$

where addition is defined *component-wise*. Notice that we cannot directly talk about the variance (or standard deviation) of the T_i since they are not numbers! We can talk about the expected size of T_i – that is, $\mathbb{E}[|T_i|] = 1$. This is obviously true since each possible outcome for T_i is one unit from the origin!

Theorem 10.8.6. *For the random walk (X_n) with states labeled by the integers $\{0, \pm 1, \pm 2, \dots\}$, $X_0 = (0, 0)$, and transition probabilities*

$$\begin{aligned} P_{(i+1,j),(i,j)} &= r, \\ P_{(i,j+1),(i,j)} &= f, \\ P_{(i-1,j),(i,j)} &= l, \\ P_{(i,j-1),(i,j)} &= b, \end{aligned}$$

we have

$$\begin{aligned}\mathbb{E}[X_n] &= (n(f-b), n(r-l)), \\ \mathbb{E}[|X_n|^2] &= n + n(n-1) [(f-b)^2 + (r-l)^2].\end{aligned}$$

Proof: We note that for each of the random variables T_i , we have

$$\mathbb{E}[T_i] = (1, 0)f + (0, 1)r + (-1, 0)b + (0, -1)l = (f-b, r-l).$$

This immediately gives

$$\mathbb{E}[X_n] = \mathbb{E}[X_0] + \mathbb{E}[T_1] + \mathbb{E}[T_2] + \cdots + \mathbb{E}[T_n] = (n(f-b), n(r-l)).$$

For the expected value of $|X_n|^2$, we use the fact that

$$\begin{aligned}|X_n|^2 &= X_n \cdot X_n \\ &= (T_1 + T_2 + \cdots + T_n) \cdot (T_1 + T_2 + \cdots + T_n) \\ &= \sum_{i=1}^n \sum_{j=1}^n T_i \cdot T_j \\ &= \sum_{i=1}^n T_i \cdot T_i + 2 \sum_{i=1}^n \sum_{j \neq i}^n T_i \cdot T_j \\ &= n + \sum_{i=1}^n \sum_{j \neq i}^n T_i \cdot T_j.\end{aligned}$$

We intentionally left off the X_0 term since it is assumed to be $(0, 0)$. In the last step, we used the fact that

$$T_i \cdot T_i = |T_i|^2 = 1.$$

This shows us that

$$\mathbb{E}[|X_n|^2] = n + \sum_{i=1}^n \sum_{j \neq i}^n \mathbb{E}[T_i \cdot T_j] = n + n(n-1)\mathbb{E}[T_1 \cdot T_2].$$

The last equality follows because the T_i are all identically distributed. A tedious computation shows that

$$\mathbb{E}[T_1 \cdot T_2] = [(f-b)^2 + (r-l)^2]. \quad \blacksquare$$

As in the one-dimensional case, the expected value of $|X_n|^2$ leads us to the conclusion that in the directionally unbiased case, the expected distance from the origin, $\mathbb{E}[|X_n|]$, is proportional to $\sqrt{n}$ after a very large number of time steps.

10.8.3 Random Walks in Three and Higher Dimensions

Once we leave two-dimensions, the computations which tell us whether random walks are recurrent or transient become much more involved (as if the past computations were not challenging). In this section, we will only present results for random walks in three and higher dimensions without discussing their proof. In fact, we will mostly limit ourselves to the totally symmetric random walk in three dimensions.

Just as in two-dimensional walks, we assume that at each node we are allowed to take a single step in one of three directions. This means that each node (or state) of our three-dimensional random walk can be labeled with a ordered triple of integers: (i, j, k) .

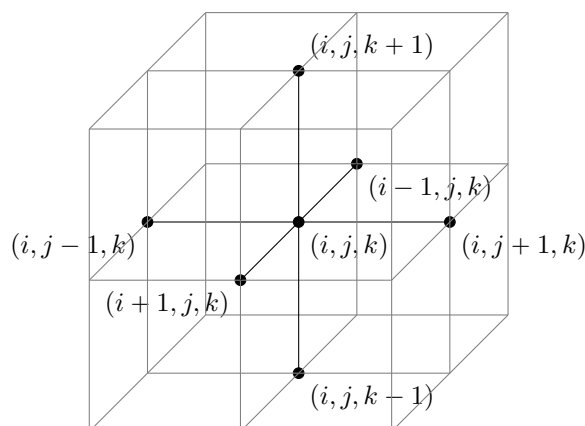


Figure 10.8.5: Lattice for Three-Dimensional Random Walks

From the lattice, we can see that there are 6 nodes that are accessible from a given node (i, j, k) . So, the transition probabilities can be listed as

$$\begin{aligned}
 P_{(i+1,j,k),(i,j,k)} &= f, \\
 P_{(i,j+1,k),(i,j,k)} &= r, \\
 P_{(i-1,j,k),(i,j,k)} &= b, \\
 P_{(i,j-1,k),(i,j,k)} &= l, \\
 P_{(i,j,k+1),(i,j,k)} &= u, \\
 P_{(i,j,k-1),(i,j,k)} &= d,
 \end{aligned}$$

where $f + r + b + l + u + d = 1$. The letters here have been chosen to make the probabilities easier to remember (f for forward, r for right, b for backward, l for left, u for up, and d for down). As long as none of these 6 probabilities are 0, each state in the process communicates with every other state, and so the Markov Chain is irreducible.

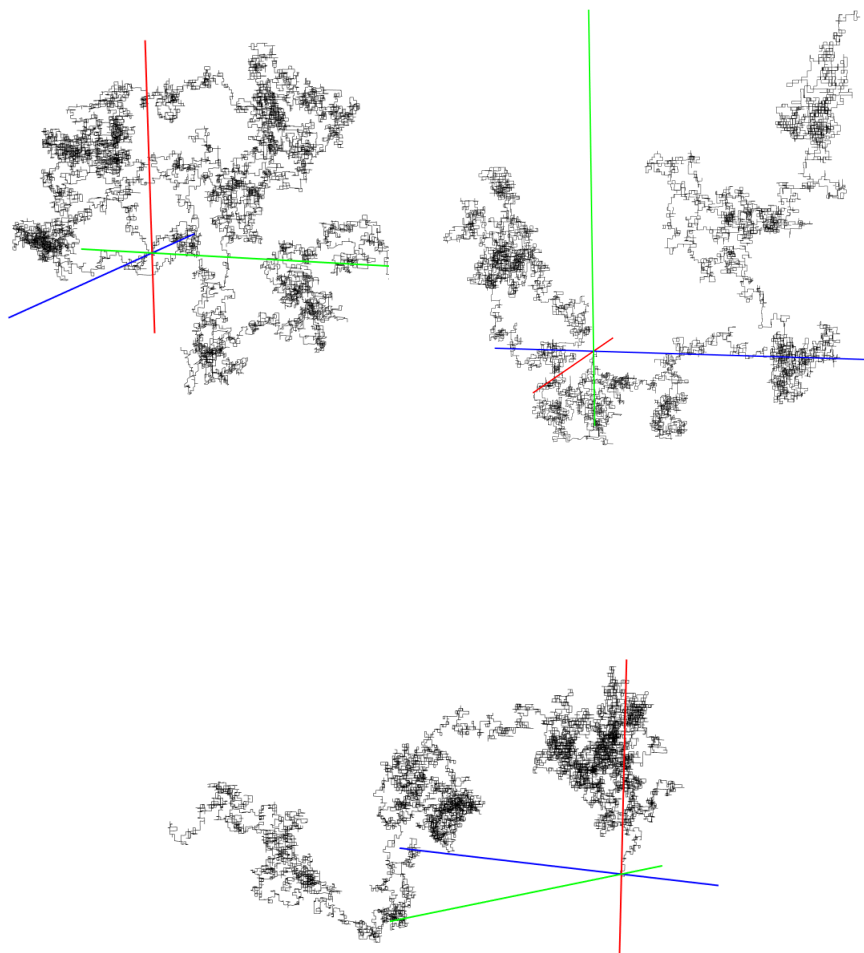


Figure 10.8.6: Three Viewpoints of a Single Symmetric 3-D Random Walk

We will mostly be interested in the case when

$$f = r = b = l = u = d = \frac{1}{6}.$$

In Figure 10.8.6, we see three different views of the same totally symmetric three-dimensional random walk which starts from the origin, $(0, 0, 0)$. It certainly appears that we eventually leave the vicinity of our starting point and never seem to return. While it is never a good idea to jump to conclusions based on a single example, it turns out that this situation is completely indicative of the long-term behavior for random walks in three (and higher) dimensions.

Theorem 10.8.7 (Totally Symmetric Random Walks in Three Dimensions). *Every state in the totally symmetric random walk in three (and higher) dimensions is transient. Moreover, the expected distance from the origin after n time steps is roughly proportional to $\sqrt{n}$ for sufficiently large n .*

Of course, if the totally symmetric case is not recurrent, then we have no hope of the biased cases being recurrent either!¹⁰ For a proof of this result see [D96, §3.2].

A famous joke by the mathematician Shizuo Kakutani sums up these results nicely: “A drunk man will find his way home, but a drunk bird may get lost forever.” The idea behind this joke is that people essentially walk in a two-dimensional space (where random walks are recurrent) while birds fly around in three dimensions. Whether or not the motion of drunk people (or animals) is accurately modeled by a random walk is left to the opinion of the reader.

10.9 Markov Chain Monte Carlo Methods

Often in simulations of real-world systems, we are faced with the problem of generating random samples from a given probability distribution (discrete or continuous). Most computer programs have routines for sampling from uniform distributions. For example, the histogram below shows the results from having the computer algebra system *Mathematica* choose 10,000 random samples uniformly (and independently) from the set $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$ (so each outcome has probability $1/10$).

¹⁰Unless some of the transition probabilities are zero. In such cases, we may have recurrent states, but the random walk will not be “truly three-dimensional.”

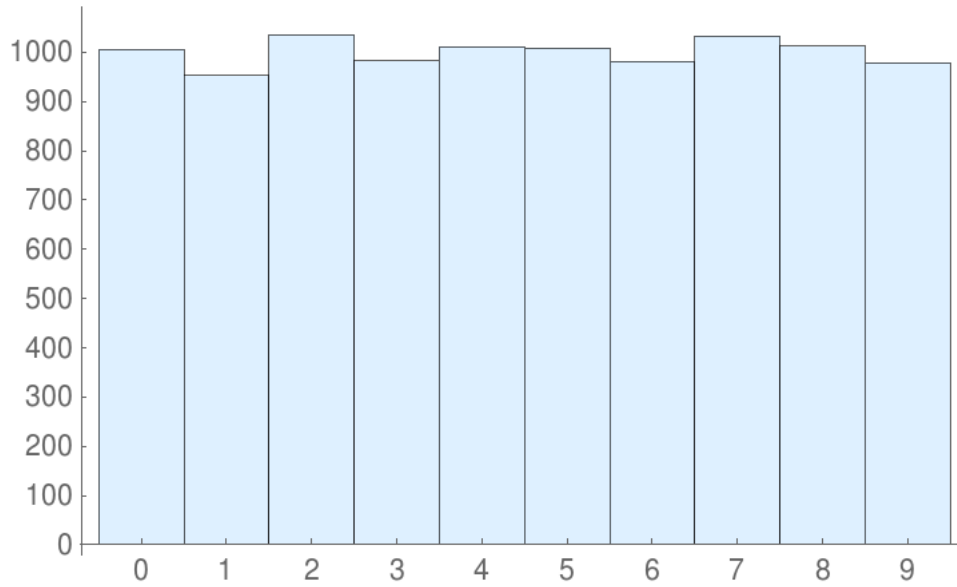


Figure 10.9.1: 10,000 Random Samples from the Digits 0 – 9

Suppose instead that we want to sample this set according to the probability distribution

$$\begin{aligned}\mathbb{P}(X = 0) &= \mathbb{P}(X = 9) = \frac{1}{30}, \\ \mathbb{P}(X = 1) &= \mathbb{P}(X = 8) = \frac{1}{15}, \\ \mathbb{P}(X = 2) &= \mathbb{P}(X = 7) = \frac{1}{10}, \\ \mathbb{P}(X = 3) &= \mathbb{P}(X = 6) = \frac{2}{15}, \\ \mathbb{P}(X = 4) &= \mathbb{P}(X = 5) = \frac{1}{6}.\end{aligned}$$

While there are several schemes that could generate samples in this simple situation, we would like a procedure that has some chance of being applicable to a wide range of situations. The idea we will develop is to design a Markov Chain whose limiting probabilities are precisely the probabilities of the distribution we want to sample from. Such techniques go under the

name **Markov Chain Monte Carlo Methods** (or MCMC Methods).¹¹

The specific MCMC Method we will develop is known as the **Metropolis–Hastings Algorithm**. We will begin by describing how this algorithm works for discrete random variables and then indicate how it can be extended to the continuous case with a couple of examples. For our setup, assume that the set of possible outcomes for our random variable X is $\{0, 1, 2, 3, \dots\}$. We also assume that there is a function $\beta(i)$ such that the probabilities of these outcomes are given by

$$\mathbb{P}(X = i) = \frac{\beta(i)}{\mathfrak{B}}$$

where $\mathfrak{B}$ is the appropriate normalization constant

$$\mathfrak{B} = \sum_{i=0} \beta(i).$$

One of the advantages of the Metropolis–Hastings Algorithm is that we do not need to know $\mathfrak{B}$ to produce the random samples! This is particularly useful in the continuous case where the normalization constant may involve an integral that is difficult to evaluate exactly.

Our job is to design an irreducible Markov Chain such that the limiting probabilities π_i satisfy

$$\pi_i = \frac{\beta(i)}{\mathfrak{B}}.$$

To do this, we start with an auxiliary Markov Chain, $(\mathcal{X}_n)$ which we can easily simulate (for example, a simple random walk). Since this Markov Chain will not be our final one, we will denote its transition probabilities as $\mathcal{P}_{ij}$ (as before, this is the probability of the auxiliary process transitioning from state j to state i). We will also need to be able to simulate Bernoulli random variables with arbitrary probability of success (which will be denoted by α). Our final product will be the Markov Chain (X_n) with transition probabilities P_{ij} . The choice of initial state X_0 for the Markov Chain turns out to be irrelevant in practice.

Assume that our Markov Chain is in state j at time step n , $X_n = j$. We first go to our auxiliary Markov Chain and let it start in state j , $\mathcal{X}_0 = j$. Suppose the next step in our auxiliary process is $\mathcal{X}_1 = i$ (which happens with probability $\mathcal{P}_{ij}$). For our actual Markov Chain, we will either let $X_{n+1} = i$

¹¹*Monte Carlo Methods* are a broad class of techniques that use random sampling to perform numerical computations.

or $X_{n+1} = j$. To decide, we simulate a Bernoulli random variable with probability of success

$$\alpha_{ij} = \min \left(\frac{\beta(i)\mathcal{P}_{ji}}{\beta(j)\mathcal{P}_{ij}}, 1 \right).$$

The function ‘min’ means to take the minimum of the two numbers listed which prevents the probability α_{ij} from being greater than 1. If this Bernoulli trial is successful, we let our Markov Chain transition to state i (so, $X_{n+1} = i$). Otherwise, the chain remains in state j , $X_{n+1} = j$.

To analyze this new Markov Chain, we need the transition probabilities. If $X_n = j$, then the probability of transitioning to a *different* state i is

$$P_{ij} = \mathcal{P}_{ij}\alpha_{ij} \quad (i \neq j)$$

since we need our auxiliary Markov chain to transition from j to i , and we need our Bernoulli trial to be successful (and these are independent events). If the Markov Chain stays in state j , then there are two different ways that could happen (which are mutually exclusive). First, the auxiliary chain could stay in state j (in which case our actual process will definitely stay in state j as well). Second, the auxiliary chain could transition to some different state k but the Bernoulli trial fails. Altogether, the probability of the process remaining in state j is

$$P_{jj} = \mathcal{P}_{jj} + \sum_{k \neq j} \mathcal{P}_{kj}(1 - \alpha_{kj}).$$

Recall that the equation for limiting probabilities is

$$\vec{\pi} = P\vec{\pi}$$

or in component form

$$\pi_i = \sum_j P_{ij}\pi_j$$

(together with the requirement that the probabilities add up to one). The claim is that these limiting probabilities are precisely

$$\pi_i = \frac{\beta(i)}{\mathfrak{B}}.$$

Rather than deriving this, we will simply verify that this is correct.

$$\begin{aligned}\sum_j P_{ij}\pi_j &= \sum_j P_{ij} \frac{\beta(j)}{\mathfrak{B}} \\ &= \frac{1}{\mathfrak{B}} \left(P_{ii}\beta(i) + \sum_{j \neq i} P_{ij}\beta(j) \right)\end{aligned}$$

Looking at the quantity in the parentheses, we find

$$\begin{aligned}P_{ii}\beta(i) + \sum_{j \neq i} P_{ij}\beta(j) &= \left(\mathcal{P}_{ii} + \sum_{j \neq i} \mathcal{P}_{ji}(1 - \alpha_{ji}) \right) \beta(i) + \sum_{j \neq i} \mathcal{P}_{ij}\alpha_{ij}\beta(j) \\ &= \left(\mathcal{P}_{ii} + \sum_{j \neq i} \mathcal{P}_{ji} - \sum_{j \neq i} \mathcal{P}_{ji}\alpha_{ji} \right) \beta(i) + \sum_{j \neq i} \mathcal{P}_{ij}\alpha_{ij}\beta(j) \\ &= \left(\mathcal{P}_{ii} + \sum_{j \neq i} \mathcal{P}_{ji} \right) \beta(i) - \sum_{j \neq i} \mathcal{P}_{ji}\alpha_{ji}\beta(i) + \sum_{j \neq i} \mathcal{P}_{ij}\alpha_{ij}\beta(j).\end{aligned}$$

The quantity in the parentheses above is just 1 since it is the total probability that our auxiliary Markov Chain transitions from state i to any other state (which must be one). So, we have

$$\begin{aligned}P_{ii}\beta(i) + \sum_{j \neq i} P_{ij}\beta(j) &= \beta(i) - \sum_{j \neq i} \mathcal{P}_{ji}\alpha_{ji}\beta(i) + \sum_{j \neq i} \mathcal{P}_{ij}\alpha_{ij}\beta(j).\end{aligned}$$

Notice, if

$$\alpha_{ij} = \frac{\beta(i)\mathcal{P}_{ji}}{\beta(j)\mathcal{P}_{ij}} \leq 1,$$

then $\alpha_{ji} = 1$ since

$$\frac{\beta(j)\mathcal{P}_{ij}}{\beta(i)\mathcal{P}_{ji}} = \frac{1}{\alpha_{ij}} \geq 1.$$

Similarly, if

$$\alpha_{ji} = \frac{\beta(j)\mathcal{P}_{ij}}{\beta(i)\mathcal{P}_{ji}} \leq 1,$$

then $\alpha_{ij} = 1$. In either case,

$$\mathcal{P}_{ij}\alpha_{ij}\beta(j) = \mathcal{P}_{ji}\alpha_{ji}\beta(i).$$

This means

$$\begin{aligned} \sum_{j \neq i} \mathcal{P}_{ij}\alpha_{ij}\beta(j) - \sum_{j \neq i} \mathcal{P}_{ji}\alpha_{ji}\beta(i) \\ = \sum_{j \neq i} \left(\mathcal{P}_{ij}\alpha_{ij}\beta(j) - \sum_{j \neq i} \mathcal{P}_{ji}\alpha_{ji}\beta(i) \right) = 0. \end{aligned}$$

Therefore,

$$P_{ii}\beta(i) + \sum_{j \neq i} P_{ij}\beta(j) = \beta(i),$$

and the limiting probabilities of our Markov Chain (X_n) are precisely

$$\pi_i = \frac{\beta(i)}{\mathfrak{B}}.$$

This derivation also shows why this algorithm is independent of the normalization constant $\mathfrak{B}$ (i.e. we only use ratios of the quantities $\beta(i)$).

Before we look at some specific examples, we discuss some practical matters in the usage of the algorithm. First, most implementations of Metropolis–Hastings allow for a *burn-in period* before beginning to accept samples. The idea is that the Markov Chain we develop is only guaranteed to have *limiting probabilities* that match the target distribution. So, an initial state for the chain, X_0 , is picked at random, and the system is allowed to run for some large number of steps before keeping any samples. This prevents a bad choice of initial state from distorting the samples.

Another common practice when using Metropolis–Hastings is *sample thinning*; instead of keeping all samples after the burn-in period, a large number of samples (perhaps the majority) are thrown away. An alternate way to think about this process is that the sampling device only keeps every n -th sample after the burn-in period (e.g. the device keeps every 20-th sample after the burn-in period). The reason for this practice is that we want to generate (approximately) *independent* samples from the target distribution. The problem with keeping all samples is that state of X_{n+1} depends on the state of X_n . Of course, X_{n+1} is dependent on all states that came before it, but after a large enough number of steps this dependence should hopefully be negligible.

Example 10.9.1. Use the Metropolis–Hastings Algorithm to set up a Markov Chain that will sample the digits 0 – 9 according to the distribution

$$\begin{aligned}\mathbb{P}(X = 0) &= \mathbb{P}(X = 9) = \frac{1}{30}, \\ \mathbb{P}(X = 1) &= \mathbb{P}(X = 8) = \frac{1}{15}, \\ \mathbb{P}(X = 2) &= \mathbb{P}(X = 7) = \frac{1}{10}, \\ \mathbb{P}(X = 3) &= \mathbb{P}(X = 6) = \frac{2}{15}, \\ \mathbb{P}(X = 4) &= \mathbb{P}(X = 5) = \frac{1}{6}.\end{aligned}$$

Answer: We take our auxiliary Markov Chain ($\mathcal{X}_n$) to be the symmetric random walk on the integers 0 – 9 (with 0 and 9 considered neighbors). In other words $\mathcal{P}_{i-1,i} = \mathcal{P}_{i+1,i} = 1/2$ for $i = 1, 2, 3, 4, 5, 6, 7$, and 8. For nodes 0 and 9, we have $\mathcal{P}_{90} = \mathcal{P}_{10} = 1/2$ and $\mathcal{P}_{89} = \mathcal{P}_{09} = 1/2$. All other transition probabilities are 0. For the numbers $\beta(i)$, we can take

$$\begin{aligned}\beta(0) &= \beta(9) = 1, \\ \beta(1) &= \beta(8) = 2, \\ \beta(2) &= \beta(7) = 3, \\ \beta(3) &= \beta(6) = 4, \\ \beta(4) &= \beta(5) = 5.\end{aligned}$$

Notice that these 10 numbers sum to 30 (which is the least common denominator of all the probabilities).

We will let the first element of our new Markov Chain be $X_0 = 0$. Since we will allow for a burn-in period, this initial choice should not make much of a difference. We will go through a few first steps in our new Markov Chain to get the idea of how the process plays out. For this example, we will not implement any sample thinning.

Step 1: Since $X_0 = 0$, we consider our auxiliary Markov Chain with starting state $\mathcal{X}_0 = 0$. The next state will either be $\mathcal{X}_1 = 9$ or $\mathcal{X}_1 = 1$ each with probability $1/2$. Suppose our auxiliary chain transitions to $\mathcal{X}_1 = 1$. Then we know $X_1 = 1$ or $X_1 = 0$. To decide which, we need the acceptance probability

$$\alpha_{10} = \min\left(\frac{\beta(1)\mathcal{P}_{01}}{\beta(0)\mathcal{P}_{10}}, 1\right) = \min\left(\frac{2(1/2)}{1(1/2)}, 1\right) = \min(2, 1) = 1.$$

So, the acceptance probability is 1, and we definitely transition from $X_0 = 0$ to $X_1 = 1$. This makes sense as 1 is twice as likely to occur in our target probability distribution as 0.

Step 2: Since $X_1 = 1$, we consider our auxiliary Markov Chain with starting state $\mathcal{X}_0 = 1$. The next state will either be $\mathcal{X}_1 = 0$ or $\mathcal{X}_1 = 2$ each with probability $1/2$. Suppose our auxiliary chain transitions to $\mathcal{X}_1 = 2$. Then we know $X_2 = 2$ or $X_2 = 1$. To decide which, we need the acceptance probability

$$\alpha_{21} = \min \left(\frac{\beta(2)\mathcal{P}_{12}}{\beta(1)\mathcal{P}_{21}}, 1 \right) = \min \left(\frac{3(1/2)}{2(1/2)}, 1 \right) = \min(3/2, 1) = 1.$$

So, the acceptance probability is 1, and we definitely transition from $X_1 = 1$ to $X_2 = 2$.

Step 3: Since $X_2 = 2$, we consider our auxiliary Markov Chain with starting state $\mathcal{X}_0 = 2$. The next state will either be $\mathcal{X}_1 = 1$ or $\mathcal{X}_1 = 3$ each with probability $1/2$. Suppose our auxiliary chain transitions to $\mathcal{X}_1 = 1$. Then we know $X_3 = 1$ or $X_2 = 2$. To decide which, we need the acceptance probability

$$\alpha_{12} = \min \left(\frac{\beta(1)\mathcal{P}_{21}}{\beta(2)\mathcal{P}_{12}}, 1 \right) = \min \left(\frac{2(1/2)}{3(1/2)}, 1 \right) = \min(2/3, 1) = \frac{2}{3}.$$

So, the acceptance probability is $\frac{2}{3}$. We have to simulate a Bernoulli($2/3$) random variable to decide whether or not we will accept the transition to state 1. Suppose this random variable returns a success. Then we transition to $X_3 = 1$.

Step 4: Since $X_3 = 1$, we consider our auxiliary Markov Chain with starting state $\mathcal{X}_0 = 1$. The next state will either be $\mathcal{X}_1 = 0$ or $\mathcal{X}_1 = 2$ each with probability $1/2$. Suppose our auxiliary chain transitions to $\mathcal{X}_1 = 0$. Then we know $X_4 = 0$ or $X_4 = 1$. To decide which, we need the acceptance probability

$$\alpha_{01} = \min \left(\frac{\beta(0)\mathcal{P}_{10}}{\beta(1)\mathcal{P}_{01}}, 1 \right) = \min \left(\frac{1(1/2)}{2(1/2)}, 1 \right) = \min(1/2, 1) = \frac{1}{2}.$$

So, the acceptance probability is $1/2$, and we have to simulate a Bernoulli($1/2$) random variable to decide whether or not we will accept the transition to state 0. Suppose this random variable returns a failure. Then we do not transition to 0, and so $X_4 = 1$.

So, the first four steps of our process are $X_0 = 0, X_1 = 1, X_2 = 2, X_3 = 1$, and $X_4 = 1$. Notice that the first few steps spends a good amount of time

in states that are far less likely than our target distribution would suggest. This is why the burn-in period is useful.

This process is easy to set up in most computer programming languages. We decided to run this procedure in *Mathematica*. The histogram below shows the results of a run of 10,100 steps in this process where we then dropped the first 100 time steps for burn-in.

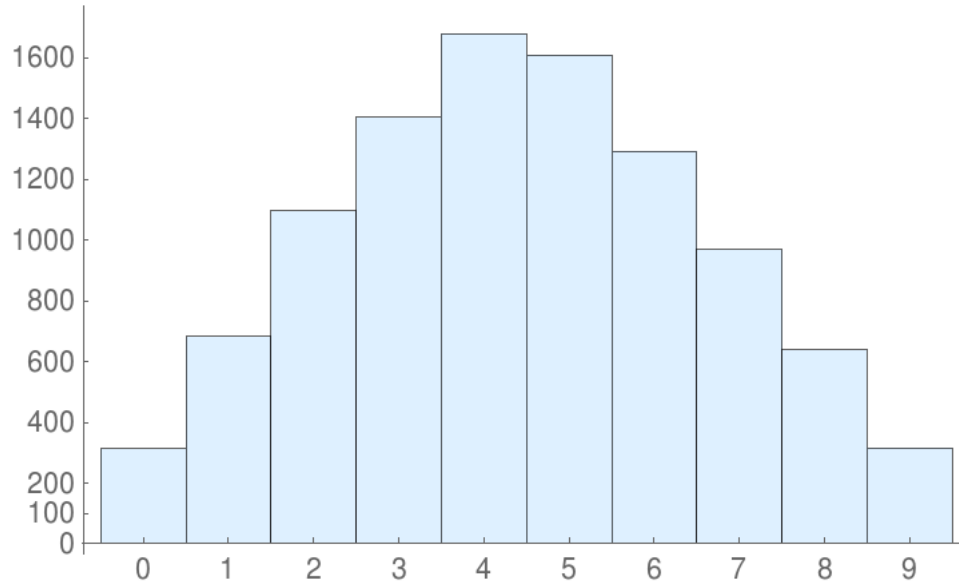


Figure 10.9.2: 10,000 Random Samples from the Target Distribution in Example 10.9.1

Notice that the Markov Chain we designed spends far more time in states 4 and 5 than in states 0 and 9. In fact, the Markov Chain visits state 4 a total of 1,623 times. Since $\mathbb{P}(X = 4) = 1/6$, we would expect the Markov chain to visit this state

$$\frac{10000}{6} = 1666.\bar{6} \text{ times.}$$

Similarly, our Markov Chain visits state 5 a total of 1,667 times (which is also close to the expected value). The other states are visited roughly close to their expected number of times. ■

Example 10.9.2. Suppose we want to generate random samples from a

binomial(20, 1/3) random variable. Set up a Markov Chain using the Metropolis–Hastings Algorithm which will generate these samples.¹²

Answer: For binomial(20, 2/3), the possible outcomes are $\{0, 1, 2, 3, \dots, 20\}$. So we take as our underlying Markov Chain a symmetric random walk on these integers (once again, we will let the endpoints, 0 and 20, be neighbors). The underlying probabilities for the binomial are

$$\begin{aligned}\mathbb{P}(X = i) &= \binom{20}{i} \left(\frac{2}{3}\right)^i \left(\frac{1}{3}\right)^{20-i} \\ &= \frac{20!}{i!(20-i)!} \frac{2^i}{3^{20}} \\ &= \frac{20!}{3^{20}} \frac{2^i}{i!(20-i)!}.\end{aligned}$$

For $\beta(i)$, we can ignore the factor of $20!/3^{20}$ (since it plays the role of a normalization). So, we can define

$$\beta(i) = \frac{2^i}{i!(20-i)!}.$$

Our acceptance probabilities become

$$\alpha_{ij} = \min\left(\frac{\beta(i)\mathcal{P}_{ji}}{\beta(j)\mathcal{P}_{ij}}, 1\right) = \min\left(\frac{2^i j!(20-j)!}{2^j i!(20-i)!} \frac{\mathcal{P}_{ji}}{\mathcal{P}_{ij}}, 1\right).$$

Remember that $\mathcal{P}_{ij}$ is the probability that our *auxiliary* Markov Chain will transition from state j to state i . Since we are using a symmetric random walk, we will only use these acceptance probabilities when i is a neighboring state to j . In that case, $\mathcal{P}_{ji}/\mathcal{P}_{ij} = 1$ (since all non-zero transition probabilities are 1/2).

Since the expected value of a binomial(20, 2/3) is $40/3 = 13.\bar{3}$, we will take $X_0 = 13$. As before, we will allow a burn-in time of 100 steps before accepting samples from our new Markov Chain. The first histogram shows the output from the Metropolis–Hastings Algorithm. The second one shows a histogram from a binomial sampler directly built into *Mathematica*. Notice the similarity in the two samples! ■

¹²Any decent software package that is designed for mathematical or statistical analysis probably has a built-in method for generating samples from most well-known random variables.

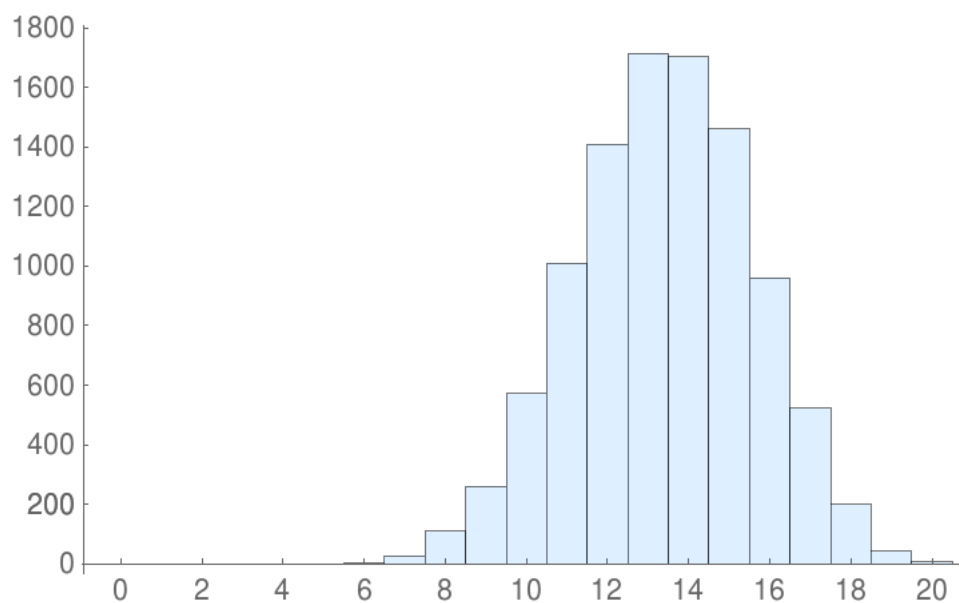


Figure 10.9.3: 10,000 Random Samples from $\text{binomial}(20, 2/3)$ by Metropolis-Hastings

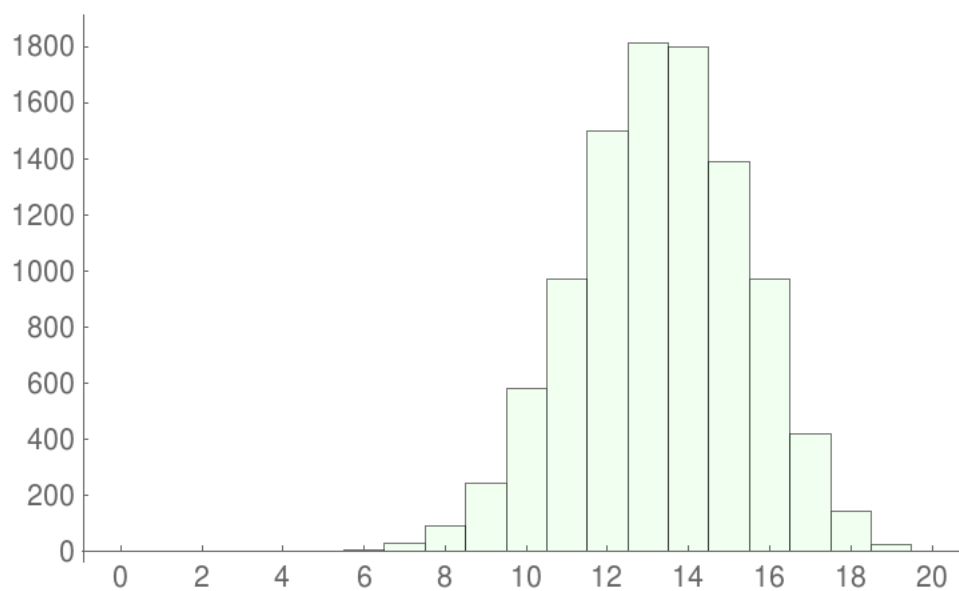


Figure 10.9.4: 10,000 Random Samples directly from $\text{binomial}(20, 2/3)$

Example 10.9.3. Suppose we want to generate random samples distributed according to $\text{Beta}(30, 50)$. Set up a Markov Chain using the Metropolis–Hastings Algorithm which will generate these samples.

Answer: Since the state space of any beta random variable is $[0, 1]$, we need our Markov Chain to have states in this range. However, we do not have the tools to deal with a continuous process. Instead, we will assume both of our Markov Chains (the auxiliary and our actual one) have states labeled with integers from 1 to 99 (so there are 99 possible states). We will interpret state i as corresponding to the fraction $i/100$. The reason for excluding states 0 and 100 will be clear soon enough.

For our auxiliary Markov Chain, we will again choose a symmetric random walk (with 1 and 99 being considered neighbors). For the numbers $\beta(i)$, we use the form of the distribution for $\text{Beta}(30, 50)$:

$$\begin{aligned} p\left(\frac{i}{100}\right) &= \frac{(i/100)^{29}(1 - i/100)^{49}}{B(30, 50)} \\ &= \frac{i^{29}(100 - i)^{49}}{100^{78}B(30, 50)}. \end{aligned}$$

So, we can take

$$\beta(i) = i^{29}(100 - i)^{49}$$

since the normalization factor is not important. Notice that $\beta(0) = \beta(100) = 0$ which is why we have excluded states 0 and 100.

The acceptance probabilities are given by

$$\begin{aligned} \alpha_{ij} &= \min\left(\frac{\beta(i)\mathcal{P}_{ji}}{\beta(j)\mathcal{P}_{ij}}, 1\right) \\ &= \min\left(\frac{i^{29}(100 - i)^{49}\mathcal{P}_{ji}}{j^{29}(100 - j)^{49}\mathcal{P}_{ij}}, 1\right) \\ &= \min\left(\left(\frac{i}{j}\right)^{29} \left(\frac{100 - i}{100 - j}\right)^{49} \frac{\mathcal{P}_{ji}}{\mathcal{P}_{ij}}, 1\right). \end{aligned}$$

Of course, $\mathcal{P}_{ij}$ is zero unless $i = j - 1$ or $i = j + 1$ (in which case our auxiliary Markov Chain has transition probabilities of $1/2$). So, in any cases where these acceptance probabilities are used, the ratio $\mathcal{P}_{ji}/\mathcal{P}_{ij} = 1$.

Since the expected value of $\text{Beta}(30, 50)$ is $3/8 = 0.375$, we will start our final Markov Chain in state 38 (since $38/100 = 0.38$ is close to the expected

value). We again generate 10,000 samples after a burn-in period of 100 steps. This gives the following results. Notice how similar the distribution is to the pdf for Beta(30, 50).

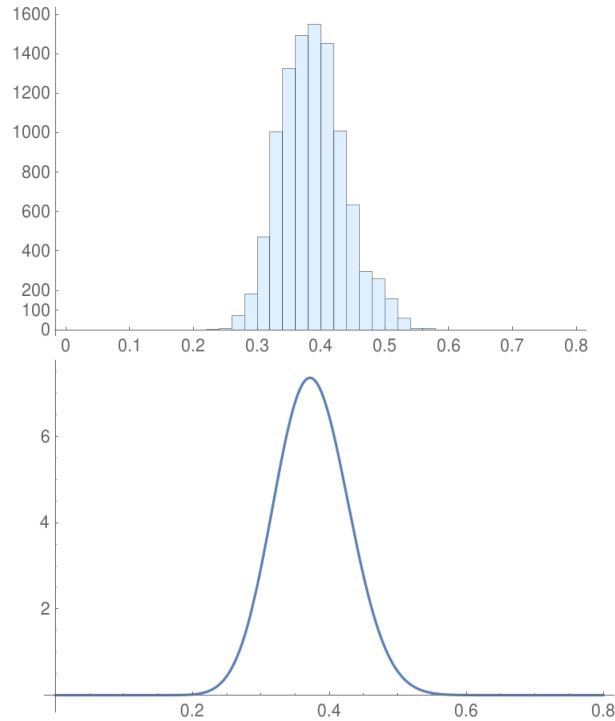


Figure 10.9.5: 10,000 Random Samples from Beta(30, 50)

Example 10.9.4. Suppose we want to generate random samples from a distribution whose sample space is $[1, 3]$ and whose pdf is of the form

$$p(x) = \frac{\mathcal{C}}{x^4}.$$

Set up a Markov Chain using the Metropolis–Hastings Algorithm which will generate these samples

Answer: While we could easily normalize the given pdf (by requiring that the integral over the sample space yields 1), we do not need to do this! Remember, the Metropolis–Hastings Algorithm does not depend on knowing any normalization constants!

Since we need to discretize the sample space, we will again take samples that are 1/100 apart (from 1.00 to 3.00). So, we will take our auxiliary Markov Chain to be a symmetric random walk on the integers from 0 to 200 (with 0 and 200 being neighbors). State i in our walk will correspond to the number $1 + i/100$ in the range $[1, 3]$. We do not have to exclude states 0 and 200 (as we did for the example above) because the pdf for our random variable is non-zero at the corresponding values (1 and 3).

Since

$$p\left(1 + \frac{i}{100}\right) = \frac{\mathcal{C}}{(1 + i/100)^4} = \frac{100^4 \mathcal{C}}{(100 + i)^4},$$

we can take

$$\beta(i) = \frac{1}{(100 + i)^4}.$$

On a practical note, we should be careful about how to choose $\beta(i)$. While we can ignore any common normalization factor, it could be that the resulting $\beta(i)$ is extremely small or extremely large. In such a case, your computer may have trouble working accurately with these numbers. Most modern computers are capable of dealing with such issues, but care should be taken nonetheless!

As above, the acceptance probabilities are

$$\begin{aligned} \alpha_{ij} &= \min\left(\frac{\beta(i)\mathcal{P}_{ji}}{\beta(j)\mathcal{P}_{ij}}, 1\right) \\ &= \min\left(\left(\frac{100 + j}{100 + i}\right)^4 \frac{\mathcal{P}_{ji}}{\mathcal{P}_{ij}}, 1\right), \end{aligned}$$

and the ratio $\mathcal{P}_{ji}/\mathcal{P}_{ij} = 1$ in any case where these probabilities are used.

Since our target pdf is a decreasing function, it seems reasonable to start our Markov Chain in state $X_0 = 0$ (which corresponds to the left-hand side of the sample space $x = 1$). We will still use a burn-in period of 100 time steps, but we will actually keep 100,000 samples for this simulation. The number of samples you produce should increase with the number of states in your auxiliary Markov Chain. As an extreme example, if your underlying Markov Chain has 20,000 possible states, then you cannot possibly visit every location in 10,000 steps!

While the random samples seem to have a frequency that roughly matches the target pdf, we still do not have great agreement after 100,000 samples.

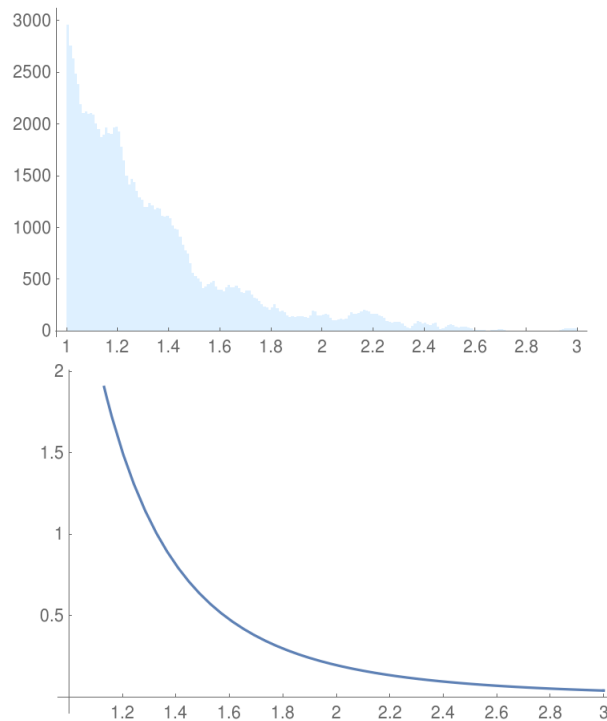


Figure 10.9.6: 100,000 Random Samples from the pdf in Example [10.9.3](#)

If we run our simulation again, but ask for 1,000,000 samples, we get closer to the target distribution. ■

You should get the impression that simulating *continuous* random variables using MCMC methods can be very tricky, and much effort has been spent on developing efficient algorithms for just that purpose.

As a final note on this subject, determining how long an appropriate burn-in period should be and how much sample thinning is appropriate for a given target distribution is a rather subtle art. All that our results tell us is that if we run our Markov Chain for long enough, the frequency of the samples will eventually match the target distribution. Much effort has been spent to study these questions, and some results are known for particular target distributions. Largely, the burn-in and sample acceptance rates are determined by trial and error – especially for more unusual probability distributions.

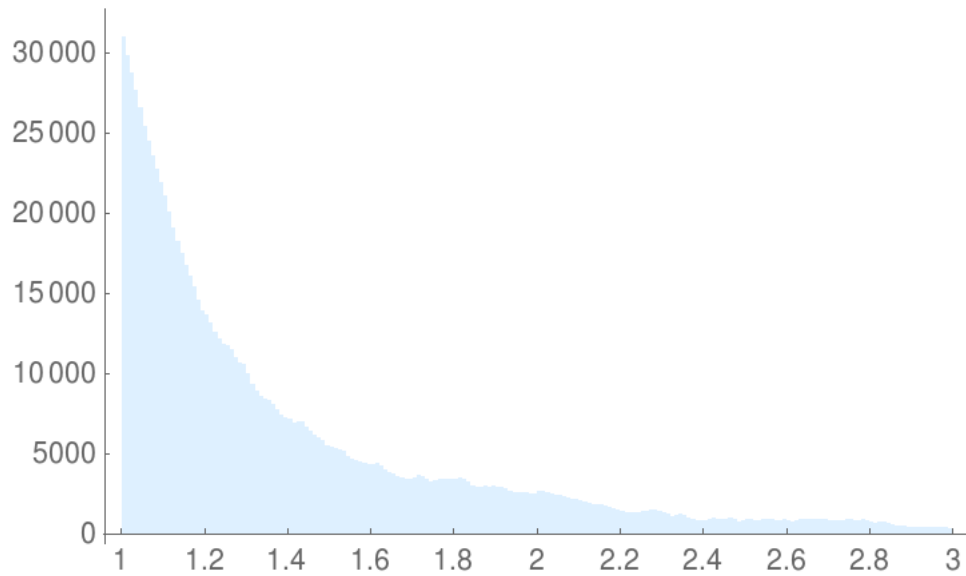
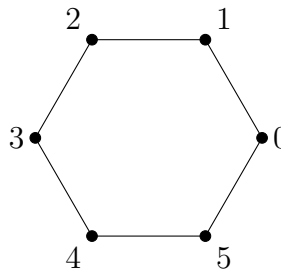


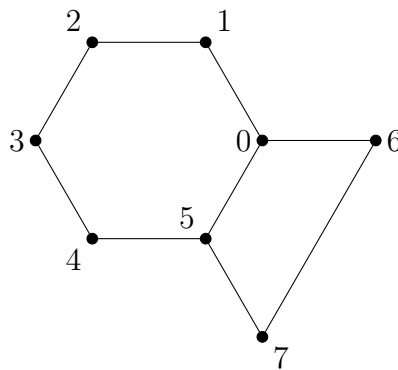
Figure 10.9.7: 1,000,000 Random Samples from the pdf in Example [10.9.3](#)

10.10 Problems

- 1.) Consider a random walk on the hexagon shown below. At each step in the process, there is a probability of $1/5$ that the walker will stay on the current node and a probability of $2/5$ of moving to either of the adjacent nodes.
 - a.) Determine the transition matrix for this process, and classify the states into communication classes and determine whether each class is transient or recurrent.
 - b.) What is the period for each class?
 - c.) If we start at node 1, what is the probability that we land on node 3 after 4 moves?



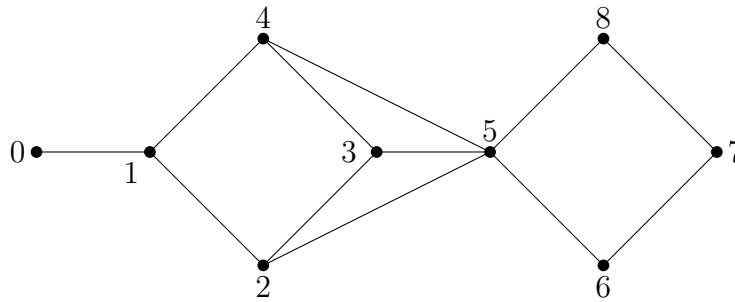
- 2.) Consider a random walk on the graph below. For nodes 1 – 4, there is a probability of $1/5$ that the walker will stay on the current node and a probability of $2/5$ of transitioning to either adjacent node. For node 0, there is a probability of $1/5$ to transition to node 6 and a probability of $2/5$ to transition to either nodes 1 or 5. For node 5, there is a probability of $1/5$ to transition to node 7 and a probability of $2/5$ to transition to either nodes 0 or 4. Node 6 transitions to node 7 with probability 1, and node 7 transitions to node 6 with probability 1.
- Determine the transition matrix for this process, and classify the states into communication classes and determine whether each class is transient or recurrent.
 - What is the period for each class?
 - If we start at node 3, what is the probability that we land on node 7 after 5 moves?



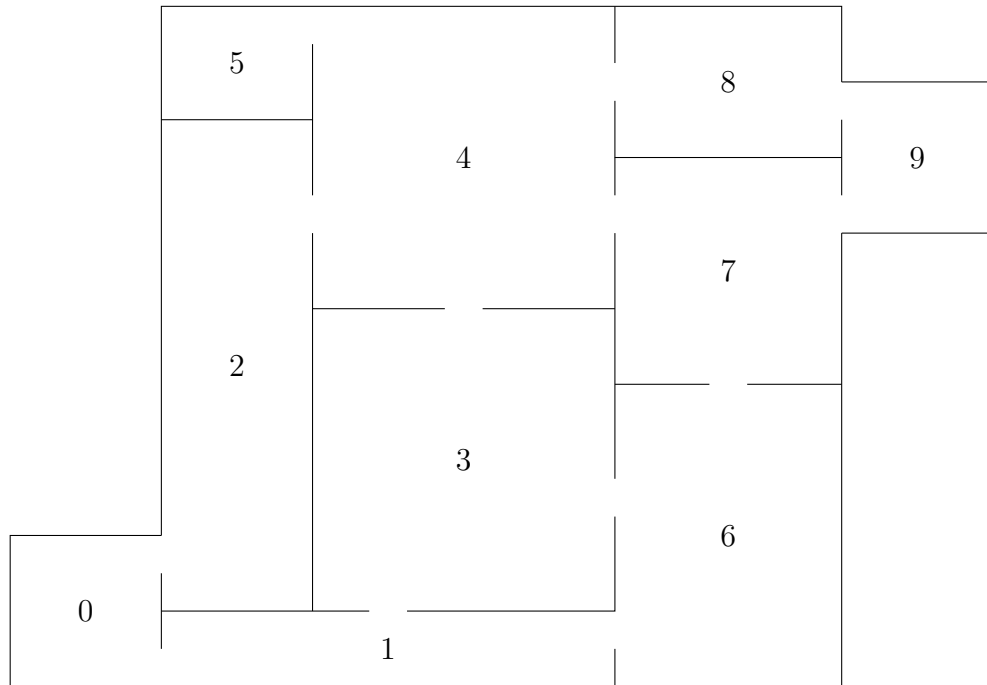
- 3.) Consider a random walk on the graph below. A walker on any of nodes 1 – 4 and 6 – 8 has an equal probability of transitioning to any node

directly connected (and no probability of staying on the current node). On node 0, there is a probability of $3/4$ of staying on node 0 and a probability of $1/4$ of transitioning to node 1. A walker on node 5 can only transition to nodes 6 or 8 each with probability $1/2$.

- a.) Determine the transition matrix for this process, and classify the states into communication classes and determine whether each class is transient or recurrent.
- b.) What is the period for each class?
- c.) If we start at node 0, what is the probability that we have not entered the loop consisting of nodes 5 – 8 after 5 steps? What about after 10 and 20 steps?



- 4.) For the maze in example [10.1.3](#), suppose that the mouse starts in room 2. What is the probability that a mouse gets to room 7 in the least number of steps possible simply by chance?
- 5.) Consider the following maze.



An untrained mouse starts in room 0 and the goal is room 9. Assume that the mouse makes room changes randomly, and each connecting room is equally likely for rooms 0 – 8. Once the mouse reaches room 9, the experiment is over.

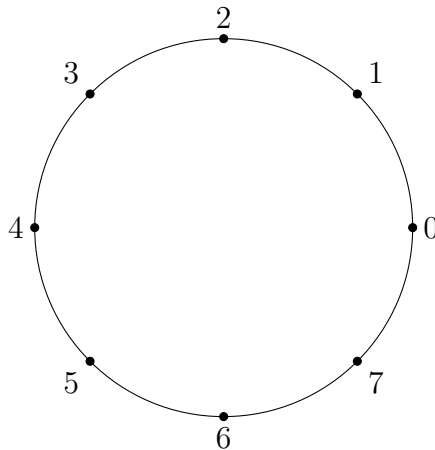
- a.) What is the minimum number of room changes before the mouse can reach room 9?
 - b.) How many room changes before the probability of reaching room 9 is 50% or greater?
 - c.) How many room changes before we can be 95% certain the mouse has reached room 9?
- 6.) The mouse in the previous problem grew frustrated with the maze and chewed new doors connecting rooms 2 and 3 and rooms 3 and 7. How does this change the answers?
 - 7.) Find the limiting probabilities for the stochastic process in problem 1.).

- 8.) A Markov Chain has 4 states (labeled 1, 2, 3, and 4) and a transition matrix

$$P = \begin{bmatrix} 1/4 & 1/3 & 0 & 1 \\ 1/4 & 0 & 0 & 0 \\ 1/4 & 1/3 & 1/2 & 0 \\ 1/4 & 1/3 & 1/2 & 0 \end{bmatrix}.$$

Explain why this process has a set of limiting probabilities. In the long run, which of the four states is least likely?

- 9.) Consider a random walk around the circle shown below. At each time step, the process must transition from its current position to one of the two adjacent nodes. For nodes 1, 2, and 3, the move counterclockwise is three times more likely than the clockwise move. For nodes 5, 6, and 7, the opposite is true. All possible transitions from nodes 0 and 4 are equally likely.



Since this process is not ergodic (why?), we cannot discuss transition probabilities for this Markov Chain. However, if we only look at the two step process between the odd nodes, we do get an irreducible Markov Process (the same is true for the even nodes). Write down the transition matrix for the odd nodes after two steps and find the limiting probabilities of the odd nodes for this new process.

- 10.) Suppose in the random walk above that a walker on node 0 has a probability of $1/3$ to remain on 0, $1/3$ to transition to 1, and $1/3$ to

transition to 7. All other transitions have the same probability. The analogous statement applies to node 4. Why is this Markov Chain ergodic? What are the limiting probabilities of the nodes?

- 11.) Consider the following process. We have 4 red and 4 green balls that are randomly placed into 1 of two urns (each urn receiving 4 balls in total). At each time step, we simultaneously pick one ball at random from each urn to place into the other urn. Let X_i be the total number of red balls in urn 1 at each time step (so $X_i = 0, 1, 2, 3$, or 4). Write down the transition matrix from this Markov Process. Explain why the process has a set of limiting probabilities, and determine these limiting probabilities.
- 12.) Rework problem 11.) but for the case that two balls from each urn are swapped at each time step.
- 13.) Consider the distribution of shapes in epithelial tissue presented in Section 10.7. Suppose that the possible shapes for daughter cells are all equally likely (e.g. a hexagon-shaped cell will produce cells with 4, 5, or 6 sides all with probability $1/3$ rather than by the binomial distribution). Numerically compute the new distribution of side lengths (again assuming no cells have more than 13 sides and the anomalous 14 sided cell collapses to a 4 sided cell).
- 14.) Suppose that two people are repeatedly playing rounds of a game. Each time a player wins, they receive \$1 from the other player, and the rounds continue until one player has lost all of their money. Player *A* starts out with \$20, Player *B* with \$15, but the game is slightly unfair. Player *A* only has a probability $p = 15/32$ of winning each round.
 - a.) What is the probability that Player *A* wins all of the money? Give your answer as a percentage rounded to two decimal places.
 - b.) How much of the total \$35 would player *A* need to start with to have a 50% chance of winning the entire amount? (Only whole dollar amounts are allowed in the game.)
- 15.) Consider the situation described in problem 14.), but now let the probability that Player *A* wins each round be some value p strictly less than $\frac{1}{2}$. If Player *A* starts out with \$20 out of the \$35 total, approximately

how small does p need to be to ensure that Player A only has a 50% chance of winning all of the money? Give p as a percentage rounded to 1 decimal place.

- 16.) **Challenge:** Suppose that in the Gambler's Ruin Problem the probability of winning a bet is dependent on the current size of the player's fortune. Specifically, assume that if a player's fortune is 0 or N , then the game is over. If a player's fortune at time i is $1, 2, 3, \dots, N-1$, then the probability of winning is given by

$$\mathbb{P}(\text{win 1 unit} | X_i = m) = \left(\frac{3}{4}\right)^m.$$

Compute P_M , the probability of reaching a fortune of N before reaching 0 starting from a fortune of size M .

Appendix A

Complex Numbers

The complex numbers, $\mathbb{C}$, are an extension of the real numbers obtained by adjoining a new element i having the property

$$i^2 = -1.$$

This new element is referred to as the *imaginary unit*.¹ Specifically, we can think of the complex numbers as

$$\mathbb{C} = \{a + bi \mid a \text{ and } b \text{ are real numbers}\}.$$

In words, a complex number is the combination of a purely real quantity (the a in $a + bi$) with a real multiple of i (the bi in $a + bi$). The number a is called the *real part* of the complex number while the coefficient b multiplying i is called the *imaginary part*. Traditionally, complex numbers are represented by the letter z . If we have a complex number $z = a + bi$, then we represent the real part of z by

$$\operatorname{Re}(z) = a$$

while the imaginary part is denoted

$$\operatorname{Im}(z) = b.$$

Note that the imaginary part of a complex number is a real number.

We can also give complex numbers a nice geometric interpretation. Recall that the real numbers, $\mathbb{R}$, can be represented on a number line by choosing two arbitrary points on a line (a point labeled 0 and a second point labeled 1). Since a complex number consists of two independent real numbers (the real and imaginary parts), it stands to reason that we should represent these numbers in a two dimensional plane. First, we chose an arbitrary point in the plane and label it as the number $0 = 0 + 0i$. We then choose an arbitrary line through this point to represent the axis of purely real numbers (choosing a second arbitrary point to represent the number $1 = 1 + 0i$). We then use the line that is perpendicular to the real axis passing through 0 to represent

¹In a few disciplines (such as electrical engineering), the imaginary unit is denoted by j since i is reserved for other uses.

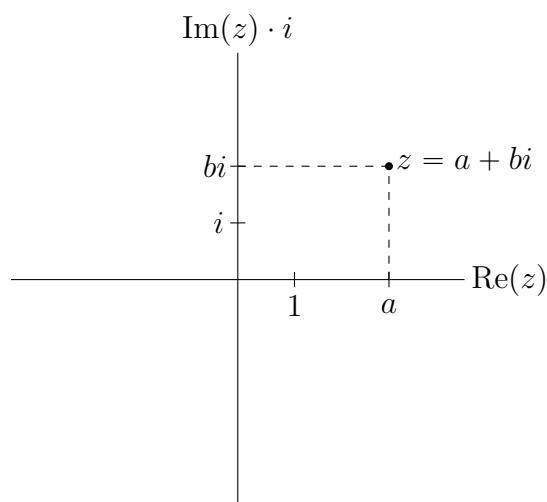


Figure A.0.1: The Complex Plane

the axis of purely imaginary numbers (choosing an arbitrary point on this line to represent the point $i = 0 + 1i$). The typical choice is to have the horizontal axis represent purely real numbers (with 1 to the right of 0) while the vertical axis represents the purely imaginary axis (with i above 0). This gives us the *complex plane*.

As the name would suggest, complex numbers are in fact numbers – meaning that we can add, subtract, multiply, and divide them. Before looking at these operations in detail, a word about terminology is in order. The names “real number,” “imaginary number,” and “complex number” are purely historical terms, and it is important not to read too much into them. In particular, it is very tempting for people to believe that imaginary numbers (and by extension, complex numbers with non-zero imaginary part) are somehow “less real” than real numbers in the sense that real numbers really exist while imaginary numbers do not. This is completely the wrong conclusion to draw! In the modern understanding of these concepts, i exists in more or less the same sense as -1 . Since most people no longer have major issues with the existence of negative numbers (as most people have been in debt at one time or another), the existence of i should not be any more or less mysterious. In fact, this persistent misunderstanding led Gauss to propose the names direct, inverse, and lateral unity for 1, -1 , and i , respectively. Unfortunately, the historical names are far too ingrained in the mathematical community to be

changed. For a delightful (and informative) review of the history of complex numbers, see the excellent book by Nahin.[\[N10\]](#)

A.1 Basic Operations

The arithmetic of complex numbers is relatively simple. With the exception of division, the usual algebraic rules (together with the fact $i^2 = -1$) suffice. Division requires the notion of complex conjugate (which we develop below).

Addition and Subtraction: To add and subtract complex numbers, we simply combine like terms:

$$(a + bi) \pm (c + di) = (a \pm c) + (b \pm d)i.$$

In particular, note that $-(a + bi) = -a - bi$.

Example A.1.1. Find the sum and difference of the following pairs of complex numbers:

- 1) $z_1 = 1 + 3i$ and $z_2 = -2 + 5i$
- 2) $z_1 = -4 + 7i$ and $z_2 = -3 - 9i$

Solution:

1)

$$\begin{aligned} z_1 + z_2 &= (1 + 3i) + (-2 + 5i) \\ &= (1 - 2) + (3 + 5)i = -1 + 8i \\ z_1 - z_2 &= (1 + 3i) - (-2 + 5i) \\ &= (1 + 2) + (3 - 5)i = 3 - 2i \end{aligned}$$

2)

$$\begin{aligned} z_1 + z_2 &= (-4 + 7i) + (-3 - 9i) \\ &= (-4 - 3) + (7 - 9)i = -7 - 2i \\ z_1 - z_2 &= (-4 + 7i) - (-3 - 9i) \\ &= (-4 + 3) + (7 + 9)i = -1 + 16i \quad \blacksquare \end{aligned}$$

Addition of complex numbers has the same nice properties as for real numbers. In particular:

$$1) \ z_1 + z_2 = z_2 + z_1,$$

$$2) \ z + 0 = z,$$

$$3) \ z + (-z) = 0.$$

Multiplication: Multiplication of complex numbers is performed using the distributive law:

$$\begin{aligned}(a + bi)(c + di) &= ac + adi + bci + bdi^2 \\ &= (ac - bd) + (ad + bc)i\end{aligned}$$

Example A.1.2. Find the product following pairs of complex numbers:

$$1) \ z_1 = 1 + 3i \text{ and } z_2 = -2 + 5i$$

$$2) \ z_1 = -4 + 7i \text{ and } z_2 = -3 - 9i$$

Solution:

1)

$$\begin{aligned}z_1 \cdot z_2 &= (1 + 3i)(-2 + 5i) \\ &= 1(-2) + 1(5i) - 2(3i) + (3i)(5i) \\ &= -2 + 5i - 6i - 15 \\ &= -17 - i\end{aligned}$$

2)

$$\begin{aligned}z_1 \cdot z_2 &= (-4 + 7i)(-3 - 9i) \\ &= -4(-3) - 4(-9i) - 3(7i) + (7i)(-9i) \\ &= 12 + 36i - 21i + 63 \\ &= 75 + 15i \blacksquare\end{aligned}$$

Multiplication of complex numbers retains the same basic properties as for real numbers:

$$1) \ z_1 \cdot z_2 = z_2 \cdot z_1,$$

$$2) \ 0 \cdot z = 0,$$

$$3) \ 1 \cdot z = z,$$

$$4) \quad z_1(z_2 + z_3) = z_1 \cdot z_2 + z_1 \cdot z_3.$$

Complex Conjugation: Complex numbers come with one new operation: complex conjugation.

$$\bar{z} = \overline{a + bi} = a - bi.$$

In other words, the complex conjugate of a complex number z is the complex number obtained from z by changing the sign of the imaginary part.

The crucial property of the conjugate is that a complex number z multiplied by its conjugate always gives a real result:

$$\begin{aligned} z \cdot \bar{z} &= (a + bi)(a - bi) \\ &= a^2 + abi - abi - b^2i^2 \\ &= a^2 + b^2. \end{aligned}$$

In other symbols,

$$z \cdot \bar{z} = (\operatorname{Re}(z))^2 + (\operatorname{Im}(z))^2.$$

Example A.1.3. Find the conjugate and compute $z \cdot \bar{z}$ for the following complex numbers:

$$1) \quad z_1 = -1 + 3i$$

$$2) \quad z_2 = 2 - 5i$$

Solution:

1)

$$\begin{aligned} \bar{z}_1 &= \overline{-1 + 3i} = -1 - 3i \\ z_1 \cdot \bar{z}_1 &= (-1)^2 + (3)^2 = 10 \end{aligned}$$

2)

$$\begin{aligned} \bar{z}_2 &= \overline{2 - 5i} = 2 + 5i \\ z_2 \cdot \bar{z}_2 &= (2)^2 + (-5)^2 = 29 \quad \blacksquare \end{aligned}$$

Conjugation interacts nicely with addition and multiplication:

$$1) \quad \overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2,$$

$$2) \quad \overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2.$$

Another nice use of conjugates is that it gives us a way to compute the real and imaginary parts of a complex number.

Theorem A.1.1. *If z is a complex number, then*

- $z + \bar{z} = 2\operatorname{Re}(z),$
- $z - \bar{z} = 2i\operatorname{Im}(z).$

Proof: Let $z = x + iy$ where x and y are real numbers. Then $\operatorname{Re}(z) = x$ and $\operatorname{Im}(z) = y$.

$$\begin{aligned}
 z + \bar{z} &= x + iy + \overline{x + iy} \\
 &= x + iy + x - iy \\
 &= 2x \\
 &= 2\operatorname{Re}(x), \\
 z - \bar{z} &= x + iy - \overline{x + iy} \\
 &= x + iy - (x - iy) \\
 &= 2iy \\
 &= 2i\operatorname{Im}(y). \blacksquare
 \end{aligned}$$

Corollary A.1.1.1. *For any complex number z , $z + \bar{z}$ is a real number.*

Division: Thanks to complex conjugation, we can replace division by a complex number with division by a real number.

$$\frac{a + bi}{c + di} = \frac{a + bi}{c + di} \cdot \frac{c - di}{c - di} = \frac{(a + bi)(c - di)}{c^2 + d^2}.$$

Example A.1.4. Compute the following:

- 1) $\frac{1 + 3i}{2 + i}$
- 2) $\frac{-4 + i}{3 - 5i}$

Solution:

1)

$$\begin{aligned}
 \frac{1+3i}{2+i} &= \frac{1+3i}{2+i} \cdot \frac{2-i}{2-i} \\
 &= \frac{(1+3i)(2-i)}{2^2+1^2} \\
 &= \frac{5+5i}{5} = 1+i
 \end{aligned}$$

2)

$$\begin{aligned}
 \frac{-4+i}{3-5i} &= \frac{-4+i}{3-5i} \cdot \frac{3+5i}{3+5i} \\
 &= \frac{(-4+i)(3+5i)}{3^2+5^2} \\
 &= \frac{-17-17i}{34} = -\frac{1}{2} - \frac{1}{2}i \quad \blacksquare
 \end{aligned}$$

Magnitude: The final operation we discuss is the idea of the magnitude (or absolute value) of a complex number.

$$|z| = \sqrt{z \cdot \bar{z}}.$$

If $z = a + bi$, then $|z| = \sqrt{a^2 + b^2}$. Just as with the absolute value of a real number, the magnitude of a complex number measures the distance of the number from the origin.

The symbol for the magnitude of a complex number is the same as the absolute value of a real number for a reason. Namely, they have the same basic properties.

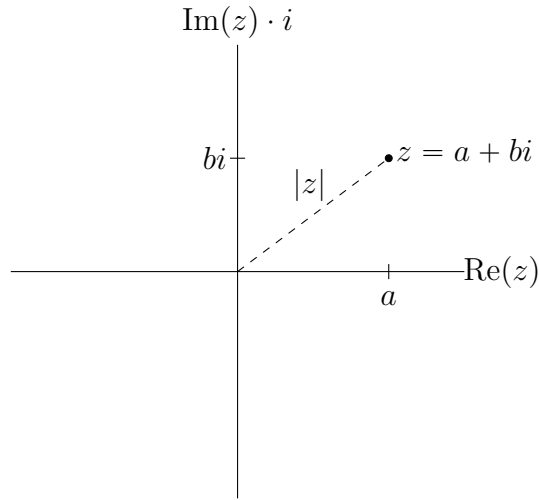


Figure A.1.1: The Magnitude of a Complex Number

- 1) $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$,
- 2) $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$,
- 3) $|z_1 + z_2| \leq |z_1| + |z_2|$,
- 4) $|z| = 0$ if and only if $z = 0$.

Note that the magnitude of a purely real complex number ($z = a + 0i$) is exactly the same as the absolute value of the corresponding real number ($|a + 0i| = \sqrt{a^2} = |a|$).

A.2 The Fundamental Theorem of Algebra

One of the most important facts about complex numbers is the following theorem and its corollary.

Theorem A.2.1 (Fundamental Theorem of Algebra). *Every polynomial equation of the form*

$$x^n + c_{n-1}x^{n-1} + c_{n-2}x^{n-2} + \cdots + c_1x + c_0 = 0$$

with complex coefficients c_i has at least one solution in the complex plane.

Corollary A.2.1.1. *Every polynomial of the form*

$$f(x) = x^n + c_{n-1}x^{n-1} + c_{n-2}x^{n-2} + \cdots + c_1x + c_0$$

factors completely over the complex numbers:

$$f(x) = (x - z_1)(x - z_2) \cdots (x - z_n).$$

For a proof see [Ah79, p.122]. Note that the roots in the factored form of f need not be distinct.

Example A.2.1. Find all solutions of the equation

$$x^3 - x^2 + 9x - 9 = 0.$$

Solution:

$$\begin{aligned} x^3 - x^2 + 9x - 9 &= x^2(x - 1) + 9(x - 1) \\ &= (x - 1)(x^2 + 9) \\ &= (x - 1)(x - \sqrt{-9})(x + \sqrt{-9}) \\ &= (x - 1)(x - 3i)(x + 3i) \end{aligned}$$

So, the solutions to this polynomial equation are

$$x = 1, 3i, -3i. \blacksquare$$

Example A.2.2. Find all solutions of the equation

$$x^4 + 15x^2 - 16 = 0.$$

Solution:

$$\begin{aligned} x^4 + 15x^2 - 16 &= (x^2 + 16)(x^2 - 1) \\ &= (x - \sqrt{-16})(x + \sqrt{-16})(x - 1)(x + 1) \\ &= (x - 4i)(x + 4i)(x - 1)(x + 1) \end{aligned}$$

So, the solutions to this polynomial equation are

$$x = 1, -1, 4i, -4i. \blacksquare$$

Of course, you should be familiar with the quadratic formula:

Lemma A.2.2 (Quadratic Formula). *The solutions to*

$$ax^2 + bx + c = 0$$

where $a \neq 0$ are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The quantity under the square root is known as the *discriminant* since it determines whether the roots are real or complex. In particular, the roots given by the quadratic formula are complex precisely when $b^2 - 4ac < 0$.

A.3 Euler's Formula and Polar Form

Many of the most important functions in mathematics can be extended to accept complex arguments. Here, we focus on the most important one – the exponential function. Suppose that $z = x + iy$. Then it is reasonable to expect that

$$e^z = e^{x+iy} = e^x e^{iy}$$

by the usual law of exponents. So, we really only need to make sense of e^{iy} . This is accomplished by Euler's Formula.

Definition A.3.1 (Euler's Formula).

$$e^{i\theta} = \cos(\theta) + i \sin(\theta)$$

Definition A.3.2 (The Complex Exponential). If $z = x + iy$, then

$$e^z = e^{x+iy} = e^x e^{iy} = e^x (\cos(y) + i \sin(y)).$$

Note that both of these facts are given as *definitions*. In more advanced treatments, the motivation for these definitions is usually discussed, but it involves concepts beyond the scope of this text. Suffice it to say that these are the only sensible definitions if certain facts about the exponential function for real numbers are expected to remain true for complex numbers. Here is a useful corollary of our definition (which follows directly from basic properties of sine and cosine).

Corollary A.3.0.1.

$$\begin{aligned}
e^{-i\theta} &= \cos(-\theta) + i \sin(-\theta) \\
&= \cos(\theta) - i \sin(\theta), \\
\overline{e^{i\theta}} &= e^{-i\theta}.
\end{aligned}$$

Example A.3.1. Compute the following:

- 1) $e^{i\pi/4}$
- 2) $e^{3-i\pi/2}$
- 3) e^{2-3i}

Solution:

1)

$$\begin{aligned}
e^{i\pi/4} &= \cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) \\
&= \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}
\end{aligned}$$

2)

$$\begin{aligned}
e^{3-i\pi/2} &= e^3 e^{-i\pi/2} \\
&= e^3 \left(\cos\left(\frac{\pi}{2}\right) - i \sin\left(\frac{\pi}{2}\right) \right) \\
&= e^3 (0 - i) \\
&= -e^3 i
\end{aligned}$$

3)

$$\begin{aligned}
e^{2-3i} &= e^2 e^{-3i} \\
&= e^2 (\cos(3) - i \sin(3)) \\
&= e^2 \cos(3) - i e^2 \sin(3) \blacksquare
\end{aligned}$$

We should note a very special consequence of Euler's Formula: $e^{i\pi} = -1$. This equation is usually presented as

$$e^{i\pi} + 1 = 0,$$

which the noted physicist Richard Feynman declared to be the most beautiful equation in all of mathematics.

The complex exponential gives us another way to represent complex numbers – the polar form of a complex number.

Theorem A.3.1 (Polar Form of a Complex Number). *Let $z = x + iy$ be any complex number other than 0, and let θ be the angle between the line segment from the origin ending at z and the positive real axis. Then*

$$z = x + iy = |z|e^{i\theta}.$$

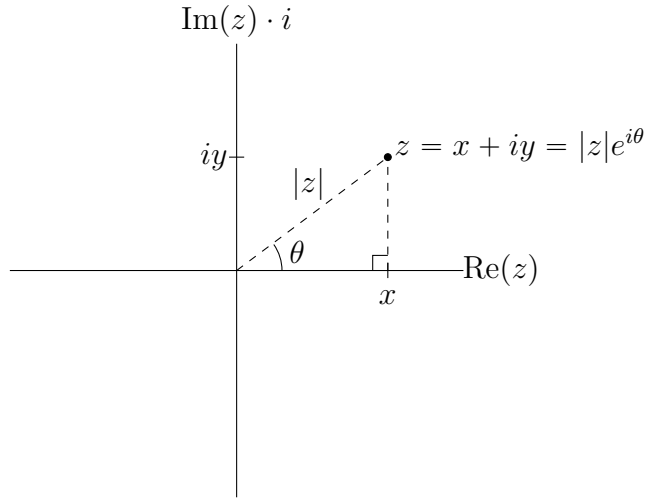


Figure A.3.1: The Polar Form of a Complex Number

Note that the angle θ satisfies

$$\tan(\theta) = \frac{y}{x}$$

when $x \neq 0$.

Proof: The fact that this is true is a simple application of right triangle trigonometry (and the Pythagorean Theorem). Note in the diagram above that

$$\begin{aligned}\cos(\theta) &= \frac{x}{\sqrt{x^2 + y^2}} = \frac{x}{|z|}, \\ \sin(\theta) &= \frac{y}{\sqrt{x^2 + y^2}} = \frac{y}{|z|}.\end{aligned}$$

Using these facts gives

$$\begin{aligned} z &= x + iy \\ &= |z| \cos(\theta) + i|z| \sin(\theta) \\ &= |z| (\cos(\theta) + i \sin(\theta)) \\ &= |z| e^{i\theta}. \end{aligned}$$

At the origin, we certainly have $|z| = 0$, but the angle θ is not defined. In fact, for *any angle* θ , $0 \cdot e^{i\theta} = 0$. As a result, the origin does not have a unique representation in polar form. ■

Example A.3.2. Find the polar form of $z = 2 + 2i$.

Solution: First, we note that

$$|2 + 2i| = \sqrt{2^2 + 2^2} = 2\sqrt{2}.$$

As for the angle θ , note that $2 + 2i$ lies on the line $y = x$ in the complex plane. Since this line is at 45° , we can take $\theta = \pi/4$. Hence,

$$2 + 2i = 2\sqrt{2}e^{i\pi/4}. \quad \blacksquare$$

One of the many uses of the polar form is that it makes it easier to take powers of complex numbers.

Example A.3.3. Compute $(2 + 2i)^{10}$.

Solution: We could multiply the given number by itself 10 times in a row, but the polar form makes this process far easier.

$$\begin{aligned} (2 + 2i)^{10} &= \left(2\sqrt{2}e^{i\pi/4}\right)^{10} \\ &= 2^{10}(\sqrt{2})^{10}e^{i10\pi/4} \\ &= 32768 \left(\cos\left(\frac{5\pi}{2}\right) + i \sin\left(\frac{5\pi}{2}\right) \right) \\ &= 32768 \left(\cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) \right) \\ &= 32768i \quad \blacksquare \end{aligned}$$

The polar form of a complex number even allows us to take roots of complex numbers. Since this process will not be important for us going forward, we leave its development to the exercises.

A.4 Exercises

1.) In the following problems, write the given complex number in the form $a + bi$.

a) $(7 + 2i) - (10 - 5i)$ b) $\frac{3}{2i}$

c) $(-2 + 3i)^2$ d) $\frac{7i-3}{i}$

e) $\frac{2+3i}{1+2i} - \frac{8+i}{6-i}$ f) $\left[\frac{2+i}{6i-(1-2i)} \right]^2$

2.) In the following problems, solve the equation for z .

a) $z = 3 + 2zi$ b) $iz = 5 + 2z$

c) $\frac{1}{z} = 2 + i$ d) $\frac{z}{1-z} = 1 - 5i$

3.) In the following problems, find all solutions (with multiplicities) to the given equation.

a) $z^2 + 2z + 2 = 0$ b) $z^4 - 16 = 0$

c) $z^3 + 8 = 0$ d) $2z^3 - z^2 + 8z - 4 = 0$

e) $z^4 - 8z^2 + 16 = 0$ f) $z^5 + 2z^3 + z = 0$

g) $z^6 - z^4 - z^2 + 1 = 0$ h) $(z^4 + 18z^2 + 18)(z^4 - 2z + 1) = 0$

4.) In the following problems, write the given complex number in the form $a + bi$.

a) $e^{-i\pi/4}$ b) $e^{2\pi i/3}$ c) $e^{3i} - e^{-3i}$

d) e^{e^i} e) $2e^{3+i\pi/6}$ f) $e^{2+i \arctan(2)}$

5.) In the following problems, write the given complex number in polar form $re^{i\theta}$.

a) $-2i$ b) $\frac{1-i}{3}$ c) $(1+i)^6$

d) $\frac{2i}{3e^{4+i}}$ e) $\frac{2+2i}{-\sqrt{3}+i}$ f) $3\sin(2) + 3i\cos(2)$

6.) Let n be a natural number. The solutions to

$$z^n - 1 = 0$$

are known as the n -th roots of unity (since $x^n = 1$). We can show that there are exactly n distinct n -th roots of unity which are given by

$$e^{i2k\pi/n} = \cos\left(\frac{2k\pi}{n}\right) + i \sin\left(\frac{2k\pi}{n}\right), \quad k = 0, 1, 2, \dots, n-1.$$

a) Find all third roots of unity (i.e. solutions to $z^3 - 1 = 0$).

b) Find all sixth roots of unity (i.e. solutions to $z^6 - 1 = 0$).

c) Find all solutions to $z^5 - 32 = 0$.

d) Find all solutions to $x^5 + 1 = 0$

7.) The *principal value of the logarithm*² for a complex number $z = re^{i\theta}$ is given by

$$\text{Log } z = \ln r + i\theta$$

where we take $-\pi < \theta \leq \pi$.

For for the following problems, find the indicated (principal value) logarithm:

a) $\text{Log}(i)$ b) $\text{Log}(-1)$

c) $\text{Log}(1 - i)$ c) $\text{Log}(\sqrt{3} + i)$

8.) We can use the principal value of the logarithm to define principal values for complex powers: For a complex number α and $z \neq 0$, we define

$$z^\alpha = e^{\alpha \text{Log}(z)}.$$

For the following problems, compute the indicated (principal value) power:

a) i^i b) $2^{\pi i}$

c) $(1 + i)^{1-i}$ d) $(\sqrt{3} + i)^{-i}$

²Since for any complex number $z = re^{i\theta}$ we also have

$$z = re^{i(\theta \pm 2\pi)} = re^{i(\theta \pm 4\pi)} = \dots,$$

there is an inherent ambiguity in what we mean by the polar angle θ . “Principal value” refers to the requirement $-\pi < \theta \leq \pi$ which remedies this problem.

Appendix B

Basic Derivatives and Integrals

Table I: General Rules

Derivative Rule	Integration Rule
Sum/Difference Rule $\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$	Sum/Difference Rule $\int [f(x) \pm g(x)]dx = \int f(x)dx \pm \int g(x)dx$
Constant Multiple Rule $\frac{d}{dx}[cf(x)] = cf'(x)$	Constant Multiple Rule $\int cf(x)dx = c \int f(x)dx$
Product Rule $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$	Integration by Parts $\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx$
Quotient Rule $\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$	(no simple rule corresponds)
Chain Rule $\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$	U-Substitution $\int f(g(x))g'(x)dx = \int f(u)du$ where $u = g(x)$

Table II: Rules for Specific Functions

Derivative Rule	Integration Rule
Constant Rule $\frac{d}{dx} [c] = 0$	Constant Rule $\int c \, dx = cx + C$
Power Rule $\frac{d}{dx} [x^p] = px^{p-1}$	Power Rule $\int x^p dx = \frac{1}{p+1} x^{p+1} + C$ for $p \neq -1$
$\frac{d}{dx} \ln x = \frac{1}{x}$	$\int \frac{dx}{x} = \ln x + C$
$\frac{d}{dx} \log_b x = \frac{1}{x \ln b}$	same as above
$\frac{d}{dx} e^x = e^x$	$\int e^x dx = e^x + C$
$\frac{d}{dx} b^x = (\ln b) b^x$	$\int b^x dx = \frac{1}{\ln b} b^x + C$
$\frac{d}{dx} \sin(x) = \cos(x)$	$\int \cos(x) dx = \sin(x) + C$
$\frac{d}{dx} \cos(x) = -\sin(x)$	$\int \sin(x) dx = -\cos(x) + C$
$\frac{d}{dx} \tan(x) = \sec^2(x)$	$\int \sec^2(x) dx = \tan(x) + C$
$\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$	$\int \sec(x) \tan(x) dx = \sec(x) + C$
$\frac{d}{dx} \csc(x) = -\csc(x) \cot(x)$	$\int \csc(x) \cot(x) dx = -\csc(x) + C$
$\frac{d}{dx} \cot(x) = -\csc^2(x)$	$\int \csc^2(x) dx = -\cot(x) + C$
$\frac{d}{dx} \arctan(x) = \frac{1}{1+x^2}$	$\int \frac{dx}{1+x^2} = \arctan(x) + C$
$\frac{d}{dx} \arcsin(x) = \frac{1}{\sqrt{1-x^2}}$	$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin(x) + C$
$\frac{d}{dx} \arccos(x) = -\frac{1}{\sqrt{1-x^2}}$	same as above (NOTE: $\arccos(x) = \frac{\pi}{2} - \arcsin(x)$)

Table III: Additional Integrals

$\int \sin^2(x) dx = \frac{x}{2} - \frac{\sin(2x)}{4} + C$
$\int \cos^2(x) dx = \frac{x}{2} + \frac{\sin(2x)}{4} + C$
$\int \tan(x) dx = \ln \sec(x) + C$
$\int \cot(x) dx = \ln \sin(x) + C$
$\int \sec(x) dx = \ln \sec(x) + \tan(x) + C$
$\int \csc(x) dx = -\ln \csc(x) + \cot(x) + C$
$\int \ln(x) dx = x \ln(x) - x + C$
$\int \log_b(x) dx = \frac{1}{\ln b} (x \ln(x) - x) + C$

Table IV: Useful Integrals

$\int \arctan(x) dx = x \arctan(x) - \frac{1}{2} \ln(1 + x^2) + C$
$\int \arcsin(x) dx = x \arcsin(x) + \sqrt{1 - x^2} + C$
$\int \arccos(x) dx = x \arccos(x) - \sqrt{1 - x^2} + C$

Appendix C

Positive Infinite Series

In Chapter 10, several results involve summing an infinite number of positive terms. Such sums are known as **positive infinite series** (or just *positive series*). In this appendix, we present the relevant facts we need about these sums. See [RA15, Chap. 10] for details and proofs of the major results below.

C.1 Definitions of Sequences and Series

A **sequence** is simply a list of numbers (real or complex) indexed by a set of increasing, sequential integers. For example,

$$a_0 = 1, a_1 = \frac{1}{2}, a_2 = \frac{1}{4}, a_3 = \frac{1}{8}, \dots, a_n = \frac{1}{2^n}, \dots$$

is a sequence beginning at index $n = 0$. Sequences of numbers often begin at index 0 or 1 (though this is not a requirement). Very often, a sequence is specified by giving the *general term* $a_n = f(n)$ (as we did above) along with the range of the index variable n . So, the sequence above can be denoted $\left(\frac{1}{2^n}\right)_{n=0}^{\infty}$. As another example, the series $\left(1 - \frac{1}{n}\right)_{n=1}^{\infty}$ has terms $0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots$. If the range of the index variable is understood (or unimportant) for a result, we often write (a_n) to stand for the sequence.

If the general term of a sequence is given by a function, $a_n = f(n)$, then we say that the **limit of the sequence** is

$$\lim_{n \rightarrow \infty} a_n = \lim_{x \rightarrow \infty} f(x)$$

provided the function f has a limit at infinity. If the limit of the sequence (a_n) is the number L , we often write

$$a_n \rightarrow L$$

to denote it. In the examples above

$$\begin{aligned} \frac{1}{2^n} &\rightarrow 0, \\ 1 - \frac{1}{n} &\rightarrow 1. \end{aligned}$$

An **infinite series** (or simply *series*, for short) is a special kind of sequence constructed by successively adding up the terms from a given sequence. If the given sequence is $(a_n)_{n=1}^{\infty}$, then the associated series

$$\sum_{n=1}^{\infty} a_n$$

represents the limit of the sequence of *partial sums* $(S_n)_{n=1}^{\infty}$ defined by

$$\begin{aligned} S_1 &= a_1, \\ S_2 &= S_1 + a_2 = a_1 + a_2, \\ S_3 &= S_2 + a_3 = a_1 + a_2 + a_3, \\ &\vdots \\ S_n &= a_1 + a_2 + a_3 + \cdots + a_{n-1} + a_n \\ &\vdots \end{aligned}$$

In other words, the infinite series associated to the sequence (a_n) is simply the limit of the sequence obtained by successively adding up more and more of the terms a_n . Notice that the series need not start at $n = 1$! As with sequences, if the range of indices is understood (or unimportant) in a result, we often suppress them.

If the partial sums have a finite limit, $S_n \rightarrow L$, then we write

$$\sum_{n=1}^{\infty} a_n = L,$$

and we say the series *converges to* L . Otherwise, we say the series *diverges*. In the special case that $S_n \rightarrow \infty$, we often say the series *diverges to positive infinity*.

Before we look at specific series, we list some basic facts about convergence of series.

Theorem C.1.1. *If $\sum a_n = L$, $\sum b_n = M$, and c is a constant, then*

- $\sum (a_n \pm b_n) = L \pm M$,
- $\sum ca_n = cL$.

Proof: This follows from basic facts about limits. ■

Theorem C.1.2 (The n -th Term Test for Divergence). *If the sequence (a_n) does not converge to 0, then $\sum a_n$ necessarily diverges.*

Proof: Since $S_n = S_{n-1} + a_n$, we must have $S_n - S_{n-1} = a_n$. If the series converges to L , then by definition $S_n \rightarrow L$. This gives us

$$\begin{aligned}\lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} (S_n - S_{n-1}) \\ &= \lim_{n \rightarrow \infty} S_n - \lim_{n \rightarrow \infty} S_{n-1} \\ &= L - L = 0. \quad \blacksquare\end{aligned}$$

WARNING: This theorem cannot be used to show that series converges! If (a_n) does not converge to zero, then the theorem states that the corresponding series must diverge. However, $a_n \rightarrow 0$ *does not guarantee* that $\sum a_n$ will converge. We will see a famous example of this fact later!

C.2 Geometric Series

Arguably the most important type of infinite series are **geometric series**:

$$\sum_{n=0}^{\infty} ar^n.$$

The number a is the *first term* of the series while r is usually called the *ratio* of the series.

It turns out that we can give a very useful formula for the n -th partial sum of a geometric series (with $a = 1$).

Lemma C.2.1. *For a geometric series $\sum_{n=0}^{\infty} r^n$ with $r \neq 1$, the n -th partial sum is given by*

$$S_n = 1 + r + r^2 + r^3 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}.$$

Proof: Since this is a finite sum for any given n , we do not have to worry about convergence issues. If we take S_n and multiply it by r , we get

$$rS_n = r + r^2 + r^3 + \cdots + r^n + r^{n+1}.$$

Subtracting S_n and rS_n gives

$$\begin{aligned} S_n - rS_n &= (1 + r + r^2 + r^3 + \cdots + r^n) - (r + r^2 + r^3 + \cdots + r^n + r^{n+1}) \\ (1 - r)S_n &= 1 - r^{n+1}. \end{aligned}$$

If $r \neq 1$, we can divide by $1 - r$ to produce the formula in the lemma. ■

Theorem C.2.2. *If $|r| < 1$, then*

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1 - r}.$$

So, the sum of a geometric series is the first term of the series divided by one minus the ratio when the ratio has magnitude less than 1. When $|r| \geq 1$, the geometric series diverges.¹

Proof: If $|r| < 1$, then $r^{n+1} \rightarrow 0$ and the partial sums from the previous lemma converge. If $|r| > 1$, then r^{n+1} definitely diverges. In the two special cases $r = \pm 1$, the corresponding geometric series are easily seen to diverge. For $r = 1$,

$$\sum_{n=0}^{\infty} a(1)^n = a + a + a + \cdots = \pm\infty.$$

For $r = -1$, we look at the sequence of partial sums:

$$\begin{aligned} S_0 &= a \\ S_1 &= a - a = 0 \\ S_2 &= a - a + a = a \\ S_3 &= a - a + a - a = 0 \\ &\vdots \end{aligned}$$

Since the partial sums alternate between a and 0, they have no limit, and so the geometric series diverges. ■

Example C.2.1. Find the sums of the following convergent geometric series.

$$1) \sum_{n=0}^{\infty} \frac{1}{2^n}, \quad 2) \sum_{n=0}^{\infty} \frac{2}{3^n}, \quad 3) \sum_{n=2}^{\infty} \frac{5^n}{7^n}$$

¹This result is also true for geometric series with complex terms! The proof is more or less the same with some modifications for the case $|r| = 1$.

Answer: For the the first series,

$$\sum_{n=0}^{\infty} \frac{1}{2^n} = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots,$$

and so the first term is 1 and the ratio is $1/2$. This gives

$$\sum_{n=0}^{\infty} \frac{1}{2^n} = \frac{1}{1 - 1/2} = \frac{1}{1/2} = 2.$$

For the second series,

$$\sum_{n=0}^{\infty} \frac{2}{3^n} = 2 \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n = 2 + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \cdots,$$

and so the first term is 2 and the ratio is $1/3$. This gives

$$\sum_{n=0}^{\infty} \frac{2}{3^n} = \frac{2}{1 - 1/3} = \frac{2}{2/3} = 3.$$

For the final series,

$$\sum_{n=2}^{\infty} \frac{5^n}{7^n} = \sum_{n=2}^{\infty} \left(\frac{5}{7}\right)^n = \frac{25}{49} + \frac{125}{343} + \frac{625}{2401} + \cdots,$$

and so the first term is $25/49$ and the ratio is $5/7$. This gives

$$\sum_{n=2}^{\infty} \frac{5^n}{7^n} = \frac{25/49}{1 - 5/7} = \frac{25/49}{2/7} = \frac{25}{14}. \blacksquare$$

C.3 Positive Series

If the sequence (a_n) consists of both positive and negative terms, then the convergence of $\sum a_n$ can be a delicate issue in general. However, if we have a sequence of *positive terms* $a_n \geq 0$, things are considerably easier. Notice that we really mean *non-negative* terms, but the use of “positive” in this case is a common misnomer in mathematics. If the terms are definitely greater than zero ($a_n > 0$), you often see the term *strictly positive* series.

Theorem C.3.1 (The Dichotomy Property for Positive Series). *For a series $\sum a_n$ of positive terms, there are only two possibilities for convergence:*

- *the sequence of partial sums, S_n , is bounded above, and so the series converges to a finite value,*
- *the sequence of partial sums is unbounded, and the series diverges to positive infinity.*

Heuristic Proof: Since $S_n = S_{n-1} + a_n$ and $a_n \geq 0$, it must be that $S_n \geq S_{n-1}$. So, the sequence of partial sums, (S_n) , is an increasing sequence. If the partial sums are unbounded, then they must stream off to positive infinity. Otherwise, all of the partial sums are bounded above by some number, $S_n \leq U$ for all n . It can be shown in this case that the sequence of partial sums must converge to a finite value. ■

While there is a lot we could say about positive series, we choose to present a few basic results (without proof) which allow us to conclude that certain kinds of positive series converge or diverge. Along the way, we will examine some particular examples.

Theorem C.3.2 (The Integral Test). *If a sequence has general term given by $a_n = f(n)$ where f is positive, decreasing, and continuous on $[1, \infty)$, then the following results are true.*

- *If $\int_1^\infty f(x) dx$ converges, then so will $\sum_{n=1}^\infty a_n$.*
- *If $\int_1^\infty f(x) dx$ diverges to positive infinity, then so will $\sum_{n=1}^\infty a_n$.*

Example C.3.1 (The Harmonic Series). The series

$$\sum_{n=1}^\infty \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

is known as the **harmonic series** and is the most famous example of a series whose terms go to zero but nonetheless fails to converge. Use the integral test to show that this series diverges.

Answer: Since $f(x) = 1/x$ is positive, continuous, and decreasing on $[1, \infty)$, we can apply the Integral Test.

$$\begin{aligned}\int_1^\infty \frac{1}{x} dx &= \lim_{R \rightarrow \infty} \int_1^R \frac{1}{x} dx \\ &= \lim_{R \rightarrow \infty} \ln(R) - \ln(1) \\ &= \lim_{R \rightarrow \infty} \ln(R) \\ &= \infty\end{aligned}$$

By the Integral Test, the harmonic series must also diverge. ■

A ***p*-series** is any series of the form

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

where p is traditionally taken to be a real number greater than zero. Notice that the harmonic series is the p -series with $p = 1$. The Integral Test along with results from Section 3.3 give us the following lemma.

Lemma C.3.3 (Convergence of p -Series). *The p -series $\sum_{n=1}^{\infty} \frac{1}{n^p}$*

- ***converges*** when $p > 1$,
- ***diverges*** when $p \leq 1$.

Theorem C.3.4 (The Ratio Test). *Suppose that (a_n) is a sequence of strictly positive terms. Also, suppose that*

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L.$$

In particular, we are assuming that this limit exists! Then the following results are true.

- *If $L < 1$, then $\sum a_n$ converges.*
- *If $L > 1$, then $\sum a_n$ diverges to positive infinity.*
- *If $L = 1$, then the test is inconclusive (that is, $\sum a_n$ could converge or diverge to infinity).*

Notice that the ratio test is only helpful when the ratio of consecutive terms a_{n+1}/a_n has a limit at infinity that is not equal to 1!

Example C.3.2. Show that the series

$$\sum_{n=0}^{\infty} \frac{n}{3^n}$$

converges.

Answer: The general term is given by $a_n = \frac{n}{3^n}$. So,

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{n+1}{3^{n+1}} \cdot \frac{3^n}{n} \\ &= \frac{n+1}{3n}. \end{aligned}$$

This shows us that

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{1}{3} \left(\frac{n+1}{n} \right) = \frac{1}{3},$$

and the series converges by the ratio test. ■

Example C.3.3. Show that the series

$$\sum_{n=0}^{\infty} \frac{2^n}{n!}$$

converges.

Answer: The general term is given by $a_n = \frac{2^n}{n!}$. So,

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} \\ &= \frac{2(n!)}{(n+1)n!} \\ &= \frac{2}{n+1}. \end{aligned}$$

This shows us that

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{2}{n+1} = 0,$$

and the series converges by the ratio test.² ■

Example C.3.4. Show that the series

$$\sum_{n=0}^{\infty} \frac{5^n}{n^2}$$

diverges.

Answer: The general term is given by $a_n = \frac{5^n}{n^2}$. So,

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{5^{n+1}}{(n+1)^2} \cdot \frac{n^2}{5^n} \\ &= \frac{5n^2}{(n+1)^2} \\ &= 5 \left(\frac{n}{n+1} \right)^2. \end{aligned}$$

This shows us that

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} 5 \left(\frac{n}{n+1} \right)^2 = 5,$$

and the series diverges by the ratio test. ■

²More advanced techniques can show that

$$\sum_{n=0}^{\infty} \frac{2^n}{n!} = e^2.$$

Appendix D

Standard Normal Probabilities

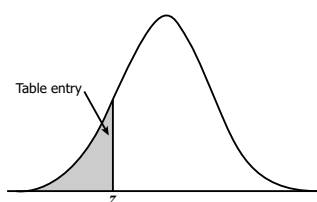
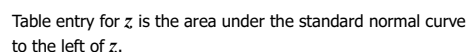


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

[illegible]

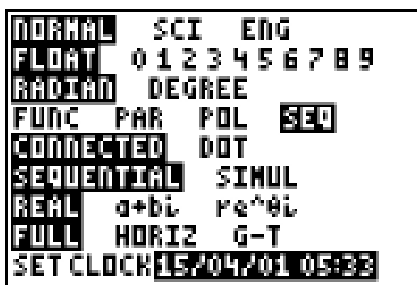
Appendix E

TI-83/84 and TI-89 Material

E.1 Cobweb Plots

E.1.1 Cobweb Plots on the TI-83/84

To create cobweb plots on a TI-83/84 the calculator must be in sequence mode. Press the MODE button and use the arrow keys to move down to the row that begins with FUNC and use the arrow keys to move across to highlight SEQ. Hit the ENTER key to save the change to sequence mode.

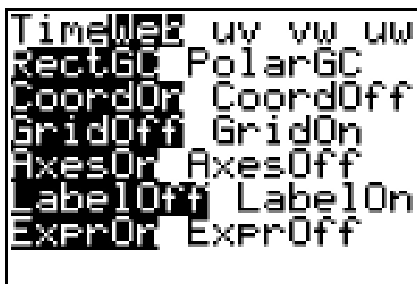


(a) TI-84 Plus Silver



(b) TI-84 Plus CE

Next we tell the calculator to use the cobweb plot format by pressing FORMAT F3 (obtained using the 2nd key and then ZOOM) and using the arrow keys to highlight WEB. Again hit ENTER to save the change.



(a) TI-84 Plus Silver



(b) TI-84 Plus CE

To enter the updating function, go to the graphing screen using the $Y =$ (or F1) key. The TI-84 uses the variables u, v , and w as the names of the possible updating functions. We can type these variables on the calculator using the 2nd key and then 7, 8 or 9 respectively. The calculator uses $nMin$ to represent the smallest value of the time variable n . The default value is 1. The TI-84 expects the updating function to be specified using $u(n)$ in terms of $u(n-1)$ instead of q_{n+1} in terms of q_n .

To set up a cobweb plot for the system $q_{n+1} = 2q_n - 1$ with initial condition $q_0 = 1.2$ we type in the formulas $u(n) = 2u(n-1) - 1$ and $u(nMin) = 1.2$. To type the time variable n use the variable key labeled X,T, θ , n . The result should look like:

```
Plot1 Plot2 Plot3
nMin=1
u(n)=2*u(n-1)-1
u(nMin)=1.2
v(n)=
v(nMin)=
w(n)=
```

(a) TI-84 Plus Silver

```
NORMAL FLOAT AUTO REAL RADIAN MP
SECOND CONDITION IF NEEDED
Plot1 Plot2 Plot3
TYPE: SEQ(n) SEQ(n+1) SEQ(n+2)
nMin=1
u(n)=2*u(n-1)-1
u(1)=1.2
v(n)=
v(1)=
v(2)=
w(n)=
```

(b) TI-84 Plus CE

We also should choose a window appropriate for the plot we are graphing. Press the WINDOW key to set the parameters for the window. $nMin$ and $nMax$ represent the smallest and largest values of the time variable n . $Xmin$, $Xmax$, $Xscl$, $Ymin$, $Ymax$, and $Yscl$ describe the size of the window and the markings on the x -axis and y -axis. Set them as shown below:

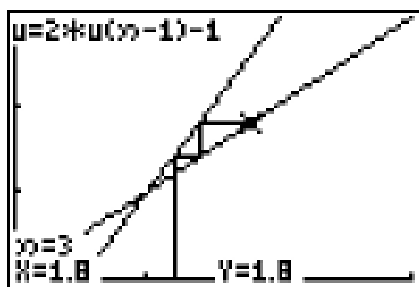
```
WINDOW
nMin=1
nMax=10
PlotStart=1
PlotStep=1
Xmin=0
Xmax=3
Xscl=1
```

(a) TI-84 Plus Silver

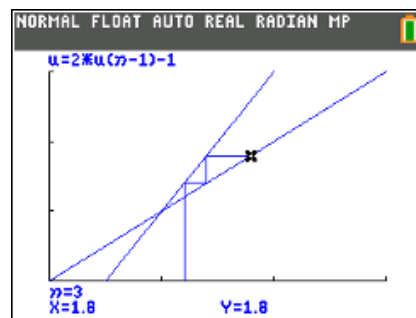
```
NORMAL FLOAT AUTO REAL RADIAN MP
WINDOW
nMin=1
nMax=10
PlotStart=1
PlotStep=1
Xmin=0
Xmax=3
Xscl=1
Ymin=0
Ymax=3
```

(b) TI-84 Plus CE

Finally we are ready to graph the cobweb plot. Press the GRAPH key and the calculator draws a graph of the updating function and a graph of $y = x$. For our example the updating function will be a line. To start generating the cobweb, press the TRACE button and then use the right arrow key to step through the cobweb plot. At the bottom of the screen the calculator displays the current value of n , and at first $X = u(n-1)$ and $Y = u(n)$ while showing the cobweb moving vertically to the the graph of the updating function and then after we press the right arrow again it shows $X = u(n)$ and $Y = u(n)$ while showing the cobweb moving horizontally to the line $y = x$. After you press the right arrow four times you should see:



(a) TI-84 Plus Silver



(b) TI-84 Plus CE

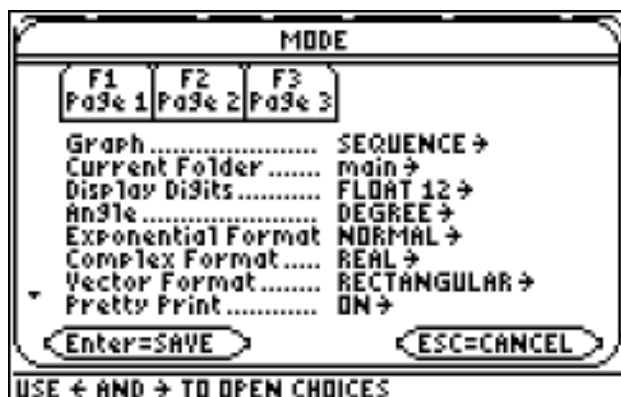
We can see more and more of the cobweb by repeatedly pressing the right arrow key.

E.1.2 Cobweb Plots on the TI-89

The TI-89 graphing calculator has built-in programming to produce cobweb plots. The first step is to change the Graph Mode. Pressing the MODE button should bring up the following options menu:



Pressing the right arrow key on the Graph option will bring up a list of different graphing environments. Select 4:SEQUENCE and then ENTER to save the change.



When you enter the graphing screen ( + F1), you should see a screen with prompts starting “u1 =.” Open the F7:Axes tab (by pressing F5 and the right arrow key twice). This brings up the following menu screen.



With Axes set to TIME, you can produce sequential plots of discrete dynamical systems. Since we want to produce cobweb plots, press the right arrow key and select 2:WEB.



The Build Web option can be set to 1:TRACE or 2:AUTO. TRACE mode will allow you to see the cobweb plot built up step-by-step (by pressing F3:TRACE and then using the right arrow key after having the calculator produce the graph of the updating function). We will set the option on AUTO which will build up the entire cobweb plot in one step. Press ENTER to save your choices.

To produce the cobweb plot for

$$q_{n+1} = 2q_n - 1$$

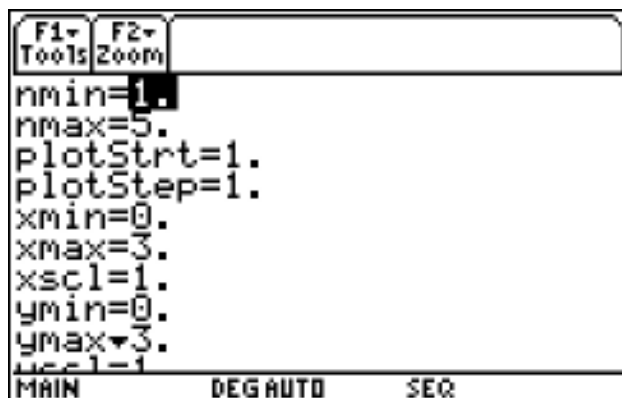
$$q_0 = 1.2,$$

we enter the graphing screen ( + F1) and enter the dynamical system in terms of u1. The TI-89 is looking for u1(n) in terms of u1(n-1) (rather than

$u_1(n+1)$ in terms of $u_1(n)$). So, we select u_1 by pressing ENTER and typing $u_1(n) = 2 \cdot u_1(n-1) - 1$. Pressing ENTER again will save this to the u_1 entry. The initial data for the system is entered just below labeled u_{i1} .

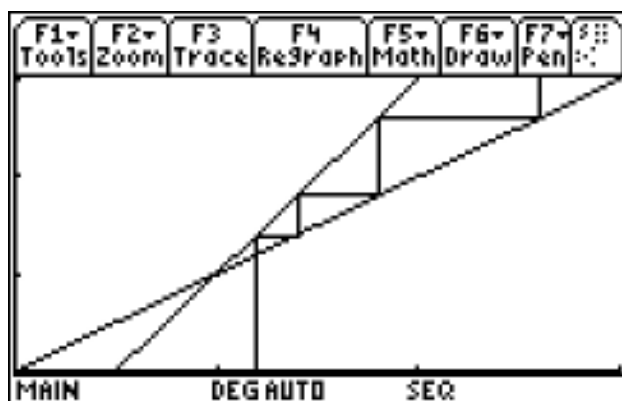


We can set the various limits for the axes by entering the WINDOW menu (pressing  + F2).



the n_{min} and n_{max} options determine how many time steps out the dynamical system will be plotted. Note that the TI-89 starts at $n = 1$ rather than $n = 0$. Start out with a modest number of steps to begin with (5 to 10 should do). The options x_{min} , x_{max} , y_{min} , and y_{max} are just as they are in the traditional graphing mode. Since the cobweb plot always includes the graph $y = x$, it is reasonable to start with the exact same values for x_{min} and y_{min} (and similarly for x_{max} and y_{max}).

Once you have entered the appropriate ranges, instruct the TI-89 to GRAPH the dynamical system (pressing  + F3). You should end up with a graph like the one below!



Remember to change the graphing mode back to FUNCTION if you need to produce a traditional graph of a function.

E.2 Piecewise Functions

E.2.1 Piecewise Functions on the TI-83/84

To enter piecewise defined functions with the TI-84 family of calculators, you define the function using both functional expressions and logical expressions. The functional expressions represent the formulas you want to include and the logical expressions will contain the conditions used to decide which functional expressions contribute for a given value of the variable x . Each logical expression will be used in the computations with a value of 1 when true and a value of 0 when false. The complete functional formula is a sum of pieces each of which is a product of a functional expression and the corresponding condition when it should be used. You should probably make sure your conditions do not overlap.

For example, you enter the formula $-1(x \leq 0) + 1(x > 0)$ to enter the piecewise function

$$f(x) = \begin{cases} -1 & x \leq 0 \\ 1 & x > 0 \end{cases}.$$

For any x less than or equal to 0, the calculator evaluates the logical expression $(x \leq 0)$ with a value of 1 for true and the logical expression $(x > 0)$ with a value of 0 for false. For such x it computes -1 times 1 plus 1 times 0 to get a value of -1 for the function as a whole.

Similarly, for any x greater than 0, the calculator evaluates the logical expression $(x \leq 0)$ with a value of 0 for false and the logical expression $(x > 0)$ with a value of 1 for true. For such x it computes -1 times 0 plus 1 times 1 to get a value of 1 for the function as a whole.

The expression and its graph are shown below on the TI-84 CE Plus. The older TI-83/84 calculators have a similar format and graph. (Details of typing these expressions are shown later.)

You can substitute any function formulas for the -1 and the 1 and you can change the logical conditions.

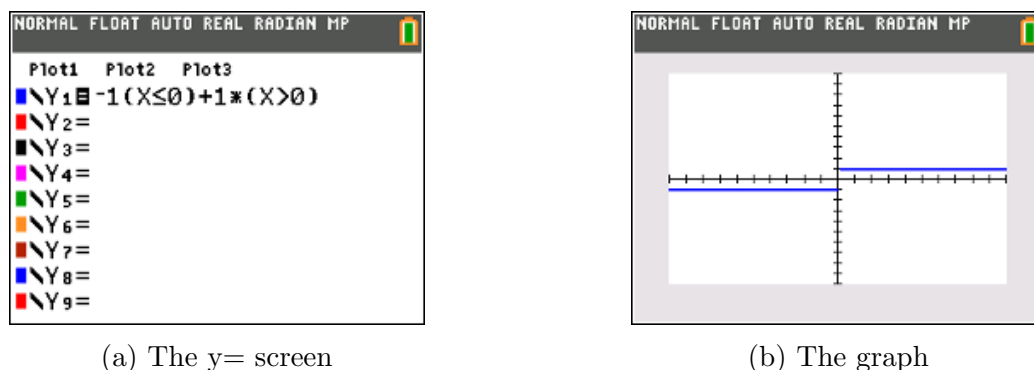


Figure E.2.1: TI-84 Plus CE

You can find the comparison symbols and the logical operators to create more complicated logical expressions in the TEST menu (press 2nd then MATH to get the TEST menu). When typing a formula for a piecewise function in the $y =$ environment, you navigate to these menus when you need one of these symbols and scroll down to the symbol you want and hit the ENTER button to put the symbol in your formula and return to the $y =$ environment.

The comparison symbols are listed in the TEST sub-menu while logical commands are found in the LOGIC sub-menu (scroll over to highlight LOGIC) as shown below on the older TI-83/84 Silver calculators.

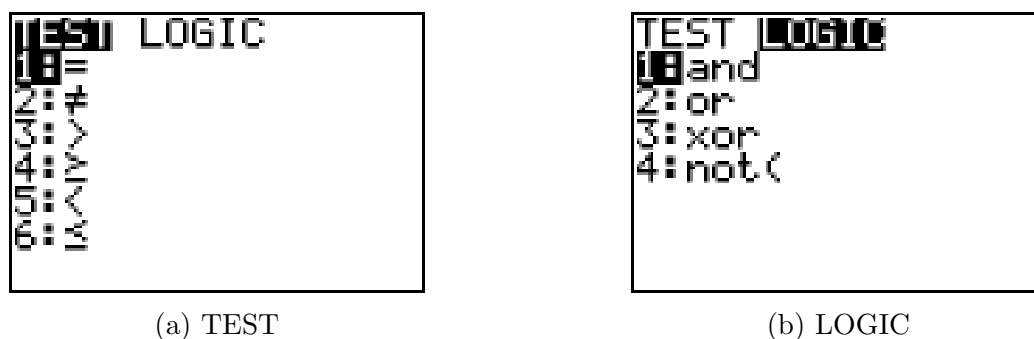


Figure E.2.2: TI-84 Plus Silver

On the newer TI-84 Plus CE models there is a third menu option that gives pre-defined conditions that can be used to more quickly set up these formulas.

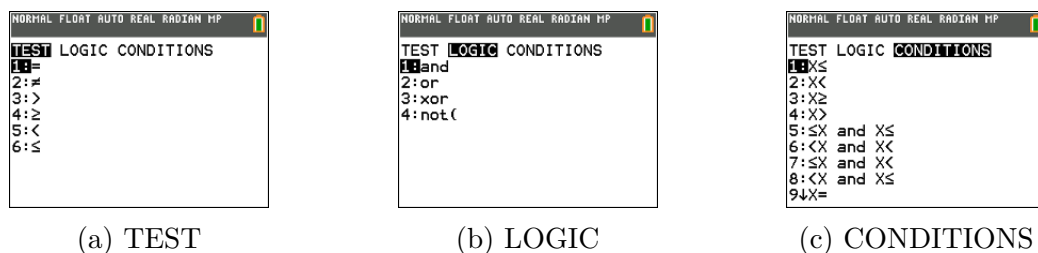


Figure E.2.3: TI-84 Plus CE

For more complicated piecewise functions, you will need to use the logical operators. A compound inequality like $0 < x < 1$ will not be evaluated as you expect. Instead use the logical operators in the LOGIC menu to rewrite compound inequalities in terms of simpler conditions: $((0 < x) \text{and} (x < 1))$.

You would enter $-x(x < 0) + 1((0 \leq x) \text{and} (x \leq 1)) + (x^2 + 2)(x > 1)$ to graph this function:

$$f(x) = \begin{cases} -x & x < 0 \\ 1 & 0 \leq x \leq 1 \\ x^2 + 2 & x > 1 \end{cases}$$

On the TI-83/84 calculators this would look like:

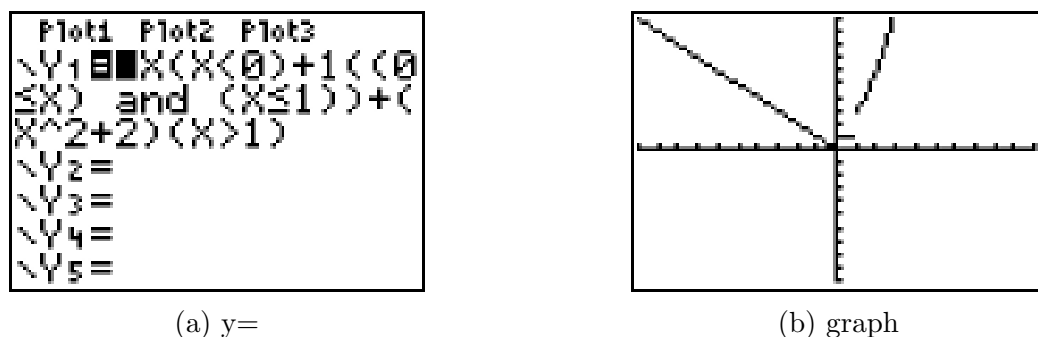


Figure E.2.4: TI-84 Plus Silver

You can also find these symbols and logical operators in the CATALOG if you prefer.

E.2.2 Piecewise Functions on the TI-89

Piecewise function on the TI-89 are inputted using the when function which can be obtained from the CATALOG.

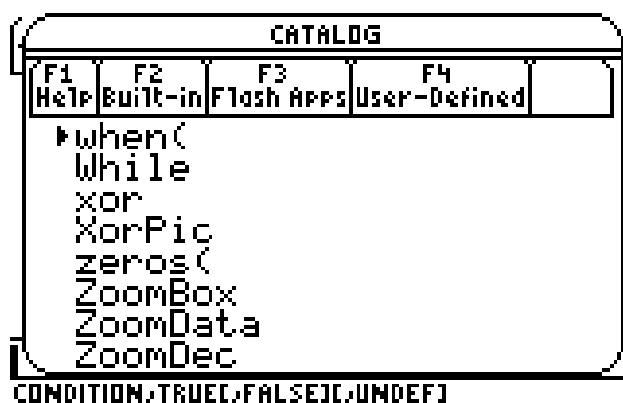


Figure E.2.5: TI-89 when Function

Press “W” to scroll through the CATALOG faster. The usual syntax for this function is

`when(condition to check, value if true, value if false).`¹

As a simple example, the piecewise function

$$f(x) = \begin{cases} -1, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

can be graphed by entering

`when( $x < 0$ , -1, 1).`

¹The “value if false” argument is optional. There is a fourth (optional) argument that is used to specify a default value if the condition cannot be evaluated.



Figure E.2.6: Sample Piecewise Function in the TI-89

Notice that the TI-89 will actually display a nicer version of the piecewise function after you enter it.

Inputting a more complicated piecewise function often requires the use of *nested* when statements. As an example, the function

$$g(x) = \begin{cases} -x, & x < 0 \\ 1, & 0 \leq x \leq 1 \\ x^2, & x > 1 \end{cases}$$

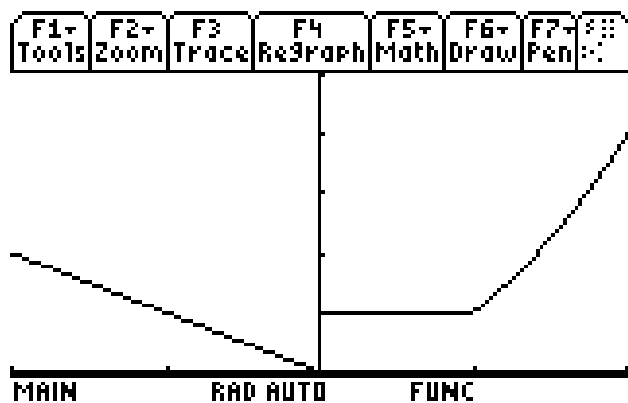
can be input as

$$\text{when}(x < 0, -x, \text{when}(x > 1, x^2, 1)).$$



Figure E.2.7: Sample Piecewise Function in the TI-89

Notice that we use the simpler condition $x > 1$ for the second when statement since the greater than and less than symbols are directly on the TI-89 keyboard (using the 2nd key and period or 0, respectively). The graph of this function makes it clear that we entered it correctly.

Figure E.2.8: Graph of $y = g(x)$

You can also enter g as

$$\text{when}(x < 0, -x, \text{when}(0 \leq x \leq 1, 1, x^2))$$

where less-than-or-equal-to can be found in the CATALOG (before the entries starting with A).

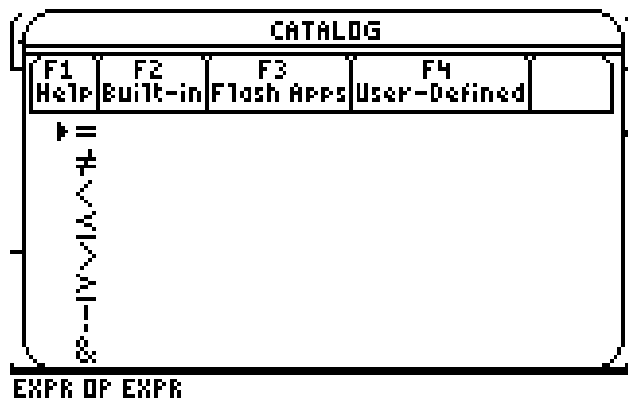


Figure E.2.9: TI-89 CATALOG Symbols

This gives us a slightly different looking version of the function in the TI-89.



Figure E.2.10: Sample Piecewise Function in the TI-89

The graph will verify that this is exactly the same function as before.

E.3 Producing Vector Plots on the TI-89

Unfortunately, vector plots cannot be easily produced on TI-83/84 calculators. As such, we only present information for the TI-89.

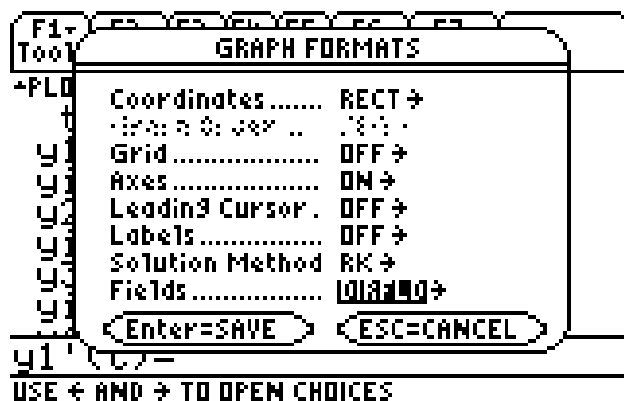
First, we have to get the TI-89 in differential equations mode by first pressing MODE and the selecting option 6:DIFF EQUATIONS under the Graph options.



After saving the new configuration by pressing ENTER, go into the equation editing environment by selecting Y= (pressing $\blacklozenge$ + F1). Next, press F1 to open the Tools menu, and then scroll down to option 9:Format...

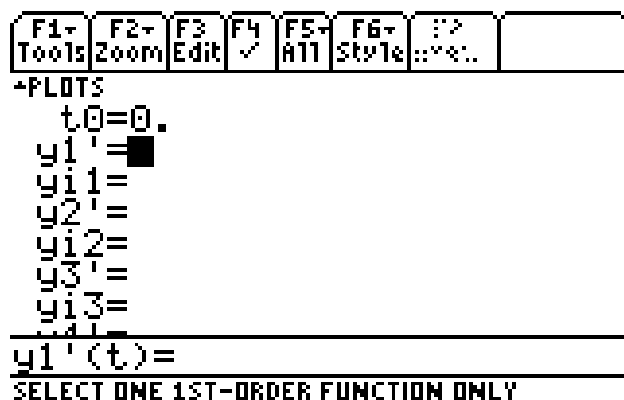


Press ENTER. In the GRAPH FORMATS menu that pops up, scroll down to Fields and select 2:DIRFLD (for “direction field”).



Once you press ENTER, the TI-89 is ready to plot vector fields.

The t_0 prompt on the $Y =$ screen is used to indicate the time when initial data is given. You can safely leave this setting as $t_0 = 0$ for our purposes.



The variables y_1 , y_2 , etc. are the names of the unknown functions in the system of differential equations. So, to plot a system like

$$\begin{aligned}\frac{dx}{dt} &= x(x - y) \\ \frac{dy}{dt} &= x + y + 2,\end{aligned}$$

we enter the following:

F1+ Tools	F2+ Zoom	F3 Edit	F4 ✓	F5+ All	F6+ Style	F7 S"et..	
--------------	-------------	------------	---------	------------	--------------	--------------	--

```

*PLOTS
  t0=0.
✓y1'=y1·(y1 - y2)
  yi1=
✓y2'=y1 + y2 + 2
  yi2=
  y3'=
  yi3=

```

```

yi2=

```

MAIN	RAD AUTO	DE
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Notice that $y1$ is playing the role of x in our system and $y2$ the role of y . Also note that you do not need to enter $y1(t)$ and $y2(t)$ when entering the differential equations. We have intentionally left the initial data options blank (i.e. the entries for $yi1$ and $yi2$ are empty). To control the display options, press  + F2 to bring up the WINDOW menu.

F1+ Tools	F2+ Zoom	
--------------	-------------	--

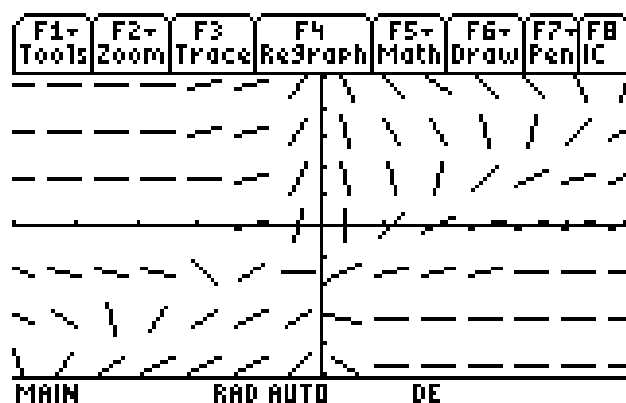
```

t0=0.
tmax=10.
tstep=.1
tplot=0.
xmin=-5.
xmax=5.
xscl=1.
ymin=-5.
ymax=5.
yscl=1

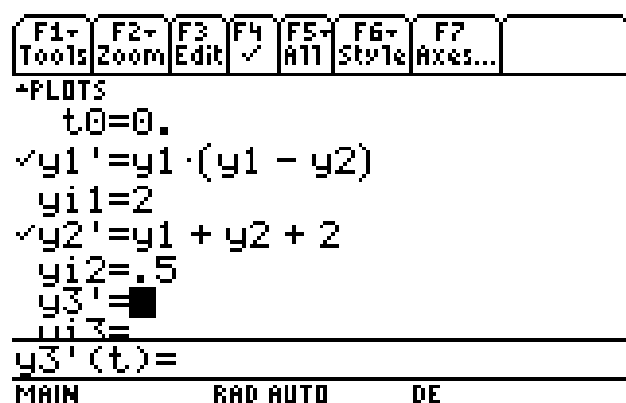
```

MAIN	RAD AUTO	DE
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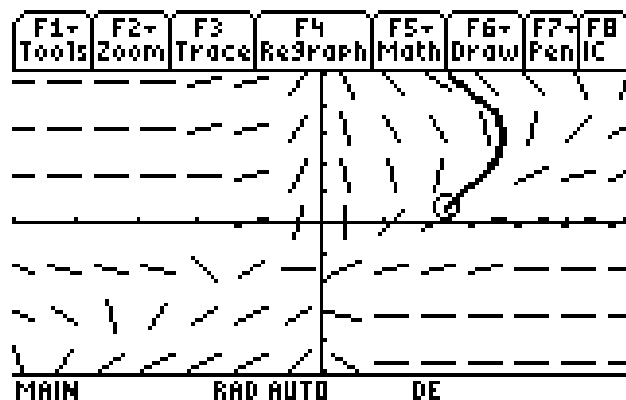
Since we are not plotting a solution curve (we did not give any initial data), the t -settings are not very important for us. The $xmin$, $xmax$, $xscl$, $ymin$, $ymax$, and $yscl$ are the ones most likely needing adjustment. The remaining options you can safely ignore. Selecting the GRAPH option ( + F3) should produce the vector plot.



If you would like to see a numerical solution curve for a particular initial condition, you can enter the initial data below each equation instead of leaving those options blank. For example, entering



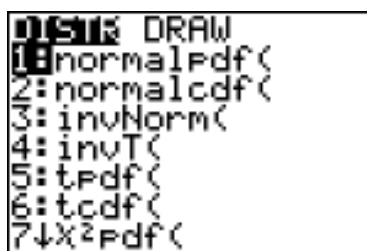
produces the same vector plot as above, but it overlays the numerical solution corresponding to the initial condition $x(0) = 2$ and $y(0) = 0.5$.



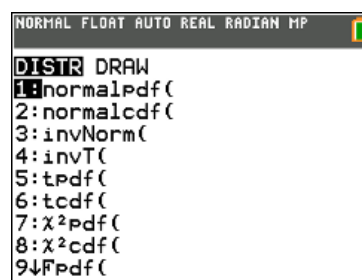
E.4 Probability

E.4.1 Probability on the TI-83/84

The built-in probability functions (pdfs and cdfs) are found in the DISTR menu (obtained by pressing the 2nd key and then the VARS key). There are menu items 1 through G. You can use the arrow keys to scroll down through the menu.



(a) TI-84 Plus Silver



(b) TI-84 Plus CE

Figure E.4.1: DISTR Menu on the TI-84

When you select a function from the menu on the older TI-83/84 calculators, the function appears in the home screen, you then type the arguments (separated by commas), and then close the parentheses. Newer TI-83/84 models and the TI-84 Plus CE have a dialog box that appears when you select one of these functions. You enter your arguments into this dialog box, and then highlight Paste and press the Enter key. The function complete with your arguments will then appear in the home screen. In either case, you press the Enter key when at the home screen to evaluate the function.

We illustrate this for the binompdf function below, and then explain the arguments for all relevant functions in the order that the corresponding random variables appear in the text.

- Choice *A* on the menu is $\text{binompdf}(n, p, x)$ which gives the probability of exactly x successes in n independent trials of a $\text{Bernoulli}(p)$ random variable. Thus it gives the value of $\mathbb{P}(X = x)$ for $X \sim \text{binomial}(n, p)$. If you omit the x you get a list of values for all the possible x 's.

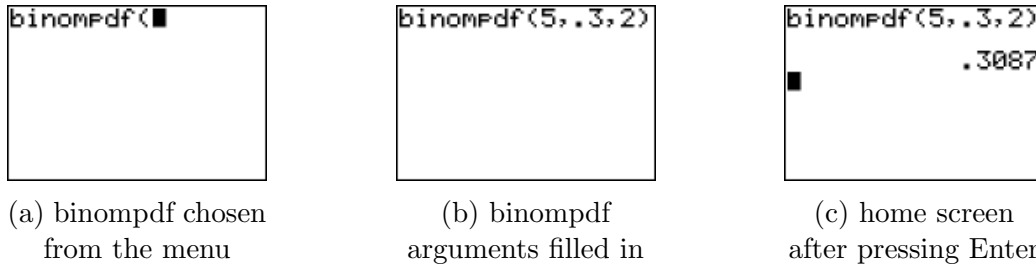


Figure E.4.2: TI-84 Plus Silver

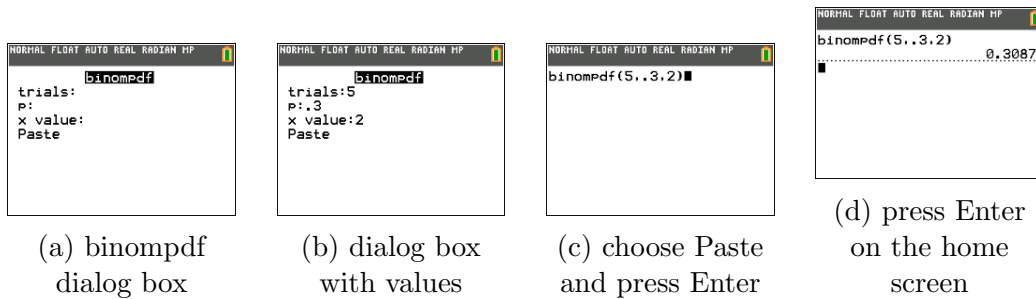


Figure E.4.3: TI-84 Plus CE

- Choice *B* on the menu is `binomcdf( $n, p, x$ )` which gives the probability of x or fewer successes in n independent trials of a $\text{Bernoulli}(p)$ random variable. Thus it gives the value of $\mathbb{P}(X \leq x)$ for $X \sim \text{binomial}(n, p)$. If you omit the x you get a list of values for all the possible x 's.
- Choice *E* on the menu is `geometpdf( $p, t$ )` which gives the probability of the first success occurring on the t th when independent trials of a $\text{Bernoulli}(p)$ random variable are repeated until a success occurs. Thus it gives the value of $\mathbb{P}(T = t)$ for $T \sim \text{geometric}(p)$.
- Choice *F* on the menu is `geometcdf( $p, t$ )` which gives the probability of the first success occurring on the t th trial or sooner when independent trials of a $\text{Bernoulli}(p)$ random variable are repeated until a success occurs. Thus it gives the value of $\mathbb{P}(T \leq t)$ for $T \sim \text{geometric}(p)$.
- Choice *C* on the menu is `poissonpdf( $\lambda, x$ )` which gives the probability $\mathbb{P}(X = x)$ for $X \sim \text{Poisson}(\lambda)$.
- Choice *D* on the menu is `poissoncdf( $\lambda, x$ )` which gives the probability

$\mathbb{P}(X \leq x)$ for $X \sim \text{Poisson}(\lambda)$.

- Choice 1 on the menu is `normalpdf( $x, \mu, \sigma$ )` which gives the value of the pdf (i.e. the height of the curve) at x for a normal distribution with a mean of μ and a standard deviation of σ . The default values are $\mu = 0$ and $\sigma = 1$ and those values will be used if you omit the second and third arguments. We will not need this value computationally in our work. If you want to graph a normal curve you can use this in the $y =$ editor.
- Choice 2 on the menu is `normalcdf(lower, upper,  $\mu, \sigma$ )` which gives the value of the cdf between the x -values lower and upper (i.e. the area under the normal curve to the right of lower and to the left of upper) for a normal distribution with a mean of μ and a standard deviation of σ . The default values are $\mu = 0$ and $\sigma = 1$ and those values will be used if you omit the last two arguments. Thus it gives the probability $\mathbb{P}(X \leq x)$ for $X \sim N(\mu, \sigma^2)$. You can use `-1E99` to represent $-\infty$ and `1E99` to represent ∞ .
- Choice 3 on the menu is `invNorm(area,  $\mu, \sigma$ )` which gives the value of x so that `normalcdf( $-\infty, x, \mu, \sigma$ ) = area`, i.e. the value of x so that $\mathbb{P}(X \leq x) = \text{area}$ for $X \sim N(\mu, \sigma^2)$. On the newer TI-84 Plus CE the dialog box includes an additional argument of either LEFT, RIGHT or CENTER. The choice of LEFT is what is described above. If you choose CENTER the answer is two numbers $-x$ and x so that `normalcdf( $-x, x, \mu, \sigma$ ) = area` and if you choose RIGHT you get an x so that `normalcdf( $x, \infty, \mu, \sigma$ ) = area`.

E.4.2 Probabilty on the TI-89

The built-in probability functions for the TI-89 can be accessed through the Stats/List Editor App.

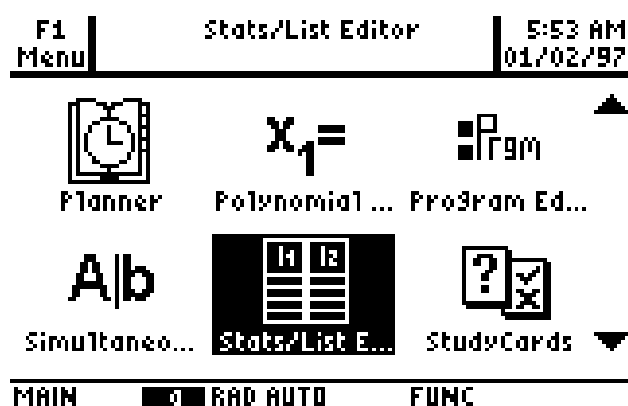


Figure E.4.4: TI-89 Stats/List Editor App

This brings up a list editor from which you can access various probability tools by selecting F5:Distr.

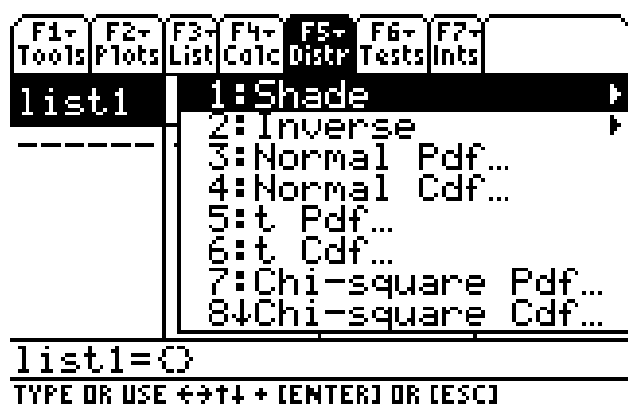


Figure E.4.5: TI-89 Stats/List Editor Distr Menu

Selecting any of the options in this menu will bring up a dialog box allowing you to obtain information from the chosen distribution. For example, choosing 4:Normal Cdf brings up the following dialog box.

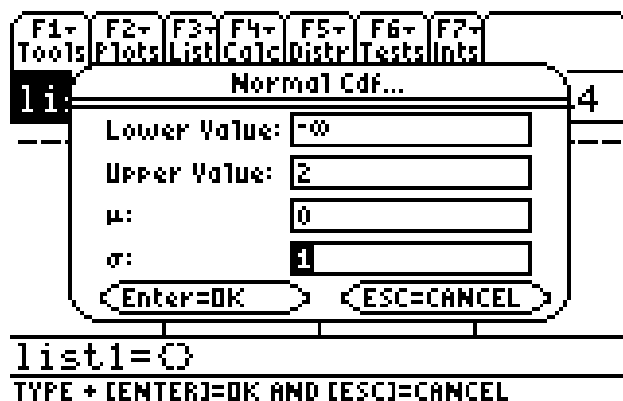


Figure E.4.6: TI-89 Dialog Box for 4:Normal Cdf

Note that the infinity symbol can be obtained by pressing  + CATALOG. After entering the appropriate information and preeing ENTER, we obtain output about the chosen distribution.

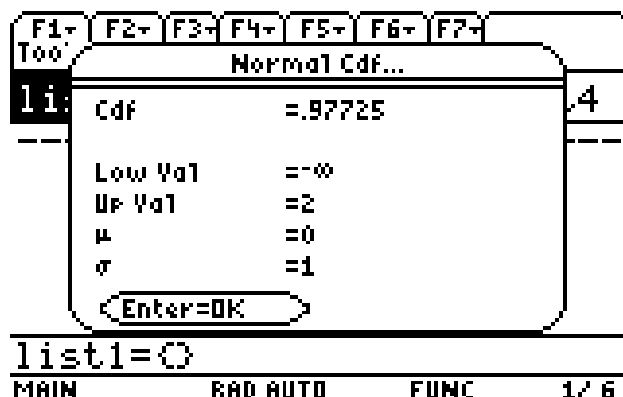


Figure E.4.7: TI-89 Output from the 4:Normal Cdf Dialog

Here are the functions we will use, listed in the order they appear in the text:

- Choice B:Binomial Pdf brings up a dialog box for the pdf of the binomial random variable. The first input field is for the total number of independent trials, n , for a Bernoulli(p) random variable where p is the probability of success (which is the second input field). The final input field is for the total number of successes, x . Thus it gives the value of

$\mathbb{P}(X = x)$ for $X \sim \text{binomial}(n, p)$. If you omit the x you get a list of values for all the possible x 's.

- Choice C:Binomial Cdf gives the probability of having a number of successes between the Lower Value and Upper Value in n independent trials of a Bernoulli(p) random variable. The parameters n and p are the same as for B:Binomial Pdf.
- Choice F:Geometric Pdf brings up a dialog box for the pdf of the geometric random variable which gives the probability of the first success occurring on the x th when independent trials of a Bernoulli(p) random variable are repeated until a success occurs. Thus it gives the value of $\mathbb{P}(T = x)$ for $T \sim \text{geometric}(p)$.
- Choice G:Geometric Cdf is for the corresponding cdf for the geometric random variable. This will compute the probability of the first success occurring between Lower Value and Upper Value when performing independent trials of a Bernoulli(p) random variable.
- Choice D:Poisson Pdf brings up a dialog box for the pdf of the Poisson random variable with parameter λ . This gives the probability $\mathbb{P}(X = x)$ for $X \sim \text{Poisson}(\lambda)$.
- Choice E:Poisson Cdf is for the corresponding cdf for the Poisson random variable. It computes the probability of obtaining an outcome from a Poisson random variable with parameter λ between Lower Value and Upper Value.
- Choice 3:Normal Pdf on the menu brings up a dialog box for the pdf (i.e. the height of the curve) at x for a normal distribution with a mean of μ and a standard deviation of σ . The default values are $\mu = 0$ and $\sigma = 1$ and those values will be used if you omit the second and third arguments. We will not need this value computationally in our work.
- Choice 4:Normal Cdf on the menu gives the value of the cdf between Lower Value and Upper Value (i.e. the area under the normal curve to the right of Lower and to the left of Upper) for a normal distribution with a mean of μ and a standard deviation of σ . Thus it gives the probability $\mathbb{P}(X \leq x)$ for $X \sim N(\mu, \sigma^2)$.

- The inverse normal dialog box can be found under the sub-menu 2:Inverse. In this sub-menu. 1:Inverse Normal... brings up the dialog box for this function. The parameters μ and σ are the mean and standard deviation of the normal random variable. The first input field, Area, is the total probability for which you want to find a corresponding x -value. The function returns the (approximate) x -value such that

$$\mathbb{P}(-\infty < X \leq x) = \text{Area}$$

for $X \sim N(\mu, \sigma^2)$.

E.5 Matrices

E.5.1 Matrix Operations on the TI-83/84

The built-in matrix manipulation tools are found in the MATRIX menu (obtained by pressing the 2nd key and then the x^{-1} key). In these calculators, matrices have names like [A]. You can see the list of recognized matrix names when either NAMES or EDIT is highlighted on the MATRIX menu. (If MATH is highlighted you see instead a list of matrix functions or commands.)

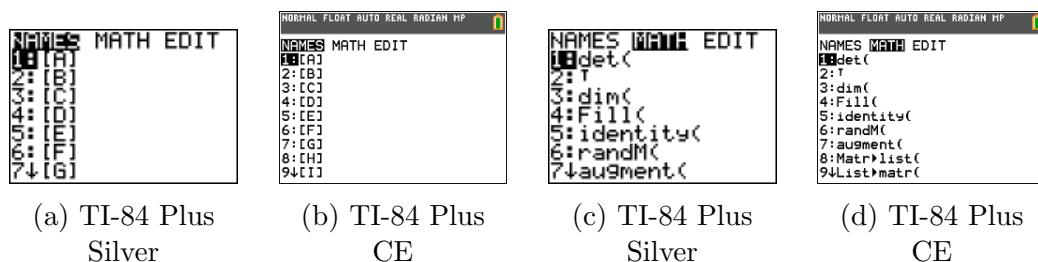


Figure E.5.1: NAMES and MATH Menus

To create a matrix, highlight EDIT and select the name. (See figures A below.) After you press enter, the current dimensions and values of the matrix are displayed. The default dimensions are 1×1 . You change dimensions by over-typing the defaults and pressing enter. (See figures B where enter has not yet been pressed for the number of columns.) Then you will see a display with the correct number of rows and columns. The default entries are all 0. Use the arrow keys to move to any entry and type a new value. (See figures C.) You will see the new value in the matrix after you press the enter key. To get out of matrix entry mode use QUIT (obtained by pressing 2nd and then MODE). If you display the name of a matrix on the home screen (by choosing it from NAMES in the MATRIX menu), the calculator will display the matrix. (See figures D.) On older TI 83/84 models the matrix will be displayed in a square bracketed list notation.

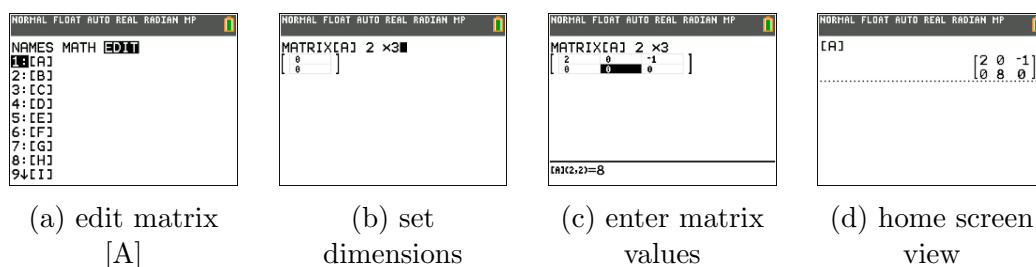


Figure E.5.2: TI-84 Plus CE

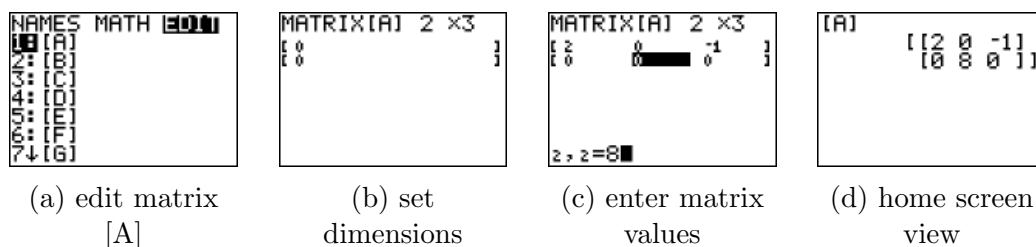


Figure E.5.3: TI-84 Plus Silver

You can also enter a matrix in the square bracketed list notation. The square brackets are typed using 2nd $\times$ and 2nd $-$. Note that you use commas within the rows as you enter the matrix, but when the calculator displays the matrix there are no commas.

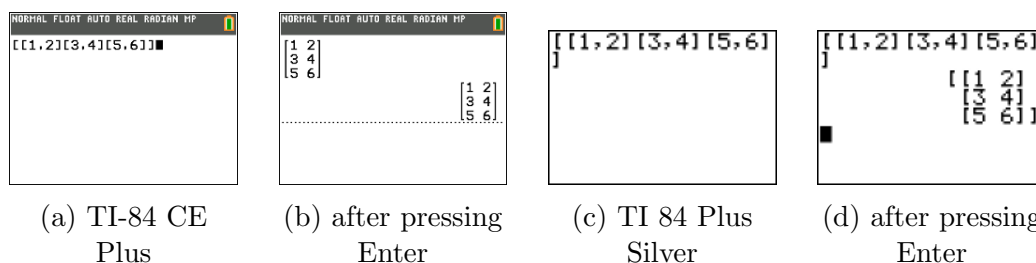


Figure E.5.4: Bracket matrix entry

Matrix addition and subtraction can be done using the same keys we use for arithmetic with numbers. Matrix multiplication can be done either with the multiplication key or by writing the two matrix names next to each other.

See the examples below using our 2 by 3 matrix [A] from above and the 3 by 3 matrices [B], [C] and [D] shown below.

$$[B] = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 2 & 2 \end{bmatrix}, \quad [C] = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 3 & 2 & 1 \end{bmatrix}, \quad \text{and} \quad [D] = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

Remember that to get the names of the matrices onto the screen to perform operations use the MATRIX menu with NAMES highlighted and select the name you wish to appear and hit enter. Be sure you have already defined the matrix as shown above using EDIT.

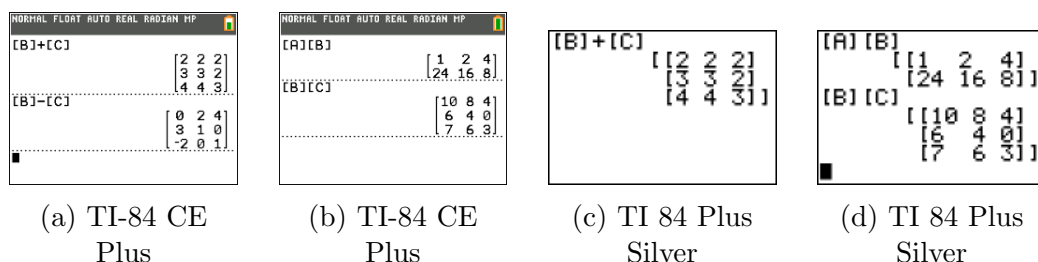


Figure E.5.5: Matrix Operations

If the matrices you are trying to add or multiply do not have compatible dimensions you will see a dimension mismatch error message:

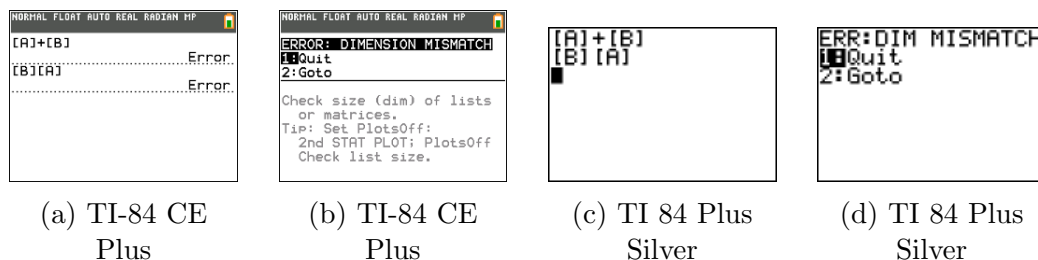


Figure E.5.6: Mismatched Dimensions

If a square matrix is invertible, you can find its inverse using the x^{-1} key. If you are trying to invert a non-square matrix you will get an invalid dimension error message, and if you are trying to invert a singular matrix you will get a singular matrix error message. On the CE Plus you will see a

notation on the home screen that your request resulted in a error but on the older TI-83/84 models you only see that there is no result shown.

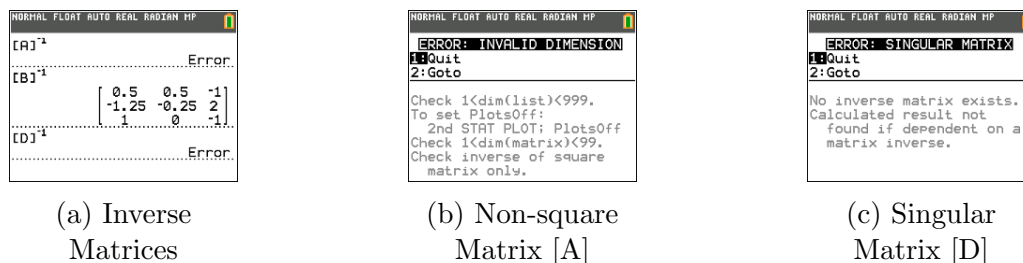


Figure E.5.7: Inverses on TI-84 CE Plus

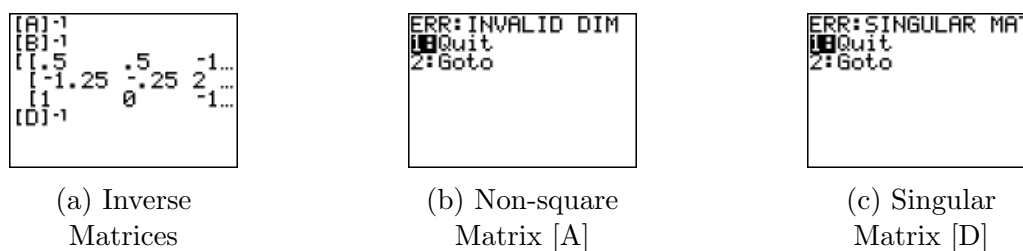


Figure E.5.8: Inverses on TI 84 Plus Silver

To find determinants of square matrices you use the determinant command, `det`, which is the first entry in the MATRIX menu when MATH is highlighted. You can also find it in the CATALOG. If you try to take the determinant of a non-square matrix you will see the invalid dimension error message.

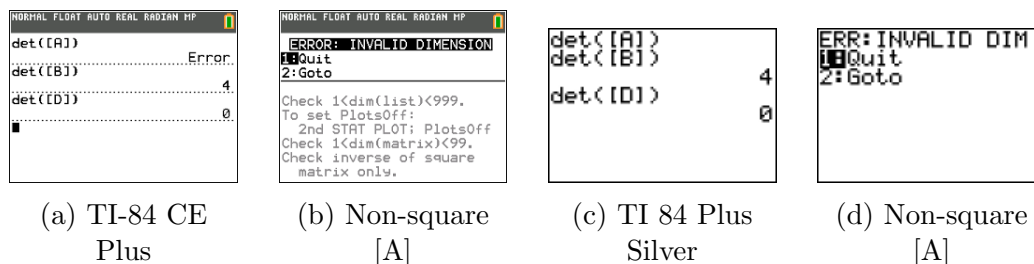
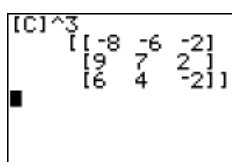
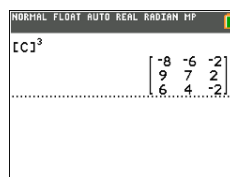


Figure E.5.9: Matrix Determinants

You can raise a square matrix to a power using the $\wedge$ key. If your matrix is not square you will again get the invalid dimension error message.



(a) TI-84 Plus
Silver



(b) TI-84 Plus
CE

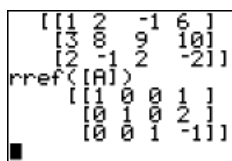
Figure E.5.10: Matrix Powers

To row reduce matrices you use the command `rref` which is found in the MATRIX menu when MATH is highlighted. You can also find it in the CATALOG. For example, if we redefine our matrix $[A]$ to correspond to the augmented matrix for the system of equations below, we can row reduce to find the solution to the system ($x = 1, y = 2, z = -1$).

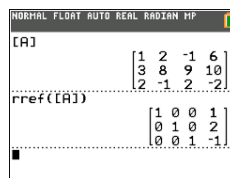
$$x + 2y - z = 6$$

$$3x + 8y + 9z = 10$$

$$2x - y + 2z = -2$$



(a) TI-84 Plus
Silver



(b) TI-84 Plus
CE

Figure E.5.11: Row Reduction

E.5.2 Matrix Operations on the TI-89

Matrices can be entered into the TI-89 in many ways, but we will focus on using the Data/Matrix Editor App.

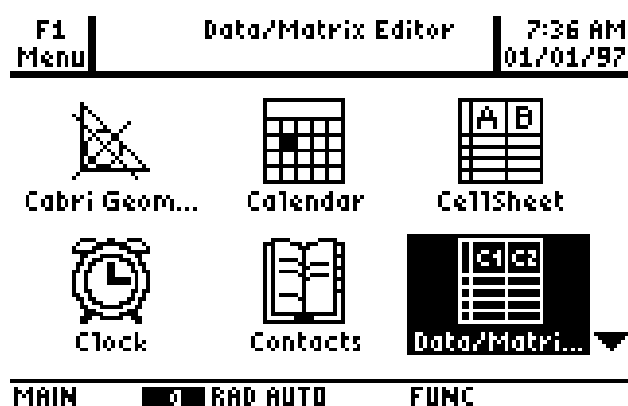


Figure E.5.12: TI-89 Data/Matrix Editor App

To create a new matrix, select 3:New... This will bring up a dialog box. For Type, you should select 2:Matrix. Leave the Folder option on its default value of main. The Variable field is where you give the name of the matrix (usually a single letter is easiest to remember). Finally, enter the row and column dimensions in the fields provided.

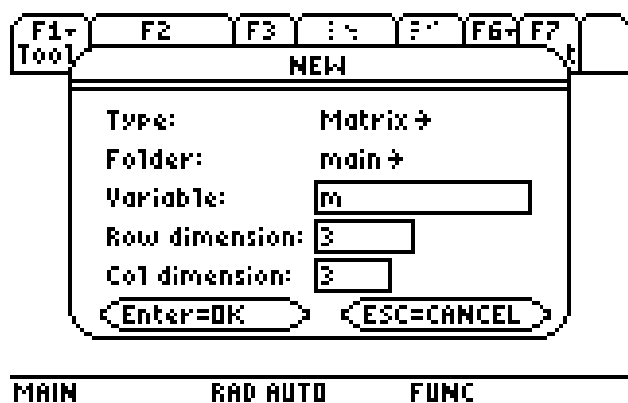


Figure E.5.13: TI-89 New Matrix Dialog

This will bring up a data entry screen with zero entries for the matrix of the specified size. Simply input the correct entries for your matrix.

F1+ Tools	F2+ Plot Setup	F3+ Cell	F4+ Answer	F5+ Copy	F6+ Util	F7+ Stat	
MAT 3x3							
	c1	c2		c3			
1	1	2		3			
2	3	2		1			
3	1	2		2			
4							
r1c1=1							
MAIN		RAD AUTO		FUNC			

Figure E.5.14: TI-89 Data Entry Screen

After data entry, your matrix can be accessed from the HOME screen using whatever variable name you specified.

F1+ Tools	F2+ Algebra	F3+ Calc	F4+ Other	F5+ PrgmIO	F6+ Clean Up	
■ M						$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 2 & 2 \end{bmatrix}$
M						
MAIN		RAD AUTO		FUNC		1/30

Figure E.5.15: Matrix on the TI-89 Home Screen

Many simple matrix operations can be carried out from the HOME screen just as for real numbers. The following figure shows matrix addition, subtraction, and multiplication for the matrices

$$m = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 2 & 2 \end{bmatrix} \quad \text{and} \quad n = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 3 & 2 & 1 \end{bmatrix}.$$

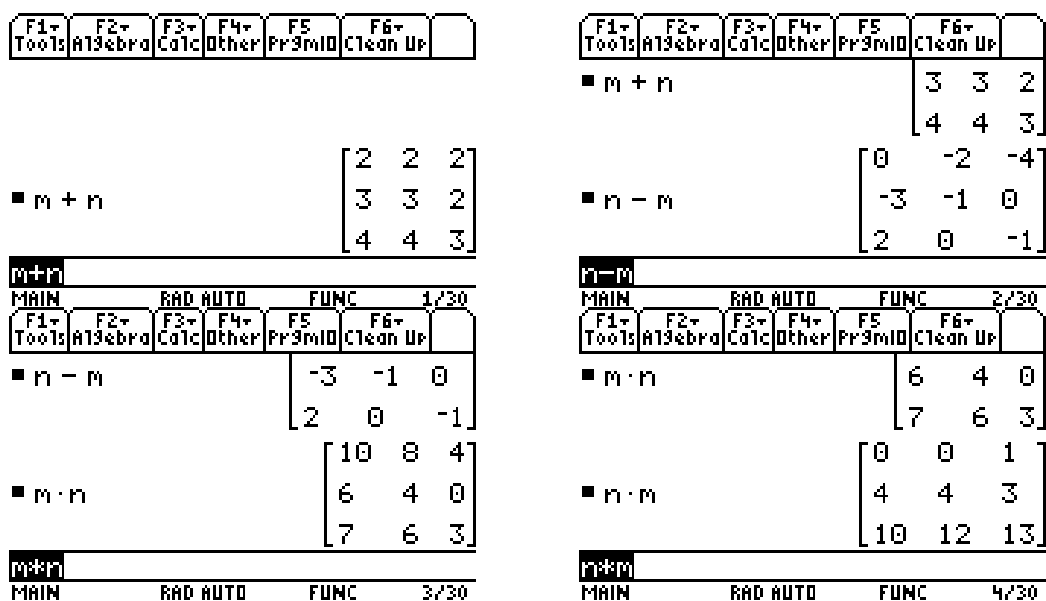


Figure E.5.16: Basic Matrix Arithmetic on the TI-89

If you attempt a binary operation between two matrices of incompatible sizes, you will get an error message warning you about mismatched dimensions.

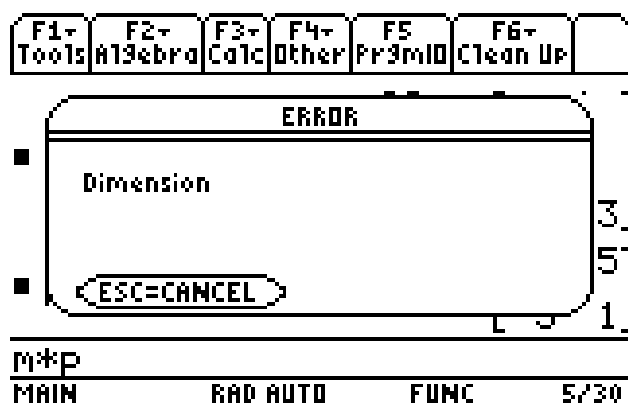


Figure E.5.17: TI-89 Dimension Error

You can compute positive integer powers of a square matrix by using the usual exponentiation command.

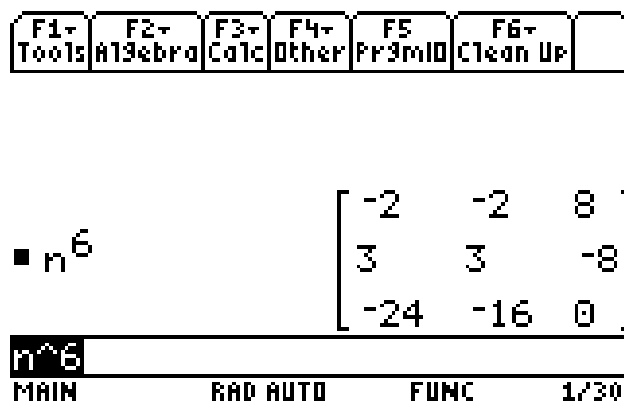


Figure E.5.18: Powers of a Square Matrix on the TI-89

If a square matrix is invertible, you can find the inverse by raising the matrix to the power -1 .

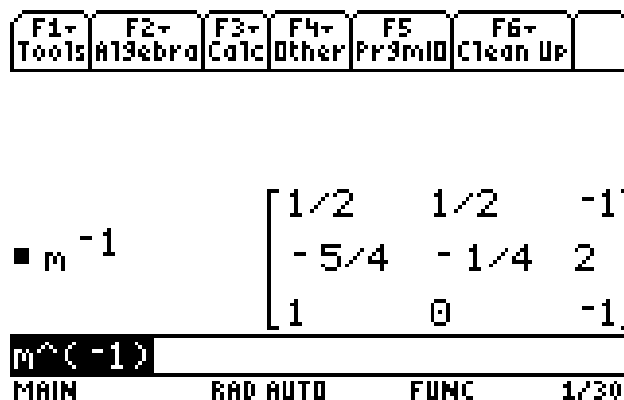


Figure E.5.19: Inverting a Square Matrix on the TI-89

If you attempt to invert a matrix that has no inverse, you will receive an error message that the matrix is singular.

Suppose that we want to edit an existing matrix. For instance, if we want to change the matrix m above to

$$m = \begin{bmatrix} 1 & 2 & -1 & 6 \\ 3 & 8 & 9 & 10 \\ 2 & -1 & 2 & -2 \end{bmatrix},$$

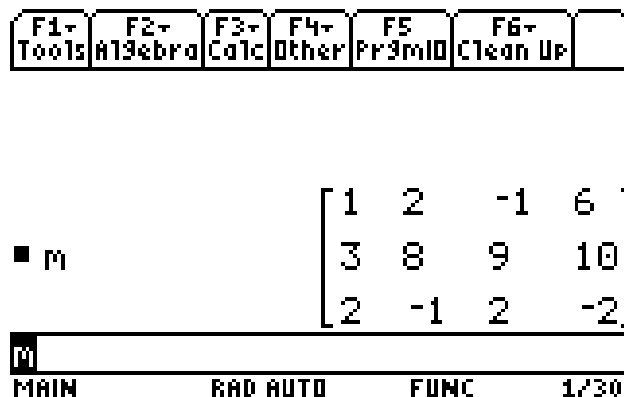
we first select 2:Open from the main Data/Matrix Editor application menu. This will bring up a menu where you can reselect m . Once we have reopened

m , we must resize it (since our new m is a 3×4 matrix). This can be accomplished by opening the F6:Util menu and selecting option 6:Resize Matrix.



Figure E.5.20: Resizing a Matrix on the TI-89

Once you have entered the new data, the new m can be accessed from the HOME screen as usual.

Figure E.5.21: Resized Matrix m

If you want to completely clear out all currently defined matrices (along with any other stored variables), you can choose F6:Clean Up from the HOME screen and then select 1:Clear a-z...

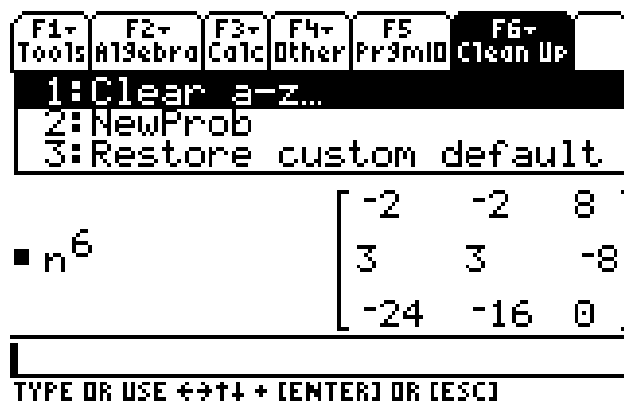
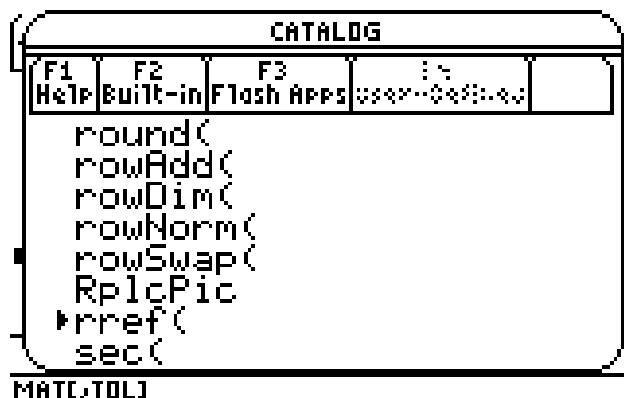


Figure E.5.22: Clearing Stored Variables on the TI-89

The TI-89 can also row reduce matrices. For example, to row reduce the newly defined matrix m , we use the row reduction command, `rref`, which can be easily accessed from the CATALOG. Hit the “R” key to scroll through faster.

Figure E.5.23: TI-89 Catalog: `rref`

Selecting this function and inputting the variable name of the matrix to be row reduced returns the equivalent matrix in reduced row echelon form.

F1→ Tools	F2→ Matrix	F3→ Calc	F4→ Other	F5 Pr3mID	F6→ Clean Up	
■ m				3 8	9 10	
				2 -1	2 -2	
■ rref(m)				1 0	0 1	
				0 1	0 2	
				0 0	1 -1	
rref(m)						
MAIN				RAD AUTO	FUNC	2/30

Figure E.5.24: Row Reduction on the TI-89 Catalog

The determinant of a square matrix can be computed similarly by selecting the det command from the CATALOG.

F1→ Tools	F2→ Matrix	F3→ Calc	F4→ Other	F5 Pr3mID	F6→ Clean Up	
■ det(n)						2
det(n)						
MAIN						1/30

Figure E.5.25: Determinants on the TI-89 Catalog

The TI-89 is capable of computing eigenvalues and eigenvectors numerically. Since a real matrix is capable of having complex eigenvalue/eigenvector pairs, you must change the mode to allow complex numbers. Press MODE and change the Complex Format from 1:REAL to 2:RECTANGULAR.

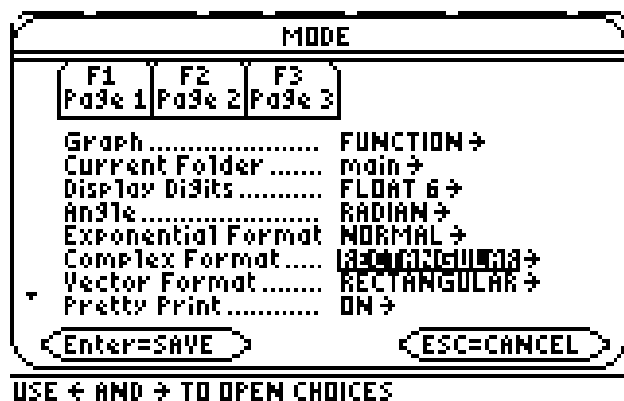


Figure E.5.26: Complex Format on the TI-89

Once the calculator is in complex mode, eigenvalues can be computed using the eigVl command from the CATALOG while the corresponding eigenvectors are computed using eigVc. For example, we can compute the eigenvalues of the matrix

$$p = \begin{bmatrix} 3 & 5 \\ -2 & 1 \end{bmatrix}$$

as shown below.

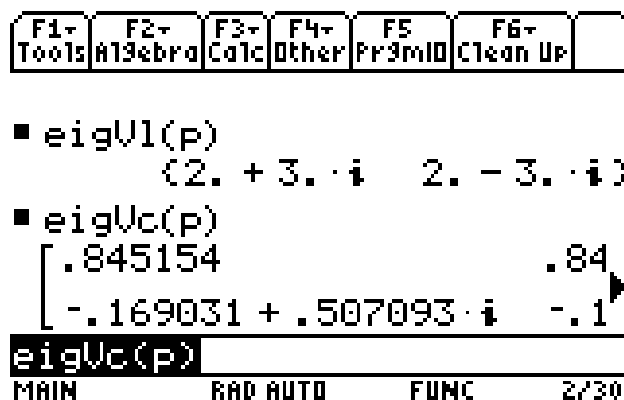


Figure E.5.27: Eigenvalues/Eigenvectors on the TI-89

The matrix p clearly has eigenvalues $2 \pm 3i$. The eigVc command returns a matrix whose *columns* are (approximate) representative eigenvectors for the corresponding eigenvalue from eigVl. For instance, an approximate eigenvec-

tor for $2 + 3i$ is

$$\vec{v}_1 = \begin{bmatrix} 0.845154 \\ -0.169031 + 0.507093i \end{bmatrix}.$$

For the eigenvector corresponding to $2 - 3i$, we scroll over to find

$$\vec{v}_2 = \begin{bmatrix} 0.845154 \\ -0.169031 - 0.507093i \end{bmatrix}.$$

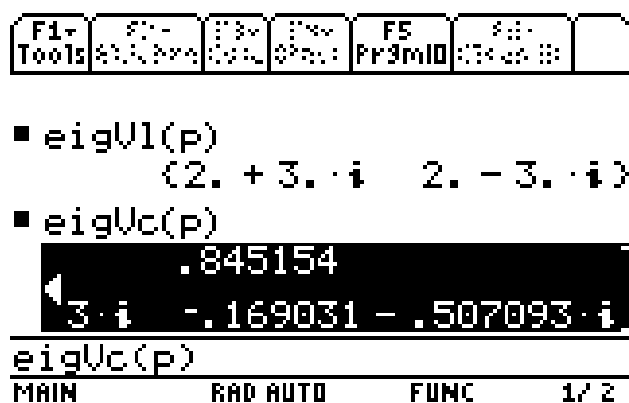


Figure E.5.28: Eigenvectors on the TI-89

Appendix F

Answers to Selected Exercises

F.1 Chapter 1

- 1.) a.) $q_1 = 24, q_2 = 36, q_3 = 54, q_4 = 81$
b.) $q_n = \left(\frac{3}{2}\right)^n q_0$ so $q_n = \left(\frac{3}{2}\right)^n 16$ when $q_0 = 16$
c.) $q_n = \frac{2}{3}q_{n+1}$
d.) $q_0 = 16$
- 3.) a.) $p_1 = \frac{1}{4}, p_2 = \frac{1}{16}, p_3 = \frac{1}{256}, p_4 = \frac{1}{65536}$
b.) $p_n = \left(\frac{1}{2}\right)^{2^n}$ so $p_{10} = \left(\frac{1}{2}\right)^{1024}$
c.) Yes, $p_n = \sqrt{p_{n+1}}$
- 5.) 11 time periods
- 7.) $q^* = \frac{1}{2}$ unstable (no spiraling)
- 9.) $L^* = -1$ unstable (no spiraling) and $L^* = 0$ asymptotically stable (no spiraling)
- 11.) $P^* = \frac{1}{2}$ asymptotically stable (no spiraling), $P^* = 1$ unstable (no spiraling) and $P^* = -1$ unstable (no spiraling)
- 13.) $p^* = 0$ asymptotically stable (no spiraling) and $p^* = 1$ unstable (no spiraling)
- 15.) $r = 2s$ and $p^* = \frac{1}{3}$. For any reasonable case we need $-1 < 1 - r - s < 1$ which says $0 < r + s < 2$ and then we get an asymptotically stable equilibrium. So here for asymptotically stable we must have $0 < s < \frac{2}{3}$.

To get spiraling we need $-1 < 1 - r - s < 0$ or equivalently $1 < r + s < 2$ which here means $\frac{1}{3} < s < \frac{2}{3}$.

17.) No equilibrium point since $\frac{cu}{1-c} > V_c$. Cobweb plots show Wenckebach phenomenon.

19.) No equilibrium point since $\frac{cu}{1-c} > V_c$. Cobweb plots show 2:1 AV block.

21.) $q^* = 0$ unstable (no spiraling) and $q^* = \frac{1}{3}$ asymptotically stable (no spiraling)

23.) a.) $q^* = 0$ unstable (no spiraling) and $q^* = \frac{1}{3}$ asymptotically stable (no spiraling)

b.) $q_n = \frac{q_{n+1}}{2 - 3q_{n+1}}$

c.) $q^* = 0$ asymptotically stable (no spiraling) and $q^* = \frac{1}{3}$ unstable (no spiraling)

25.) a.) $0 < \alpha \leq \frac{27}{4}$

b.) For $0 < \alpha < 4$ the only equilibrium is 0.

For $\alpha = 4$ there are two equilibria, 0 and $\frac{1}{2}$.

For $4 < \alpha \leq 27/4$ there are three equilibria, 0 and $\frac{1}{2} \pm \sqrt{\frac{1}{2} - \frac{1}{\alpha}}$.

c.) In all cases, the equilibrium of 0 is stable (no spiral).

For $\alpha = 4$ the second equilibrium, $\frac{1}{2}$, is also stable (no spiral).

For $4 < \alpha < \frac{27}{4}$ the equilibrium $\frac{1}{2} - \sqrt{\frac{1}{2} - \frac{1}{\alpha}}$ is unstable (no spiral).

For $4 < \alpha \leq \frac{9}{2}$ the third equilibrium $\frac{1}{2} + \sqrt{\frac{1}{2} - \frac{1}{\alpha}}$ is stable (no spiral).

For $\frac{9}{2} < \alpha \leq \frac{49}{10}$ the third equilibrium $\frac{1}{2} + \sqrt{\frac{1}{2} - \frac{1}{\alpha}}$ is stable (spiral).

For $\frac{49}{10} < \alpha \leq \frac{27}{4}$ the third equilibrium $\frac{1}{2} + \sqrt{\frac{1}{2} - \frac{1}{\alpha}}$ is unstable (spiral).

F.2 Chapter 2

1.) $-\frac{1}{2} \ln |1 - x^2| + C$

3.) $\frac{4\sqrt{2} + \pi\sqrt{2} - 8}{8}$

5.) $-\frac{\sqrt{2}}{2}$

7.) $\ln |x + 3| - \ln |x + 2| - \frac{2}{x + 3} + C$

9.) $\tan x + C$

11.) $\frac{1}{2} \ln |2x - 1| - \ln |x + 1| + x + C$

13.) $\frac{4832}{15}$

15.) $\frac{5e^4 - 1}{4}$

17.) $\frac{1}{4} \ln |x| - \frac{1}{8} \ln (x^2 + 4) - \frac{1}{2} \arctan \frac{x}{2} + C$

19.) $\arctan x - \frac{1}{2} \ln (x^2 + 1) + C$

F.3 Chapter 3

1.) integral: $\frac{20}{3}$ area: $\frac{28}{3}$ They are different because some of the area is below the x -axis.

3.) 72

5.) $\frac{71}{6}$

7.) $\frac{1}{3}$

9.) $\frac{121\pi}{10}$

11.) $\frac{2\pi}{35}$

13.) $\frac{128\pi}{15}$

15.) $\int_0^h \pi\left(\frac{r}{h}y\right)^2 dy$

17.) $\frac{e^3}{3}$

19.) $\frac{2}{e}$

21.) $\frac{3}{2}$

23.) $8\sqrt{3}$

25.) 1

F.4 Chapter 4

1.) dependent: v
independent: t

verify: $\frac{dv}{dt} = 10e^{-3t} = 10 - 3v$ and $v(0) = \frac{10}{3} - \frac{10}{3}e^0 = 0$

- 3.) dependent: x
independent: t

verify: $t \frac{dx}{dt} + x = t \left(\frac{3}{4}t^2 - \frac{7}{4}t^{-2} \right) + \left(\frac{1}{4}t^3 + \frac{7}{4}t^{-1} \right) = t^3$ and $x(1) = \frac{1}{4} + \frac{7}{4} = 2$

5.) $x = \frac{3}{4 - t^3}$

7.) $x = \sqrt{t^2 + 25}$

9.) $x = \frac{e^t}{e^t - \frac{5}{6}}$

11.) $x = \frac{1 + e^{-t^2}}{2}$

- 13.) $x = -2$ is a source (unstable) and $x = 3$ is a sink (asymptotically stable)

- 15.) $x = 1$ is a source (unstable), $x = -\frac{3}{2}$ is a node (semi-stable) and both $x = -5$ and $x = 4$ are sinks (asymptotically stable)

- 17.) $x = 2n\pi$ for any integer n is a source (unstable) and $x = (2n + 1)\pi$ for any integer n is a sink (asymptotically stable)

- 19.) there will be 10000 bacteria after $\frac{8 \ln 20}{\ln 3}$ hours

- 21.) there will be $20\mu g$ left after $\frac{12 \ln 5}{\ln 2}$ hours

- 23.) the rod will be 100°C after $\frac{5 \ln \frac{87}{7}}{\ln \frac{5}{4}}$ hours

- 25.) the growth rate is $\frac{1000}{3}$, the carrying capacity is 1000 and if the initial

population is 300 then the population will be 600 after $\frac{3 \ln 3.5}{1000}$ time units

27.) $C(20) = 35e^{-20/3} \approx 0.045$ g/L

29.) $C(400) = \frac{25 - \sqrt{5}}{100} \approx 0.228$ lbs/gal

30.) the carrying capacity is $\frac{54}{5}$ and the growth rate is $\ln 7$

F.5 Chapter 5

- 1.) $(1, 1)$ is an asymptotically stable equilibrium and $(-1, -1)$ is an unstable equilibrium.
- 3.) $\left(3, \frac{1}{2}\right)$ is a stable center point equilibrium and $(0, 0)$ is an unstable saddle point equilibrium.
- 5.) Every point $(c, 0)$ is an equilibrium. If $c < 6$ then the equilibrium is asymptotically stable.
- 6.) Stable equilibrium is 500 mice and 30 owls. If initially there are 600 mice and 20 owls then the populations of mice and owls will both start out increasing.
- 7.) Stable equilibrium is 1200 grasshoppers and 60 frogs. If initially there are 1300 grasshoppers and 70 frogs then the population of grasshoppers will start out decreasing and the population of frogs will start out increasing.
- 8.) Largest stable number of susceptible people is $\frac{\beta}{\alpha} = 500$.
- 9.) The equilibrium is 500 susceptible and approximately 186 infected, so the disease is endemic.
- 10.) Here $N < \frac{\beta}{\alpha}$ so the only equilibrium is $(3000, 0)$ In the long run the number of infected people goes to 0 so the disease is not endemic.

- 11.) We have competitive exclusion and in the long run there are 250 of species a and none of species b .
- 12.) We have weak competition with an equilibrium of approximately 57.14 of species a and approximately 107.14 of species b .
- 13.) We have competitive exclusion and in the long run there are 300 of species b and none of species a .
- 14.) We have strong competition and both $(a, b) = (100, 0)$ and $(a, b) = (0, 300)$ are asymptotically stable. Without initial values we cannot predict which species will win out but we do know that in the long run only one will survive.
- 15.) If $N_R < \frac{\gamma}{\delta}$ there are two equilibria: $(0, 0)$, an unstable saddle point, and the asymptotically stable $(N_R, 0)$. If $N_R > \frac{\gamma}{\delta}$ there are three equilibria. $(0, 0)$ and $(N_R, 0)$ are unstable saddle points. The third equilibrium is an asymptotically stable $\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta} \left(1 - \frac{\gamma}{\delta N_R}\right)\right)$. If N_R is not close to $\frac{\gamma}{\delta}$ there will likely be spiral behavior.

F.6 Chapter 6

- 1.) $\mathbb{P}(X = 5) = \frac{1}{3}$, $\mathbb{E}(X) = 3$ and $\sigma(X) = \frac{2\sqrt{7}}{3}$
- 3.) $\mathbb{P}(X = 1) = \frac{16}{31}$, $\mathbb{P}(X = 2) = \frac{8}{31}$, $\mathbb{P}(X = 3) = \frac{4}{31}$, $\mathbb{P}(X = 4) = \frac{2}{31}$,
 $\mathbb{P}(X = 5) = \frac{1}{31}$
 $\mathbb{E}(X) = \frac{57}{31} \approx 1.8387$, $\text{Var}(X) = \frac{1122}{961} \approx 1.1675$ and $\sigma(X) = \frac{\sqrt{1122}}{31} \approx 1.0805$
- 5.) $S = \{3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18\}$, $\mathbb{P}(R = 5) = \frac{1}{36}$
- 7.) $\mathbb{P}(\text{both green}) = \frac{1}{4}$, $\mathbb{P}(\text{one green and one yellow}) = \frac{3}{10}$
- 9.) $\mathbb{P}(R = 5 | \text{at least one } 1) = \frac{2}{11}$, not independent

- 11.) $\mathbb{P}(\text{one ball is yellow}|\text{both balls have odd numbers}) = \frac{2}{5}$, not independent
- 13.) $\mathbb{P}(\text{knew right answer}|\text{gave right answer}) = \frac{70}{76} \approx .9211$
- 15.) $\mathbb{P}(I) = \frac{15}{985} \approx 1.5228\%$
- 17.) binomial, .3352
- 19.) the question asks for the expected number of tails in 12 tosses of a fair coin which is 6
- 21.) $\mathbb{E}(T) = \frac{1}{.45} = \frac{20}{9}$, $\sigma(T) = \frac{\sqrt{.55}}{.45} \approx 1.648$, $\mathbb{P}(T \geq 5) = 1 - \mathbb{P}(T < 5) \approx .0915$
- 23.) $\mathbb{P}(0 \text{ accidents}) \approx .0498$, $\mathbb{P}(1 \text{ accident}) \approx .1494$,
 $\mathbb{P}(5 \text{ accidents}|\text{at least 4 accidents}) \approx .2858$
- 25.) $\mathbb{P}(w) = q = \sqrt{0.36} = 0.6$, $\mathbb{P}(W) = p = 1 - q = 0.4$, $\mathbb{P}(Ww) = 2pq = 0.48$
- 27.) $\mathbb{P}(D) = p = 0.35$, $\mathbb{P}(d) = q = 0.65$, $\mathbb{P}(dd) = q^2 = 0.4225$
- 29.) $\mathbb{P}(AAAA) = p^4$, $\mathbb{P}(AAaA) = 4p^3q$, $\mathbb{P}(AAaa) = 6p^2q^2$, $\mathbb{P}(Aaaa) = 4pq^3$, and $\mathbb{P}(aaaa) = q^4$

F.7 Chapter 7

- 1.) Sample space: $[1, \infty)$, to normalize let $A = 3$, and $\mathbb{P}(2 \leq X \leq 4) = \frac{7}{64}$
- 3.) Sample space: $(0, 1]$, to normalize let $A = \frac{1}{2}$, and $\mathbb{P}\left(\frac{1}{2} \leq R \leq 1\right) = 1 - \frac{1}{\sqrt{2}}$
- 5.) Sample space: $[0, 5]$, to normalize let $A = \frac{6}{125}$, the cdf is $F(y) =$

$$\frac{15y^2 - 2y^3}{125}, \text{ and}$$

$$\mathbb{E}(Y) = \frac{5}{2}, \text{ Var}(Y) = \frac{5}{4} \text{ and } \sigma(Y) = \frac{\sqrt{5}}{2}$$

7.) a.) To normalize, let $k = 20$

b.) $\mathbb{P}\left(0 \leq X \leq \frac{1}{4}\right) = \frac{1}{64}$

c.) $\mathbb{E}(X) = \frac{2}{3}$

d.) $\text{Var}(X) = \frac{2}{63}$

9.) a.) To normalize, let $A = \frac{2}{\pi}$

b.) $\mathbb{P}(0 \leq X \leq 1) = \frac{1}{2}$

c.) The integrals do not converge.

11.) $\mathbb{E}(X) = 4, \text{ Var}(X) = 8 \text{ and } \sigma(X) = 2\sqrt{2}$

13.) a.) $[b, a + b]$

b.) $F(y) = \frac{y - b}{a}$

c.) $F'(y) = \frac{1}{a}$

d.) Uniform

15.) $e^{-\frac{8}{5}}$

17.) $\frac{16}{25}$

19.) 4.01%

21.) For standard normal: $e^{\frac{t^2}{2}}$

For normal with mean μ and standard deviation σ : $e^{t\mu + \frac{t^2\sigma^2}{2}}$

F.8 Chapter 8

1.) Original: $\left[\begin{array}{cc|c} 2 & -1 & 7 \\ 3 & 2 & 0 \end{array} \right]$ Reduced: $\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -3 \end{array} \right]$ Solution: $x = 2, y = -3$

3.) Original: $\left[\begin{array}{ccc|c} 1 & 1 & 3 & 10 \\ 1 & 2 & 1 & 16 \\ 2 & 3 & 4 & 26 \end{array} \right]$ Reduced: $\left[\begin{array}{ccc|c} 1 & 0 & 5 & 4 \\ 0 & 1 & -2 & 6 \\ 0 & 0 & 0 & 0 \end{array} \right]$
 Solution: $x = 4 - 5t, y = 6 + 2t, z = t$

5.) Original: $\left[\begin{array}{ccc|c} 1 & -2 & 3 & 4 \\ 2 & 1 & 1 & 8 \\ 3 & 4 & -5 & 8 \end{array} \right]$ Reduced: $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right]$ Solution: $x = 3, y = 1, z = 1$

7.) Original: $\left[\begin{array}{cccc|c} 1 & 1 & 0 & 5 & 3 \\ 1 & 2 & -1 & 8 & 4 \\ 2 & 3 & -1 & 13 & 7 \\ 1 & 2 & -1 & 8 & 4 \end{array} \right]$ Reduced: $\left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 2 \\ 0 & 1 & -1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$
 Solution: $x = 2 - s - 2t, y = 1 + s - 3t, z = s, w = t$

9.) a.) $\begin{bmatrix} 3 \\ -9 \\ 6 \end{bmatrix}$

b.) $\begin{bmatrix} -2 \\ -1 \\ \frac{7}{2} \end{bmatrix}$

c.) $\begin{bmatrix} 11 \\ -5 \\ -8 \end{bmatrix}$

d.) $\begin{bmatrix} 0 \\ 14 \\ -15 \end{bmatrix}$

e.) $\sqrt{14}, \sqrt{69}$

11.) linearly dependent:

$$5 \cdot \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} + 2 \cdot \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix} - \begin{bmatrix} 1 \\ 18 \\ -5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

13.) a.) Cannot be done.

b.) $\begin{bmatrix} 3 & 1 & -7 \\ -21 & -10 & 6 \end{bmatrix}$

c.) $\begin{bmatrix} -2 & 14 & -13 \\ 1 & -19 & 14 \end{bmatrix}$

d.) $\begin{bmatrix} 9 & -24 \\ -12 & 33 \end{bmatrix}$

e.) $\begin{bmatrix} 3 & -1 & -3 \\ 3 & 7 & 2 \\ -2 & -1 & 3 \end{bmatrix}$

15.) a.) Invertible. $\begin{bmatrix} \frac{3}{8} & -\frac{1}{4} \\ \frac{1}{16} & \frac{1}{8} \end{bmatrix}$

b.) Invertible. $\begin{bmatrix} \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{2}{3} & -1 & \frac{4}{3} \\ 1 & -1 & 1 \end{bmatrix}$

c.) Invertible. $\begin{bmatrix} -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{4}{3} & -\frac{5}{3} & -\frac{1}{3} \\ -\frac{4}{3} & \frac{8}{3} & \frac{1}{3} \end{bmatrix}$

d.) Not invertible.

17.) $\lambda_1 = -1$, associated eigenvectors: $c \begin{bmatrix} 1 \\ 1 \end{bmatrix}$; $\lambda_2 = 2$, associated eigenvectors: $c \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

19.) $\lambda_1 = 3 + i$, associated eigenvectors: $c \begin{bmatrix} 1 + i \\ 1 \end{bmatrix}$;

$$\lambda_2 = 3 - i, \text{ associated eigenvectors: } c \begin{bmatrix} 1 - i \\ 1 \end{bmatrix}.$$

$$21.) \lambda_1 = -4, \text{ associated eigenvectors: } c \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix};$$

$$\lambda_2 = 0, \text{ associated eigenvectors: } c \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix};$$

$$\lambda_3 = 1, \text{ associated eigenvectors: } c \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

$$23.) \lambda_1 = 2, \text{ associated eigenvectors: } c \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix};$$

$$\lambda_2 = i, \text{ associated eigenvectors: } c \begin{bmatrix} 1 + i \\ 0 \\ 1 \end{bmatrix};$$

$$\lambda_3 = -i, \text{ associated eigenvectors: } c \begin{bmatrix} 1 - i \\ 0 \\ 1 \end{bmatrix}.$$

$$25.) \lambda_1 = -2, \text{ associated eigenvectors: } c \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix};$$

$$\lambda_2 = 0, \text{ associated eigenvectors: } c \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix};$$

$$\lambda_3 = \lambda_4 = 1, \text{ associated eigenvectors: } c \begin{bmatrix} 0 \\ 2 \\ 3 \\ 1 \end{bmatrix} + d \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

$$17\text{diag.) } M = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, D = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}, M^{-1} = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}, M^5 = \begin{bmatrix} 65 & -66 \\ 33 & -34 \end{bmatrix}.$$

$$\begin{aligned} 19\text{diag.}) \quad M &= \begin{bmatrix} 1+i & 1-i \\ 1 & 1 \end{bmatrix}, D = \begin{bmatrix} 3+i & 0 \\ 0 & 3-i \end{bmatrix}, M^{-1} = \begin{bmatrix} -\frac{i}{2} & \frac{1}{2} + \frac{i}{2} \\ \frac{i}{2} & \frac{1}{2} - \frac{i}{2} \end{bmatrix}, \\ M^5 &= \begin{bmatrix} 304 & -632 \\ 316 & -328 \end{bmatrix}. \end{aligned}$$

$$\begin{aligned} 21\text{diag.}) \quad M &= \begin{bmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, D = \begin{bmatrix} -4 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, M^{-1} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 1 & 0 & -1 \end{bmatrix}, \\ M^5 &= \begin{bmatrix} 513 & -512 & -513 \\ 1025 & -1024 & -1025 \\ 512 & -512 & -512 \end{bmatrix}. \end{aligned}$$

$$\begin{aligned} 23\text{diag.}) \quad M &= \begin{bmatrix} -1 & 1+i & 1-i \\ -1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}, D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & -i \end{bmatrix}, M^{-1} = \begin{bmatrix} 0 & -1 & 0 \\ -\frac{i}{2} & \frac{1}{2} + i & \frac{1}{2} + \frac{i}{2} \\ \frac{i}{2} & \frac{1}{2} - i & \frac{1}{2} - \frac{i}{2} \end{bmatrix}, \\ M^5 &= \begin{bmatrix} 1 & 29 & -2 \\ 0 & 32 & 0 \\ 1 & -34 & -1 \end{bmatrix}. \end{aligned}$$

$$\begin{aligned} 25\text{diag.}) \quad M &= \begin{bmatrix} 0 & 0 & 0 & 1 \\ -1 & 1 & 2 & 0 \\ 0 & 0 & 3 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}, D = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, M^{-1} = \begin{bmatrix} 0 & -\frac{1}{2} & \frac{1}{6} & \frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{3} & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \\ M^5 &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -16 & 6 & 16 \\ 0 & 0 & 1 & 0 \\ 0 & 16 & -5 & -16 \end{bmatrix}. \end{aligned}$$

F.9 Chapter 9

1.)

$$\begin{aligned} x(t) &= c_1 e^{-3t} + c_2 e^{2t} \\ y(t) &= -c_1 e^{-3t} + 4c_2 e^{2t} \end{aligned}$$

3.)

$$x(t) = -13e^{-3t/2} + 14e^{-t/2}$$

$$y(t) = -13e^{-3t/2} + 10e^{-t/2}$$

5.)

$$x(t) = 2e^{2t} \cos(5t) - 3e^{2t} \sin(5t)$$

$$y(t) = 3e^{2t} \cos(5t) + 2e^{2t} \sin(5t)$$

7.)

$$x(t) = c_2 + c_3 e^{3t}$$

$$y(t) = c_1 e^{-4t} - c_2$$

$$z(t) = c_1 e^{-4t} + c_2 + c_3 e^{3t}$$

9.)

$$x(t) = e^{2t} - \cos(4t) - 3 \sin(4t)$$

$$y(t) = e^{2t} - \cos(4t) + 2 \sin(4t)$$

$$z(t) = e^{2t} + \cos(4t) - 2 \sin(4t)$$

11.)

$$C_A(t) = -\frac{14}{3}e^{-\frac{7t}{50}} + \frac{5}{3}e^{-\frac{2t}{25}} + 3$$

$$C_B(t) = \frac{7}{3}e^{-\frac{7t}{50}} + \frac{5}{3}e^{-\frac{2t}{25}} + 1$$

$$\lim_{t \rightarrow \infty} C_A(t) = 3 \text{ g/L}$$

$$\lim_{t \rightarrow \infty} C_B(t) = 1 \text{ g/L}$$

13.)

$$\lim_{t \rightarrow \infty} C_A(t) = \frac{2}{9} \text{ kg/L}$$

$$\lim_{t \rightarrow \infty} C_B(t) = \frac{1}{9} \text{ kg/L}$$

$$\lim_{t \rightarrow \infty} C_C(t) = \frac{1}{5} \text{ kg/L}$$

15.)

$$\begin{aligned}\begin{bmatrix} x' \\ v' \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ -21 & -10 \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix} + \begin{bmatrix} 0 \\ 25 \sin(t) \end{bmatrix} \\ \begin{bmatrix} x(0) \\ v(0) \end{bmatrix} &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ x(t) &= -\frac{3}{8}e^{-7t} + \frac{7}{8}e^{-3t} + \sin(t) - \frac{1}{2}\cos(t)\end{aligned}$$

	Equilibrium	Classification
	$(-2, -2)$	Asymptotically Stable Node
17.)	$(0, 0)$	Unstable Saddle Point
	$(0, 1)$	Asymptotically Stable Spiral Point
	$(3, -2)$	Unstable Node

	Equilibrium	Classification
	$(-5, -3)$	Unstable Saddle Point
19.)	$(0, 0)$	Asymptotically Stable Node
	$(-5, 1)$	Unstable Saddle Point

21.) a.) See Theorem 5.5.3

b.) Eigenvalues: $-\frac{31}{100}, -\frac{11}{100}, -\frac{1}{100}$ c.) Eigenvalues: $-\frac{23}{50}, -\frac{1}{100}, \frac{19}{100}$

23.) a.)

$$\vec{x}(t) = c_1 \begin{bmatrix} 3 \\ -1 \end{bmatrix} e^{-2t} + c_2 \left(t \begin{bmatrix} 3 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) e^{-2t}$$

b.)

$$\vec{x}(t) = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^t + c_2 \left(t \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right) e^t$$

25.) a.)

$$\vec{x}_p(t) = \begin{bmatrix} 6te^{2t} + e^{-t} - e^{2t} \\ 3te^{2t} + e^{-t} - e^{2t} \end{bmatrix}$$

b.)

$$\vec{x}_p(t) = \begin{bmatrix} 3te^{-2t} + te^{2t} \\ -3te^{-2t} + 3te^{2t} - e^{-2t} + e^{2t} \end{bmatrix}$$

F.10 Chapter 10

1.) a.) $P = \begin{bmatrix} \frac{1}{5} & \frac{2}{5} & 0 & 0 & 0 & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} & \frac{2}{5} & 0 & 0 & 0 \\ 0 & \frac{2}{5} & \frac{1}{5} & \frac{2}{5} & 0 & 0 \\ 0 & 0 & \frac{2}{5} & \frac{1}{5} & \frac{2}{5} & 0 \\ 0 & 0 & 0 & \frac{2}{5} & \frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & 0 & 0 & 0 & \frac{2}{5} & \frac{1}{5} \end{bmatrix}$, all states communicate and all states are recurrent

b.) all states have period 1

c.) $\mathbb{P}(X_4 = 3 | X_0 = 1) = \frac{104}{625}$

3.) a.) $P = \begin{bmatrix} \frac{3}{4} & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{4} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix}$, $\{0, 1, 2, 3, 4\}$ is transient and $\{5, 6, 7, 8\}$ is recurrent

b.) Each state in $\{0, 1, 2, 3, 4\}$ has period 1, and each state in $\{5, 6, 7, 8\}$ has period 2.

c.)

$$\begin{aligned} \mathbb{P}(X_5 \neq 5, 6, 7, 8 | X_0 = 0) &= \frac{2099}{2592} \approx 80.98\% \\ \mathbb{P}(X_{10} \neq 5, 6, 7, 8 | X_0 = 0) &= \frac{318787727}{644972544} \approx 49.43\% \\ \mathbb{P}(X_{20} \neq 5, 6, 7, 8 | X_0 = 0) &\approx 17.16\% \end{aligned}$$

- 5.) a.) 4 room changes
 b.) 17 room changes
 c.) 62 room changes
- 7.) The limiting probability for each state is $1/6$.
- 9.) All states in the original process have period 2. The transition probabilities for the odd nodes (1, 3, 5, and 7) after two steps is

$$P = \begin{bmatrix} 5/16 & 1/16 & 0 & 1/8 \\ 9/16 & 9/16 & 3/8 & 0 \\ 0 & 3/8 & 9/16 & 9/16 \\ 1/8 & 0 & 1/16 & 5/16 \end{bmatrix}.$$

Nodes 3 and 5 both having limiting probabilities of $9/20$ while nodes 1 and 7 have limiting probabilities of $1/20$.

- 11.) $P = \begin{bmatrix} 0 & 1/16 & 0 & 0 & 0 \\ 1 & 3/8 & 1/4 & 0 & 0 \\ 0 & 9/16 & 1/2 & 9/16 & 0 \\ 0 & 0 & 1/4 & 3/8 & 1 \\ 0 & 0 & 0 & 1/16 & 0 \end{bmatrix}$, all states communicate and have period 1, the limiting vector of probabilities is

$$\vec{\pi} = \begin{bmatrix} 1/70 \\ 8/35 \\ 18/35 \\ 8/35 \\ 1/70 \end{bmatrix}$$

- 13.) Pentagons: 36.79%, Hexagons: 36.79%, Heptagons: 18.39%, Octagons: 6.13%, Nonagons: 1.53% (all other shapes combined account for $\approx 0.37\%$ of cells)
- 15.) $p \approx 49.6\%$

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